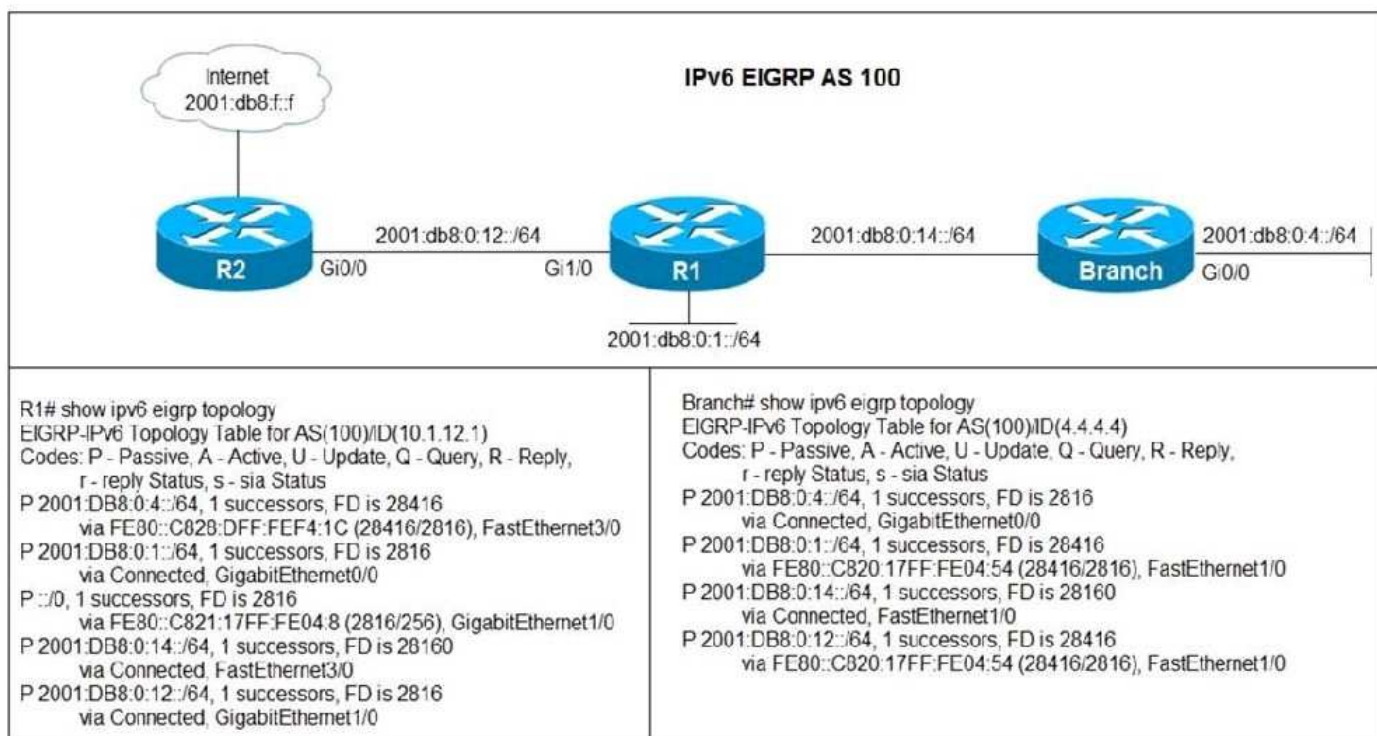


Normal Questions

QUESTION 1

Refer to the exhibit. Users in the branch network of 2001:db8:0:4::/64 report that they cannot access the Internet.

Which command is issued in IPv6 router EIGRP 100 configuration mode to solve this issue?



- A. Issue the eigrp stub command on R1.
- B. Issue the no eigrp stub command on R1.
- C. Issue the eigrp stub command on R2.
- D. Issue the no eigrp stub command on R2.

Correct Answer: B

Section: Layer 3 Technologies Explanation

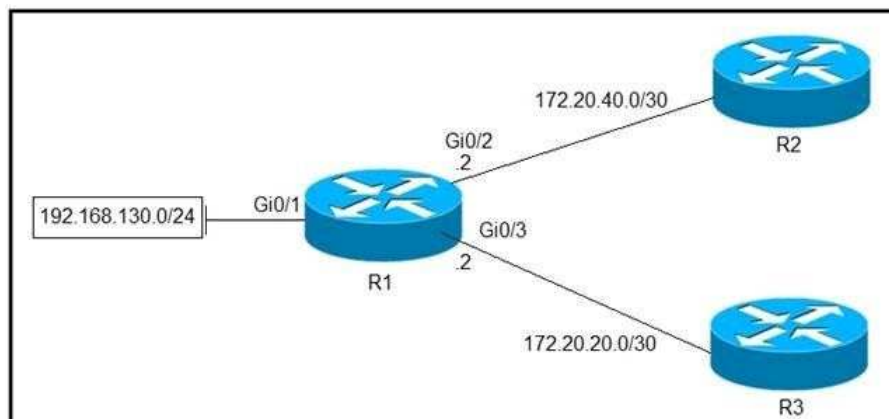
Explanation

Explanation/Reference:

::/0 is the EIGRP route learnt by R1 from R2 (through interface R1's g1/0) but the route cannot be found in Branch router. One of the reason is that R1 is configured as a Stub router and therefore the route does not re-advertise to Branch router. If this is the case, you can solve the problem by removing the Stub settings in R1.

QUESTION 2

Refer to the exhibit. Which configuration configures a policy on R1 to forward any traffic that is sourced from the 192.168.130.0/24 network to R2?



- A. **access-list 1 permit 192.168.130.0 0.0.0.255**
!
interface Gi0/2
ip policy route-map test
!
route-map test permit 10
match ip address 1
set ip next-hop 172.20.20.2
- B. **access-list 1 permit 192.168.130.0 0.0.0.255**
!
interface Gi0/1
ip policy route-map test
!
route-map test permit 10
match ip address 1
set ip next-hop 172.20.40.2
- C. **access-list 1 permit 192.168.130.0 0.0.0.255**
!
interface Gi0/2
ip policy route-map test
!
route-map test permit 10
match ip address 1
set ip next-hop 172.20.20.1
- D. **access-list 1 permit 192.168.130.0 0.0.0.255**
!
interface Gi0/1
ip policy route-map test
!
route-map test permit 10
match ip address 1
set ip next-hop 172.20.40.1

Correct Answer: D

Section: Layer 3 Technologies Explanation
Explanation

Explanation/Reference:

Since the packets are incoming at R1's g0/1, PBR should be configured in this interface.

Next Hop address should be configured as R2. Since the network is 172.20.40.0/30, the only valid IP is 172.20.40.1 and 172.20.40.2. However, as the diagram shows ".2" in R1's g0/2, therefore the only remaining IP for R2 is ".1" i.e. 172.20.40.1. As a result, the next hop configured for PBR should be set ip next-hop 172.20.40.1

QUESTION 3

R2 has a locally originated prefix 192.168.130.0/24 and has these configurations:

```
ip prefix-list test seq 5 permit 192.168.130.0/24
!
route-map OUT permit 10
match ip address prefix-list test
set as-path prepend 65000
```

What is the result when the route-map OUT command is applied toward an eBGP neighbor R1 (1.1.1.1) by using the neighbor 1.1.1.1 route-map OUT out command?

- A. R1 sees 192.168.130.0/24 as two AS hops away instead of one AS hop away.
 B. R1 does not accept any routes other than 192.168.130.0/24
 C. R1 does not forward traffic that is destined for 192.168.30.0/24
 D. Network 192.168.130.0/24 is not allowed in the R1 table

Correct Answer: A

Section: Layer 3 Technologies Explanation
Explanation

Explanation/Reference:

B seems to be partially true since R2 will only send the route 192.168.130.0/24 to R1 if there is no other route map statement in "OUT".

However A is the best answer instead of B since:

- If there is no special configuration in R1, R1 is willing to accept all routes from R2. It is R2 which send the route 192.168.130.0/24 only.
- Moreover R1 may accept any route from other neighbors.

QUESTION 4

Which method changes the forwarding decision that a router makes without first changing the routing table or influencing the IP data plane?

- A. nonbroadcast multiaccess
- B. packet switching
- C. policy-based routing
- D. forwarding information base

Correct Answer: C

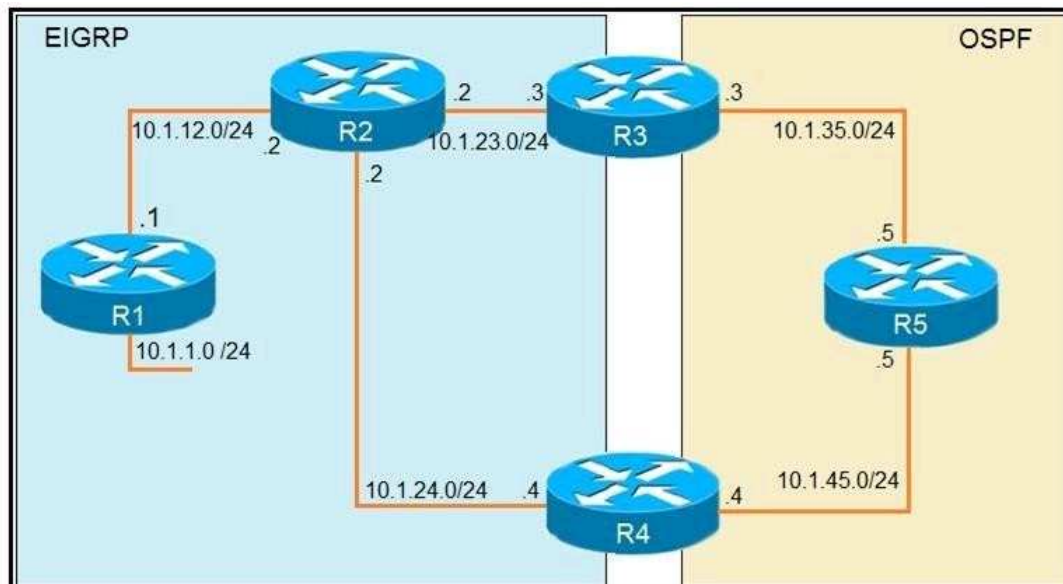
Section: Layer 3 Technologies Explanation

Explanation/Reference:

QUESTION 5

Refer to the exhibit. The output of the trace route from R5 shows a loop in the network.

Which configuration prevents this loop?



```
R1
router eigrp 1
 redistribute connected
 network 10.1.12.1 0.0.0.0
```

```
R3
router ospf 1
 redistribute eigrp 1 subnets
 network 10.1.35.3 0.0.0.0 area 0
```

```
R4
router eigrp 1
 redistribute ospf 1 metric 2000000 1 255 1 1500
!
router ospf 1
 network 10.1.45.4 0.0.0.0 area 0
```

```
R5#traceroute 10.1.1.1
```

```
Type escape sequence to abort.
Tracing the route to 10.1.1.1
```

```
 1 10.1.35.3 80 msec 44 msec 20 msec
 2 10.1.23.2 44 msec 104 msec 64 msec
 3 10.1.24.4 44 msec 64 msec 40 msec
 4 10.1.45.5 24 msec 40 msec 20 msec
 5 10.1.35.3 92 msec 144 msec 148 msec
 6 10.1.23.2 108 msec 76 msec 80 msec
 <output truncated>
```


- A. R3
router ospf 1
redistribute eigrp 1 subnets route-map SET-TAG
!
route-map SET-TAG permit 10
set tag 1
- R4
router eigrp 1
redistribute ospf 1 metric 2000000 1 255 1 1500 route-map FILTER-TAG
!
route-map FILTER-TAG deny 10
match tag 1
!
route-map FILTER-TAG permit 20
- B. R3
router eigrp 1
redistribute OSPF 1 route-map SET-TAG
!
route-map SET-TAG permit 10
set tag 1
- R4
router eigrp 1
redistribute ospf 1 metric 2000000 1 255 1 1500 route-map FILTER-TAG
network 10.1.24.4 0.0.0.0
!
route-map FILTER-TAG deny 10
match tag 1
!
route-map FILTER-TAG permit 20
- C. R3
router ospf 1
redistribute eigrp 1 subnets route-map SET-TAG
!
route-map SET-TAG permit 10
set tag 1
- R4
router eigrp 1
redistribute ospf 1 metric 2000000 1 255 1 1500 route-map FILTER-TAG
!
route-map FILTER-TAG permit 10
match tag 1
- D. R3
router ospf 1
redistribute eigrp 1 subnets route-map SET-TAG
!
route-map SET-TAG deny 10
set tag 1
- R4
router eigrp 1
redistribute ospf 1 metric 2000000 1 255 1 1500 route-map FILTER-TAG
!
route-map FILTER-TAG deny 10
match tag 1

Explanation

Explanation/Reference:

In this case, you need to tag the route redistributed from EIGRP to OSPF in R3 so that tagged route should not be redistributed back in R4.
B is wrong since both routers R3 and R4 are redistributing OSPF into EIGRP.
C is wrong since it permits the tagged route to be redistributed from OSPF to EIGRP in R4 and implicitly deny all other routes.
D is wrong since it denies the tagged route and implicitly deny all other routes.

QUESTION 6

Refer to the exhibit. An engineer configures a static route on a router, but when the engineer checks the route to the destination, a different next hop is chosen.

What is the reason for this?

```
Router#show running-config | include ip route
ip route 192.168.2.2 255.255.255.255 209.165.200.225 130
Router#show ip route

<output omitted>

Gateway of last resort is not set

    192.168.1.0/32 is subnetted, 1 subnets
C       192.168.1.1 is directly connected, Loopback0
    192.168.2.0/32 is subnetted, 1 subnets
O       192.168.2.2[110/11] via 192.168.12.2, 00:52:09, Ethernet0/0
    192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, Ethernet0/0
L       192.168.12.1/32 is directly connected, Ethernet0/0
    209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.0/24 is directly connected, Ethernet0/1
        209.165.200.226/32 is directly connected, Ethernet0/1
```

- A. Dynamic routing protocols always have priority over static routes.
- B. The metric of the OSPF route is lower than the metric of the static route.
- C. The configured AD for the static route is higher than the AD of OSPF.
- D. The syntax of the static route is not valid, so the route is not considered.

Correct Answer: C

Section: Layer 3 Technologies Explanation

Explanation/Reference:

The configured static route has an AD of 130 which is higher than the AD 110 of OSPF.

QUESTION 7

Refer to the exhibit. An engineer is trying to generate a summary route in OSPF for network 10.0.0.0/8, but the summary route does not show up in the routing table.

Why is the summary route missing?

```

Router#show ip route
<output omitted>
Gateway of last resort is not set

    192.168.1.0/32 is subnetted, 1 subnets
O       192.168.1.1 [110/11] via 192.168.12.1, 16:56:40, Ethernet0/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, Loopback0
L       192.168.2.2/32 is directly connected, Loopback0
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, Ethernet0/1
L       192.168.3.1/32 is directly connected, Ethernet0/1
    192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, Ethernet0/0
L       192.168.12.2/32 is directly connected, Ethernet0/0
Router#show running-config | section ospf
router ospf 1
 summary-address 10.0.0.0 255.0.0.0
 redistribute static subnets
 network 192.168.3.0 0.0.0.255 area 0
 network 192.168.12.0 0.0.0.255 area 0
Router#

```

- A. The summary-address command is used only for summarizing prefixes between areas.
- B. The summary route is visible only in the OSPF database, not in the routing table.
- C. There is no route for a subnet inside 10.0.0.0/8, so the summary route is not generated.
- D. The summary route is not visible on this router, but it is visible on other OSPF routers in the same area.

Correct Answer: C

Section: Layer 3 Technologies Explanation
Explanation

Explanation/Reference:

summary-address is for summarizing imported route.

Although there is a command "redistribute static subnets", there are no static route covered by "10.0.0.0/8". Therefore no summarized external route will be created.

QUESTION 8

Refer to the exhibit. An engineer is trying to block the route to 192.168.2.2 from the routing table by using the configuration that is shown. The route is still present in the routing table as an OSPF route.

Which action blocks the route?

```

Router#show access-lists
Standard IP access list 1
    10 permit 192.168.2.2 (1 match)
Router#
Router#show route-map
route-map RM-OSPF-DL, permit, sequence 10
Match clauses:
    ip address (access-lists): 1
Set clauses:
Policy routing matches: 0 packets, 0 bytes
Router#
Router#show running-config | section ospf
router ospf 1
 network 192.168.1.1 0.0.0.0 area 0
 network 192.168.12.0 0.0.0.255 area 0
 distribute-list route-map RM-OSPF-DL in
Router#

```

- A. Use an extended access list instead of a standard access list.
- B. Change sequence 10 in the route-map command from permit to deny.
- C. Use a prefix list instead of an access list in the route map.
- D. Add this statement to the route map: route-map RM-OSPF-DL deny 20.

Correct Answer: B

Section: Layer 3 Technologies Explanation

Explanation

Explanation/Reference:

For A & C, other than changing to extended access list or prefix list, you also need to change the rules to deny the route to 192.168.2.2
For D, the route to 192.168.2.2 is already permitted by seq 10 and therefore adding seq 20 is not helping.

B is the only way to block the route to 192.168.2.2. However this will actually block all OSPF routes from entering into the routing table.

QUESTION 9

What is a prerequisite for configuring BFD?

- A. Jumbo frame support must be configured on the router that is using BFD.
- B. All routers in the path between two BFD endpoints must have BFD enabled.
- C. Cisco Express Forwarding must be enabled on all participating BFD endpoints.
- D. To use BFD with BGP, the timers 3 9 command must first be configured in the BGP routing process.

Correct Answer: C

Section: Layer 3 Technologies Explanation

Explanation

Explanation/Reference:

QUESTION 10

Drag and drop the OSPF adjacency states from the left onto the correct descriptions on the right.

Select and Place:

Init	Each router compares the DBD packets that were received from the other router.
2-way	Routers exchange information with other routers in the multiaccess network.
Down	The neighboring router requests the other routers to send missing entries.
Exchange	The network has already elected a DR and a backup BDR.
ExStart	The OSPF router ID of the receiving router was not contained in the hello message.
Loading	No hellos have been received from a neighbor router.

Correct Answer:

	Exchange
	ExStart
	Loading
	2-way
	Init
	Down

Section: (none)

Explanation

Explanation/Reference:

QUESTION 11

Refer to the exhibit. R2 is a route reflector, and R1 and R3 are route reflector clients. The route reflector learns the route to 172.16.25.0/24 from R1, but it does not advertise to R3.

What is the reason the route is not advertised?

R1 #show ip bgp summary

BGP router identifier 192.168.1.1, local AS number 65000

<output omitted>

Neighbor	V	AS	MsgRcvd	MsgSent	Tblver	InQ	OutQ	Up/Down	State/PfxRcd
192.168.2.2	4	65000	28	28	22	0	0	00:21:31	0

R1#show ip bgp

BGP table version is 22, local router ID is 192.168.1.1

Status codes: s suppressed, d damped, h history, * valid, > best, i – internal,
r RIB-failure, s stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, C RIB-compressed,

Origin codes: i – IGP, e – EGP, ? – incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	172.16.25.0/24	209.165.200.225	0		32768	?

R1#

R2 #show ip bgp summary

BGP router identifier 192.168.2.2, local AS number 65000

<output omitted>

Neighbor	V	AS	MsgRcvd	MsgSent	Tblver	InQ	OutQ	Up/Down	State/PfxRcd
192.168.1.1	4	65000	29	28	3	0	0	00:22:07	1
192.168.3.3	4	65000	7	8	3	0	0	00:02:55	0

R2#show ip bgp

BGP table version is 3, local router ID is 192.168.2.2

Status codes: s suppressed, d damped, h history, * valid, > best, i – internal,
r RIB-failure, s stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, C RIB-compressed,

Origin codes: i – IGP, e – EGP, ? – incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
* i	172.16.25.0/24	209.165.200.225	0	100	0	?

R2#

R3 #show ip bgp summary

BGP router identifier 192.168.3.3, local AS number 65000

BGP table version is 4, main routing table version 4

Neighbor	V	AS	MsgRcvd	MsgSent	Tblver	InQ	OutQ	Up/Down	State/PfxRcd
192.168.2.2	4	65000	8	7	4	0	0	00:03:08	0

R3#

- A. R2 does not have a route to the next hop, so R2 does not advertise the prefix to other clients.
- B. Route reflector setup requires full IBGP mesh between the routers.
- C. In route reflector setup, only classful prefixes are advertised to other clients.
- D. In route reflector setups, prefixes are not advertised from one client to another.

Correct Answer: A

Section: Layer 3 Technologies Explanation

Explanation

Explanation/Reference:

In the RR (i.e. R2), the route 172.16.25.0/24 is not selected as best (not marked with ">"). This is probably due to inaccessible next hop. Since RR does not have any BGP route selected as best for the network prefix, it is not advertised to other client.

QUESTION 12

```

Router#sh ip route ospf
<output omitted>
Gateway is last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
o   E2    10.0.0.0 [110/20] via 192.168.12.2, 00:00:10, Ethernet0/0
o   192.168.3.0/24 [110/20] via 192.168.12.2, 00:00:50, Ethernet0/0
Router#

Router#show ip bgp
<output omitted>
      Network          Next Hop      Metric      LocPrf      Weight      Path
>*   192.168.1.1/32      0.0.0.0          0              32768        ?
>*   192.168.3.0         192.168.12.2     20             32768        ?
>*   192.168.12.0        0.0.0.0          0              32768        ?

Router#show running-config | section router bgp
router bgp 65000
  bgp log-neighbor-changes
  redistribute ospf 1
Router#

```

Refer to the exhibit. An engineer is trying to redistribute OSPF to BGP, but not all of the routes are redistributed.

What is the reason for this issue?

- A. By default, only internal routes and external type 1 routes are redistributed into BGP
- B. Only classful networks are redistributed from OSPF to BGP
- C. BGP convergence is slow, so the route will eventually be present in the BGP table
- D. By default, only internal OSPF routes are redistributed into BGP

Correct Answer: D

Section: Layer 3 Technologies

Explanation

Explanation/Reference:

B is wrong since "subnet" keyword is only needed when redistributing into OSPF i.e. redistributing under OSPF router configuration.

QUESTION 13

```

R200#show ip bgp summary
BGP router identifier 10.1.1.1, local AS number 65000
BGP table version is 26, main routing table version 26
1 network entries using 132 bytes of memory
1 path entries using 52 bytes of memory
2/1 BGP path/bestpath attribute entries using 296 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 2) using 28 bytes of memory
BGP using 508 total bytes of memory
BGP activity 24/23 prefixes, 24/23 paths, scan interval 60 secs
Neighbor    V    AS MsgRcvd MsgSent   TblVer  InQ  OutQ  Up/Down  State/PfxRcd
192.0.2.2    4 65100 20335    20329     0  0    0 00:02:04  Idle (PfxCt)
R200#

```

Refer to the exhibit. In which circumstance does the BGP neighbor remain in the idle condition?

- A. if prefixes are not received from the BGP peer
- B. if prefixes reach the maximum limit
- C. if a prefix list is applied on the inbound direction
- D. if prefixes exceed the maximum limit

Correct Answer: D

Section: Layer 3 Technologies

Explanation

Explanation/Reference:

Idle (PfxCt) means the session is in the Idle state because the neighbor has sent more prefixes than the configured maximum-prefixes limit.

QUESTION 14

Which attribute eliminates LFAs that belong to protected paths in situations where links in a network are connected through a common fiber?

- A. shared risk link group-disjoint
- B. linecard-disjoint
- C. lowest-repair-path-metric
- D. interface-disjoint

Correct Answer: A

Section: Layer 3 Technologies **Explanation**

Explanation/Reference:

QUESTION 15

```
* Jun 28 14:41:57: %BGP-5-ADJCHANGE: neighbor 192.168.2.2 Down User reset
* Jun 28 14:41:57: %BGP_SESSION-5-ADJCHANGE: neighbor 192.168.2.2 IPv4 Unicast
topology base removed from session User reset
* Jun 28 14:41:57: %BGP-5-ADJCHANGE: neighbor 192.168.2.2 Up
R1#show clock
*15:42:00.506 CET Fri Jun 28 2019
```

Refer to the exhibit. An engineer is troubleshooting BGP on a device but discovers that the clock on the device does not correspond to the time stamp of the log entries.

Which action ensures consistency between the two times?

- A. Configure the service timestamps log uptime command in global configuration mode.
- B. Configure the logging clock synchronize command in global configuration mode.
- C. Configure the service timestamps log datetime localtime command in global configuration mode.
- D. Make sure that the clock on the device is synchronized with an NTP server.

Correct Answer: C

Section: Layer 3 Technologies **Explanation**

Explanation/Reference:

QUESTION 16

Refer to the exhibit. What is the result of applying this configuration?

```
R1#show policy-map control-plane
Control Plane
  Service-policy input: CoPP-BGP
    Class-map: BGP (match all)
      2716 packets, 172071 bytes
      5 minute offered rate 0000 bps, drop rate 0000 bps
      Match: access-group name BGP
      drop

    Class-map: class-default (match-any)
      5212 packets, 655966 bytes
      5 minute offered rate 0000 bps, drop rate 0000 bps
      Match: any
```

- A. The router can form BGP neighborships with any other device.
- B. The router cannot form BGP neighborships with any other device.
- C. The router cannot form BGP neighborships with any device that is matched by the access list named "BGP".
- D. The router can form BGP neighborships with any device that is matched by the access list named "BGP".

Correct Answer: C

Section: Layer 3 Technologies **Explanation**

Explanation/Reference:

C is the answer since any packets matching the class "BGP" i.e. the ACL "BGP" will be dropped.

QUESTION 17

Which command displays the IP routing table information that is associated with VRF-Lite?

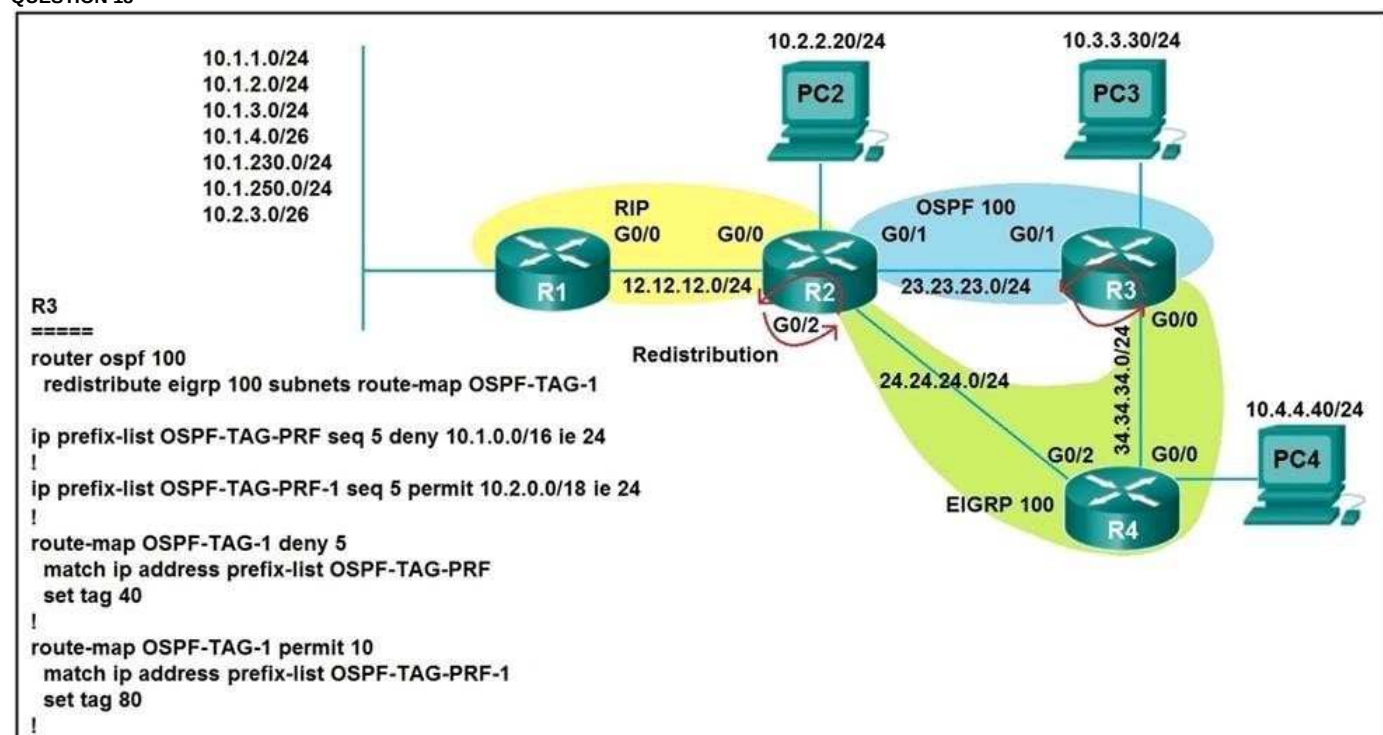
- A. show ip vrf
- B. show ip route vrf
- C. show run vrf
- D. show ip protocols vrf

Correct Answer: B

Section: Layer 3 Technologies Explanation
Explanation

Explanation/Reference:

QUESTION 18



Refer to the exhibit. Which subnet is redistributed from EIGRP to OSPF routing protocols?

- A. 10.2.2.0/24
- B. 10.1.4.0/26
- C. 10.1.2.0/24
- D. 10.2.3.0/26

Correct Answer: A

Section: Layer 3 Technologies Explanation
Explanation

Explanation/Reference:

- i. Routes matching Prefix List OSPF-TAG-PRF is denied by Route Map i.e. not redistributed.
- ii. Routes match Prefix List OSPF-TAG-PRF-1 is permitted by Route Map i.e. redistributed. 10.2.0.0/18 le 24 means the route must start with the first 18 bits matching "10.2.0.0" (i.e. with first three octets "10.2.0" - "10.2.0.63") and have prefix length in the range of 18 - 24.
- iii. Routes not matching any of the above will be denied implicitly.

By just checking the choices, only A satisfy the requirement.

For the working:

- Connected networks of R1 and R2 will be redistributed from RIP into EIGRP.
- The redistributed EIGRP routes + connected network of R4 will be redistributed from EIGRP into OSPF if "ii" is met.

All connected networks of R1 will meet either i or iii (i.e. denied)

The connected network of R2 (192.168.2.2.0/24) meets ii (i.e. permitted.)

The connected network of R4 (10.4.4.0/24) meets iii (i.e. denied)

QUESTION 19

Which configuration adds an IPv4 interface to an OSPFv3 process in OSPFv3 address family configuration?

- A. router ospfv3 1 address-family ipv4
- B. Router(config-router)#ospfv3 1 ipv4 area 0
- C. Router(config-if)#ospfv3 1 ipv4 area 0
- D. router ospfv3 1 address-family ipv4 unicast

Correct Answer: C

Section: Layer 3 Technologies Explanation
Explanation

Explanation/Reference:

QUESTION 20


```

R1(config)#route-map ADD permit 20
R1(config-route-map)#set tag 1

R1(config)#router ospf1
R1(config-router)#redistribute rip subnets route-map ADD

```

Refer to the exhibit. Which statement about R1 is true?

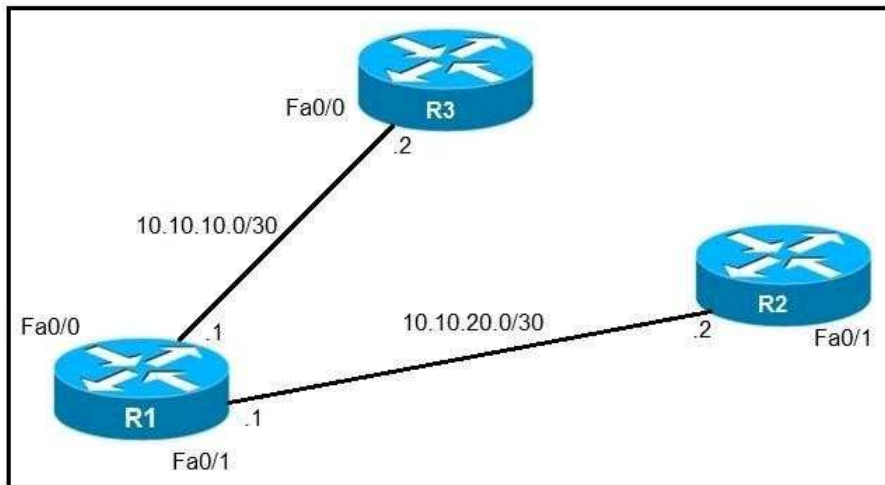
- A. OSPF redistributes RIP routes only if they have a tag of one.
- B. RIP learned routes are distributed to OSPF with a tag value of one.
- C. R1 adds one to the metric for RIP learned routes before redistributing to OSPF.
- D. RIP routes are redistributed to OSPF without any changes.

Correct Answer: B

Section: Layer 3 Technologies Explanation
Explanation

Explanation/Reference:

QUESTION 21



Refer to the exhibit. An IP SLA was configured on router R1 that allows the default route to be modified in the event that Fa0/0 loses reachability with the router R3. The route has changed to flow through router R2.

Which debug command is used to troubleshoot this issue?

- A. debug ip flow
- B. debug ip sla error
- C. debug ip routing
- D. debug ip packet

Correct Answer: B

Section: Layer 3 Technologies Explanation
Explanation

Explanation/Reference:

To examine the change of the static routing entry in the routing table, you can use debug ip routing. However since the change has been made, you can use "debug ip sla error" to find out the reason why IP SLA becomes down e.g. the IP SLA destination does not respond to control message.

QUESTION 22

Which configuration enabled the VRF that is labeled "Inet" on FastEthernet0/0?

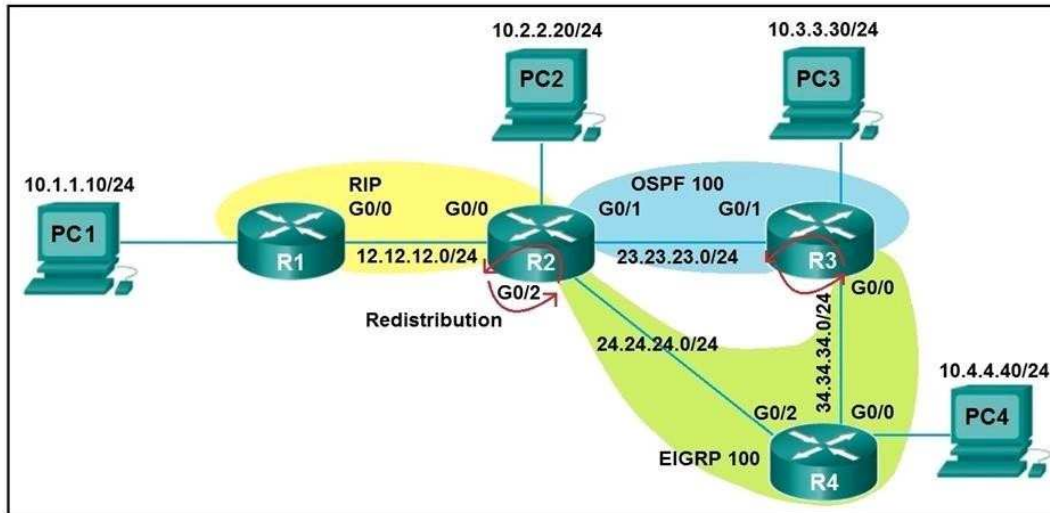
- A. R1(config)# ip vrf Inet
R1(config-vrf)#ip vrf FastEthernet0/0
- B. R1(config)#ip vrf Inet FastEthernet0/0
- C. R1(config)# ip vrf Inet
R1(config-vrf)#interface FastEthernet0/0
R1(config-if)#ip vrf forwarding Inet
- D. R1(config)#router ospf 1 vrf Inet
R1(config-router)#ip vrf forwarding FastEthernet0/0

Correct Answer: C

Section: Layer 3 Technologies Explanation
Explanation

Explanation/Reference:

QUESTION 23



Refer to the exhibit. After redistribution is enabled between the routing protocols; PC2, PC3, and PC4 cannot reach PC1.

Which action can the engineer take to solve the issue so that all the PCs are reachable?

- A. Set the administrative distance 100 under the RIP process on R2.
- B. Filter the prefix 10.1.1.0/24 when redistributed from OSPF to EIGRP.
- C. Filter the prefix 10.1.1.0/24 when redistributed from RIP to EIGRP.
- D. Redistribute the directly connected interfaces on R2.

Correct Answer: A

Section: Layer 3 Technologies Explanation
Explanation

Explanation/Reference:

In the diagram, it seems that:

- R2 is performing mutual redistribution between RIP and EIGRP.
- R3 is performing mutual redistribution between OSPF and EIGRP.

If all PC2, PC3 and PC4 connecting to different routers cannot reach PC1, the most probable reason is that the RIP route for 10.1.1.0/24 learnt from R1 is no longer entered into R2's routing table due to:

- i. R2 redistribute the RIP route for the network into EIGRP.
- ii. R5 redistribute the EIGRP route for the network into OSPF.
- iii. R2 learns the OSPF external route for the network and use it instead of RIP route due to lower AD.
- iv. No more RIP route to 10.1.1.0/24 in R2 and the route will not be redistributed into EIGRP and then OSPF.
- v. R2 re-enter the RIP route to 10.1.1.0/24 in the routing table.
- vi. The above is repeated.

1. The recommended way is to set a higher AD value for OSPF external route.

2. Another way (usually not recommended) but can also work is to set a lower AD for RIP (e.g. a value less than 110) so that RIP route is preferred than the redistributed OSPF route for 10.1.1.0/24.

Since there is no choice for the 1st method, A (i.e. the 2nd method) is therefore the answer.

B is wrong since the route 10.1.1.0/24 is redistributing from EIGRP into OSPF by a single router R3. Hence the route will not be redistributed back from OSPF into EIGRP by default i.e. no need to filter.

C is wrong since if the route is not redistributed from RIP to EIGRP, R3 and R4 will not have the route. In this case, PC3 and PC4 will not be able to access PC1.

QUESTION 24

```

router bgp 100
!
neighbor 10.222.1.1 route-map SET-WEIGHT in
neighbor 10.222.1.1 remote-as 1
!
ip as-path access-list 200 permit ^690$
ip as-path access-list 200 permit ^1800
!
route-map SET-WEIGHT permit 10
match as-path 200
set local-preference 250
set weight 200

```

Refer to the exhibit. A router is receiving BGP routing updates from multiple neighbors for routes in AS 690.

What is the reason that the router still sends traffic that is destined to AS 690 to a neighbor other than 10.222.1.1?

- A. The local preference value in another neighbor statement is higher than 250.
- B. The local preference value should be set to the same value as the weight in the route map.
- C. The route map is applied in the wrong direction.
- D. The weight value in another neighbor statement is higher than 200.

Correct Answer: D

Section: Layer 3 Technologies Explanation
Explanation

Explanation/Reference:

For the fact "The router still sends traffic that is destined to AS 690 to a neighbor other than 10.222.1.1".
You want the route to networks in AS 690 to use the BGP path with 10.222.1.1 as next hop, but actually it is NOT.

Although the weight set i.e. 200 is the first factor to evaluate for BGP path selection, since the router has multiple neighbors, another neighbor of this router may be configured as follows:

```
neighbor 10.222.3.3 remote-as 1
```

```
neighbor 10.222.3.3 weight 300
```

Then the BGP path with next hop 10.222.3.3 will be selected as best instead.

Remarks:

Actually the AS path list 200 shown in the question is wrong or incomplete.

The current router of AS 100 is forming eBGP neighbor with the router 10.222.1.1 of AS 1.

Hence all BGP routes learnt from 10.222.1.1 should start with the AS number "1".

Therefore the two rules in the AS path list 200 should not match any BGP route received from "10.222.1.1" and no route can be learnt from 10.222.1.1. Hence it will not be used as next hop for any routes.

As a result, a better answer (not available in the choices) is the AS Path List is NOT configured properly.

QUESTION 25

Which command allows traffic to load-balance in an MPLS Layer 3 VPN configuration?

- A. multi-paths eibgp 2
- B. maximum-paths 2
- C. maximum-paths ibgp 2
- D. multi-paths 2

Correct Answer: C

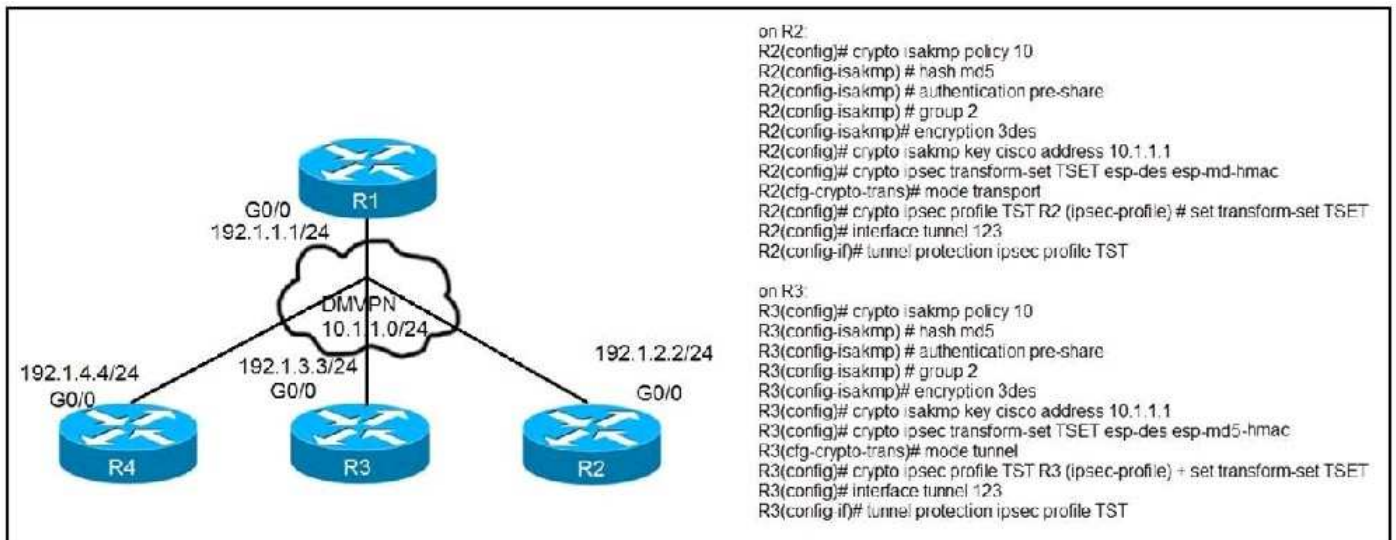
Section: VPN Technologies Explanation
Explanation

Explanation/Reference:

QUESTION 26

Refer to the exhibit.

After applying IPsec, the engineer observed that the DMVPN tunnel went down, and both spoke-to-spoke and hub were not establishing.
Which two actions resolve the issue? (Choose two.)



- A. Change the mode from mode tunnel to mode transport on R3.
- B. Remove the crypto isakmp key cisco address 10.1.1.1 on R2 and R3.
- C. Configure the crypto isakmp key cisco address 192.1.1.1 on R2 and R3.
- D. Configure the crypto isakmp key cisco address 0.0.0.0 on R2 and R3.
- E. Change the mode from mode transport to mode tunnel on R2.

Correct Answer: AD

Section: VPN Technologies Explanation
Explanation

Explanation/Reference:

D instead of C is required for spoke-to-spoke tunnel.

QUESTION 27

Which statement about route distinguishers in an MPLS network is true?

- A. Route distinguishers allow multiple instances of a routing table to coexist within the edge router.
- B. Route distinguishers are used for label bindings.
- C. Route distinguishers make a unique VPNv4 address across the MPLS network.
- D. Route distinguishers define which prefixes are imported and exported on the edge router.

Correct Answer: C

Section: VPN Technologies

Explanation

Explanation/Reference:

QUESTION 28

Which statement about MPLS LDP router ID is true?

- A. If not configured, the operational physical interface is chosen as the router ID even if a loopback is configured.
- B. The loopback with the highest IP address is selected as the router ID.
- C. The MPLS LDP router ID must match the IGP router ID.
- D. The force keyword changes the router ID to the specified address without causing any impact.

Correct Answer: B

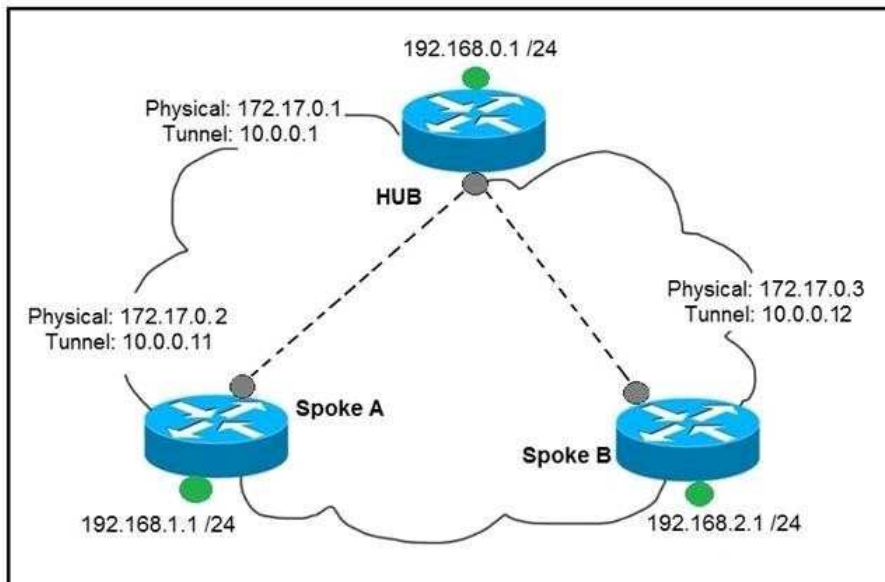
Section: VPN Technologies Explanation

Explanation

Explanation/Reference:

QUESTION 29

Refer to the exhibit. Which interface configuration must be configured on the spoke A router to enable a dynamic DMVPN tunnel with the spoke B router?



- A. **interface Tunnel0**
description mGRE – DMVPN Tunnel
ip address 10.0.0.11 255.255.255.0
ip nhrp map multicast dynamic
ip nhrp network-id 1
tunnel source 10.0.0.1
tunnel destination FastEthernet 0/0
tunnel mode gre multipoint
- B. **interface Tunnel0**
ip address 10.0.0.11 255.255.255.0
ip nhrp network-id 1
tunnel source FastEthernet 0/0
tunnel mode gre multipoint
ip nhrp nhs 10.0.0.1
ip nhrp map 10.0.0.1 172.17.0.1

- C. **interface Tunnel0**
ip address 10.1.0.11 255.255.255.0
ip nhrp network-id 1
tunnel source 1.1.1.10
ip nhrp map 10.0.0.11 172.17.0.2
tunnel mode gre
- D. **interface Tunnel0**
ip address 10.0.0.11 255.255.255.0
ip nhrp map multicast static
ip nhrp network-id 1
tunnel source 10.0.0.1
tunnel mode gre multipoint

Correct Answer: B

Section: VPN Technologies Explanation
Explanation

Explanation/Reference:

The configuration required in a Spoke are:

- GRE multipoint mode
- no tunnel desintation
- NHS Protocol and NBMA mapping

QUESTION 30

Which list defines the contents of an MPLS label?

- A. 20-bit label; 3-bit traffic class; 1-bit bottom stack; 8-bit TTL
- B. 32-bit label; 3-bit traffic class; 1-bit bottom stack; 8-bit TTL
- C. 20-bit label; 3-bit flow label; 1-bit bottom stack; 8-bit hop limit
- D. 32-bit label; 3-bit flow label; 1-bit bottom stack; 8-bit hop limit

Correct Answer: A

Section: VPN Technologies Explanation
Explanation

Explanation/Reference:

QUESTION 31

Refer to the exhibit. What does the imp-null tag represent in the MPLS VPN cloud?

```
Router# show tag-switching tdp bindings
(...)
tib entry: 10.10.10.1/32, rev 31
    local binding: tag: 18
    remote binding: tsr: 10.10.10.1:0, tag: imp-null
    remote binding: tsr: 10.10.10.2:0, tag: 18
    remote binding: tsr: 10.10.10.6:0, tag: 21
tib entry: 10.10.10.2/32, rev 22
    local binding: tag: 17
    remote binding: tsr: 10.10.10.2:0, tag: imp-null
    remote binding: tsr: 10.10.10.1:0, tag: 19
    remote binding: tsr: 10.10.10.6:0, tag: 22
```

- A. Pop the label
- B. Impose the label
- C. Include the EXP bit
- D. Exclude the EXP bit

Correct Answer: A

Section: VPN Technologies Explanation
Explanation

Explanation/Reference:

QUESTION 32

Drag and drop the MPLS VPN concepts from the left onto the correct descriptions on the right.

Select and Place:

route distinguisher	propagates VPN reachability information
route target	distributes labels for traffic engineering
Resource Reservation Protocol	uniquely identifies a customer prefix
multiprotocol BGP	controls the import/export of customer prefixes

Correct Answer:

	multiprotocol BGP
	Resource Reservation Protocol
	route distinguisher
	route target

Section: (none)

Explanation

Explanation/Reference:

QUESTION 33

Which transport layer protocol is used to form LDP sessions?

- A. UDP
- B. SCTP
- C. TCP
- D. RDP

Correct Answer: C

Section: VPN Technologies Explanation

Explanation

Explanation/Reference:

Both LDP and TDP uses TCP to form sessions. However their Hello packets use UDP.

QUESTION 34

Drag and drop the MPLS terms from the left onto the correct definitions on the right.

Select and Place:

PE	device that forwards traffic based on labels
P	path that the labeled packet takes
CE	device that is unaware of MPLS labeling
LSP	device that removes and adds the MPLS labeling

Correct Answer:

P
LSP
CE
PE

Section: (none)
Explanation

Explanation/Reference:

QUESTION 35

On R1:

```
R1(config)# interface tunnel 1
R1(config-if)# ip address 10.1.1.1 255.255.255.0
R1(config-if)# tunnel source 192.1.1.1
R1(config-if)# tunnel mode gre multipoint
R1(config-if)# ip nhrp network-id 111
```

On R2:

```
R2(config)# interface tunnel 1
R2(config-if)# ip address 10.1.1.2 255.255.255.0
R2(config-if)# tunnel source FastEthernet0/0
R2(config-if)# tunnel mode gre multipoint
R2(config-if)# ip nhrp network-id 222
R2(config-if)# ip nhrp nhs 10.1.1.1
R2(config-if)# ip nhrp map 10.1.1.1 192.1.1.1
```

On R3:

```
R3(config)# interface tunnel 1
R3(config-if)# ip address 10.1.1.3 255.255.255.0
R3(config-if)# tunnel source FastEthernet0/0
R3(config-if)# tunnel mode gre multipoint
R3(config-if)# ip nhrp network-id 333 R3(config-if)# ip nhrp nhs 10.1.1.1
R3(config-if)# ip nhrp map 10.1.1.1 192.1.1.1
```

On R4:

```
R4(config)# interface tunnel 1
R4(config-if)# ip address 10.1.1.4 255.255.255.0
R4(config-if)# tunnel source FastEthernet0/0
R4(config-if)# tunnel mode gre multipoint
R4(config-if)# ip nhrp network-id 444
R4(config-if)# ip nhrp nhs 10.1.1.1
R4(config-if)# ip nhrp map 10.1.1.1 192.1.1.1
```

Refer to the exhibits. Phase-3 tunnels cannot be established between spoke-to-spoke in DMVPN.

Which two commands are missing? (Choose two.)

- A. The ip nhrp redirect command is missing on the spoke routers.
- B. The ip nhrp shortcut command is missing on the spoke routers.
- C. The ip nhrp redirect command is missing on the hub router.
- D. The ip nhrp shortcut command is missing on the hub router.
- E. The ip nhrp map command is missing on the hub router.

Correct Answer: BC

Section: VPN Technologies Explanation

Explanation

Explanation/Reference:

QUESTION 36

Which protocol is used to determine the NBMA address on the other end of a tunnel when mGRE is used?

- A. NHRP
- B. IPsec
- C. MP-BGP
- D. OSPF

Correct Answer: A

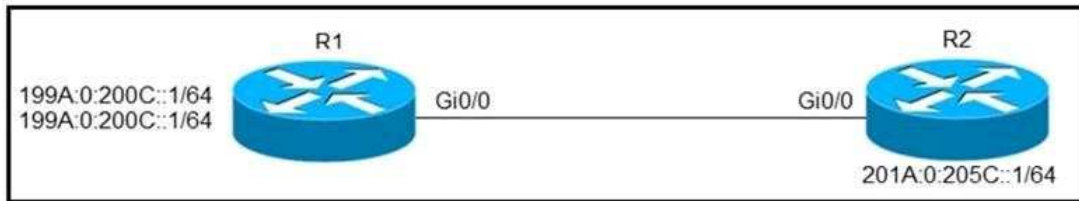
Section: VPN Technologies Explanation

Explanation

Explanation/Reference:

QUESTION 37

Refer to the exhibit. Which configuration denies Telnet traffic to router 2 from 198A:0:200C::1/64?



- A. `ipv6 access-list Deny_Telnet sequence 10 deny tcp host 198A:0:200C::1/64 host 201A:0:205C::1/64 eq telnet`
`!`
`int Gi0/0`
`ipv6 traffic-filter Deny_Telnet in`
`!`
- B. `ipv6 access-list Deny_Telnet sequence 10 deny tcp host 198A:0:200C::1/64 host 201A:0:205C::1/64 eq telnet`
`!`
`int Gi0/0`
`ipv6 access-map Deny_Telnet in`
`!`
- C. `ipv6 access-list Deny_Telnet sequence 10 deny tcp host 198A:0:200C::1/64 host 201A:0:205C::1/64`
`!`
`int Gi0/0`
`ipv6 access-map Deny_Telnet in`
`!`
- D. `ipv6 access-list Deny_Telnet sequence 10 deny tcp host 198A:0:200C::1/64 host 201A:0:205C::1/64`
`!`
`int Gi0/0`
`ipv6 traffic-filter Deny_Telnet in`
`!`

Correct Answer: A

Section: Infrastructure Security

Explanation

Explanation/Reference:

The IPv6 access-list "Deny_Telnet" should have other rules not shown in the choice. Otherwise all traffic will be blocked due to implicit deny.

QUESTION 38


```

access-list 100 deny tcp any any eq 465
access-list 100 deny tcp any eq 465 any
access-list 100 permit tcp any any eq 80
access-list 100 permit tcp any eq 80 any
access-list 100 permit udp any any eq 443
access-list 100 permit udp any eq 443 any

```

Refer to the exhibit. During troubleshooting it was discovered that the device is not reachable using a secure web browser.

What is needed to fix the problem?

- A. permit tcp port 443
- B. permit udp port 465
- C. permit tcp port 465
- D. permit tcp port 22

Correct Answer: A

Section: Infrastructure Security

Explanation

Explanation/Reference:

Secure web server listen on TCP port 443 for HTTPS traffic.

According to the ways that existing rules are configured, the complete answer is:

access-list 100 permit tcp any any eq 443

access-list 100 permit tcp any eq 443 any

QUESTION 39

Drag and drop the packet types from the left onto the correct descriptions on the right.

Select and Place:

data plane packets	user-generated packets that are always forwarded by network devices to other end-station devices
control plane packets	network device generated or received packets that are used for the creation of the network itself
management plane packets	network device generated or received packets; packets that are used to operate the network
services plane packets	user-generated packets that are forwarded by network devices to other end-station devices, but that require higher priority than the normal traffic by the network devices

Correct Answer:

	data plane packets
	control plane packets
	management plane packets
	services plane packets

Section: (none)

Explanation

Explanation/Reference:

According to Cisco documentations:

Data plane packets – End-station, user-generated packets that are always **forwarded** by network devices to other end-station devices.

Control plane packets – Network device generated or received packets that are used for the **creation and operation** of the network itself.

Management plane packets – Network device generated or received packets, or management station generated or received packets that are used to **manage** the network.

Services plane packets – A special case of data plane packets, services plane packets are also user-generated packets that are also forwarded by network devices to other end-station devices, but that require high-touch handling by the network device (e.g. **QoS**, MPLS, VPNs ...etc) to forward the packet.

Therefore the question allows that not all items are used, two descriptions should be placed with the item "control plane packets" and there is no description for item "managment plane packets".

QUESTION 40

Drag and drop the addresses from the left onto the correct IPv6 filter purposes on the right.

Select and Place:

<pre>permit ip 2001:d8b:800:200c::/117 2001:0DBB:800:2010::/64 eq 443</pre>	<pre>Permit NTP from this source 2001:0D8B:0800:200c::1f</pre>
<pre>permit ip 2001:D88:800:200C::e/126 2001:0DBB:800:2010::/64 eq 514</pre>	<pre>Permit syslog from this source 2001:0D88:0800:200c::1c</pre>
<pre>permit ip 2001:d8b:800:200c::800 /117 2001:0DBB:800:2010::/64 eq 80</pre>	<pre>Permit HTTP from this source 2001:0D8B:0800:200c::0fff</pre>
<pre>permit ip 2001:D8B:800:200C::c/126 2001:0DBB:800:2010::/64 eq 123</pre>	<pre>Permit HTTPS from this source 2001:0D8B:0800:200c::07ff</pre>

Correct Answer:

	<pre>permit ip 2001:D8B:800:200C::c/126 2001:0DBB:800:2010::/64 eq 123</pre>
	<pre>permit ip 2001:D88:800:200C::e/126 2001:0DBB:800:2010::/64 eq 514</pre>
	<pre>permit ip 2001:d8b:800:200c::800 /117 2001:0DBB:800:2010::/64 eq 80</pre>
	<pre>permit ip 2001:d8b:800:200c::/117 2001:0DBB:800:2010::/64 eq 443</pre>

Section: (none)

Explanation

Explanation/Reference:

In this question, you can answer by mapping the port number with the concerned protocol.

Remarks :

- The ACL rules in this question are wrong since you cannot specify port number for "ip" protocol. The actual question probably uses "tcp" or "udp" as the protocol in the rule.
- There are some problems in the source addresses in the rules:

i. /177 means 128 - 117 = 11 bits don't care

2001:d8b:800:200c::/117 = range from 2001:d8b:800:200c:: to 2001:d8b:800:200c::7ff
OK

ii. /126 means 128 - 126 = 2 bits don't care

2001:d8b:800:200c::e/126 is not proper since the last two bits is "10" instead of "00".
However it will be changed to 2001:d8b:800:200c::c/126 automatically after configuration.
2001:d8b:800:200c::c/126 = range from 2001:d8b:800:200c::c to 2001:d8b:800:200c::f
The actual source IPv6 in the purpose on the right should be "c" instead of "1c".

iii. /177 means 128 - 117 = 11 bits don't care

2001:d8b:800:200c::800/117 = range from 2001:d8b:800:200c::800 to 2001:d8b:800:200c::fff
OK

iv. /126 means 128 - 126 = 2 bits don't care

2001:d8b:800:200c::c/126 = range from 2001:d8b:800:200c::c to 2001:d8b:800:200c::f
Therefore, the actual source IPv6 in the purpose should be "f" instead of "1f" in order to match the answer on the right.

QUESTION 41

Refer to the exhibit. An engineer is trying to configure local authentication on the console line, but the device is trying to authenticate using TACACS+.

Which action produces the desired configuration?

```
R1#show running-config | include aaa
aaa new-model
aaa authentication login default group tacacs+ local
aaa authentication login Console local
R1#show running-config | section line
line con 0
logging synchronous
R1#
```

- Add the aaa authentication login default none command to the global configuration.
- Replace the capital "C" with a lowercase "c" in the aaa authentication login Console local command.
- Add the aaa authentication login default group tacacs+ local-case command to the global configuration.
- Add the login authentication Console command to the line configuration

Correct Answer: D

Section: Infrastructure Security

Explanation

Explanation/Reference:

The default AAA login authentication uses TACAS+ first. Hence you need to apply the authentication list "Console" for authentication against local username / password only.

QUESTION 42

Refer to the exhibit. An engineer is trying to connect to a device with SSH but cannot connect. The engineer connects by using the console and finds the displayed output when troubleshooting.

Which command must be used in configuration mode to enable SSH on the device?

```
R1#show ip ssh
SSH Disabled -- version 1.99
%Please create RSA keys to enable SSH (and of atleast 768 bits for SSH v2).
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size: 1024 bits
IOS Keys in SECSH format (ssh-rsa, base64 encoded) : NONE
R1#
```

- A. no ip ssh disable
- B. ip ssh enable
- C. ip ssh version 2
- D. crypto key generate rsa

Correct Answer: D

Section: Infrastructure Security
Explanation

Explanation/Reference:

SSH is disabled if no RSA key is found.

QUESTION 43

Which statement about IPv6 ND inspection is true?

- A. It learns and secures bindings for stateless autoconfiguration addresses in Layer 3 neighbor tables.
- B. It learns and secures bindings for stateless autoconfiguration addresses in Layer 2 neighbor tables.
- C. It learns and secures bindings for stateful autoconfiguration addresses in Layer 3 neighbor tables.
- D. It learns and secures bindings for stateful autoconfiguration addresses in Layer 2 neighbor tables.

Correct Answer: B

Section: Infrastructure Security
Explanation

Explanation/Reference:**QUESTION 44**

While troubleshooting connectivity issues to a router, these details are noticed:

- Standard pings to all router interfaces, including loopbacks, are successful.
- Data traffic is unaffected.
- SNMP connectivity is intermittent.
- SSH is either slow or disconnects frequently.

Which command must be configured first to troubleshoot this issue?

- A. show policy-map control-plane
- B. show policy-map
- C. show interface | inc drop
- D. show ip route

Correct Answer: A

Section: Infrastructure Security
Explanation

Explanation/Reference:

The occurrence of intermittent or slow is not related to blocking and all affected traffic are management packets. Hence it should be caused by the policing in CoPP.

QUESTION 45

```
TAC+: TCP/IP open to 171.68.118.101/49 failed --
Destination unreachable; gateway or host down
AAA/AUTHN (2546660185): status = ERROR
AAA/AUTHN/START (2546660185): Method=LOCAL
AAA/AUTHN (2546660185): status = FAIL
As1 CHAP: Unable to validate Response. Username chapuser: Authentication failure
```

Refer to the exhibit. Why is user authentication being rejected?

- A. The TACACS+ server expects "user", but the NT client sends "domain/user".

- B. The TACACS+ server refuses the user because the user is set up for CHAP.
- C. The TACACS+ server is down, and the user is in the local database.
- D. The TACACS+ server is down, and the user is not in the local database.

Correct Answer: D

Section: Infrastructure Security

Explanation

Explanation/Reference:

The message mentioned the following

- "TAC+" about the failure to open TCP connection.
- "LOCAL" but the status is FAIL

QUESTION 46

```
Cat3850-Stack-2# show policy-map
```

```
Policy Map LIMIT_BGP
```

```
Class BGP
drop
```

```
Policy Map SHAPE_BGP
```

```
Class BGP
Average Rate Traffic Shaping
cir 10000000 (bps)
```

```
Policy Map POLICE_BGP
```

```
Class BGP
police cir 1000k bc 1500
conform-action transmit
exceed-action transmit
```

```
Policy Map COPP
```

```
Class BGP
police cir 1000k bc 1500
conform-action transmit
exceed-action drop
```

Refer to the exhibit. Which control plane policy limits BGP traffic that is destined to the CPU to 1 Mbps and ignores BGP traffic that is sent at higher rate?

- A. policy-map SHAPE_BGP
- B. policy-map LIMIT_BGP
- C. policy-map POLICE_BGP
- D. policy-map COPP

Correct Answer: D

Section: Infrastructure Security

Explanation

Explanation/Reference:

If the question says about buffering traffic sent at higher rate, B is the answer.

QUESTION 47

Which statement about IPv6 RA Guard is true?

- A. It does not offer protection in environments where IPv6 traffic is tunneled.
- B. It cannot be configured on a switch port interface in the ingress direction.
- C. Packets that are dropped by IPv6 RA Guard cannot be spanned.
- D. It is not supported in hardware when TCAM is programmed.

Correct Answer: A

Section: Infrastructure Security

Explanation

Explanation/Reference:

QUESTION 48

An engineer is trying to copy an IOS file from one router to another router by using TFTP.

Which two actions are needed to allow the file to copy? (Choose two.)

- A. Copy the file to the destination router with the copy tftp: flash: command
- B. Enable the TFTP server on the source router with the tftp-server flash: <filename> command
- C. TFTP is not supported in recent IOS versions, so an alternative method must be used
- D. Configure a user on the source router with the username tftp password tftp command
- E. Configure the TFTP authentication on the source router with the tftp-server authentication local command

Correct Answer: AB
Section: Infrastructure Services
Explanation

Explanation/Reference:

QUESTION 49

Refer to the exhibit. Users report that IP addresses cannot be acquired from the DHCP server. The DHCP server is configured as shown. About 300 total nonconcurrent users are using this DHCP server, but none of them are active for more than two hours per day.

Which action fixes the issue within the current resources?

```
R1#show running-config | section dhcp
ip dhcp excluded-address 192.168.1.1 192.168.1.49
ip dhcp pool DHCP
  network 192.168.1.0 255.255.255.0
  default-router 192.168.1.1
  dns-server 8.8.8.8
  lease 0 12
```

- A. Modify the subnet mask to the network 192.168.1.0 255.255.254.0 command in the DHCP pool
- B. Configure the DHCP lease time to a smaller value
- C. Configure the DHCP lease time to a bigger value
- D. Add the network 192.168.2.0 255.255.255.0 command to the DHCP pool

Correct Answer: B
Section: Infrastructure Services
Explanation

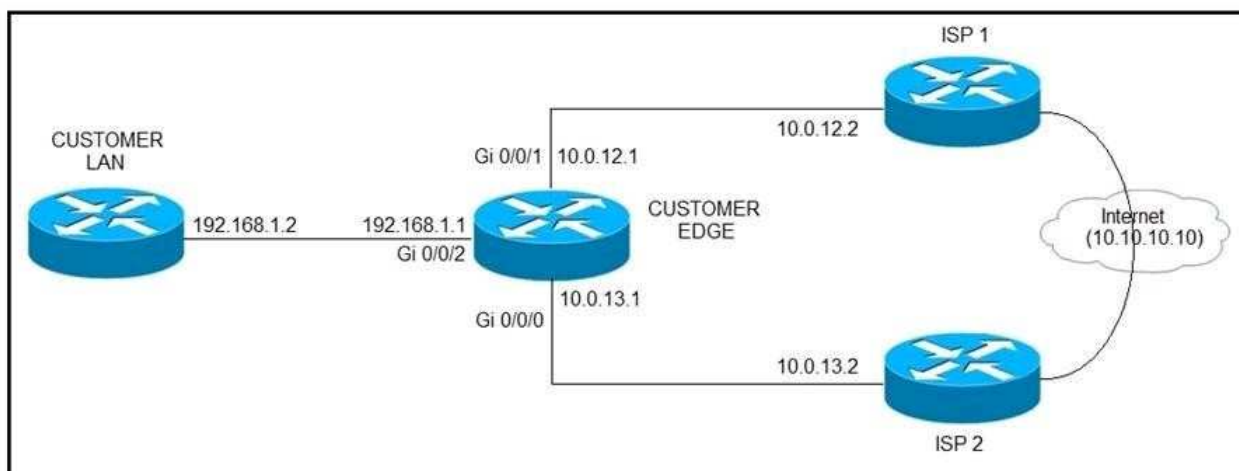
Explanation/Reference:

The current settings in the configuration shown is 0 day 12 hours.

QUESTION 50

Refer to the exhibit. ISP 1 and ISP 2 directly connect to the Internet. A customer is tracking both ISP links to achieve redundancy and cannot see the Cisco IOS IP SLA tracking output on the router console.

Which command is missing from the IP SLA configuration?



- A. Start-time 00:00
- B. Start-time 0
- C. Start-time immediately
- D. Start-time now

Correct Answer: D
Section: Infrastructure Services
Explanation

Explanation/Reference:

QUESTION 51

Refer to the exhibit. An administrator noticed that after a change was made on R1, the timestamps on the system logs did not match the clock.

What is the reason for this error?

```

service timestamps debug datetime msec
service timestamps log datetime
clock timezone MST -7 0
clock summer-time MST recurring
ntp authentication-key 1 md5 00101A0B0152181206224747071E 7
ntp server 10.10.10.10

```

R1#show clock

*06:13:44.045 MST Sun Dec 30 2018

R1#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R1(config) #logging host 10.10.10.20

R1(config) #end

R1#

*Dec 30 13:15:28: %SYS-5-CONFIG_I: Configured from console by console

R1#

*Dec 30 13:15:28: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 10.10.10.20 port 514 started – CLI initiated

- A. An authentication error with the NTP server results in an incorrect timestamp.
- B. The keyword localtime is not defined on the timestamp service command.
- C. The NTP server is in a different time zone.
- D. The system clock is set incorrectly to summer-time hours.

Correct Answer: B

Section: Infrastructure Services

Explanation

Explanation/Reference:

QUESTION 52

Drag and drop the DHCP messages from the left onto the correct uses on the right.

Select and Place:

DHCPACK	server-to-client communication, refusing the request for configuration parameters
DHCPINFORM	client-to-server communication, indicating that the network address is already in use
DHCPNAK	server-to-client communication with configuration parameters, including committed network address
DHCPDECLINE	client-to-server communication, asking for only local configuration parameters that the client has already externally configured as an address

Correct Answer:

	DHCPNAK
	DHCPDECLINE
	DHCPACK
	DHCPINFORM

Section: (none)

Explanation

Explanation/Reference:

QUESTION 53

A network engineer is investigating a flapping (up/down) interface issue on a core switch that is synchronized to an NTP server. Log output currently does not show the

time of the flap.

Which command allows the logging on the switch to show the time of the flap according to the clock on the device?

- A. service timestamps log uptime
- B. clock summer-time mst recurring 2 Sunday mar 2:00 1 Sunday nov 2:00
- C. service timestamps log datetime localtime show-timezone
- D. clock calendar-valid

Correct Answer: C

Section: Infrastructure Services

Explanation

Explanation/Reference:

QUESTION 54

When provisioning a device in Cisco DNA Center, the engineer sees the error message "Cannot select the device. Not compatible with template".

What is the reason for the error?

- A. The template has an incorrect configuration.
- B. The software version of the template is different from the software version of the device.
- C. The changes to the template were not committed.
- D. The tag that was used to filter the templates does not match the device tag.

Correct Answer: D

Section: Infrastructure Services

Explanation

Explanation/Reference:

QUESTION 55

While working with software images, an engineer observes that Cisco DNA Center cannot upload its software image directly from the device.

Why is the image not uploading?

- A. The device must be resynced to Cisco DNA Center.
- B. The software image for the device is in install mode.
- C. The device has lost connectivity to Cisco DNA Center.
- D. The software image for the device is in bundle mode

Correct Answer: B

Section: Infrastructure Services

Explanation

Explanation/Reference:

QUESTION 56

An engineer configured the wrong default gateway for the Cisco DNA Center enterprise interface during the install.

Which command must the engineer run to correct the configuration?

- A. sudo maglev-config update
- B. sudo maglev install config update
- C. sudo maglev reinstall
- D. sudo update config install

Correct Answer: A

Section: Infrastructure Services

Explanation

Explanation/Reference:

QUESTION 57

```
R1(config) # do show running-config | section line|username
username cisco secret 5 $1$yb/o$L3G5cXODxpYMSJ70PzEyo0
line con 0
  logging synchronous
line vty 0 4
  login local
  transport input telnet
R1(config) # logging console 7
R1(config) # do debug aaa authentication
R1(config) #
```

Refer to the exhibit. An administrator that is connected to the console does not see debug messages when remote users log in.

Which action ensures that debug messages are displayed for remote logins?

- A. Enter the transport input ssh configuration command.
- B. Enter the terminal monitor exec command.
- C. Enter the logging console debugging configuration command.
- D. Enter the aaa new-model configuration command.

Correct Answer: D

Section: Infrastructure Services

Explanation

Explanation/Reference:

In order to debug login through "debug aaa authentication", the router must be configured with AAA commands for authentication. "terminal monitor" is only needed if the administrator is connected to the router through telnet / SSH.

QUESTION 58

```
snmp-server community ciscotest1
snmp-server host 192.168.1.128 ciscotest
snmp-server enable traps bgp
```

Refer to the exhibit. Network operations cannot read or write any configuration on the device with this configuration from the operations subnet.

Which two configurations fix the issue? (Choose two.)

- A. Configure SNMP rw permission in addition to community ciscotest.
- B. Modify access list 1 and allow operations subnet in the access list.
- C. Modify access list 1 and allow SNMP in the access list.
- D. Configure SNMP rw permission in addition to version 1.
- E. Configure SNMP rw permission in addition to community ciscotest 1.

Correct Answer: BE

Section: Infrastructure Services

Explanation

Explanation/Reference:

The configuration show should be :

```
snmp-server community ciscotest 1
```

i.e. the "1" is a separate parameter meaning the access list number / word.

Hence after the choice E, the command should be:

```
snmp-server community ciscotest rw 1
```

QUESTION 59

```
config t
flow record v4_r1
match ipv4 tos
match ipv4 protocol
match ipv4 source address
match ipv4 destination address
match transport source-port
match transport destination-port
collect counter bytes long
collect counter packets long
!
flow exporter EXPORTER-1
destination 172.16.10.2
transport udp 90
exit
!
flow monitor FLOW-MONITOR-1
record v4_r1
exit
!
ip cef
!
interface Ethernet0/0.1
ip address 172.16.6.2 255.255.255.0
ip flow monitor FLOW-MONITOR-1 input
!
```

Refer to the exhibit. Why is the remote NetFlow server failing to receive the NetFlow data?

- A. The flow exporter is configured but is not used.
- B. The flow monitor is applied in the wrong direction.
- C. The flow monitor is applied to the wrong interface.
- D. The destination of the flow exporter is not reachable.

Correct Answer: A
Section: Infrastructure Services
Explanation

Explanation/Reference:
The export should be used as follows:
flow monitor FLOW-MONITOR-1
record v4_r1
exporter EXPORTER-1

QUESTION 60

Which Cisco VPN technology can use multipoint tunnel, resulting in a single GRE tunnel interface on the hub, to support multiple connections from multiple spoke devices?

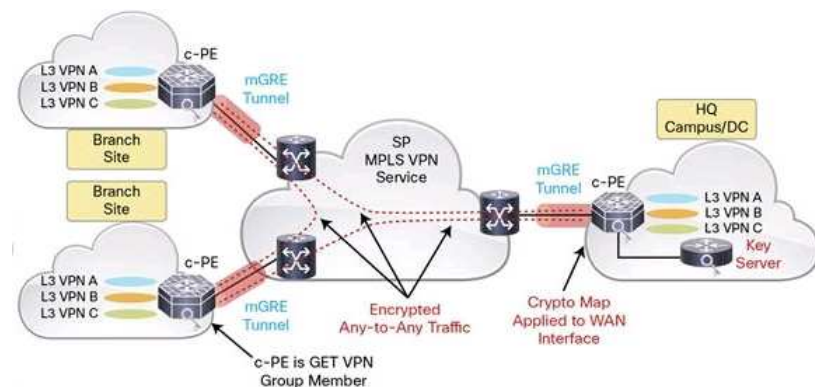
- A. DMVPN
- B. GETVPN
- C. Cisco Easy VPN
- D. FlexVPN

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

Features	IPSEC	Easy VPN	DMVPN	GETVPN
Encapsulation	Tunneled IPSEC	Tunneled IPSEC	Tunneled IPSEC	Non-Tunneled IPSEC
Scalability	Low	High for Hub	High for device	high for device
Authentication	Low	High with PSK or PKI	High with PKI	High with PSK or PKI
Suitable Topologies	Point to point	Hub to spoke	Hub to spoke	Full mesh
Transport Networks	Any	Any	Any	Private WAN, MPLS
Routing	NA	Reverse route injection	Dynamic routing on tunnels	Dynamic routing on IP WAN, MPLS
Failover	NA	Stateful Hub Crypto Failover	Route Distribution Model	Route Distribution model+stateful
Encryption	IPSEC	Peer to Peer Protection	Peer to Peer Protection	Group Protection
Qos	Supported, Limited	Not-Supported	Supported	Supported
Multicast	No	No	Yes	Yes

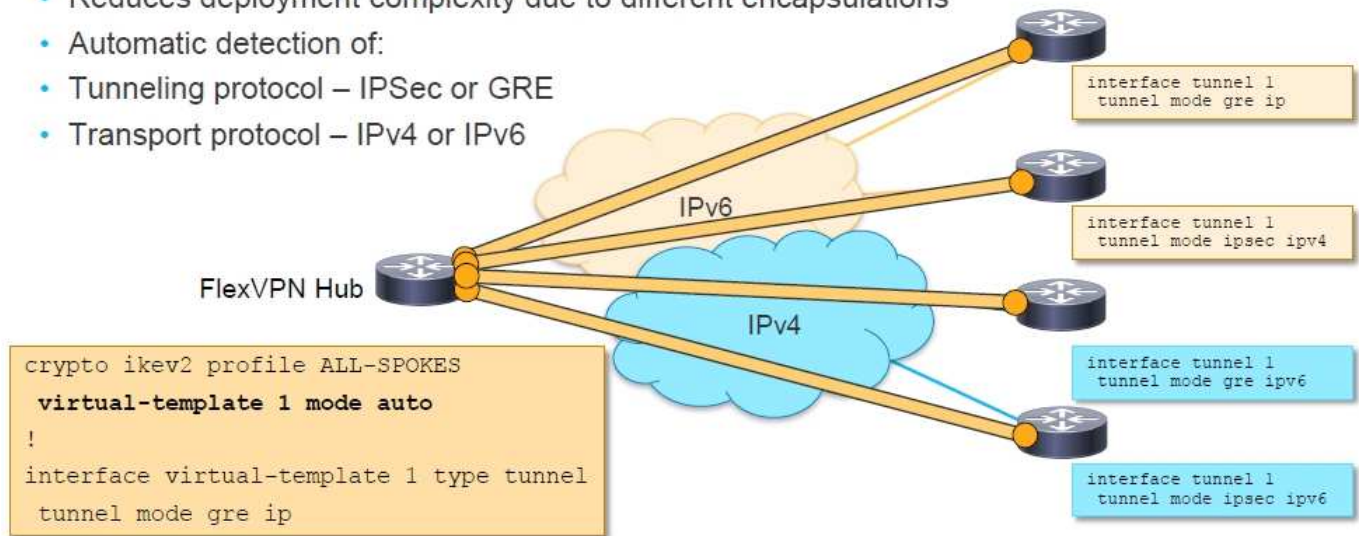
The following shows an example of GETVPN. although it uses mGRE, there is no classification of Hub and Spoke router. It is tunnelless VPN and therefore cannot connect networks of private addressing through Internet.



FlexVPN is a IKEv2-based unified VPN technology that combines site-to-site, remote-access, hub-spoke and spoke-to-spoke topologies. It can build VPN easily using "smart" defaults using Point-to-Point tunnel interface (i.e. it does not use mGRE).

Auto Tunnel Mode

- Reduces deployment complexity due to different encapsulations
- Automatic detection of:
- Tunneling protocol – IPsec or GRE
- Transport protocol – IPv4 or IPv6



QUESTION 61

Which next hop is going to be used for 172.17.1.0/24 ?

Router(config-if)#do show ip bgp

BGP table version is 4, local router ID is 156.12.1.6

Status codes: s suppressed, d damped, h history, * valid, > best, i – internal, r RIB-failure, S Stale Origin codes: i – IGP, e – EGP, ? – incomplete

Network	Next Hop	Metric	LocPrf	Weight	Path
*>i 10.1.1.0/24	192.168.1.2	0	0	10000	i
*>i 10.2.2.0/24	192.168.3.2	0	0	10000	i
*>i 172.17.1.0/24	10.0.0.1	0	0	32768	i
*i	10.0.0.2	0	0	32768	i

- A. 10.0.0.1
- B. 192.168.1.2
- C. 10.0.0.2
- D. 192.168.3.2

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

This the the next hop of the entry with ">" i.e. selected as best.

QUESTION 62

Which label operations are performed by a label edge router?

- A. SWAP and POP
- B. SWAP and PUSH
- C. PUSH and PHP
- D. PUSH and POP

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 63

Refer to the exhibit. A network engineer executes the show ipv6 ospf database command and is presented with the output that is shown. Which flooding scope is referenced in the link-state type?

OSPFv3Router with ID (2.2.2.2) (Process ID 1)

Router Link States (Area 0)

ADV Router	Age	Seq#	Fragment ID	Link count	Bits
2.2.2.2	1694	0x80000002	0	1	B
4.4.1.4	1695	0x80000002	0	1	None

Inter Area Prefix Link States (Area 0)

ADV Router	Age	Seq#	Prefix
2.2.2.2	1692	0x30000001	2001:DB8 :0:123::/64

Link (Type 8) Link States (Area 0)

ADV Router	Age	Seq#	Link ID	InterFace
2.2.2.2	1696	0x80000002	6	Se1/0
4.4.1.4	1699	0x80000002	6	Se1/0

- A. link-local
- B. area
- C. AS (OSPF domain)
- D. reserved

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 64

Refer to the exhibit.

```
R1#show policy-map control-plane
Control Plane
Class-map: NMS (match-all)
 500461 packets, 24038351 bytes
 5 minute offered rate 1390000 bps, drop rate 0 bps
police:
  cir 50000 bps, bc 5000 bytes
conformed 50444 packets, 24031001 bytes; actions:
transmit
exceeded 990012 packets, 94030134 bytes; actions
drop conformed 4000 bps, exceed 0 bps
R1#
```

A company is evaluating multiple network management system tools. Trending graphs generated by SNMP data are returned by the NMS and appear to have multiple gaps. While troubleshooting the issue, an engineer noticed the relevant output.

What solves the gaps in the graphs?

- A. Remove the exceed-rate command in the class map.
- B. Remove the class map NMS from being part of control plane policing.
- C. Configure the CIR rate to a lower value that accommodates all the NMS tools
- D. Separate the NMS class map in multiple class maps based on the specific protocols with appropriate CoPP actions

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Since the access list matching NMS control all management protocols, if you want to allow more traffic for SNMP, you should separate the SNMP traffic into a new class that allows more traffic rate.

QUESTION 65

Refer to the following output:

```
Router#show ip nhrp detail
1.1.2/8 via 10.2.1.2, Tunnel1 created 00:00:12, expire 01:59:47 TypE. dynamic, Flags: authoritative unique nat registered
used NBMA address: 10.12.1.2
```

What does the authoritative flag mean in regards to the NHRP information?

- A. It was obtained directly from the next-hop server.
- B. Data packets are process switches for this mapping entry.
- C. NHRP mapping is for networks that are local to this router.
- D. The mapping entry was created in response to an NHRP registration request.
- E. The NHRP mapping entry cannot be overwritten

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 66

Which security feature can protect DMVPN tunnels?

- A. IPsec
- B. TACACS+
- C. RTBH
- D. RADIUS

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 67

Which two methods use IPsec to provide secure connectivity from the branch office to the headquarters office? (Choose two.)

- A. DMVPN
- B. MPLS VPN
- C. Virtual Tunnel Interface (VTI)
- D. SSL VPN
- E. PPPoE

Correct Answer: AC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 68

Which SNMP verification command shows the encryption and authentication protocols that are used in SNMPV3?

- A. show snmp group
- B. show snmp user
- C. show snmp
- D. show snmp view

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 69

Which two protocols can cause TCP starvation? (Choose two.)

- A. TFTP
- B. SNMP
- C. SMTP
- D. HTTPS
- E. FTP

Correct Answer: AB

Section: (none)

Explanation

Explanation/Reference:

UDP traffic can cause TCP starvation.

QUESTION 70

A network engineer needs to verify IP SLA operations on an interface that shows an indication of excessive traffic. Which command should the engineer use to complete this action?

- A. show frequency
- B. show track
- C. show reachability
- D. show threshold

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 71

Which protocol does VRF-Lite support?

- A. IS-IS
- B. ODR
- C. EIGRP
- D. IGRP

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Other supported protocols include RIPv2, OSPF, BGP ... etc

QUESTION 72

Which two statements about redistributing EIGRP into OSPF are true? (Choose two.)

- A. The redistributed EIGRP routes appear as type 3 LSAs in the OSPF database.
- B. The redistributed EIGRP routes appear as type 5 LSAs in the OSPF database.
- C. The administrative distance of the redistributed routes is 170.
- D. The redistributed EIGRP routes appear as OSPF external type 1.
- E. The redistributed EIGRP routes as placed into an OSPF area whose area ID matches the EIGRP autonomous system number.
- F. The redistributed EIGRP routes appear as OSPF external type 2 routes in the routing table.

Correct Answer: BF

Section: (none)

Explanation

Explanation/Reference:

QUESTION 73

Users were moved from the local DHCP server to the remote corporate DHCP server. After the move, none of the users were able to use the network. Which two issues will prevent this setup from working properly? (Choose two.)

- A. Auto-QoS is blocking DHCP traffic.
- B. The DHCP server IP address configuration is missing locally.
- C. 802.1X is blocking DHCP traffic.
- D. The broadcast domain is too large for proper DHCP propagation.
- E. The route to the new DHCP server is missing.

Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 74

Which command is used to check IP SLA when an interface is suspected to receive lots of traffic with options?

- A. show track
- B. show threshold
- C. show timer
- D. show delay

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 75

What is an advantage of using BFD?

- A. It detects local link failure at layer 1 and updates routing table.
- B. It detects local link failure at layer 2 and updates routing protocols.
- C. It has sub-second failure detection for layer 1 and layer 3 problems.
- D. It has sub-second failure detection for layer 1 and layer 2 problems.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 76

Which component of MPLS VPNs is used to extend the IP address so that an engineer is able to identify to which VPN it belongs?

- A. VPNv4 address family

- B. RD
- C. RT
- D. LDP

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 77

Refer to the exhibit:

```
ip access-list extended FILTER
deny tcp 192.168.10.0 0.0.0.255 192.168.100.0 0.0.0.255 eq 22
deny tcp 192.168.10.0 0.0.0.255 192.168.100.0 0.0.0.255 eq 23
deny tcp 192.168.10.0 0.0.0.255 192.168.100.0 0.0.0.255 eq 80
deny tcp 192.168.10.0 0.0.0.255 192.168.100.0 0.0.0.255 eq 443
permit tcp host 192.168.10.10 host 192.168.100.10 eq ssh
permit ip any any
!
interface GigabitEthernet0/1
ip address 192.168.10.1 255.255.255.0
ip access-group FILTER in
!
```

The ACL is placed on the inbound GigabitEthernet 0/1 interface of the router. Host 192.168.10.10 cannot SSH to host 192.168.100.10 even though the flow is permitted. Which action resolves the issue without opening full access to this router?

- A. Move the SSH entry to the beginning of the ACL.
- B. Temporarily move the permit ip any any line to the beginning of the ACL to see if the flow works.
- C. Temporarily remove the ACL from the interface to see if the flow works.
- D. Run the show access-list FILTER command to view if the SSH entry has any hit statistic associated with it.

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

The SSH traffic is matched to the first rule and denied. Hence you need to move the SSH rule for 192.168.10.10 to 192.168.100.10 before the first rule.

QUESTION 78

Refer the exhibit:

```
R3#show policy-map control-plane
Control Plane

Service-policy output: R3_CoPP

Class-map: mgmt (match-all)
 361 packets, 73858 bytes
 5 minute offered rate 0 bps, drop rate 0 bps
 Match: access-group 120
  police:
   dir 8000 bps, bc 1500 bytes, be 1500 bytes
   conformed 8 packets, 1506 bytes; actions:
    transmit
   exceeded 353 packets, 72352 bytes; actions:
    drop
   violated 0 packets, 0 bytes; actions:
    drop
   conformed 0 bps, exceed 0 bps, violate 0 bps

Class-map: class-default (match-any)
 124 packets, 10635 bytes
 5 minute offered rate 0 bps, drop rate 0 bps
 Match: any
R3#show access-lists 120
Extended IP access list 120
 10 permit udp any any eq snmptrap (361 matches)
```


Which action resolves intermittent connectivity observed with the SNMP trap packets?

- A. Decrease the committed burst Size of the mgmt class map.
- B. Increase the CIR of the mgmt class map.
- C. Add a new class map to match TCP traffic.
- D. Add one new entry in the ACL 120 to permit the UDP port 161.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since the class map "mgmt" is using access-list 120 which permits for SNMP trap only, you should increase the CIR.

QUESTION 79

What destination addresses does EIGRP use when feasible? (Choose two.)

- A. IP address 224.0.0.9
- B. IP address 224.0.0.10
- C. IP address 224.0.0.8
- D. MAC address 01:00:5E:00:00:0A
- E. MAC address 0C:15:C0:00:00:01

Correct Answer: BD

Section: (none)

Explanation

Explanation/Reference:

Remarks:

OSPF uses 224.0.0.5 (224.0.0.6 is used in DR/BDR)

Their corresponding MAC address are 01:00:5E:00:00:05 and 01:00:5E:00:00:06.

QUESTION 80

How are packets forwarded in an MPLS domain?

- A. Using the destination IP address of the packet.
- B. Using the source IP address of the packet.
- C. Using a number that has been specified in a label.
- D. Using the MAC address of the frame.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 81

Refer to the exhibit:

```
aaa new-model
aaa authentication login default none
aaa authentication login telnet local
!
username cisco password 0 ocsic
!
line vty 0
 password LetMeIn
 login authentication telnet
 transport input telnet
line vty 1
 password LetMeIn
 transport input telnet
```

Drag and drop the credentials from the left onto the remote login information on the right to resolve a failed login attempt to vtys. (Not all credentials are used, choose the best.)

Select and Place:

no password	vty 0	username
ocsic		password
no username		
LetMeIn	vty 1	username
cisco		password
LetMeIn		

Correct Answer:

	vty 0	cisco
		ocsic
LetMeIn	vty 1	no username
LetMeIn		no password

Section: (none)

Explanation

Explanation/Reference:

Since AAA is enabled:

- line vty 0 uses local username / password authentication since the authentication list "telnet" is specified.
- line vty 1 uses the default authentication list which is configured with "none" since no authentication list is specified.

QUESTION 82

You just discovered that a ping packet sent from one of the devices to another took a different path in the return than it did on its way to the destination. What behavior caused this?

- A. Windowing
- B. Global synchronization
- C. MSS
- D. Asymmetric routing

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 83

How long is the default NHRP cache timer?

- A. 2 hours
- B. 1 hour
- C. 30 minutes
- D. 15 minutes

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The default length of time is 7200 seconds (2 hours). However it may vary depending on the Cisco IOS version.

QUESTION 84

The OSPF dead interval defaults to how many times the hello interval?

- A. Two
- B. Three
- C. Four
- D. Five

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 85

How are customer routes isolated on PE routers in an MPLS Layer 3 VPN?

- A. By using VRF
- B. By using VDCs
- C. By using MP-BGP
- D. By using LDP

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 86

The OSPF database of a router shows LSA types 1, 2, 7, and 3 for the default route only. Which type of area is this router connected to?

- A. stub area
- B. totally stubby area
- C. NSSA totally stub
- D. NSSA

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 87

Which three IP SLA performance metrics can you use to monitor enterprise-class networks? (Choose three.)

- A. Packet loss
- B. Delay
- C. bandwidth
- D. Connectivity
- E. Reliability
- F. traps

Correct Answer: ABD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 88

A network administrator notices that the BGP state drops and logs are generated for missing BGP hello keepalives. What is the potential problem?

- A. Incorrect neighbor options.
- B. Hello timer mismatch.
- C. BGP path MTU enabled.
- D. MTU mismatch.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 89

Which access list entry checks for an ACK within a packet header?

- A. access-list 49 permit ip any any eq 21 tcp-ack
- B. access-list 49 permit tcp any any eq 21 tcp-ack
- C. access-list 149 permit tcp any any eq 21 established
- D. access-list 49 permit tcp any any eq 21 established

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Access list range 1 - 99 are standard access list and therefore "tcp" protocol cannot be specified. You can only specify TCP flag in a rule using "tcp" as protocol.

Moreover, the keyword for matching ACK only is "ack" (not tcp-ack). Hence only C is the correct answer.

QUESTION 90

You are implementing WAN access for an enterprise network while running applications that require a fully meshed network, which two design standards are appropriate for such an environment? (Choose two.)

- A. A centralized DMVPN solution to simplify connectivity for the enterprise.
- B. A dedicated WAN distribution layer to consolidate connectivity to remote sites.
- C. A collapsed core and distribution layer to minimize costs.
- D. Multiple MPLS VPN connections with static routing.
- E. Multiple MPLS VPN connections with dynamic routing.

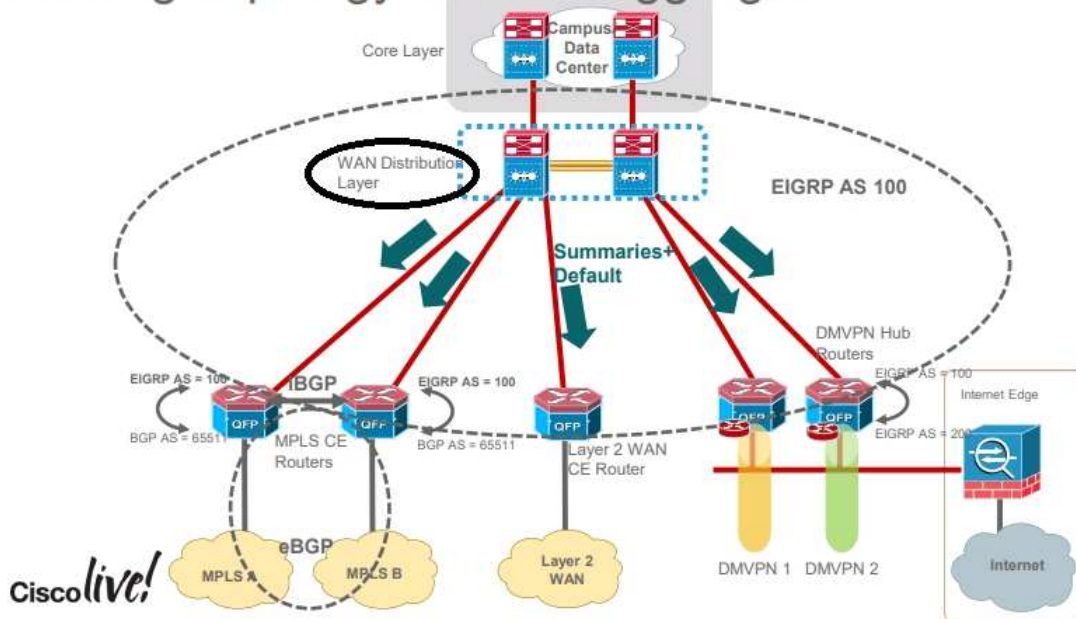
Correct Answer: AB

Section: (none)

Explanation

Explanation/Reference:

Routing Topology at WAN Aggregation

**QUESTION 91**

Which task do you need to perform first when you configure IP SLA to troubleshoot a network connectivity issue?

- A. Specify the test frequency.
- B. Enable the ICMP echo operation.
- C. Schedule the ICMP echo operation.
- D. Verify the ICMP echo operation.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 92

Which two features are provided by EIGRP for IPv6? (Choose two.)

- A. Backbone areas
- B. SPF algorithm
- C. Partial updates
- D. Area border router
- E. Scaling

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

Both are the general characteristics of EIGRP i.e. both for IPv4 and IPv6

QUESTION 93

Which IP SLA operation can be used to simulate voice traffic on a network?

- A. TCP connect
- B. UDP-jitter
- C. ICMP-echo
- D. ICMP-jitter

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 94

What happens when two EIGRP peers have mismatched K values?

- A. The two devices are unable to correctly perform equal-cost routing.
- B. The two devices fail to perform EIGRP graceful shutdown when one device goes down.
- C. The two devices fail to form an adjacency.
- D. The two devices are unable to correctly perform unequal-cost load balancing.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 95

Which location within the network is preferred when using a dedicated router for Cisco IP SLA operations?

- A. user edge
- B. provider edge
- C. access edge
- D. distribution edge

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

In this implementation, a service provider wants to provide SLA information to its customers.

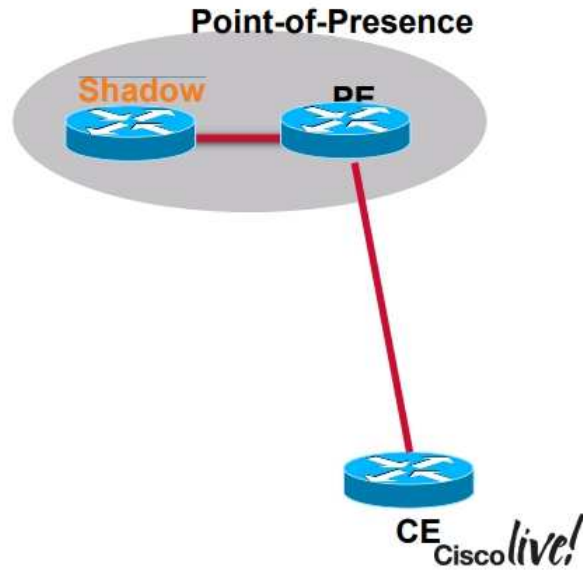
Why Use a Shadow Router?

- A **shadow router** is a dedicated box for IP SLAs.
But why?
- If your Provider Edge (PE) router is already overloaded (> 60% CPU at interrupt level)
- If your PE lacks memory
- If your PE is a distributed platform
- If you want to separate measurement from forwarding
- Upgrade freely for the latest and greatest IP SLA features without disturbing the traffic, then...
- Use a **shadow router (router dedicated to IPSLA)**

Cisco *live!*

Shadow Router Configuration

- A shadow router is typically a dedicated router located near a ideal measurement point.
- A point-of-presence (POP) is an ideal location.
- It can be connected to the PE via various methods: direct IP connection, tunnels, dot1q



QUESTION 96

Which protocol is used in a DMVPN network to map physical IP addresses to logical IP addresses?

- A. BGP
- B. LLDP
- C. EIGRP
- D. NHRP

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 97

An engineer is configuring a network and needs packets to be forwarded to an interface for any destination address that is not in the routing table. What should be configured to accomplish this task?

- A. set ip next-hop
- B. set ip default next-hop
- C. set ip next-hop recursive
- D. set ip next-hop verify-availability

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 98

Which protocol does MPLS use to support traffic engineering?

- A. Tag Distribution Protocol (TDP)
- B. Resource Reservation Protocol (RSVP)
- C. Border Gateway Protocol (BGP)
- D. Label Distribution Protocol (LDP)

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 99

Which IGPs are supported by the MPLS LDP autoconfiguration feature?

- A. ISIS and RIPv2
- B. RIPv2 and OSPF
- C. OSPF and ISIS
- D. OSPF and EIGRP

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

The MPLS LDP Autoconfiguration feature enables you to globally enable Label Distribution Protocol (LDP) on every interface associated with an Interior Gateway Protocol (IGP) instance. This feature is supported on Open Shortest Path First (OSPF) and Intermediate System-to-Intermediate System (IS-IS) IGPS.

QUESTION 100

What does the PE router convert the IPv4 prefix to within an MPLS VPN?

- A. 48-bit route combining the IP and PE router-id.
- B. VPN-IPv4 prefix combined with the 64-bit route distinguisher.
- C. eBGP path association between the PE and CE sessions.
- D. Prefix that combines the ASN, PE router-id, and IP prefix.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 101

An engineer needs dynamic routing between two routers and is unable to establish OSPF adjacency. The output of the show ip ospf neighbor command shows that the neighbor state is EXSTART/EXCHANGE. Which action should be taken to resolve this issue?

- A. match the passwords
- B. match the hello timers
- C. match the MTUs
- D. match the network types

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 102

Refer to the exhibit. AAA server 10.1.1.1 is configured with the default authentication and accounting settings, but the switch cannot communicate with the server. Which action resolves this issue?

```
Global RADIUS shared secret:*****
retransmission count: 5
timeout value: 10
following RADIUS servers are configured:
  myradius.network.users.com:
    available for authentication on port: 1814
    available for accounting on port: 1813
  10.1.1.1:
    available for authentication on port: 1814
    available for accounting on port: 1813
    RADIUS shared secret: *****
  10.2.2.3
    available for authentication on port: 1814
    available for accounting on port: 1813
    RADIUS shared secret: *****
```

- A. Correct the timeout value
- B. Match the authentication port
- C. Correct the shared secret
- D. Match the accounting port

Correct Answer: B

Section: (none)

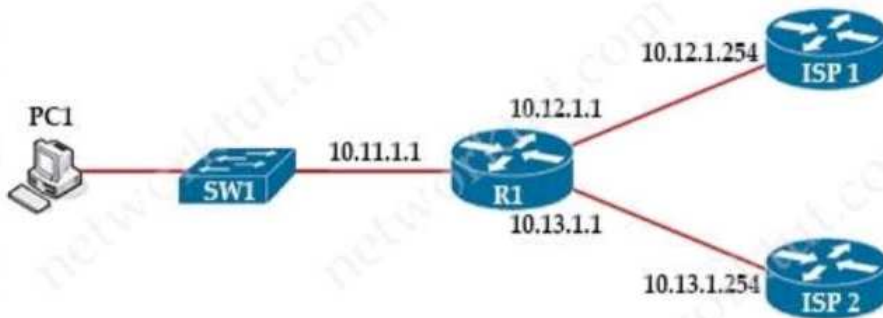
Explanation

Explanation/Reference:

Port number 1812 (instead of 1814) is usually used for AAA authentication.

QUESTION 103

Refer to the exhibit. An engineer is monitoring reachability of the configured default routes to ISP1 and ISP2. The default route from ISP1 is preferred if available. How is this issue resolved?



R1

```

ip sla 100
  icmp-echo 10.12.1.254
!
track 10 ip sla 100 reachability
!
ip route 0.0.0.0 0.0.0.0 10.12.1.254 track 10
ip route 0.0.0.0 0.0.0.0 10.13.1.254 10
!

```

R1#show ip route

--Output Omitted--

Gateway of last resort is 10.13.1.254 to network 0.0.0.0

```

S* 0.0.0.0/0 [10/0] via 10.13.1.254
    10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.11.1.0/24 is directly connected, GigabitEthernet0/1
L    10.11.1.1/32 is directly connected, GigabitEthernet0/1
C    10.12.1.0/24 is directly connected, GigabitEthernet0/0
L    10.12.1.1/32 is directly connected, GigabitEthernet0/0
C    10.13.1.0/24 is directly connected, GigabitEthernet0/2
L    10.13.1.1/32 is directly connected, GigabitEthernet0/2

```

- A. Use the icmp-echo command to track both default routes
- B. Start IP SLA by matching numbers for track and ip sla commands
- C. Start IP SLA by defining frequency and scheduling it
- D. Use the same AD for both default routes

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 104

Refer to the exhibit. The network administrator configured VRF lite for customer A. The technician at the remote site misconfigured VRF on the router. Which configuration will resolve connectivity for both sites of customer A?

```

ip vrf customer_a
  rd 1:1
  route-target export 1:1
  route-target import 1:1
!
interface FastEthernet0.1
  encapsulation dot1Q 2
  ip vrf forwarding customer_a
  ip address 192.168.4.1 255.255.255.0
!
router ospf 1
  log-adj adjacency-changes
!
  router ospf 2 vrf customer_a
  log-adj adjacency-changes
  network 192.168.4.0 0.0.0.255 area 0
!
end

```

- A. ip vrf customer_a
rd 1:2
route-target both 1:1
- B. ip vrf customer_a
rd 1:2
route-target both 1:2
- C. ip vrf customer_a
rd 1:1
route-target import 1:1
route-target export 1:2
- D. ip vrf customer_a
rd 1:1
route-target export 1:2
route-target import 1:2

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The exhibit shows the router's configuration at one site.

You need to choose the configuration for the router at the other site. Hence, the same RT i.e. 1:1 is required for both import and export (or a single command using "both").

QUESTION 105

After some changes in the routing policy, it is noticed that the router in AS 45123 is being used as a transit AS router for several service providers. Which configuration ensures that the branch router in AS 45123 advertises only the local networks to all SP neighbors?

- A. ip as-path access-list 1 permit ^45123\$
!
router bgp 45123
neighbor SP-Neighbors filter-list 1 out
- B. ip as-path access-list 1 permit ^45123
!
router bgp 45123
neighbor SP-Neighbors filter-list 1 out
- C. ip as-path access-list 1 permit ^\$
!
router bgp 45123
neighbor SP-Neighbors filter-list 1 out
- D. ip as-path access-list 1 permit
!
router bgp 45123
neighbor SP-Neighbors filter-list 1 out

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

BGP route generated for local networks has no AS (i.e. matched by ^\$) in the BGP table.

The local AS i.e. 45123 is only added when sending to eBGP neighbor.

QUESTION 106

Refer to the exhibit. An engineer is trying to get a packet destined for 192.168.32.100 forwarded through 10.1.1.1, but it was forwarded through 10.1.1.2. What action forwards the packets through 10.1.1.1?

```
Router#show ip route
...
D 192.168.32.0/19 [90/25789217] via 10.1.1.1
R 192.168.32.0/24 [120/4] via 10.1.1.2
O 192.168.32.0/26 [110/229840] via 10.1.1.3
```

- A. Configure EIGRP to receive 192.168.32.0 route with lower metric
- B. Configure EIGRP to receive 192.168.32.0 route with lower administrative distance
- C. Configure EIGRP to receive 192.168.32.0 route with equal or longer prefix than /24
- D. Configure EIGRP to receive 192.168.32.0 route with longer prefix than /19

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

The destination 192.168.32.100 will not be matched to OSPF route since 192.168.32.0/26 has the range 192.168.32.0 - 192.168.32.63.

The destination 192.168.32.100 is matched to RIP route 192.168.32.0/24 since it is more specific than the EIGRP route 192.168.32.0/19.

If more specific EIGRP route /24 or longer exists in the network but are summarized or filtered, you need to:

- For summarized, leak those more specific EIGRP routes in the summary address configuration OR
- For filtered, permit those more specific EIGRP routes in the route filtering.

Even the more specific EIGRP route has the same prefix length 24, it will be entered into the routing table replacing RIP route since EIGRP has a lower AD than RIP.

QUESTION 107

In the following SNMP attributes for SNMPv2c and SNMPv3, which are the attributes that matches best for SNMPv2c? (Choose 3)

- A. Community String
- B. username and password
- C. authentication
- D. no encryption
- E. privileged
- F. read-only

Correct Answer: ADF

Section: (none)

Explanation

Explanation/Reference:

	SNMPv2c	SNMPv3
community string		
username and password		
authentication		
no encryption		
privileged		
read-only		

QUESTION 108

In the following SNMP attributes for SNMPv2c and SNMPv3, which are the attributes that matches best for SNMPv3? (Choose 3)

- A. Community String
- B. username and password
- C. authentication
- D. no encryption
- E. privileged
- F. read-only

Correct Answer: BCE

Section: (none)

Explanation

Explanation/Reference:

	SNMPv2c	SNMPv3
community string		
username and password		
authentication		
no encryption		
privileged		
read-only		

QUESTION 109

What of the followings are best matched for Label Switch Router? (Choose two)

- A. assigns labels to unlabeled packets
- B. handles traffic between multiple VPNs

- C. reads the labels and forwards the packet based on the labels
- D. performs penultimate hop popping

Correct Answer: CD
Section: (none)
Explanation

Explanation/Reference:

Label Switch Router
assigns labels to unlabeled packets
handles traffic between multiple VPNs
reads the labels and forwards the packet based on the labels
performs penultimate hop popping

QUESTION 110

What of the followings are best matched for Label Edge Router? (Choose two)

- A. assigns labels to unlabeled packets
- B. handles traffic between multiple VPNs
- C. reads the labels and forwards the packet based on the labels
- D. performs penultimate hop popping

Correct Answer: AB
Section: (none)
Explanation

Explanation/Reference:

Label Switch Router
assigns labels to unlabeled packets
handles traffic between multiple VPNs
reads the labels and forwards the packet based on the labels
performs penultimate hop popping

QUESTION 111

Refer to the exhibit. An IT staff member comes into the office during normal office hours and cannot access devices through SSH Which action should be taken to resolve this issue?

```

interface Ethernet0/0
ip address 10.1.1.1 255.255.255.0
ip access-group 101 in
!
time-range Office-hour
periodic weekdays 08:00 to 17:00
!
access-list 101 permit tcp 10.0.0.0 0.0.0.0 172.16.1.0 0.0.0.255 eq ssh time-range Office-hour
  
```

- A. Modify the access list to use the correct IP address.
- B. Configure the correct time range.
- C. Modify the access list to correct the subnet mask
- D. Configure the access list in the outbound direction.

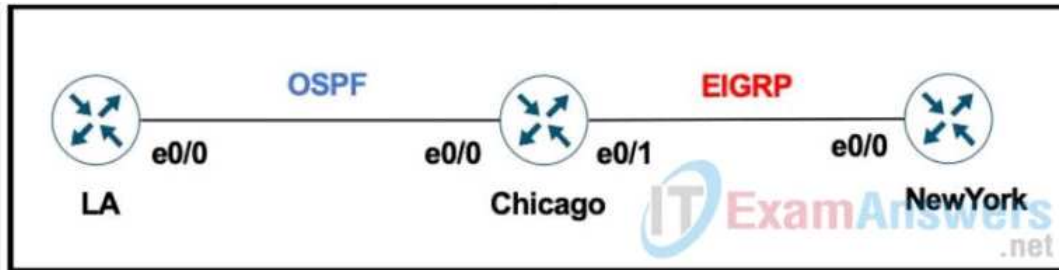
Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

If the staff wants to make SSH connection to the router, source and destination addresses settings in the access list should be swapped.

QUESTION 112

Refer to The exhibit.



The network administrator must mutually redistribute routes at the Chicago router to the LA and NewYork routers. The configuration of the Chicago router is this:

```
router ospf 1
 redistribute eigrp 100
router eigrp 100
 redistribute ospf 1
```

After the configuration, the LA router receives all the NewYork routes, but NewYork router does not receive any LA routes. Which set of configurations fixes the problem on the Chicago router?

- A. router ospf 1
redistribute eigrp 100 metric 20
- B. router eigrp 100
redistribute ospf 1 metric 10 10 10 10
- C. router eigrp 100
redistribute ospf 1 subnets
- D. router ospf 1
redistribute eigrp 100 subnets

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Metric has to be configured when redistributing routes from other routing protocols into EIGRP

QUESTION 113

Refer to the exhibit. BGP is flapping after the Copp policy is applied. What are the two solutions to fix the issue? (Choose two)

```
policy-map COPP-7600
class COPP-CRITICAL-7600
 police cir 2000000 bc 62500
 conform-action transmit
 exceed-action transmit
!
class class-default
 police cir 200000 bc 6250
 conform-action transmit
 exceed-action drop
!
class-map match-all COPP-CRITICAL-7600
 match access-group name COPP-CRITICAL-7600
!
ip access-list extended COPP-CRITICAL-7600
 permit ip any any eq http
 permit ip any any eq https
```

- A. Configure BGP in the COPP-CRITICAL-7600 ACL
- B. Configure a higher value for CIR under the default class to allow more packets during peak traffic
- C. Configure a higher value for CIR under the class COPP-CRITICAL-7600
- D. Configure a three-color policer instead of two-color policer under class COPP-CRITICAL-7600
- E. Configure IP CEF to CoPP policy and BGP to work

Correct Answer: AB

Section: (none)

Explanation

Explanation/Reference:

In the original configuration, BGP is matched by class-default.

QUESTION 114

Refer to the exhibit. Which two actions should be taken to access the server? (Choose two.)

```

Router#show access-lists
Standard IP access list 1
    10 permit 192.168.2.2 (1 match)
Router#
Router#show route-map
route-map RM-OSPF-DL, deny, sequence 10
  Match clauses:
    ip address (access-lists): 1
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
Router#
Router#show running-config | section ospf
router ospf 1
  network 192.168.1.1 0.0.0.0 area 0
  network 192.168.12.0 0.0.0.255 area 0
  distribute-list route-map RM-OSPF-DL in
Router#

```

- A. Modify the access list to add a second line of permit ip any
- B. Modify the access list to deny the route to 192.168.2.2.
- C. Modify distribute list seq 10 to permit the route to 192.168.2.2.
- D. Add a sequence 20 in the route map to permit access list 1.
- E. Add a floating static route to reach to 192.168.2.2 with administrative distance higher than OSPF

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

In the setup:

- the route 192.168.2.2 will be prevented from entering routing table due to Route Map statement sequence 10.
- all other routes will also be prevented from entering since there are no more Route Map statement to permit them.

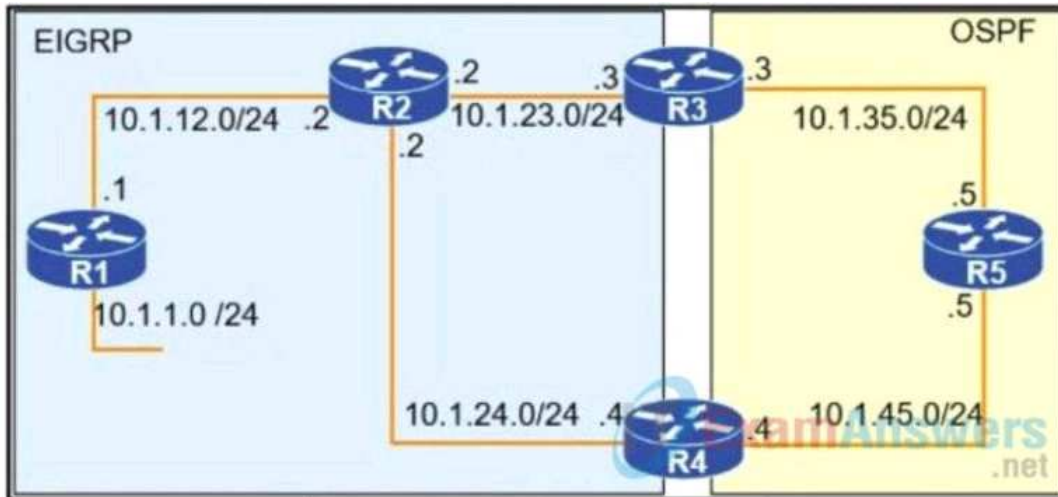
Both C and E adds the route to 192.168.2.2 since:

- C allows the route 192.168.2.2 by changing the route map seq 10 from deny to permit.
- E adds the route 192.168.2.2 as static route manually.

No choice can allow other routes e.g. A is wrong since standard access list allows "permit any" (not "permit ip any").

QUESTION 115

Refer to the exhibit.



```

R1
router eigrp 1
 redistribute connected
 network 10.1.12.1 0.0.0.0
 default-metric 1000000 10 255 1 1500

R3
router eigrp 1
 network 10.1.23.3 0.0.0.0
!
router ospf 1
 redistribute eigrp 1 subnets
 network 10.1.35.3 0.0.0.0 area 0

```

To provide reachability to network 10.1.1.0/24 from R5, the network administrator redistributes EIGRP into OSPF on R3 but notices that R4 is now taking a sub-optimal path through R5 to reach 10.1.1.0/24 network. Which action fixes the issue while keeping the reachability from R5 to 10.1.1.0/24 network?

- Change the administrative distance of the external EIGRP to 90.
- Apply the outbound distribution list on R5 toward R4 in OSPF.
- Change the administrative distance of OSPF to 200 on R5.
- Redistribute OSPF into EIGRP on R4.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

For 10.1.1.0/24:

- R4 has a EIGRP external route (AD 170) since R1 is redistributing the connected network.
- R4 also has a OSPF external route (AD 110) since R3 is redistributing EIGRP to OSPF

By default R4 will prefer OSPF since it has a lower AD unless either:

- increase the AD of OSPF external; OR
- decrease the AD of EIGRP external (i.e. answer A).

QUESTION 116

An engineer configured a company's multiple area OSPF head office router and Site A cisco routers with VRF lite. Each site router is connected to a PE router of an MPLS backbone.

Head Office & Site A

```

ip cef
ip vrf abc
rd 101:101
!
interface FastEthernet0/0
ip vrf forwarding abc
ip address 172.16.16.X 255.255.255.252
!
router ospf 1 vrf abc
log-adjacency-changes
network 172.16.16.0 0.0.0.255 area 1

```

After finishing both site router configurations, none of the LSA 3,4 5, and 7 are installed at Site A router. Which configuration resolves this issue?

- configure capability vrf-lite on Site A and its connected PE router under router ospf 1 vrf abc

- B. configure capability vrf-lite on Head Office and its connected PE router under router ospf 1 vrf abc
- C. configure capability vrf-lite on both PE routers connected to Head Office and Site A routers under routtr ospf 1 vrf abc
- D. configure capability vrf-lite on Head Office and Site A routers under router ospf 1 vrf abc

Correct Answer: D

Section: (none)

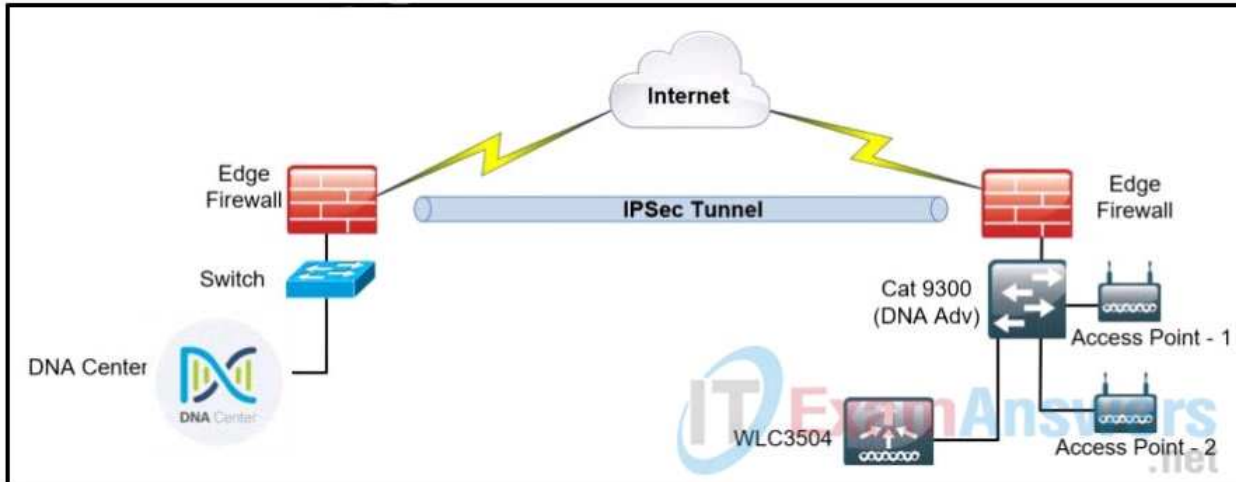
Explanation

Explanation/Reference:

You need to configure "capability vrf-lite" in each customers' router that is configured with VRF Lite.

QUESTION 117

Refer to the exhibit. A network administrator is discovering a Cisco Catalyst 9300 and a Cisco WLC 3504 in Cisco DNA Center. The Catalyst 9300 is added successfully However the WLC is showing [error "uncontactable" when the administrator tries to add it in Cisco DNA Center. Which action discovers WLC in Cisco DNA Center successfully?



- A. Copy the .cert file from the Cisco DNA Center on the USB and upload it to the WLC 3504.
- B. Delete the WLC 3504 from Cisco DNA Center and add it to Cisco DNA Center again.
- C. Add the WLC 3504 under the hierarchy of the Catalyst 9300 connected devices.
- D. Copy the .pem file from the Cisco DNA Center on the USB and upload it to the WLC 3504.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 118

During the maintenance window, an administrator accidentally deleted the telnet-related configuration that permits a Telnet connection from the inside network (Eth 0/0) to the outside of the network between Friday-Sunday night hours only. Which configuration resolves the issue?

- A.

```
interface Ethernet0/0
ip address 10.1.1.1 255.255.255.0
ip access-group 101 in
!
access-list 101 permit udp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255
eq telnet time-range changewindow
!
time-range changewindow
periodic Friday Saturday Sunday 22:00 to 05:00
```
- B.

```
interface Ethernet0/0
ip address 10.1.1.1 255.255.255.0
ip access-group 101 in
!
access-list 101 permit tcp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255
eq telnet time-range changewindow
!
time-range changewindow
periodic 22:00 to 05:00
```

- C. interface Ethernet0/0
ip address 10.1.1.1 255.255.255.0
ip access-group 101 in
!
access-list 101 permit tcp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255
eq telnet time-range changewindow
!
time-range changewindow
periodic Friday Saturday Sunday 22:00 to 05:00
- D. interface Ethernet0/0
ip address 10.1.1.1 255.255.255.0
ip access-group 101 in
!
access-list 101 permit udp 10.1.1.0 0.0.0.255 172.16.1.0 0.0.0.255
eq telnet time-range changewindow
!
time-range changewindow
periodic 22:00 to 05:00

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

A is wrong. Although time-range is correct, telnet is not UDP.

QUESTION 119

Refer to the exhibit. An engineer has configured DMVPN on a spoke router. What is the WAN IP address of another spoke router within the DMVPN network?

```
Spoke# show dmvpn
Tunnel0, Type:Spoke, NHRP Peers:2,
# Ent Peer NBMA Addr Peer Tunnel Add State UpDn Tm Attrb
-----
1 172.18.16.2 192.168.1.1 UP 01:05:35 S
1 172.18.46.2 192.168.1.4 UP 00:00:25 D
```

- A. 172.18.46.2
B. 192.168.1.4
C. 172.18.16.2
D. 192.168.1.1

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The second column is "Peer NBMA Addr" i.e. the WAN IP.

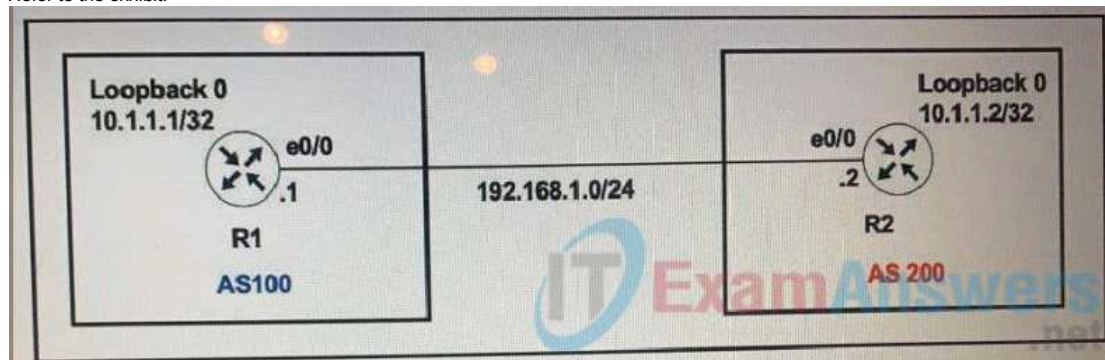
In the last column:

S is static entry and therefore it is the Hub router's entry.

D is static entry and therefore it is the Spoke router's entry.

QUESTION 120

Refer to the exhibit.



```

R1
router bgp 100
neighbor 10.1.1.2 remote-as 200

```

```

R2
router bgp 200
neighbor 10.1.1.1 remote-as 100

```

The neighbor is not coming up. Which two sets of configurations bring the neighbors up? (Choose two.)

A. R2

```

ip route 10.1.1.1 255.255.255.255 192.168.1.1
!
router bgp 200
neighbor 10.1.1.1 disable-connected-check
neighbor 10.1.1.1 update-source loopback 0

```

B. R2

```

ip route 10.1.1.1 255.255.255.255 192.168.1.1
!
router bgp 200
neighbor 10.1.1.1 ttl-security hops 1
neighbor 10.1.1.1 update-source loopback 0

```

C. R1

```

ip route 10.1.1.2 255.255.255.255 192.168.1.2
!
router bgp 100
neighbor 10.1.1.2 disable-connected-check
neighbor 10.1.1.2 update-source Loopback0

```

D. R1

```

ip route 10.1.1.2 255.255.255.255 192.168.1.2
!
router bgp 100
neighbor 10.1.1.1 ttl-security hops 1
neighbor 10.1.1.2 update-source loopback 0

```

E. R2

```

ip route 10.1.1.2 255.255.255.255 192.168.1.2
!
router bgp 100
neighbor 10.1.1.2 ttl-security hops 1
neighbor 10.1.1.2 update-source loopback 0

```

Correct Answer: AC

Section: (none)

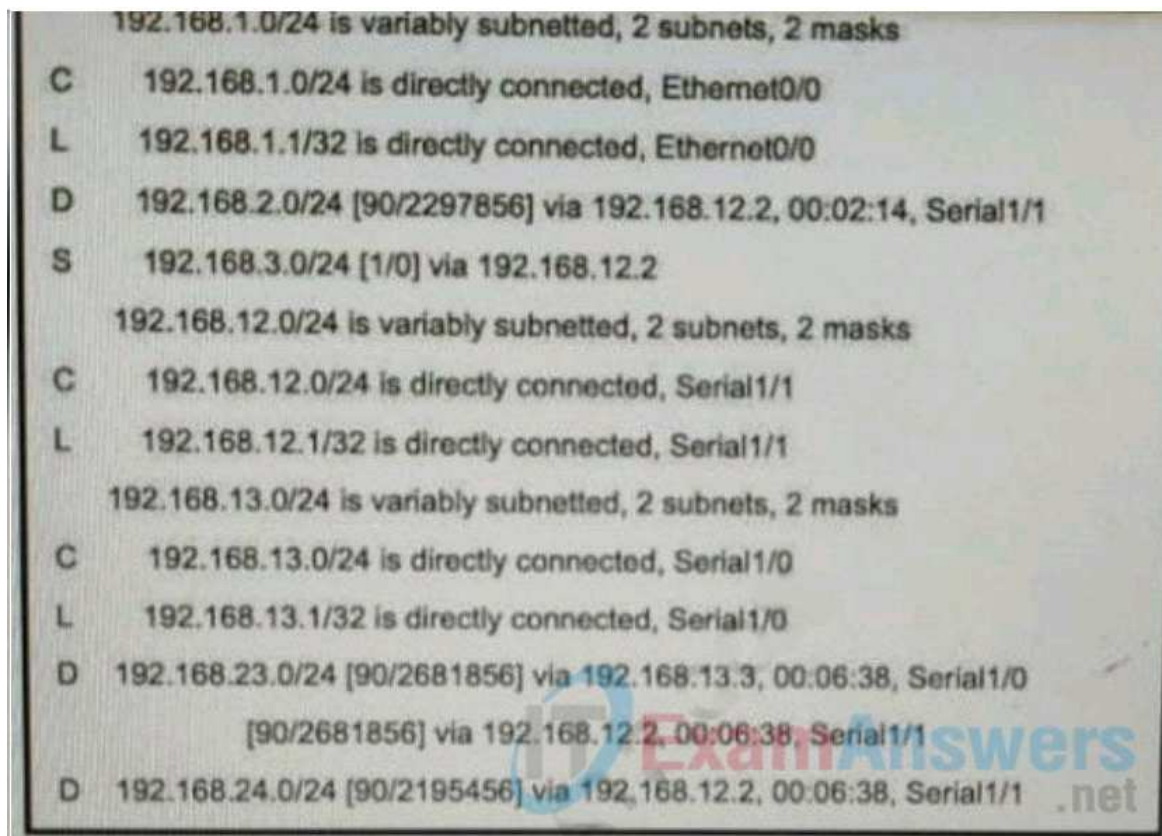
Explanation

Explanation/Reference:

Since R1 and R2 is connecting to each other with no router in between, in order to form neighbor with loopback address, you can just disable connected network checking i.e. without the need to use "ebgp-multihop" which modifies the TTL value.

QUESTION 121

Refer to the exhibit. All the serial links between R1, R2, and R3 have the same bandwidth. Users on 192.168.1.0/24 (R1) network report slow responses on network 192.168.3.0/24 (R3). A traceroute is run on the path, it shows that the packet is getting forwarded via R2 to R3 although the link between R1 and R3 are up. What can the network administrator do to fix the slowness.



- A. Change the Administrative Distance of EIGRP to 5
- B. Add a static route on R1 using the next hop of R3
- C. Remove the static route on R1.
- D. Redistribute the R1 route to EIGRP

Correct Answer: C
Section: (none)
Explanation

Explanation/Reference:

The packet to 192.168.3.0/24 matches the static route and send to the next hop 192.168.12.2 (probably R2).
B can be an option but this is not the best answer.

if EIGRP is properly configured in all routers, you can just use the choice C (i.e. remove the static route). The EIGRP route for network 192.168.3.0/24 learnt from R3 should be able to enter into R1's routing table.

QUESTION 122

Refer to the exhibit. An IP SLA is configured to use the backup default route when the primary is down, but it is not working as desired. Which command fixes the issue?

```
R1(config)# ip route 0.0.0.0 0.0.0.0 1.1.1.1
R1(config)# ip route 0.0.0.0 0.0.0.0 2.2.2.2 10
R1(config)# ip sla 1
R1(config)# icmp-echo 1.1.1.1 source-interface FastEthernet0/0
R1(config)# ip sla schedule 1 life forever start-time now
R1(config)# track 1 ip sla 1 reachability
```

- A. R1(config)# ip route 0.0.0.0 0.0.0.0 2.2.2.2 10 track 1
- B. R1(config)# ip route 0.0.0.0 0.0.0.0 2.2.2.2
- C. R1(config)# ip sla track 1
- D. R1(config)# ip route 0.0.0.0 0.0.0.0 1.1.1.1 track 1

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 123

What are two functions of IPv6 Source Guard? (Choose two.)

- A. It uses the populated binding table for allowing legitimate traffic.

- B. It works independent from IPv6 neighbor discovery.
- C. It denies traffic from unknown sources or unallocated addresses.
- D. It denies traffic by inspecting neighbor discovery packets for specific pattern.
- E. It blocks certain traffic by inspecting DHCP packets for specific sources.

Correct Answer: AC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 124

What is the output of the following command:

`show ip vrf`

- A. Show's default RD values
- B. Displays IP routing table information associated with a VRF
- C. Show's routing protocol information associated with a VRF.
- D. Displays the ARP table (static and dynamic entries) in the specified VRF

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 125

An engineer configured access list NON-CISCO in a policy to influence routes. What are the two effects of this route map configuration? (Choose two.)

route-map PBR, deny, sequence 5

Match clauses:

ip address (access-list): NON-CISCO

Set clauses:

Policy routing matches: 0 packets, 0 bytes

route-map PBR, permit, sequence 10

Match clauses:

Set clauses:

ip next-hop 192.168.1.5

Policy routing matches: 388213827 packets, 222009685077 bytes

- A. Packets are not evaluated by sequence 10.
- B. Packets are evaluated by sequence 10.
- C. Packets are forwarded to the default gateway.
- D. Packets are forwarded using normal route lookup.
- E. Packets are dropped by the access list.

Correct Answer: BC

Section: (none)

Explanation

Explanation/Reference:

Packets matching NON-CISCO, since it is a deny Route Map statement, will be routed through normal routing table. All other packets (no match clause in seq 10) will be routed to the next hop 192.168.1.5..

A and E are wrong.

B, C and D can be correct.

However in the counters of the above output, there is no packet matching sequence 5 i.e. no packets are using normal route lookup (D seems wrong).

Hence:

- all packets are evaluated by sequence 10 (choice B)

- all packets are sent to the next hop 192.168.1.5 (no choice matching).

However if 192.168.1.5 is considered as "the default gateway", C may be the answer.

QUESTION 126

Refer to the exhibit. An engineer identifier a Layer 2 loop using DNAC. Which command fixes the problem in the SFD9300-1 switch?

Filter

Priority	Issue Type	Device Role	Category	Issue Count	Site Count (Area)	Device Count
P2	Layer 2 loop symptoms	DISTRIBUTION	Connectivity	43	1	2

Layer 2 loop symptoms

2

Open Issues

1

Area
1 Buildings, 0 Floors

2

DISTRIBUTION

Filter

Issue	Site	Device	Device Type	Issue Count
Host flaps observed in 1 VLAN(s)	USA/SF	SF-D9300-1	Cisco Catalyst 9300 Switch	24
Host flaps observed in 1 VLAN(s)	USA/SF	SF-D9300-2	Cisco Catalyst 9300 Switch	24

Potential Loop Details

EQ Find

	Device	Role	Port in loop	Duplex	VLAN in loop
●	SF-D9300-1	DISTRIBUTION	GigabitEthernet1/0/13	Full	30-33
	SF-D9300-2	DISTRIBUTION	GigabitEthernet1/0/13	Full	30-33
●	SF-D9300-1	DISTRIBUTION	GigabitEthernet1/0/23	Full	30-33
	SF-A3850-1	ACCESS	GigabitEthernet1/0/23	Full	30-33

```

interface GigabitEthernet1/0/13
 switchport trunk allowed vlan 30-33
 switchport mode trunk
!
interface GigabitEthernet1/0/23
 switchport trunk allowed vlan 30-33
 switchport mode trunk

```

- A. spanning-tree portfast bpduguard
- B. no spanning-tree uplinkfast
- C. spanning-tree loopguard default
- D. spanning-tree backbonefast

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

The loop is probably formed due to unidirectional link.

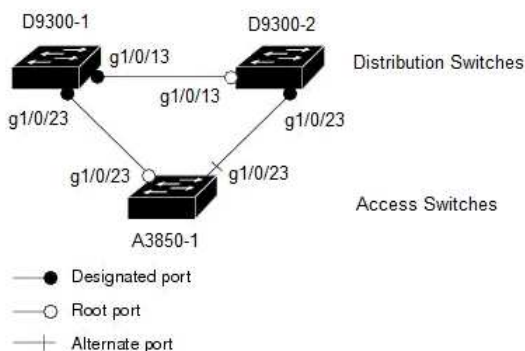
This can be prevented by loopguard.

portfast bpdugard is not applicable since the ports are connecting to another switch and therefore can receives BPDU

Remarks :

According to the output in DNA, assume there are only three switches, the conneciton will be:

(Actually connection between D9300-2 and AS 3850-1 is not shown in the output).



There is a common scenerio where you should configured loopguard in ports g1/0/13 and g1/0/23 in all the three switches.

Note that:

For D9300-1 and D9300-2, since they are distribution switches, if all their ports are connecting to other switches, you can configure loopguard as default.

For AS3850-1, since it is a access switches, you should only configure loopguard in the interfaces connecting to the distribution switches.

QUESTION 127

Refer to the exhibit.

```

permit tcp any any
deny ipv6 any any log
!
interface gi0/0
ipv6 traffic-filter inbound out

```

A network administrator configured an IPv6 access list to allow TCP return frame only, but it is not working as expected. Which changes resolve this issue?

- A.

```

ipv6 access-list inbound
permit tcp any any established
deny ipv6 any any log
!
interface gi0/0
ipv6 traffic-filter inbound out

```
- B.

```

ipv6 access-list inbound
permit tcp any any syn
deny ipv6 any any log
!
interface gi0/0
ipv6 traffic-filter inbound out

```
- C.

```

ipv6 access-list inbound
permit tcp any any established
deny ipv6 any any log
!
interface gi0/0
ipv6 traffic-filter inbound in

```
- D.

```

ipv6 access-list inbound
permit tcp any any syn
deny ipv6 any any log
!
interface gi0/0
ipv6 traffic-filter inbound in

```

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Option A:

This setup permits established TCP traffic, which is correct for allowing return traffic. However, it applies the ACL on the outbound direction (out). This would filter outgoing traffic based on these rules, which is not the intent stated in the question.

OptionB:

This configuration permits TCP packets with the SYN flag, typically used for initiating new connections, and applies the filter to outgoing traffic. This is incorrect for allowing return traffic only and also incorrectly oriented for incoming return traffic.

Option C

This option correctly permits established TCP traffic, filtering for packets that are part of an existing connection. It applies the ACL to incoming traffic (in), which is correct for filtering return traffic entering the interface.

Option D:

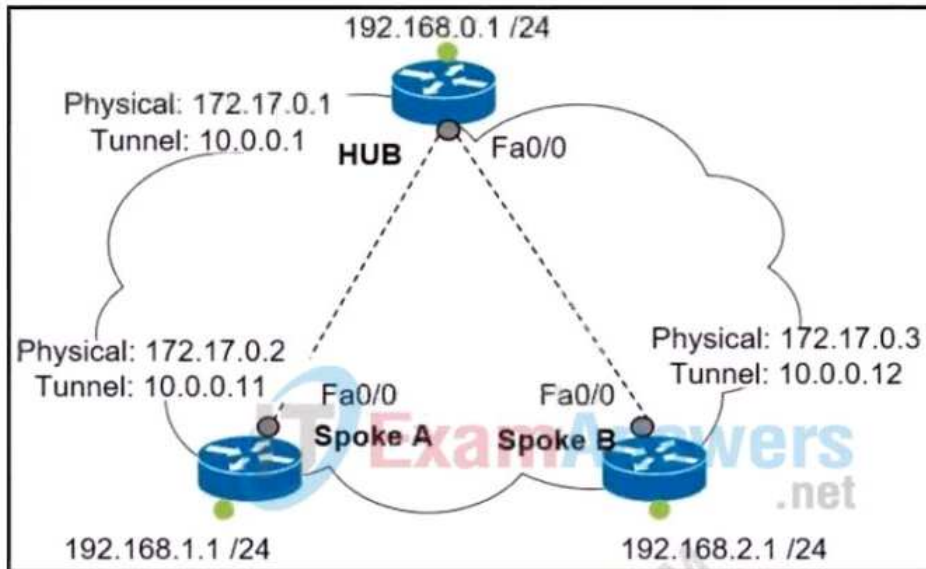
This configuration allows TCP SYN packets which are used to initiate connections, not return traffic. It applies the filter to incoming traffic, but the rule itself does not fulfill the requirement to allow only return traffic.

Conclusion:

The correct answer is Option C. It uses the established keyword to allow only TCP return traffic (i.e., packets part of an established TCP connection) and applies this rule to incoming traffic (in) on interface gi0/0. This setup ensures that only return traffic for already established sessions is allowed into the network, aligning with the requirement described.

QUESTION 128

Refer to the exhibit. Which interface configuration must be configured on the HUB router to enable **MVPN** with mGRE mode?



- A. `interface Tunnel0`
`description mGRE – DMVPN Tunnel`
`ip address 10.1.0.1 255.255.255.0`
`ip nhrp map multicast dynamic`
`ip nhrp network-id 1`
`tunnel source 172.17.0.1`
`ip nhrp map 10.0.0.11 172.17.0.2`
`ip nhrp map 10.0.0.12 172.17.0.3`
`tunnel mode gre`
- B. `interface Tunnel0`
`description mGRE – DMVPN Tunnel`
`ip address 10.0.0.1 255.255.255.0`
`ip nhrp map multicast dynamic`
`ip nhrp network-id 1`
`tunnel source 10.0.0.1`
`tunnel mode gre multipoint`
- C. `interface Tunnel0`
`description mGRE – DMVPN Tunnel`
`ip address 10.0.0.1 255.255.255.0`
`ip nhrp network-id 1`
`tunnel source 172.17.0.1`
`tunnel mode gre multipoint`
- D. `interface Tunnel0`
`description mGRE – DMVPN Tunnel`
`ip address 10.0.0.1 255.255.255.0`
`ip nhrp map multicast dynamic`
`ip nhrp network-id 1`
`tunnel source 10.0.0.1`
`tunnel destination 172.17.0.2`
`tunnel mode gre multipoint`

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

MVPN with mgre mode means DMVPN that supports Multicast.

For the tunnel interface with mgre:

- It uses physical / WAN / NBMA address as tunnel source i.e. 172.17.0.1
- It does NOT have any tunnel destination.
- It has tunnel mode "gre multipoint"

Only C matches all the above three.

Moreover:

- a Hub router has no NHRP mapping for the Spoke routers. Mappings are learnt through NHRP registration.
- For multicast, "ip nhrp map multicast dynamic" is required but it is default settings in newer IOS versions. (Choice B looks OK for this but its tunnel source is wrong).

QUESTION 129

Which is statement about IPv6 inspection is true?

- A. It learns and secures bindings for stateless autoconfiguration addresses in Layer 3 neighbor tables
- B. It learns and secures bindings for stateful autoconfiguration addresses in Layer 3 neighbor tables
- C. It learns and secures bindings for stateful autoconfiguration addresses in Layer 2 neighbor tables
- D. It learns and secures binding for stateless autoconfiguration addresses in Layer 2 neighbor tables.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Same as Q43. The only difference is "IPv6 ND inspection" vs "IPv6 inspection". But both mean the same feature.

QUESTION 130

Which feature drops packets if the source address is not found in the snooping table?

- A. IPv6 Source Guard
- B. IPv6 Destination Guard
- C. IPv6 Prefix Guard
- D. Binding Table Recovery

Correct Answer: A

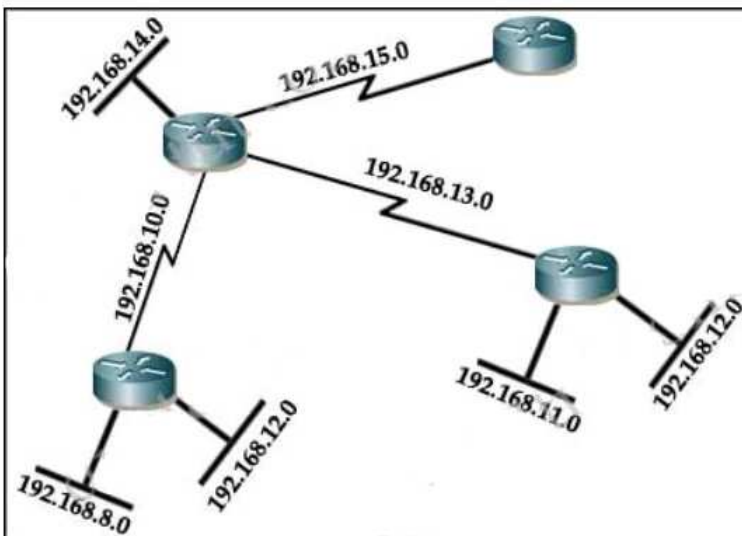
Section: (none)

Explanation

Explanation/Reference:

QUESTION 131

Given the network diagram, which address would successfully summarize only the networks seen?



- A. 192.168.0.0/24
- B. 192.168.8.0/20
- C. 192.168.8.0/21
- D. 192.168.12.0/20
- E. 192.168.16.0/21
- F. These networks cannot be summarized.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Among the networks shown in the diagram:
 - the smallest number in the third octets is 8.
 - the largest number in the third octets is 15.

Hence they are in the range 8 - 15. Since 8 is a number of 2^3 and 15 is smaller than 2^4 , therefore the summarized network is simply: 192.168.8.0 with prefix length $24 - 3 = 29$.

QUESTION 132

Refer to the exhibit. An engineer receives this error message when trying to access another router in-band from the serial interface connected to the console of R1. Which configuration is needed on R1 to resolve this issue?

```

R1#show run | begin line
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
  transport preferred telnet
  transport output none
  stopbits 0 4
line vty 0 4
  login
  transport referred telnet
  transport input none
  transport output telnet
R1#

R1#ssh -1 cisco 192.168.12.2
% ssh connections not permitted from this terminal
R1#

```

- A. R1(config)#**line console 0**
R1(config-line)# **transport preferred ssh**
- B. R1(config)#**line vty 0**
R1(config-line)# **transport output ssh**
- C. R1(config)#**line vty 0**
R1(config-line)# **transport output ssh**
R1(config-line)# **transport preferred ssh**
- D. R1(config)#**line console 0**
R1(config-line)# **transport output ssh**

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

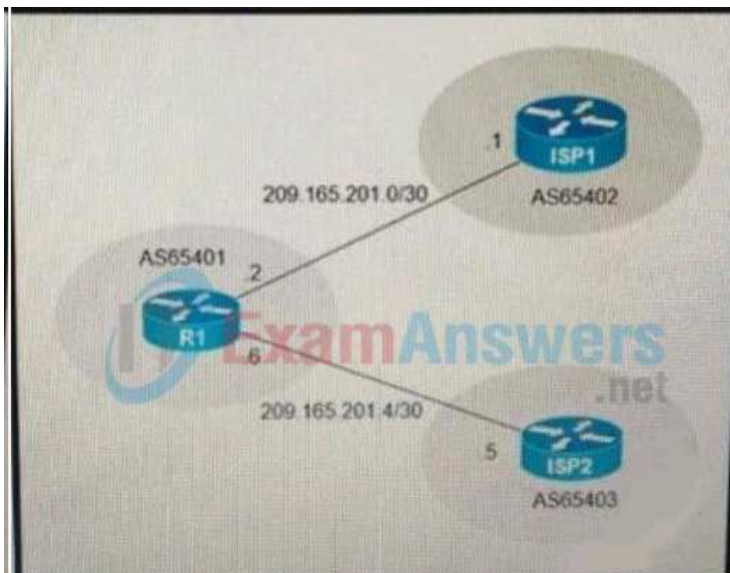
In simple term:

- the engineer is connected to R1 through console port.
- the engineer then wants to make SSH connection to another router through R1's serial interface.

However the settings in R1's console is "transport output none" and therefore he cannot make SSH connection from R1 when having CLI session through R1's console port.

QUESTION 133

Refer to the exhibit. A company with autonomous system number AS65401 has obtained IP address block 209.165.200.224/27 from ISP1. The company needed more IP addresses and was assigned block 209.165.202.128/27 from ISP2. An engineer in ISP1 reports they are receiving ISP2 routes from AS65401. Which configuration on R1 resolves the issue?



```

R1#
interface GigabitEthernet0/0
 ip address 209.165.201.2 255.255.255.252
!
interface GigabitEthernet0/1
 ip address 209.165.201.6 255.255.255.252
!
router bgp 65401
 bgp log-neighbor-changes
 redistribute static
 neighbor 209.165.201.1 remote-as 65402
 neighbor 209.165.201.5 remote-as 65403
!
ip route 209.165.200.224 255.255.255.224 Null0
ip route 209.165.202.128 255.255.255.224 Null0
!

```

- A.
- ```

access-list 10 deny 209.165.202.128 0.0.0.31
access-list 10 permit any
!
router bgp 65401
 neighbor 209.165.201.1 distribute-list 10 out

```
- B.
- ```

access-list 10 deny 209.165.202.128 0.0.0.31
access-list 10 permit any
!
router bgp 65401
 neighbor 209.165.201.1 distribute-list 10 in

```
- C.
- ```

ip route 209.165.200.224 255.255.255.224 209.165.201.1
ip route 209.165.202.128 255.255.255.224 209.165.201.5

```
- D.
- ```

ip route 0.0.0.0 0.0.0.0 209.165.201.1
ip route 0.0.0.0 0.0.0.0 100 209.165.201.5

```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The problem occurs since R1 is advertising routes for ISP2's IP addresses to ISP1.

The answer A filter 209.165.202.128/27 (IP obtained from ISP2) to prevent it from being advertised to 209.165.201.1 i.e. ISP1's BGP router.

QUESTION 134

Refer to the exhibit. Which two actions restrict access to router R1 by SSH? (Choose two.)

```
R1#show policy-map control-plane
Control Plane
Service-policy input: CoPP
Class-map: PERMIT (match-all)
  50 packets, 3811 bytes
  5 minute offered rate 0000 bps
  Match: access-group 100
Class-map: ANY (match-all)
  210 packets, 19104 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
  Match: access-group 199
  drop
Class-map: class-default (match-any)
  348 packets, 48203 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
  Match: any

R1#show access-list 100
Extended IP access list 100
  10 permit udp any any eq 23 (100 matches)
  20 permit tcp any any eq telnet (5 matches)
  30 permit tcp any any eq telnet any (10 matches)

R1#show access-list 199
Extended IP access list 199
  10 deny tcp any any eq telnet any (50 matches)
  50 permit ip any any (1 match)

R1#show running-config | section line vty
line vty 0 4
login
transport input telnet ssh
transport output telnet ssh
```

- A. Configure transport input ssh on line vty and remove sequence 30 from access list 100.
- B. Configure transport output ssh on line vty and remove sequence 20 from access list 100.
- C. Remove class-map ANY from service-policy CoPP
- D. Configure transport output ssh on line vty and remove sequence 10 from access list 199.
- E. Remove sequence 10 from access list 100 and add sequence 20 deny tcp any any eq telnet to access list 199

Correct Answer: AB

Section: (none)

Explanation

Explanation/Reference:

The question should mean only allow SSH access to R1 (i.e. telnet is not allowed).

The choices available is very strange since there is many unrelated actions included:

The first part of A will allow SSH connection only. (Second part is unrelated)

The second part of B prevent telnet to R1 from being matched to ACL 100. Telnet packets will be dropped in CoPP due to seq 50 in ACL 199. (First part is unrelated)

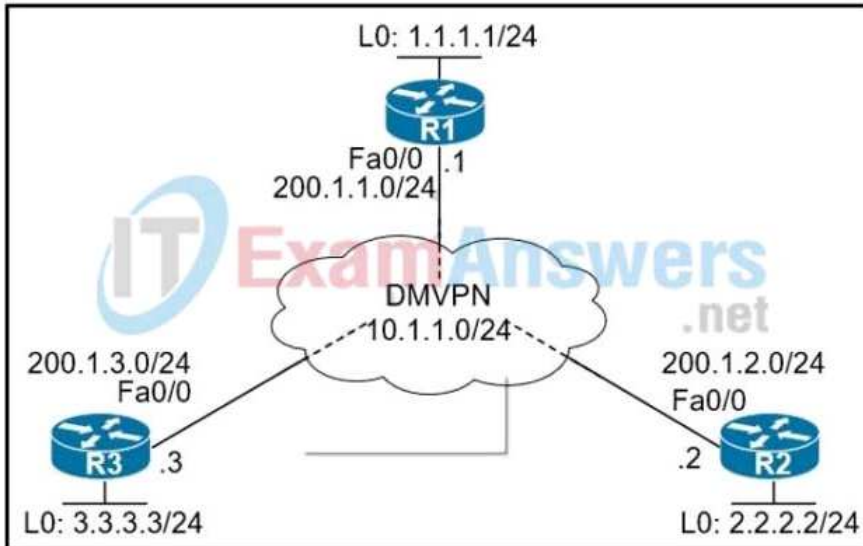
C will remove the only drop actions in input CoPP and therefore will not block anything. (not the answer).

Both first and second part of the choice D has nothing to do with telnet connection to R1.

Both first and second part of the choice E has nothing to do with telnet connection to R1. For denying telnet in 199, this only means that telnet packets to R1 will not be dropped by Route Map.

QUESTION 135

Refer to the exhibits. When DMVPN is configured, which configuration allows spoke-to-spoke communication using loopback as tunnel source?



R2:

=====

```
R2(config)# crypto isakmp policy 10
R2(config-isakmp)# hash md5
R2(config-isakmp)# authentication pre-share
R2(config-isakmp)# group 2
R2(config-isakmp)# encryption 3des
R2(config)# crypto ipsec transform-set TSET esp-des esp-md5-hmac
R2(cfg-crypto-trans)# mode transport
R2(config)# crypto ipsec profile TST
R2(ipsec-profile)# set transform-set TSET
R2(config)# interface tunnel 123
R2(config-if)# tunnel protection ipsec profile TST
```

- A. Configure crypto isakmp key cisco address 0.0.0.0 on the hub
- B. Configure crypto isakmp key Cisco address 200.1.0.0 255.255.0.0 on the hub
- C. Configure crypto isakmp key cisco address 200.1.0.0 255.255.0.0 on the spokes.
- D. Configure crypto isakmp key cisco address 0.0.0.0 on the spokes.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

You can use a loopback as tunnel source. In this case, the NBMA address being used in configuration will be that of loopback interface instead of the physical WAN interface (e.g. NHS mapping setting in Spoke routers). However, before any tunnel is formed, all routers should have routes to each other loopback interfaces.

In this question, using loopback as tunnel sources is not important in choosing the answer.

QUESTION 136

Refer to the exhibit. A client is concerned that passwords are visible when running this show archive log config all. Which router configuration is needed to resolve this issue?

```

MASS-RTR#show running-config
!
hostname MASS-RTR
!
aaa new-model
!
aaa authentication login default local
aaa authorization exec default local
aaa authorization commands 15 default local
!
username admin privilege 15 password 7 0236244818115F3348
username cisco privilege 15 password 7 0607072C494A5B
archive
 log config
  logging enable
  logging size 1000
!
interface GigabitEthernet0/0
 ip address dhcp
 duplex auto
 speed auto
!
line vty 0 4
!

```

```

MASS-RTR#show archive log config all

```

idx	sess	user@line	Logged command
1	1	console@console	interface GigabitEthernet0/0
2	1	console@console	no shutdown
3	1	console@console	ip address dhcp
4	2	admin@vty0	username cisco privilege 15 password cisco
5	2	admin@vty0	!config: USER TABLE MODIFIED

- A. MASS-RTR(config-archive-log-cfg)#password encryption aes
- B. MASS-RTR(config)#aaa authentication arap
- C. MASS-RTR(config)#service password-encryption
- D. MASS-RTR(config-archive-log-cfg)#hidekeys

Correct Answer: D

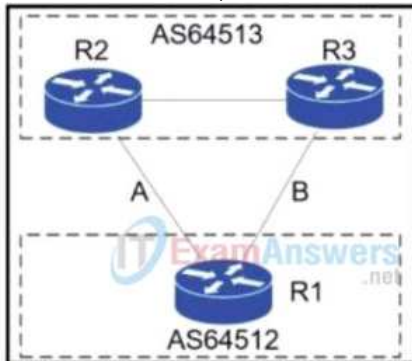
Section: (none)

Explanation

Explanation/Reference:

QUESTION 137

Refer to the exhibit. A network engineer for AS64512 must remove the inbound and outbound traffic from link A during maintenance without closing the BGP session so that there is still a backup link over link B toward the ASN. Which BGP configuration on R1 accomplishes this goal?



- A.
 - route-map link-a-in permit 10
 - set weight 200
 - route-map link-a-out permit 10
 - set as-path prepend 64512
 - route-map link-b-in permit 10
 - set weight 100
 - route-map link-b-out permit 10
- B.
 - route-map link-a-in permit 10
 - set weight 200
 - route-map link-a-out permit 10
 - route-map link-b-in permit 10
 - set weight 100
 - route-map link-b-out permit 10
 - set as-path prepend 64512

- C. route-map link-a-in permit 10
 set local-preference 200
 route-map link-a-out permit 10
 route-map link-b-in permit 10
 route-map link-b-out permit 10
 set as-path prepend 64512
- D. route-map link-a-in permit 10
 route-map link-a-out permit 10
 set as-path prepend 64512
 route-map link-b-in permit 10
 set local-preference 200
 route-map link-b-out permit 10

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

route-map link-a-out with AS prepend makes routers in AS64513 prefer to use the route with shorter AS path list learnt from link B.
 route-map link-b-in set a higher local preference for routes learnt from link B.

Since there is only one router in AS64512, set weight can achieve the result. However choice A set the wrong weights for link A and link B making the router to prefer link A.

QUESTION 138

Refer to the exhibit. A user cannot SSH to the router. What action must be taken to resolve this issue?

```

router# show running-config
Building configuration...

!
<output omitted ----!>
!
hostname R1
!
ip domain-name cisco.com
!
crypto key generate rsa modulus 2048
!
username admin privilege 15 secret cisco123
!
access-list 1 permit 10.1.1.0 0.0.0.255
access-list 1 deny any log
!
line vty 0 15
access-class 1 in
login local
!
<output omitted ----!>
!
end
  
```

- A. Configure transport input ssh
 B. Configure transport output ssh
 C. Configure ip ssh version 2
 D. Configure ip ssh source-interface loopback0

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Without further details, since SSH RSA keys are generated, A seems to be the best answer.

QUESTION 139

An engineer configured a leak-map command to summarize EIGRP routes and advertise specifically loopback 0 with an IP of 10.1.1.1.255.255.255.252 along with the summary route. After finishing configuration, the customer complained not receiving summary route with specific loopback address. Which two configurations will fix it? (Choose two.)

```

router eigrp 1
!
route-map Leak-Route deny 10
!
interface Serial 0/0
ip summary-address eigrp 1 10.0.0.0 255.0.0.0 leak-map Leak-Route

```

- A. Configure access-list 1 permit 10.1.1.0.0.0.3.
- B. Configure access-list 1 permit 10.1.1.1.0.0.0.252
- C. Configure access-list 1 and match under route-map Leak-Route.
- D. Configure route-map Leak-Route permit 10 and match access-list 1.
- E. Configure route-map Leak-Route permit 20.

Correct Answer: AD

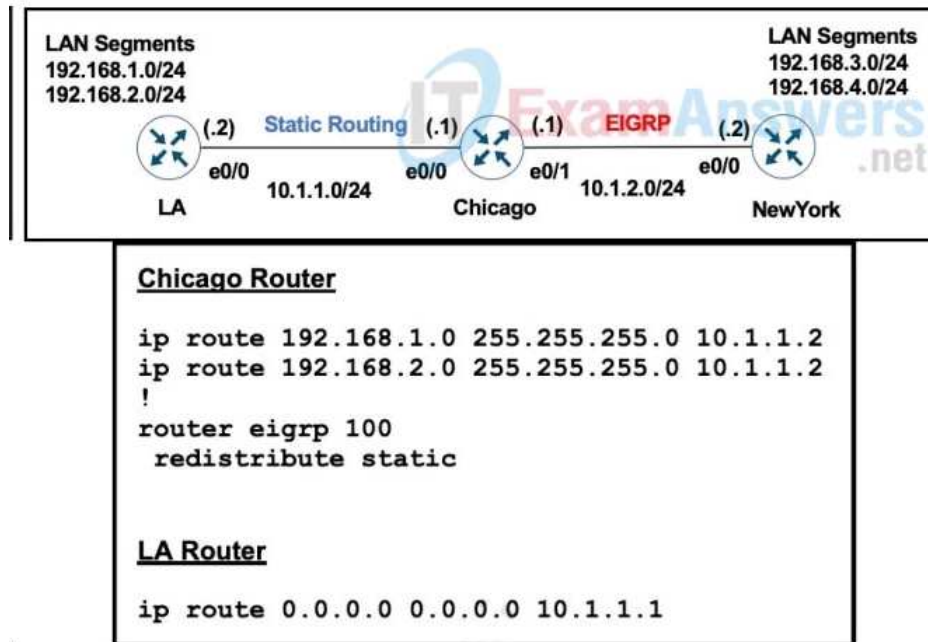
Section: (none)

Explanation

Explanation/Reference:

QUESTION 140

Refer to the exhibits.



A user on the 192.168.1.0/24 network can successfully ping 192.168.3.1, but the administrator cannot ping 192.168.3.1 from the LA router. Which set of configurations fixes the issue?

- A. **Chicago Router**

```

router eigrp 100
 redistribute static metric 10 10 10 10 10

```
- B. **Chicago Router**

```

router eigrp 100
 redistribute connected

```
- C. **Chicago Router**

```

ip route 192.168.3.0 255.255.255.0 10.1.2.2
ip route 192.168.4.0 255.255.255.0 10.1.2.2

```
- D. **LA Router**

```

ip route 192.168.3.0 255.255.255.0 10.1.1.1
ip route 192.168.4.0 255.255.255.0 10.1.1.1

```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

The reason is that New York router has no EIGRP route to 10.1.1.0/24 which is used as source for ping from LA. Other than B, configuring "network 10.1.1.0 0.0.0.255" under EIGRP configuration in Chicago can also solve the problem.

QUESTION 141

What is the role of a route distinguisher via a VRF-Lite setup implementation?

- A. It extends the IP address to identify which VFP instance it belongs to.
- B. It manages the import and export of routes between two or more VRF instances
- C. It enables multicast distribution for VRF-Lite setups to enhance EGP routing protocol capabilities
- D. It enables multicast distribution for VRF-Lite setups to enhance IGP routing protocol capabilities

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 142

Refer to the exhibit. The server for the finance department is not reachable consistently on the 200.30.40.0/24 network and after every second month it gets a new IP address. Which two actions must be taken to resolve this issue? (Choose two.)

```
ip dhcp pool 1
network 200.30.30.0/24
default-router 200.30.30.100
lease 40
!
ip dhcp pool 2
network 200.30.40.0/24
default-router 200.30.40.100
lease 40
!
```

- A. Configure the server to use DHCP on the network with default gateway 200.30.40.100.
- B. Configure the server with a static IP address and default gateway.
- C. Configure the router to exclude a server IP address.
- D. Configure the server to use DHCP on the network with default gateway 200.30.30.100.
- E. Configure the router to exclude a server IP address and default gateway.

Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

The solution requires the reservation for the server.

Although excluding the server IP from the pool meets the requirement, usually the default gateway should also be excluded.

QUESTION 143

Refer to the exhibit. Which routes from OSPF process 5 are redistributed into EIGRP?

```
router eigrp 1

redistribute ospf 5 match external route-map OSPF-TO-EIGRP
metric 10000 2000 255 1 1500
route-map OSPF-TO-EIGRP
match ip address TO-OSPF
```

- A. E1 and E2 subnets matching access list TO-OSPF
- B. E1 and E2 subnets matching prefix list TO-OSPF
- C. only E2 subnets matching access list TO-OSPF
- D. only E1 subnets matching prefix list TO-OS1

Correct Answer: A

Section: (none)

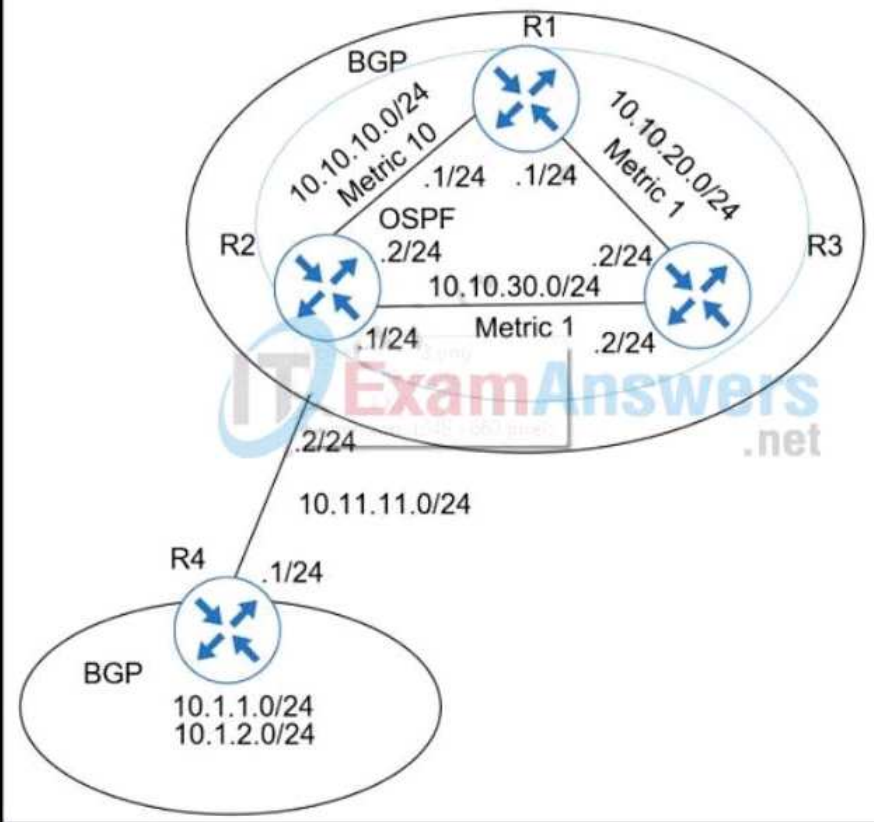
Explanation

Explanation/Reference:

QUESTION 144

Refer to the exhibit. A user has set up an IP SLA probe to test if a non SLA host web server on IP address 10.1.1.1 accepts HTTP sessions prior to deployment. The probe is failing. Which action should the network administrator recommend for the probe to succeed?

```
ip sla 10
tcp connect 10.1.1.1 80
ip sla schedule 10 life 30 start time now
```



- A. Re-issue the ip sla schedule command.
- B. Add icmp-echo command for the host.
- C. Add the control disable option to the tcp connect
- D. Modify the ip sla schedule frequency to forever.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 145

Refer to the exhibit. R1 is connected with R2 via GigabitEthernet0/0, and R2 cannot ping R1. What action will fix the issue ?

```
*Jun 24 08:54:51.530: IF-EvD(GigabitEthernet0/0): IP Routing reports state transition from DOWN to DOWN
*Jun 24 08:54:52.525: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down
*Jun 24 08:54:52.528: IF-EvD(GigabitEthernet0/0): IP Routing reports state transition from DOWN to DOWN
*Jun 24 08:54:53.215: IF-EvD(GigabitEthernet0/0): IP Routing reports state transition from DOWN to DOWN
*Jun 24 08:54:54.998: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Jun 24 08:54:55.006: IF-EvD(GigabitEthernet0/0): IP Routing reports state transition from DOWN to UP
*Jun 24 08:54:55.998: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

- A. Fix route dampening configured on the router.
- B. Replace the SFP module because it is not supported.
- C. Fix IP Event Dampening configured on the interface.
- D. Correct the IP SLA probe that failed.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

The IP Event Dampening feature introduces a configurable exponential decay mechanism to suppress the effects of excessive interface flapping events on routing protocols and routing tables in the network. This feature allows the network operator to configure a router to automatically identify and selectively dampen a local interface that is flapping. Dampening an interface removes the interface from the network until the interface stops flapping and becomes stable.

The logging message in the exhibit is about IP Event Dampening. Therefore the interface g0/0 is probably removed from network causing ping to fail. For example, you can reduce the half-life period or increase the reuse threshold.

QUESTION 146

Refer to the exhibit. The administrator can see the traps for the failed login attempts, but cannot see the traps of successful login attempts. What command is needed to resolve the issue?

```
login block-for 15 attempts 10 within 120
login on-failure log
login on-success log
archive
log config
logging enable
logging size 300
notify syslog

snmp-server enable traps syslog
snmp-server host 172.16.17.1 public syslog
```

- A. Configure logging history 2
- B. Configure logging history 3
- C. Configure logging history 4
- D. Configure logging history 5

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

The following shows an example of successful login due to "login on-success log":

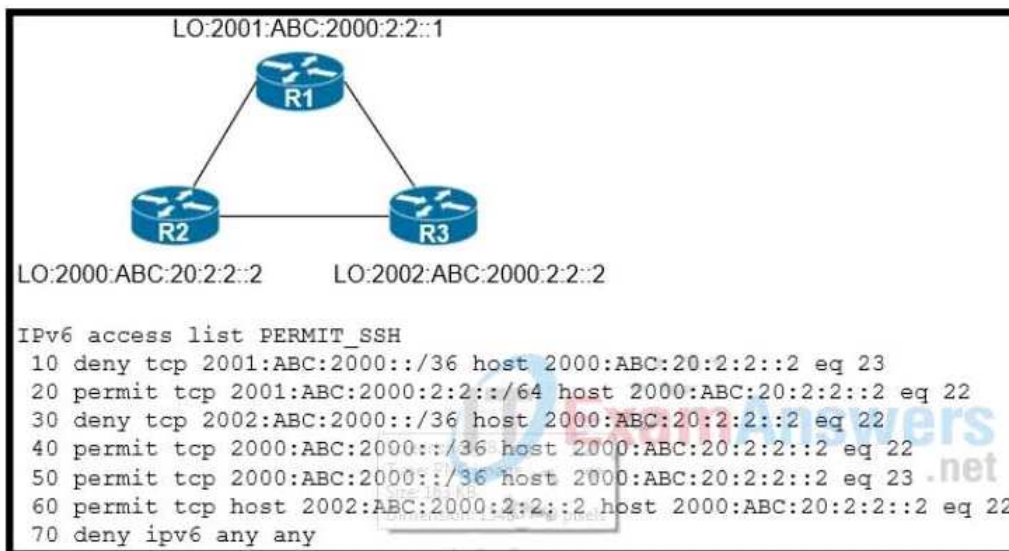
```
%SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: payala] [Source: UNKNOWN] [localport: 22] at 20:03:59 CST Fri Feb 9 2018
```

The following shows an example of failed login due to "login on-failure log"

```
%SEC_LOGIN-4-LOGIN_FAILED: Login failed [user: root] [Source: UNKNOWN] [localport: 22] [Reason: Login Authentication Failed] at 20:08:47 CST Fri Feb 9 2018
```

QUESTION 147

Refer to the exhibit. An IPv6 network was newly deployed in the environment and the help desk reports that R3 cannot SSH to the R2's Loopback interface. Which action resolves the issue?



- A. Modify line 10 of the access list to permit instead of deny.
- B. Remove line 60 from the access list.
- C. Modify line 30 of the access list to permit instead of deny.
- D. Remove line 70 from the access list.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

The question assumes that:

- R3 is using loopback to make SSH connection to R2.
- The IPv6 access list is applied to R2 as inbound access list.

The SSH traffic can be matched by both 30 and 60 and therefore it will be denied by line 30 since it is matched first.

Usually the fix is to move line 60 to a line before 30. However since it is not a choice, the only choice that can fix the SSH connection is C (although it has effects on

other traffic e.g. other IPv6 addresses in 2002:ABC:2000::/36 will also be able to SSH to R2).

QUESTION 148

An engineer configured SNMP notifications sent to the management server using authentication and encrypting data with DES. An error in the response PDU is received as "UNKNOWNUSERNAME. WRONGDIGEST". Which action resolves the issue?

- A. Configure the correct authentication password using SNMPv3 authPriv.
- B. Configure the correct authentication password using SNMPv3 authNoPriv.
- C. Configure correct authentication and privacy passwords using SNMPv3 authNoPriv.
- D. Configure correct authentication and privacy passwords using SNMPv3 authPriv.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

- i. If authentication password is wrong, messages "unknownUserName / wrongDigests" is usually shown.
- ii. Otherwise, if privacy (encryption) password is wrong, usually the message is discarded and there will be no response.

Therefore the message "UNKNOWNUSERNAME. WRONGDIGEST" means incorrect authentication password but the privacy password may or may not be correct.

Since both authentication and encrypting is required, D seems to be a better answer than A.

QUESTION 149

What are two functions of LDP? (Choose two.)

- A. It is defined in RFC 3038 and 3039.
- B. It requires MPLS Traffic Engineering.
- C. It advertises labels per Forwarding Equivalence Class.
- D. It must use Resource Reservation Protocol.
- E. It uses Forwarding Equivalence Class

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

Forwarding Equivalence Class (FEC)—A set of packets with similar characteristics that might be bound to the same MPLS label. An FEC tends to correspond to a label switched path (LSP); however, an LSP might be used for multiple FECs

In MPLS, the Layer 3 header is then mapped into a fixed length, unstructured value called a label. Many different headers can map to the same label, as long as those headers always result in the same choice of the next hop. A label represents a forwarding equivalence class—that is, a set of packets that, however different they may be, are indistinguishable by the forwarding function.

Remarks:

LDP is defined by the IETF (RFC 5036).

QUESTION 150

Refer to the exhibit. An engineer wanted to set a tag of 30 to route 10.1.80.65/32 but it failed How is the issue fixed?

```
R1
ip prefix-list ccnp1 seq 5 permit 10.1.48.0/24 le 24
ip prefix-list ccnp2 seq 5 permit 10.1.80.0/24 le 32
ip prefix-list ccnp3 seq 5 permit 10.1.64.0/24 le 24

route-map ospf-to-eigrp permit 10
  match ip address prefix-list ccnp1
  set tag 30
route-map ospf-to-eigrp permit 20
  match ip address prefix-list ccnp2
  set tag 20
route-map ospf-to-eigrp permit 30
  match ip address prefix-list ccnp3
  set tag 10
```

- A. Modify route-map ospf-to-eigrp permit 30 and match prefix-list ccnp2.
- B. Modify route-map ospf-to-eigrp permit 10 and match prefix-list ccnp2.
- C. Modify prefix-list ccnp3 to add 10.1.64.0/20 le 24
- D. Modify prefix-list ccnp3 to add 10.1.64.0/20 ge 32

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

The route 10.1.80.65/32 can only match to prefix list "ccnp2".
Therefore, it matches route map statement 20 and set with a tag of 20.

For using the action in route map statement 10 i.e. set tag 30, you need to modify that statement to use prefix list "ccnp2".

QUESTION 151

Refer to the exhibit. A junior engineer updated a branch router configuration. Immediately after the change, the engineer receives calls from the help desk that branch personnel cannot reach any network destinations. Which configuration restores service and continues to block 10.1.1.100/32?

```
BRANCH-RTR#
router eigrp 100
 network 10.4.31.0 0.0.0.7
 network 10.100.100.1 0.0.0.0
 distribute-list route-map FILTER-IN in FastEthernet0/0
 eigrp router-id 10.100.100.1
!
ip prefix-list 102 seq 10 permit 10.1.1.100/32
!
route-map FILTER-IN deny 10
 match ip address prefix-list 102
!
```

- A. route-map FILTER-IN deny 5
- B. ip prefix-list 102 seq 15 permit 0.0.0.0/32 le 32
- C. ip prefix-list 102 seq 5 permit 0.0.0.0/32 le 32
- D. route-map FILTER-IN permit 20

Correct Answer: D

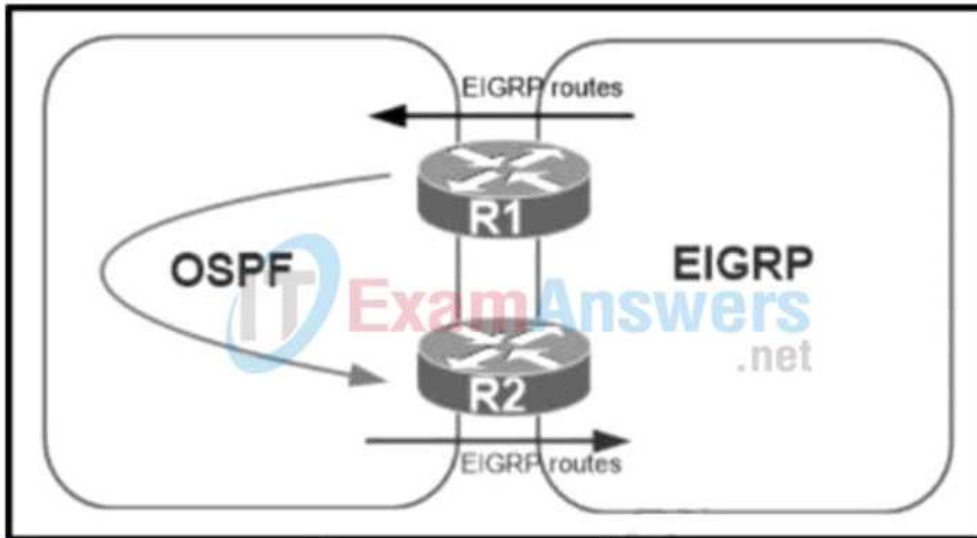
Section: (none)

Explanation

Explanation/Reference:

QUESTION 152

Refer to the exhibit. A network administrator configured mutual redistribution on R1 and R2 routers, which caused instability in the network. Which action resolves the issue?



- A. Set a tag in the route map when redistributing EIGRP into OSPF on R1, and match the same tag on R2 to allow when redistributing OSPF into EIGRP.
- B. Apply a prefix list of EIGRP network routes in OSPF domain on R1 to propagate back into the EIGRP routing domain.
- C. Set a tag in the route map when redistributing EIGRP into OSPF on R1, and match the same tag on R2 to deny when redistributing OSPF into EIGRP.
- D. Advertise summary routes of EIGRP to OSPF and deny specific EIGRP routes when redistributing into OSPF.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 153

Which two statements about VRF-Lite configurations are true? (Choose two.)

- A. They support the exchange of MPLS labels
- B. Different customers can have overlapping IP addresses on different VPNs
- C. They support a maximum of 512,000 routes
- D. Each customer has its own dedicated TCAM resources
- E. Each customer has its own private routing table

F. They support IS-IS

Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 154

Refer to Exhibit. Which statement about redistribution from BGP into OSPF process 10 is true?

```
router ospf 10
  router-id 192.168.1.1
  log-adjacency-changes
  redistribute bgp 1 subnets route-map BGP-TO-OSPF
!
route-map BGP-TO-OSPF deny 10
  match ip address 50
route-map BGP-TO-OSPF permit 20
!
access-list 50 permit 172.16.1.0 0.0.0.255
```

- A. Network 172.16.1.0/24 is not redistributed into OSPF
- B. Network 10.10.10.0/24 is not redistributed into OSPF
- C. Network 172.16.1.0/24 is redistributed with administrative distance of 1.
- D. Network 10.10.10.0/24 is redistributed with administrative distance of 20.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Remarks:

The default cost or metric of a redistributed route is 1 for BGP and 20 for all other protocols.

Hence if there is a BGP route 10.10.10.0/24, after redistribution, the corresponding OSPF route have a metric of 1 (NOT Administrative Distance).

QUESTION 155

Which option is the best for protecting CPU utilization on a device?

- A. fragmentation
- B. COPP
- C. ICMP redirects
- D. ICMP unreachable messages

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 156

Refer to the exhibit. R1 is being monitored using SNMP and monitoring devices are getting only partial information. What action should be taken to resolve this issue?

```

R1#show policy-map control-plane
Control Plane

Service-policy output: CoPP

Class-map: SNMP-Out (match-all)
  124 packets, 3693 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
  Match: access-group name SNMP
  police:
    cir 8000 bps, bc 1500 bytes
    conformed 0 packets, 0 bytes; actions:
      transmit
    exceeded 0 packets, 0 bytes; actions:
      drop
    conformed 0000 bps, exceeded 0000 bps

Class-map: class-default (match-any)
  10 packets, 1003 bytes
  5 minute offered rate 0000 bps, drop rate 0000 bps
  Match: any
R1#show ip access-list SNMP
Extended IP access list SNMP
  10 permit udp any eq snmp any

```

- A. Modify the CoPP policy to increase the configured exceeded limit for SNMP.
- B. Modify the access list to include snmptrap.
- C. Modify the CoPP policy to increase the configured CIR limit for SNMP.
- D. Modify the access list to add a second line to allow udp any any eq snmp

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

For all traffic not matched by the access list SNMP (e.g. SNMP Trap), it will be sent out through the class "class-default" anyway. Hence, the only traffic that will be dropped causing partial response is excessive traffic matched by the access list. In order to avoid dropping, CIR should be increased.
Remarks: The counters shown in the diagram does not match the question since they do show any sign about packet dropping.

QUESTION 157

An engineer configured a Cisco router to send reliable and encrypted notifications for any events to the management server. It was noticed that the notification messages are reliable but not encrypted. Which action resolves the issue?

- A. Configure all devices for SNMPv3 informs with priv.
- B. Configure all devices for SNMPv3 informs with auth.
- C. Configure all devices for SNMPv3 traps with auth.
- D. Configure all devices for SNMPv3 traps with priv.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

For reliable, you need SNMP informs. For encrypted, you need SNMP with priv.

QUESTION 158

Refer to the exhibit. A network administrator is using the DNA Assurance Dashboard panel to troubleshoot an OSPF adjacency that failed between Edge_NYC interface GigabitEthernet1/3 with Neighbor Edge_SNJ. The administrator observes that the neighborship is stuck in exstart state. How does the administrator fix this issue?

Priority	Issue Type	Device Role	Category	Issue Count	Site Count (Building)	Device Count	Last Occurred Time
P1	Network Device Interface Connectivity - OSPF Adjacency Failure	ACCESS	Connectivity	12	1	2	Jan 9, 2020 3:14 pm

- A. Configure to match the OSPF interface speed and duplex settings on both routers.
- B. Configure to match the OSPF interface MTU settings on both routers.
- C. Configure to match the OSPF interface unique IP address and subnet mask on both routers.
- D. Configure to match the OSPF interface network types on both routers.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 159

What does IPv6 Source Guard utilize to determine if IPv6 source addresses should be forwarded?

- A. ACE
- B. ACLS
- C. DHCP
- D. Binding Table

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 160

How are MPLS Layer 3 VPN services deployed?

- A. The RD and RT values must match under the VRF
- B. The RD and RT values under a VRF must match on the remote PE router
- C. The import and export RT values under a VRF must always be the same
- D. The label switch path must be available between the local and remote PE routers.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 161

When configuring Control Plane Policing on a router to protect it from malicious traffic, an engineer observes that the configured routing protocols start flapping on that device. Which action in the Control Plane Policy prevents this problem in a production environment while achieving the security objective?

- A. Set the conform-action and exceed-action to transmit initially to test the ACLs and transmit rates and apply the Control Plane Policy in the output direction
- B. Set the conform-action and exceed-action to transmit initially to test the ACLs and transmit rates and apply the Control Plane Policy in the input direction
- C. Set the conform-action to transmit and exceed-action to drop to test the ACLs and transmit rates and apply the Control Plane Policy in the input direction
- D. Set the conform-action to transmit and exceed-action to drop to test the ACLs and transmit rates and apply the Control Plane Policy in the output direction

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

For protection from malicious traffic, you need input CoPP.

Setting both conform-action and exceed-action to transmit will allow packets to be accepted even if the CIR has exceeded.

After applying, check the counter for exceed-action and tune the CIR so that there is no more exceed action (i.e. no increment in that counter). Finally you can set the exceed-action to drop.

QUESTION 162

Refer to the exhibit. A network administrator configured NTP on a Cisco router to get synchronized time for system and logs from a unified time source The configuration did not work as desired Which service must be enabled to resolve the issue?

Configuration output:

```
clock timezone PST -8
clock summer-time PDT recurring
service timestamps debug datetime
service timestamps log datetime
logging buffered 16000 debugging
ntp clock-period 17179272
ntp server 161.181.92.152
```

Debug output:

```
router#show clock
14:12:26.312 PDT Thu Apr 27 2019
router#config t
Enter configuration commands, one per line. End with CNTL/Z.
router(config)#exit

router#
Apr 27 21:12:28: %SYS-5-CONFIG_I: Configured from console by vty0
```

- A. Enter the service timestamps log datetime localtime global command.
- B. Enter the service timestamps log datetime synchronize global command.
- C. Enter the service timestamps log datetime console global command.
- D. Enter the service timestamps log datetime clock-period global command

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 163

Refer to the exhibit. R1 and R2 cannot establish an EIGRP adjacency. Which action establishes EIGRP adjacency?

```

R1
interface Loopback0
  ip address 172.16.1.1 255.255.255.255
interface FastEthernet0/0
  ip address 192.168.12.1 255.255.255.0
router eigrp 100
  no auto-summary
  network 192.168.12.0
  network 172.16.0.0
  neighbor 192.168.12.2 FastEthernet0/0

R2
interface Loopback0
  ip address 172.16.2.2 255.255.255.255
interface FastEthernet0/0
  ip address 192.168.12.2 255.255.255.0
router eigrp 100
  network 192.168.12.0
  network 172.16.0.0
  neighbor 192.168.12.1 FastEthernet0/0
  passive-interface FastEthernet0/0

```

- A. Remove the current autonomous system number on one of the routers and change to a different value.
- B. Remove the passive-interface command from the R2 configuration so that it matches the R1 configuration.
- C. Add the no auto-summary command to the R2 configuration so that it matches the R1 configuration.
- D. Add the passive-interface command to the R1 configuration so that it matches the R2 configuration.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Instead of just suppressing multicast Hello packets, configuring the passive-interface command suppresses all incoming and outgoing routing updates and hello messages. EIGRP neighbor adjacencies cannot be established or maintained over an interface that is configured as passive.

QUESTION 164

Which two components are needed for a service provider to utilize the L3 VPN MPLS application? (Choose two.)

- A. The P routers must be configured for MP-iBGP toward the PE routers
- B. The P routers must be configured with RSVP.
- C. The PE routers must be configured for MP-iBGP with other PE routers
- D. The PE routers must be configured for MP-eBGP to connect to CEs
- E. The P and PE routers must be configured with LDP or RSVP

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 165

An engineer configured two routers connected to two different service providers using BGP with default attributes. One of the links is presenting high delay, which causes slowness in the network. Which BGP attribute must the engineer configure to avoid using the high-delay ISP link if the second ISP link is up?

- A. LOCAL_PREF
- B. MED
- C. WEIGHT
- D. AS-PATH

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

A is chosen assume the engineer wants to prefer ISP link for traffic outgoing to Internet.

D should be chosen instead if the engineer wants to prefer ISP link for traffic incoming to his own AS. (B is not suitable in this case since the eBGP neighbors are two different ISPs which have different ASes).

C should be used for traffic outgoing to Internet if only there is only one BGP router that is connecting to the two ISPs.

QUESTION 166

Refer to Exhibit. Which two configurations allow clients to get dynamic ip addresses assigned?

```
ip dhcp excluded-address 172.16.16.1 172.16.16.2
!
ip dhcp pool 0
network 172.16.16.0 255.255.255.0
domain-name cisco.com
dns-server 172.16.16.2
lease 30

interface Ethernet0/0
ip address 10.1.1.1 255.255.255.252
ip access-group 100 in

access-list 100 deny  udp any any
access-list 100 permit ip any any
```

- A. Configure access-list 100 permit udp any any eq 61 as the first line
- B. Configure access-list 100 permit udp any any eq 86 as the first line
- C. Configure access-list 100 permit udp any any eq 68 as the first line
- D. Configure access-list 100 permit udp any any eq 69 as the first line
- E. Configure access-list 100 permit udp any any eq 67 as the first line

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

DHCP messages to DHCP server uses UDP source port 68 and UDP destination port 67
DHCP messages to DHCP client uses UDP source port 67 and UDP destination port 68

In the above configuration, it is a DHCP server and therefore allow port 67 ie. E is sufficient. However since choose 2, C is also chosen.

QUESTION 167

Refer to the exhibit. The administrator configured route advertisement to a remote low resources router to use only the default route to reach any network but failed. Which action resolves this issue?

```
ip prefix-list DefaultRouteOnly seq 5 deny 0.0.0.0/0 le 32
ip prefix-list DefaultRouteOnly seq 10 permit 0.0.0.0/0

router eigrp ccnp
address-family ipv4 unicast autonomous-system 1
topology base
distribute-list prefix DefaultRouteOnly out Tunnel0
```

- A. Change the direction of the distribute-list command from out to in.
- B. Remove the line with the sequence number 5 from the prefix list.
- C. Remove the prefix keyword from the distribute-list command.
- D. Remove the line with the sequence number 10 from the prefix list.

Correct Answer: B

Section: (none)

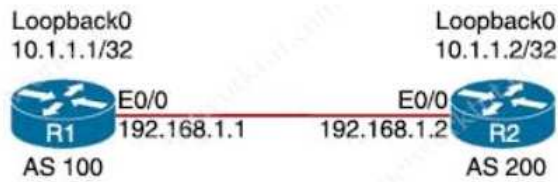
Explanation

Explanation/Reference:

After removing 5, all routes except default route will be denied by implicit deny.

QUESTION 168

Refer to the exhibit. The neighbor relationship is not coming up. Which two sets of configurations bring the neighbors up? (Choose two)



R1 router bgp 100 neighbor 10.1.1.2 remote-as 200	R2 router bgp 200 neighbor 10.1.1.1 remote-as 100
--	--

- A. R2
ip route 10.1.1.2 255.255.255.255 192.168.1.2
router bgp 100
neighbor 10.1.1.2 ttl-security hops 1
neighbor 10.1.1.2 update-source loopback 0
- B. R1
ip route 10.1.1.2 255.255.255.255 192.168.1.2
router bgp 100
neighbor 10.1.1.2 disable-connected-check
neighbor 10.1.1.2 update-source loopback0
- C. R2
ip route 10.1.1.1 255.255.255.255 192.168.1.1
router bgp 200
neighbor 10.1.1.1 disable-connected-check
neighbor 10.1.1.1 update-source loopback0
- D. R2
ip route 10.1.1.1 255.255.255.255 192.168.1.1
router bgp 200
neighbor 10.1.1.1 ttl-security hops 1
neighbor 10.1.1.1 update-source loopback 0

Correct Answer: BC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 169

Refer to the exhibit. Which option describes why the EIGRP neighbors of this router are not learning routes that are received from OSPF?

```

router eigrp 1
 redistribute ospf 100
 network 10.10.10.0 0.0.0.255
 auto-summary
!
router ospf 100
 network 172.16.0.0 0.0.255.255 area 100
 redistribute eigrp 1

```

- A. The subnet defined in OSPF is not part of area 0
- B. Default metrics are not configured under EIGRP
- C. There is no overlap in the subnets advertised
- D. The routing protocols do not have the same AS number

Correct Answer: B

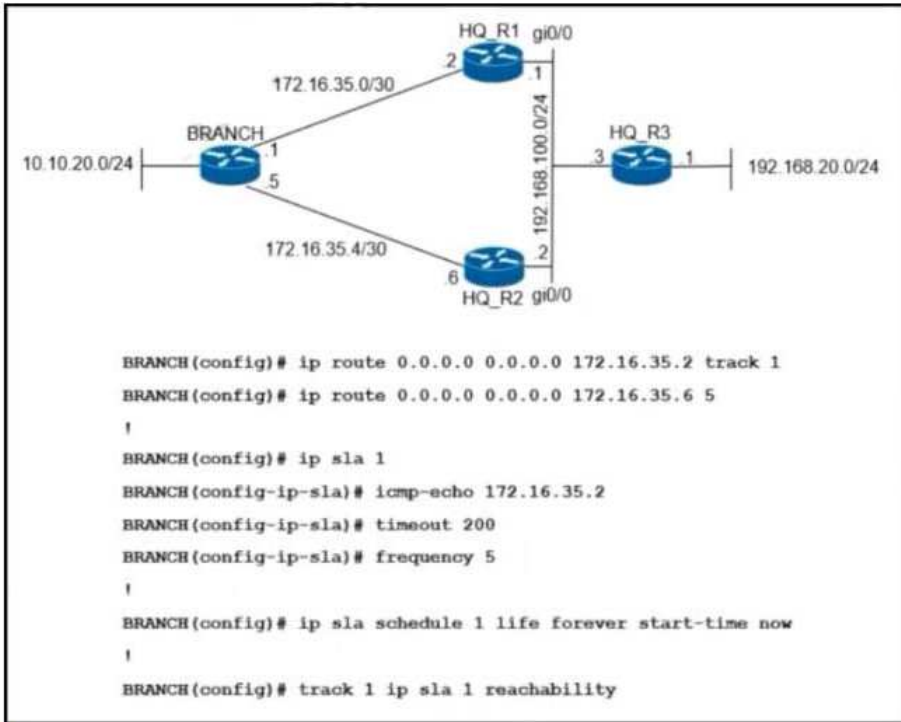
Section: (none)

Explanation

Explanation/Reference:

QUESTION 170

Refer to the exhibit. An engineer has successfully set up a floating static route from the BRANCH router to the HQ network using HQ_R1 as the primary default gateway. When the g0/0 goes down on HQ_R1, the branch network cannot reach the HQ network 192.168.20.0/24. Which set of configurations resolves the issue?



- A. HQ_R3(config)# ip sla responder
HQ_R3(config)# ip sla responder icmp-echo 172.16.35.1
- B. BRANCH(config)# ip sla 1
BRANCH(config-ip-sla)# icmp-echo 192.168.100.2
- C. HQ_R3(config)# ip sla responder
HQ_R3(config)# ip sla responder icmp-echo 172.16.35.5
- D. BRANCH(config)# ip sla 1
BRANCH(config-ip-sla)# icmp-echo 192.168.100.1

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Since the LAN interface is down, monitoring HQ_R1 WAN interface will continue to be up. Hence you should setup IP SLA in Branch router to monitor HQ_R1 LAN interface i.e. 192.168.100.1.

QUESTION 171

An engineer configured a DHCP server for Cisco IP phones to download its configuration from a TFTP server, but the IP phones failed to load the configuration. What must be configured to resolve the issue?

- A. BOOTP port 67
- B. DHCP option 66
- C. BOOTP port 68
- D. DHCP option 69

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

There are two DHCP options for providing TFTP server IP address to DHCP client.

- Option 66 which supports the providing of only one IP address (It is a standard specified in RFC 2132)
- Option 150 which supports the providing of multiple IP addresses i.e. multiple TFTP servers. (Cisco Proprietary).

Cisco IP phone usually uses DHCP option 150 provided by DHCP server to get TFTP server for downloading its phone configuration. But some may use option 66.

QUESTION 172

Refer to the exhibit. Which action restores the routes from neighbors while still filtering 1.1.1.0/24?

```

access-list 1 permit 1.1.1.0 0.0.0.255
!
route-map FILTER1 deny 10
match ip address 1
!
router eigrp 1
distribute-list route-map FILTER1 in

```

- A. Add a second line in the access list to permit any.
- B. Modify the route map to permit the access list instead of deny it
- C. Modify the access list to deny instead of permit it.
- D. Add a second sequence in the route map permit 20

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 173

A DMVPN single hub topology is using IPsec + mGRE with OSPF. What should be configured on the hub to ensure it will be the designated router?

- A. tunnel interface of the hub with ip nhrp ospf dr
- B. OSPF priority to 0
- C. route map to set the metrics of learned routes to 110
- D. OSPF priority greater than 1

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 174

An engineer configured policy-based routing for a destination IP address that does not exist in the routing table. How is the packet treated through the policy for configuring the set ip default nexthop command?

- A. Packets are not forwarded to the specific next hop.
- B. Packets are forwarded based on the routing table.
- C. Packets are forwarded based on a static route.
- D. Packets are forwarded to the specific next hop.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 175

When determining a device's support, what is the minimum time spacing required for a BFD control packet to receive once a BFD control packet is received?

- A. Detect Mult
- B. Required Min Echo RX Interval
- C. Desired Min TX Interval
- D. Required Min RX Interval

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

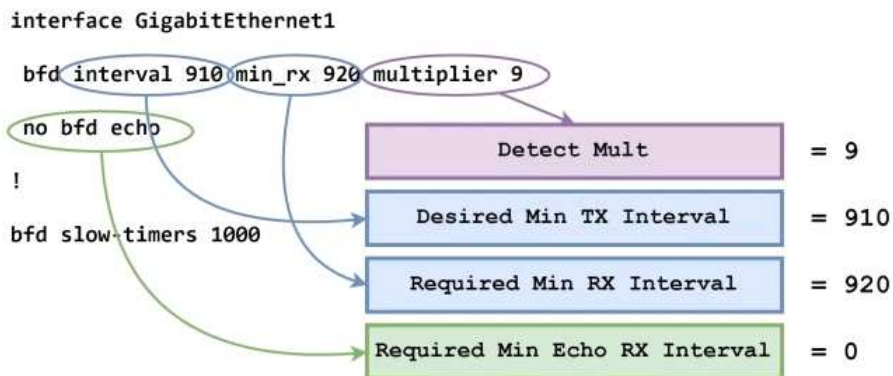
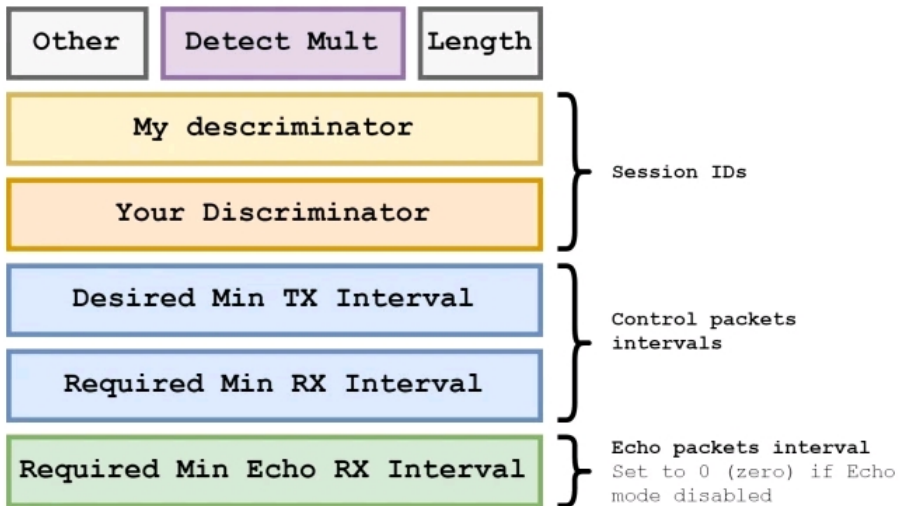
Detect Mult: Detect time multiplier.

Desired Min TX Interval: This is the minimum interval, in microseconds, that the local system would like to use when transmitting BFD Control packets.

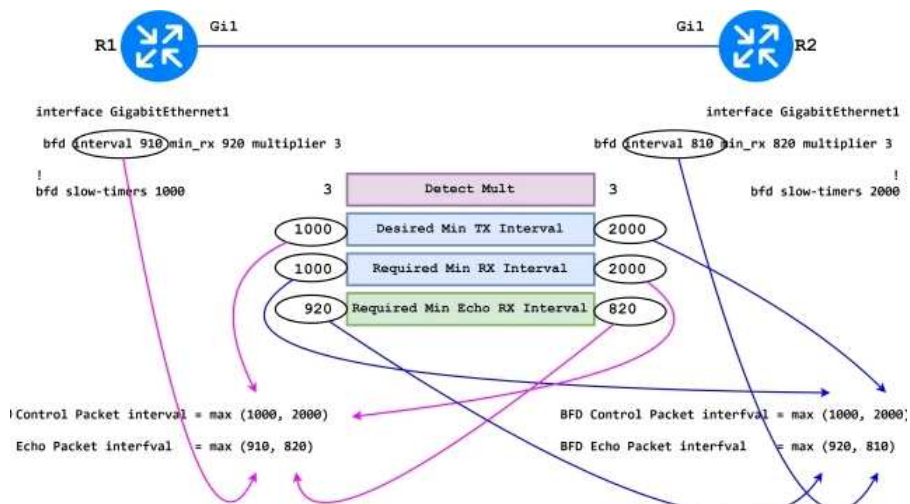
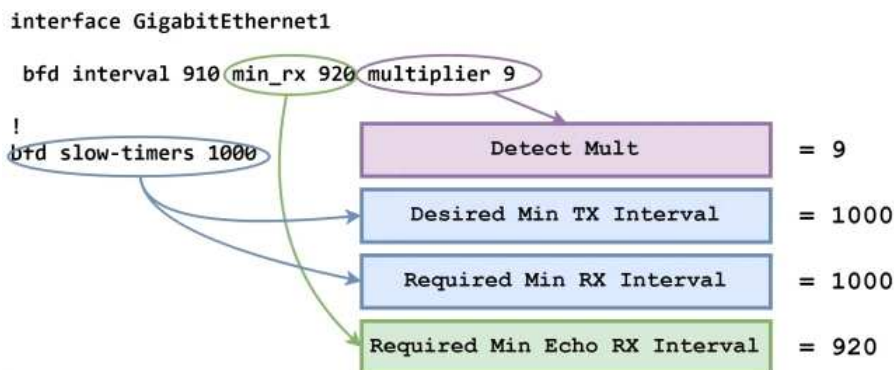
Required Min RX Interval: This is the **minimum interval**, in microseconds, between received **BFD Control packets** that this system is **capable of supporting**.

Required Min Echo RX Interval: This is the minimum interval, in microseconds, between received BFD Echo packets that this system is capable of supporting. (0 means do not support BFD Echo).

Remarks:



slow timers settings are used for control packets when using BFD Echo:



QUESTION 176

Refer to the exhibit. The remote server is failing to receive the NetFlow data. Which action resolves the issue?

```

config t
flow record v4_r1
match ipv4 tos
match ipv4 protocol
match ipv4 source address
match ipv4 destination address
match transport source-port
match transport destination-port
collect counter bytes long
collect counter packets long
!
flow exporter EXPORTER-1
destination 172.16.10.2
transport udp 2055
exit
!
flow monitor FLOW-MONITOR-1
exporter EXPORTER-1
record v4_r1
exit
!
flow monitor v4_r1
!
ip cef
!
interface Ethernet0/0.1
ip address 172.16.6.2 255.255.255.0
ip flow monitor v4_r1 input
!

```

- A. Modify the flow transport command transport udp 2055 to move under flow monitor profile.
- B. Modify the interface command to ip flow monitor FLOW-MONITOR-1 input.
- C. Modify the udp port under flow exporter profile to ip transport udp 4739.
- D. Modify the flow record command record v4_r1 to move under flow exporter

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 177

What are two MPLS label characteristics? (Choose two)

- A. The label edge router swaps labels on the received packets.
- B. Labels are imposed in packets after the Layer 3 header.
- C. LDP uses TCP for reliable delivery of information.
- D. An MPLS label is a short identifier that identifies a forwarding equivalence class.
- E. A maximum of two labels can be imposed on an MPLS packet

Correct Answer: CD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 178

What are two characteristics of VRF instance? (Choose two.)

- A. All VRFs share customers routing and CEF tables.
- B. An interface must be associated to one VRF.
- C. Each VRF has a different set of routing and CEF tables
- D. It is defined by the VPN membership of a customer site attached to a P device.
- E. A customer site can be associated to different VRFs

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

A is wrong

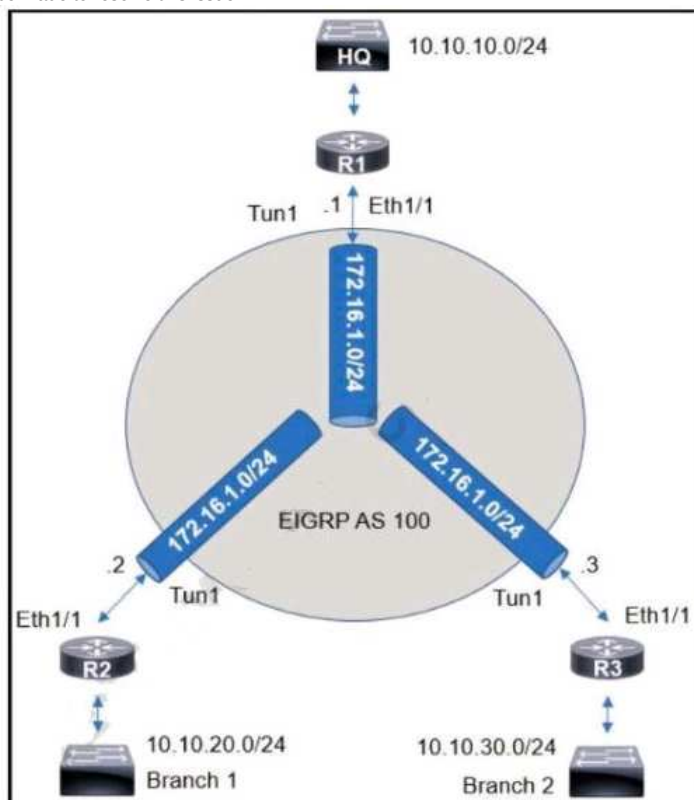
B is wrong since an interface can remain in global

D P device has no information about VRF or VPN.

E can be used in a special case in which a customer site has two CE routers which wants to import / export routes with different remote sites.

QUESTION 179

An engineer sets up a DMVPN connection to connect branch 1 and branch 2 to HQ. branch 1 and branch 2 cannot communicate with each other. Which change must be made to resolve this issue?



- A. R1(config)#int eth1/1
R1(config-if)#no ip split-horizon eigrp 100
- B. R2(config)#router eigrp 100
R2(config-router)#neighbor 172.16.1.3
- C. R3(config)#router eigrp 100
R3(config-router)#neighbor 172.16.1.2
- D. R1(config)#int tunnel 1
R1(config-if)#no ip split-horizon eigrp 100

Correct Answer: D

Section: (none)

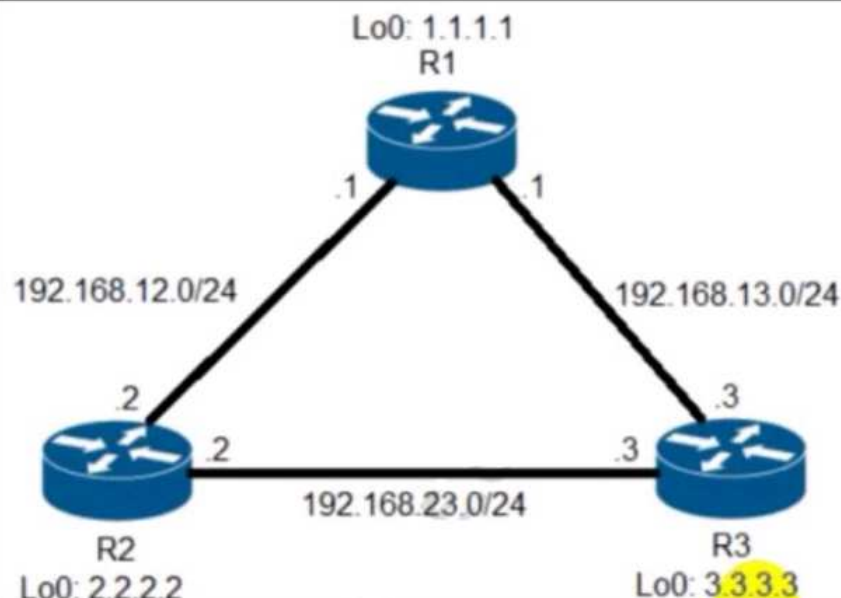
Explanation

Explanation/Reference:

Since EIGRP is running within DMVPN, split horizon is disabled in the tunnel interface.

QUESTION 180

Refer to the exhibit. R2 has two paths to reach 192.168.13.0/24. but traffic is sent only through R3. Which action allows traffic to use both paths?



```
R2#show ip protocols | include eigrp|Maximum
```

```
Routing Protocol is "eigrp 1"
  Maximum path: 4
  Maximum hopcount 100
  Maximum metric variance 1
```

```
R2#show ip eigrp topology 192.168.13.0/24
```

```
EIGRP-IPv4 Topology Entry for AS(1)/ID(2.2.2.2) for 192.168.13.0/24
State is Passive, Query origin flag is 1, 1 Successor(s), FD is 1075200
Descriptor Blocks:
```

```
192.168.23.3 (FastEthernet0/1), from 192.168.23.3, Send flag is 0x0
  Composite metric is (1075200/281600), route is Internal
  Vector metric:
```

```
  Minimum bandwidth is 2500 Kbit
  Total delay is 2000 microseconds
  Reliability is 255/255
  Load is 255/255
  Minimum MTU is 1500
  Hop count is 1
  Originating router is 3.3.3.3
```

```
192.168.12.1 (FastEthernet0/0), from 192.168.12.1, Send flag is 0x0
  Composite metric is (2611200/281600), route is Internal
  Vector metric:
```

```
  Minimum bandwidth is 1000 Kbit
  Total delay is 2000 microseconds
  Reliability is 255/255
  Load is 1/255
  Minimum MTU is 1500
  Hop count is 1
  Originating router is 1.1.1.1
```

```
R2#show ip route 192.168.13.0
```

```
Routing entry for 192.168.13.0/24
```

```
Known via "eigrp 1", distance 90, metric 1075200, type internal
```

```
Redistributing via eigrp 1
```

```
Last update from 192.168.23.3 on FastEthernet0/1, 00:00:57 ago
```

```
Routing Descriptor Blocks:
```

```
* 192.168.23.3, from 192.168.23.3, 00:00:57 ago, via FastEthernet0/1
```

```
Route metric is 1075200, traffic share count is 1
```

```
Total delay is 2000 microseconds, minimum bandwidth is 2500 Kbit
```

```
Reliability 255/255, minimum MTU 1500 bytes
```

```
Loading 255/255, Hops 1
```

- A. Configure the bandwidth 2000 command under interface FastEthernet0/0 on R2.
- B. Configure the variance 4 command under the EIGRP process on R2.
- C. Configure the delay 1 command under interface FastEthernet0/0 on R2.
- D. Configure the variance 2 command under the EIGRP process on R2

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

The metric of the two routes are 1075200 and 2611200, therefore you need the value 3 or more when configuring variance.

QUESTION 181

Refer to the exhibits. An engineer filtered messages based on severity to minimize log messages. After applying the filter, the engineer noticed that it filtered required messages as well. Which action must the engineer take to resolve the issue?

Filtered

```
00:00:46: %LINK-3-UPDOWN: Interface Port-channel1, changed state to up
00:00:47: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
00:00:47: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
```

Desired

```
00:00:46: %LINK-3-UPDOWN: Interface Port-channel1, changed state to up
00:00:47: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
00:00:47: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
00:00:48: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to down
00:00:48: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to down 2 *Mar 1 18:46:11: %SYS-5-CONFIG_I: Configured from console by vty2
```

- A. Configure syslog level 2.
- B. Configure syslog level 3.
- C. Configure syslog level 4.
- D. Configure syslog level 5.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Since the highest value (i.e. least severity level) found in desired log is 5, you need to configure syslog level 5.

QUESTION 182

An engineer must configure a Cisco router to initiate secure connections from the router to other devices in the network but kept failing. Which two actions resolve the issue? (Choose two.)

- A. Configure a source port for the SSH connection to initiate
- B. Configure a TACACS+ server and enable it
- C. Configure transport input ssh command on the console
- D. Configure a domain name
- E. Configure a crypto key to be generated

Correct Answer: DE

Section: (none)

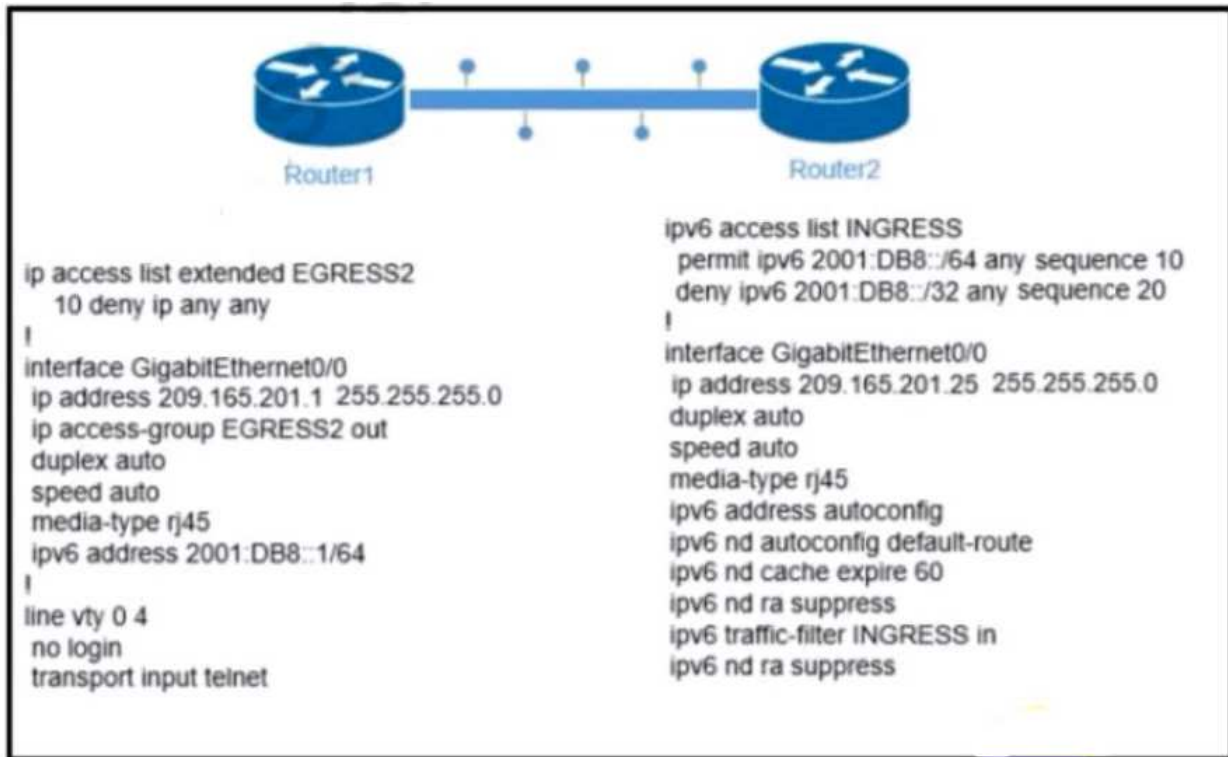
Explanation

Explanation/Reference:

Actually it is the "other devices" that require the above configuration. An SSH client does not need the above configuration. C is wrong in the "other devices" since the command should be configured under line vty.

QUESTION 183

Refer to the exhibit. The engineer configured and connected Router2 to Router1. The link came up but could not establish a Telnet connection to Router1 IPv6 address of 2001:DB8::1. Which configuration allows Router2 to establish a Telnet connection to Router1?



- A. ipv6 unicast-routing
- B. permit ICMPv6 on access list INGRESS for Router2 to obtain IPv6 address
- C. permit ip any any on access list EGRESS2 on Router1
- D. IPv6 address on GigabitEthernet0/0

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since R2's g0/0 is configured with "ipv6 address autoconfig", R2 needs to accept ICMP RA (Type 134) messages in order to configure the interface's IPv6 address.

Remarks :

Alternatively, you can just allow RA as follows:

```
permit icmp any any router-advertisement
```

QUESTION 184

Which IPv6 first-hop security feature helps to minimize denial of service attacks?

- A. IPv6 Router Advertisement Guard
- B. IPv6 Destination Guard
- C. DHCPv6 Guard
- D. IPv6 MAC address filtering

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

The Destination Guard feature helps in minimizing denial-of-service (DoS) attacks. It performs address resolutions only for those addresses that are active on the link, and requires the FHS binding table to be populated with the help of the IPv6 snooping feature.

The feature enables the filtering of IPv6 traffic based on the destination address, and blocks the NDP resolution for destination addresses that are not found in the binding table. By default, the policy drops traffic coming for an unknown destination.

QUESTION 185

Refer to the exhibit. The OSPF neighbor relationship is not coming up What must be configured to restore OSPF neighbor adjacency?

```

OSPF: Send DBD to 10.100.1.2 on GigabitEthernet0/1 seq 0x9E6 opt
0x52 flag 0x7
    len 32
OSPF: Retransmitting DBD to 10.100.1.2 on GigabitEthernet0/1
[10]
OSPF: Send DBD to 10.100.1.2 on GigabitEthernet0/1 seq 0x9E6 opt
0x52 flag 0x7
    len 32
OSPF: Retransmitting DBD to 10.100.1.2 on GigabitEthernet0/1
[11]
%OSPF-5-ADJCHG: Process 1, Nbr 10.100.1.2 on GigabitEthernet0/1
from EXSTART to
    DOWN, Neighbor Down: Too many retransmissions

```

- A. OSPF on the remote router
- B. matching hello timers
- C. use router ID
- D. matching mtu values

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Since the logging messages is about retransmitting DBD, this happens during Exstart state. The neighbor will stuck in Exstart. It is usually a MTU problem.

QUESTION 186

In which two ways does the IPv6 First-Hop Security Binding Table operate? (Choose two.)

- A. by the recovery mechanism to recover the binding table in the event of a device reboot
- B. by IPv6 routing protocols to securely build neighborships without the need of authentication
- C. by IPv6 HSRP to make sure neighbors are authenticated before being used as gateways
- D. by various IPv6 guard features to validate the data link layer address
- E. by storing hashed keys for IPsec tunnels for the built-in IPsec features

Correct Answer: AD

Section: (none)

Explanation

Explanation/Reference:

Binding Table Recovery

This feature helps in recovering the missing binding table entries when the resolution for a destination address fails in the destination guard. It does so by querying the DHCP server or the destination host, depending on the configuration.

QUESTION 187

An engineer is troubleshooting on the console session of a router and turns on multiple debug commands. The console screen is filled with scrolling debug messages that none of the commands can be verified if entered correctly or display any output. Which action allows the engineer to see entered console commands while still continuing the analysis of the debug messages?

- A. Configure the logging synchronous command
- B. Configure the no logging console debugging command globally
- C. Configure the logging synchronous level all command
- D. Configure the term no mon command globally

Correct Answer: A

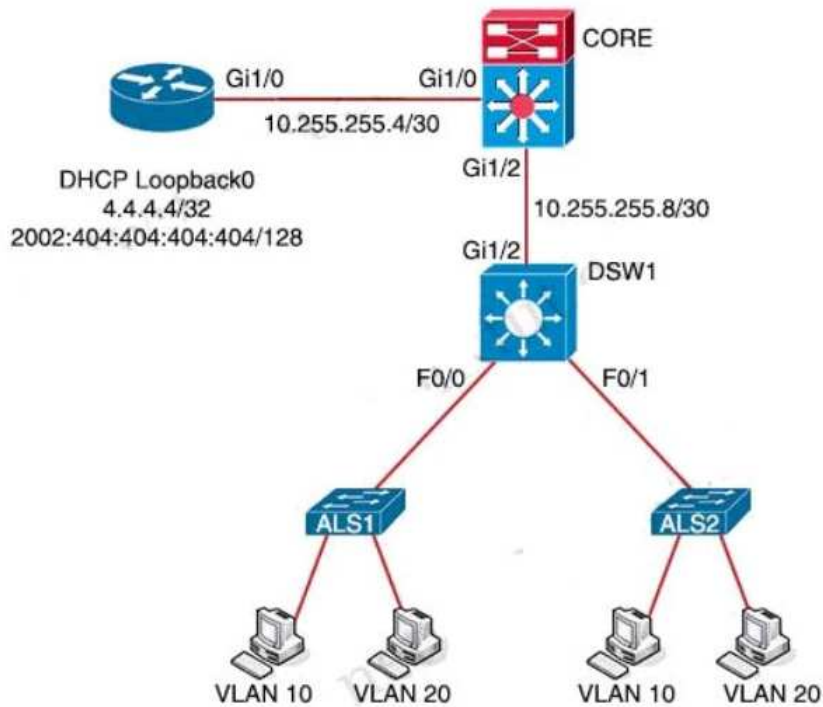
Section: (none)

Explanation

Explanation/Reference:

QUESTION 188

Refer to the exhibit. Router DHCP is configured to lease IPv4 and IPv6 addresses to clients on ALS1 and ALS2. Clients on ALS2 receive IPv4 and IPv6 addresses. Clients on ALS1 receive IPv4 addresses. Which configuration on DSW1 allows clients on ALS1 to receive IPv6 addresses?



```
DSW1#sh run int f0/0
Building configuration...
Current configuration : 174 bytes
!
interface FastEthernet 0/0
ip address 10.4.10.1 255.255.255.0
ip helper-address 4.4.4.4
duplex auto
speed auto
ipv6 address 2002:A04:A01::A04:A01/120
ipv6 enable
end
```

- A. DSW1(dhcp-config)# default-router 2002:A04:A01::A04:A01
- B. DSW1(config)# ipv6 dhcp relay destination 2002:404:404::404:404 GigabitEthernet 1/2
- C. DSW1(config)#ipv6 route 2002:404:404::404:404/128 FastEthernet 1/0
- D. DSW1(config-if)# ipv6 helper address 2002:404:404::404:404

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

For configuration DHCP relay agent:

For IPv4: the Interface Configuration mode command "ip helper-address" is used. (already configured in DSW1).

For IPv6: the Interface Configuration mode command "ipv6 dhcp relay destination" is used.

QUESTION 189

What are two purposes of using IPv4 and VPNv4 address-family configurations in a Layer 3 MPLS VPN? (Choose two.)

- A. RD is prepended to the IPv4 route to make it unique.
- B. The VPNv4 address consists of a 64-bit route distinguisher that is prepended to the IPv4 prefix.
- C. MP-BGP is used to allow overlapping IPv4 addresses between customers to advertise through the network.
- D. The IPv4 address is needed to tag the MPLS label.
- E. The VPNv4 address is used to advertise the MPLS VPN label.

Correct Answer: AB

Section: (none)

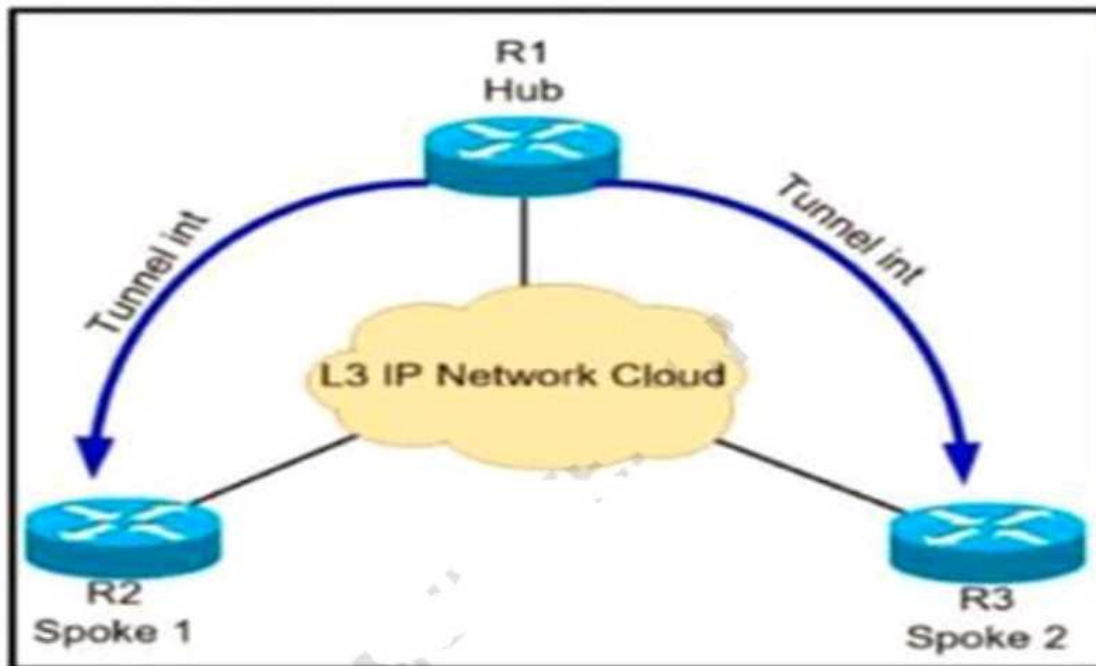
Explanation

Explanation/Reference:

C is only partially correct since it requires the prepending of the 64-bit RD to the 32-bit IPv4 address which result in 96-bit VPNv4. Hence VRF configured with RD is required.

QUESTION 190

Refer to Exhibit. A network administrator has successfully configured DMVPN topology between a hub and two spoke routers. Which two configuration commands should establish direct communications between spoke 1 and spoke 2 without going through the hub? (Choose two).



- A. At the hub router, configure the ip nhrp shortcut command.
- B. At the spoke routers, configure the ip nhrp spoke-tunnel command.
- C. At the hub router, configure the ip nhrp redirect command
- D. At the spoke routers, configure the ip nhrp shortcut command.
- E. At the hub router, configure the ip nhrp spoke-tunnel command

Correct Answer: CD

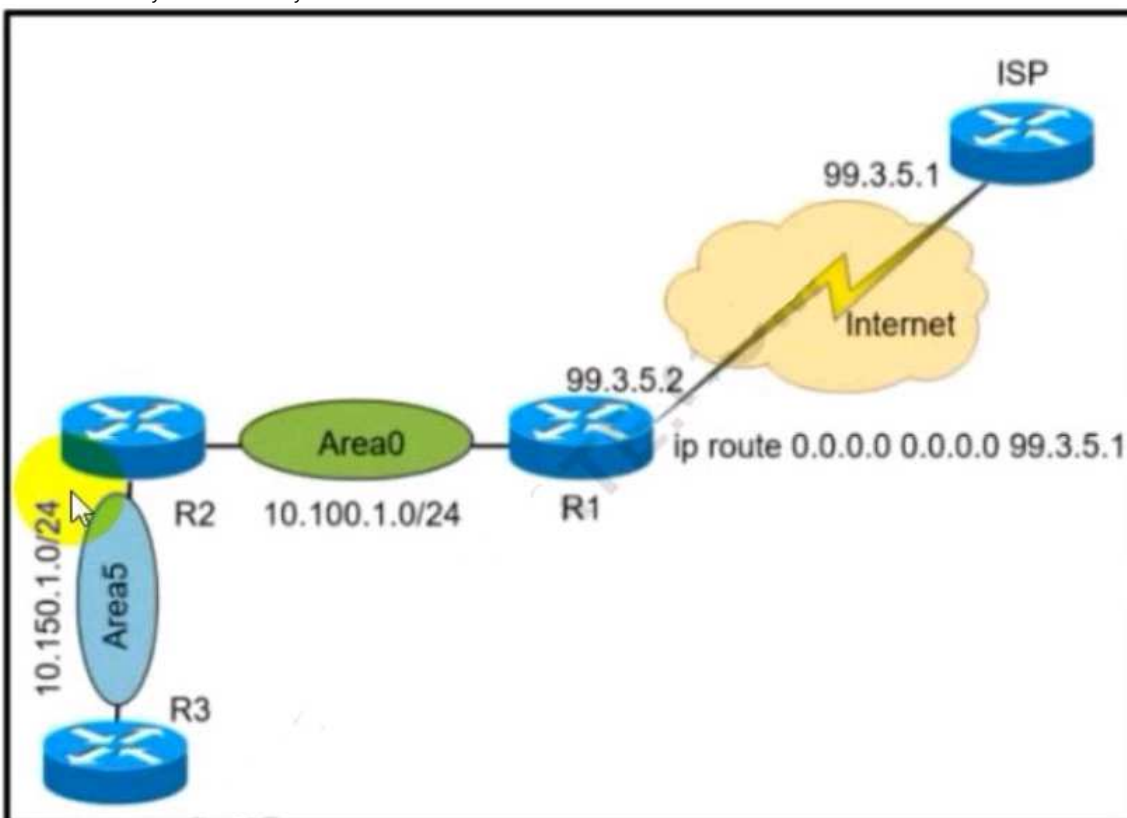
Section: (none)

Explanation

Explanation/Reference:

QUESTION 191

Refer to the exhibit. A network administrator redistributed the default static route into OSPF toward all internal routers to reach to Internet. Which set of commands restores reachability to the Internet by internal routers?



- A. router ospf 1
default-information originate
- B. router ospf 1
network 0.0.0.0 0.0.0.0 area 0

- C. router ospf 1
redistribute connected 0.0.0.0
- D. router ospf 1
redistribute static subnets

Correct Answer: A

Section: (none)

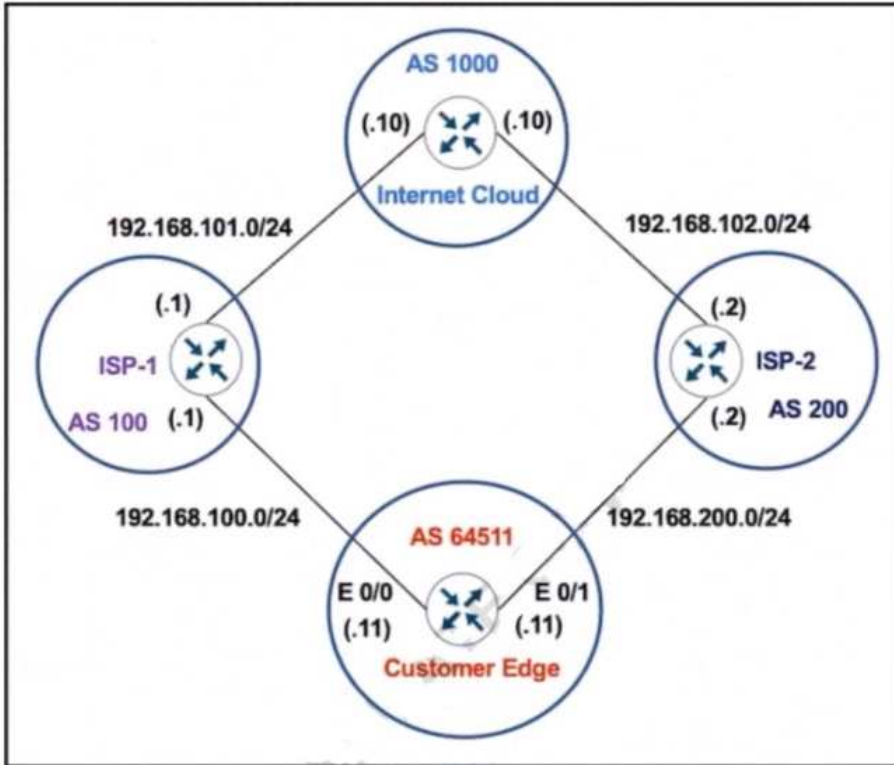
Explanation

Explanation/Reference:

redistribute static subnets command is used to redistribute static routes in OSPF, However for OSPF, static default route (ip route 0.0.0.0 0.0.0.0...) is cannot be redistributed using this method. Hence you must use default-information originate.

QUESTION 192

Refer to the exhibit.



The network administrator has configured the Customer Edge router (AS 64511) to send only summarized routes toward ISP-1 (AS 100) and ISP-2 (AS 200).

```
router bgp 64511
network 172.16.20.0 mask 255.255.255.0
network 172.16.21.0 mask 255.255.255.0
network 172.16.22.0 mask 255.255.255.0
network 172.16.23.0 mask 255.255.255.0
aggregate-address 172.16.20.0 255.255.252.0
```

After this configuration, ISP-1 and ISP-2 continue to receive the specific routes and the summary route. Which configuration resolves the issue?

- A. router bgp 64511
aggregate-address 172.16.20.0 255.255.252.0 summary-only
- B. router bgp 64511
neighbor 192.168.100.1 summary-only
neighbor 192.168.200.2 summary-only
- C. interface E 0/0
ip bgp suppress-map BLOCK_SPECIFIC
!
interface E 0/1
ip bgp suppress-map BLOCK_SPECIFIC
!
ip prefix-list PL_BLOCK_SPECIFIC permit 172.16.20.0/22 ge 24
!
route-map BLOCK_SPECIFIC permit 10
match ip address prefix-list PL_BLOCK_SPECIFIC
- D. ip prefix-list PL_BLOCK_SPECIFIC deny 172.16.20.0/22 ge 22
ip prefix-list PL_BLOCK_SPECIFIC permit 172.16.20.0/22
!
route-map BLOCK_SPECIFIC permit 10
match ip address prefix-list PL_BLOCK_SPECIFIC
!
router bgp 64511
aggregate-address 172.16.20.0 255.255.252.0 suppress-map BLOCKSPECIFIC

Correct Answer: A

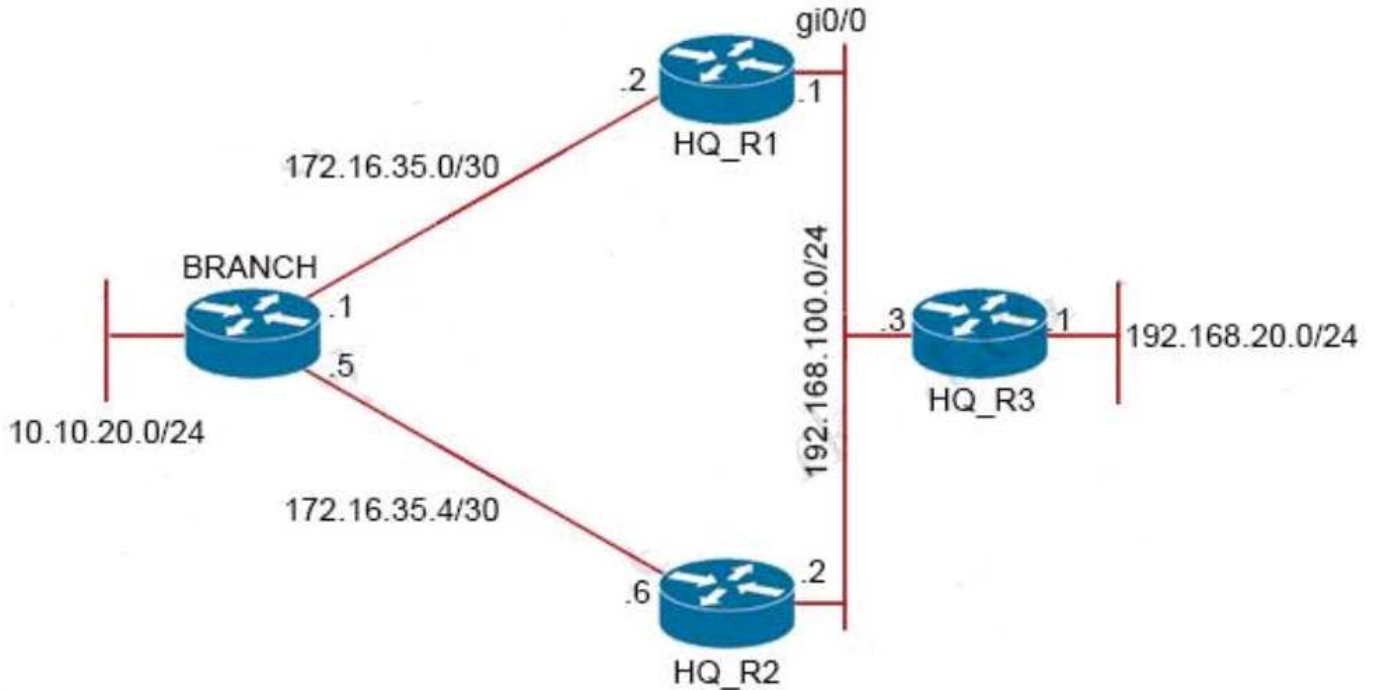
Section: (none)

Explanation

Explanation/Reference:

QUESTION 193

Refer to the exhibit. Traffic from the branch network should route through HQ_R1 unless the path is unavailable. An engineer tests this functionality by shutting down interface on the BRANCH router toward HQ_R1 router but 192.168.20.0/24 is no longer reachable from the branch router. Which set of configurations resolves the issue?



```
BRANCH(config)# ip route 0.0.0.0 0.0.0.0 172.16.35.2 track 1
BRANCH(config)# ip route 0.0.0.0 0.0.0.0 172.16.35.6 5
!
BRANCH(config)# ip sla 1
BRANCH(config-ip-sla)# icmp-echo 172.16.35.6
BRANCH(config-ip-sla)# timeout 200
BRANCH(config-ip-sla)# frequency 5
!
BRANCH(config)# ip sla schedule 1 life forever start-time now
!
BRANCH(config)# track 1 ip sla 1 reachability
```

- A. HQ_R1(config)# ip sla responder
HQ_R1(config)# ip sla responder icmp-echo 172.16.35.2
- B. BRANCH(config)# ip sla 1
BRANCH(config-ip-sla)# icmp-echo 172.16.35.1
- C. HQ_R2(config)# ip sla responder
HQ_R2(config)# ip sla responder icmp-echo 172.16.35.5
- D. BRANCH(config)# ip sla 1
BRANCH(config-ip-sla)# icmp-echo 172.16.35.2

Correct Answer: D

Section: (none)

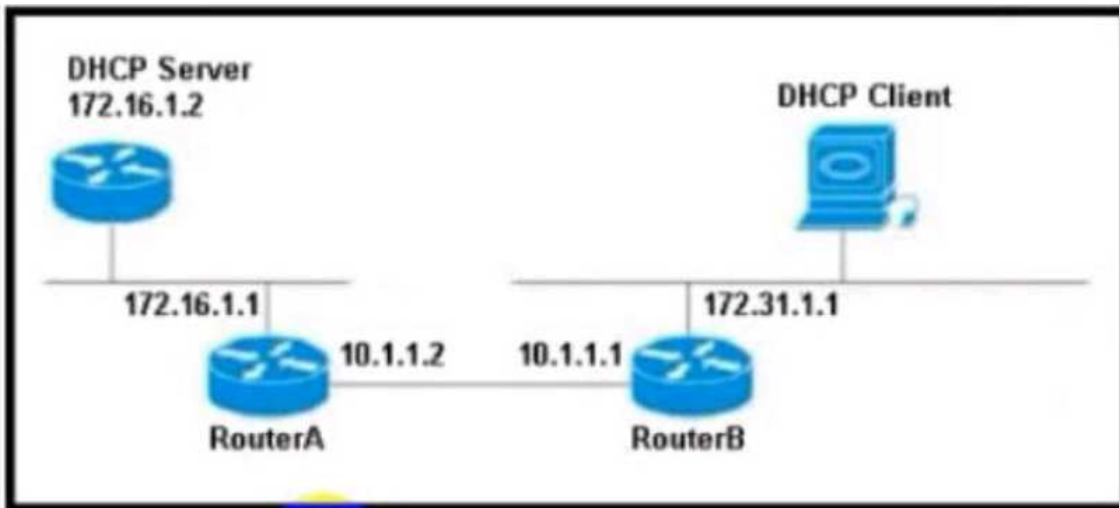
Explanation

Explanation/Reference:

The destination of the IP SLA icmp-echo should be HQ_R1.

QUESTION 194

Refer to the exhibit.



The DHCP client is unable to receive an IP address from the DHCP server RouterB is configured as follows:

```
Interface fastethernet 0/0
description Client DHCP ID 394482431
Ip address 172.31.1.1 255.255.255.0
!
ip route 172.16.1.0 255 255 255.0 10.1.1.2
```

Which command is required on the fastethernet 0/0 interface of RouterB to resolve this issue?

- A. RouterB(config-if)# ip helper-address 172.31.1.1
- B. RouterB(config-if)# ip helper-address 255.255.255.255
- C. RouterB(config-if)# ip helper-address 172.16.1.1
- D. RouterB(config-if)# ip helper-address 172.16.1.2

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

The DHCP server and the DHCP relay agent can be connected through one or more routers since they exchange messages through unicast and can therefore be routed through those routers.

QUESTION 195

What are two functions of MPLS Layer 3 VPNs? (Choose two.)

- A. LDP and BGP can be used for Pseudowire signaling
- B. It is used for transparent point-to-multipoint connectivity between Ethernet links/sites.
- C. BGP is used for signaling customer VPNv4 routes between PE nodes.
- D. A packet with node segment ID is forwarded along with shortest path to destination.
- E. Customer traffic is encapsulated in a VPN label when it is forwarded in MPLS network.

Correct Answer: CE

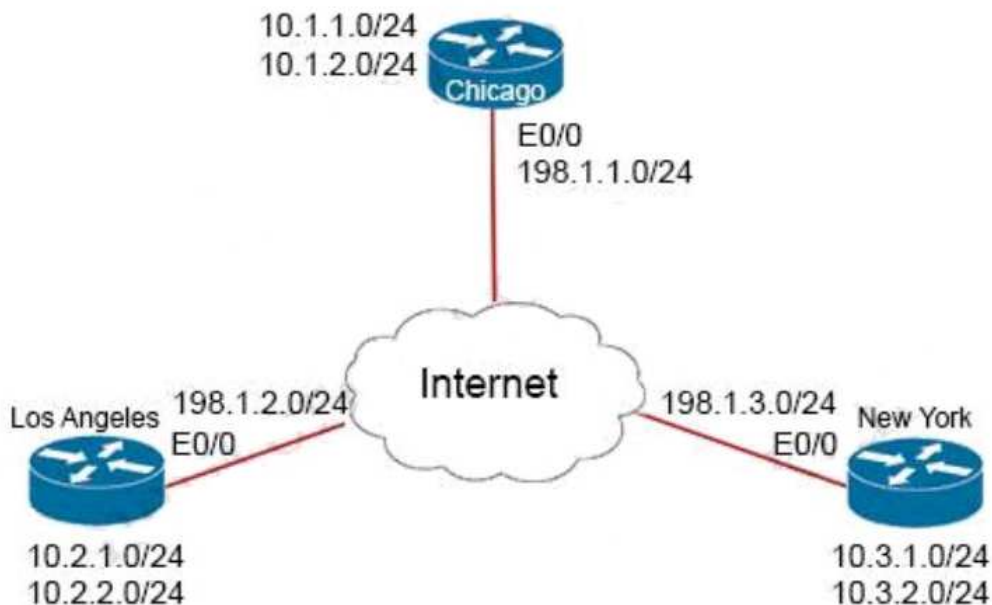
Section: (none)

Explanation

Explanation/Reference:

QUESTION 196

Refer to the exhibit. The Los Angeles and New York routers are receiving routes from Chicago but not from each other Which configuration fixes the issue?



Chicago

```
interface Tunnel 1
ip address 192.168.1.1 255.255.255.0
tunnel source E0/0
tunnel mode gre multipoint
ip nhrp network-id 1
ip nhrp map multicast dynamic
no ip next-hop-self eigrp 111
tunnel protection ipsec profile IPSec-PROFILE
!
router eigrp 111
network 192.168.1.0
network 10.0.0.0
```

- A. interface Tunnel1
no ip split-horizon eigrp 111
- B. interface Tunnel1
ip next-hop-self eigrp 111
- C. interface Tunnel1
tunnel mode Isec ipv4
- D. interface Tunnel1
tunnel protection ipsec profile IPSec-PROFILE

Correct Answer: A

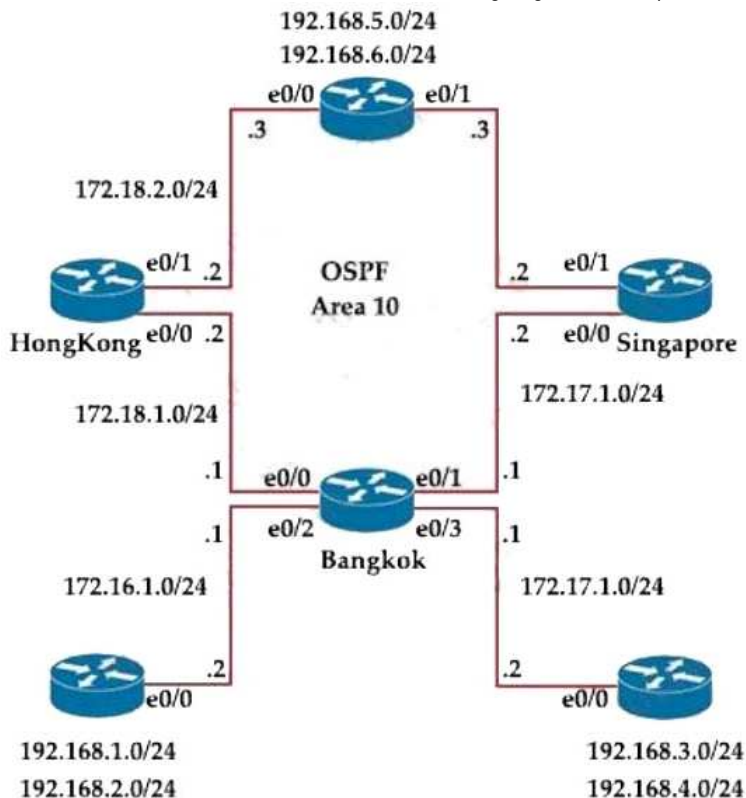
Section: (none)

Explanation

Explanation/Reference:

QUESTION 197

Refer to the exhibit. Bangkok is using ECMP to reach to the 192.168.5.0/24 network. The administrator must configure Bangkok in such a way that Telnet traffic from 192.168.3.0/24 and 192.168.4.0/24 networks uses the HongKong router as the preferred route. Which set of configurations accomplishes this task?



- A. access-list 101 permit ip 192.168.3.0 0.0.0.255 192.168.5.0 0.0.0.255
 access-list 101 permit ip 192.168.4.0 0.0.0.255 192.168.5.0 0.0.0.255
 !
 route-map PBR1 permit 10
 match ip address 101
 set ip next-hop 172.18.1.2
 !
 interface Ethernet0/3
 ip policy route-map PBR1
- B. access-list 101 permit tcp 192.168.3.0 0.0.0.255 192.168.5.0 0.0.0.255 eq 23
 access-list 101 permit tcp 192.168.4.0 0.0.0.255 192.168.5.0 0.0.0.255 eq 23
 !
 route-map PBR1 permit 10
 match ip address 101
 set ip next-hop 172.18.1.2
 !
 interface Ethernet0/1
 ip policy route-map PBR1
- C. access-list 101 permit tcp 192.168.3.0 0.0.0.255 192.168.5.0 0.0.0.255 eq 23
 access-list 101 permit tcp 192.168.4.0 0.0.0.255 192.168.5.0 0.0.0.255 eq 23
 !
 route-map PBR1 permit 10
 match ip address 101
 set ip next-hop 172.18.1.2
 !
 interface Ethernet0/3
 ip policy route-map PBR1
- D. access-list 101 permit ip 192.168.3.0 0.0.0.255 192.168.5.0 0.0.0.255
 access-list 101 permit ip 192.168.4.0 0.0.0.255 192.168.5.0 0.0.0.255
 !
 route-map PBR1 permit 10
 match ip address 101
 set ip next-hop 172.18.1.2
 !
 interface Ethernet0/1
 ip policy route-map PBR1

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

ECMP is Equal-Cost Multi-Path (i.e. load balancing routes)

QUESTION 198

Refer to the exhibit. Which action resolves the authentication problem?

```

13:35:07.826: AAA/BIND (00000055): Bind i/
13:35:07.826: AAA/AUTHEN/LOGIN (00000055): Pick method list 'default'
13:35:07.826: TPLUS: Queuing AAA Authentication request 85 for processing
13:35:07.826: TPLUS (00000055) login timer started 1020 sec timeout
13:35:07.826: TPLUS: processing authentication start request id 85
13:35:07.826: TPLUS: Authentication start packet created for 85()
13:35:07.826: TPLUS: Using server 10.106.60.182
13:35:07.826: TPLUS (00000055)/0/NB_WAIT/225FE2DC: Started 5 sec timeout
13:35:07.830: TPLUS (00000055)/0/NB_WAIT: socket event 2
13:35:07.830: TPLUS (00000055)/0/NB_WAIT: wrote entire 38 bytes request
13:35:07.830: TPLUS (00000055)/0/READ: socket event 1
13:35:07.830: TPLUS (00000055)/0/READ: Would block while reading
13:35:07.886: TPLUS (00000055)/0/READ: socket event 1
13:35:07.886: TPLUS (00000055)/0/READ: read entire 12 header bytes (expect 8 bytes data)
13:35:07.886: TPLUS (00000055)/0/READ: socket event 1
13:35:07.886: TPLUS (00000055)/0/READ: read entire 18 bytes response
13:35:07.886: TPLUS (00000055)/0/225FE2DC: Processing the reply packet
13:35:07.886: TPLUS: received bad AUTHEN packet: length = 6, expected 43974
13:35:07.886: TPLUS: Invalid AUTHEN packet (check keys).
  
```

- A. Configure the user name on the TACACS+ server
 B. Configure the UDP port 1812 to be allowed on the TACACS+ server
 C. Configure the TCP port 49 to be reachable by the router
 D. Configure the same password between the TACACS+ server and router.

Correct Answer: D

Section: (none)

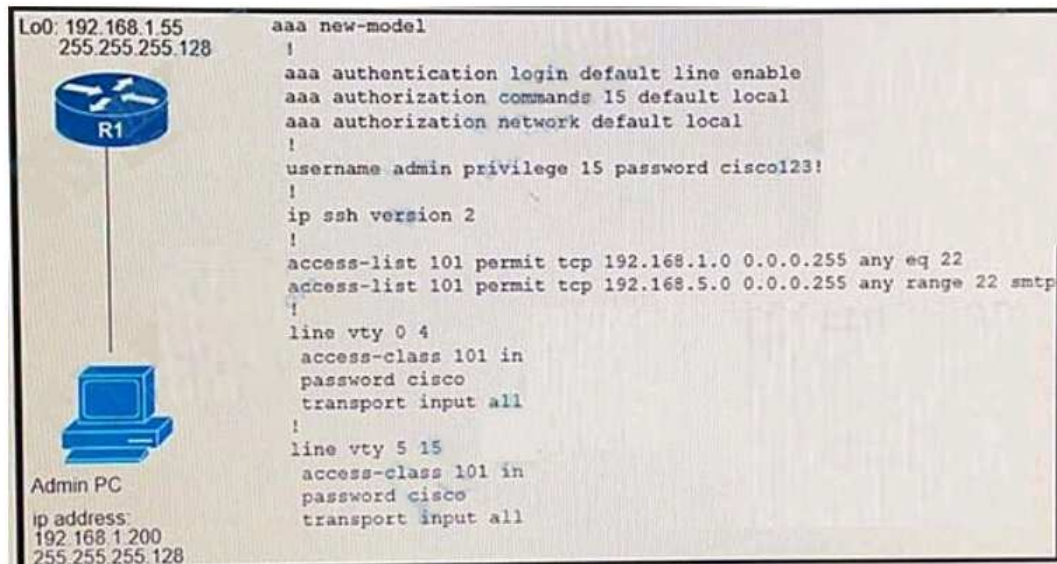
Explanation

Explanation/Reference:

The last message shows that authentication key is invalid.

QUESTION 199

Refer to the exhibit. The administrator successfully logs into R1 but cannot access privileged mode commands. What should be configured to resolve the issue?



- A. aaa authorization reverse-access
- B. secret cisco 123! at the end of the username command instead of password cisco123!
- C. matching password on vty lines as cisco123!
- D. enable secret or enable password commands to enter into privileged mode

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

The configuration set line vty to use password "cisco" to login. Hence the telnet user needs to use enable password / secret to access privileged mode. If enable password / secret is not configured, you need to configure one first.

Remarks :

Alternatively, you can configure authentication list using "local" and apply that list to line vty for logging in with the username "admin" and password "cisco123!" to enter privileged mode directly.

QUESTION 200

Drag and drop the MPLS VPN device types from me left onto the definitions on the right.

Select and Place:

Customer (C) device	device in the core of the provider network that switches MPLS packets
CE device	device that attaches and detaches the VPN labels to the packets in the provider network
PE device	device in the enterprise network that connects to other customer devices
Provider (P) device	device at the edge of the enterprise network that connects to the SP network

Correct Answer:

	Provider (P) device
	PE device
	Customer (C) device
	CE device

Section: (none)

Explanation

Explanation/Reference:

QUESTION 201

Drag and Drop the IPv6 First-Hop Security features from the left onto the definitions on the right.

Select and Place:

IPv6 Binding Table	Block reply and advertisement messages from unauthorized DHCP servers and relay agents
IPv6 DHCPv6 Guard	Create a binding table that is based on NS and NA messages
IPv6 Source Guard	Filter inbound traffic on Layer 2 switch port that are not in the IPv6 binding table
IPv6 ND Inspection	Block a malicious host and permit the router from a legitimate route
IPv6 RA Guard	Create IPv6 neighbors connected to the device from information sources such as NDP snooping

Correct Answer:

	IPv6 DHCPv6 Guard
	IPv6 ND Inspection
	IPv6 Source Guard
	IPv6 RA Guard
	IPv6 Binding Table

Section: (none)

Explanation

Explanation/Reference:

QUESTION 202

Drag and drop the actions from the left into the correct order on the right to configure a policy to avoid following packet forwarding based on the normal routing path.

Select and Place:

Configure route map instances.	step 1
Configure set commands.	step 2
Configure fast switching for PBR.	step 3
Configure ACLs.	step 4
Configure match commands.	step 5
Configure PBR on the interface.	step 6

Correct Answer:

	Configure ACLs.
	Configure route map instances.
	Configure match commands.
	Configure set commands.
	Configure PBR on the interface.
	Configure fast switching for PBR.

Section: (none)

Explanation

Explanation/Reference:

The last step can be optionally enabled in some Cisco devices / IOS versions by the interface configuration mode command:
ip route-cache policy

However, you must enable PBR in the interface before enabling fast-switching PBR.

QUESTION 203

Drag and drop the MPLS concepts from the left onto the descriptions on the right

Select and Place:

label edge router	allows an LSR to remove the label before forwarding the packet
lable switch router	accepts unlabeled packets and imposes labels
forwarding equivalence class	group of packets that are forwarded in the same manner
penultimate hop popping	receives labeled packets and swaps labels

Correct Answer:

	penultimate hop popping
	label edge router
	forwarding equivalence class
	lable switch router

Section: (none)

Explanation

Explanation/Reference:

QUESTION 204

Refer to the exhibit.

```
Jan 9 15:29:29.713: DHCP_SNOOPING: process new DHCP packet, message type: DHCPINFORM, input interface: Po2, MAC da: ffff.ffff.ffff, DHCP yiaddr: 0.0.0.0, DHCP siaddr: 0.0.0.0, DHCP giaddr: 0.0.0.0
Jan 9 15:29:29.713: DHCP_SNOOPING_SW: bridge packet get invalid mat entry: FFFF.FFFF.FFFF, packet is flooded to ingress VLAN: (1)
Jan 9 15:29:29.722: DHCP_SNOOPING_SW: bridge packet send packet to cpu port: Vlan1.
Jan 9 15:29:31.509: DHCP_SNOOP(hlrm_set_if_input): Setting if_input to Po2 for pak. Was V11
Jan 9 15:29:31.509: DHCP_SNOOP(hlrm_set_if_input): Setting if_input to V11 for pak. Was Po2
Jan 9 15:29:31.509: DHCP_SNOOP(hlrm_set_if_input): Setting if_input to Po2 for pak. Was V11
Jan 9 15:29:31.517: DHCP_SNOOPING: received new DHCP packet from input interface (Port-channel2)
```

A network administrator enables DHCP snooping on the Cisco Catalyst 3750-X switch and configures the uplink port (Port-channel2) as a trusted port. Clients are not receiving an IP address, but when DHCP snooping is disabled, clients start receiving IP addresses. Which global command resolves the issue?

A. no ip dhcp snooping information option

- B. ip dhcp snooping
- C. ip dhcp relay information trust portchannel2
- D. ip dhcp snooping trust

Correct Answer: A

Section: (none)

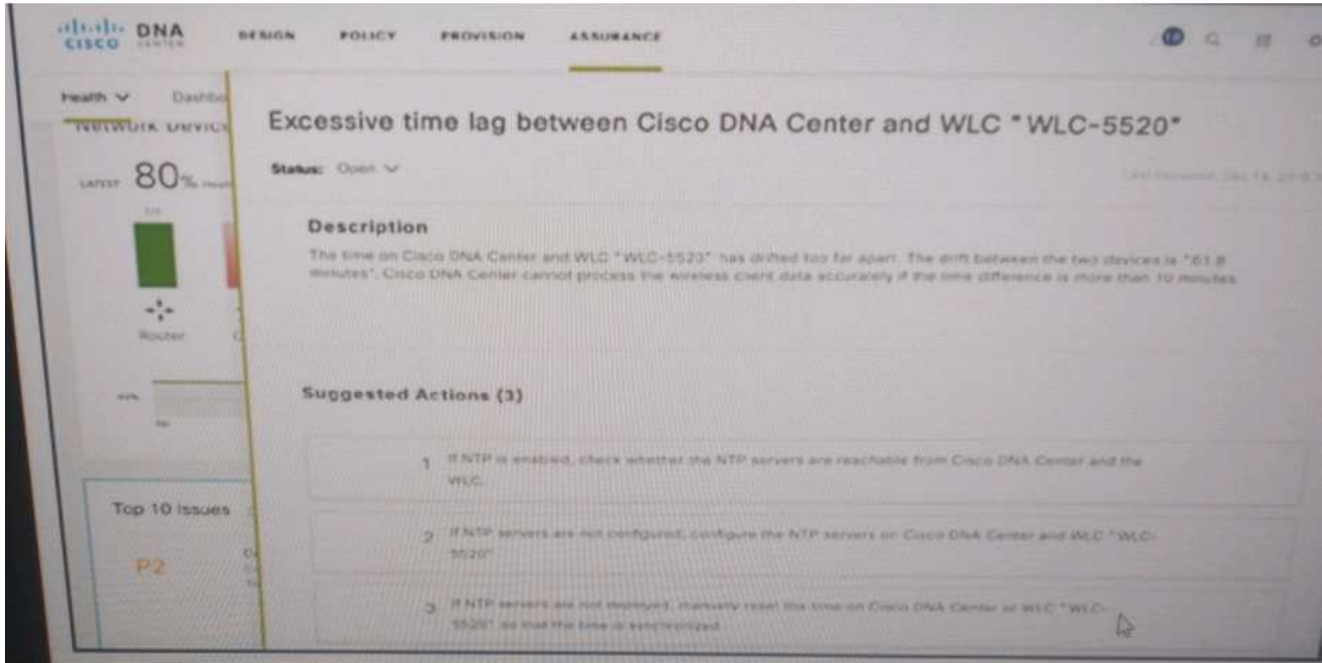
Explanation

Explanation/Reference:

A switch with DHCP Snooping enabled will drop packets on untrusted ports that contain Option 82 or have a non-zero giaddr (e.g. 0.0.0.0).

QUESTION 205

Refer to the exhibit. NTP is configured across the network infrastructure and Cisco DNA center. An NTP issue was reported on the Cisco DNA Center at 17:15. Which action resolves the issue?



- A. Check and resolve reachability between the WLC and the NTP server
- B. Reset the NTP server to resolve any synchronization issues for all devices
- C. Check and resolve reachability between Cisco DNA Center and the NTP server.
- D. Check and configure NTP on the WLC and synchronize with Cisco DNA Center.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

C seems unlikely since, unless the WLC-5520 is the only device managed by DNA, there will be many issues since "Excessive time lag" will occur for all other devices.

D should be wrong since the question mentioned that NTP has been configured.

QUESTION 206

Which two protocols work in the control plane of P routers across the MPLS cloud? (Choose two)

- A. LSP
- B. RSVP
- C. ECMP
- D. LDP
- E. MPLS OAM

Correct Answer: BD

Section: (none)

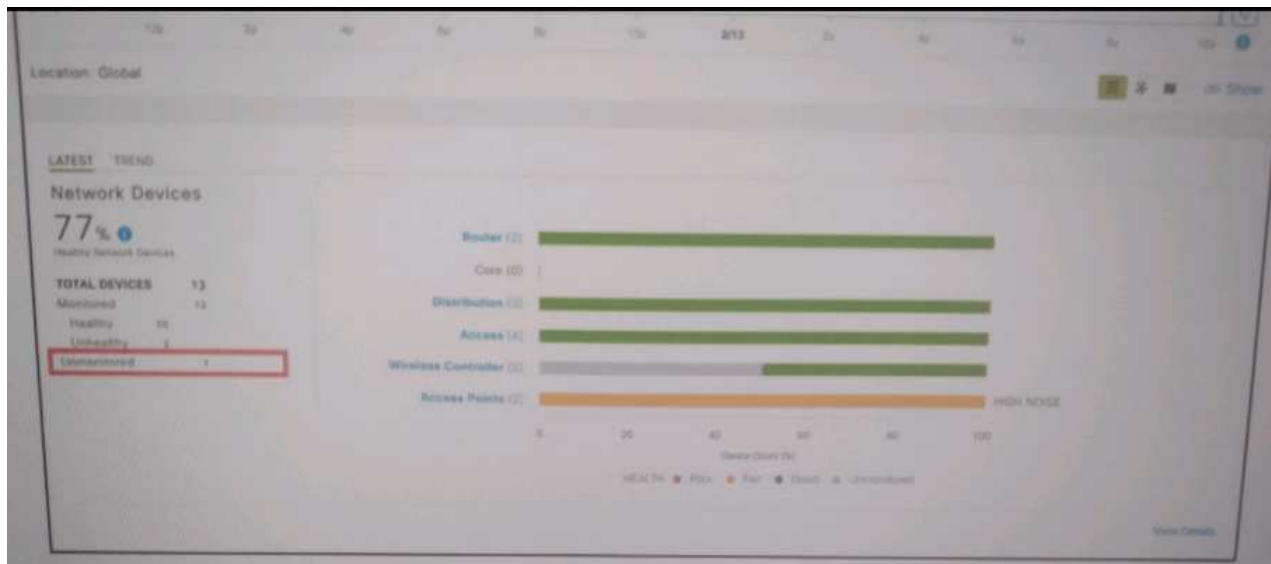
Explanation

Explanation/Reference:

RSVP is optional and is used for Traffic Engineering / QoS.

QUESTION 207

Refer to the exhibit.



A network administrator added one router in the Cisco DNA Center and checked its discovery and health from the Network Health Dashboard. The network administrator observed that the router is still showing up as unmonitored. What must be configured on the router to mount it in the Cisco DNA Center?

- A. Configure router with Netflow data.
- B. Configure router with the telemetry data.
- C. Configure router with routing to reach Cisco DNA Center
- D. Configure router with SNMPv2c or SNMPv3 traps.

Correct Answer: B

Section: (none)

Explanation

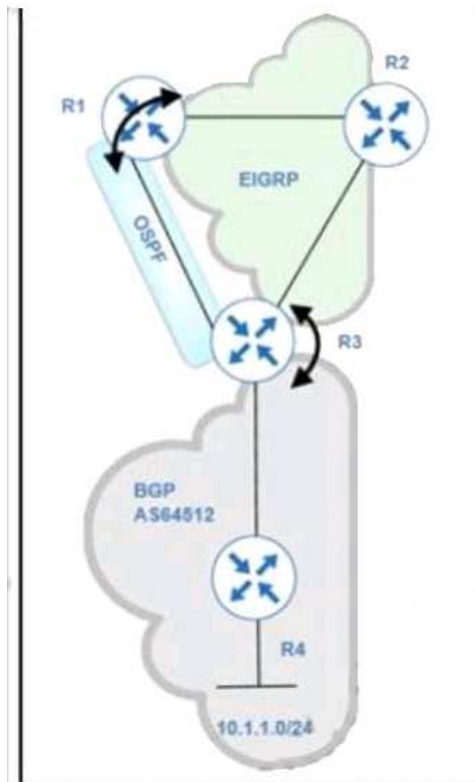
Explanation/Reference:

Unmonitored: Unmonitored devices are devices for which Assurance did not receive any telemetry data during the specified time range. Unmonitored devices are included in the Network Health Score computation. They are used as part of the total number of devices against which the health device percentage is calculated.

Telemetry data includes Syslog, SNMP Traps and NetFlow.

QUESTION 208

Refer to the exhibit.



Routing protocols are mutually redistributed on R3 and R1. Users report intermittent connectivity to services based on the 10.1.1.0/24 prefix. Significant routing update changes are noticed on R3 when the show ip route profile command is run. How must the services be stabilized?

- A. The issue with using BGP must be resolved by using another protocol and redistributing it into EIGRP on R3
- B. The routing loop must be fixed by reducing the admin distance of iBGP from 200 to 100 on R3.
- C. The routing loop must be fixed by reducing the admin distance of OSPF from 110 to 80 on R3
- D. The issue with using iBGP must be fixed by running eBGP between R3 and R4.

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

R3 learnt routes 10.1.1.0/24 by the following:

- from iBGP peer R4 and therefore is a iBGP route having AD of 200.
- from OSPF as a OSPF External router (since the iBGP route is redistributed into EIGRP by R3 and then redistributed into OSPF by R1.)

R3 will prefer the OSPF External route (AD of 110) than iBGP route (AD of 200). Since the iBGP route is no longer entered into the routing table, the route 10.1.1.0/24 will not be redistributed into EIGRP and OSPF. After OSPF route is missing, iBGP route is entered into the routing table again.

The above flapping can be resolved if R3 prefers iBGP route than OSPF External route. e.g. changing iBGP from 200 to 100.

QUESTION 209

Which configuration feature should be used to block rogue router advertisements instead of using the IPv6 Router Advertisement Guard feature?

- A. VACL blocking broadcast frames from nonauthorized hosts
- B. PVLANS with promiscuous ports associated to route advertisements and isolated ports for nodes
- C. PVLANS with community ports associated to route advertisements and isolated ports for nodes
- D. IPv4 ACL blocking route advertisements from nonauthorized hosts.

Correct Answer: B

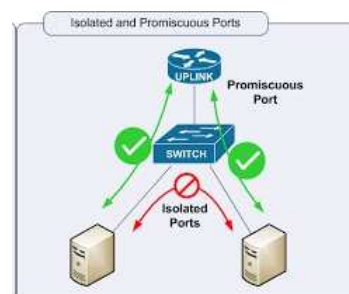
Section: (none)

Explanation

Explanation/Reference:

RA is IPv6 and IPv6 does not have broadcast.

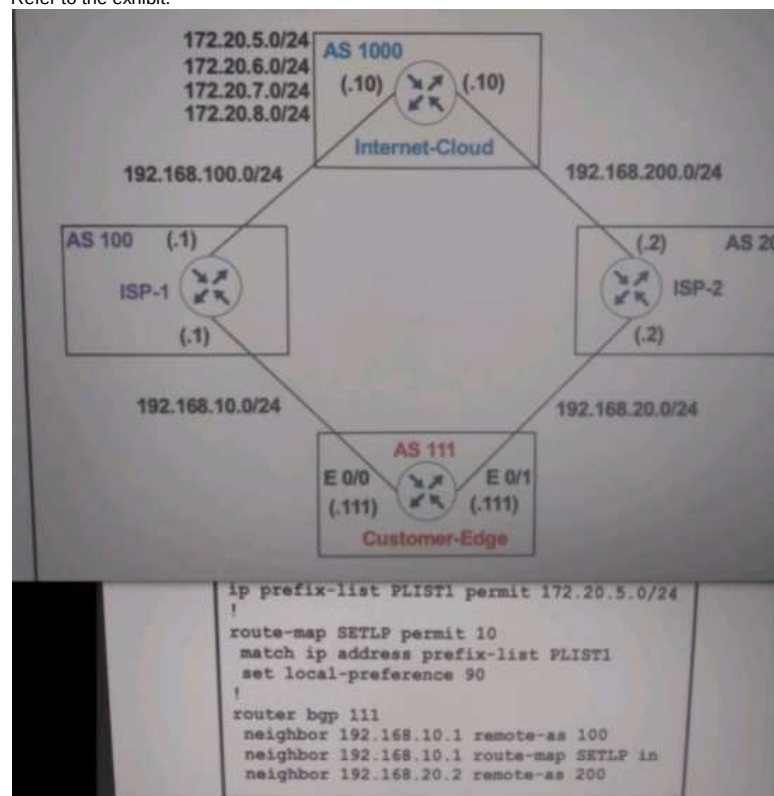
PVLANS means Private VLAN.



The promiscuous port can communicate with all interfaces, including the community and isolated host ports, that belong to those secondary VLANs associated to the promiscuous port and associated with the primary VLAN. Hence a router connected to this port can send RA to all hosts connected to isolated ports.

QUESTION 210

Refer to the exhibit.



AS 111 wanted to use AS 200 as the preferred path for 172.20.5.0/24 and AS100 as the backup. After the configuration, AS 100 is not used for any other routes.

Which configuration resolves the issue?

- A. route-map SETLP permit 10
match ip address prefix-list PLIST1
set local-preference 110
route-map SETLP permit 20
- B. route-map SETLP permit 10
match ip address prefix-list PLIST1
set local-preference 99
route-map SETLP permit 20
- C. router bgp 111
no neighbor 192.168.10.1 route-map SETLP in
neighbor 192.168.20.2 route-map SETLP in
- D. router bgp 111
no neighbor 192.168.10.1 route-map SETLP in
neighbor 192.168.10.1 route-map SETLP out

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since the Route Map is applied to the neighbor in AS 100 which is used as backup for 172.20.5.0/24. Moreover AS 100 can be used for other routes.

- i) a Local Preference lower than 100 should be set for 172.20.5.0/24. (The original configuration achieves this)
- ii) a permit with no match command should be configured in the Route Map as the last statement. (Missing configuration to achieve this)

Since there is no choice to keep the original settings of Local Preference 90, B is the best choice since it is still lower than 100.

QUESTION 211

Refer to the exhibit. The network administrator configured the branch router for IPv6 on the E0/0 interface. The neighboring router is fully configured to meet requirements, but the neighbor relationship is not coming up. Which action fixes the problem on the branch router to bring the IPv6 neighbors up?

```
Ipv6 unicast-routing
!
Router ospfv3 4
  Router-id 192.168.1.1
!
Interface E 0/0
  Ipv6 enable
  Ip address 10.1.1.1 255.255.255.0
  Ospfv3 4 area 0 ipv4
  No shut
!
Interface Loopback0
  Ipv6 enable
  Ipv4 172.16.1.1 255.255.255.0
  Ospfv3 4 area 0 ipv4
```

- A. Enable the IPv4 address family under the router ospfv3 4 process by using the address-family ipv4 unicast command
- B. Disable IPv6 on the E0/0 interface using the no ipv6 enable command
- C. Enable the IPv4 address family under the E0/0 interface by using the address-family ipv4 unicast command
- D. Disable OSPF for IPv4 using the no ospfv3 4 area 0 ipv4 command under the E0/0 interface

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

There are many issues in this questions. However we shall first look at the choices:

A can be configured. This allows you to configure passive interface under "address-family ipv4 unicast".

B is a valid command. This disable IPv6 but it will not allow the router to form IPv6 OSPF neighbor and will not affect IPv4 OSPF neighbor.

C is wrong since "address-family ipv4 unicast" cannot be configured under interface configuration.

D is a valid command. This disable the OSPF process for IPv4. But it will not allow the router to form IPv6 OSPF neighbor or IPv4 OSPF neighbor..

Only A seems to be a reasonable choice but it has nothing to do with what the question is asking i.e. forming IPv6 neighbor.

The correct answer is to configure "**ospfv3 4 ipv6 area 0**" or "**ipv6 ospf 4 area 0**" in the interface but there is no choice mentioning this.

Remarks:

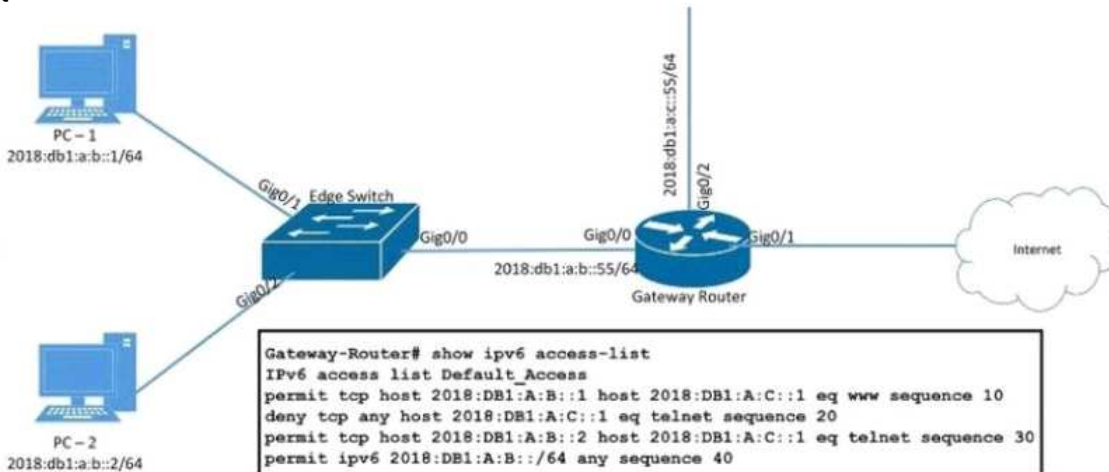
Major issues in this question:

- For OSPFv3 for IPv4, unlike EIGRP, you do not need to configure address-family ipv4. OSPFv3 will start running just by enabling it in the Interface.
- The Interface command for enabling OSPF in the exhibit is wrong, the correct command should be "**ospfv3 4 ipv4 area 0**".
- The question is about IPv6 neighbor but the only choice concerning IPv6 is B which disable IPv6. Moreover, OSPFv3 requires IPv6 to be enabled.

```
Router(config)#int g0/1
Router(config-if)#ospfv3 4 ipv4 area 0
% OSPFv3: IPv6 routing not enabled
Router(config-if)#ipv6 unicast-routing
Router(config)#int g0/1
Router(config-if)#ospfv3 4 ipv4 area 0
% OSPFv3: IPv6 is not enabled on this interface
```

```
Router(config-if)#ipv6 enable
Router(config-if)#ospfv3 4 ipv4 area 0
Router(config-if)#
```

QUESTION 212



Which should be configured to allow telnet connection from the PC 2018:DB1:A:B::2 to the server 2018:DB1:A:C::1 ?

- A. R2(config)#ipv6 access-list input1
R2(config-ipv6-acl)#sequence 15 permit tcp host 2018:DB1:A:B::2 host 2018:DB1:A:C::1 eq telnet
- B. R2(config)#ipv6 access-list input1
R2(config-ipv6-acl)#sequence 25 permit tcp host 2018:DB1:A:B::2 host 2018:DB1:A:C::1 eq telnet
- C. R2(config)#ipv6 access-list input1
R2(config-ipv6-acl)#no sequence 20
R2(config-ipv6-acl)#sequence 5 permit tcp host 2018:DB1:A:B::2 host 2018:DB1:A:C::1 eq telnet
- D. R2(config)#ipv6 access-list input1
R2(config-ipv6-acl)#permit tcp host 2018:DB1:A:B::2 host 2018:DB1:A:C::1 eq telnet

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

All telnet traffic from any host to the server 2018:DB1:A:C::1 is blocked by the rule sequence 20. Therefore the rule allowing telnet from the PC to the server in sequence 30 will not be evaluated.

Hence you should configure the rule in sequence 30 to be located before sequence 20 e.g. 15.

C is not the best answer since remove sequence 20 will allow all other devices to telnet the server. The only rule that requires removal is sequence 30 if you have added that rule to another location e.g. 15.

QUESTION 213

How does an MPLS Layer 3 VPN function?

- A. set of sites use multiprotocol BGP at the customer site for aggregation
- B. set of sites interconnect privately over the Internet for security
- C. multiple customer sites interconnect through service provider network to create secure tunnels between customer edge devices
- D. multiple customer sites interconnect through a service provider network using customer edge to provider edge connectivity

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

A is wrong since MP-BGP is run in PE router. A CE router can run normal BGP or IGP.

B is wrong since MPLS network is required instead of Internet.

C is wrong since no encryption is provided by default through service provider' MPLS network.

QUESTION 214

A customer reports to the support desk that they cannot print from their PC to the local printer id:401987778. Which tool must be used to diagnose the issue using Cisco DNA Center Assurance?

- A. application trace
- B. path trace
- C. ACL trace
- D. device trace

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 215

**** Missing a network diagram in which some Ethernet interface e.g. e0/0, e0/1 are actually 100Mbps or 1Gbps links. ****

```

L    172.1.12.3/32 is directly connected, Ethernet0/0
C    172.1.13.0/24 is directly connected, Ethernet0/1
L    172.1.13.3/32 is directly connected, Ethernet0/1
O    192.168.1.0/24 [110/2] via 172.1.12.1, 00:04:44, Ethernet0/0
O    192.168.2.0/24 [110/2] via 172.1.12.1, 00:04:44, Ethernet0/0
O    192.168.3.0/24 [110/2] via 172.1.13.2, 00:04:44, Ethernet0/1
O    192.168.4.0/24 [110/2] via 172.1.13.2, 00:04:44, Ethernet0/1
    192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.5.0/24 is directly connected, Loopback0
L    192.168.5.1/32 is directly connected, Loopback0
    192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.6.0/24 is directly connected, Loopback1
L    192.168.6.1/32 is directly connected, Loopback1

```

Refer to the exhibit. SanFrancisco and Boston routers are choosing slower links to reach each other despite the direct links being up Which configuration fixes the issue?

A. **Boston Router**

```

router ospf 1
auto-cost reference-bandwidth 1000

```

B. **SanFrancisco Router**

```

router ospf 1
auto-cost reference-bandwidth 1000

```

C. **All Routers**

```

router ospf 1
auto-cost reference-bandwidth 100

```

D. **All Routers**

```

router ospf 1
auto-cost reference-bandwidth 1000

```

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

By default, OSPF uses 100 as reference bandwidth in metric calculation and therefore cannot distinguish among 100Mbps and 1Gbps. A router may therefore prefer to reach destination through one 100Mbps link and two 1Gbps links.

To solve this, you should configure all concerned routers with 1000 or more as reference bandwidth.

QUESTION 216

```

NY
router ospf 1
network 192.168.12.0 0.0.0.255 area 0
network 172.16.2.0 0.0.0.255 area 0
!
interface E 0/0
ip ospf authentication message-digest
ip ospf message-digest-key 1 md5 Cisco123

```

*** seems LA router configuration is missing ***

*** probably the LA router is missing OSPF authentication settings ***

Refer to the exhibit. The neighbor relationship is not coming up Which two configurations bring the adjacency up? (Choose two)

A. **NY**

```

interface E 0/0
no ip ospf message-digest-key 1 md5 Cisco123
ip ospf authentication-key Cisco123

```

B. **LA**

```

router ospf 1
area 0 authentication message-digest

```

- C. NY
 router ospf 1
 area 0 authentication message-digest
- D. LA
 interface E 0/0
 ip ospf authentication-key Cisco123
- E. LA
 interface E 0/0
 ip ospf message-digest-key 1 md5 Cisco123

Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

For the choices, the most probably reason is the LA router is missing MD5 authentication configuration.

QUESTION 217

```
router ospf 1
 redistribute eigrp 1 subnets route-map EIGRP->OSPF
!
router eigrp 1
 network 10.0.106.0 0.0.0.255
!
route-map EIGRP->OSPF permit 10
 match ip address WAN_PREFIXES
route-map EIGRP->OSPF permit 20
 match ip address LOCAL_PREFIXES
route-map EIGRP->OSPF permit 30
 match ip address VPN_PREFIXES
!
ip prefix-list LOCAL_PREFIXES seq 5 permit 172.16.0.0/12 le 24
ip prefix-list VPN_PREFIXES seq 5 permit 192.168.0.0/16 le 24
ip prefix-list WAN_PREFIXES seq 5 permit 10.0.0.0/8 le 24
!
```

Refer to the exhibit. The network administrator configured redistribution on an ASBR to reach to all WAN networks but failed Which action resolves the issue?

- A. The route map must have the keyword prefix-list to evaluate the prefix list entries
- B. The OSPF process must have a metric when redistributing prefixes from EIGRP.
- C. The route map EIGRP->OSPF must have the 10.0.106.0/24 entry to exist in one of the three prefix lists to pass
- D. EIGRP must redistribute the 10.0.106.0/24 route instead of using the network statement

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The keyword "prefix-list" are missing when specifying the Prefix List in the match command in all Route Map statements in the exhibit.

e.g. match ip address **prefix-list** EIGRP->OSPF

QUESTION 218

```
ipv6 access-list INTERNET
 permit ipv6 2001:DB8:AD59:BA21::/64 2001:DB8:C0AB:BA14::/64
 permit tcp 2001:DB8:AD59:BA21::/64 2001:DB8:C0AB:BA13::/64 eq telnet
 permit tcp 2001:DB8:AD59:BA21::/64 any eq http
 permit ipv6 2001:DB8:AD59::/48 any
 deny ipv6 any any log
```

Refer to the exhibit. When monitoring an IPv6 access list, an engineer notices that the ACL does not have any hits and is causing unnecessary traffic to pass through the interface Which command must be configured to resolve the issue?

- A. access-class INTERNET in
- B. ipv6 traffic-filter INTERNET in
- C. ipv6 access-class INTERNET in
- D. ip access-group INTERNET in

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 219

An engineer configured Reverse Path Forwarding on an interface and noticed that the routes are dropped when a route lookup fails on that interface for a prefix that is available in the routing table. Which interface configuration resolves the issue?

- A. ip verify unicast source reachable-via rx
- B. ip verify unicast source reachable-via any
- C. ip verify unicast source reachable-via allow-default
- D. ip verify unicast source reachable-via 12-src

Correct Answer: B

Section: (none)

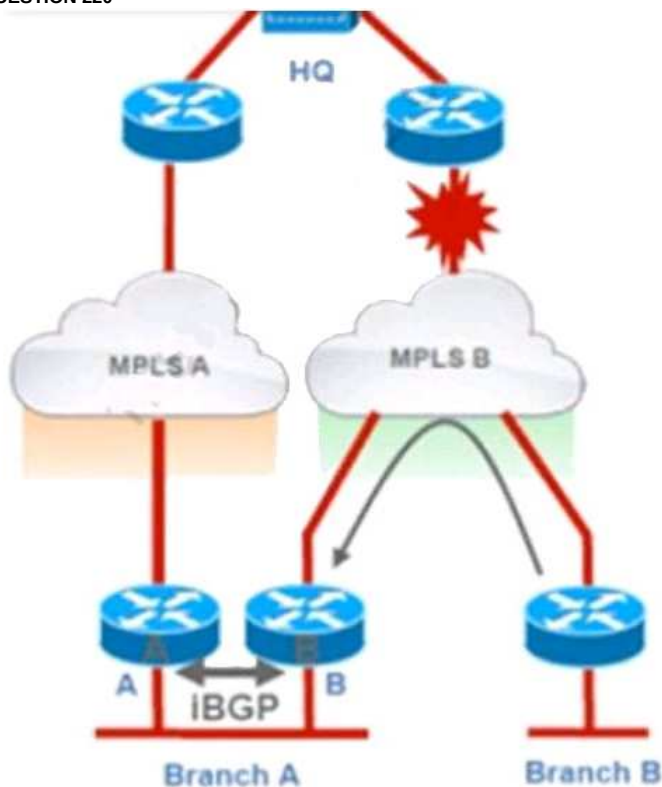
Explanation

Explanation/Reference:

Seems the wordings "the routes are dropped" should be "the packets are dropped".

Since the prefix is available in the routing table, the reason may be due to interface not matched. Hence you can fix this by using "any" instead of "rx".

QUESTION 220



Refer to the exhibit. Troubleshoot and ensure that branch B only ever uses the MPLS B network to reach HQ. Which action achieves this requirement?

- A. Introduce an AS path filter on branch A routers so that only local prefixes are advertised into BGP
- B. increase the local preference for all HQ prefixes received at branch B from the MPLS B network to be higher than the local preferences used on the MPLS A network
- C. Introduce AS path prepending on the branch A MPLS B network connection so that any HQ advertisements from branch A toward the MPLS B network are prepended three times
- D. Modify the weight of all HQ prefixes received at branch B from the MPLS B network to be higher than the weights used on the MPLS A network

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Local Preference and AS prepend can only configure a preferred path. If the preferred path is down, the other non-preferred path will be used.

If you do not want to use a non-preferred / path even if the preferred path is down, you should configure filtering.

QUESTION 221

Debug output:

username: USER55

password:

Aug 26 12:39:23.813: TPLUS: Queuing AAA Authentication request 4950 for processing

Aug 26 12:39:23.813: TPLUS(00001356) login timer started 1020 sec timeout

Aug 26 12:39:23.813: TPLUS: processing authentication continue request id 4950

Aug 26 12:39:23.813: TPLUS: Authentication continue packet generated for 4950

Aug 26 12:39:23.813: TPLUS(00001356)/0/WRITE/3A72C8D0: Started 5 sec timeout

|

|----- output omitted -----|

|

Aug 26 12:40:01.241: TAC+: using previously set server 192.168.1.3 from group tacacs+

Aug 26 12:40:01.241: TAC+: Opening TCP/IP to 192.168.1.3/49 timeout=5

Aug 26 12:40:01.249: TAC+: Opened TCP/IP handle 0x3BE31D1C to 192.168.1.3/49

Aug 26 12:40:01.249: TAC+: Opened 192.168.1.3 index=1

Aug 26 12:40:01.250: TAC+: 192.168.1.3 (3653537180) AUTHOR/START queued

Aug 26 12:40:01.449: TAC+: (3653537180) AUTHOR/START processed

Aug 26 12:40:01.449: TAC+: (-641430116): received author response status = FAIL

Aug 26 12:40:01.450: TAC+: Closing TCP/IP 0x3BE31D1C connection to 192.168.1.3/49

Refer to the exhibit. A network administrator logs into the router using TACACS+ username and password credentials, but the administrator cannot run any privileged commands. Which action resolves the issue?

- A. Configure TACACS+ synchronization with the Active Directory admin group
- B. Configure the username from a local database
- C. Configure full access for the username from TACACS+ server
- D. Configure an authorized IP address for this user to access this router

Correct Answer: C**Section: (none)****Explanation****Explanation/Reference:**

Command Sets

Filter: Match if: Go

☐	Command Set Name	Description
☐	DenyAllCommands	
☐	Full-Access	

|

You can also create a custom command set e.g. "Show-Access" in the following diagram and configure the commands for this set.

General

Name:

Description:

☐ Permit any command that is not in the table below

Grant	Command	Arguments
Permit	show	
Permit	enable	
Permit	exit	

Grant:
Command:
Arguments:

Select Command/Arguments from Command Set:

QUESTION 222

Router# show ip route

```

2.0.0.0/24 is subnetted, 1 subnets
C    2.2.2.0 is directly connected, Ethernet0/0
C    3.0.0.0/8 is directly connected, Serial1/0
O E2 200.1.1.0/24 [110/20] via 2.2.2.2, 00:16:17, Ethernet0/0
O E1 200.2.2.0/24 [110/104] via 2.2.2.2, 00:00:41, Ethernet0/0
    131.108.0.0/24 is subnetted, 2 subnets
O    131.108.2.0 [110/74] via 2.2.2.2, 00:16:17, Ethernet0/0
O IA  131.108.1.0 [110/84] via 2.2.2.2, 00:16:17, Ethernet0/0

```

Router# show ip bgp

```

Network        Next Hop        Metric LocPrf Weight Path
*> 2.2.2.0/24   0.0.0.0         0      32768 ?
*> 131.108.1.0/24 2.2.2.2        84      32768 ?
*> 131.108.2.0/24 2.2.2.2        74      32768 ?

```

Refer to the exhibit. The OSPF routing protocol is redistributed into the BGP routing protocol, but not all the OSPF routes are distributed into BGP. Which action resolves the issue?

- A. Include the word external in the redistribute command
- B. Use a route-map command to redistribute OSPF external routes defined in an access list
- C. Include the word internal external in the redistribute command
- D. Use a route-map command to redistribute OSPF external routes defined in a prefix list.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

In summary, since the question does not mention whether you reconfigure redistribution from start again or just configure an additional command, therefore answer C is more safe than A.

Detailed explanations:

By default only OSPF internal routes are redistributed into BGP, you need to specify both "internal" and "external" in all to redistribute all e.g.:

```

router bgp 1
 redistribute ospf 1 match internal external

```

Note that if redistribute is configured without "match" i.e. redistribute OSPF internal only, if you add an additional redistribute command with "match external", both internal and external will be included (i.e. the case for this question).

```
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 1
Router(config-router)#redistribute ospf 1          <--- assume this already exist
Router(config-router)#redistribute ospf 1 match external
Router(config-router)#end
Router#
```

```
Router#sh run | section router bgp
router bgp 1
  bgp log-neighbor-changes
  redistribute ospf 1 match internal external 1 external 2
Router#
```

If the question asks for an additional command, configuring "match external" is enough

However if no redistribute has been configured before (e.g. you issue "no redistribute ospf 1" and configure the redistribute command again), if you configure "match external" only, only external will be included (i.e. the case for this question).

```
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 1
Router(config-router)#no redistribute ospf 1
Router(config-router)#redistribute ospf 1 match external
Router(config-router)#end
Router#
```

```
Router#sh run | section router bgp
router bgp 1
  bgp log-neighbor-changes
  redistribute ospf 1 match external 1 external 2
Router#
```

If you configure a route map, both internal and external has to be matched in the Route Map statements.

QUESTION 223

Refer to the exhibit. Which action resolves the failed authentication attempt to the router?



- A. Configure aaa authorization login command on line vty 0 4
- B. Configure aaa authorization login command on line console 0
- C. Configure aaa authorization console global command
- D. Configure aaa authorization console command on line vty 0 4

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

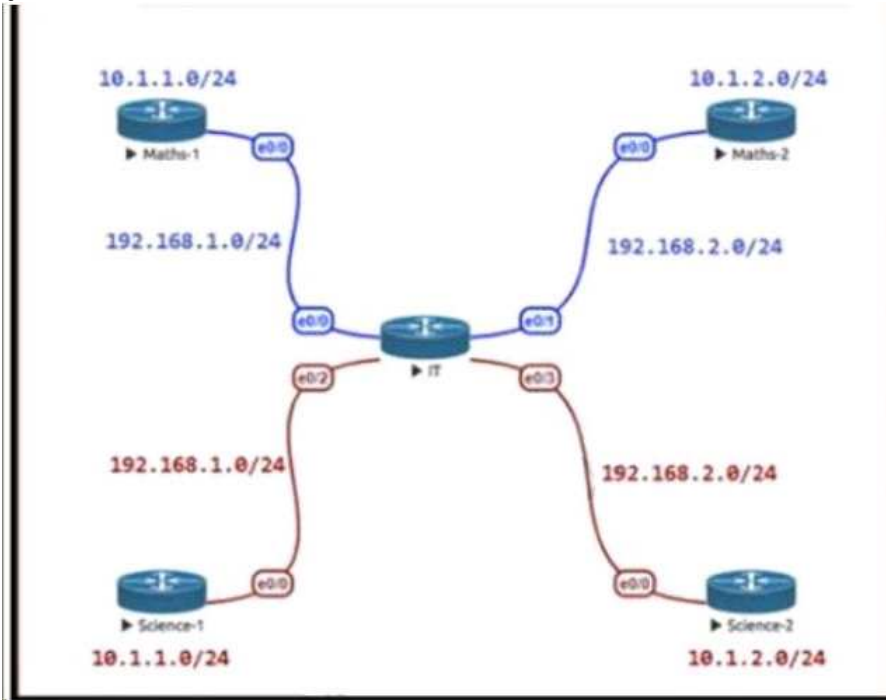
Unlike vty, you need the global command "aaa authorization console" in order to apply AAA Authorization in line console. The following shows the output if you have not configured this command:

```
Router(config)#line con 0
Router(config-line)#authorization exec authorlist1
%Authorization without the global command 'aaa authorization console' is useless
```

Remarks:

Although the diagram is missing some lines or the question is missing some description, we can only assume that the above debug messages are generated when a user is logging in through console.

QUESTION 224



Refer to the exhibit. The Math and Science departments connect through the corporate IT router but users in the Math department must not be able to reach the Science department and vice versa. Which configuration accomplishes this task?

- A. vrf definition Science
 - address-family ipv4
 - !
 - interface E 0/2
 - vrf forwarding Science
 - ip address 192.168.1.1 255.255.255.0
 - no shut
 - !
 - interface E 0/3
 - vrf forwarding Science
 - ip address 192.168.2.1 255.255.255.0
 - no shut
- B. vrf definition Science
 - address family ipv4
 - !
 - interface E 0/2
 - ip address 192.168.1.1 255.255.255.0
 - no shut
 - !
 - interface E 0/3
 - ip address 192.168.2.1 255.255.255.0
 - no shut
- C. vrf definition Science
 - !
 - interface E 0/2
 - ip address 192.168.1.1 255.255.255.0
 - no shut
 - !
 - interface E 0/3
 - ip address 192.168.2.1 255.255.255.0
 - no shut
- D. vrf definition Science
 - address-family ipv4
 - !
 - interface E 0/2
 - ip address 192.168.1.1 255.255.255.0
 - vrf forwarding Science
 - no shut
 - !
 - interface E 0/3
 - ip address 192.168.2.1 255.255.255.0
 - vrf forwarding Science
 - no shut

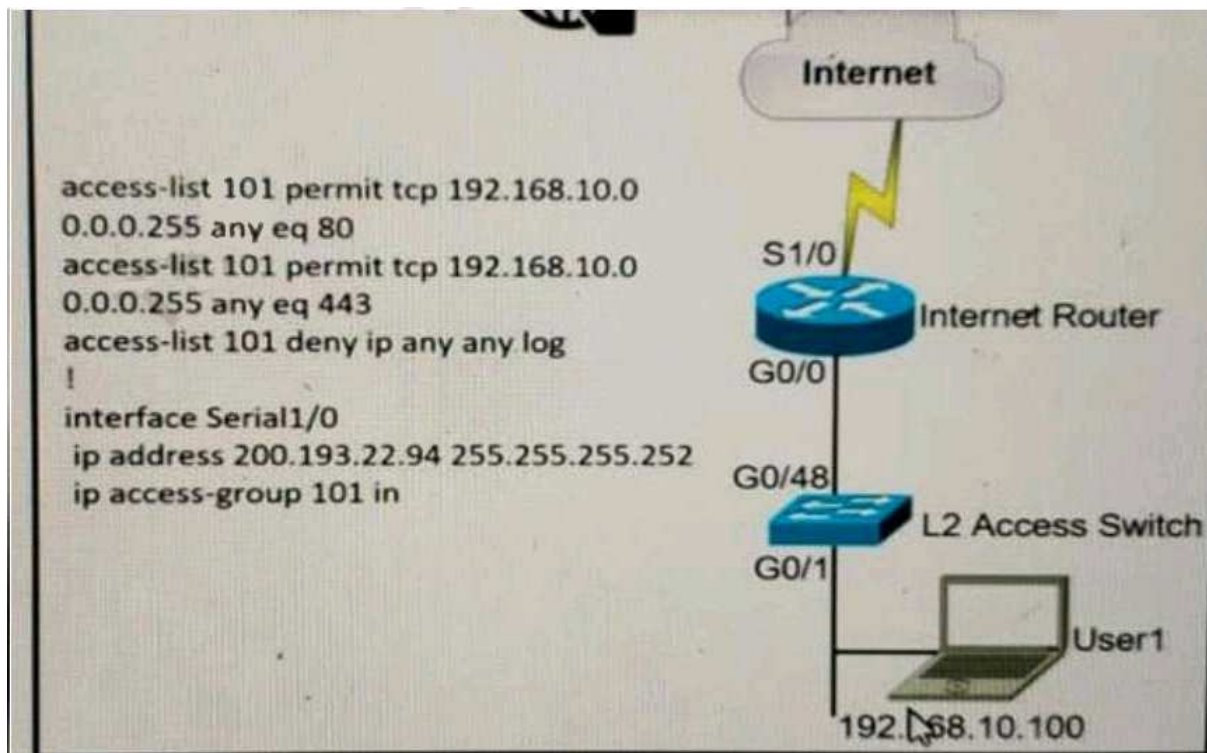
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 225



A network administrator is tasked to permit http and https traffic only toward the internet from the User1 laptop to adhere to company's security policy. The administrator can still ping to www.cisco.com Which interface should the access list 101 be applied to resolve this issue?

- A. Interface G0/48 in the incoming direction
- B. Interface G0/0 in the outgoing direction.
- C. Interface S1/0 in the outgoing direction.
- D. Interface G0/0 in the incoming direction.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

A little strange in the question:

the access list is applied to s1/0 inbound and no ICMP is allowed in access-list 100. Therefore no ping reply should be successful pass the inbound access list even the administrator issue ping in the router.

Anyway, since ports 80 and 443 are configured as destination port in the access list, the access list should be applied to:

- g0/0 inbound; OR
- s1/0 outbound

Neither of the above can stop ping from the router.

In most cases, g0/0 inbound is better since it blocks traffic before routing.

QUESTION 226

Drag and drop the LDP features from the left onto the descriptions on the right

Select and Place:

implicit null label	provides ways of improving load balancing by eliminating the need for DPI at transit LSRs
explicit null label	LSR receives an MPLS header with the label set to 3
inbound label binding filtering	packet is encapsulated in MPLS with the option of copying the IP precedence to EXP bits
entropy label	controls the amount of memory used to store LDP label bindings advertised by other devices

Correct Answer:

	entropy label
	implicit null label
	explicit null label
	inbound label binding filtering

Section: (none)

Explanation

Explanation/Reference:

By default, Cisco IOS Software copies the three most significant bits of the DSCP or the IP precedence of the IP packet to the EXP field in the MPLS header. This is not directly related to Explicit Null Label or any choices in the question.

However, Explicit Null Label can ensure the using of EXP for packet preference along the LSP until the final MPLS router LER is reached.

The MPLS LDP Inbound Label Binding Filtering feature can be used to control the amount of memory used to store Label Distribution Protocol (LDP) label bindings advertised by other devices

QUESTION 227

```

Router#show ip eigrp interfaces
EIGRP-IPv4 Interfaces for AS(1)
Interface      Xmit Queue  PeerQ      Mean  Pacing Time  Multicast  F
Peers  Un/Reliable  Un/Reliable  SRTT  Un/Reliable  Flow T
Lo0           0       0/0       0/0    0    0/0        0       0
Fa0/0         1       0/0       0/0    7    0/2       50      0

Router#show running-config | section eigrp
router eigrp 1
 network 172.16.0.0 0.0.0.255
 network 192.168.2.2 0.0.0.0
 network 192.168.12.2 0.0.0.0

Router#show running-config interface Fa0/3
Building configuration...

Current configuration : 93 bytes
!
interface FastEthernet0/3
 ip vrf forwarding CLIENT1
 ip address 172.16.0.1 255.255.0

```

Refer to the exhibit. While troubleshooting an EIGRP neighbor adjacency problem, the network engineer notices that the interface connected to the neighboring router is not participating in the EIGRP process. Which action resolves the issues?

- A. Configure the network command to network 172.16.0.1 0.0.0.0
- B. Configure the network command under EIGRP address family vrf CLIENT1
- C. Configure EIGRP metrics on interface FastEthernet0/3
- D. Configure the network command under EIGRP address family ipv4

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

The interface f0/3 cannot be found in the EIGRP interfaces shown in the first output even its is included in the network comamnd shown in the second output.

From the third output, the interface f0/3 is assigned to vrf CLIENT1. Therefore you must configure address family vrf CLIENT1 in order to include the interface. in the EIGRP.

QUESTION 228

Refer to the exhibit.

Branch-Router#

*Nov 29 15:20:22.415: OSPF-1 HELLO Fa1/1: Rcv hello from 3.3.3.3 area 1 10.2.1.3

*Nov 29 15:20:23.195: OSPF-1 HELLO Fa1/1: Send hello to 224.0.0.5 area 1 from 10.2.1.1

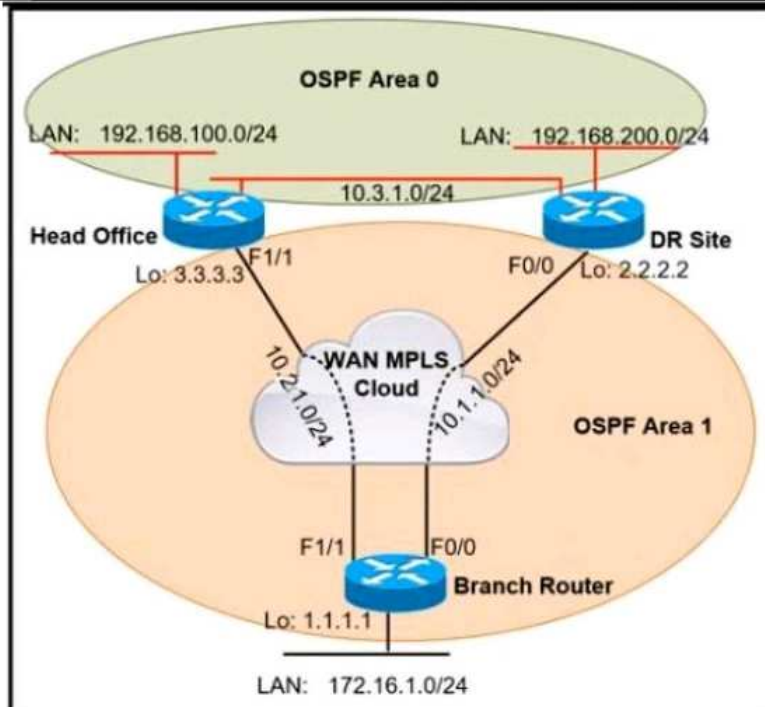
Branch-Router#

*Nov 29 15:20:27.955: OSPF-1 HELLO Fa0/0: Rcv hello from 2.2.2.2 area 1 10.1.1.2

*Nov 29 15:20:27.955: OSPF-1 HELLO Fa0/0: Mismatched hello parameters from 10.1.1.2

*Nov 29 15:20:27.955: OSPF-1 HELLO Fa0/0: Dead R 40 C 40, Hello R 10 C 10 Mask R 255.255.255.0 C 255.255.255.240

*Nov 29 15:20:28.311: OSPF-1 HELLO Fa0/0: Send hello to 224.0.0.5 area 1 from 10.1.1.1



A network administrator reviews the branch router console log to troubleshoot the OSPF adjacency issue with the DR router. Which action resolves this issue?

- A. Configure the WAN interface for DR site in the related OSPF area.
- B. Stabilize the DR site flapping link to establish OSPF adjacency.
- C. Advertise the branch WAN interface matching subnet for the DR site.
- D. Configure matching hello and dead intervals between sites-

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

For the last message:

- Received Dead is 40 and Configured Dead is 40 (matched)
- Received Hello is 10 and Configured Hello is 10 (matched)
- Received mask is 255.255.255.128 and Configured mask is 255.255.255.0 (mis-matched)

Therefore the network subnet masks are not matched between Branch router's WAN interface and DR router's WAN interface.

QUESTION 229

How is VPN routing information distributed in an MPLS network?

- A. The top level of the customer data packet directs it to the correct CE device.
- B. It is established using VPN IPsec peers.
- C. It is controlled through the use of RD.
- D. It is controlled using of VPN target communities.

Correct Answer: D

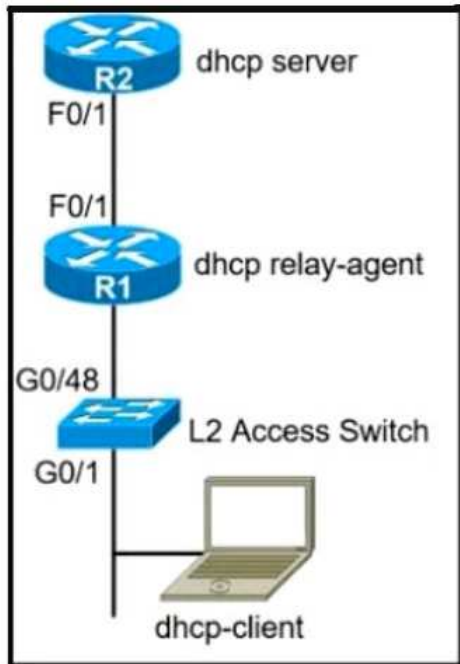
Section: (none)

Explanation

Explanation/Reference:

QUESTION 230

Refer to the exhibit.



The network administrator can see the DHCP discovery packet in R1, but R2 is not replying to the DHCP request. The R1 related interface is configured with the DHCP helper address. If the PC is directly connected to the Fa0/1 interface on R2, the DHCP server assigns an IP address from the DHCP pool to the PC.

Which two commands resolve this issue? (Choose two.)

- A. service dhcp command on R1
- B. ip dhcp relay information trust-all command on R2
- C. ip dhcp relay information enable command on R1
- D. service dhcp-relay command on R1
- E. ip dhcp option 82 command on R2

Correct Answer: AB

Section: (none)

Explanation

Explanation/Reference:

From the facts shown in the question, R2's DHCP service is working. The reason for PC not obtaining IP address is:

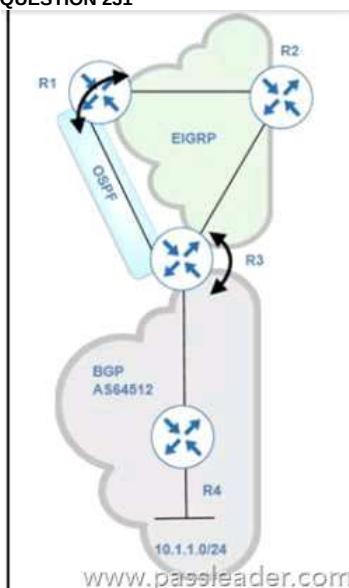
- There is no DHCP pool configured for the remote network in DHCP service in R2 (No choice is related to this)
- R1 is not working properly as DHCP relay agent even if helper-address is configured since the DHCP service (for DHCP server service and relay-agent service) is disabled in R1. This can be solved by A.
- The switch (between PC and R1) may have added dhcp option 82 to the DHCP message that is not accepted by R1 or R2. This can be solved by B.

C is NOT a valid command

D is NOT a valid command

E is NOT a valid command

QUESTION 231



Refer to the exhibit. BGP and EIGRP are mutually redistributed on R3, and EIGRP and OSPF are mutually redistributed on R1. Users report packet loss and interruption of service to applications hosted on the 10.1.1.0/24 prefix.

An engineer tested the link from R3 to R4 with no packet loss present but has noticed frequent routing changes on R3 when running the debug ip route command. Which action stabilizes the service?

- A. Tag the 10.1.1.0/24 prefix and deny the prefix from being redistributed into OSPF on R1.
- B. Reduce frequent OSPF SPF calculations on R3 that cause a high CPU and packet loss on traffic traversing R3.
- C. Place an OSPF distribute-list outbound on R3 to block the 10.1.1.0/24 prefix from being advertised back to R3.
- D. Repeat the test from R4 using ICMP ping on the local 10.1.1.0/24 prefix, and fix any Layer 2 errors on the host or switch side of the subnet.

Correct Answer: A

Section: (none)

Explanation

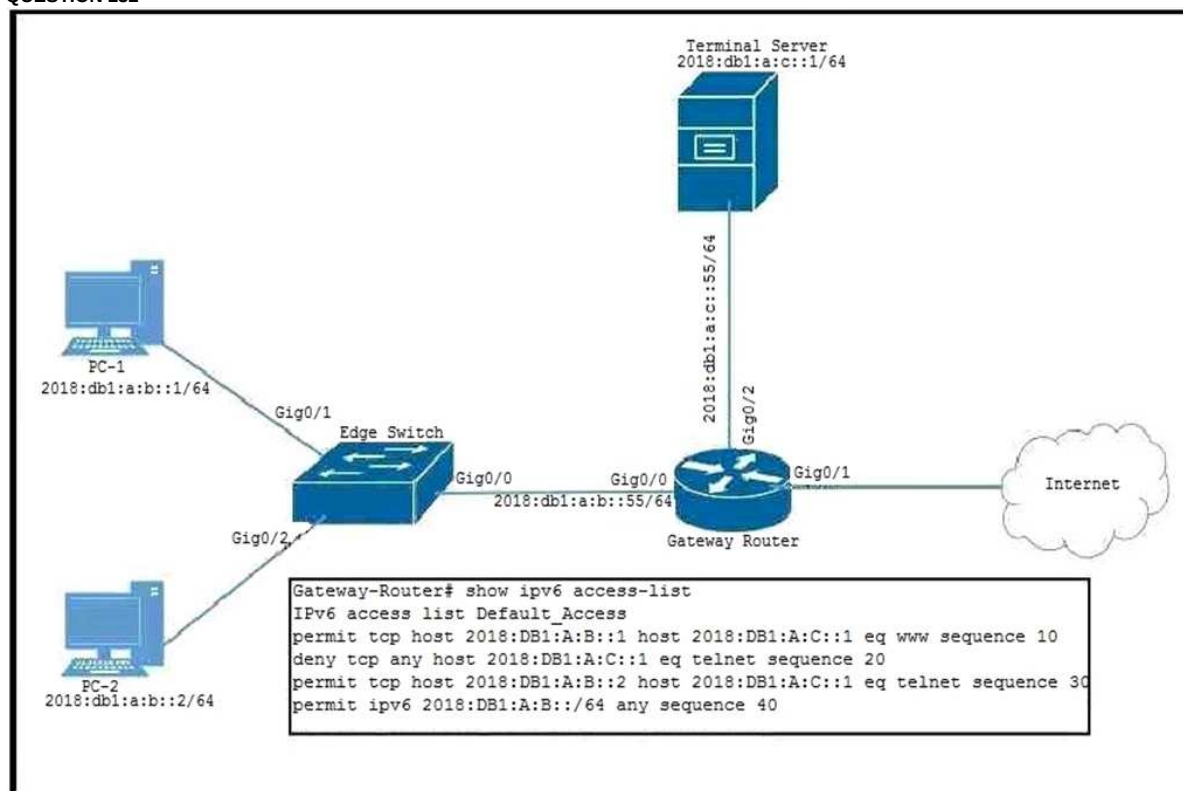
Explanation/Reference:

The problem occurs since:

- R3 learns 10.1.1.0/24 from R4 through iBGP and therefore has the route to this network with AD 200.
- R3 redistributes this route to EIGRP and then R1 redistributes the route to OSPF. R3 learn the same route from OSPF which has an AD 110 and is preferred.
- Then the route will no longer be redistributed into EIGRP and the OSPF route is not learnt.

The above repeats and causes route flapping for this network in R3.

QUESTION 232



Refer to the exhibit PC-2 failed to establish a Telnet connection to the terminal server. Which configuration resolves the issue?

- A. Gateway-Router(config)#ipv6 access-list Default_Access
Gateway-Router(config-ipv6-acl)#sequence 25 permit tcp host 2018:DB1:A:B::2 host 2018:DB1:A:C::1 eq telnet
- B. Gateway-Router(config)#ipv6 access-list Default_Access
Gateway-Router(config-ipv6-acl)#permit tcp host 2018:DB1:A:B::2 host 2018:DB1:A:C::1 eq telnet
- C. Gateway-Router(config)#ipv6 access-list Default_Access
Gateway-Router(config-ipv6-acl)#no sequence 20
Gateway-Router(config-ipv6-acl)#sequence 6 permit tcp host 2018:DB1:A:B::2 host 2018:DB1:A:C::1 eq telnet
- D. Gateway-Router(config)#ipv6 access-list Default_Access
Gateway-Router(config-ipv6-acl)#sequence 15 permit tcp host 2018:DB1:A:B::2 host 2018:DB1:A:C::1 eq telnet

Correct Answer: D

Section: (none)

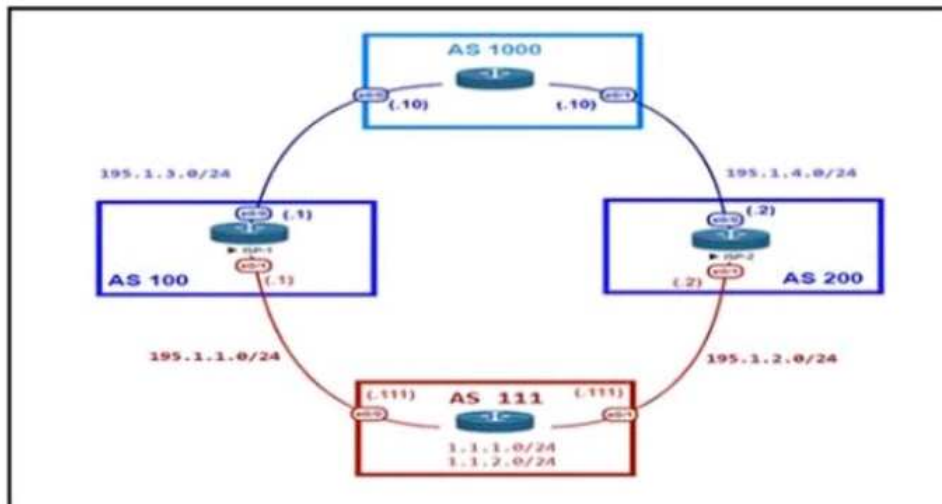
Explanation

Explanation/Reference:

The problem occurs because although the rule sequence 30 allows the access, there is a rule that denies all TCP traffic to the terminal server in sequence 20.

Although C and D allow the access, C will also allow other hosts to telnet to the terminal server. Therefore D is the answer.

QUESTION 233



AS111

Router bgp 111

```
Neighbor 195.1.1.1 remote-as 100
Neighbor 195.1.1.1 allowas-in
Neighbor 195.1.2.2 remote-as 200
Neighbor 195.1.2.2 allowas-in
```

Refer to the exhibit. AS111 is receiving its own routes from AS200 causing a loop in the network. Which configuration provides loop prevention?

- A. router bgp 111
neighbor 195.1.2.2 as-override
no neighbor 195.1.1.1 allowas-in
- B. router bgp 111
neighbor 195.1.1.1 as-override
neighbor 195.1.2.2 as-override
- C. router bgp 111
no neighbor 195.1.1.1 allowas-in
no neighbor 195.1.2.2 allowas-in
- D. router bgp 111
neighbor 195.1.1.1 as-override
no neighbor 195.1.2.2 allowas-in

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

"allowas-in" accept a route even if it contains an AS number of the router's own AS.

It is often used in scenario similar as the following. CE1 and CE2 having the same AS 650001 can configure this so that route from each other can be learnt through AS 1234.



"as-override" has similar effect but it is configured on PE1 and PE2 so that the AS number 650001 in the routes learnt from CE1 or CE2 will be changed to AS 1234.

If the BGP routes sent by a router to an AS can be received back from another AS, "allowas-in" will cause the router to learn the internal networks as eBGP routes and this causes loop.

QUESTION 234

IPv6 is enabled in the infrastructure to support customers with an IPv6 network over WAN and to connect the head office to branch offices in the local network. One of the customers is already running IPv6 and wants to enable IPv6 over the DMVPN network infrastructure between the headend and branch sites. Which configuration command must be applied to establish an mGRE IPv6 tunnel neighborhood?

- A. tunnel protection mode ipv6
- B. ipv6 unicast-routing

- C. ipv6 nhrp holdtime 30
- D. tunnel mode gre multipoint ipv6

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Configuring mGRE Tunnel Support over IPv6

1. enable
2. configure terminal
3. ipv6 unicast-routing
4. interface tunnel tunnel-number
5. description description-string
6. ipv6 address ip-address mask
7. ipv6 nhrp map multicast dynamic
8. ipv6 nhrp network-id network-id
9. ipv6 nhrp holdtime seconds (optional)
10. ipv6 nhrp nhs ipv6-nhs-address
11. tunnel source ip-address | ipv6-address | interface-type | interface-number
12. tunnel mode gre multipoint ipv6
13. end

QUESTION 235

```
interface loopback0
ip address 4.4.4.4 255.255.255.0
!
interface FastEthernet1/0
Description **** WAN link ****
ip address 10.0.0.1 255.255.255.0
!
interface FastEthernet1/1
Description **** LAN Network ****
ip address 192.168.1.1 255.255.255.0
!
!
router ospf 1
router-id 4.4.4.4
log-adjacency-changes
network 4.4.4.4 0.0.0.0 area 0
network 10.0.0.1 0.0.0.0 area 0
network 192.168.1.1 0.0.0.0 area 10
!
```

Refer to the exhibit. Which set of commands restore reachability to loopback0?

- A. interface loopback0
ip address 4.4.4.4 255.255.255.0
ip ospf interface area 10
- B. interface loopback0
ip address 4.4.4.4 255.255.255.0
ip ospf network broadcast
- C. interface loopback0
ip address 4.4.4.4 255.255.255.0
ip ospf interface type network
- D. interface loopback0
ip address 4.4.4.4 255.255.255.0
ip ospf network point-to-point

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

The route for 4.4.4.4 should be advertised properly but as a host route i.e. 4.4.4.4/32.

A and C are incorrect since there is no such command

R1(config-if)#ip ospf interface ?

% Unrecognized command

For C and D, both can configure the router to advertise 4.4.4.0/24 route instead of a host route.

However, D is often preferred.

QUESTION 236

A network administrator performed a Compact Flash Memory upgrade on a Cisco Catalyst 6509 Switch. Everything is functioning normally except SNMP, which was configured to monitor the bandwidth of key interfaces but the interface indexes are changed. Which global configuration resolves the issue?

- A. snmp ifindex persist
- B. snmp ifindex permanent

- C. snmp-server ifindex permanent
- D. snmp-server ifindex persist

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

QUESTION 237

Which mechanism must be chosen to optimize the reconvergence time for OSPF at company location 408811049 that is less CPU-intensive than reducing the hello and dead timers?

- A. BFD
- B. OSPF demand circuit
- C. Dead Peer Detection keepalives
- D. SSO

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

QUESTION 238

```
admin@linux:~$ scp script.py admin@198.51.100.64:script.py
Password:
Administratively disabled.
admin@linux:~$ Connection to 198.51.100.64 closed by remote
host.
```

Refer to the exhibit. A network administrator has developed a Python script on the local Linux machine and is trying to transfer it to the router. However, the transfer fails. Which action resolves this issue?

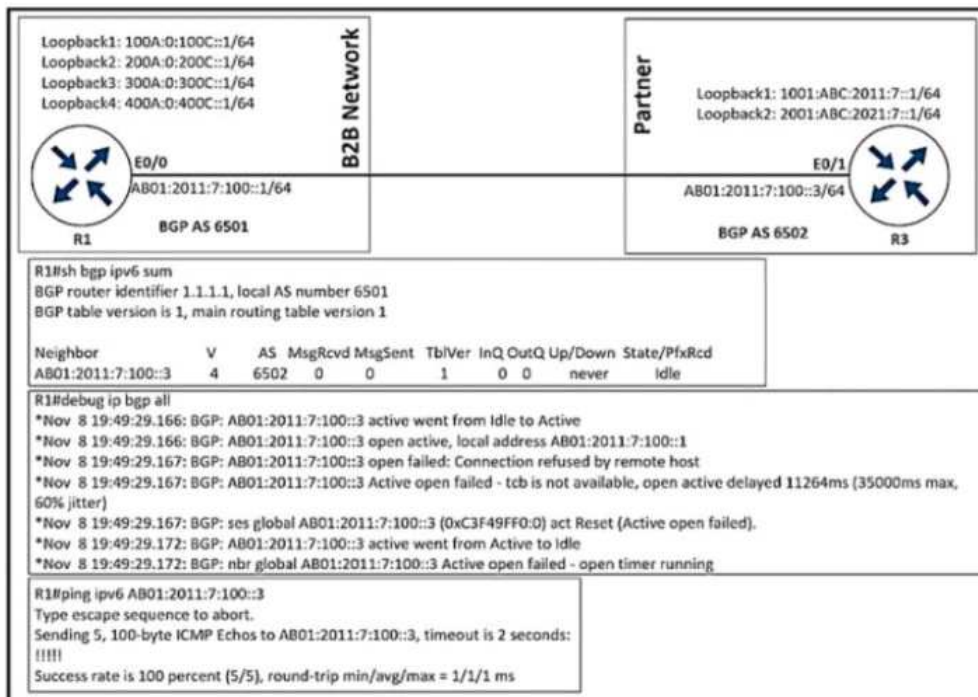
- A. The SSH service must be enabled with the crypto key generate rsa command.
- B. The SCP service must be enabled with the ip scp server enable command.
- C. The SSH access must be allowed on the VTY lines using the transport input ssh command.
- D. The Python interpreter must first be enabled with the guestshell enable command.

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

QUESTION 239

Refer to the exhibit.



An engineer configured BGP between routers R1 and R3. The BGP peers cannot establish neighbor adjacency to be able to exchange routes. Which configuration resolves this issue?

- A. R3
 router bgp 6502
 address-family ipv6
 neighbor AB01:2011:7:100::1 activate
- B. R1
 router bgp 6501
 address-family ipv6
 neighbor AB01:2011:7:100::3 activate
- C. R3
 router bgp 6502
 neighbor AB01:2011:7:100::1 ebgp-multihop 255
- D. R1
 router bgp 6501
 neighbor AB01:2011:7:100::3 ebgp-multihop 255

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

From R1's output, BGP seems to be configured properly.

From R1's debugging message, it seems that R3 has not been configured with BGP and therefore R1 cannot form TCP connection with R3. Therefore A should be the answer.

QUESTION 240

A network administrator is troubleshooting a high utilization issue on the route processor of a router that was reported by NMS. The administrator logged into the router to check the control plane policing and observed that the BGP process is dropping a high number of routing packets and causing thousands of routes to recalculate frequently. Which solution resolves this issue?

- A. Shape the pir for BGP, conform-action set-prec-transmit, and exceed action set-frde-transmit.
- B. Shape the cir for BGP, conform-action transmit, and exceed action transmit.
- C. Police the pir for BGP, conform-action set-prec-transmit, and exceed action set-clp-transmit.
- D. Police the cir for BGP, conform-action transmit, and exceed action transmit

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

set-clp-transmit

Sets the ATM Cell Loss Priority (CLP) bit from 0 to 1 on the ATM cell and transmits the packet.

set-frde-transmit

Sets the Frame Relay Discard Eligibility (DE) bit from 0 to 1 on the Frame Relay frame and transmits the packet.

In the above, frequent route recalculation is needed since many BGP packets are dropped. In order to prevent this, you should configure the router not to drop any BGP packet. If policing is configured, this means that packets should be transmitted for both conform and exceed traffic.

Note that choice B is wrong since shaping does not support conform / exceed action.

QUESTION 241

EIGRP AS 100 10.1.1.1/30  R1 10.1.1.2/30  R2 Ge0/0 Ge0/1	<pre> R1# debug eigrp packets (UPDATE, REQUEST, QUERY, REPLY, HELLO, UNKNOWN, PROBE, ACK, STUB, SIAQUERY, SIAREPLY) EIGRP Packet debugging is on R1# EIGRP: Sending HELLO on Gi0/0 - paklen 20 AS 100, Flags 0x0:(NULL), Seq 0/0 interfaceQ 0/0 iidxbQ un/rely 0/0 R1# EIGRP: Sending HELLO on Gi0/0 - paklen 20 AS 100, Flags 0x0:(NULL), Seq 0/0 interfaceQ 0/0 iidxbQ un/rely 0/0 </pre>
---	---

Refer to the exhibit. Which action resolves the adjacency issue?

- A. Match the hello interval timers-
- B. Configure the same EIGRP process IDs.
- C. Match the authentication keys.
- D. Configure the same autonomous system numbers.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

From the debugging messages, there is no message about Hello packet received by R1. This is probably due to the fact the the Hello packet sent by the other router R2 has a different AS number.

QUESTION 242

Refer to the exhibit.

```
*Sep 26 19:50:43.504: SNMP: Packet received via UDP from
192.168.1.2 on GigabitEthernet0/1SrParseV3SnmplibMessage: No
matching Engine ID.

SrParseV3SnmplibMessage: Failed.
SrDoSnmplib: authentication failure, Unknown Engine ID

*Sep 26 19:50:43.504: SNMP: Report, reqid 29548, errstat 0,
erridx 0
internet.6.3.15.1.1.4.0 = 3
*Sep 26 19:50:43.508: SNMP: Packet sent via UDP to 192.168.1.2
process_mgmt_req_int: UDP packet being de-queued
```

Which two commands provide the administrator with the information needed to resolve the issue? (Choose two.)

- A. debug snmpv3 engine-id
- B. debug snmp engine-id
- C. show snmpv3 user
- D. debug snmp packet
- E. show snmp user

Correct Answer: DE

Section: (none)

Explanation

Explanation/Reference:

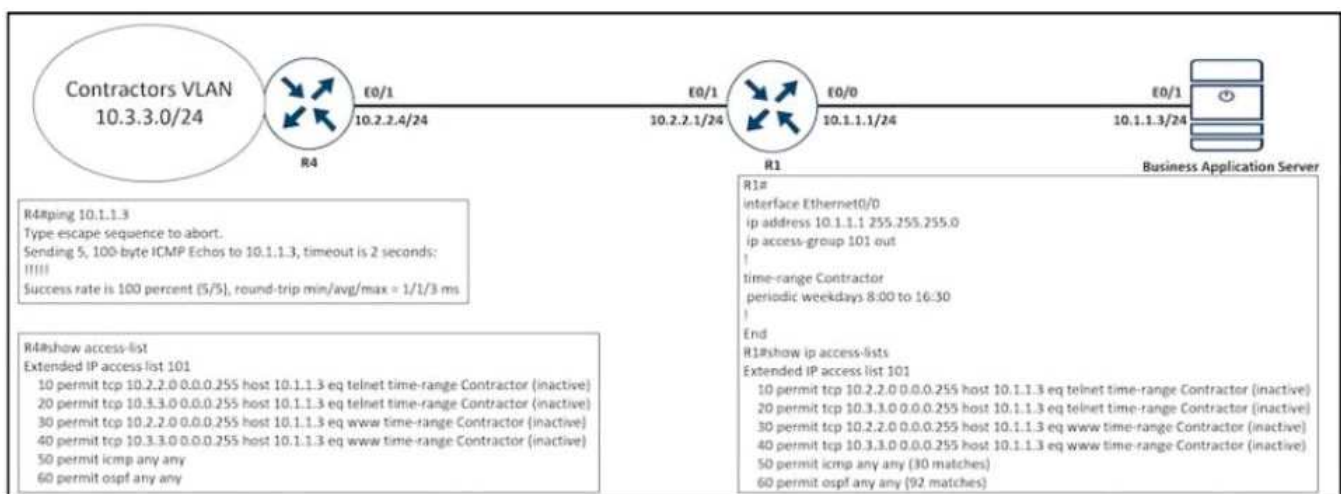
There are 3 values in the SNMPv3 header that must match for the communication to take place: snmpEngineID, snmpEngineTime, snmpEngineBoots.

The error received indicates a problem with the EngineID value. The value can be found from:

- debug snmp packets
- show snmp user

QUESTION 243

Refer to the exhibit:



Contractors' LAN 10.3.3.0/24 failed to access the server through telnet and HTTP during weekend. Which of the following can solved the problem?

- A. **R1**
no access-list 101 permit tcp 10.3.3.0 0.0.0.255 host 10.1.1.3 eq telnet time-range Contractor
- B. **R1**
time-range Contractor
no periodic weekdays 8:00 to 16:30
periodic daily 8:00 to 16:30

- C. **R4**
no access-list 101 permit tcp 10.3.3.0 0.0.0.255 host 10.1.1.3 eq telnet time-range Contractor
- D. **R4**
time-range Contractor
no periodic weekdays 17:00 to 23:59
periodic daily 8:00 to 16:30

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Although there is a similar ACL in R4, the configuration does not show that it has been applied to any interface. Therefore change needs to be made in R1 only.

QUESTION 244

Refer to the exhibit:

```
P 172.29.0.0/16, 1 successors, FD is 307200, serno 2
    via 192.168.254.2 (307200/281600), FastEthernet0/1
    via 192.168.253.2 (410200/352300), FastEthernet0/0
```

When the FastEthernet0/1 goes down, the route to 172.29.0.0/16 via 192.168.253.2 is not installed in the RIB. Which action solves the issue?

- A. Configure feasible distance greater than the successors feasible distance.
- B. Configure feasible distance greater than the reported distance.
- C. Configure reported distance greater than the successors feasible distance
- D. Configure reported distance greater than the feasible distance.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since Reported Distance of the route via 192.168.253.2 is 352300 which is greater than the FD 307200, it will not be used immediately when f0/1 goes down.

QUESTION 245

Users report issues with reachability between areas as soon as an engineer configured summary routes between areas in a multiple area OSPF Autonomous system. Which action resolves the issue?

- A. Configure the area range command on the ABR.
- B. Configure the summary-address command on the ASBR.
- C. Configure the area range command on the ASBR.
- D. Configure the summary-address command on the ABR.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The problem involves routes for inter-area and therefore you should fix it in ABR. For summarizing routers between areas in an ABR, area range command is used.

QUESTION 246

```
Route-map PBR, permit, sequence 10
Match clauses:
  ip address (access-lists): FILTER_ACL
Set clauses:
  ip next-hop verify-availability 209.165.202.129 1 track 100 [down]
  ip next-hop verify-availability 209.165.202.131 2 track 200 [up]
Policy routing matches: 0 packets, 0 bytes
route-map PBR, deny, sequence 20
Match clauses:
Set clauses:
  ip next-hop 209.165.201.30
Policy routing matches: 275364861 packets, 12200235037 bytes
```

Refer to the exhibit. An engineer has configured policy-based routing and applied the configuration to the correct interface. How is the configuration applied to the traffic that matches the access list?

- A. It is dropped.
- B. It is sent to 209.165.202.131.
- C. It is sent to 209.165.202.129.
- D. It is forwarded using the routing table lookup.

Correct Answer: B

Section: (none)

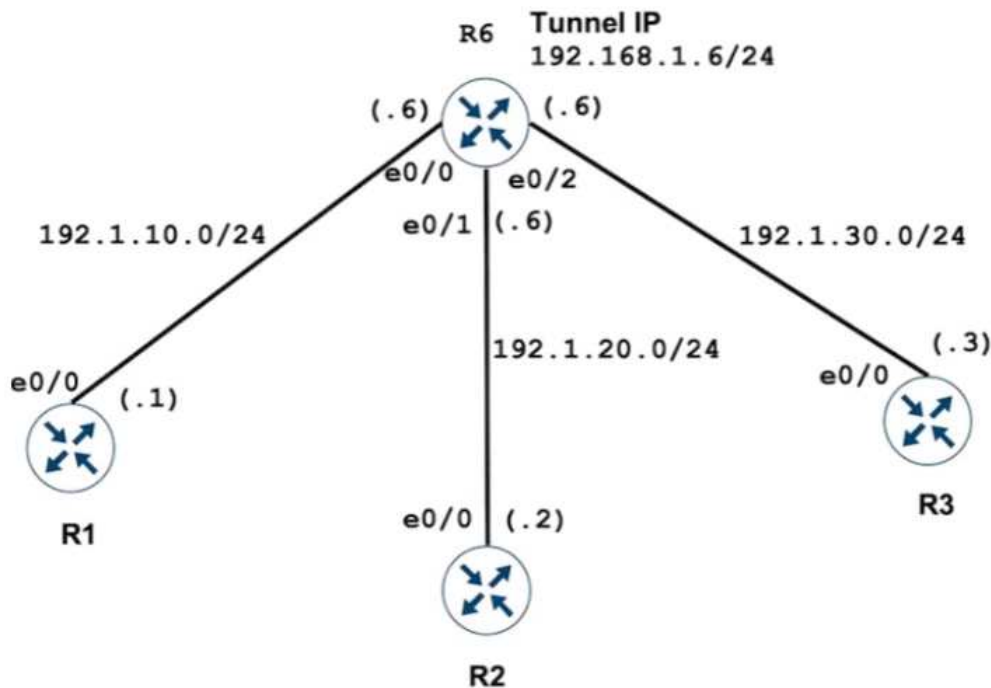
Explanation

Explanation/Reference:

The second one is used since the first next-hop entry is shown as "down".

QUESTION 247

Refer to the exhibit.



An engineer must establish multipoint GRE tunnels between hub router R6 and branch routers R1, R2, and R3. Which configuration accomplishes this task on R1?

- A.

```
interface Tunnel 1
ip address 192.168.1.1 255.255.255.0
tunnel source e0/0
tunnel mode gre multipoint
ip nhrp nhs 192.168.1.6
ip nhrp map 192.168.1.6 192.1.10.1
ip nhrp map 192.168.1.2 192.1.20.2
ip nhrp map 192.168.1.3 192.1.30.3
```
- B.

```
interface Tunnel 1
ip address 192.168.1.1 255.255.255.0
tunnel source e0/0
tunnel mode gre multipoint
ip nhrp network-id 1
ip nhrp nhs 192.168.1.6
ip nhrp map 192.168.1.6 192.1.10.1
ip nhrp map 192.168.1.2 192.1.20.2
ip nhrp map 192.168.1.3 192.1.30.3
```
- C.

```
interface Tunnel 1
ip address 192.168.1.1 255.255.255.0
tunnel source e0/0
tunnel mode gre multipoint
ip nhrp network-id 1
ip nhrp nhs 192.168.1.6
ip nhrp map 192.168.1.6 192.1.10.6
```
- D.

```
interface Tunnel 1
ip address 192.168.1.1 255.255.255.0
tunnel source e0/0
tunnel mode gre multipoint
ip nhrp nhs 192.168.1.6
ip nhrp map 192.168.1.6 192.1.10.6
```

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

R6 is hub router. It has tunnel IP 192.168.1.6.

The physical IP address of the interface connecting to R1 is 192.1.10.6.

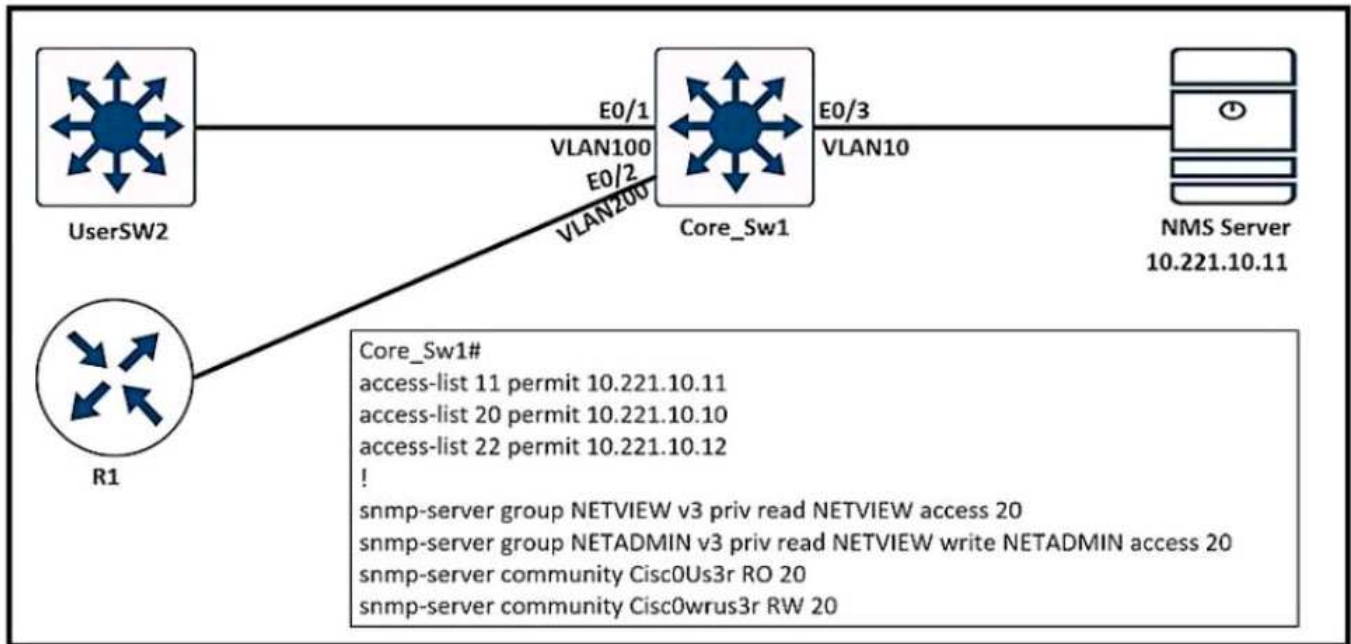
In A and B, the mapping for the Hub router is NOT correct (NBMA / physical address of Hub is 192.168.10.6)

Moreover, in most cases, a spoke router do not need mapping for other spoke routers.

Note that for D, "ip nhrp network-id" command which enables NHRP is missing. Hence C is the answer.

QUESTION 248

Refer to the exhibit.



An engineer configured SNMP communities on the Core_Sw1, but the SNMP server cannot obtain information from core_Sw1. Which configuration resolves this issue?

- A. access-list 20 permit 10.221.10.11
- B. snmp-server group NETADMIN v3 priv read NETVIEW write NETADMIN access 22
- C. snmp-server group NETVIEW v2c priv read NETVIEW access 20
- D. access-list 20 permit 10.221.10.12

Correct Answer: A

Section: (none)

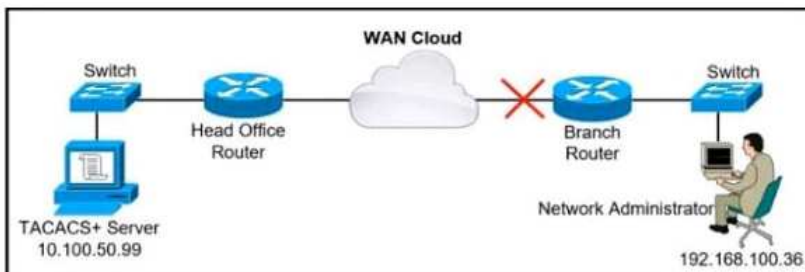
Explanation

Explanation/Reference:

The access list 20 which is used in SNMP settings only allows the host 10.221.10.10 but it is not the IP of the NMS server.

QUESTION 249

Refer to the exhibit.



A network administrator is trying to access a branch router using TACACS+ username and password credentials, but the administrator cannot log in to the router because the WAN connectivity is down. The branch router has following AAA configuration:

```

aaa new-model
aaa authorization commands 15 default group tacacs+
aaa accounting commands 15 default stop-only group tacacs+
aaa accounting commands 15 default stop-only group tacacs+
tacacs-server host 10.100.50.99
tacacs-server key Cisco123

```

Which command will resolve this problem when WAN connectivity is down?

- A. aaa authentication login default group tacacs+ console
- B. aaa authentication login console group tacacs+ enable
- C. aaa authentication login default group tacacs+ local
- D. aaa authentication login default group tacacs+ enable

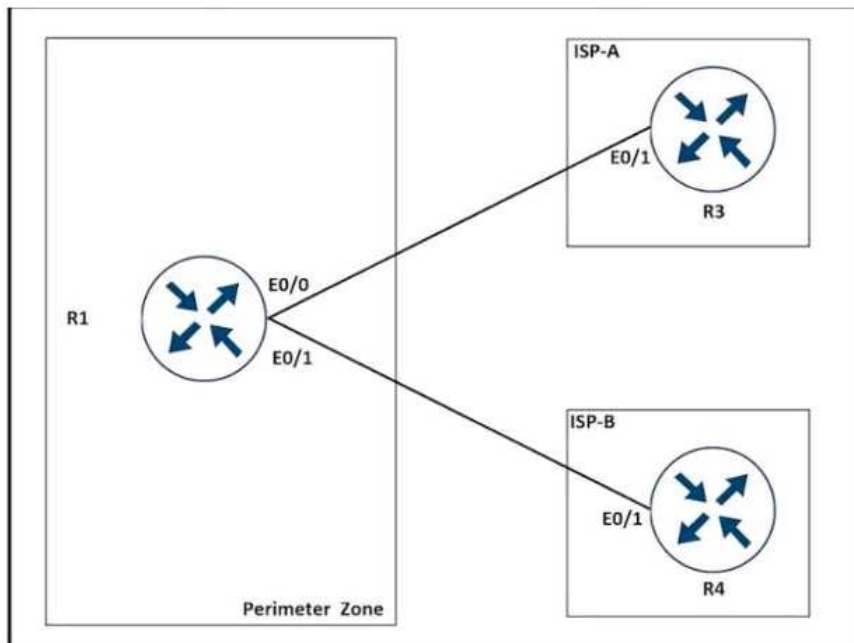
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 250



Refer to the exhibits.

A network is under a cyberattack. A network engineer connected to R1 by SSH and enabled the terminal monitor via SSH session to find the source and destination of the attacks. The session was flooded with messages, which made it impossible for the engineer to troubleshoot the issue. Which command resolves this issue on R1?

- A. #terminal no monitor
- B. (config)#terminal no monitor
- C. (config)#no terminal monitor
- D. #no terminal monitor

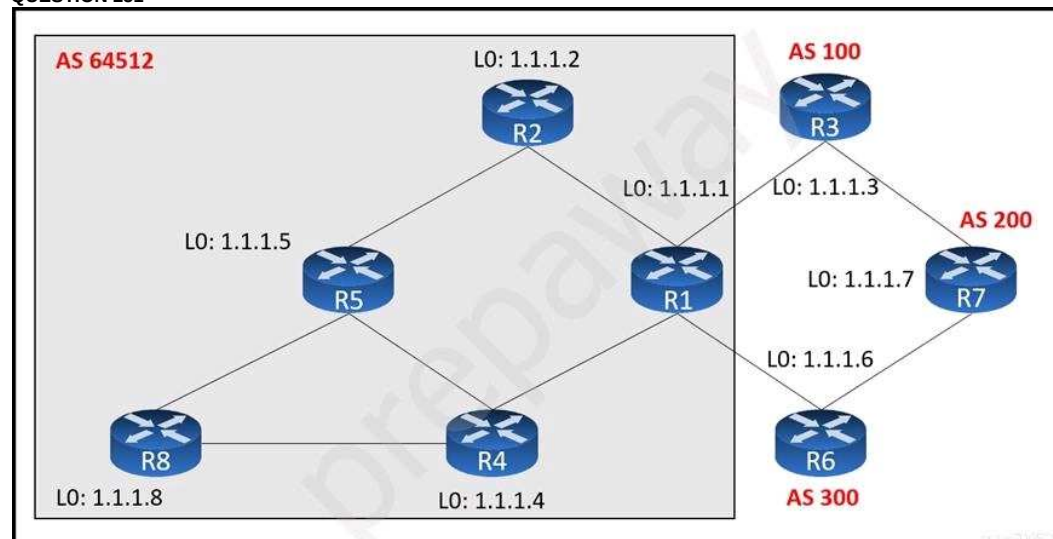
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 251



Refer to the exhibit. An engineer configured R2 and R5 as route reflectors and noticed that not all routes are sent to R1 to advertise to the eBGP peers. Which iBGP routers must be configured as route reflectors to advertise all routes to restore reachability across all networks?

- A. R1 and R4
- B. R1 and R5
- C. R4 and R5
- D. R2 and R5

Correct Answer: C

Section: (none)

Explanation

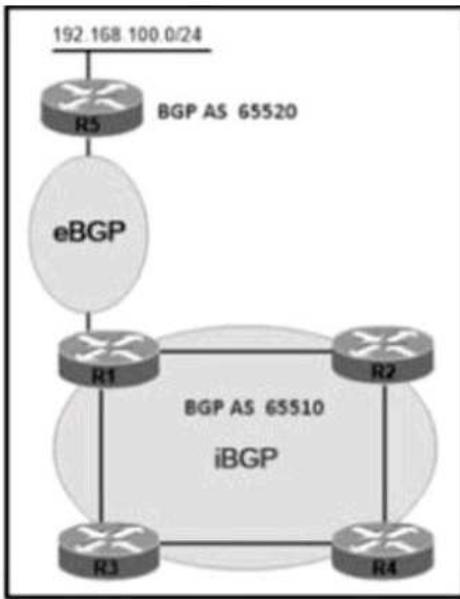
Explanation/Reference:

Actually choices A, C and D can be correct answers if all other routers are properly configured as route reflector client. B is properly wrong since the two route reflectors are not neighbors to each other (assuming that neighbor can only be established between directly connected router). This is because all router reflectors must be fully meshed.

C is chosen since it is the suggested answer found in Internet. One benefit of C is that R8 can be a non-client.

QUESTION 252

Refer to the exhibit.



AS65535 iBGP is configured for directly connected neighbors. R4 cannot ping or traceroute network 192.168.100.0/24. Which action resolves this issue?

- A. Configure R4 as a route reflector server and configure R2 and R3 as route reflector clients
- B. Configure R4 as a route reflector server and configure R1 as a route reflector client
- C. Configure R1 as a route reflector server and configure R4 as a route reflector client.
- D. Configure R1 as a route reflector server and configure R2 and R3 as route reflector clients.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Without any action, eBGP route learnt by R1 will be send to R2 and R3. But R2 or R3 will not send that route learnt from iBGP peer R1 to R4.

For A, eBGP route learnt by R1 will send to R2 or R3 as iBGP route but they will not send the iBGP route to its route reflector R4.

For B, no neighbor has configured between R1 and R4.

For C, no neighbor has configured between R1 and R4.

For D, eBGP route learnt by R1 will send to its clients R2 or R3 as iBGP route but they will not send the iBGP route to R4.

Actually all of the choices can NOT resolve the problem if you are not allowed to add additional neighbor.

If adding additional non-directly directed neighbor is allowed, both choices B and C can solve the problem.

(However, there will still be other issue since R2 and R3 cannot learn route from each other. But this is not related to the question).

B is chosen above because:

- this solves the problem about R4 ping to 192.168.100.0/24 (if adding additional neighbor is allowed).

- the using of a router with eBGP connection as route reflector (i.e. C) is less common.

QUESTION 253

What is a characteristic of Layer 3 MPLS VPNs?

- A. LSP signaling requires the use of unnumbered IP links for traffic engineering.
- B. Traffic engineering supports multiple IGP instances
- C. Traffic engineering capabilities provide QoS and SLAs.
- D. Authentication is performed by using digital certificates or preshared keys.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 254

What does the MP-BGP OPEN message contain?

- A. MPLS labels and the IP address of the router that receives the message
- B. the version number and the AS number to which the router belongs
- C. IP routing information and the AS number to which the router belongs
- D. NLRI, path attributes, and IP addresses of the sending and receiving routers

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

```

Frame 31: 99 bytes on wire (792 bits), 99 bytes captured (792 bits) on interface 0
Ethernet II, Src: c2:01:18:48:00:00 (c2:01:18:48:00:00), Dst: c2:02:1e:6c:00:00 (c2:02:1e:6c:00:00)
Internet Protocol Version 4, Src: 192.168.12.1 (192.168.12.1), Dst: 192.168.12.2 (192.168.12.2)
Transmission Control Protocol, Src Port: 42513 (42513), Dst Port: 179 (179), Seq: 1, Ack: 1, Len: 45
Border Gateway Protocol - OPEN Message
  Marker: ffffffffffffffffffffffffffffffffff
  Length: 45
  Type: OPEN Message (1)
  Version: 4
  My AS: 1
  Hold Time: 180
  BGP Identifier: 1.1.1.1 (1.1.1.1)
  Optional Parameters Length: 16
  Optional Parameters
    Optional Parameter: Capability
      Parameter Type: Capability (2)
      Parameter Length: 6
      Capability: Multiprotocol extensions capability
    Optional Parameter: Capability
      Parameter Type: Capability (2)
      Parameter Length: 2
      Capability: Route refresh capability
    Optional Parameter: Capability
      Parameter Type: Capability (2)
      Parameter Length: 2
      Capability: Route refresh capability

```

QUESTION 255

What statement about route distinguishes in an MPLS network is true?

- A. Route distinguishes make a unique VPNv4 address across the MPLS network.
- B. Route distinguishers allow multiple instances of a routing table to coexist within the edge router.
- C. Route distinguishers are used for label bindings
- D. Route distinguishers define which prefixes are imported and exported on the edge router

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:**QUESTION 256**

The network administrator configured R1 for Control Plane Policing so that the inbound Telnet traffic is policed to 100 kbps. This policy must not apply to traffic coming in from 10.1.1.1/32 and 172.16.1.1/32. The administrator has configured this:

```

access-list 101 permit tcp host 10.1.1.1 any eq 23
access-list 101 permit tcp host 172.16.1.1 any eq 23

```

```

!
class-map COPP-TELNET
  match access-group 101
!
policy-map PM-COPP
  class COPP-TELNET
    police 100000 conform transmit exceed drop
!
control-plane
  service-policy input PM-COPP

```

The network administrator is not getting the desired results. Which set of configurations resolves this issue?

- A. no access-list 101
access-list 101 deny tcp host 10.1.1.1 any eq 23
access-list 101 deny tcp host 172.16.1.1 any eq 23
access-list 101 permit ip any any
- B. no access-list 101
access-list 101 deny tcp host 10.1.1.1 any eq 23
access-list 101 deny tcp host 172.16.1.1 any eq 23
access-list 101 permit ip any any
!
interface E0/0
service-policy input PM-CoPP
- C. control-plane
no service-policy input PM-CoPP
!
interface Ethernet 0/0
service-policy input PM-CoPP
- D. control-plane
no service-policy input PM-CoPP
service-policy input PM-CoPP

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The existing access list configuration police the telnet traffic from the two host but no policing others.
Other configuration settings are proper.

Hence, you just need to change the access list.

Note that B is wrong since it will police all traffic (except telnet from those two hosts) entering the router for forwarding.

Remarks :

Note that after the change in A, other traffic e.g. SSH to the router will also be policed. However since there is no other better answer, A is chosen.

QUESTION 257

Which mechanism provides traffic segmentation within a DMVPN network?

- A. RSVP
- B. BGP
- C. MPLS
- D. IPsec

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Traffic Segmentation within DMVPN can be setup by configuring multiprotocol label switching (MPLS) by using the mpls ip command.

Remarks:

Traffic Segmentation in DMVPN is specified in RFC 2547 VPNs over DMVPN (also known as 2547oDMVPN). It specifies how you can use a single DMVPN overlay and multiplex the different user segments (VNs) over this single DMVPN

QUESTION 258

R1 and R2 are configured as eBGP neighbor , R1 is in AS100 and R2 is in AS200. R2 is advertising these networks to R1:

172.16.16.0/20
172.16.3.0/24
172.16.4.0/24
192.168.1.0/24
192.168.2.0/24
172.16.0.0/16

The network administrator on R1 must improve convergence by blocking all subnets of 172.16.0.0/16 major network with a mask lower than 23 from coming in, Which set of configurations accomplishes the task on R1?

- A.

```
ip prefix-list PL-1 deny 172.16.0.0/16 le 23
ip prefix-list PL-1 permit 0.0.0.0/0 le 32
!
router bgp 100
neighbor 192.168.100.2 remote-as 200
neighbor 192.168.100.2 prefix-list PL-1 in
```
- B.

```
ip prefix-list PL-1 deny 172.16.0.0/16 ge 23
ip prefix-list PL-1 permit 0.0.0.0/0 le 32
!
router bgp 100
neighbor 192.168.100.2 remote-as 200
neighbor 192.168.100.2 prefix-list PL-1 in
```
- C.

```
access-list 1 deny 172.16.0.0 0.0.254.255
access-list 1 permit any
!
router bgp 100
neighbor 192.168.100.2 remote-as 200
neighbor 192.168.100.2 distribute-list 1 in
```
- D.

```
ip prefix-list PL-1 deny 172.16.0.0/16
ip prefix-list PL-1 permit 0.0.0.0/0
!
router bgp 100
neighbor 192.168.100.2 remote-as 200
neighbor 192.168.100.2 prefix-list PL-1 in
```

Correct Answer: A

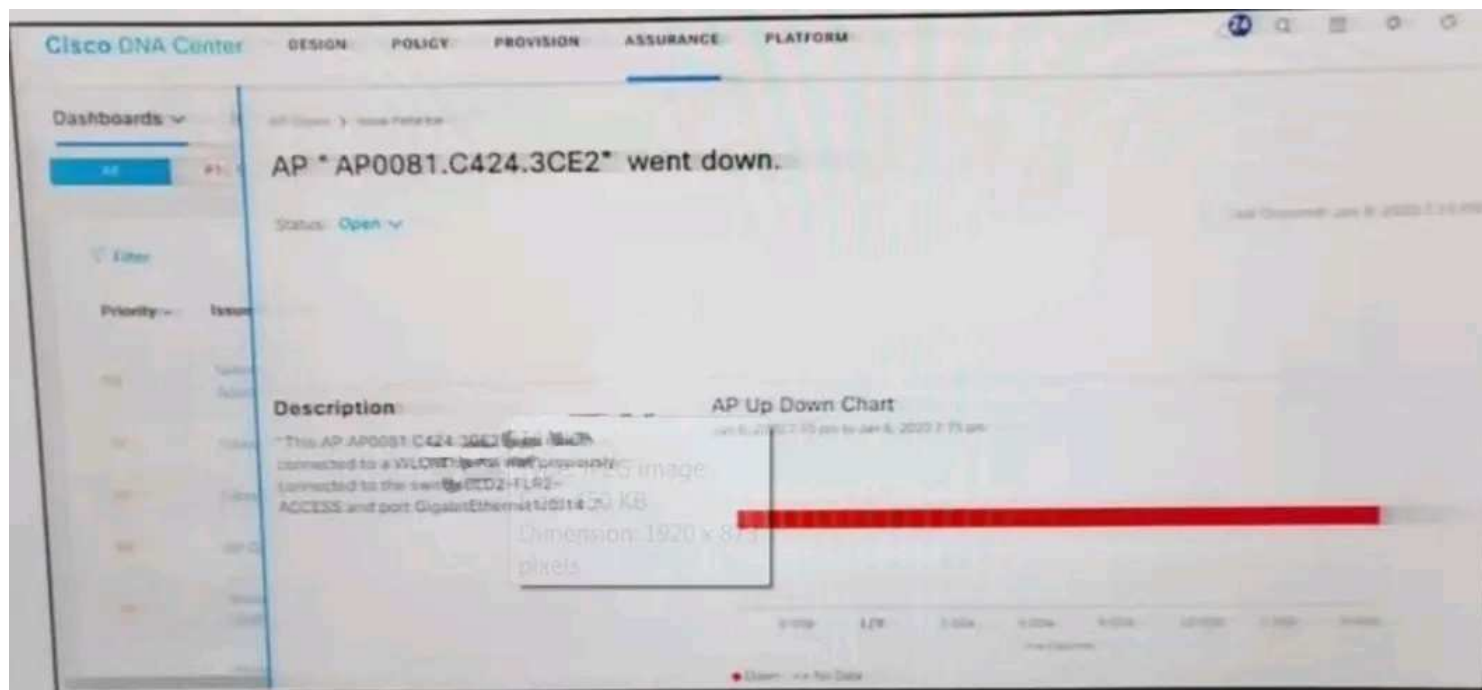
Section: (none)

Explanation

Explanation/Reference:

QUESTION 259

Refer to the exhibit. The AP status from Cisco DNA Center Assurance Dashboard shows some physical connectivity issues from access switch interface G1/0/14. Which command generates the diagnostic data to resolve the physical connectivity issues?



- A. test cable-diagnostics tdr interface GigabitEthernet1/0/14
- B. Check cable-diagnostics tdr interface GigabitEthernet1/0/14
- C. show cable-diagnostics tdr interface GigabitEthernet1/0/14
- D. Verify cable-diagnostics tdr interface GigabitEthernet1/0/14

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

S3750G#test cable-diagnostics tdr interface gigabitethernet 1/0/14

TDR test started on interface Gi1/0/14

A TDR test can take a few seconds to run on an interface

Use 'show cable-diagnostics tdr' to read the TDR results.

S3750G#show cable-diagnostics tdr interface gigabitethernet 1/0/14

TDR test last run on: February 09 22:24:28

Interface	Speed	Local pair	Pair length	Remote pair	Pair status
Gi1/0/14	1000M	Pair A	N/A	Pair B	Normal
		Pair B	N/A	Pair A	Normal
		Pair C	N/A	Pair D	Normal
		Pair D	N/A	Pair C	Normal

S3750G#

Since the question asks to generate the diagnostic data, therefore test cable-diagnostics should be used.

Remarks :

"A time-domain reflectometer (TDR) is an electronic instrument used to characterize and locate faults in metallic cables (for example, twisted wire pairs, coaxial cables)1."

This feature is only supported in the ethernet interfaces of some Cisco switch models.

QUESTION 260

Refer to the exhibit. An engineer configures the router 10.1.100.10 for EIGRP autosummarization so that R1 should receive the summary route of 10.0.0.0/8. However, R1 receives more specific /24 routes. Which action resolves this issue?

```

R1#sh ip route
      10.0.0.0/8 is variably subnetted, 3 subnets, 1 masks
D       10.1.2.0/24 [90/409600] via 10.1.100.10, 00:08:45,
FastEthernet0/0
D       10.1.1.0/24 [90/409600] via 10.1.100.10, 00:08:45,
FastEthernet0/0
C       10.1.100.0/24 is directly connected, FastEthernet0/0

```

- A. Router R1 should configure ip summary address eigrp (AS number) 10.0.0.0 255.0.0.0 for the R1 Fast Ethernet 0/0 connected interface.
- B. Router R1 should configure ip route 10.0.0.0 255.0.0.0 null 0 for the routes that are received on R1.
- C. Router 10.1.100.10 should configure ip route 10.0.0.0 255.0.0.0 null 0 for the routes that are summarized toward R1.

D. Router 10.1.100.10 should configure ip summary address eigrp (AS number) 10.0.0.0 255.0.0.0 for the R1 Fast Ethernet 0/0 connected interface.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

In order for R1 to receive a summary route, it should be configured in the neighbor router 10.1.100.10 that sends the 10 network routes to R1. On the other hand, configuration settings in R1 can only configure summary route to be sent to other routers.

EIGRP auto-summary will only summarize routes from subnets on its own interfaces, not the ones you learn from other routers. Moreover, it only effective when sending over another network boundary. In this case, since R1 and its neighbor router 10.1.100.10 is connected through 10.1.00.0/24, if the neighbor generates the auto-summarized route 10.0.0.0/8, it will not send to R1 since they are connected through a subnet within 10.0.0.0.8. Hence you need to force the neighbor router to send the route through "ip addumary addresss" in the interface.

QUESTION 261

Refer to the exhibit. Although summarization is configured for R1 to receive 10.0.0.0/8. more specific routes are received by R1. How should the 10.0.0.0/8 summary route be received from the neighbor, attached to R1 via Fast Ethernet0/0 interface?

```
R1#sh ip route
      10.0.0.0/8 is variably subnetted, 3 subnets, 1 masks
D       10.1.2.0/24 [90/409600] via 10.1.100.10, 00:08:45,
FastEthernet0/0
D       10.1.1.0/24 [90/409600] via 10.1.100.10, 00:08:45,
FastEthernet0/0
C       10.1.100.0/24 is directly connected, FastEthernet0/0
```

- A. R1 should configure the ip summary-address eigrp <AS number> 10.0.0.0.255.0.0.0 command under the Fast Ethernet 0/0 interface.
- B. The summarization condition is not met Router 10.1.100.10 requires a route for 10 0.0.0/8 that points to null 0
- C. The summarization condition is not met. The network 10.1.100.0/24 should be changed to 172.16.0.0/24.
- D. R1 should configure the ip summary-address eigrp <AS number> 10.0.0.0 0.0.0.255 command under the Fast Ethernet 0/0 interface.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Similar to Q260. EIGRP auto-summary will only summarize routes from subnets on its own interfaces, not the ones you learn from other routers. Moreover, it only effective when sending over another network boundary. In this case, since R1 and its neighbor router 10.1.100.10 is connected through 10.1.00.0/24, if the neighbor generates the auto-summarized route 10.0.0.0/8, it will not send to R1 since they are connected through a subnet within 10.0.0.0.8.

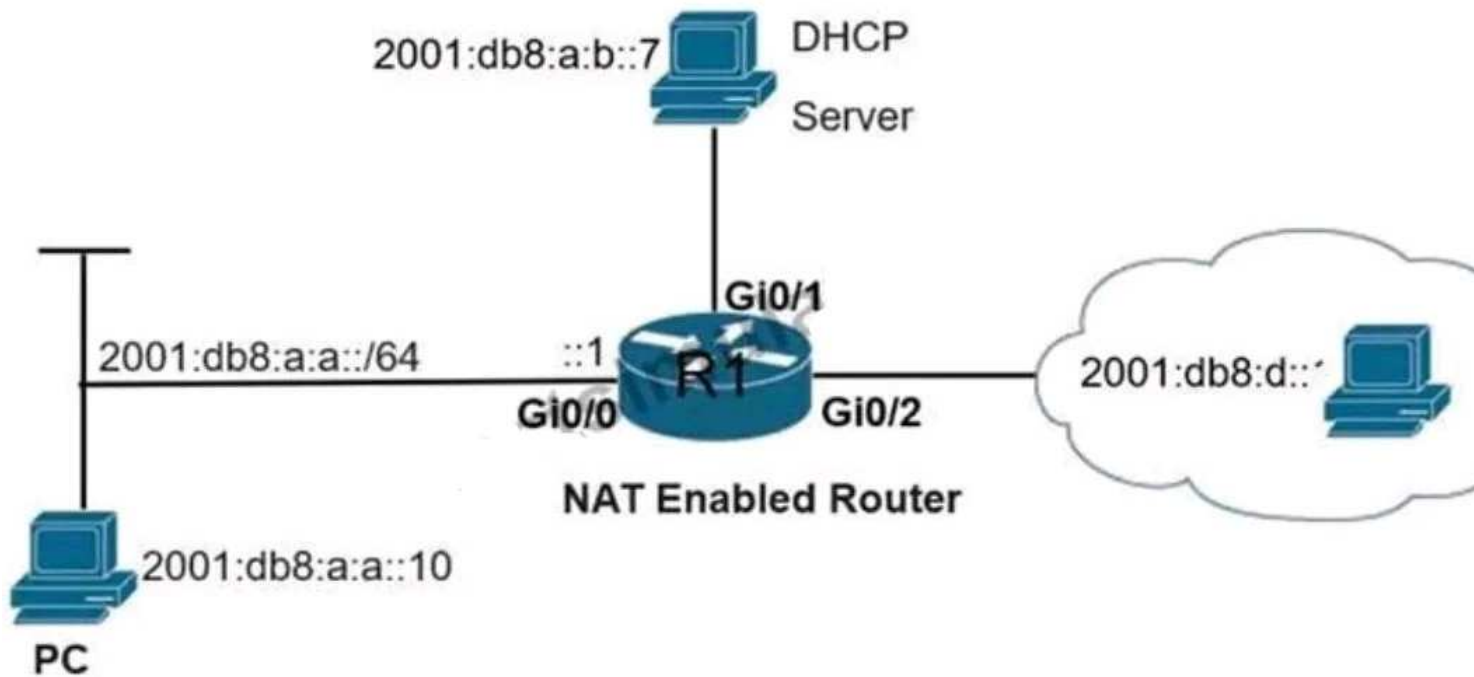
C means that the two connecting interfaces: interface of the neighbor router connecting to R1 and the interface of R1 connecting to the neighbor router should be changed to e.g. 172.16.0.10/24 and 172.16.0.1/24 respectively. Then since the two routers are connected through another network i.e. not within 10.0.0.0/8, then the auto-summarized route 10.0.0.0/8 generated by the neighbor router can send to R1.

Remarks :

The answer assumes that, after the change, the neighbor router still have a connected network within 10.0.0.0/8 advertised by EIGRP.

QUESTION 262

Refer to the exhibit. An engineer is troubleshooting a failed Telnet session from PC to the DHCP server. Which action resolves the issue?



```
C:\PC> ping 2001:db8:a:b::7
Pinging 2001:db8:a:b::7 with 32 bytes of data:
Reply from 2001:db8:a:b::7: time=46ms
Reply from 2001:db8:a:b::7: time=40ms
Reply from 2001:db8:a:b::7: time=40ms
Reply from 2001:db8:a:b::7: time=40ms
Ping statistics for 2001:db8:a:b::7:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 40ms, Maximum = 46ms, Average = 41ms
```

```
R1# telnet 2001:db8:a:b::7
Trying 2001:DB8:A:B::7 ... Open
User Access Verification
Password:
```

```
R1# show ipv6 access-list TSHOOT
IPv6 access list TSHOOT
deny tcp any host 2001:DB8:A:B::7 eq telnet (6 matches) sequence 10
permit tcp host 2001:DB8:A:A::10 host 2001:DB8:A:B::7 eq telnet sequence 20
permit tcp host 2001:DB8:A:A::10 host 2001:DB8:D::1 eq www sequence 30
permit ipv6 2001:DB8:A:A::/64 any (67 matches) sequence 40
```

- A. Remove sequence 30 and add it back to the IPv6 traffic filter as sequence 5.
- B. Remove sequence 20 and add it back to the IPv6 traffic filter as sequence 5.
- C. Remove sequence 10 to add the PC source IP address and add it back as sequence 10.
- D. Remove sequence 20 for sequence 40 in the access list to allow Telnet.

Correct Answer: B
 Section: (none)
 Explanation

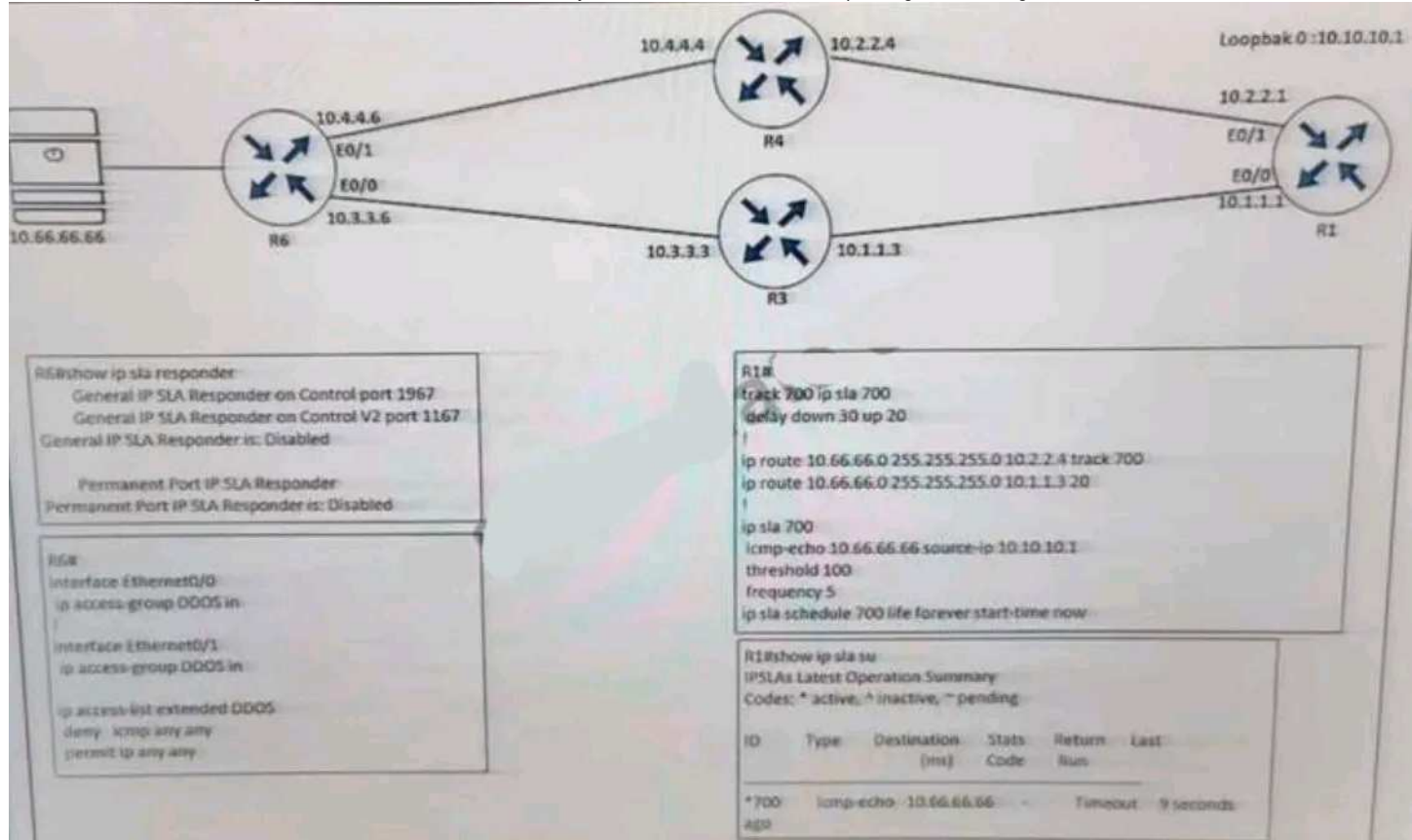
Explanation/Reference:

The rule with sequence 10 block all telnet access from all hosts except R1 itself.

In order to allow the PC 2001:DB8:A::10 to telnet, you need to move the rule with sequence 20 that permit the telnet access from PC before 10.

QUESTION 263

Refer to the exhibit. R1 is configured with IP SLA to check the availability of the server behind R6 but it kept failing. Which configuration resolves the issue?



- A. R6(config)# ip sla responder
- B. R6(config)# ip sla responder udp-echo ip address 10.10.10.1 port 5000
- C. R6(config)# ip access-list extended DDOS
R6(config ext-nac)# 5 permit icmp host 10.66.66.66 host 10.10.10.1
- D. R6(config)# ip access-list extended DDOS
R6(config ext-nac)# 5 permit icmp host 10.10.10.1 host 10.66.66.66

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

In R6, the access list DDOS with the following rules applied to both f0/0 and f0/1 inbound direction blocked all ICMP traffic included the IPSLA icmp-echo generated by R1.

```
deny icmp any any
permit ip any any
```

Since R1 uses the source IP 10.10.10.1 for the IPSLA operation and the target is 10.66.66.66, therefore you need to add the following rule at the top of R6's access list.

```
permit icmp host 10.10.10.1 host 10.66.66.66
```

Moreover, since the first existing rule in an access list has a default sequence number of 10.

In order to add the new rule as the first rule, it must be smaller than 10.

QUESTION 264

Which OSI model is used to insert an MPLS label?

- A. between Layer 5 and Layer 6
- B. between Layer 1 and Layer 2
- C. between Layer 3 and Layer 4
- D. between Layer 2 and Layer 3

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:**QUESTION 265**

The network administrator configured R1 to authenticate Telnet connections based on Cisco ISE using TACACS+. ISE has been configured with an IP address of 192.168.1.5 and with a network device pointing toward R1(192.168.1.1) with a shared secret password of Cisco123.


```

aaa new-model
!
tacacs server ISE1
  address ipv4 192.168.1.5
  key Cisco123
!
aaa group server tacacs+ TAC-SERV
  server name ISE1
!
aaa authentication login telnet group TAC-SERV

```

The administrator cannot authenticate to R1 based on ISE. Which configuration fixes the issue?

- A. ip tacacs-server host 192.168.1.5 key Cisco123
- B. line vty 0 4
login authentication TAC-SERV
- C. line vty 0 4
login authentication telnet
- D. tacacs-server host 192.168.1.5 key Cisco123

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

In R1, the aaa authentication list is NOT a default list, it has the list name "telnet". Therefore in order to use it for telnet (or other vty access), you need to apply the list name "telnet" to line vty.

QUESTION 266

Refer to the exhibit. After a security audit, the administrator implemented an ACL in the route reflector. The RR became unreachable from any router in the network. Which two actions resolve the issue? (Choose two.)

```

RR# show running-config
!
interface Ethernet0/1
  no ip address
  ipv6 address 2001:DB8:1:12::2/64
  ipv6 traffic-filter ACL in
!
ipv6 access-list ACL
  sequence 10 permit tcp any any eq 22
  sequence 20 permit tcp any eq 22 any
  sequence 30 permit tcp any any eq bgp
  sequence 40 permit tcp any eq bgp any
  sequence 50 permit udp any any eq ntp
  sequence 60 permit udp any eq ntp any
  sequence 70 permit udp any any eq snmp
  sequence 80 deny ipv6 any any log

RR# show ipv6 cef ::/0
::/0
  nexthop 2001:DB8:1:12::1 Ethernet0/1

*Feb 23 00:23:17.211: %IPV6_ACL-6-ACCESSLOGDP: list ACL/80
denied icmpv6 2001:DB8:1:12::1 -> FF02::1:FF00:2 (135/0), 7321
packets

```

- A. Enable the ND proxy feature on the default gateway.
- B. Configure a link-local address on the Ethernet0/1 interface.
- C. Permit ICMPv6 neighbor discovery traffic in the ACL.
- D. Remove the ACL entry 80.
- E. Change the next hop of the default route to the link-local address of the default

Correct Answer: CD

Section: (none)

Explanation

Explanation/Reference:

In IPv6, ARP is not longer used for finding MAC address between devices on the same network. Instead IPv6 Neighbor Discovery which is based on ICMP is used.

Normally, an IPv6 access list will implicitly add the rules required by IPv6 ND before the implicit deny. However, if an explicitly deny all rule is added, those implicit IPv6 ND will no longer be effective.

In this question, an explicitly deny all rule is added in sequence 80 as the last rule. Hence you need to:

- add a permit rule for IPv6 ND explicitly before sequence 80, OR
- just remove the sequence 80 so that the implicitly IPv6 ND rules and implicitly deny rules will be used.

QUESTION 267

A company is expanding business by opening 35 branches over the Internet. A network engineer must configure DMVPN at the branch routers to connect with the hub router and allow NHRP to add spoke routers securely to the multicast NHRP mappings automatically Which configuration meets this requirement at the hub router?

- A. interface Tunnel0
ip address 10.0.0.1 255.255.255.0
ip nhrp authentication KEY1
ip nhrp nhs dynamic ip nhop network-id 10
tunnel mode mgre auto
- B. interface Tunnel0
ip address 10.0.0.1 255.255.255.0
ip nhrp authentication KEY1
ip nhrp registration no-unique
ip nhrp network-id 10
tunnel mode gre nmba
- C. interface Tunnel0
ip address 10.0.0.1 255.255.255.0
ip nhrp authentication KEY1
ip nhrp map multicast dynamic
ip nhrp network-id 10
tunnel mode gre multipoint
- D. interface Tunnel0
ip address 10.0.0.1 255.255.255.0
ip nhrp authentication KEY1
ip nhrp map multicast 224.0.0.0
ip nhrp network-id 10
tunnel mode gre ipv4

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Assuming the hub router's IOS version does not enable multicast by default, you need to add the following command in the tunnel interface.

ip nhrp map multicast dynamic

Remarks :

Actually only C has a valid "tunnel mode" command

```
R1(config-if)#tunnel mode mgre?
% Unrecognized command
R1(config-if)#tunnel mode gre ?
ip          over IP
ipv6        over IPv6
multipoint  over IPv4 (multipoint)
```

```
R1(config-if)#tunnel mode gre
```

QUESTION 268

What is a function of IPv6 Source Guard?

- A. It works with address glean or ND to find existing addresses.
- B. It inspects ND and DHCP packets to build an address binding table.
- C. It denies traffic from known sources and allocated addresses.
- D. It notifies the ND protocol to inform hosts if the traffic is denied by it.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

IPv6 source guard is an interface feature between the populated binding table and data traffic filtering. This feature enables the device to deny traffic when it is originated from an address that is not stored in the binding table.

However, IPv6 source guard does not inspect ND or DHCP packets; rather, it works in conjunction with IPv6 neighbor discovery (ND) inspection or IPv6 address glean, both of which detect existing addresses on the link and store them into the binding table.

When traffic is denied, the IPv6 address glean feature is notified so that it can try to recover the traffic by querying the DHCP server or by using IPv6 ND. The data-glean function prevents the device and end user from getting deadlocked, whereupon a valid address fails to be stored into the binding table.

Depending on the meaning, A or D may be correct.

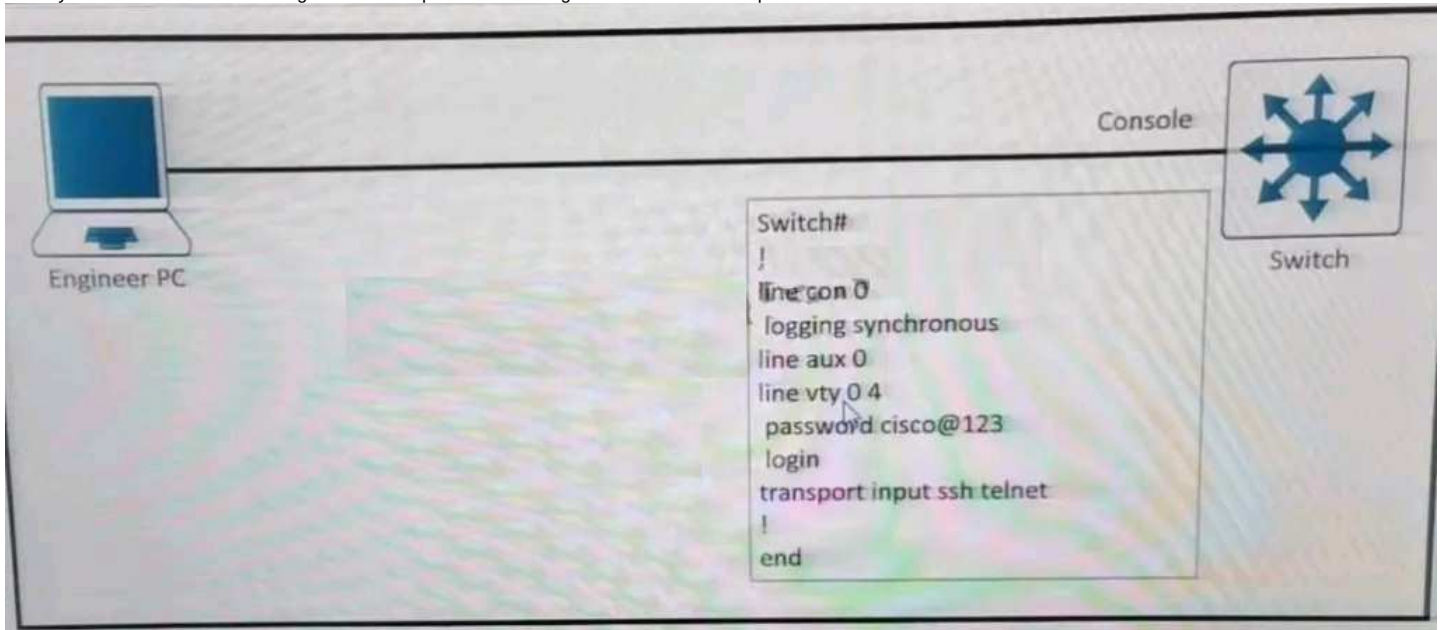
A is correct if it means that "address glean or ND find existing addresses for used by IPv6 Guard" instead of "IPv6 Guard" participate in finding existing address.

D is correct if it means that "IPv6 Guard inform IPv6 address glean to use ND to find the new / updated address of a denied host".

Seems the first one is more likely and therefore A is chosen.

QUESTION 269

Refer to the exhibit. An engineer must block access to the console ports for all corporate remote Cisco devices based on the recent corporate security policy but the security team still can connect through the console port. Which configuration on the console port resolves the issue?



- A. transport input telnet
- B. login and password
- C. no exec
- D. exec 0.0

Correct Answer: C

Section: (none)

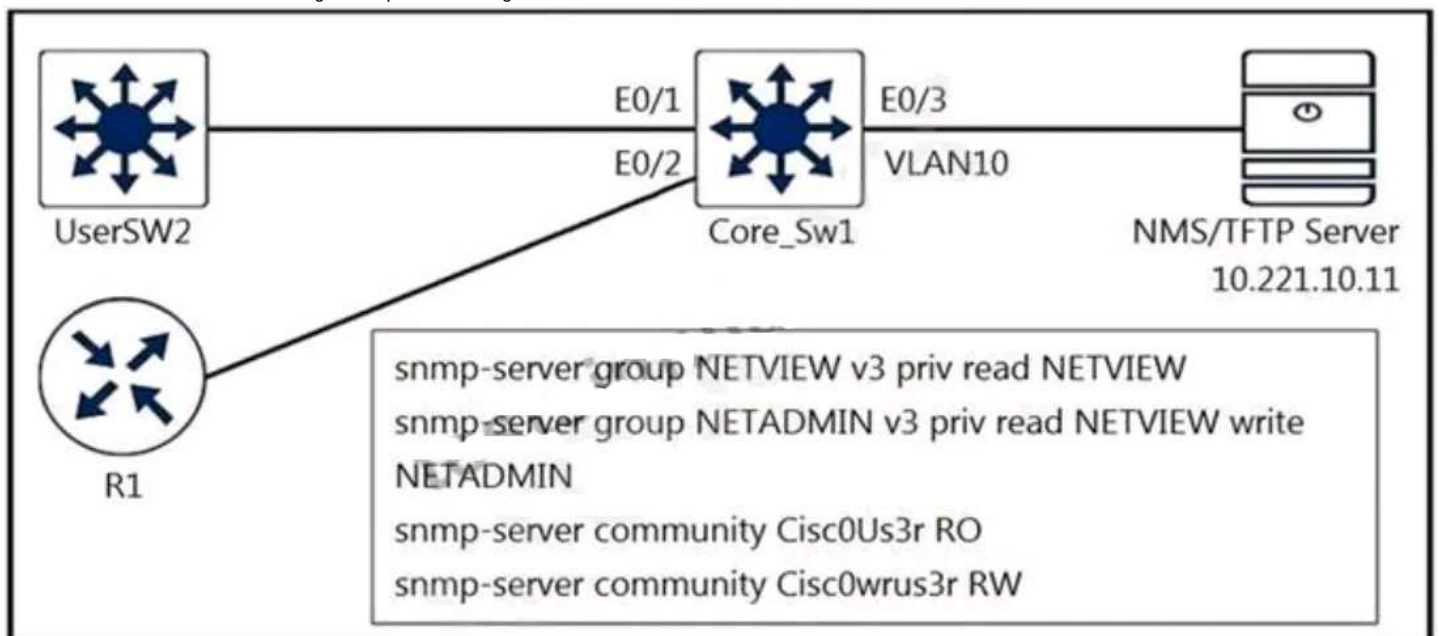
Explanation

Explanation/Reference:

In order to block access through console port, you can enter "no exec" under line con 0. This prevents the console port to create a user exec session for access.

QUESTION 270

Refer to the exhibit. A junior engineer configured SNMP to network devices. Malicious users have uploaded different configurations to the network devices using SNMP and TFTP servers. Which configuration prevents changes from unauthorized NMS and TFTP servers?



- A. access-list 20 permit 10.221.10.11
access-list 20 deny any log
!
snmp-server group NETVIEW v3 priv read NETVIEW access 20
snmp-server group NETADMIN v3 priv read NETVIEW write NETADMIN access 20
snmp-server community Cisc0Us3r RO 20
snmp-server community Cisc0wrus3r RW 20
snmp-server tftp-server-list 20
- B. access-list 20 permit 10.221.10.11
access-list 20 deny any log
!
snmp-server group NETVIEW v3 priv read NETVIEW access 20

```
snmp-server group NETADMIN v3 priv read NETVIEW write NETADMIN access 20
snmp-server community Cisc0wrus3r RO 20
snmp-server community Cisc0Us3r RW 20
snmp-server tftp-server-list 20
```

- C. access-list 20 permit 10.221.10.11
access-list 20 deny any log
- D. access-list 20 permit 10.221.10.11

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

In this question, it said that other malicious users can use their own NMS/TFTP server to upload configuration to the device (i.e. other than the official one with 10.221.10.11).

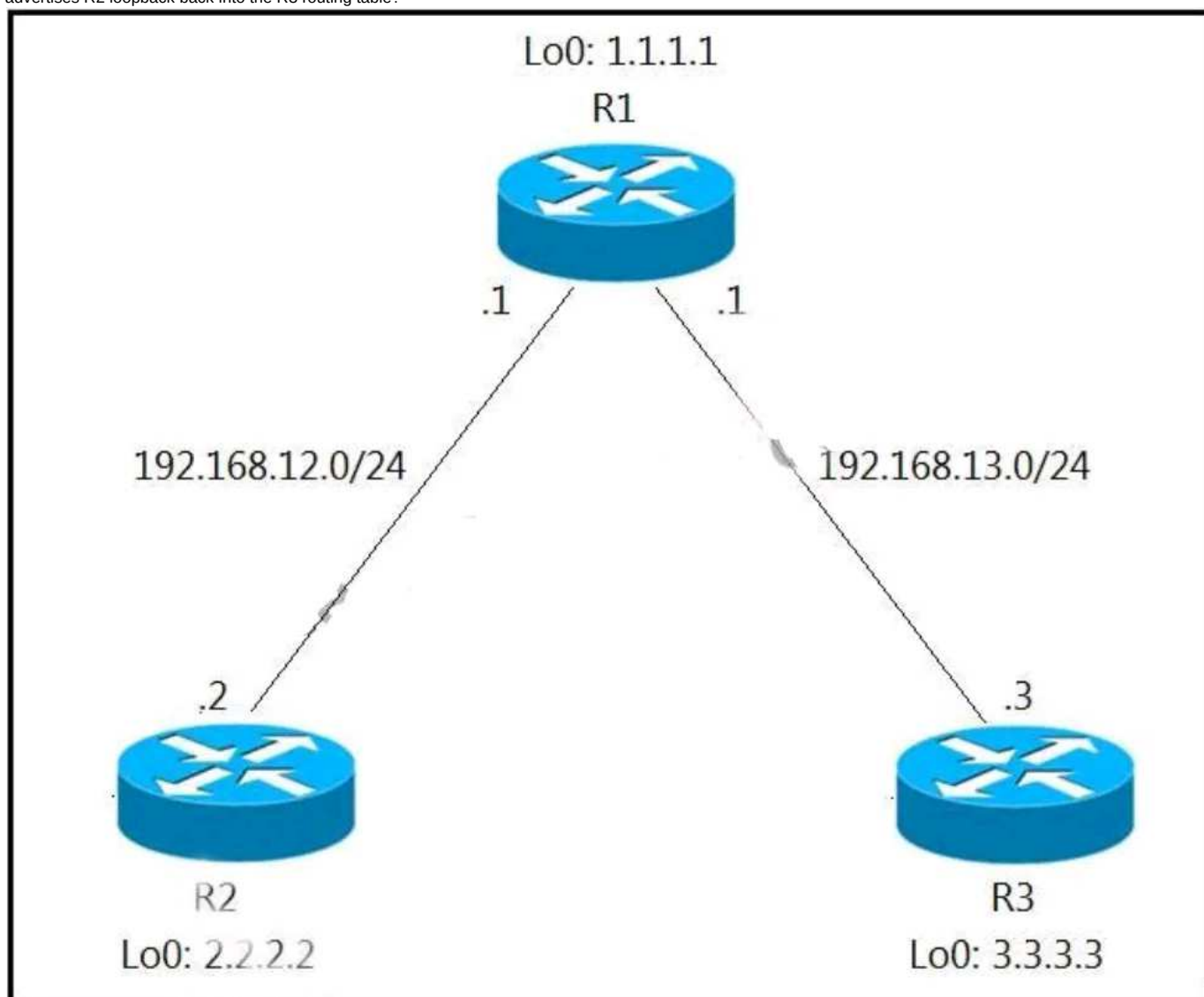
The simplest way is to add an access list to all SNMP settings to allow 10.221.10.11 only.

Note that B may look like A but the community strings configured for SNMP version 2 are wrong.

The RO community and the RW community are exchanged. Hence B is not the answer.

QUESTION 271

Refer to the exhibit. An engineer has configured R1 as EIGRP stub router. After the configuration, router R3 failed to reach to R2 loopback address. Which action advertises R2 loopback back into the R3 routing table?



- A. Add a static route for R2 loopback address in R1 and redistribute it to advertise to R3.
- B. Use a leak map on R1 that matches the required prefix and apply it with the distribute list command toward R3.
- C. Use a leak map on R3 that matches the required prefix and apply it with the EIGRP stub feature.
- D. Add a static null route for R2 loopback address in R1 and redistribute it to advertise to R3.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Since R1 is configured with EIGRP Stub:

- learnt routes from R2 (i.e. route for R2's lo0) will not be advertised to R3.
- learnt routes from R3 (i.e. route for R3's lo0) will not be advertised to R2.

Among the choices:

For A, it can generate a EIGRP external route for R2's lo0 in R1. However, whether it can be sent to R3 depends on the EIGRP Stub settings in R1. The default Stub setting is "eigrp stub connected summary". You need to add "static" e.g. "eigrp stub connected summary static" before the redistributed route sourced from static can be sent.

For B, distribute list can filter the EIGRP routes (i.e. blocking some of the routes) sent. However, since R2's lo0 has been blocked by Stub feature in R1, setting filter in distribute can NOT help to send additional route.

For C, configure leak map for Stub is a good way to send routes which were blocked by Stub feature. However, the choice configure it in R3 which is NOT useful since R1 is the Stub router that blocks the R2's lo0 route.

For D, it is similar to A. However, even if the R2's lo0 route is sent to R3. If R3 sends a packet to R2's lo0 through R1, R1 will drop it due to the null route for R2's lo0.

Therefore, the best one is A but additional configuration may be needed.

Additional Reference :

About how to setup leak map in R1' stub feature.

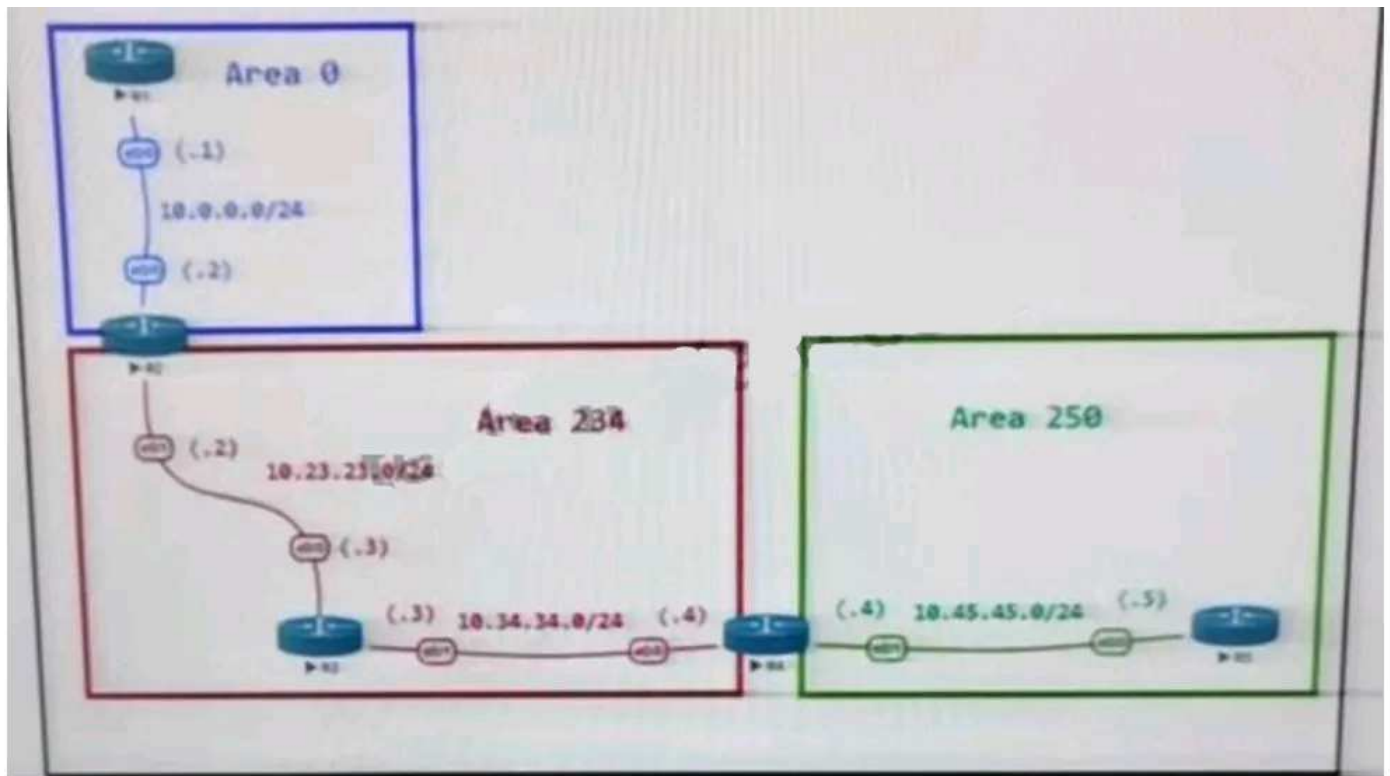
In this example, we leak both R2's lo0 and R3's lo0 so that the two routers can reach the loopback of each other.

```
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip access-list standard stub_leak
R1(config-std-nacl)#permit host 2.2.2.2
R1(config-std-nacl)#permit host 3.3.3.3
R1(config-std-nacl)#exit
R1(config)#route-map stub_leak_map
R1(config-route-map)#match ip address stub_leak
R1(config-route-map)#exit
R1(config)#router eigrp 1
R1(config-router)#eigrp stub leak-map stub_leak_map
R1(config-router)#end
R1#

R3#ping 2.2.2.2 source 3.3.3.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2.2.2.2, timeout is 2 seconds:
Packet sent with a source address of 3.3.3.3
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/4/6 ms
R3#
```

QUESTION 272

Refer to the exhibit. The network administrator configured the network to connect two disjointed networks and all the connectivity is up except the virtual link which causes area 250 to be unreachable. Which two configurations resolve this issue? (Choose two.)



ABR Configurations

R2

```
router ospf 1
router-id 0.0.0.22
area 234 virtual-link 10.34.34.4
network 10.0.0.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.255 area 0
network 10.22.22.0 0.0.0.255 area 234
network 10.23.23.0 0.0.0.255 area 234
```

R4

```
router ospf 1
router-id 0.0.0.44
area 234 virtual-link 10.23.23.2
network 10.34.34.0 0.0.0.255 area 234
network 10.45.45.0 0.0.0.255 area 250
```

Virtual Link Status

R2 -> sh ip ospf virtual-links

Virtual Link OSPF_VL0 to router 10.34.34.4 is down

Run as demand circuit

DoNotAge LSA allowed.

Transit area 234

Topology-MTID	Cost	Disabled	Shutdown	Topology Name
---------------	------	----------	----------	---------------

0	65535	no	no	Base
---	-------	----	----	------

Transmit Delay is 1 sec, State DOWN,

- R4
router ospf 1
no area 234 virtual-link 10.23.23.2
area 234 virtual-link 0.0.0.22
- R4
router ospf 1
no area area 234 virtual-link 10.23.23.2
area 0 virtual-link 0.0.0.22
- R2
router ospf 1
no area area 234 virtual-link 10.34.34.4
area 0 virtual-link 0.0.0.44
- R2
router ospf 1


```
router-id 10.23.23.2
```

```
E. R2
router ospf 1
no area 234 virtual-link 10.34.34.4
area 234 virtual-link 0.0.0.44
```

Correct Answer: AE

Section: (none)

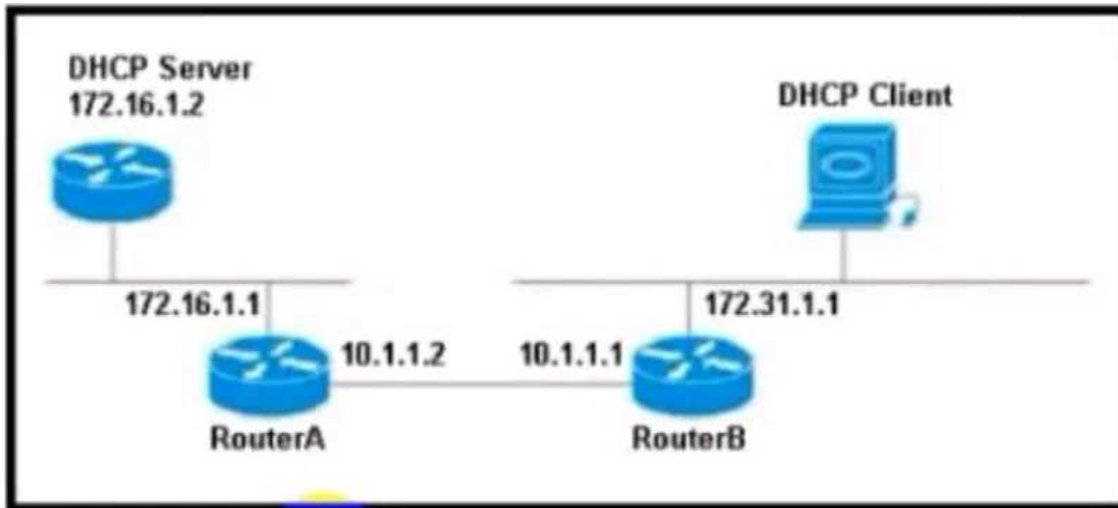
Explanation

Explanation/Reference:

Instead of the physical IP address in area 234, you need to use the Router ID to form virtual link.

QUESTION 273

Refer to the exhibit.



The DHCP client is unable to receive an IP address from the DHCP server.
RouterB is configured as follows:

```
Interface fastethernet 0/0
description Client DHCP ID 394482431
Ip address 172 31 11 255 255.255 0
!
ip route 172.16.1.0 255 255 255.0 10.1.1.2
```

Which command is required on the fastethernet 0/0 interface of RouterB to resolve this issue?

- A. RouterB(config-if)# ip helper-address 172.31.1.1
- B. RouterB(config-if)# ip helper-address 255.255.255.255
- C. RouterB(config-if)# ip helper-address 172.16.1.1
- D. RouterB(config-if)# ip helper-address 172.16.1.2

Correct Answer: D

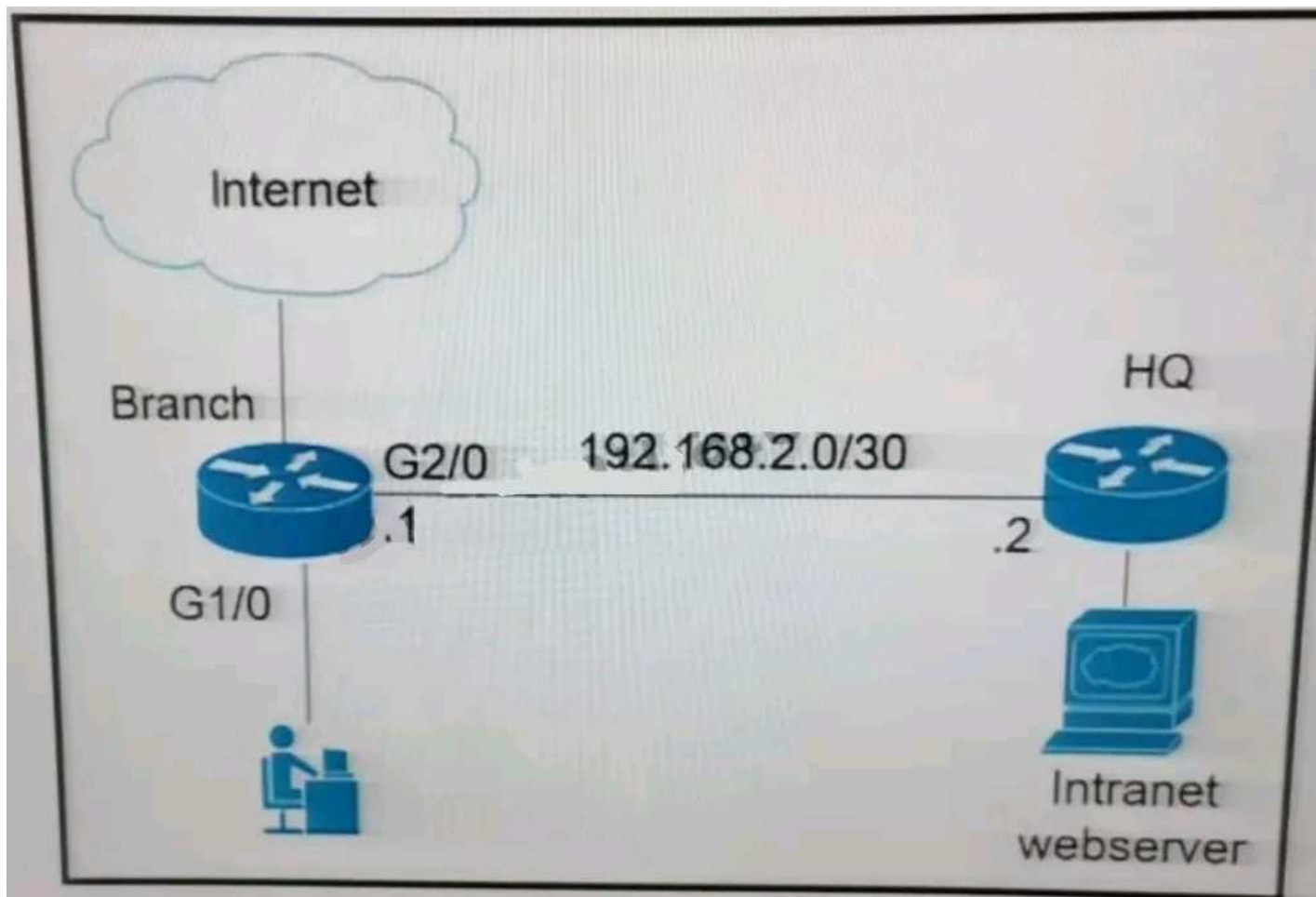
Section: (none)

Explanation

Explanation/Reference:

QUESTION 274

Refer to the exhibit. The branch router is configured with a default route toward the internet and has no routes configured for the HQ site that is connected through interface G2/0. The HQ router is fully configured and does not require changes. Which configuration on the branch router makes the intranet website (TCP port 80) available to the branch office users?



- A. `access-list 100 permit tcp any host intranet-webserver-ip eq 80`
`!`
`route-map pbr permit 10`
`match ip address 100`
`set ip next-hop 192.168.2.2`
`!`
`interface G2/0`
`ip policy route-map pbr`
- B. `access-list 101 permit tcp any any eq 80`
`access-list 102 permit tcp any host intranet-webserver-ip`
`!`
`route-map pbr permit 10`
`match ip address 101 102`
`set ip next-hop 192.168.2.2`
`!`
`interface G1/0`
`ip policy route-map pbr`
- C. `access-list 101 permit tcp any any eq 80`
`access-list 102 permit tcp any host intranet-webserver-ip`
`!`
`route-map pbr permit 10`
`match ip address 101`
`set ip next-hop 192.168.2.2`
`route-map pbr permit 20`
`match ip address 102`
`set ip next-hop 192.168.2.2`
`!`
`interface G2/0`
`ip policy route-map pbr`
- D. `access-list 100 permit tcp host intranet-webserver-ip eq 80 any`
`!`
`route-map pbr permit 10`
`match ip address 100`
`set ip next-hop 192.168.2.2`
`!`
`interface G1/0`
`ip policy route-map pbr`

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

A and C applies PBR to the wrong interface.

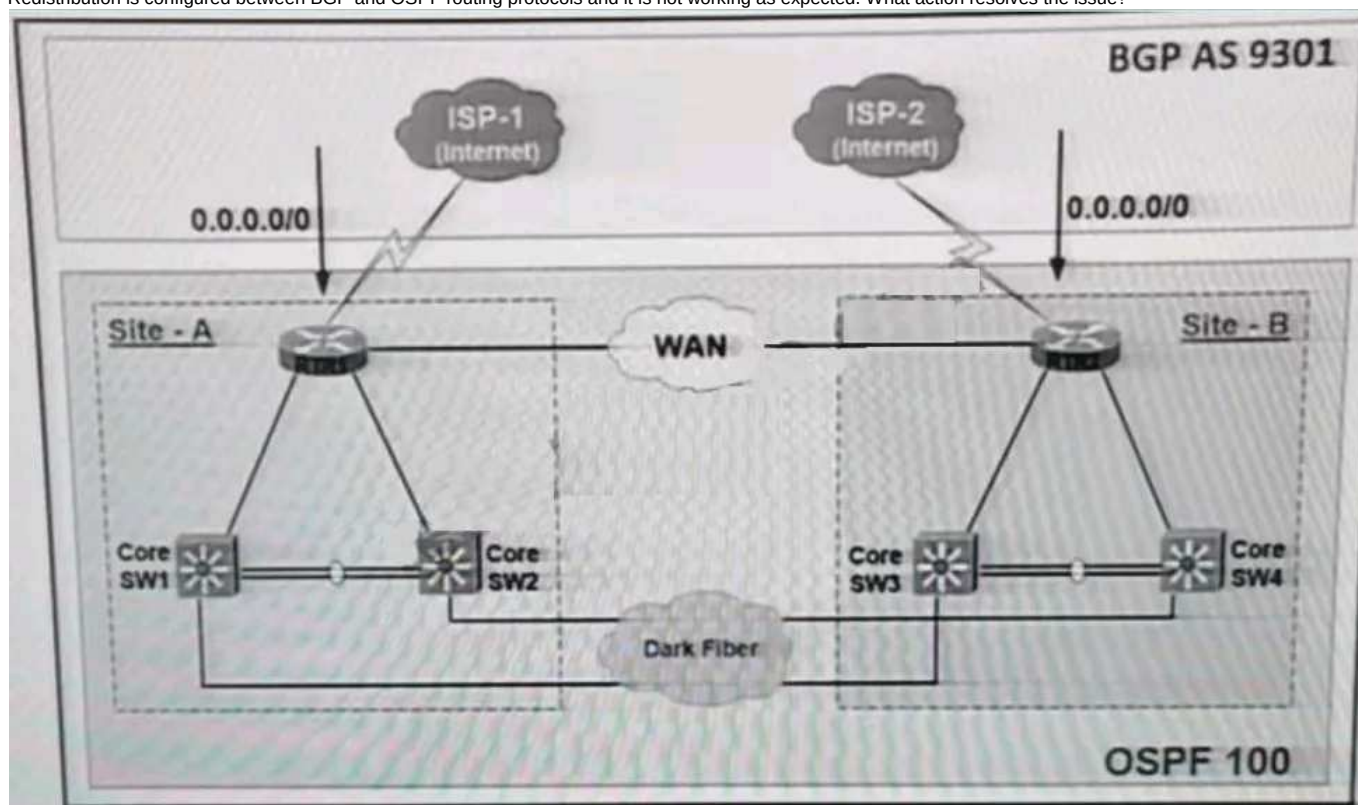
D is wrong since the ACL matches packets using the intranet web server IP and port as source.

B can use PBR to send traffic to intranet web server using PBR but it also sends port 80 traffic to other destination to HQ as well. (since match ip address 101 102

means matching either ACL 101 or ACL 102).
However, since no other choice can send packets to intranet web server, B is chosen as answer.

QUESTION 275

Refer to the exhibit. The Internet traffic should always prefer Site-A ISP-1 if the link and BGP connection are up; otherwise, all Internet traffic should go to ISP-2. Redistribution is configured between BGP and OSPF routing protocols and it is not working as expected. What action resolves the issue?



- A. Set metric-type 2 at Site-A RTR1, and set metric-type 1 at Site-B RTR2
- B. Set OSPF cost 100 at Site-A RTR1, and set OSPF Cost 200 at Site-B RTR2
- C. Set OSPF cost 200 at Site-A RTR1 and set OSPF Cost 100 at Site-B RTR2
- D. Set metric-type 1 at Site-A RTR1, and set metric-type 2 at Site-B RTR2

Correct Answer: D

Section: (none)

Explanation

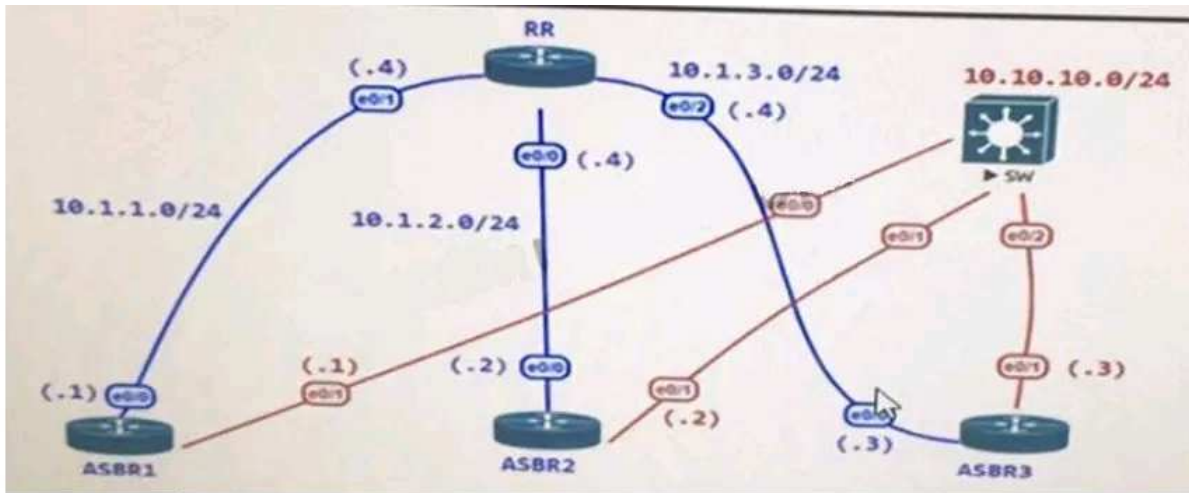
Explanation/Reference:

Since both ASBR is in the same area Area 100, you can set RTR1 (on the left side of the diagram) to redistributed as E1 and set RTR2 (on the right side of the diagram) to redistributed as E2.

It is because E1 route (the one that includes internal cost) is always preferred than E2 route regardless of redistribute metric / import cost.

QUESTION 276

Refer to the exhibit. The administrator configured the network device for end-to-end reachability, but the ASBRs are not propagation routes to each other. Which set of configuration resolves this issue?



```

RR
router bgp 100
neighbor 10.1.1.1 remote-as 100
neighbor 10.1.2.2 remote-as 100
neighbor 10.1.3.3 remote-as 100

ASBR2
router bgp 100
neighbor 10.1.1.4 remote-as 100

ASBR3
router bgp 100
neighbor 10.1.2.4 remote-as 100

ASBR4
router bgp 100
neighbor 10.1.3.4 remote-as 100

```

- A. router bgp 100
 - neighbor 10.1.1.1 route-reflector-client
 - neighbor 10.1.2.2 route-reflector-client
 - neighbor 10.1.3.3 route-reflector-client
- B. router bgp 100
 - neighbor 10.1.1.1 next-hop-self
 - neighbor 10.1.2.2 next-hop-self
 - neighbor 10.1.3.3 next-hop-self
- C. router bgp 100
 - neighbor 10.1.1.1 update-source Loopback0
 - neighbor 10.1.2.2 update-source Loopback0
 - neighbor 10.1.3.3 update-source Loopback0
- D. router bgp 100
 - neighbor 10.1.1.1 ebgp-multihop
 - neighbor 10.1.2.2 ebgp-multihop
 - neighbor 10.1.3.3 ebgp-multihop

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Since RR, ASBR1, ASBR2 and ASBR3 are in the same AS, one ASBR can send through to another ASBR through RR if Route-Reflector is setup in RR. Although the router named is RR, no route-reflector setting has been configured yet.

QUESTION 277

Refer to the exhibit. R1 is configured with uRPF, and ping to R1 is failing from a source present in the R1 routing table via the GigabitEthernet 0/0 interface. Which action resolves the issue?

```

R1#show ip interface GigabitEthernet0/0 | include drops
0 verification drops
0 suppressed verification drops

R1#show ip interface GigabitEthernet0/1 | include drops
5 verification drops
0 suppressed verification drops

```

- A. Remove the access list from the interface GigabitEthernet 0/0
- B. Modify the uRPF mode from strict to loose
- C. Enable Cisco Express Forwarding to ensure that uRPF is functioning correctly
- D. Add a floating static route to the source on R1 to the GigabitEthernet 0/1

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since verification drops are detected in g0/1, the packets enters R1 from this interface and therefore does not match the next hop interface g0/0 specified in the routing table.

B allows the packet without verifying the incoming interface to match the next hop interface of the routing entry.

Remarks:

D does not help since a floating static route does not appear in the routing table unless the primary route is gone.

QUESTION 278

The network administrator configured CoPP so that all HTTP and HTTPS traffic from the administrator device located at 172.16.1.99 toward the router CPU is limited to 500 kbps. Any traffic that exceeds this limit must be dropped.

```

access-list 100 permit ip host 172.16.1.99 any
!
class-map CM-ADMIN
match access-group 100
!
policy-map PM-COPP
class CM-ADMIN
police 500000 conform-action transmit
!
interface E0/0
service-policy input PM-COPP

```

CoPP failed to capture the desired traffic and the CPU load is getting higher. Which two configurations resolve the issue? (Choose two.)

- A. interface E0/0
no service-policy input PM-COPP
!
control-plane
service-policy input PM-COPP
- B. policy-map PM-COPP
class CM-ADMIN
no police 500000 conform-action transmit
police 500 conform-action transmit
!
control-plane
service-policy input PM-COPP
- C. no access-list 100
access-list 100 permit tcp host 172.16.1.99 any eq 80
- D. no access-list 100
access-list 100 permit tcp host 172.16.1.99 any eq 80
access-list 100 permit tcp host 172.16.1.99 any eq 443
- E. policy-map PM-COPP
class CM-ADMIN
no police 500000 conform-action transmit
police 500 conform-action transmit

Correct Answer: AD

Section: (none)

Explanation

Explanation/Reference:

There are three problems in the existing setup:

- Packets will not be policed if the traffic enters the router through an interface other than e0/0.
- all traffic entering e0/0 but not targeted to the router will also be policed.
- all traffic from 172.16.1.99 whether HTTP/HTTPS or not will be policed.

The policy map for CoPP should be applied to the "control-plane".

Moreover, the access-list should be changed to so that policing will be enforced on HTTP (port 80) and HTTPS (port 443) only

Remarks :

The change of 500000 to 500 is wrong since without any unit (e.g. "k"), the value represents bps. Hence 500000 is the correct setup for 500kbps.

```
R1(config-pmap-c)#police ?
<8000-2000000000> Target Bit Rate (bits per second) (postfix k, m, g
                  optional; decimal point allowed)
cir               Committed information rate
rate              Specify police rate, PCR for hierarchical policies or SCR
                  for single-level ATM 4.0 policer policies
```

```
R1(config-pmap-c)#
```

QUESTION 279

An engineer creates a Cisco DNA Center cluster with three nodes, but all the services are running on one host node. Which action resolves this issue?

- A. Restore the link on the switch interface that is connected to a cluster link on the Cisco DNA Center
- B. Click the master host node with all the services and select services to be moved to other hosts
- C. Enable service distribution from the Systems 360 page.
- D. Click system updates, and upgrade to the latest version of Cisco DNA Center.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Cisco DNA Center provides a mechanism for distributed processing and database replication among multiple nodes. Clustering provides both sharing of resources and features, as well as enabling high availability.

1. Click the setting button and then choose System Settings.
The System 360 tab is displayed by default.
2. In the Hosts area, click Enable Service Distribution.

After you click EnableService Distribution, Cisco DNA Center entersinto maintenance mode. In this mode, Cisco DNA Center is unavailable until the redistribution of services is completed. You should take this into account when scheduling an HA deployment.

QUESTION 280

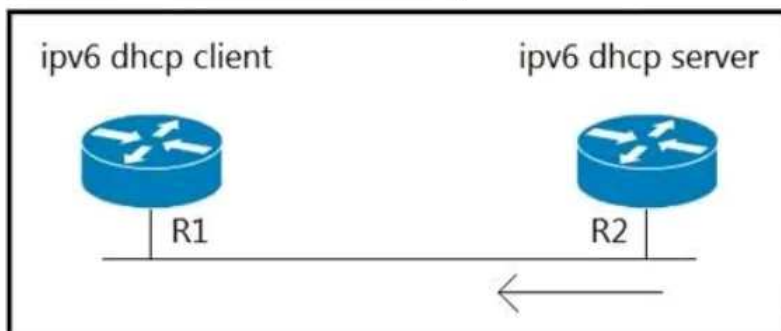
Refer to the exhibit. A network administrator is troubleshooting IPv6 address assignment for a DHCP client that is not getting an IPv6 address from the server. Which configuration retrieves the client IPv6 address from the DHCP server?

ipv6 dhcp server:

```
ipv6 unicast-routing
!
int e0/1
ipv6 enable
ipv6 add 2001:11::1/64
ipv6 nd other-config-flag
no shut
ipv6 dhcp server IPv6Pool
!
ipv6 dhcp pool IPv6Pool
dns-server 2002:555::1
domain-name my.net
```

ipv6 dhcp client:

```
interface Ethernet0/1
no ip address
ipv6 address dhcp
ipv6 enable
no shut
```



- A. ipv6 address autoconfig command on the interface
- B. ipv6 dhcp server automatic command on DHCP server
- C. ipv6 dhcp relay-agent command on the interface
- D. service dhcp command on DHCP server

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

There are two ways to use DHCPv6:

StateLess

In this setup, only information such as DNS servers, domain name is obtained from the DHCP server. Part of the setup includes:

DHCP server:

```
ipv6 dhcp pool poolname
dns-server <ipv6-address>
domain-name <domain>
```

```

int e0/1
  ipv6 dhcp server <server pool name>
  ipv6 nd other-config-flag

```

```

DHCP client
int e0/1
  ipv6 address autoconfig

```

Stateful

In this setup, all information including IPv6 address is obtained from the DHCP server. Part of the setup includes:

```

DHCP server:
  ipv6 dhcp pool poolname
    address prefix <specify address prefix>
    dns-server <ipv6-address>
    domain-name <domain>

```

```

int e0/1
  ipv6 dhcp server <server pool name>

```

```

DHCP client
int e0/1
  ipv6 address dhcp

```

In the question, since the DHCP pool is not configured with any address prefix and there is no choice for adding it. Therefore we assume that the setup is for Stateless setup.

For Stateless setup, the DHCP client interface settings is wrong. Since the command required is :

```

  ipv6 address autoconfig

```

QUESTION 281

Refer to the exhibit. While monitoring VTY access to a router, an engineer notices that the router does not have any filter and anyone can access the router with username and password even though an ACL is configured. Which command resolves this issue?

```

ipv6 access-list INTERNET
  permit ipv6 2001:DB8:AD59:BA21::/64 2001:DB8:C0AB:BA14::/64
  permit tcp 2001:DB8:AD59:BA21::/64 2001:DB8:C0AB:BA13::/64 eq telnet
  permit tcp 2001:DB8:AD59:BA21::/64 any eq http
  permit ipv6 2001:DB8:AD59::/48 any
  deny ipv6 any any log

```

- A. access-class INTERNET in
- B. ip access-group INTERNET in
- C. ipv6 traffic-filter INTERNET in
- D. ipv6 access-class INTERNET in

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

For applying IPv6 access-list, you need C (for applying to interface) or D (for applying to line vty)

For limiting VTY access only, applying the access list to line vty is preferred.

QUESTION 282

Refer to the exhibit. Which command must be configured to make VRF CCNP work?

```

R1 (config)# ip vrf CCNP
R1 (config-vrf)# rd 1:100
R1 (config-vrf)# exit
R1 (config)# interface Loopback0
R1 (config-if)# ip address 10.1.1.1 255.255.255.0
R1 (config-if)# ip vrf forwarding CCNP
R1 (config-if)# exit
R1 (config)# exit
R1# ping vrf CCNP 10.1.1.1
% Unrecognized host or address, or protocol not running.

```

- A. interface Loopback0
vrf forwarding CCNP

- B. interface Loopback0
ip address 10.1.1.1 255.255.255.0
- C. interface Loopback0
ip address 10.1.1.1 255.255.255.0
vrf forwarding CCNP
- D. interface Loopback0
ip address 10.1.1.1 255.255.255.0
ip vrf forwarding CCNP

Correct Answer: B

Section: (none)

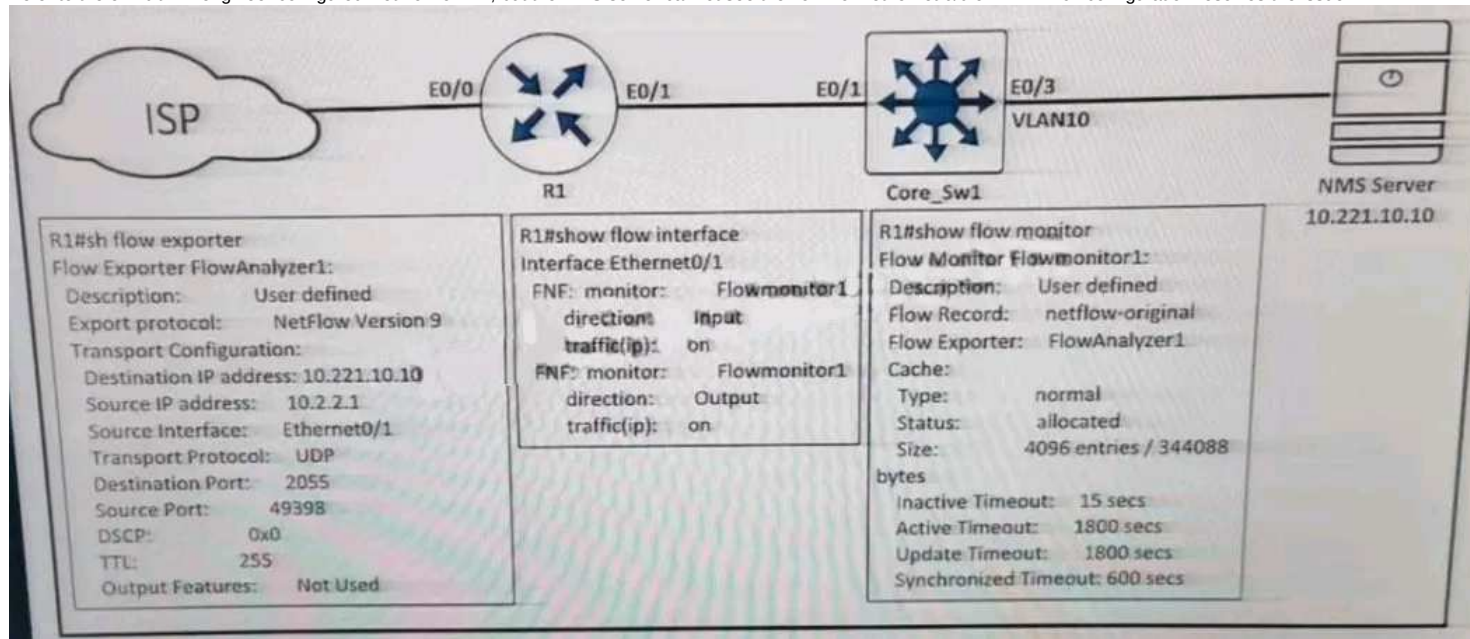
Explanation

Explanation/Reference:

Since configuring a VRF to an interface will remove the IP address setting in that interface, you need to configure the IP address again after setting up an VRF.

QUESTION 283

Refer to the exhibit. An engineer configured NetFlow on R1, but the NMS server cannot see the flow from ethernet 0/0 of R1. Which configuration resolves the issue?



- A. flow monitor Flowmonitor1
source Ethernet0/0
- B. interface Ethernet0/1
ip flow monitor Flowmonitor1 input
ip flow monitor Flowmonitor1 output
- C. interface Ethernet0/0
ip flow monitor Flowmonitor1 input
ip flow monitor Flowmonitor1 output
- D. flow exporter FlowAnalyzer1
source Ethernet0/0

Correct Answer: C

Section: (none)

Explanation

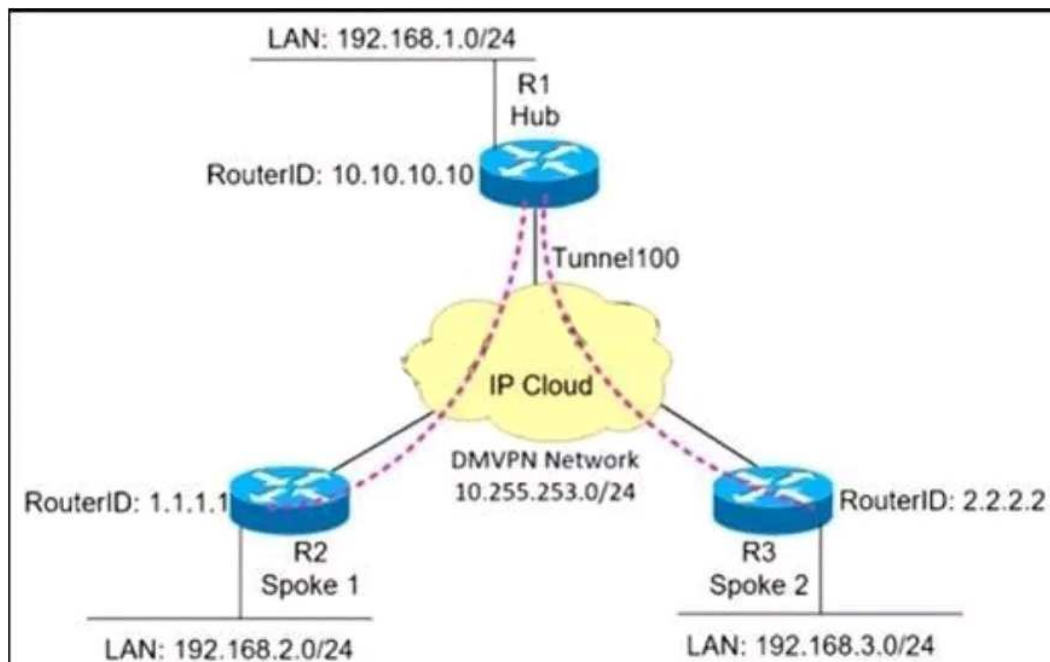
Explanation/Reference:

From the R1's show flow interface output (the middle one), only interface e0/1 is shown.

The reason why the flow from ethernet 0/0 cannot be seen is due to the fact that e0/0 has not been enabled for monitoring.

QUESTION 284

Refer to the exhibit. A network administrator sets up an OSPF routing protocol for a DMVPN network on the hub router. Which configuration required to establish a DMVPN tunnel with multiple spokes?



```
*Mar 1 17:19:04.051: %OSPF-5-ADJCHG: Process 100, Nbr 1.1.1.1 on Tunnel100 from LOADING to FULL, Loading Done
*Mar 1 17:19:06.375: %OSPF-5-ADJCHG: Process 100, Nbr 1.1.1.1 on Tunnel100 from FULL to DOWN, Neighbor Down: Adjacency forced to
reset
*Mar 1 17:19:06.627: %OSPF-5-ADJCHG: Process 100, Nbr 2.2.2.2 on Tunnel100 from LOADING to FULL, Loading Done
*Mar 1 17:19:10.123: %OSPF-5-ADJCHG: Process 100, Nbr 2.2.2.2 on Tunnel100 from FULL to DOWN, Neighbor Down: Adjacency forced to
reset
*Mar 1 17:19:14.499: %OSPF-5-ADJCHG: Process 100, Nbr 10.10.10.10 on Tunnel100 from LOADING to FULL, Loading Done
*Mar 1 17:19:19.139: %OSPF-5-ADJCHG: Process 100, Nbr 10.10.10.10 on Tunnel100 from EXSTART to DOWN, Neighbor Down: Interface
down or detached
*Mar 1 17:01:51.975: %OSPF-4-NONEIGHBOR: Received database description from unknown neighbor 192.168.1.1
*Mar 1 17:01:57.783: OSPF: Rcv LS UPD from 192.168.1.1 on Tunnel100 length 88 LSA count 1
*Mar 1 17:01:57.155: OSPF: Send UPD to 10.255.253.1 on Tunnel100 length 100 LSA count 2
```

- A. ip ospf network point-to-multipoint on both spoke routers
- B. ip ospf network point-to-point on the hub router
- C. ip ospf network point-to-multipoint on One spoke router
- D. ip ospf network point-to-point on both spoke routers

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Since DMVPN network is partially meshed, you need to setup special OSPF network type for the concerned interface.

The logging messages are shown since the R1's interface is using Point-to-Point which only allows one neighbor. Therefore, the neighboring keeps swapping between R2 and R3.

In order to solve the problems, you can e.g. configure the OSPF network type to point-to-multipoint in all routers.

There are many problems in this question.

Actually the hub routers and 2 spoke routers must be configured with "point-to-multipoint". However no choice mentioned setting up "point-to-multipoint" in the hub routers. The nearest one is A.

Remarks :

There are other problems in the logging messages and diagrams shown in the diagram

- The first four logging message is probably sourced from the Hub router. The 5th and 6th logging message is probably sourced from a Spoke router (i.e. the logging messages shown are from different routers).

- The last three logging message is probably sourced from the Hub router. However, there is no mention about the IP address 192.168.1.1 in the diagram. It should be a network connected by one of the spoke router. You need to assume that either there is a hidden Spoke router not shown in the diagram OR the network 192.168.1.0/24 is also a network connected by one of the spoke routers.

QUESTION 285

Refer to the exhibit. An engineer is troubleshooting a TACACS problem. Which action resolves the issue?

```

*17:40:07.826: AAA/BIND(00000055): Bind i/f
*17:40:07.826: AAA/AUTHEN/LOGIN (00000055): Pick method list 'default'
*17:40:07.826: TPLUS: Queuing AAA Authentication request 85 for processing
*17:40:07.826: TPLUS: TPLUS(00000055) login timer started 1020 sec timeout
*17:40:07.826: TPLUS: processing authentication start request id 85
*17:40:07.826: TPLUS: Authentication start packet created for 85()
*17:40:07.826: Using server 10.106.60.182
*17:40:07.826: TPLUS(00000055)/0/NB_WAIT/225FE2DC: Started 5 sec timeout
*17:40:07.830: TPLUS(00000055)/0/NB_WAIT: socket event 2
*17:40:07.830: TPLUS(00000055)/0/NB_WAIT: wrote entire 38 bytes request
*17:40:07.830: TPLUS(00000055)/0/READ: socket event 1
*17:40:07.830: TPLUS(00000055)/0/READ: Would block while reading
*17:40:07.886: TPLUS(00000055)/0/READ: socket event 1
*17:40:07.886: TPLUS(00000055)/0/READ: read entire 12 header bytes (expect 6 bytes data)
*17:40:07.886: TPLUS(00000055)/0/READ: socket event 1
*17:40:07.886: TPLUS(00000055)/0/READ: read entire 18 bytes response
*17:40:07.886: TPLUS(00000055)/0/225FE2DC: Processing the reply packet
*17:40:07.886: TPLUS: received bad AUTHEN packet: length = 6, expected 43974
*17:40:07.886: TPLUS: Invalid AUTHEN packet (check keys).

```

- A. Configure a matching TACACS server IP.
- B. Configure a matching preshared key.
- C. Generate authentication from a relative source interface.
- D. Apply a configured AAA profile to the VTY.

Correct Answer: B

Section: (none)

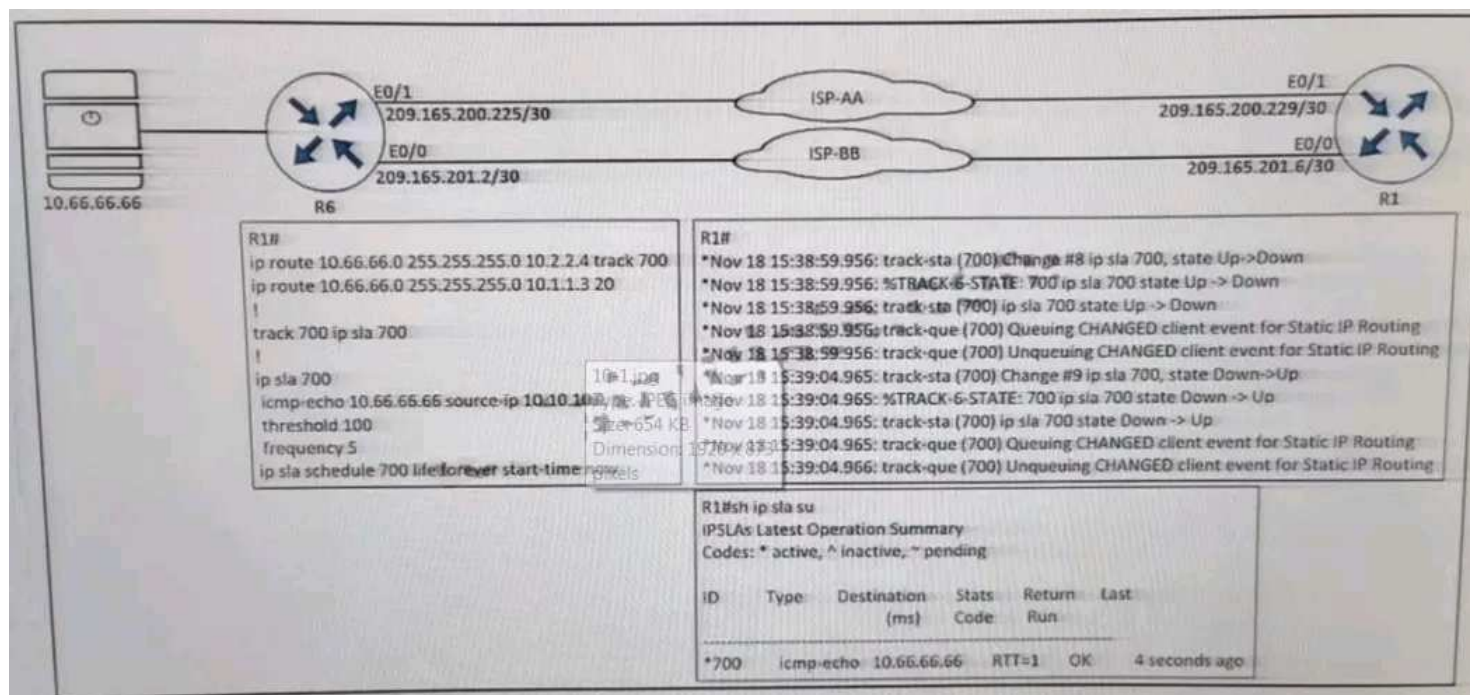
Explanation

Explanation/Reference:

The last two messages (especially the words "check keys") shows what the problem is.

QUESTION 286

Refer to the exhibit. An engineer configured IP SLA on R1 to avoid the ISP link flapping problem. but it is not working as designed IP SLA should wait 30 seconds before switching traffic to a secondary connection and then revert to the primary link after waiting 20 seconds, when the primary link is available and stabilized. Which configuration resolves the issue?



- A. R1(config)#ip sla 700
R1(config-ip-sla)#delay down 30 up 20
- B. R1(config)#ip sla 700
R1(config-ip-sla)#delay down 20 up 30
- C. R1(config)#track 700 ip sla 700
R1(config-track)#delay down 30 up 20
- D. R1(config)#track 700 ip sla 700
R1(config-track)#delay down 20 up 30

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Since we just have configuration of R1 (the router on the left) and do not have the configuration of R6 (the router on the right), we just assume that all setup for using ISP-A and ISP-B is configured properly in both routers.

The problem found in the logging messages is that after IP SLA failed and switched to ISP-BB at 15:38:59.956, it is switched back to ISP-AA when IP SLA is successful again at 15:39:04.965. Even if you disconnect ISP-AA and connect it again, it should wait 20 seconds. However, this period of time is not waited before switching back.

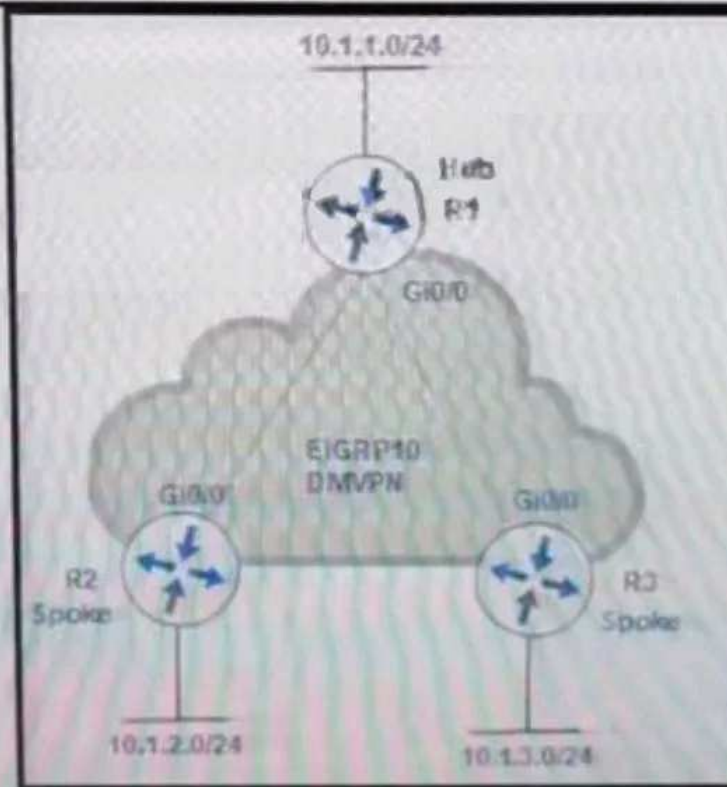
For adding delay, you should add it in track object (not under config-ip-sla which is the mode you select the IP SLA operation).

QUESTION 287

Refer to the exhibit. An engineer configures DMVPN and receives the hub location prefix of 10.1.1.0/24 on R2 and R3. The R3 prefix of 10.1.3.0/24 is not received on R2, and the R2 prefix 10.1.2.0/24 is not received on R3. Which action resolves the issue?


```
R2#show ip route eigrp | include 10.1.1.0/24
D    10.1.1.0/24
```

```
R3#show ip route eigrp | include 10.1.1.0/24
D    10.1.1.0/24
```



- A. Split horizon prevents the routes from being advertised between spoke routers it should be disabled with the command `no ip split-horizon eigrp 10` on the tunnel interface of R1
- B. There is no spoke-to-spoke connection DMVPN configuration should be modified to enable a tunnel connection between R2 and R3 and neighbor relationship confirmed by use of the `show ip eigrp neighbor` command
- C. Split horizon prevents the routes from being advertised between spoke routers it should be disabled with the `no ip split-horizon eigrp 10` command on the Gi0/0 interface of R1.
- D. There is no spoke-to-spoke connection DMVPN configuration should be modified with a manual neighbor relationship configured between R2 and R3 and confirmed by use of the `show ip eigrp neighbor` command.

Correct Answer: A

Section: (none)

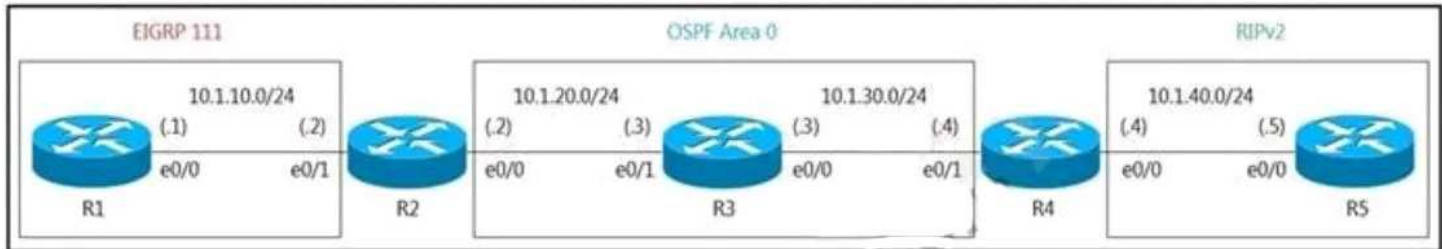
Explanation

Explanation/Reference:

When using distance vector routing protocol, the most probable reason is the split horizon enabled in all routing interfaces. Hence you need to disable it in the tunnel interface on the hub router i.e. R1.

QUESTION 288

Refer to the exhibit. R5 should not receive any routes originated in the EIGRP domain. Which set of configuration changes removes the EIGRP routes from the R5 routing table to fix the issue?



```

R2
route-map E20 permit 10
set tag 111
!
router ospf 1
redistribute ospf 1 metric 10 10 10 10 10
!
router ospf 1
redistribute eigrp 111 route-map E20 subnets

R4
router rip
router ospf 1
redistribute rip subnets

```

- A. R4
route-map O2R deny 10
match tag 111
route-map O2R permit 20
!
router rip
redistribute ospf 1 route-map O2R metric 1
- B. R2
route-map E20 deny 20
- R4
route-map O2R deny 10
match tag 111
!
router rip
redistribute ospf 1 route-map O2R metric 1
- C. R4
route-map O2R permit 10
match tag 111
route-map O2R deny 20
!
router rip
redistribute ospf 1 route-map O2R metric 1
- D. R4
route-map O2R deny 10
match tag 111
!
router rip
redistribute ospf 1 route-map O2R metric 1

Correct Answer: A

Section: (none)

Explanation

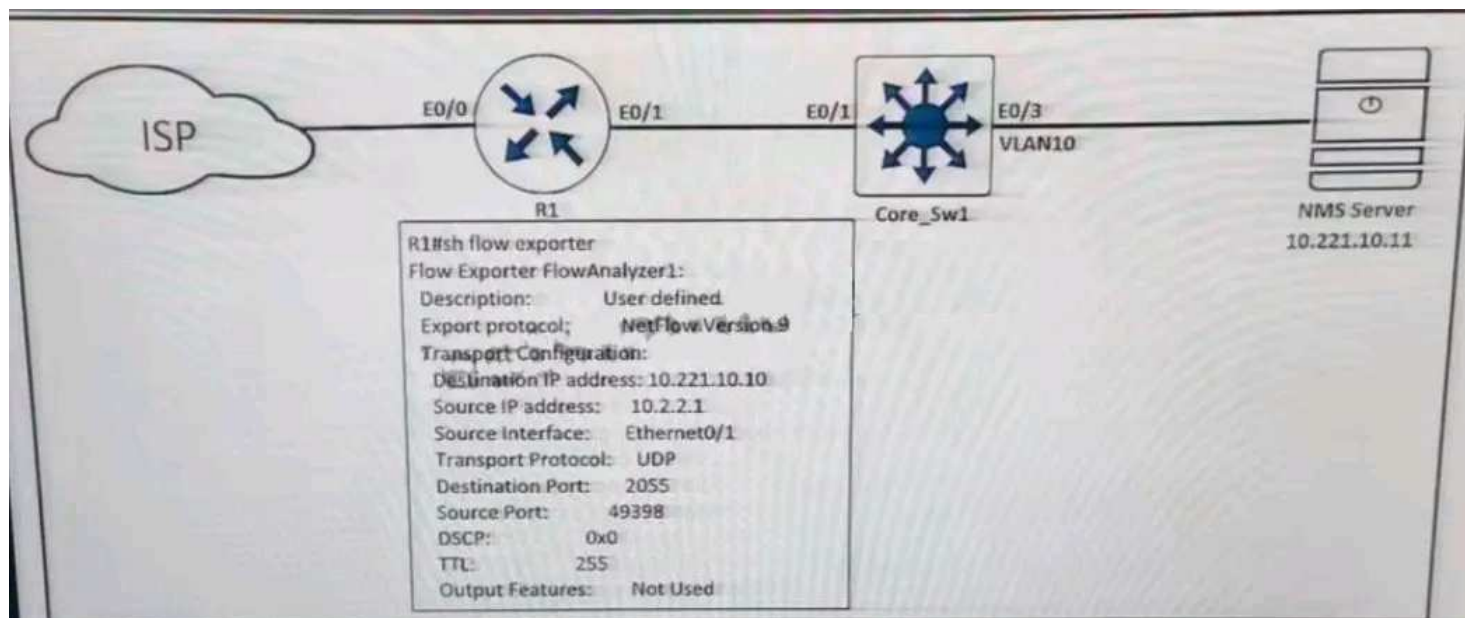
Explanation/Reference:

Since R2 redistribute EIGRP into OSPF with the tag "111" in the existing configuration setup, you can configure R4 to prevent i.e. deny routes with tag "111" from redistributing to RIPv2. Then R5 will not receive routes in EIGRP networks.

D is wrong since there is an implicit deny all at the end of any route-map. Hence no route will be distributed with choice D settings.

QUESTION 289

Refer to the exhibit. An engineer configured NetFlow on R1, but the NMS server cannot see the flow from R1. Which configuration resolves the issue?



- A. flow monitor Flowmonitor1
destination 10.221.10.11
- B. flow exporter FlowAnalyzer1
destination 10.221.10.11
- C. interface Ethernet0/1
flow-destination 10.221.10.11
- D. interface Ethernet0/0
flow-destination 10.221.10.11

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

The NMS server has an IP address of 10.221.10.11 but the current configuration (as shown in the report). is setting the destination to 10.221.10.10

QUESTION 290

If HTTP packet from 10.1.1.0/24 to a destination in 10.100.100.0/24 must be directed to a next-hop of 10.2.2.2, two hops away from the router. Which configuration meets the requirements?

- A. access-list 100 permit tcp 10.1.1.0 0.0.0.255 host 10.100.100.100 eq http
!
route-map POLICY deny 10
match ip address 100
set ip next-hop recursive 10.2.2.2
route-map POLICY permit 20
- B. access-list 100 permit tcp 10.1.1.0 0.0.0.255 host 10.100.100.100 eq http
!
route-map POLICY permit 10
match ip address 100
set ip next-hop 10.2.2.2
route-map POLICY permit 20
- C. access-list 100 permit tcp 10.1.1.0 0.0.0.255 host 10.100.100.100 eq http
!
route-map POLICY permit 10
match ip address 100
set ip next-hop 10.2.2.2
- D. access-list 100 permit tcp 10.1.1.0 0.0.0.255 host 10.100.100.100 eq http
!
route-map POLICY permit 10
match ip address 100
set ip next-hop recursive 10.2.2.2

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Since the next hop is not directly connected by the router, "set ip next-hop recursive" is required. A is wrong since the route map entry 10 is "deny"

QUESTION 291

For the following ACL

```

ip access-list extended COPP
permit ospf any host 224.0.0.5 log
permit ospf any host 224.0.0.6 log-input
  
```

Problem occurs when it is applied as CoPP. What action resolves the issue?

- A. Apply the ACL on the OSPF physical interface in outward direction
- B. Apply the ACL on the OSPF physical interface in inward direction
- C. Add one more ACL line to permit 224.0.0.6 in the inward direction
- D. Remove the log and log-input keywords from ACL

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

ACL with logging is not allowed for use in CoPP.

QUESTION 292

For the following ACL and Route Map for EIGRP route inbound filtering

```
ip access-list standard EIGRP
deny 10.10.10.0 0.0.0.0
permit 0.0.0.0 0.0.0.0
```

```
route-map EIGRP-MAP permit 10
match ip address EIGRP
```

A route 10.10.10.128/25 is found in the routing table. What action resolves the issue?

- A. The ACL entry should be modified to deny 10.10.10.0 0.0.0.255
- B. The ACL entry should be modified to deny 10.10.10.0 255.255.255.0
- C. A deny 20 with no match condition should be added to Route Map
- D. A deny 20 with a new ACL that deny all should be added to Route Map

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

In order to deny route with 10.0.0.X as network prefix, you need an ACL to deny 10.10.10.0 0.0.0.255

Remarks:

Actually the question has problem since 10.10.10.128/25 will NOT be learnt:

- Since the ACL entry 0.0.0.0 0.0.0.0 means exactly the network prefix 0.0.0.0. Therefore the route is not matched by ACL and will not be permitted by permit 10 in Route Map.
- There is no permit 20 with no match condition in the Route Map.

However, ignoring the above, A should be the most reasonable choice.

QUESTION 293

For a EIGRP router with the following settings

```
ip route 2.2.2.2 255.255.255.255 10.10.10.1
!
router eigrp 100
network 1.1.1.1 0.0.0.0
network 10.10.10.0 0.0.0.255
eigrp stub static
eigrp stub connected
```

The router is not advertising the route 2.2.2.2/32 learnt from another router. Which action resolves the issue?

- A. Redistribute the connected metric in EIGRP
- B. Redistribute the static metric in EIGRP
- C. Add the eigrp stub connected static command
- D. Remove the eigrp stub connected command

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since the router is a Stub router, it will not advertise learnt route.

However since eigrp stub static is configured, you can redistribute the static route into EIGRP for advertising.

Remarks :

- The wording "Redistribute the static metric" in B means the command "redistribute static metric"
- The two stub commands entered will be joined automatically as "eigrp stub connected static".

QUESTION 294

Refer to the exhibit. A network administrator successfully logs in to a switch using SSH from a RADIUS server. When the network administrator uses a console port to access the switch the RADIUS server returns shell:privilege=15 and the switch asks to enter the enable command. When the command is entered, it gets rejected. Which command set is used to troubleshoot and resolve this issue?

```

aaa new-model
aaa group server radius RADIUS-SERVERS
aaa authentication login default group RADIUS-SERVERS local
aaa authentication enable default group RADIUS-SERVERS enable
aaa authorization exec default group RADIUS-SERVERS if-authenticated
aaa authorization network default group RADIUS-SERVERS if-authenticated
aaa accounting send stop-record authentication failure
aaa session-id common
!
line con 0
logging synchronous
stopbits 1
line vty 0 4
logging synchronous
transport input ssh

```

- A. line con 0
aaa authorization console
authorization exec
!
line vty 0 4
transport input ssh
- B. line con 0
aaa authorization console
!
line vty 0 4
authorization exec
- C. line con 0
aaa authorization console priv15
!
line vty 0 4
authorization exec
- D. line con 0
aaa authorization console
authorization priv15
!
line vty 0 4
transport input ssh

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

All choices are wrong since "aaa authorization console" is a Global Configuration mode command.

After entering the Global Configuration mode command "aaa authorization console", you can enter "authorization exec ..." to choose the authorization list that you want to apply to the console. All choices are also wrong in this command since you must specify the use of default list or the name of the authorization list.

```

R1(config)#line con 0
R1(config-line)#authorization exec ?
WORD      Use an authorization list with this name
default   Use the default authorization list

```

Despite all the problems, A seems to be the best answer since it includes both commands mentioned and applies the second command correctly in "line con 0".

QUESTION 295

Refer to the exhibit.

```

R1(config)#ip prefix-list EIGRP seq 10 deny 0.0.0.0/0 le 32
R1(config)#ip prefix-list EIGRP seq 20 permit 10.0.0.0/8
R1(config)#router eigrp 10
R1(config-router)#distribute-list prefix EIGRP in Ethernet0/0

R1#show ip route eigrp

```

A prefix list is created to filter routes inbound to an EIGRP process except for network 10 prefixes. After the prefix list is applied no network 10 prefixes are visible in the routing table from EIGRP. Which configuration resolves the issue?

- A. ip prefix-list EIGRP seq 20 permit 10.0.0.0/8 ge 9
- B. ip prefix-list EIGRP seq 10 permit 0.0.0.0/0 le 32
- C. ip prefix-list EIGRP seq 5 permit 10.0.0.0/8 ge 9
no ip prefix-list EIGRP seq 20 permit 10.0.0.0/8
- D. ip prefix-list EIGRP seq 20 permit 10.0.0.0/8 ge 9
ip prefix-list EIGRP seq 10 permit 0.0.0.0/0 le 32

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Since all prefixes are denied by the rule in seq 10, you need to move the one in seq 20 to a sequence number before 10.

Remarks :

- With C, the final prefix list allows all 10 subnet prefixes but 10.0.0.0/8 is not allowed for this configuration.

A better choice (if available) is to remove sequence 10 instead of seq 20 in choice C. This will keep the permit for 10.0.0.0/8 and let the implicit deny to block other routes.

- Only C is valid commands for applying to the existing configuration. For other choices, error similar to the following will be shown since you need to use "no" to remove an existing sequence number before you can change it.

%Insertion failed - seq # exists with different policy: 10

QUESTION 296

How does an MPLS Layer 3 VPN differentiate the IP address space used between each VPN?

- A. by RD
- B. by address family
- C. by MP-BGP
- D. by RT

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 297

An engineer is implementing a coordinated change with a server team. As part of the change, the engineer must configure interface GigabitEthernet2 in an existing VRF "RED" then move the interface to an existing VRF "BLUE" when the server team is ready. The engineer configured interface GigabitEthernet2 in VRF "RED"

```

interface GigabitEthernet2
description Migration ID: B410A60D0806G06
vrf forwarding RED
ip address 10.0.0.0 255.255.255.254
negotiation auto

```

Which configuration completes the change?

- A. interface GigabitEthernet2
no ip address
vrf forwarding BLUE
- B. interface GigabitEthernet2
no vrf forwarding RED
vrf forwarding BLUE
ip address 10.0.0.0 255.255.255.254
- C. interface GigabitEthernet2
no vrf forwarding RED
vrf forwarding BLUE
- D. interface GigabitEthernet2
no ip address
ip address 10.0.0.0 255.255.255.254
vrf forwarding BLUE

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since any change in VRF in an interface will remove the IP address, you need to configure the IP address again after the change.

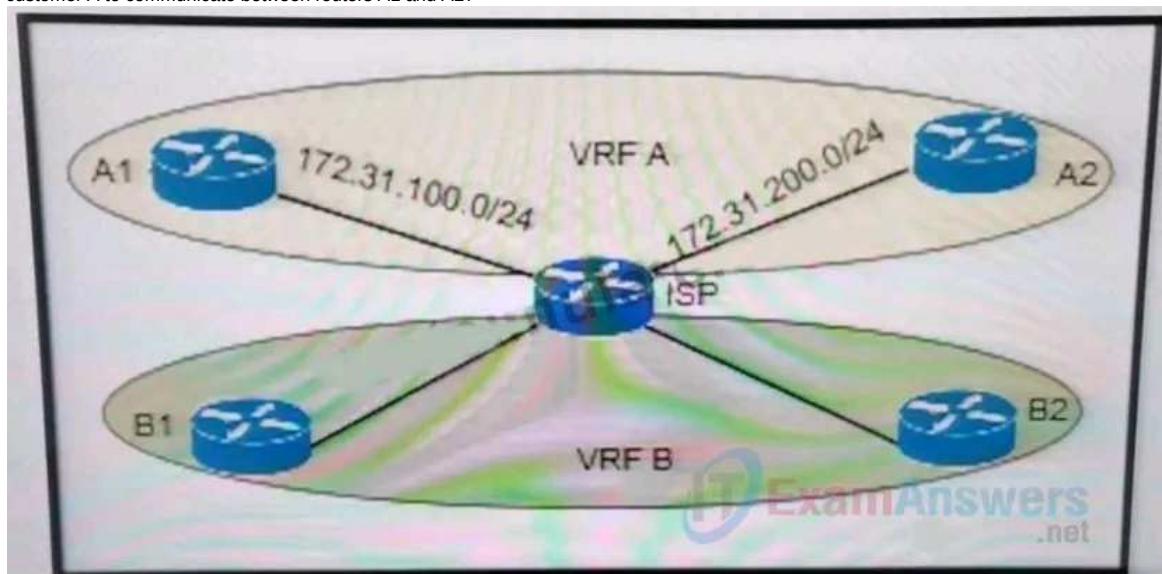
Remarks :

Actually IP address 10.0.0.0 as well as prefix length /31 is NOT valid for assigning to an normal Ethernet interface (Although it may be used with cautious in a point-to-

point interface) However, since there is no other choices, we need to ignore this fact in answering this question

QUESTION 298

Refer to the exhibit. The ISP router is fully configured for customer A and customer B using the VRF-Lite feature. What is the minimum configuration required for customer A to communicate between routers A1 and A2?



A. A1:
interface fa0/0
description To->ISP
ip vrf forwarding A
ip add 172.31.100.1 255.255.255.0
no shut
!
router ospf 100
net 172.31.100.1 0.0.0.255 area 0

A2:
interface fa0/0
description To->ISP
ip vrf forwarding A
ip add 172.31.200.1 255.255.255.0
no shut
!
router ospf 100
net 172.31.200.1 0.0.0.255 area 0

B. A1:
interface fa0/0
description To->ISP
ip add 172.31.200.1 255.255.255.0
no shut
!
router ospf 100
net 172.31.200.1 0.0.0.255 area 0

A2:
interface fa0/0
description To->ISP
ip add 172.31.100.1 255.255.255.0
no shut
!
router ospf 100
net 172.31.100.1 0.0.0.255 area 0

C. A1:
interface fa0/0
description To->ISP
ip vrf forwarding A
ip add 172.31.100.1 255.255.255.0
no shut
!
router ospf 100 vrf A
net 172.31.200.1 0.0.0.255 area 0

A2:
interface fa0/0
description To->ISP
ip vrf forwarding A
ip add 172.31.200.1 255.255.255.0
no shut
!
router ospf 100 vrf A
net 172.31.100.1 0.0.0.255 area 0

D. A1:
interface fa0/0
description To->ISP
ip add 172.31.100.1 255.255.255.0
no shut

```

!
router ospf 100
net 172.31.100.1 0.0.0.255 area 0

A2:
interface fa0/0
description To->ISP
ip add 172.31.200.1 255.255.255.0
no shut
!
router ospf 100
net 172.31.200.1 0.0.0.255 area 0

```

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Although ISP router is configured with VRF, customer's routers do not need to setup VRF especially when minimal configuration is required.

A is wrong since interface is assigned with VRF but OSPF is not configured with VRF.

B is wrong since incorrect IP network / address is configured for the routers (e.g. A1's IP should be 172.31.100.1)

C is wrong since the OSPF network command for each router does not match the IP address of the interface of that router.

D is the answer.

QUESTION 299

What are the two prerequisites to enable BFD on Cisco routers? (Choose two)

- A. A supported IP routing protocol must be configured on the participating routers.
- B. OSPF Demand Circuit must run BFD on all participating routers.
- C. ICMP must be allowed on all participating routers.
- D. UDP port 1985 must be allowed on all participating routers.
- E. Cisco Express Forwarding and IP Routing must be enabled on all participating routers.

Correct Answer: AE

Section: (none)

Explanation

Explanation/Reference:

QUESTION 300

Refer to the exhibit.

R2# show ip ospf neighbor						
Neighbor ID	Pri	State	Dead Time	Address	Interface	
192.168.99.2	1	EXCHANGE/	-	00:00:36	192.168.99.1	Serial0/1
router-6#						
R3# show ip ospf neighbor						
Neighbor ID	Pri	State	Dead Time	Address	Interface	
192.168.99.1	1	EXSTART/	-	00:00:33	192.168.99.2	Serial0/1

An OSPF neighbor relationship between R2 and R3 is showing stuck in EXCHANGE/EXSTART state. The neighbor is established between R1 and R2. The network engineer can ping from R2 to R3 and vice versa, but the neighbor is still down. Which action resolves the issue?

- A. Restore the Layer 2/Layer 3 connectivity issue in the ISP network.
- B. Match MTU on both router interfaces or ignore MTU.
- C. Administrative "shut then no shut" both router interfaces.
- D. Enable OSPF on the interface, which is required.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 301

Refer to the exhibit



R1 and R2 have been configured with the following settings for forming neighbor relationship with M05:

```

R1
router bgp 100
neighbor 172.16.1.2 remote-as 200
neighbor 172.16.1.1 password cisco123

```

```
R2
router bgp 200
neighbor 172.16.1.1 remote-as 100
neighbor 172.16.1.1 password cisco
```

The neighbor relationship is not coming up. Which configuration resolves the issue?

- A. R1
router bgp 100
neighbor 172.16.1.2 password cisco
- B. R1
router bgp 100
neighbor 172.16.1.2 password MD5 cisco
- R2
router bgp 200
neighbor 172.16.1.1 password MD5 cisco
- C. R2
router bgp 200
neighbor 172.16.1.1 password cisco123
- D. R1
router bgp 100
neighbor 172.16.1.2 password MD5 cisco123
- R2
router bgp 200
neighbor 172.16.1.1 password MD5 cisco123

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The are two problems in current configuration settings:

- R1, the IP address in the neighbor command for configuration password is wrong.
- The two MD5 passwords configured in the two routers do not match.

Choice A can fix both of the above problems.

Note that MD5 is the default algorithm for BGP authentication.

If you want to use other algorithm you need to setup keychain.

QUESTION 302

Refer to the exhibit.

**** missing diagram ****

The diagram shows R1 is having router ID 10.255.255.1.

It also includes a message similar to the following at the bottom

```
%OSPF-4-FL00D_WAR: . . . . . type-5 adv-rtr 10.255.255.1 . . . .
*****
```

After configuring OSPF in R1, some external destinations in the network became unreachable. Which action resolves the issue?

- A. Change the R1 router ID from 10.255.255.1 to a unique value and clear the process.
- B. Increase the SPF delay interval on R1 to synchronize routes.
- C. Disconnect the router with the OSPF router ID 0.0.0.0 from the network.
- D. Clear the OSPF process on R1 to flush stale LSAs sent by other routers.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The OSPF flood war occurs when the router repeatedly receives its own Link State Advertisement (LSA) and flushes it from the network or sends a new version of it. This is meant to detect issues with Type-2 LSAs when duplicate IP addresses are present in the network, or with Type-5 LSAs when there is a duplicate router ID in different OSPF Areas.

In a typical scenario, there is one router in the network that originates the LSA and a second router that flushes the LSA.

QUESTION 303

Which two solutions are used to overcome a flapping link that causes a frequent label binding exchange between MPLS routers? (Choose two)

- A. create targeted hellos to protect the session
- B. increase a hold-timer to protect the session
- C. increase input queue on links to protect the session
- D. create link dampenirg on links to protect the session
- E. increase a session delay to protect the session.

Correct Answer: AB

Section: (none)

Explanation

Explanation/Reference:

MPLS LDP Session Protection maintains Label Distribution Protocol (LDP) bindings when a link fails. MPLS LDP Session Protection uses LDP Targeted Hellos to

protect LDP sessions.

On the other hand, you can change the time for which an Label Distribution Protocol (LDP) session is maintained in the absence of LDP messages from the session peer, use the `session holdtime` command in MPLS LDP configuration mode.

Remarks:

Session Delay feature only delay the setup of a session instead of preventing them from disconnect.

QUESTION 304

Refer to the exhibit.

missing

A command "aaa authentication login default local" is entered

A network engineer must configure a login authentication method at line console 0 on Cisco ISR router. During configuration the engineer got a call and router console timed out. When the call was completed, the engineer pressed "enter" on the console. The engineer was asked for a username to log in to the router even though no username and password were configured previously on the router. What action resolves the issue?

- A. Telnet to the router to login in
- B. Power cycle to reload the router and reconfigure it
- C. Configure the user and password in the RADIUS server
- D. Use CTRL+C to bypass authentication in the router.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

The entered default authentication list applies to all login authentication including login through console or vty. You need to enter the username and password configured in the router locally for login.

If username and password has been configured locally, you can use it for login.

However if no username and password has been configured locally, since the configuration has not been saved in this case, you can reboot the router.

QUESTION 305

An engineer creates a default static route on a router with a next hop of 10.1.1.1. On inspection, the engineer finds the router has two VRFs, Red and Blue. The next hop is valid for both VRFs and exists in each assigned VRF. Which configuration achieves connectivity?

- A. `ip route vrf Red 0.0.0.0 255.255.255.255 10.1.1.1`
`ip route vrf Blue 0.0.0.0 255.255.255.255 10.1.1.1`
- B. `ip route 0.0.0.0 0.0.0.0 10.1.1.1`
- C. `ip route vrf Red 0.0.0.0 0.0.0.0 10.1.1.1`
`ip route vrf Blue 0.0.0.0 0.0.0.0 10.1.1.1`
- D. `ip route vrf Blue 0.0.0.0 255.255.255.255 10.1.1.1`
`ip route vrf Red 0.0.0.0 255.255.255.255 10.1.1.1`

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Since the default static route has been entered, you just need to add the default static route in both Red and Blue VRFs.

QUESTION 306

What are two characteristics of IPv6 Source Guard? (Choose two)

- A. requires the user to configure a static binding
- B. recovers missing binding table entries
- C. used in service provider deployments to protect DDoS attacks
- D. requires that validate prefix be enabled
- E. requires IPv6 snooping on Layer 2 access or trunk ports

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

For A, Source Guard only looks at information found in the binding table, and it doesn't fill the binding table. Although you can enter static binding, features such as ND inspection or IPv6 snooping is used to fill the table.

For B, when a traffic is denied, IPv6 Source Guard can use ND or DHCPgleaning to find missing entries. However, straightly speaking, it is not IPv6 Source Guard itself to recover the missing entries. ND or DHCPgleaning is provided by IPv6 First-Hop Security Binding Table recovery mechanism feature.

For D, IPv6 Prefix Guard is a feature included in IPv6 Source Guard but does not need to be enabled when using IPv6 Source Guard only. Hence the command "validate prefix" is optional.

For C, since IPv6 Source Guard allows traffic for known hosts, it is not suitable for service provider in much a lot of customer's hosts or Internet hosts are involved. However, the Prefix Guard which deny traffic based on IPv6 Prefix can be useful in service provider environment. Since Prefix Guard works within IPv6 Source Guard feature, therefore C seems to be correct.

QUESTION 307

An engineer is creating a policy that overrides normal routing behavior. If the route to a destination of 10.100.100.0/24 is withdrawn from the routing table, the policy must direct traffic to a next-hop of 10.1.1.1. If the route is present in the routing table, then normal forwarding must occur. Which configuration meets the requirements?

- A. access-list 100 permit ip any any
!
route-map POLICY permit 10
match ip address 100
set ip next-hop recursive 10.1.1.1
- B. access-list 100 permit ip any 10.100.100.0 0.0.0.255
!
route-map POLICY permit 10
match ip address 100
set ip next-hop recursive 10.1.1.1
!
route-map POLICY permit 20
- C. access-list 100 permit ip any 10.100.100.0 0.0.0.255
!
route-map POLICY permit 10
match ip address 100
set ip next-hop 10.1.1.1
!
route-map POLICY permit 20
- D. access-list 100 permit ip any 10.100.100.0 0.0.0.255
!
route-map POLICY permit 10
match ip address 100
set ip default next-hop 10.1.1.1

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

set ip default next-hop is used only when a route for the destination is not present.

Remarks :

Note that unlike data traffic filtering or route filtering, no permit all is required at the bottom of the route map.

QUESTION 308

Refer to the exhibit.

*** missing diagram ***

A diagram showing R1 with 3 paths to 5.5.5.0/24

EIGRP is used as the routing protocol.

Only two EIGRP routes for 5.5.5.0/24 are entered into the routing table.

An engineer investigates a routing issue on R1 and finds that traffic destined to 5.5.5.0/24 does not take all of the paths. Which action resolves the issue?

- A. Decrease the variance value in EIGRP.
- B. Stop advertising 192.168.13.0/24 in EIGRP.
- C. Remove the adjacency of R3 from EIGRP.
- D. Increase the variance value in EIGRP.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

To include all three paths, you need to increase the variance value so the route with the highest metric can also be included.

QUESTION 309

Refer to the exhibit.

```
admin@linux:~$ scp script.py admin@198.51.100.64:script.py
Password:
Administratively disabled.
admin@linux:~$ Connection to 198.51.100.64 closed by remote
host.
```

A network administrator has developed a Python script on the local Linux machine and is trying to transfer it to the router. However, the transfer fails. Which action resolves this issue?

- A. The SSH service must be enabled with the crypto key generate rsa command.
- B. The SCP service must be enabled with the ip scp server enable command.
- C. The Python interpreter must first be enabled with the guestshell enable command.
- D. The SSH access must be allowed on the VTY lines using the transport input ssh command.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Remarks :

For the wrong choices A and D, there will be no prompt for password and no connection can be made (therefore no message about "Connection to x.x.x.x closed by remote").

QUESTION 310

Refer to the exhibit.

```
BRANCH-RTR#  
router eigrp 100  
network 10.4.31.0 0.0.0.7  
network 10.100.100.1 0.0.0.0  
distribute-list route-map FILTER-IN in FastEthernet0/0  
eigrp router-id 10.100.100.1  
!  
ip prefix-list 102 seq 10 permit 10.1.1.100/32  
!  
route-map FILTER-IN deny 10  
match ip address prefix-list 102  
!
```

A junior engineer updated a branch router configuration. Immediately after the change, the engineer receives calls from the help desk that branch personnel cannot reach any network destinations. Which configuration restores service and continues to block 10.1.1.100/32?

- A. route-map FILTER-IN deny 5
- B. ip prefix-list 102 seq 15 permit 0.0.0.0/32 le 32
- C. ip prefix-list 102 seq 5 permit 0.0.0.0/32 le 32
- D. route-map FILTER-IN permit 20

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

For blocking one or more specific prefixes only, you must add a permit all at the end of a route-map in order to allow all remaining prefixes (i.e. preventing them from blocked by implicit deny).

QUESTION 311

Refer to the exhibit.

The diagram shows a network setup with an FTP Server and a Switch. The FTP Server has IP 10.0.0.2/24 and is connected to the Switch via E0/1. The Switch has VLAN2 and IP 10.0.0.1/24. The FTP Server shows a successful ping to 10.0.0.1. The Switch shows a failed copy command from the FTP server.

FTP Server:

```
10.0.0.2/24  
FTP Server  
Username: cisco  
Password: cisco  
File to download: IOS.bin  
C:\Users\FTPServer>ping 10.0.0.1  
Pinging 10.0.0.1 with 32 bytes of data:  
Reply from 10.0.0.1: bytes=32 time=1ms TTL=64  
Reply from 10.0.0.1: bytes=32 time=1ms TTL=64  
Reply from 10.0.0.1: bytes=32 time=1ms TTL=64  
Reply from 10.0.0.1: bytes=32 time=1ms TTL=64  
Ping statistics for 10.0.0.1:  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

Switch:

```
Switch#  
!  
Interface VLAN2  
ip address 10.0.0.1 255.255.255.0  
!  
ip ftp source-interface vlan 2  
Switch#copy ftp://cisco:cisco@10.0.0.2/IOS.bin flash:/  
Destination filename [IOS.bin]?  
Accessing ftp://cisco:cisco@10.0.0.2/IOS.bin...  
%Error opening ftp://cisco:cisco@10.0.0.2/IOS.bin (No such file or directory)
```

An engineer cannot copy the IOS.bin
Which action resolves the issue?

- A. Allow file permissions to download the file from the FTP server.
- B. Add the IOS.bin file, which does not exist on FTP server.
- C. Make memory space on the switch flash or USB drive to download the file.
- D. Use the copy flash:/ ftp://cisco@10.0.0.2/IOS.bin command.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Both "Allow file permissions ..." and "Add the IOS.bin ..." can be the answer.

However, if you think that the box under the FTP server having the words "File to download: IOS.bin" means the file exists in the server, then "Allow file permissions ..." should be the answer.

Remarks:

Depending on the TFTP server software, the response may be different:

- For some TFTP server software, when the file does not exist or permission of the file is not granted, the response will be the same as follows:

```
Switch#copy tftp://server1/exist_but_no_permission.txt flash:
```

```
Destination filename [exist_but_no_permission.txt]?  
Accessing tftp://server1/exist_but_no_permission.txt...
```

```
%Error opening tftp://server1/exist_but_no_permission.txt (Timed out)
```


Switch#

```
Switch#copy tftp://server1/not_exist.txt flash:
Destination filename [not_exist.txt]?
Accessing tftp://server1/not_exist.txt...
%Error opening tftp://server1/not_exist.txt (Timed out)
Switch#
```

- For other TFTP server software, if the file does not exist or permission of the file is not granted, the response will be the same as follows:

```
Switch#copy tftp://server2/exist_but_no_permission.txt flash:
Destination filename [exist_but_no_permission.txt]?
Accessing tftp://server2/exist_but_no_permission.txt...
%Error opening tftp://server2/exist_but_no_permission.txt (No such file or directory)
Switch#
```

```
Switch#copy tftp://server2/not_exist.txt flash:
Destination filename [not_exist.txt]?
Accessing tftp://server2/not_exist.txt...
%Error opening tftp://server2/not_exist.txt (No such file or directory)
Switch#
```

QUESTION 312

What is an advantage of implementing BFD?

- A. BFD provides faster updates for any flapping route.
- B. BFD provides millisecond failure detection
- C. BFD is deployed without the need to run any routing protocol
- D. BFD provides better capabilities to maintain the routing table

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 313

What is the purpose of the DHCPv6 Guard?

- A. It filters messages between a DHCPv6 server and a DHCPv6 client (or relay agent).
- B. It shows that clients of a DHCPv5 server are affected.
- C. It block DHCPv6 messages from relay agents to a DHCPv6 server.
- D. It allows DHCPv6 reply and advertisements from (rouge) DHCPv6 servers.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

DHCPv6 Guard feature blocks DHCP reply and advertisement messages that originate from unauthorized DHCP servers and relay agents that forward DHCP packets from servers to clients.

Client messages or messages sent by relay agents to servers are not blocked. Only unauthorizaed messages to client is blocked.

If D is change to "It prevents DHCPv6 reply and advertisements from rouge DHCPv6 servers.". Then it will be more specific and will be the answer.

QUESTION 314

Which function does LDP provide in an MPLS topology?

- A. It enables a MPLS topology to connect multiple VPNs to P routers.
- B. It provides hop-by-hop forwarding in an MPLS topology for LSRs.
- C. It exchanges routes for MPLS VPNs across different VRFs.
- D. It provides a means for LSRs to exchange IP routes.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 315

Refer to the exhibit.

```

ip sla 1
 icmp-echo 8.8.8.8
 threshold 1000
 timeout 2000
 frequency 5
ip sla schedule 1 life forever start-time now
!
track 1 ip sla 1
!
ip route 0.0.0.0 0.0.0.0 203.0.113.1 name ISP1 track 1
ip route 0.0.0.0 0.0.0.0 198.51.100.1 2 name ISP2

```

The administrator noticed that the connection was flapping between the two ISPs instead of switching to ISP2 when the ISP1 failed. Which action resolves the issue?

- A. Include a valid source-interface keyword in the icmp-echo statement.
- B. Reference the track object 1 on the default route through ISP2 instead of ISP1.
- C. Modify the static routes to refer both to the next hop and the outgoing interface.
- D. Modify the threshold to match the administrative distance of the ISP2 route.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Flapping occurs since when ISP-1 is down and the IP SLA fails, ISP-2 will be used but this will make IP SLA success again since there is not setting to force IP SLA to continue to use ISP-1.

QUESTION 316

Refer to the exhibit.

```

Tunnel source 199.1.1.1, destination 200.1.1.3
Tunnel protocol/transport GRE/IP
Key disabled, sequencing disabled
Checksumming of packets disabled
Tunnel TTL 255, Fast tunneling enabled
Tunnel transport MTU 1476 bytes
Tunnel transmit bandwidth 8000 (kbps)
Tunnel receive bandwidth 8000 (kbps)

```

An engineer must establish a point-to-point GRE VPN between R1 and the remote site. Which configuration accomplishes the task for the remote site?

- A. Interface Tunnel1
tunnel source 199.1.1.1
tunnel destination 200.1.1.3
ip address 192.168.1.3 255.255.255.0
- B. Interface Tunnel1
tunnel source 200.1.1.3
tunnel destination 199.1.1.1
ip address 192.168.1.1 255.255.255.0
- C. Interface Tunnel1
tunnel source 200.1.1.3
tunnel destination 199.1.1.1
ip address 192.168.1.3 255.255.255.0
- D. Interface Tunnel
lunnel source 199.1.1.1
tunnel destination 200.1.1.3
ip address 192.168.1.1 255.255.255.0

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Seems the diagram is missing something:

- assume the missing part of the diagram shows the tunnel IP is 192.168.1.1,
then you should choose an answer with tunnel IP address 192.168.1.3. in the remote site router i.e. C.
- assume the missing part of the diagram shows the tunnel IP is 192.168.1.3,
then you should choose an answer with tunnel IP address 192.168.1.1. in the remote site router i.e. B

The answer chosen here assume that the last octet of the tunnel source and tunnel IP are matched i.e. the missing part of the diagram is showing a tunnel IP

192.168.1.1.

QUESTION 317

How is a VPN forwarding table used in an MPLS network?

- A. learns VPN labels through LDP or RSVP
- B. learns PE route next hops through eBGP
- C. learns customer routes through MP-BGP from the connected CE
- D. stores VPN routes with associated labels

Correct Answer: D

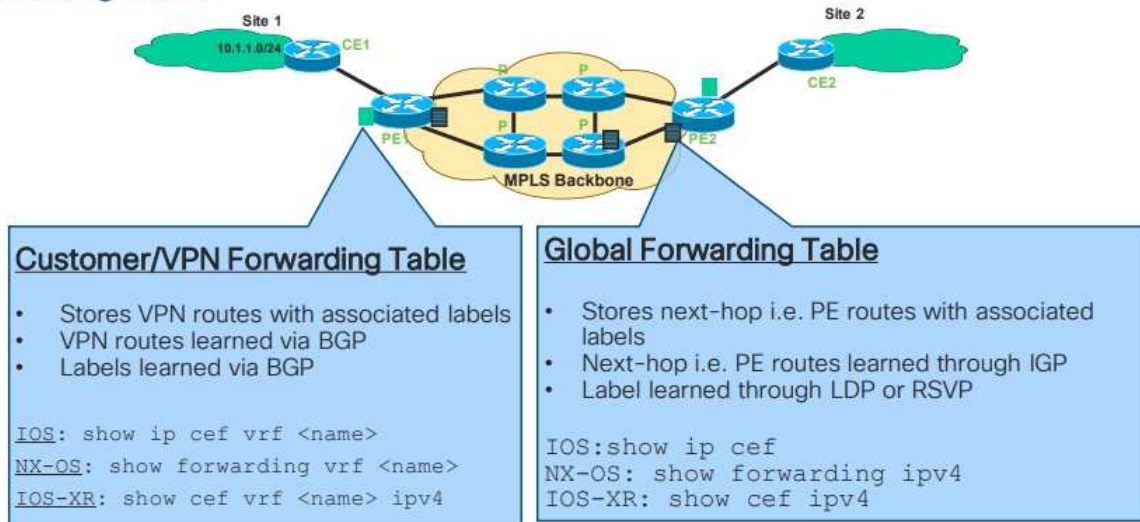
Section: (none)

Explanation

Explanation/Reference:

IP/VPN Technology Overview

Forwarding Plane



QUESTION 318

Drag and drop the IPv6 first hop security device roles from the left onto the corresponding descriptions on the right.

Select and Place:

host	Receives router advertisements from valid routers, and no router solicitation are received.
router	Receives router solicitation and sends router advertisements.
monitor	Receives valid and rogue router advertisements and all router solicitation.
switch	Received router advertisements are trusted and are flooded to synchronize states.

Correct Answer:

	router
	host
	monitor
	switch

Section: (none)

Explanation

Explanation/Reference:

For RA-guard, the device roles are:

"router"

Interface / Port that connects to the router. This interface / port allows incoming (i.e. can receive) RA packets from the router.

"host"

Interface / Port that connects to the other devices e.g. PC. This interface / port allows incoming (i.e. can receive) RS packets from those devices. It can send RA packets out to those devices but do not allow incoming RA packets from those devices

"switch"

behaves similar to the device-role "host" and usually use for a trunk port.

"monitor"

This device monitors network traffic. It behaves similar to the device-role host, except that RS packets are also sent out to the connected device for capturing traffic.

QUESTION 319

Refer to Exhibit. PC2 is directly connected to R1. A user at PC2 cannot Telnet to 2001:db8:a:b::10. The user can ping 2001:db8:a:b::10 and receive DHCP-related information from the DHCP server. Which action resolves the issue?

```
R1#sh ipv6 access-list GUARD
IPv6 access list GUARD
deny tcp any host 2001:DB8:A:B::10 eq telnet (6 matches) sequence 10
permit tcp host 2001:DB8:A:A::20 host 2001:DB8:A:B:10 eq telnet sequence 20
permit tcp host 2001:DB8:A:A::2 host 2001:DB8:D::1 eq www sequence 30
permit ipv6 2001:DB8:A:A::/64 any (67 matches) sequence 40
```

- A. Remove sequence 10 and put it back as sequence 25.
- B. Remove sequence 20 and put it back as sequence 45.
- C. Remove sequence 30 and put it back as sequence 5.
- D. Remove sequence 40 and put it back as sequence 15.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Seems some information is missing. We can only assume that PC2 has an IPv6 address 2001:DB8:a:A::20.

The current IPv6 ACL will block all telnet to 2001:DB8:a:A::10 due to sequence 10.

Although the sequence 20 should allow PC2 to telnet to 2001:DB8:a:A::10, the traffic was blocked by sequence 10 without matching sequence 20.

QUESTION 320

Refer to the exhibit. The remote server is failing to receive the NetFlow data Which action resolves the issue?

```

config t
flow record v4_r1
match ipv4 tos
match ipv4 protocol
match ipv4 source address
match ipv4 destination address
match transport source-port
match transport destination-port
collect counter bytes long
collect counter packets long
!
flow exporter EXPORTER-1
destination 172.16.10.2
transport udp 2055
exit
flow monitor FLOW-MONITOR-1
exporter EXPORTER-1
record v4_r1
exit
!
flow monitor v4_r1
!
ip cef
!
interface Ethernet0/0.1
ip address 172.16.6.2 255.255.255.0
ip flow monitor v4_r1 input
!

```

- A. Modify the flow transport command transport udp 2055 to move under flow monitor profile.
- B. Modify the interface command to ip flow monitor FLOW-MONITOR-1 Input.
- C. Modify the udp port under flow exporter profile to ip transport udp 4739.
- D. Modify the flow record command record v4_r1 to move under flow exporter profile.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

The existing configuration applies an empty flow monitor.

QUESTION 321

What is a function of an end device configured with DHCPv6 guard?

- A. If it is configured as a server, only prefix assignments are permitted.
- B. If it is configured as a relay agent, only prefix assignments are permitted.
- C. If it is configured as a client, messages are switched regardless of the assigned role.
- D. If it is configured as a client, only DHCP requests are permitted.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 322

The network administrator configured the router for Control Plane Policing to limit OSPF traffic to be policed to 1 Mbps. Any traffic that exceeds this limit must also be allowed at this point for traffic analysis. The router configuration is:

```

access-list 100 permit ospf any any
!
class-map CM-OSPF
match access-group 100
!
policy-map PM-COPP

```

```

class CM-OSPF
police 1000000 conform-action transmit
!
control-plane
service-policy output PM-COPP

```

The Control Plane Policing failed to monitor and police OSPF traffic. Which configuration resolves this issue?

- A. policy-map PM-COPP


```

class CM-OSPF
no police 1000000 conform-action transmit
police 1000000 conform-action transmit exceed-action transmit
!
control-plane
no service-policy output PM-COPP

```
- B. policy-map PM-COPP


```

class CM-OSPF
no police 1000000 conform-action transmit
police 1000000 conform-action transmit exceed-action transmit

```
- C. control-plane


```

no service-policy output PM-COPP
service-policy input PM-COPP

```
- D. no access-list 100


```

access-list 100 deny ospf any any
access-list 100 permit ip any any
!
policy-map PM-COPP
class CM-OSPF
no police 1000000 conform-action transmit
police 1000000 conform-action transmit exceed-action drop
!
control-plane
no service-policy output PM-COPP
service-policy input PM-COPP

```

Correct Answer: B

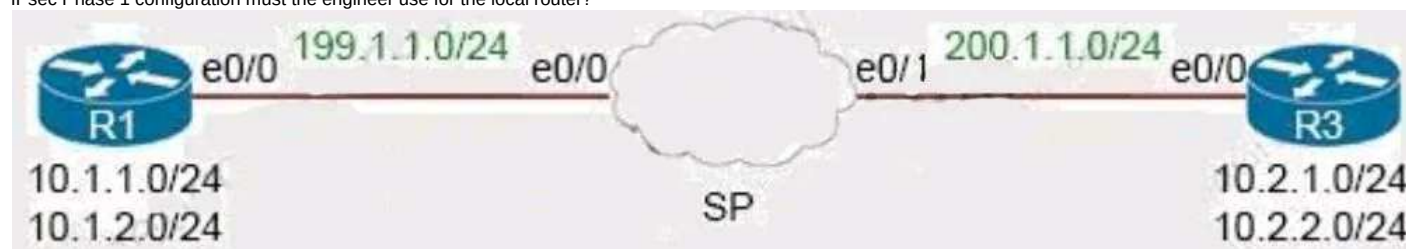
Section: (none)

Explanation

Explanation/Reference:

QUESTION 323

Refer to the exhibit. An engineer must configure a LAN-to-LAN IPsec VPN between R1 and the remote router. Assume that the remote router is using SHA, Which IPsec Phase 1 configuration must the engineer use for the local router?



- A. crypto isakmp policy 5


```

authentication pre-share
encryption 3des
hash sha
group 2
!
crypto isakmp key cisco123 address 200.1.1.3

```
- B. crypto isakmp policy 5


```

authentication pre-share
encryption 3des
hash md5
group 2
!
crypto isakmp key cisco123 address 200.1.1.3

```
- C. crypto isakmp policy 5


```

authentication pre-share
encryption 3des
hash md5
group 2
!
crypto isakmp key cisco123 address 199.1.1.1

```
- D. crypto isakmp policy 5


```

authentication pre-share
encryption 3des
hash md5
group 2
!
crypto isakmp key cisco123! address 199.1.1.1

```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 324

A CoPP policy is applied for receiving SSH traffic from the WAN interface on a Cisco ISR4321 router. However, the SSH response from the router is abnormal and stuck during the high link utilization. The problem is identified as SSH traffic does not match in the ACL. Which action resolves the issue?

- A. Rate-limit SSH traffic to ensure dedicated bandwidth.
- B. Apply CoPP on the control plane interface.
- C. Increase the IP precedence value of SSH traffic to 6.
- D. Apply CoPP on the WAN interface inbound direction.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

CoPP policy should apply to control plane not WAN interface.

QUESTION 325

How are MPLS Layer 3 VPN services deployed?

- A. The RD and RT values must match under the VRF
- B. The RD and RT values under a VRF must match on the remote PE router
- C. The import and export RT values under a VRF must always be the same.
- D. The label switch path must be available between the local and remote PE routers.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 326

Which feature minimizes DoS attacks on an IPv6 network?

- A. IPv6 Binding Security Table
- B. IPv6 Router Advertisement Guard
- C. IPv6 Prefix Guard
- D. IPv6 Destination Guard

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 327

Refer to the exhibit. An administrator is configuring a GRE tunnel to establish an EIGRP neighbor to a remote router. The other tunnel endpoint is already configured. After applying the configuration as shown, the tunnel started flapping. Which action resolves the issue?

```

R2(config)# int tun0
*Jun 23 00:42:06.179: %LINEPROTO-5-UPDOWN: Line protocol on
Interface Tunnel0, changed state to down

R2(config-if)# ip address 192.168.12.2 255.255.255.0
R2(config-if)# tunnel source lo0
R2(config-if)# tunnel destination 10.255.255.1

*Jun 23 00:42:15.845: %LINEPROTO-5-UPDOWN: Line protocol on
Interface Tunnel0, changed state to up

R2(config-if)# router eigrp E
R2(config-router)# address-family ipv4 autonomous-system 1
R2(config-router-af)# net 192.168.12.2 0.0.0.0

*Jun 23 00:43:05.730: %DUAL-5-NBRCHANGE: EIGRP-IPv4 1: Neighbor
192.168.12.1 (Tunnel0) is up: new adjacency
*Jun 23 00:43:05.993: %ADJ-5-PARENT: Midchain parent maintenance
for IP midchain out of Tunnel0 - looped chain attempting to stack
*Jun 23 00:43:15.193: %TUN-5-RECURDOWN: Tunnel0 temporarily
disabled due to recursive routing

*Jun 23 00:43:15.193: %LINEPROTO-5-UPDOWN: Line protocol on
Interface Tunnel0, changed state to down

```

- A. Modify the network command to use the Tunnel0 interface netmask
- B. Advertise the Loopback0 interface from R2 across the tunnel
- C. Stop sending a route matching the tunnel destination across the tunnel
- D. Readdress the IP network on the Tunnel0 on both routers using the /31 netmask

Correct Answer: C

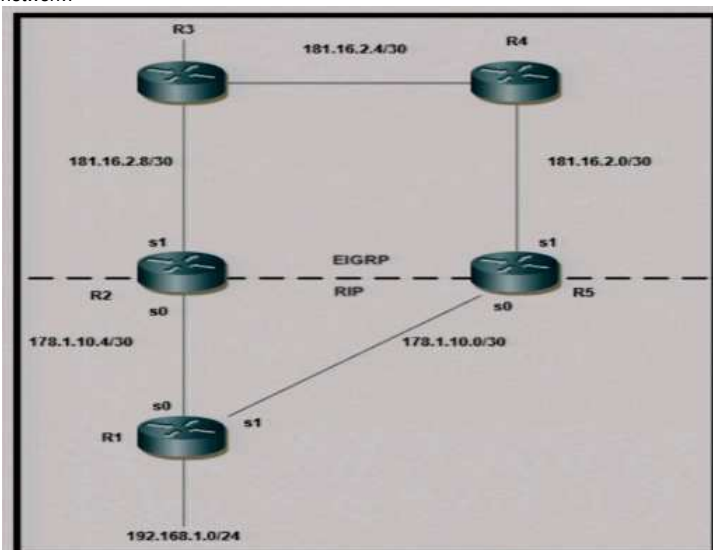
Section: (none)

Explanation

Explanation/Reference:

QUESTION 328

Refer to the exhibit Mutual redistribution is enabled between RIP and EIGRP on R2 and R5. Which configuration resolves the routing loop for the 192.1631 .024 network?



A. R2:
router eigrp 10
network 181.16.0.0
redistribute rip metric 1 1 1 1 1
distribute-list 1 in s1
!
router rip
network 178.1.0.0
redistribute eigrp 10 metric 2
!
access-list 1 deny 192.168.1.0
access-list 1 permit any

R5:
router eigrp 10
network 181.16.0.0
redistribute rip metric 1 1 1 1 1
distribute-list 1 in s0
!
router rip
network 178.1.0.0
redistribute eigrp 10 metric 2
!
access-list 1 deny 192.168.1.0
access-list 1 permit any

B. R2:
router eigrp 7
network 181.16.0.0
redistribute rip metric 1 1 1 1 1
distribute-list 1 in s1
!
router rip
network 178.1.0.0
redistribute eigrp 10 metric 2
!
access-list 1 deny 192.168.1.0
access-list 1 permit any

R5:
router eigrp 7
network 181.16.0.0
redistribute rip metric 1 1 1 1 1
distribute-list 1 in s1
!
router rip
network 178.1.0.0
redistribute eigrp 10 metric 2
!
access-list 1 deny 192.168.1.0
access-list 1 permit any

C. R2:
router eigrp 10
network 181.16.0.0
redistribute rip metric 1 1 1 1 1
distribute-list 1 in s0
!
router rip
network 178.1.0.0
redistribute eigrp 10 metric 2
!
access-list 1 deny 192.168.1.0
access-list 1 permit any

R5:
router eigrp 10
network 181.16.0.0
redistribute rip metric 1 1 1 1 1
distribute-list 1 in s1
!
router rip
network 178.1.0.0
redistribute eigrp 10 metric 2
!
access-list 1 deny 192.168.1.0
access-list 1 permit any

D. R2:
router eigrp 10
network 181.16.0.0
redistribute rip metric 1 1 1 1 1
distribute-list 1 in s0
!
router rip
network 178.1.0.0
redistribute eigrp 10 metric 2
!
access-list 1 deny 192.168.1.0
access-list 1 permit any

R5:
router eigrp 10
network 181.16.0.0
redistribute rip metric 1 1 1 1 1
distribute-list 1 in s0

```

!
router rip
network 178.1.0.0
redistribute eigrp 10 metric 2
!
access-list 1 deny 192.168.1.0
access-list 1 permit any

```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

One of the way to avoid loop is to prevent R2 or R5 from learning 192.168.1.0/24 from EIGRP routers R3/R4.

Therefore both R2 and R5 should filter the route in the inbound direction through their s1 interface.

Since no mentioned about EIGRP AS number, the AS number can be any as long as both R2 and R5 has the same AS number.

QUESTION 329

```

access-list 1 permit 219.65.100.15
access-list 2 permit 219.65.100.16
!
interface ethernet 1
ip policy route-map Texas
!
route-map Texas permit 10
match ip address 1
set ip precedence priority
set ip next-hop 219.65.100.17
!
route-map Texas permit 20
match ip address 2
set ip next-hop 219.65.100.18

```

Refer to the exhibit. Packets arriving from source 219.65.100.15 must be sent with the precedence bit set to 1, and packets arriving from source 219.65.100.16 must be sent with the precedence but set to 5. Which action resolves the issue?

- A. set ip precedence critical in route-map Texas permit 20
- B. set ip precedence immediate in route-map Texas pennit 10
- C. set ip precedence critical in route-map Texas permit 10
- D. set ip precedence priority in route-map Texas permit 20

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

routine	Match packets with routine precedence (0)
priority	Match packets with priority precedence (1)
immediate	Match packets with immediate precedence (2)
flash	Match packets with flash precedence (3)
flash-override	Match packets with flash override precedence (4)
critical	Match packets with critical precedence (5)
internet	Match packets with internetwork control precedence (6)
network	Match Packets with network control precedence (7)

from source 219.65.100.15 -> precedence bit set to 1 : already configured properly

QUESTION 330

- C 192.168.10.0/24 is directly connected, Serial1/0/0
- 172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
- C 172.16.160.0/19 is directly connected, Loopback1
- C 172.16.128.0/19 is directly connected, Loopback2
- C 172.16.224.0/19 is directly connected, Loopback3
- C 172.16.192.0/19 is directly connected, Loopback4
- D 172.16.0.0/16 is a summary, 00:01:23, Null0

Refer to the above routinig entries in R2. An engineer must configure EIGRP between R1 and R2 with no summary route. Which configuration resolves the issue?

- A. R1(config)#router eigrp 1
R1(config-router)#no auto-summary
- B. R2(config)#router eigrp 1
R2(config-router)#no auto-summary
- C. R1(config)#router eigrp 1
R1(config-router)#auto-summary
- D. R2(config)#router eigrp 1
R2(config-router)#auto-summary

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since R2 has a EIGRP route 172.16.0.0/16 to Null0, R2 is probably configured with EIGRP auto-summary.

R1 will probably learn this summarized route with R2 as next hop.

QUESTION 331

```

R2#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address                      Interface      Hold Uptime    SRTT    RTO  Q  Seq
                               (sec)          (ms)                Cnt Num
1   192.168.10.1                  Ser1/0         12 00:00:39    1    5000  2    0
*Jan  1 15:40:21.295: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 192.168.10.1 (Serial1/0) is down: retry limit exceeded
*Jan  1 15:40:51.567: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 192.168.10.1 (Serial1/0) is up: new adjacency
*Jan  1 15:42:11.107: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 192.168.10.1 (Serial1/0) is down: retry limit exceeded
*Jan  1 15:42:14.879: %DUAL-5-NBRCHANGE: IP-EIGRP(0) 100: Neighbor 192.168.10.1 (Serial1/0) is up: new adjacency

R1#show ip eigrp neighbors
IP-EIGRP neighbors for process 100

```

```

R1 Configuration:
key chain cisco
key 2
  key-string abc
!
interface Loopback0
ip address 10.10.1.1 255.255.255.0
!
interface Serial1/0
ip address 192.168.10.1 255.255.255.0
ip authentication mode eigrp 100 md5
ip authentication key-chain eigrp 100 cisco
serial restart-delay 0
!
router eigrp 100
network 10.10.1.0 0.0.0.255
network 192.168.10.0
no auto-summary

```

```

R2 configuration:
key chain cisco
key 1
  key-string 123
key 2
  key-string abc
!
interface Loopback0
ip address 10.10.2.2 255.255.255.0
!
interface Serial1/0
ip address 192.168.10.2 255.255.255.0
ip authentication mode eigrp 100 md5
ip authentication key-chain eigrp 100 cisco
no fair-queue
!
router eigrp 100
network 10.10.2.0 0.0.0.255
network 192.168.10.0
no auto-summary

```

Refer to the exhibit. R1 and R2 are configured for EIGRP peering using authentication and the neighbors failed to come up. Which action resolves the issue?

- A. Configure a matching authentication type on both routers.
- B. Configure a matching key-chain name on both routers.
- C. Configure a matching key-id number on both routers.
- D. Configure a matching lowest key-id on both routers.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

A router will use the key with lowest ID for sending.

- R1 sends key ID 2 and messages authenticated with "abc" to R2, R2 therefore uses the string "abc" of key ID 2 for authentication. Since pass string matches, it is successful.

- R2 sends key ID 1 and messages authenticated with "123" to R1, R1 cannot find any string of key ID 1 for authentication. The packet is rejected.

QUESTION 332

Which method provides failure detection in BFD?

- A. short duration, low overhead
- B. long duration, low overhead
- C. long duration, high overhead
- D. short duration, high overhead

Correct Answer: A

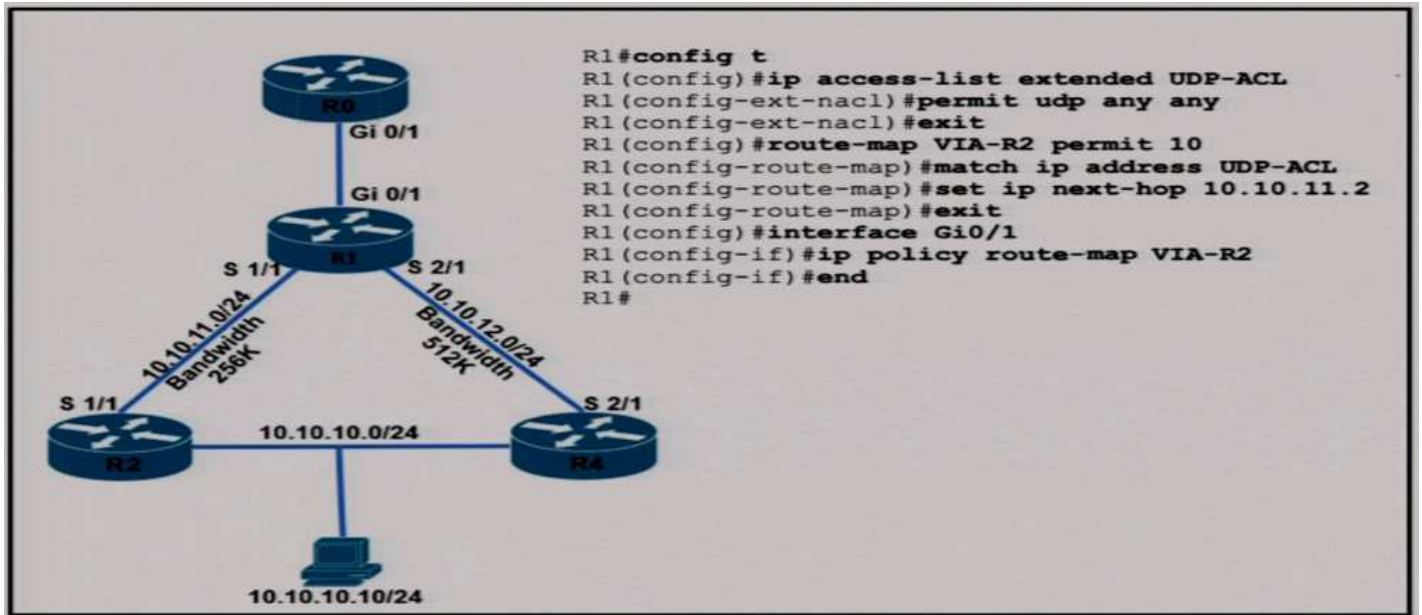
Section: (none)

Explanation

Explanation/Reference:

BFD provides a low-overhead, short-duration method of detecting failures in the forwarding path between two adjacent routers, including the interfaces, data links, and forwarding planes.

QUESTION 333



Refer to the exhibit. TCP traffic should be reaching host 10.10.10.10/24 via R2 (left of the diagram). Which action resolves the issue?

- A. TCP traffic will reach the destination via R2 without any changes.
- B. Set IP next.hop to 10.10.12.2 under the route-map permit 10 to allow TCP traffic.
- C. Allow TCP in the access list with no changes to the route map.
- D. Add a permit 20 statement in the route map to allow TCP traffic.

Correct Answer: C

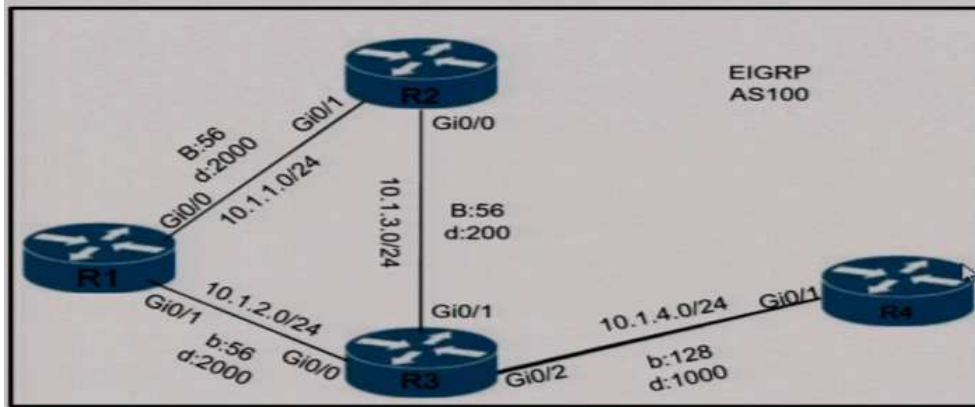
Section: (none)

Explanation

Explanation/Reference:

At present, R1 (the router will s1/1 and s2/1) will send UDP traffic received from R0 (the top router) to R2 by PBR. For including TCP traffic, you should add permit statement for "tcp" into the access list.

QUESTION 334



Refer to the exhibit A loop occurs between R1, R2, and R3 while EIGRP is run with poison reverse enabled. Which action prevents the loop between R1, R2, and R3?

- A. Configure route filtering
- B. Configure route tagging
- C. Configure R3 as stub receive-only.
- D. Enable split horizon

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

B is wrong since route tagging is used when redistribution is involved.

C is wrong since others will not received routes learnt by R3

D is wrong since one interface is split horizon is used when partially meshed network is involved.

QUESTION 335

A network administrator added a new spoke site with dynamic IP on the DMVPN network Which configuration command passes traffic on the DMVPN tunnel from the spoke router?

- A. ip nhrp registration no-unique
- B. ip nhrp registration no-registration
- C. ip nhrp registration ignore
- D. ip nhrp registration dynamic

Correct Answer: A

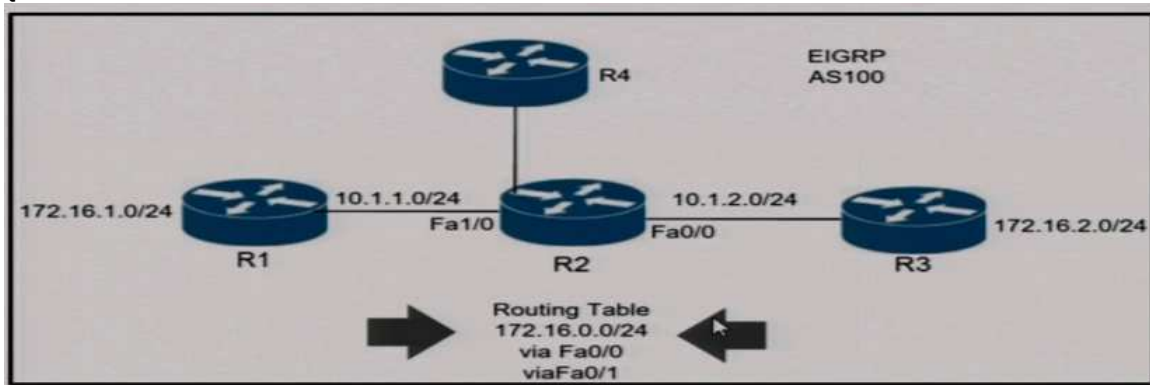
Section: (none)

Explanation

Explanation/Reference:

A allows the NHS to accept new physical / NBMA IP address from the Spoke router after its dynamic IP address changes.

QUESTION 336



Refer to the exhibit R4 is experiencing packet drop when trying to reach 172.16.2.7 behind R2. Which action resolves the issue?

- A. Enable auto summarization on all three routers R1 , R2. and R3.
- B. Insert a /16 floating static route on R2 toward R3 with metric 254.
- C. Disable auto summarization on R2
- D. Insert a /24 floating static route on R2 toward R3 with metric 254.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

R1 and R3 seems to enable auto-summarization and both send 172.16.0.0/16 to R2.

Although then disabling auto-summarization in R2 and/or R3, you can add a 172.16.2.7/24 route towards R3.

Whether it is floating or not is not important since only one such route exists.

QUESTION 337

```
ipv6 dhcp pool DHCPPOOL
address prefix 2001:0:1:4::/64 lifetime infinite infinite

interface FastEthernet0/0
ip address 10.0.0.1 255.255.255.240
duplex auto
speed auto
ipv6 address 2001:0:1:4::1/64
ipv6 enable
ipv6 nd ra suppress
ipv6 ospf 1 area 1
ipv6 dhcp server DHCPPOOL
```

Refer to the exhibit Reachability between servers in a network deployed with DHCPv6 is unstable. Which command must be removed from the configuration to make DHCPv6 function?

- A. ipv6 address 2001:0:1:4::1/64
- B. ipv6 dhcp server DHCPPOOL
- C. ipv6 nd ra suppress
- D. address prefix 2001:0:1:4::1/64 lifetime infinite infinite

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

For some nodes, they need to receive RA with M flag set to 1 from the router to tell them to use DHCPv6.

QUESTION 338

An engineer failed to run diagnostic commands on devices using Cisco DNA Center. Which action in Cisco DNA Center resolve the issue?

- A. Enable APIs
- B. Enable Secure Shell
- C. Enable CDP.
- D. Enable Command Runner.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Command Runner lets you run diagnostic CLI commands on selected devices and view the resulting command output.

QUESTION 339



Refer to the exhibit. An engineer is trying to add an encrypted user password that should not be visible in the router configuration. Which two configuration commands resolve the issue? (Choose two.)

- A. username Admin password 5 Cisco@123
- B. username Admin password Cisco@123
- C. password encryption aes
- D. username Admin secret Cisco@123
- E. no service password-encryption
- F. service password-encryption

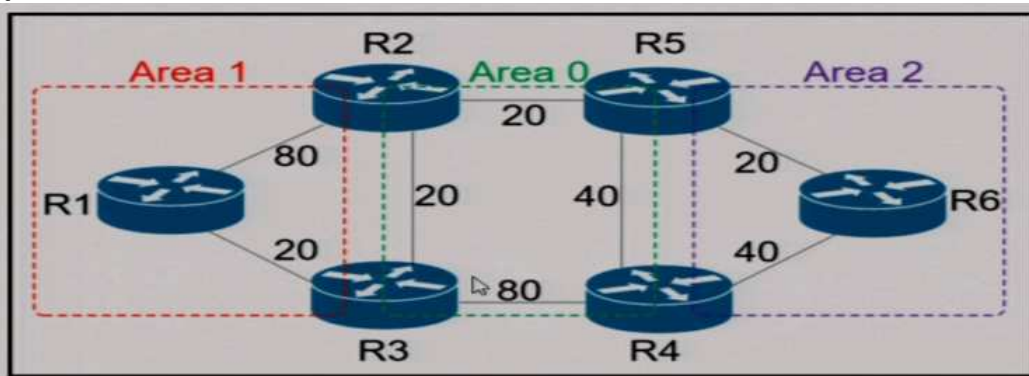
Correct Answer: DF

Section: (none)

Explanation

Explanation/Reference:

QUESTION 340



Refer to the exhibit. R6 should reach R1 via R5>R2>R1. Which action resolves the issue?

- A. Decrease the cost to 2 between R6-R5-R2
- B. Increase the cost to 61 between R2-R3-R1.
- C. Increase the cost to 61 between R2 and R3
- D. Decrease the cost to 41 between R2 and R1.

Correct Answer: C

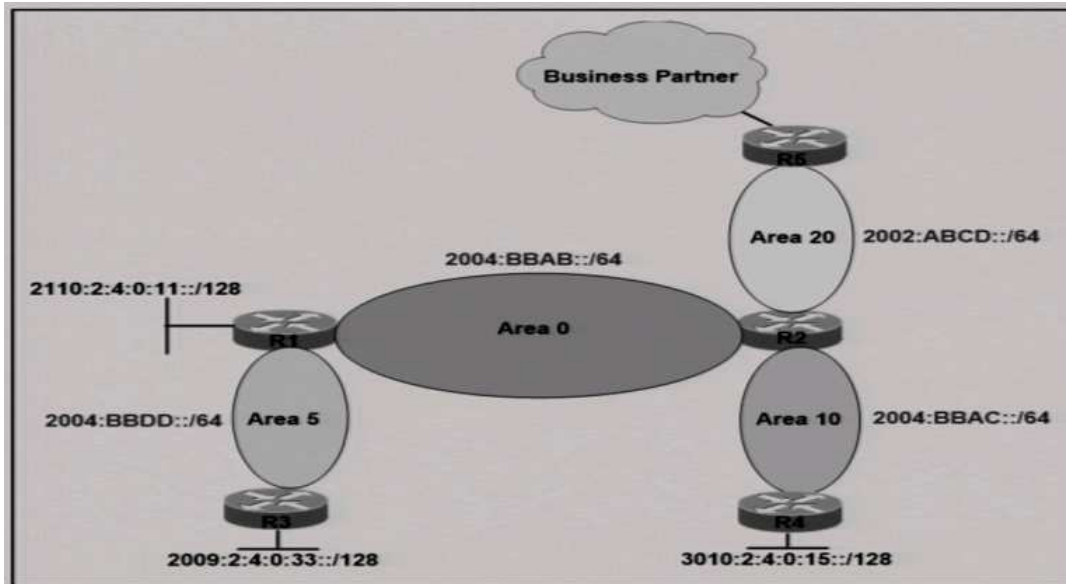
Section: (none)

Explanation

Explanation/Reference:

In existing configuration, R6 will send traffic to R1 through shortest path R6>R5>R2>R3>R1. Increase the cost between R2 and R3 will cause R2 to send to R1 without going through R3.

QUESTION 341



```
R2#sh ipv6 route ospf
O 2002:ABCD::/64 [110/1]
  via FastEthernet0/1, directly connected
O 2004:BBAB::/64 [110/1]
  via FastEthernet0/0, directly connected
O 2004:BBAC::/64 [110/1]
  via FastEthernet1/0, directly connected
O 3010:2:4:0:15::/128 [110/1]
  via FE80::C804:1DFF:FE20:8, FastEthernet0/0
```

Refer to the exhibit. A network engineer applied a filter for LSA traffic on OSPFv3 interarea routes on the area 5 ABR to protect advertising the internal routes of area 5 to the business partner network. All other areas should receive the area 5 internal routes. After the respective route filtering configuration is applied on the ABR, area 5 routes are not visible on any of the areas. How must the filter list be applied on the ABR to resolve this issue?

- A. in the "out" direction for area 20 on router R2
- B. in the "out" direction for area 5 on router R1
- C. in the "in" direction for area 20 on router R2
- D. in the "in" direction for area 5 on router R1

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 342

```
ip sla 10
  icmp-echo 10.1.1.10
  timeout 2000
  threshold 2000
  frequency 40
ip sla schedule 10 life forever start-time now
!
track 1 ip sla 10 reachability
```

Refer to the exhibit. An engineer configured IP SLA to monitor a next hop on a router for reachability. When the next hop is unreachable, the router is executing tracking and failing over another route, but packet loss is experienced because the reachability is flapping. Which action resolves the issue?

- A. Increase the frequency of the sla probe to 60.
- B. Increase the timeout of the sla probe to 6000.
- C. Append delay up 0 down 0 to the track command.
- D. Append delay up 60 down 60 to the track command.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 343

```

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

D    10.0.0.0/8 [90/409600] via 172.16.1.200, 00:00:28, Ethernet0/0
D    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.16.1.0/24 is directly connected, Ethernet0/0
L      172.16.1.100/32 is directly connected, Ethernet0/0
D    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/24 is directly connected, Loopback0
L      192.168.1.100/32 is directly connected, Loopback0
R1#

```

Refer to the exhibit. The R2 loopback interface is advertised with RIP and EIGRP using default values. Which configuration changes make R1 reach the R2 loopback using RIP?

- A. R1(config)#router rigrp 1
R1(config-router)#distance eigrp 120 120
- B. R1(config)#router rip
R1(config-router)#distance 100
- C. R1(config)#router oigrp 1
R1(config-router)#distance eigrp 130 120
- D. R1(config)#router rip
R1(config-route)#distrance 90

Correct Answer: C

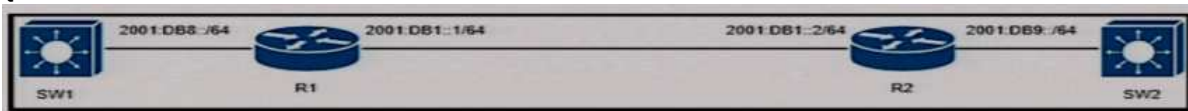
Section: (none)

Explanation

Explanation/Reference:

RIP (default 120) needs to have an administrative distance smaller than EIGRP (default 90). Choice C sets AD for internal and external EIGRP to 130 and 120 respectively.

QUESTION 344



Refer to the exhibit. An engineer must advertise routes into IPv6 MP-BGP and failed. Which configuration resolves the issue on R1?

- A. router bgp 64900
no bgp default ipv4-unicast
address-family ipv6 unicast
redistribute ospf
network 2001:DB9::/64
- B. router bgp 65000
no bgp default ipv4-unicast
address-family ipv6 multicast
network 2001:DB8::/64
- C. router bgp 65000
no bgp default ipv4-unicast
address-family ipv6 unicast
network 2001:DB8::/64
- D. router bgp 64900
no bgp default ipv4-unicast
address-family ipv6 multicast
neighbor 2001:DB8:7000::2 translate-update ipv6 multicast

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 345

```

crypto isakmp policy 1
 authentication pre-share
 crypto isakmp key cisco47 address 0.0.0.0
!
crypto ipsec transform-set trans2 esp-des esp-md5-hmac
 mode transport
!
crypto ipsec profile vpnprof
 set transform-set trans2
!
interface Tunnel0
 bandwidth 1000
 ip address 10.0.0.1 255.255.255.0
 ip mtu 1400
 ip nhrp authentication donttell
 ip nhrp map multicast dynamic
 ip nhrp network-id 99
 ip nhrp holdtime 300
 no ip split-horizon eigrp 1
 ip tcp adjust-mss 1360
 delay 1000
 tunnel source GigabitEthernet 0/0/0
 tunnel mode gre multipoint
 tunnel key 100000
 tunnel protection ipsec profile vpnprof
!
interface FastEthernet0/0/0
 ip address 172.17.0.1 255.255.255.0
!
interface FastEthernet0/0/1
 ip address 192.168.0.1 255.255.255.0
!
router eigrp 1
 network 10.0.0.0 0.0.0.255
 network 192.168.0.0 0.0.0.255
!

```

Refer to the exhibit. A network administrator must configure DMVPN tunnels between the hub and spoke with dynamic spoke-to-spoke tunnel capabilities using EIGRP. Which tunnel interface command must the network administrator configure to establish an EIGRP peer?

- A. no ip next-hop-self eigrp 1
- B. no ip nhrp next-hop-self
- C. ip nhrp next-hop-self
- D. ip next-hop-self eigrp 1

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 346

```

access-list 120 permit tcp any gt 1024 eq bgp log
access-list 120 permit tcp any eq bgp gt 1024 established
access-list 120 permit tcp any gt 1024 eq 639
access-list 120 permit tcp any eq 639 gt 1024 established
access-list 120 permit tcp any eq 646
access-list 120 permit udp any eq 646
access-list 120 permit ospf any
access-list 120 permit ospf any host 224.0.0.5
access-list 120 permit ospf any host 224.0.0.6
access-list 120 permit eigrp any
access-list 120 permit eigrp any host 224.0.0.10
access-list 120 permit udp any any eq pim-auto-rp

```

Refer to the exhibit. The control plane is heavily impacted after the above CoPP configuration is applied to the router. Which command removal lessens the impact on the control plane?

- A. access-list 120 permit udp any any eq pim-auto-rp
- B. access-list 120 permit tcp any gt 1024 eq bgp log
- C. access-list 120 permit ospf any
- D. access-list 120 permit eigrp any host 224.0.0.10

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Assume that all entries with missing destination is "any" e.g.:

```

access-list 120 permit ospf any any
access-list 120 permit eigrp any any

```

Then the followings are redundant.

```

access-list 120 permit ospf any host 224.0.0.5
access-list 120 permit ospf any host 224.0.0.6
access-list 120 permit eigrp any host 224.0.0.10

```

QUESTION 347

Which two components are required for MPLS Layer 3 VPN configuration? (Choose two.)

- A. Use a unique RD per customer VRF
- B. Use OSPF between PE and CE
- C. Use MP-BGP for customer routes
- D. Use pseudowire for Layer 2 routes
- E. Use LDP for customer routes

Correct Answer: AC

Section: (none)

Explanation

Explanation/Reference:

QUESTION 348

A customer reports that traffic is not passing on an EIGRP enabled multipoint interface on a router configured as below:

```
interface Serial0/0/0
  no ip address

interface Serial0/0/0.9 multipoint
  ip address 10.1.1.1 255.255.255.248
  ip split-horizon eigrp 1
```

Which action resolves the issue?

- A. Disable poison reverse.
- B. Enable split horizon
- C. Enable poison reverse
- D. Disable split horizon

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 349

```
snmp-server community Public RO 90
snmp-server community Private RW 90
R1#show access-list 90
Standard IP access list 90
 permit 10.11.110.11
 permit 10.11.111.12

Nov 6 06:45:11: %SNMP-3-AUTHFAIL: Authentication failure for SNMP req from host
10.11.110.12
Nov 6 06:45:12: %SNMP-3-AUTHFAIL: Authentication failure for SNMP req from host
10.11.110.12
```

Refer to the exhibit A network administrator notices these console messages from host 10.11.110.12 originated from interface E1/0. The administrator considers this an unauthorized attempt to access SNMP on R1 . Which action prevents the attempts to reach R1 E1/0?

- A. Configure IOS control plane protection using ACL 90 on interface E1/0
- B. Add a permit statement including the host 10.11.110.12 into ACL 90
- C. Configure IOS management plane protection using ACL 90 on interface E1/0.
- D. Create an inbound ACL on interface E1/0 to deny SNMP from host 10.11.110.12

Correct Answer: D

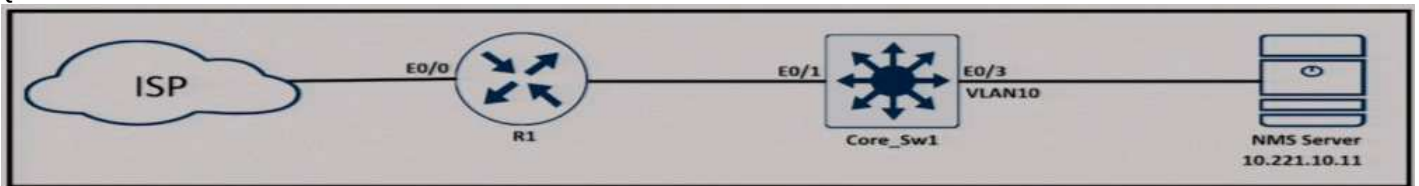
Section: (none)

Explanation

Explanation/Reference:

Control / Management plane protection is configured through control-plane

QUESTION 350



Refer to the exhibit. During ISP router maintenance, the network produced many alerts because of the flapping interface. Which configuration on R1 resolves the issue?

- A. sump trap ip verify drop-rate
- B. ip verify drop-rate notify hold-down 60
- C. no snmp trap link-status
- D. snmp trap link-status down

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 351


```

CPE# ping 10.0.2.4
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.2.4, timeout is 2
seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/1/1 ms
CPE# copy flash:/packages.conf tftp://10.0.2.4/
Address or name of remote host [10.0.2.4]?
Destination filename [packages.conf]?
%Error opening tftp://10.0.2.4/packages.conf (Undefined error)

```

Refer to the exhibit. The administrator is trying to overwrite an existing file on the TFTP server that was previously uploaded by another router. However, the attempt to update the file fails. Which action resolves this issue?

- A. Make the package.conf file executable by all on the TFTP server.
- B. Make sure to run the TFTP service on the TFTP server.
- C. Make the TFTP folder writable by all on the TFTP server.
- D. Make the package.conf file writable by all on the TFTP server.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Since the question mentioned that the file is an existing file on the TFTP server, then the file should be writable.

Remarks:

If the file is a new file, the TFTP folder has to be writable in order to create a new file in the folder.

If the TFTP service is not running, the error "(Timed out)" is usually shown.

QUESTION 352

A newly installed spoke router is configured for DMVPN with the ip mtu 1400 command. Which configuration allows the spoke to use fragmentation with the maximum negotiated TCP MTU over GRE?

- A. ip tcp payload-mtu 1360
crypto ipsec fragmentation after-encryption
- B. ip tcp adjust-mtu 1360
crypto ipsec fragmentation mtu-discovery
- C. ip tcp payload-mtu 1360
crypto ipsec fragmentation mtu-discovery
- D. ip tcp adjust-mtu 1360
crypto ipsec fragmentation after-encryption

Correct Answer: D

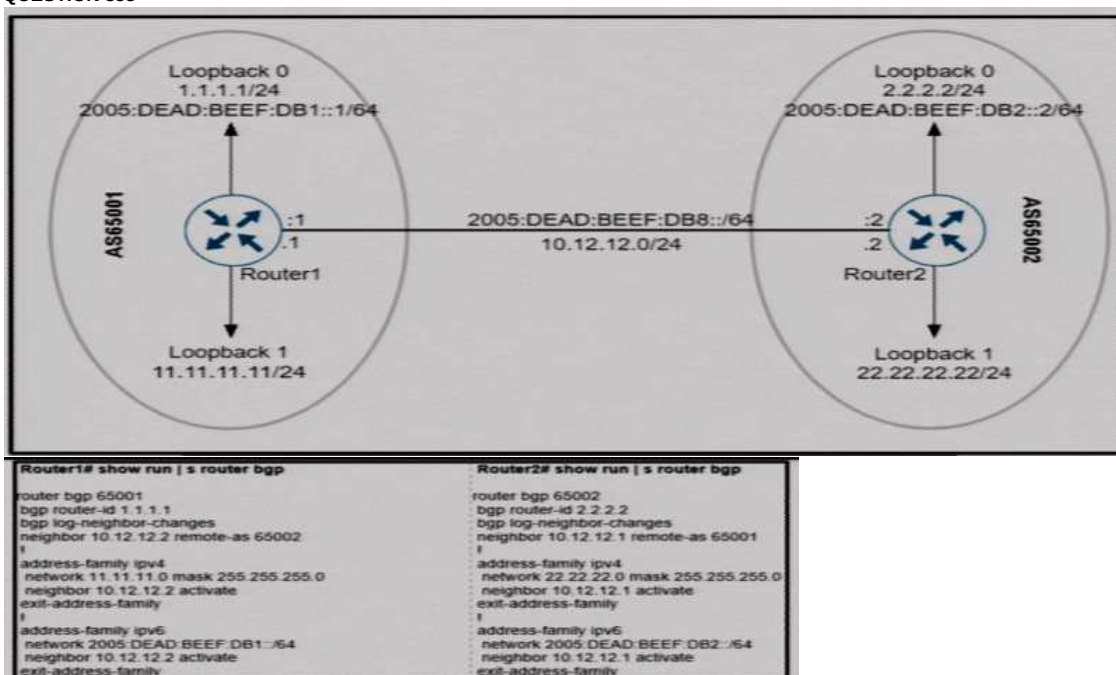
Section: (none)

Explanation

Explanation/Reference:

Only D is the combination for two valid commands.

QUESTION 353



```

Router1# show bgp ipv6 unicast
BGP table version is 4, local router ID is 1.1.1.1
Status codes: s - suppressed, d - damped, h - history, * - valid, > - best, i - internal,
               r - RIB-failure, S - Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop      Metric LocPrf Weight Path
  -----
*> 2005:DEAD:BEEF:DB1::/64
  * 2005:DEAD:BEEF:DB2::/64
    .FFFF:10.12.12.2  0          0 65002 i

Router1# show bgp ipv6 unicast 2005:DEAD:BEEF:DB2::/64
BGP routing table entry for 2005:DEAD:BEEF:DB2::/64, version 0
Paths: (1 available, no best path)
Not advertised to any peer
65002
 .FFFF:10.12.12.2 (inaccessible) from 10.12.12.2 (2.2.2.2)
  Origin IGP, metric 0, localpref 100, valid, external

```

Refer to the exhibit. Which configuration allows Router1 to reach 2005:DEAD:BEEF:DB2::/64?

- A. Router1
- ```

route-map Router1V6 permit 10
 set ipv6 next-hop 2006:DEAD:BEEF:DB1::1
 exit
!
router bgp 65001
 address-family ipv6
 network 2005:DEAD:BEEF:DB1::/64
 neighbor 10.12.12.2 activate
 neighbor 10.12.12.2 route-map Router1V6 out
 exit-address-family

```
- B. Router2
- ```

route-map Router2V6 permit 10
 set ipv6 next-hop 2006:DEAD:BEEF:DB8::2
 exit
!
router bgp 65002
 address-family ipv6
  network 2005:DEAD:BEEF:DB2::/64
  neighbor 10.12.12.2 activate
  neighbor 10.12.12.2 route-map Router2V6 out
 exit-address-family

```
- C. Router2
- ```

route-map Router2V6 permit 10
 set ipv6 next-hop 2006:DEAD:BEEF:DB8::2
 exit
!
router bgp 65002
 address-family ipv6
 network 2005:DEAD:BEEF:DB2::/64
 neighbor 10.12.12.2 activate
 neighbor 10.12.12.2 route-map Router out
 exit-address-family

```
- D. Router1
- ```

route-map Router1V6 permit 10
 set ipv6 next-hop 2006:DEAD:BEEF:DB8::2
 exit
!
router bgp 65001
 address-family ipv6
  network 2005:DEAD:BEEF:DB1::/64
  neighbor 10.12.12.2 activate
  neighbor 10.12.12.2 route-map Router1V6 out
 exit-address-family

```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

2005:DEAD:BEEF:DB2::/64 is the loopback0 of Router2, R1 needs to send use R2 as nexthop to send packets to that network.

The above can be solved by :

- using route-map to add next-hop to the inbound route in R1
- using route-map to add next-hop to the outbound route in R2

Since all choices applies the route-map in out direction, R2 should be the router that requires the configuration.

C is wrong since the wrong route-map name is applied to the neighbor.

QUESTION 354

What are the two goals of micro BFD sessions? (Choose two)

- A. Each member link of a link aggregation group must run BFD
- B. Any member link on a link aggregation group must run BFD
- C. The high bandwidth member link of a link aggregation group must run BFD
- D. Continuity for each member link of a link aggregation group must be verified
- E. Run the BFD session with 3x3 ms hello timer.

Correct Answer: AD

Section: (none)

Explanation

Explanation/Reference:

Micro BFD sessions are supported for both LACP and non-LACP based-port channels. If BFD detects a fault in the bidirectional path between two forwarding engines that includes interfaces and data links, the member link is removed from the forwarding table. This mechanism delivers faster failure detection as the BFD sessions are created on individual port-channel interface.

The goal of micro BFD sessions are:

- Run BFD session over each LAG member link.

- Verify link continuity for each member link.
- Allow BFD to control the LAG member link to be part of the L2 load-balancing table of the LAG interface in the presence or absence of LACP.

Note that Micro BFD sessions are not supported on partial member links of bundled port-channel.

QUESTION 355

What action is performed for untagged outgoing labels in an MPLS router?

- A. Convert the incoming MPLS packet to an untagged packet and then do a FIB lookup.
- B. Convert the untagged packet to a labeled packet and forward it to the next router.
- C. Convert the incoming MPLS packet to an IP packet and forward it to the next router.
- D. Convert the incoming MPLS packet to an untagged packet and then do a RIB lookup.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

The following outlines the types of **outgoing labels** that can be associated with a packet:

Untagged – The incoming MPLS packet is converted to an IP packet and forwarded to the destination (MPLS to IP Domain transition).

Implicit-null or POP label – This label is assigned when the top label of the incoming MPLS packet is removed and the resulting MPLS or IP packet is forwarded to the next-hop downstream router.

Explicit-null Label – This label is assigned to preserve the EXP value of the top label of an incoming packet.

Aggregate – In this label, the incoming MPLS packet is converted to an IP packet (by removing all labels if label stack is found on incoming packet), and an FIB (CEF) lookup is performed to identify the outgoing interface to destination

QUESTION 356

What is a characteristic of an MPLS LSP tunnel?

- A. bidirectional tunnel
- B. unidirectional tunnel
- C. hop-by-hop tunnel
- D. secured bidirectional tunnel

Correct Answer: C

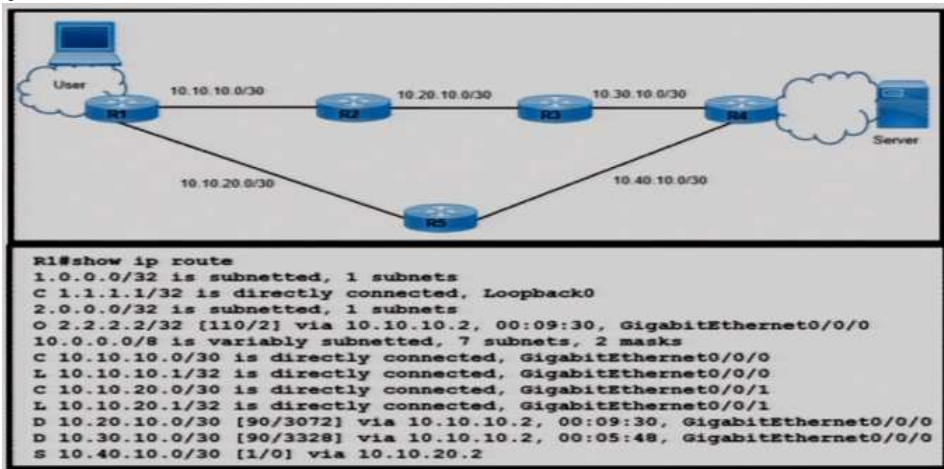
Section: (none)

Explanation

Explanation/Reference:

Due to the forwarding of packets through an LSP being opaque to higher network layers, an LSP is also sometimes referred to as an MPLS tunnel.

QUESTION 357



Refer to the exhibit. Routers R1, R2, R3, and R4 use EIGRP. However, traffic always prefers R1 to R5 backup links in nonfailure scenarios. Which configuration resolves the issue?

- A. R1(config)#int gigabitEthernet 0/0/0
R1(config-if)#bandwidth 10000000
- B. R1(config)#no ip route 10.40.10.0 255.255.255.252 10.10.20.2
R1(config)#ip route 10.40.10.0 255.255.255.252 10.10.20.2 115
- C. R1(config)#no ip route 10.40.10.0 255.255.255.252 10.10.20.2
R1(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.2
- D. R1(config)#int gigabitEthernet 0/0/0
R1(config-if)#bandwidth 10000

Correct Answer: B

Section: (none)

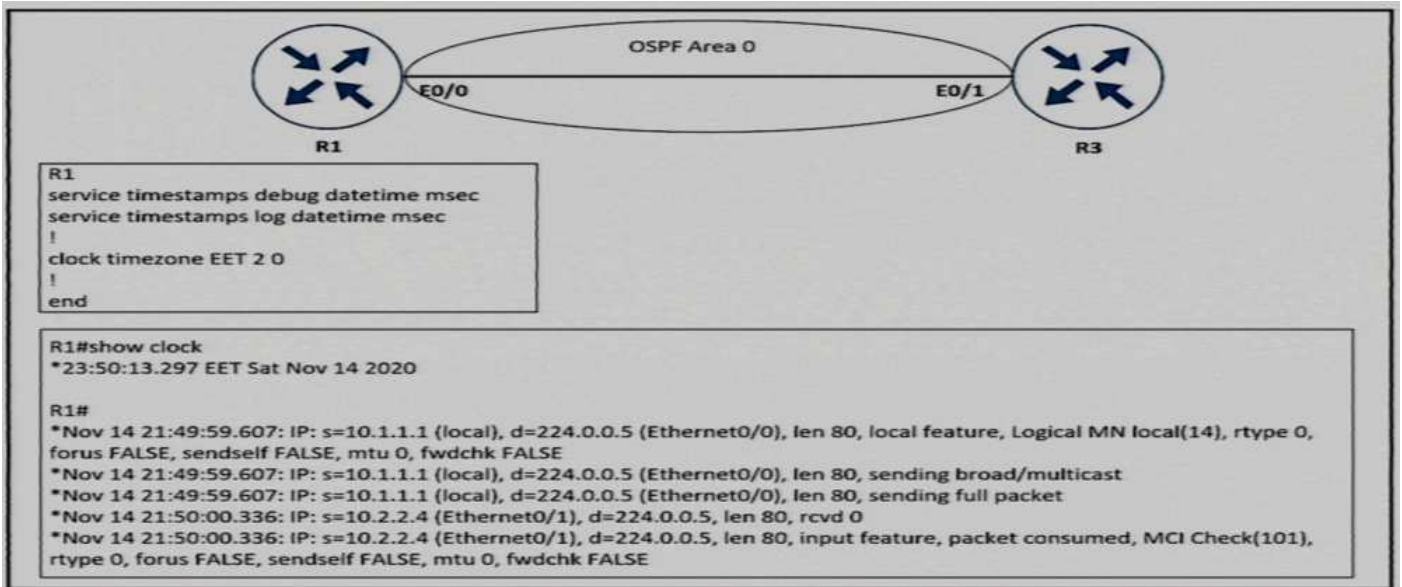
Explanation

Explanation/Reference:

From the diagram, R1 only uses R5 for 10.40.10.0/30 (which only involves two IP addresses of R4 and R5). Moreover, there is no information about the IP addressing of the server. From the routing table, OSPF is also running.

Ignoring all the above, the most reason way is to remove the static route to 10.10.20.2 and configure it as a floating static route for backup i.e. AD greater than 90. For considering the OSPF running also, AD should be greater than 110.

QUESTION 358



Refer to the exhibit. An engineer cannot determine the time of the problem on R1 due to a mismatch between the router local clock and logs. Which command synchronizes the time between new log entries and the local clock on R1?

- A. service timestamps log datetime msec show-timezone
- B. service timestamps debug datetime localtime msec
- C. service timestamps debug datetime msec show-timezone
- D. service timestamps log datetime localtime msec

Correct Answer: B

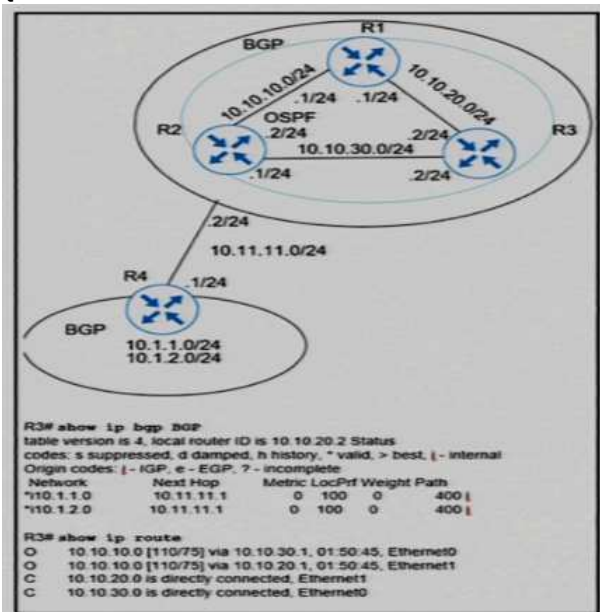
Section: (none)

Explanation

Explanation/Reference:

The logging messages shown in the diagram is debugging messages (no syslog level shown). Assume the question wants to find the correct time for debugging messages, B is therefore chosen.

QUESTION 359



Refer to the exhibit. Which action installs routes into the routing table of R1 and R3?

- A. BGP Synchronization has been enabled. Disable synchronization.
- B. BGP weight is set to 0. Configure a higher weight for these routes.
- C. IGP does not have a route to R4. Rectify route reachability.
- D. BGP has not been redistributed into the IGP. Redistribute BGP.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Since R3 has BGP table, it is running BGP. However, the next hop 10.11.11.1 of BGP routes cannot be accessed by R3's routing table.

QUESTION 360

```
R2#show ip route
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 12 subnets, 3 masks
C    10.1.3.0/30 is directly connected, FastEthernet0/1
C    10.1.2.0/30 is directly connected, FastEthernet0/0
C    10.1.1.0/30 is directly connected, FastEthernet1/0
O E2 10.19.0.0/24 [110/20] via 10.1.3.2, 00:02:04, FastEthernet0/1
D    10.55.13.0/24 (90/4096001 via 10.1.2.2. 00:01:00. FastEthernet0/0
D    10.37.100. 0/24 (90/4096001 via 10.1.2.2. 00:01:00. FastEthernet0/0
C    10.100.10.0/29 is directly connected, FastEthernet2/0.10
D    10.55.72.0/24 (90/409600] via 10.1.2.2. 00:01:01. FastEthernet0/0
C    10.100.20.0/29 is directly connected. FastEthernet2/0.20
O E2 10.144.1.0/24 /110/201 via 10.1.3.2. 00:12:51. FastEthernet0/1
D    10.55.144.0/24 (90/4096001 via 10.1.2.2. 00:01:01. FastEthernet0/0
O E2 10.123.187.0/24 (110/20] via 10.1.3.2. 00:12:51, FastEthernet0/1
```

```
R2#sh ip eigrp topology
```

```
IP-EIGRP Topology Table for AS(100)/ID(10.100.20.2)
```

```
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
```

```
r- reply Status, s - sia Status
```

```
P 10.1.3.0/30, 1 successors, FD is 281600 via Connected, FastEthernet0/1
P 10.1.2.0/30, 1 successors, FD is 281600 via Connected, FastEthernet0/0
P 10.1.1.0/30, 1 successors, FD is 28160 via Connected, FastEthernet1/0
P 10.55.13.0/24, 1 successors, FD is 409600 via 10.1.2.2 (409600/128256). FastEthernet0/0
P 10.37.100.0/24, 1 successors, FD is 409600 via 10.1.2.2 (409600/128256). FastEthernet0/0
P 10.55.72.0/24. 1 successors. FD is 409600 via 10.1.2.2 (409600/128256), FastEthernet0/0
P 10.55.144.0/24. 1 successors, FD is 409600 via 10.1.2.2 (409600/128256), FastEthernet0/0
P 10.123.187.0/24. 0 successors, FD is Inaccessible via 10.1.2.2 (409600/128256), FastEthernet0/0
```

Refer to the exhibit. Router R2 should be learning the route for 10.123.187.0/24 via EIGRP. Which action resolves the issue without introducing more issues?

- A. Use distribute-list to filter the external routes in OSPF.
- B. Redistribute the route in EIGRP with metric, delay, and reliability.
- C. Remove route redistribution in R2 for this route in OSPF.
- D. Use distribute-list to modify the route as an internal EIGRP route.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

From the output:

- Routing table has a OSPF external route 10.123.187.0/24

- EIGRP Topology table has a route 10.123.187.0/24 with 0 successor and FD inaccessible. This occurs if EIGRP route 10.123.187.0/24 cannot be entered into the routing table.

The reason is probably EIGRP route 10.123.187.0/24 is an EIGRP external route with AD 170 which is less preferred than OSPF external router having AD 110.

Note that if R2 performs redistribution from EIGRP to OSPF, only the source route will be entered in the routing table. The redistributed route will not be entered i.e. the OSPF route in the routing table is NOT due to redistribution in R2.

distribute-list does not affect OSPF LSA advertisement but can prefer an OSPF route from entering into the local router's routing table.

QUESTION 361

While examining log messages from a router, an administrator sees this error message:

```
%DUAL-5-NBRCHANGE: IP-EIGRP(0) 1: Neighbor 10.4.1.5 (Ethernet1/0) is down: K-value mismatch
```

Both routes are using default values. Which action resolves this issue?

- A. Configure metric weights between the peers to calculate the valid K-value.
- B. Resolve metric differences between the peers to calculate the valid K-value.
- C. Upgrade the peer device IOS to support the goodbye message that establishes adjacency
- D. Match authentication parameters to support the goodbye message that establishes adjacency.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

The message appears if the two neighboring routers have been configured with different K values.

If the K-values are the same, the issue might be caused by the EIGRP graceful shutdown feature. In that case, a router sends an EIGRP Hello packet with the K-values set to 255 so that the K-values mismatch occurs intentionally. This is to indicate to the neighbor EIGRP router that it is going down. On the neighbor router supporting this feature, you would see this goodbye message received:

```
%DUAL-5-NBRCHANGE: IP-EIGRP(0) 1: Neighbor 10.1.1.1 (Ethernet0/0) is down: Interface Goodbye received
```

However, if the neighbor router runs an older code version, it does not recognize this as a graceful shutdown message, but as a mismatch in K-values and the "K-value mismatch" message will be shown.

QUESTION 362

Which IPv6 security feature blocks all traffic from an IPv6 host when initially connecting to a switch port except for traffic to gain an IPv6 address and source IPv6 neighbors?

- A. IPv6 RA Guard
- B. IPv6 Source Guard

- C. IPv6 Destination Guard
- D. IPv6 DHCP Guard

Correct Answer: B
Section: (none)
Explanation

Explanation/Reference:

Source Guard is an ingress feature and filters incoming data packets alone. If Source Guard is enabled, every ingress packet on a switch port or Layer 2 VLAN is checked against entries in the IPv6 binding table

Initially, before the entry of a host is entered into IPv6 binding table, Source Guard blocks all IPv6 traffic on the target except for Dynamic Host Configuration Protocol (DHCP) or Neighbor Discovery Protocol (NDP) packets that are used for IPv6 Snooping processes to populate the binding table.

QUESTION 363

```
ip prefix-list DMZ-STATIC seq 5 permit 10.1.1.0/24
!
route-map DMZ permit 10
 match ip address prefix-list DMZ-STATIC
!
router ospf 1
 network 0.0.0.0 0.0.0.0 area 0
 redistribute static route-map DMZ
!
ip route 10.1.1.0 255.255.255.0 10.20.20.1
```

Refer to the exhibit. The static route is not present in the routing table of an adjacent OSPF neighbor router. Which action resolve the issue?

- A. Configure the subnets keyword in the redistribution command.
- B. Configure a permit 20 statement to the route map to redistribute the static route.
- C. Configure the next-hop interface at the end of the static route for it to get redistributed.
- D. Configure the next hop of 10.20.20.1 in the prefix list DMZ-STATIC

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

In older IOS version, "subnets" is not included by default.

QUESTION 364

```
R1#
router ospf 1
 redistribute rip subnets
 network 131.108.1.0 0.0.0.255 area 2
 network 131.108.2.0 0.0.0.255 area 2
 distribute-list 1 out
!
access-list 1 permit 132.108.4.0 0.0.0.255
```

Refer to the exhibit. The R1 OSPF neighbor is not receiving type 5 external LSAs for 132.108.2.0/24 and 132.108.3.0/24 networks. Which configuration command resolves the issue?

- A. access-list 1 permit 132.108.0.0 0.0.3.255
- B. access-list 1 permit 132.108.0.0 0.0.1.255
- C. access-list 1 permit 132.108.4.0 0.0.3.255
- D. access-list 1 permit 132.108.2.0 0.0.0.255

Correct Answer: A
Section: (none)
Explanation

Explanation/Reference:

132.108.0.0 0.0.3.255 will include 132.108.0.0/24, 132.108.1.0/24, 132.108.2.0/24 and 132.108.3.0/24

Remarks :

In OSPF, distribute-list ... out has no effect normally except for external routes redistributing into OSPF. In this case, it can filter routes redistributed by RIP. In the existing access-list, only routes matching 192.168.4.X can be redistributed into OSPF.

QUESTION 365

Which two functions are the functions of LDP? (Choose two.)

- A. uses TCP for targeted LDP session
- B. uses P2P VPWS services
- C. uses pseudowire signaling
- D. uses MP2MP VPLS services
- E. used for MPLS-TE Fast Reroute

Correct Answer: AC
Section: (none)
Explanation

Explanation/Reference:

Actually the MPLS Pseudowire uses LDP for sending signals.

QUESTION 366

A customer requested a GRE tunnel through the provider network between two customer sites using loopback to hide internal networks. Which configuration on R2

establishes the tunnel with R1?

- A. R2(config)#interface Tunnel1
R2(config-if)#ip address 172.20.1.2 255.255.255.0
R2(config-if)#ip mtu 1500
R2(config-if)#ip tcp adjust-mss 1360
R2(config-if)#tunnel source 192.168.20.1
R2(config-if)#tunnel destination 10.10.1.1
- B. R2(config)#interface Tunnel1
R2(config-if)#ip address 172.20.1.2 255.255.255.0
R2(config-if)#ip mtu 1400
R2(config-if)#ip tcp adjust-mss 1360
R2(config-if)#tunnel source 192.168.20.1
R2(config-if)#tunnel destination 192.168.10.1
- C. R2(config)#interface Tunnel1
R2(config-if)#ip address 172.20.1.2 255.255.255.0
R2(config-if)#ip mtu 1400
R2(config-if)#ip tcp adjust-mss 1360
R2(config-if)#tunnel source 10.10.2.2
R2(config-if)#tunnel destination 10.10.1.1
- D. R2(config)#interface Tunnel1
R2(config-if)#ip address 172.20.1.2 255.255.255.0
R2(config-if)#ip mtu 1500
R2(config-if)#ip tcp adjust-mss 1360
R2(config-if)#tunnel source 10.10.2.2
R2(config-if)#tunnel destination 10.10.1.1

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Although the question does not mention IPsec, the common practice for setting MTU for a GRE/IPsec tunnel is 1400 since 1400 allows for all the typical GRE/IPsec overhead, which is often actually closer to about 80 bytes.

Among B and C, there is no way to determine which is correct without further information or a diagram. We can only assume that classful network is used in the provider's IP addressing. In this case, the source and destination being in different networks are more logical.

QUESTION 367

In a DMVPN network, the Spoke1 user observed that the voice traffic is coming to Spoke2 users via the hub router. Which command is required on both spoke routers to communicate directly to one another?

- A. ip nhrp shortcut
- B. ip nhrp redirect
- C. ip nhrp nhs multiast
- D. ip nhrp map dynamic

Correct Answer: A

Section: (none)

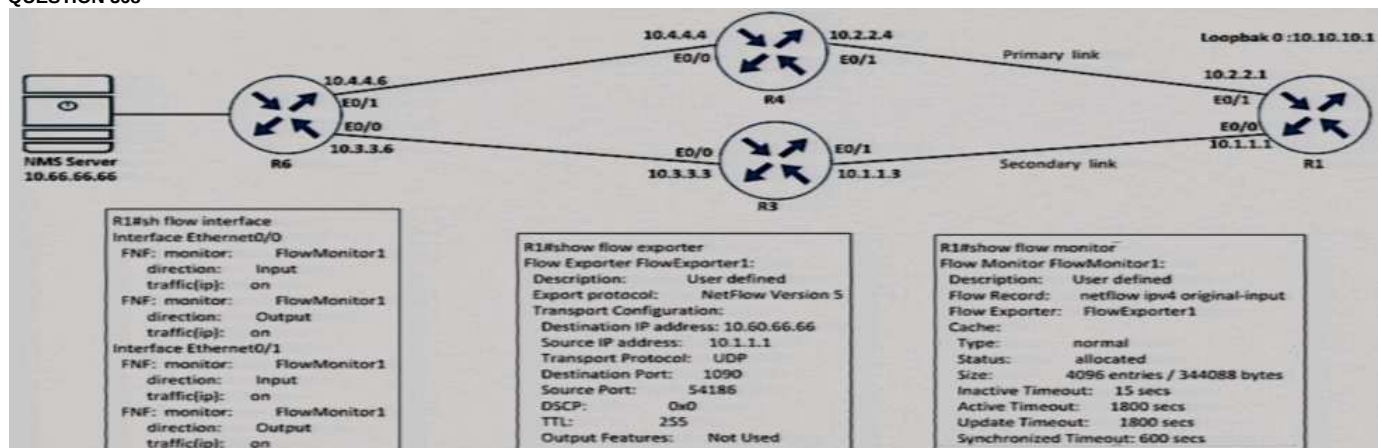
Explanation

Explanation/Reference:

"ip nhrp redirect" command in the Hub router sends the message to the Spokes about a better route to the destination.

"ip nhrp shortcut" allows the Spoke routers to install the route mentioned in the redirect message.

QUESTION 368



Refer to the exhibit. An engineer configured NetFlow on R1, but the flow do not reach the NMS server from R1. Which configuration resolves this issue?

- A. R1(config)#interface Ethernet0/1
R1(config-if)#ip flow monitor FlowMonitor1 input
R1(config-if)#ip flow monitor FlowMonitor1 output
- B. R1(config)#interface Ethernet0/0
R1(config-if)#ip flow monitor FlowMonitor1 input
R1(config-if)#ip flow monitor FlowMonitor1 output
- C. R1(config)#flow monitor FlowMonitor1
R1(config-flow-monitor)#destination 10.66.66.66
- D. R1(config)#flow monitor FlowExporter1
R1(config-flow-exporter)#destination 10.66.66.66

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

From the output in "show flow exporter" (the middle output diagram), the destination IP address is incorrect. It should be "10.66.66.66" instead of "10.60.66.66".

QUESTION 369

What is an MPLS LDP targeted session?

- A. LDP session established by exchanging multicast hello packets
- B. label distribution session between non-directly connected neighbors
- C. session between neighbors that are connected no more than one hop away
- D. LDP session established between LSRs by exchanging TCP hello packets

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 370

What is the purpose of an OSPF sham-link?

- A. to correct OSPF backdoor routing when OSPF is used as the PE-PE connection protocol in an MPLS VPN network
- B. to correct OSPF backdoor routing when OSPF is used as the PE-CE connection protocol in an MPLS VPN network
- C. to allow inter-area routing when OSPF is used as the PE-CE connection protocol in a MPLS VPN network
- D. to allow intra-area routing when OSPF is used as the PE-CE connection protocol in an MPLS VPN network

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 371

The network administrator configured CoPP so that all routing protocol traffic toward the router CPU is limited to 1 mbps. All traffic that exceeds this limit must be dropped. The router is running BGP and OSPF. Management traffic for Telnet and SSH must be limited to 500 kbps.

```
access-list 100 permit tcp any any eq 179
access-list 100 pennit tcp any any range 22 23
access-list 100 permit ospf any any
!
class-mnmap CM-ROUTING
 match access-group 100
class-map CM-MGMT
 match access-group 100
!
policy-map PM-COPP
 class CM-ROUTING
  police 1000000 conform-action transmit
 class CM-MGMT
  police 500000 conform-action transmit
!
control-plane
 service-policy output PM-COPP
```

No traffic is filtering through CoPP, which is resulting in high CPU utilization. Which configuration resolves the issue?

- A. no access-list 100
access-list 100 permit tcp any any eq 179
access-list 100 permit ospf any any
access-list 101 permit tcp any any range 22 23
!
class-map CM-MGMT
no match access-group 100
match access-group 101
- B. control-plane
no service-policy output PM-COPP
service-policy input PM-COPP
- C. no access-list 100
access-list 100 permit tcp any any eq 179
access-list 100 permit ospf any any
access-list 101 permit tcp any any range 22 23
!
class-map CM-MGMT
no match access-group 100
match access-group 101
!
control-plane
no service-policy output PM-COPP
service-policy input PM-COPP
- D. no access-list 100
access-list 100 permit tcp any any eq 179
access-list 100 permit ospf any any
access-list 100 permit tcp any any range 22 23
access-list 100 permit ospf any any

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

There are two problem in the existing configuraion.

- The two classes are matching the same ACL and therefore cannot distinguish the routing and management traffic.
- The policy is applied to output direction incorrectly.

The first problem can be corrected by choice A and the second problem can be corrected by choice B. Therefore you need choice C which combines the changes required.

QUESTION 372

```
May 5 15:19:30.749: OSPF: Send DBD to 192.168.95.11 on GigabitEthernet3/1 seq 0x2AC opt 0x50 flag 0x7 len 32
May 5 15:19:30.749: OSPF: Retransmitting DBD to 192.168.95.11 on GigabitEthernet3/1 [1]
May 5 15:19:35.509: OSPF: Send DBD to 192.168.95.11 on GigabitEthernet3/1 seq 0x2AC opt 0x50 flag 0x7 len 32
May 5 15:27:29.904: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.99.22 on Tunnel0 from LOADING to FULL, Loading Done
May 5 15:28:28.176: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.99.22 on Tunnel5 from LOADING to FULL, Loading Done
May 5 15:30:02.028: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.99.22 on Tunnel0 from LOADING to FULL, Loading Done
May 5 15:30:34.720: %CRYPTO-4-IKE_DEFAULT_POLICY_ACCEPTED: IKE default policy was matched and is being used.
May 5 15:30:44.009: %CRYPTO-4-IKE_DEFAULT_POLICY_ACCEPTED: IKE default policy was matched and is being used.
May 5 15:19:30.749: OSPF: Send DBD to 192.168.95.11 on GigabitEthernet3/1 seq 0x2AC opt 0x50 flag 0x7 len 32
May 5 15:19:30.749: OSPF: Retransmitting DBD to 192.168.95.11 on GigabitEthernet3/1 [1]
May 5 15:31:09.441: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.99.22 on Tunnel0 from LOADING to FULL, Loading Done
May 5 15:31:27.341: %CRYPTO-4-IKE_DEFAULT_POLICY_ACCEPTED: IKE default policy was matched and is being used.
May 5 15:31:42.137: %CRYPTO-4-IKE_DEFAULT_POLICY_ACCEPTED: IKE default policy was matched and is being used.
May 5 15:32:14.777: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.99.22 on Tunnel5 from LOADING to FULL, Loading Done
May 5 15:19:30.749: OSPF: Send DBD to 192.168.95.11 on GigabitEthernet3/1 seq 0x2AC opt 0x50 flag 0x7 len 32
May 5 15:19:30.749: OSPF: Retransmitting DBD to 192.168.95.11 on GigabitEthernet3/1 [1]
May 5 15:33:40.761: %CRYPTO-4-IKE_DEFAULT_POLICY_ACCEPTED: IKE default policy was matched and is being used.
May 5 15:34:32.065: %CRYPTO-4-IKE_DEFAULT_POLICY_ACCEPTED: IKE default policy was matched and is being used.
May 5 15:35:05.950: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.99.22 on Tunnel0 from LOADING to FULL, Loading Done
May 5 15:56:36.603: %PARSER-5-CFGLOG: LOGGED CMD: User gus logged command: showc: enable.
```

Refer to the exhibit. A network administrator is troubleshooting OSPF adjacency issue by going through the console logs in the router, but due to an overwhelming log messages stream, it is impossible to capture the problem. Which two commands reduce console log messages to relevant OSPF neighbor problem details so that the issue can be resolved? (Choose two.)

- A. debug condition ip
- B. debug condition all
- C. debug condition ospf neighbor
- D. debug condition interface
- E. debug condition session-id ADJCHG

Correct Answer: AD

Section: (none)

Explanation

Explanation/Reference:

Only two of the choices are valid options for debug condition:

```
Router#debug condition ?
called          called number
calling         calling
cpl             Cisco Provisioning Language debugging
glbp            interface group
interface       interface
ip              IP address
mac-address     MAC address
match-list      apply the match-list
profile         Media Services Profile
standby         interface group
username        username
vcid            VC ID
vrf             Virtual Routing and Forwarding
xconnect        Xconnect conditional debugging on segment pair
```

QUESTION 373

An engineer noticed that the router log messages do not have any information about when the event occurred. Which action should the engineer take when enabling service time stamps to improve the logging functionality at a granular level?

- A. Configure the debug uptime option.
- B. Configure the msec option.
- C. Configure the timezone option
- D. Configure the log uptime option.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

When enabling enabling service time stamps, msec option can be added to include milliseconds information in a granular (i.e. smaller and more distinct) level.

QUESTION 374

```
interface GigabitEthernet2
 no ip address
 ip helper-address 192.168.255.3
 no shutdown
!
interface GigabitEthernet2.10
 encapsulation dot1q 210
 ip address 192.168.210.1 255.255.255.0
 ip ospf 1 area 0
 no shutdown
```

Refer to the exhibit. With the partial configuration of a router-on-a-stick, clients in VLAN 10 on Gi2 cannot obtain IP configuration from the central DHCP server. The DHCP server is reachable by a successful ping from the router. Which action resolves the issue?

- A. Configure a valid IP address on the Gi2 interface so that DHCP requests can be forwarded.
- B. Configure the ip dhcp pool 1 and network 192.168.210.0 255.255.255.0 commands.
- C. Configure the ip helper-address 192.168.255.3 command on the Gi2.10 subinterface.

D. Configure the ip dhcp excluded-address 192.168.255.3 command on the Gi2.10 subinterface.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 375

An engineer configured routing between multiple OSPF domains and introduced a routing loop that caused network instability. Which action resolves the problem?

- A. Set a tag using the redistribute command toward a domain and deny inbound in the other domain by a matching tag
- B. Set a tag using the redistribute command toward a different domain and deny the matching tag when exiting from that domain
- C. Set a tag using the network command in a domain and use the route-map command to deny the matching tag when existing toward a different domain.
- D. Set a tag using the network command in a domain and use the route-map command to deny the matching tag when entering into a different domain.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The following shows a simple example in which OSPF process ID "1" represents OSPF Domain 1 and OSPF process ID "2" represents OSPF Domain 2. Since the redistributing between two OSPF domains works just like two different routing protocols, the using of tags for loop prevention is similar.



In the following example, redistribute command towards Domain 1 is configured with tag 1. Then distribute-list is configured in the other domain Domain 2 to deny that tag 1.

```
router ospf 1
 redistribute ospf 2 subnet tag 1
 distribute-list 1 route-map filter_domain2 in
!
route-map filter_domain2 deny 10
 match tag 2
route-map filter_domain2 permit 20
```

```
router ospf 2
 redistribute ospf 1 subnet tag 2
 distribute-list 1 route-map filter_domain1 in
!
route-map filter_domain1 deny 10
 match tag 1
route-map filter_domain1 permit 20
```

QUESTION 376

```
router ospfv3 1
router-id 10.1.1.1
address-family ipv4 unicast
passive-interface Loopback0
exit-address-family
address-family ipv6 unicast
passive-interface Loopback0
exit-address-family
interface Loopback0
ip address 10.1.1.1 255.255.255.255
ipv6 address 2001:DB8::1/64
ospfv3 10 ipv4 area 10
ospfv3 10 ipv6 area 0
interface GigabitEthernet2
ip address 10.10.10.1 255.255.255.0
ipv6 enable
ospfv3 10 ipv4 area 10
ospfv3 10 ipv6 area 0
```

Refer to the exhibit. An administrator must configure the router with OSPF for IPv4 and IPv6 networks under a single process. The OSPF adjacencies are not established and did not meet the requirement. Which action resolves the issue?

- A. Replace OSPFprocess 10 on the interfaces with OSPF process 1 for the IPv4 address, and remove process 10 from the global configuration.
- B. Replace OSPFprocess 10 on the interfaces with OSPF process 1, and remove process 10 from the global configuration.
- C. Replace OSPFprocess 10 on the interfaces with OSPF process 1 for the IPv6 address, and remove process 10 from the global configuration.
- D. Replace OSPFprocess 10 on the interfaces with OSPF process 1, and configure an additional router ID with IPv6 address.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since the global configuration is configured with an OSPF process of ID 1 and the interfaces are configured with a OSPF process of ID 10, two OSPF processes will be running on the router (although process ID 1 does not include any OSPF interface).

Remarks:

- Before action is taken to resolves the issue, "sh ip protocols" will show the followings:

Routing Protocol is "ospfv3 1"

```
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
```

```

Router ID 10.1.1.1
Number of areas: 0 normal, 0 stub, 0 nssa
Maximum path: 4
Routing Information Sources:
  Gateway         Distance       Last Update
Distance: (default is 110)

```

```

Routing Protocol is "ospfv3 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.1.1.1
  Number of areas: 1 normal, 0 stub, 0 nssa
  Interfaces (Area 10):
    Loopback0
    GigabitEthernet2
  Maximum path: 4
  Routing Information Sources:
    Gateway         Distance       Last Update
  Distance: (default is 110)

```

- Before action is taken to resolves the issue, "sh run | s router ospf" will show the following configuraiton in the running configuration:

```

router ospfv3 1
  router-id 10.1.1.1
  !
  address-family ipv4 unicast
    passive-interface Loopback0
  exit-address-family
  !
  address-family ipv6 unicast
    passive-interface Loopback0
  exit-address-family

router ospfv3 10
  !
  address-family ipv4 unicast
  exit-address-family
  !
  address-family ipv6 unicast
  exit-address-family

```

The router configuration for process ID 10 will remain even if you change all interfaces to OSPF process ID 1. Therefore, you also need to remove the router configuraion for process ID 10 by the Global configuration command "no router ospfv3 10"

QUESTION 377

Which failure detection mechanism is used for BFD?

- A. consistent rate
- B. Layer 2 protocol failure
- C. variable rate
- D. routing protocol failute

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

"Because the network administrator can use BFD to detect forwarding path failures at a uniform rate, rather than the variable rates for different routing protocol hello mechanisms, network profiling and planning will be easier, and reconvergence time will be consistent and predictable."

The choice "consistent rate" means "uniform rate" mentioned above.

QUESTION 378

The IPv6 network is under attack by an unknown source that is neither in the binding table nor learned through neighbor discovery. Which feature helps prevent the attack?

- A. IPv6 Prefix Guard
- B. IPv6 Snooping
- C. IPv6 Router Advertisement Guard
- D. IPv6 Destination Guard

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

IPv6 source guard is an interface feature between the populated binding table and data traffic filtering. This feature enables the device to deny traffic when it is originated from an address that is not stored in the binding table.

The IPv6 Prefix Guard feature works within the IPv6 Source Guard feature, enabling the device to deny traffic originated from nontopologically correct addresses. IPv6 prefix guard is often used when IPv6 prefixes are delegated to devices (for example, home gateways) using DHCP prefix delegation.

QUESTION 379

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PIR

Gateway of last resort is 10.255.0.1 to network 0.0.0.0

```

S* 0.0.0.0/0 [254/0] via 10.255.0.1
    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
B   10.0.0.0/8 [20/0] via 192.168.20.2, 23:07:56
D   10.0.0.0/16 [90/2816] via 192.168.90.2, 22:59:54, GigabitEthernet4
O   10.0.0.0/24 [110/2] via 192.168.110.2, 22:45:53, GigabitEthernet3
C   10.255.0.0/16 is directly connected, GigabitEthernet1
S   10.255.0.2/32 [254/0] via 10.255.0.1, GigabitEthernet1
L   10.255.4.85/32 is directly connected, GigabitEthernet1
C   192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.20.0/24 is directly connected, GigabitEthernet2
L   192.168.20.1/32 is directly connected, GigabitEthernet2
C   192.168.90.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.90.0/30 is directly connected, GigabitEthernet4
L   192.168.90.1/32 is directly connected, GigabitEthernet4
C   192.168.110.0/24 is variably subnetted, 2 subnets, 2 masks
L   192.168.110.0/30 is directly connected, GigabitEthernet3
L   192.168.110.1/32 is directly connected, GigabitEthernet3

```

Refer to the exhibit. The network administrator configured BGP as the backup route for 10.0.0.0/8 and it should work only when EIGRP 10.0.0.0/8 failed to install for the site. Which configuration resolves the issue?

- A. configure terminal
!
ip route 10.0.0.0 255.0.0.0 192.168.90.2
- B. configure terminal
!
router eigrp 1
distance eigrp 10 170
- C. configure terminal
!
router eigrp 1
distance eigrp 90 170
- D. configure terminal
!
router eigrp 1
redistribute bgp metric 10000 1 255 1 1500

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since BGP route learnt from eBGP neighbor has an AD 20, it will be preferred than the EIGRP route having an AD of 90.

Hence you need to change the AD of the EIGRP to a value smaller than 20 e.g. 10.

(Note that the first number 10 is the AD for internal routes and the second number 170 is the AD for external routes).

QUESTION 380

```

Switch(config)# ip vrf 70
Switch(config-vrf)# rd 70:1
Switch(config-vrf)# route-target export 70:1
Switch(config-vrf)# route-target import 70:1
Switch(config-vrf)# exit

Switch(config)# ip vrf 80
Switch(config-vrf)# rd 80:1
Switch(config-vrf)# route-target export 80:1
Switch(config-vrf)# route-target import 80:1

```

Refer to the exhibit. An engineer must extend VRF-Lite over a trunk to another switch for VLAN 70 (10.70.70.0/24) on port GigabitEthernet0/0 and VLAN 80 (10.80.80.0/24) on port GigabitEthernet0/1. Which configuration accomplishes this objective?

- A. interface GigabitEthernet0/0
no switchport
ip vrf forwarding 70
ip address 10.70.70.1 255.255.255.0
!
interface GigabitEthernet0/1
no switchport
ip vrf forwarding 80
ip address 10.80.80.1 255.255.255.0
- B. interface GigabitEthernet0/0
switchport mode access
switchport access vlan 70
ip vrf forwarding 70
!
interface GigabitEthernet0/1
switchport mode access
switchport access vlan 80
ip vrf forwarding 70
- C. interface GigabitEthernet0/0
switchport mode access
switchport access vlan 70
!
interface GigabitEthernet0/1


```

switchport mode access
switchport access vlan 80
D. interface GigabitEthernet0/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 70
!
interface GigabitEthernet0/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 80

```

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 381

The network administrator must configure R1 to authenticate Telnet connections based on Cisco ISE using RADIUS. ISE has been configured with an IP address of 192.168.1.5 and with a network device pointing toward R1 (192.168.1.1) with a shared secret password of Cisco123.

The administrator has configured this on R1:

```

aaa new-model
!
radius server ISE1
address ipv4 192.168.1.5
key Cisco123
!
aaa group server tacas+ RAD-SERV
server name ISE1
!
aaa authentication login default group RAD-SERV

```

The network administrator cannot authenticate to access R1 based on ISE. Which set of configurations fixes the issue?

- A. aaa group server radius RAD-SERV
server name ISE1
- B. aaa group server tacacs+ ISE1
server name RAD-SERV
- C. line vty 0 4
login authentication RAD-SERV
- D. line vty 0 4
login authentication default

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Since RADIUS is being used, the group RAD-SERV should be setup with "radius".

The following is the default settings after aaa new-model is enabled and therefore should not be needed normally:

```

line vty 0 4
login authentication default

```

QUESTION 382

Network operations report issues with receiving too many external routes, which caused CPU spike on routers with smaller memories. Which action resolves the issue?

- A. Configure the area range command when redistributing on ASBR
- B. Configure the summary-address command when redistributing on ABR
- C. Configure the area range command when redistributing on ABR
- D. Configure the summary-address command when redistributing on ASBR

Correct Answer: D

Section: (none)

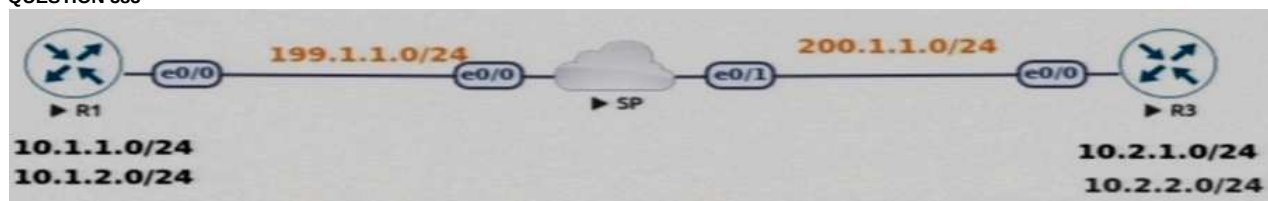
Explanation

Explanation/Reference:

area range is configured in ABR and summary-address is configured in ASBR.

Since there are too many external routes, summary-address should be configured on ASBR.

QUESTION 383



Refer to the exhibit. An engineer must configure a LAN-to-LAN IPsec VPN between R1 and the remote router. Which IPsec Phase 1 configuration must the engineer use for the local router?

- A. crypto isakmp policy 5
authentication pre-share

```

    encryption 3des
    hash md5
    group 2
    !
crypto isakmp key cisco123 address 199.1.1.1
B. crypto isakmp policy 5
   authentication pre-share
   encryption 3des
   hash md5
   group 2
   !
crypto isakmp key cisco123 address 200.1.1.3
C. crypto isakmp policy 5
   authentication pre-share
   encryption 3des
   hash md5
   group 2
   !
crypto isakmp key cisco123! address 199.1.1.1
D. crypto isakmp policy 5
   authentication pre-share
   encryption 3des
   hash sha
   group 2
   !
crypto isakmp key cisco123 address 200.1.1.3

```

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Since the remote router is R3, the local router refers to R1.

For R1 to authenticate with the remote router R3, you need to configure the pre-shared key with the IP address of R3 i.e. having IP address in 200.1.1.0/24

Both choices having address 200.1.1.3 should work. However since the hash algorithm SHA is more secure, it is chosen as suggested answer.

Remarks :

- If the question asks you to configure R3, one of the choices with "address 199.1.1.1" is correct. The correct one depends on which pre-shared key is correct.

- The isakmp policy using either SHA or MD5 should work since they are included in the default policy of the other router:

R3#sh cry isakmp default policy

```

Default IKE policy
Default protection suite of priority 65507
  encryption algorithm: AES - Advanced Encryption Standard (128 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Rivest-Shamir-Adleman Signature
  Diffie-Hellman group: #5 (1536 bit)
  lifetime:            86400 seconds, no volume limit
Default protection suite of priority 65508
  encryption algorithm: AES - Advanced Encryption Standard (128 bit keys).
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #5 (1536 bit)
  lifetime:            86400 seconds, no volume limit
Default protection suite of priority 65509
  encryption algorithm: AES - Advanced Encryption Standard (128 bit keys).
  hash algorithm:      Message Digest 5
  authentication method: Rivest-Shamir-Adleman Signature
  Diffie-Hellman group: #5 (1536 bit)
  lifetime:            86400 seconds, no volume limit
Default protection suite of priority 65510
  encryption algorithm: AES - Advanced Encryption Standard (128 bit keys).
  hash algorithm:      Message Digest 5
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #5 (1536 bit)
  lifetime:            86400 seconds, no volume limit
Default protection suite of priority 65511
  encryption algorithm: Three key triple DES
  hash algorithm:      Secure Hash Standard
  authentication method: Rivest-Shamir-Adleman Signature
  Diffie-Hellman group: #2 (1024 bit)
  lifetime:            86400 seconds, no volume limit
Default protection suite of priority 65512
  encryption algorithm: Three key triple DES
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #2 (1024 bit)
  lifetime:            86400 seconds, no volume limit
Default protection suite of priority 65513
  encryption algorithm: Three key triple DES
  hash algorithm:      Message Digest 5
  authentication method: Rivest-Shamir-Adleman Signature
  Diffie-Hellman group: #2 (1024 bit)
  lifetime:            86400 seconds, no volume limit
Default protection suite of priority 65514
  encryption algorithm: Three key triple DES
  hash algorithm:      Message Digest 5
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #2 (1024 bit)
  lifetime:            86400 seconds, no volume limit

```

R3#

QUESTION 384

```
R4#
interface FastEthernet1/0
ip address 10.1.1.14 255.255.255.252
ip access-group VENDOR in
ip authentication mode eigrp 100 md5
ip authentication key-chain eigrp 100 EIGRPKEY
speed 100
full-duplex
!
interface loopback 100
ip address 10.199.100.1 255.255.255.255
!
router eigrp 100
network 10.1.1.8 0.0.0.3
network 10.1.1.12 0.0.0.3
no auto-summary
eigrp router-id 100.4.4.4
neighbor 10.1.1.13 FastEthernet1/0
redistribute connected
!
router bgp 65001
no synchronization
bgp log-neighbor-changes
network 100.4.4.4 mask 255.255.255.255
neighbor 10.1.1.13 remote-as 65001
no auto-summary
!
ip access-list extended VENDOR
permit tcp 192.168.32.0 0.0.7.255 host 10.199.100.1 eq 22 time-range VENDOR_ACCESS
!
time-range VENDOR_ACCESS
periodic weekend 22:00 to 23:00
```

Refer to the exhibit. A network engineer received a call from the vendor for a failed attempt to remotely log in to their managed router loopback interface from 192.168.40.15. Which action must the network engineer take to resolve the issue?

- A. The source IP summarization must be updated to include the vendor source IP address.
- B. The IP access list VENDOR must be applied to interface loopback 100.
- C. The time-range configuration must be changed to use absolute instead of periodic
- D. The EIGRP configuration must be updated to include a network statement for loopback 100.

Correct Answer: A

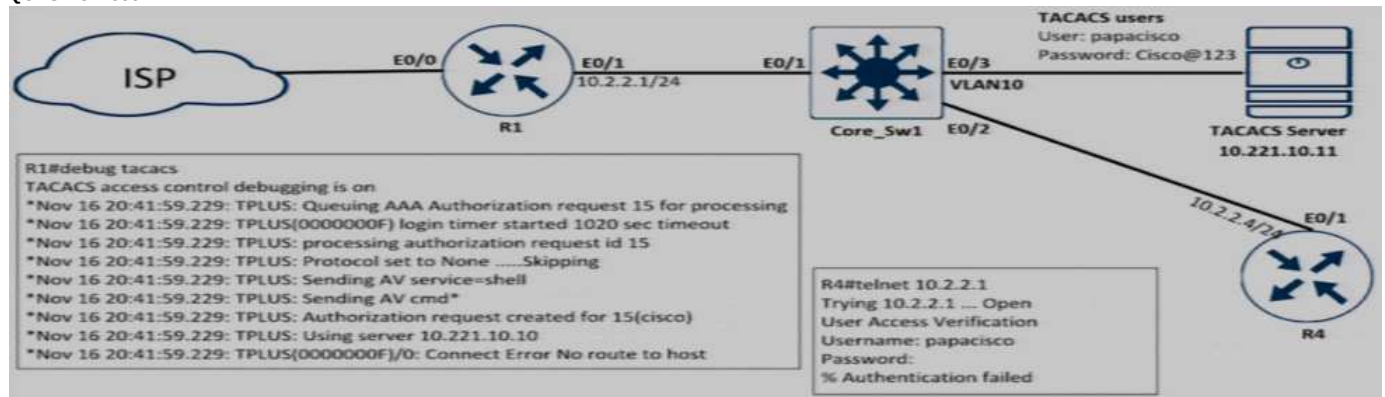
Section: (none)

Explanation

Explanation/Reference:

For the rules in the ACL, the source 192.168.32.0 0.0.7.255 means IP addresses from 192.168.32.x to 192.168.39.x. Hence, the vendor's IP address 192.168.40.15 is not included.

QUESTION 385



Refer to the exhibit. An engineer is trying to connect to R1 via Telnet with no success. Which configuration resolves the issue?

- A. tacacs server prod
address ipv4 10.221.10.11
exit
- B. ip route 10.221.10.10 255.255.255.255 ethernet 0/1
- C. ip route 10.221.0.11 255.255.255.255 ethernet 0/1
- D. tacacs server prod
address ipv4 10.221.10.10
exit

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The last message "No route to host" means the TACACS server cannot be reached. From the logging message, it shows that the IP address of the TACACS server is wrongly configured.

QUESTION 386

Drag and drop the ICMPv6 neighbor discovery messages from the left onto the correct packet types on the right.

Select and Place:

Neighbor Solicitation	ICMPv6 Type 134
Neighbor Advertisement	ICMPv6 Type 137
Router Advertisement	ICMPv6 Type 135
Redirect Message	ICMPv6 Type 133
Router Solicitation	ICMPv6 Type 136

Correct Answer:

	Router Advertisement
	Redirect Message
	Neighbor Solicitation
	Router Solicitation
	Neighbor Advertisement

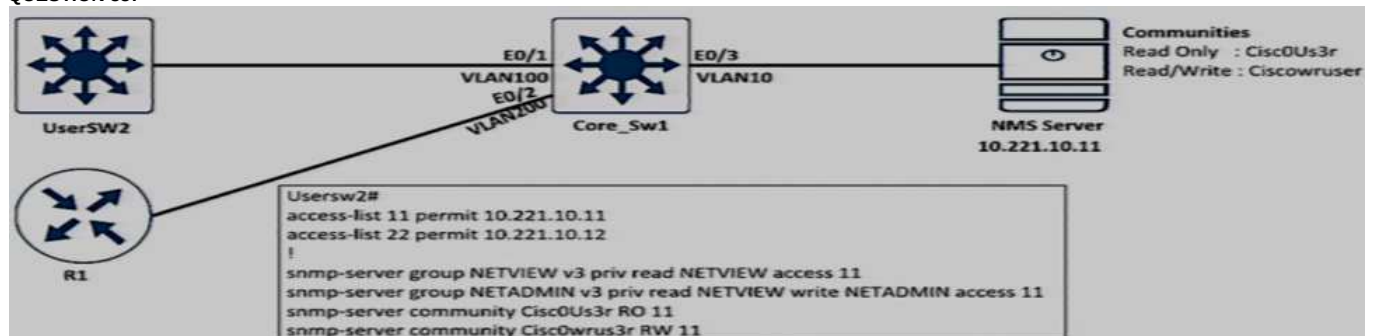
Section: (none)

Explanation

Explanation/Reference:

Type 133 - Router Solicitation
Type 134 - Router Advertisement
Type 135 - Neighbor Solicitation
Type 136 - Neighbor Advertisement
Type 137 - Redirect Message

QUESTION 387



Refer to the exhibit. An engineer configured SNMP Communities on UserSW2 switch, but the SNMP server cannot upload modified configurations to the switch. Which configuration resolves this issue?

- A. snmp-server community Ciscowruser RW 11
- B. snmp-server group NETVIEW v2c priv read NETVIEW access 11
- C. snmp-server community Cisc0Us3r RW 11
- D. snmp-server group NETADMIN v3 priv read NETVIEW write NETADMIN access 22

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

From the network topology, the R/W community is "Ciscowruser" but the configuration shows that it is configured as "Cisc0wrus3r".

If the network topology is correct, the configuration should be changed for using "Cisc0wrus3r".

QUESTION 388

How is a preshared key "Test" for all the remote VPN routers configured in a DMVPN using GRE over IPsec set up?

- A. authentication pre-share Test address 0.0.0.0 0.0.0.0
- B. set pre-share Test address 0.0.0.0 0.0.0.0
- C. crypto ipsec key Test address 0.0.0.0 0.0.0.0
- D. crypto isakmp key Test address 0.0.0.0 0.0.0.0

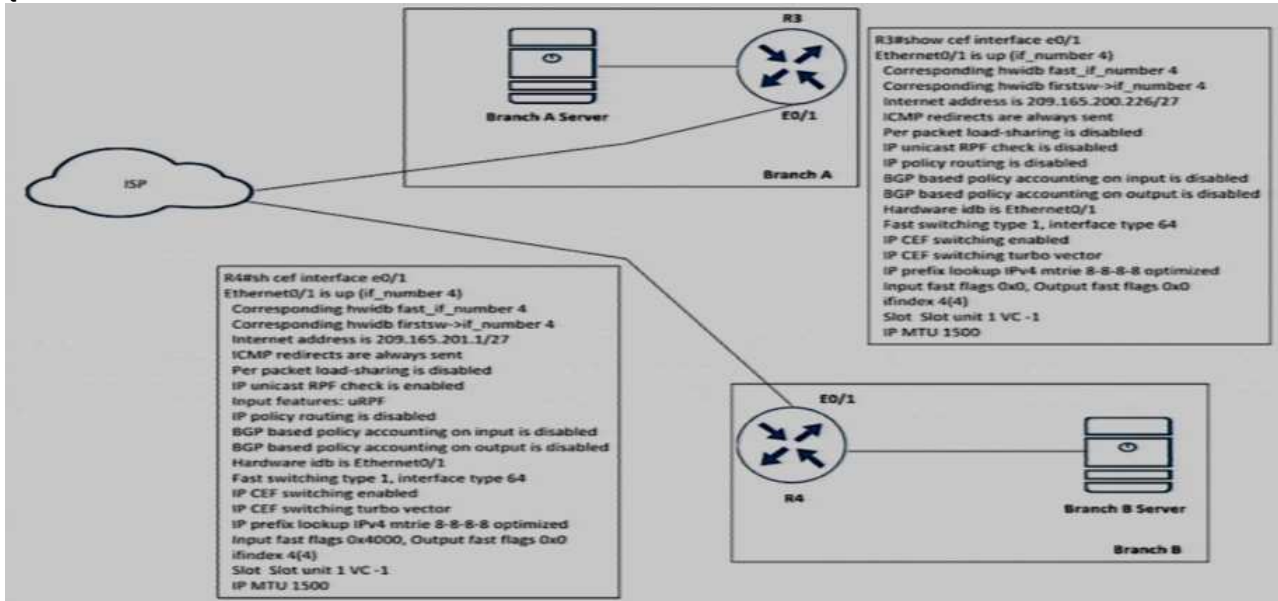
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 389



Refer to the exhibit. A shoe retail company implemented the uRPF solution for an antispoofing attack. A network engineer received the call that the branch A server is under an IP spoofing attack. Which configuration must be implemented to resolve the attack?

- A. R4
interface ethernet0/1
ip unicast RPF check reachable-via any allow-default allow-self-ping
- B. R4
interface ethernet0/1
ip verify unicast source reachable-via any allow-default allow-self-ping
- C. R3
interface ethernet0/1
ip unicast RPF check reachable-via any allow-default allow-self-ping
- D. R3
interface ethernet0/1
ip verify unicast source reachable-via any allow-default allow-self-ping

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

From the output, R3 has not enabled IP unicast RPF and R4 has enabled IP unicast RPF.

QUESTION 390

```
R1# configure terminal
R1(config)# hostname CPE1
CPE1(config)# ip domain-name example.com
CPE1(config)# crypto key generate rsa
The name for the keys will be: CPE1.example.com
Choose the size of the key modulus in the range of 360 to 4096
for your
  General Purpose Keys. Choosing a key modulus greater than 512
may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...
[OK] (elapsed time was 2 seconds)

CPE1(config)# service password-encryption
CPE1(config)# username csadmin secret Secur3p4s$w0rd
CPE1(config)# line vty 0 4
CPE1(config-line)# transport input telnet ssh
CPE1(config-line)# login local
CPE1(config-line)# end
CPE1# copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
CPE1# ssh 10.0.0.1
% No user specified nor available for SSH client
```

Refer to the exhibit. An administrator must harden a router, but the administrator failed to test the SSH access successfully to the router. Which action resolves the issue?

- A. SSH syntax must be `ssh -l user ip` to log in to the remote device
- B. Configure enable `secret` to log in to the device

- C. Configure SSH on the remote device to log in using SSH
- D. SSH must be allowed with the transport output ssh command

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

ssh -l csadmin 10.0.0.1 must be used unless the current exec session is already login using the username "csadmin".

Since that username / password pair has just been configured, the current exec session is probably a console / telnet connection in which no username has been used for login.

QUESTION 391

The network administrator configured the router for Control Plane Policing so that inbound SSH traffic is policed to 500 kbps. This policy must apply to traffic coming in from 10.10.10.0/24 and 192.168.10.0/24 networks.

```
access-list 100 permit ip 10.10.10.0 0.0.0.255 any
access-list 100 permit tcp 192.168.10.0 0.0.0.255 any eq 23
!
class-map CLASS-SSH
match access-group 100
!
policy-map PM-COPP
class CLASS-SSH
police 500000 conform-action transmit
!
interface E0/0
service-policy input PM-COPP
!
interface E0/1
service-policy input PM-COPP
```

The Control Plane Policing is not applied to SSH traffic and SSH is open to use any bandwidth available. Which configuration resolves this issue?

- A. no access-list 100
access-list 100 permit tcp 10.10.10.0 0.0.0.255 any eq 22
access-list 100 permit tcp 192.168.10.0 0.0.0.255 any eq 22
!
policy-map PM-COPP
class CLASS-SSH
no police 500000 conform-action transmit
police 500000 conform-action transmit exceed-action drop
- B. no access-list 100
access-list 100 permit tcp 10.10.10.0 0.0.0.255 any eq 22
access-list 100 permit tcp 192.168.10.0 0.0.0.255 any eq 22
- C. no access-list 100
access-list 100 permit tcp 10.10.10.0 0.0.0.255 any eq 22
access-list 100 permit tcp 192.168.10.0 0.0.0.255 any eq 22
!
interface E0/0
no service-policy input PM-COPP
!
interface E0/1
no service-policy input PM-COPP
!
control-plane
service-policy input PM-COPP
- D. interface E0/0
no service-policy input PM-COPP
!
interface E0/1
no service-policy input PM-COPP
!
control-plane
service-policy input PM-COPP

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

There are two problems in the existing configuration:

- The ACL 100 matches all traffic from 10.10.10.0/24 and telnet traffic from 192.168.10.0/24. Therefore the policy map cannot limit SSH traffic from 192.168.10.0/24
- The policy is applied to physical interface E0/0 and E0/1. Hence, it is not a Control Plane Policy and will not have effect if traffic is entering from other interfaces.

The correct modification to handle the issue is the one that fixes both the above 2 problems.

QUESTION 392


```

R1#show ip route ospf

    10.0.0.0/24 is subnetted, 7 subnets

O E2    10.4.9.0 [110/200] via 10.4.17.6, 00:06:43,
FastEthernet0/0

                [110/200] via 10.4.15.5, 00:06:43,
FastEthernet0/1

O IA    10.4.27.0 [110/2] via 10.4.15.5, 00:06:44,
FastEthernet0/1

O E2    10.4.49.0 [110/200] via 10.4.17.6, 00:06:43,
FastEthernet0/0

```

Refer to the exhibit. An engineer configures two ASBRs, 10.4.17.6 and 10.4.15.5, in an OSPF network to redistribute routes from EIGRP. However, both ASBRs show the EIGRP routes as equal costs even though the next-hop router 10.4.17.6 is closer to R1. How should the network traffic to the EIGRP prefixes be sent via 10.4.17.6?

- A. The administrative distance should be raised to 120 from the ASBR 10.4.15.5.
- B. The ASBR 10.4.17.6 should assign a tag to match and assign a lower metric than R1.
- C. The administrative distance should be raised to 120 from the ASBR 10.4.17.6.
- D. The redistributed prefixes should be advertised as Type 1.

Correct Answer: D

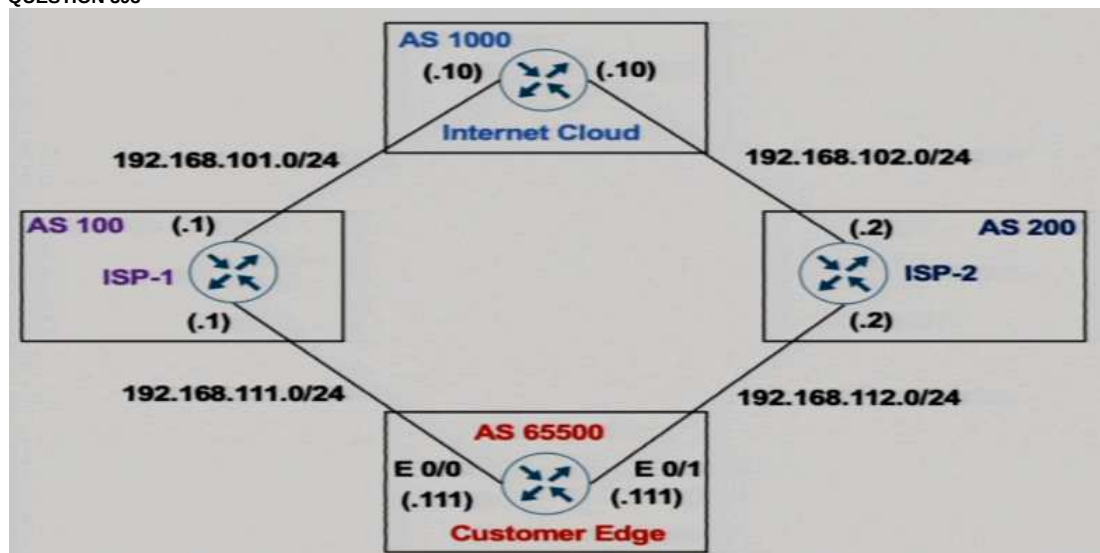
Section: (none)

Explanation

Explanation/Reference:

The simplest way to solve this issue is to use E1 type for the external route since the internal cost to the two next hops will be included in the external routes.

QUESTION 393



Refer to the exhibit. The Customer Edge router (AS 65500) wants to use AS 100 as the preferred ISP for all external routes.

Customer Edge

```

route-map SETLP

```

```

  set local-preference 111

```

```

!

```

```

router bgp 65500

```

```

  neighbor 192.168.111.1 remote-as 100

```

```

  neighbor 192.168.111.1 route-map SETLP out

```

```

  neighbor 192.168.112.2 remote-as 200

```

This configuration failed to send routes to AS 100 as the preferred path. Which set of configurations resolves the issue?

- A. route-map SETLP


```

        set local-preference 111
      
```

```

      !
      router bgp 65500
        neighbor 192.168.111.1 remote-as 100
        neighbor 192.168.111.1 route-map SETLP out
      
```
- B. route-map SETLP


```

        set local-preference 111
      
```

```

      !
      router bgp 65500
        neighbor 192.168.111.1 remote-as 100
        neighbor 192.168.111.1 route-map SETLP in
      
```
- C. route-map SETPP


```

        set as-path prepend 111 111
      
```

```

      !
      router bgp 65500
        neighbor 192.168.111.1 remote-as 100
        neighbor 192.168.111.1 route-map SETPP out
      
```

```

D. route-map SETPP
   set as-path prepend 100 100
   !
router bgp 65500
 neighbor 192.168.111.1 remote-as 100
 neighbor 192.168.111.1 route-map SETPP in

```

Correct Answer: B

Section: (none)

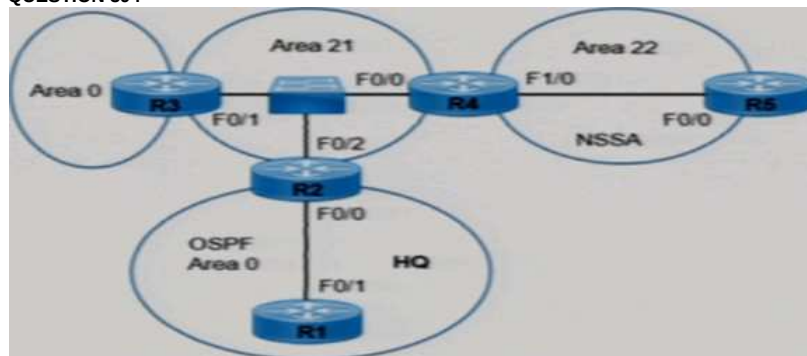
Explanation

Explanation/Reference:

In order to affect the attributes of the BGP route learnt, the route map should be applied with "in" direction.

Note that AS-path prepend will make the learnt BGP route less preferred and therefore using AS-path prepend on the BGP routes learnt from R1 is not suitable for this case.

QUESTION 394



```

R2(config)# router ospf 1
R2(config-router)# area 21 virtual-link 3.3.3.3

R3(config)# router ospf 1
*Apr  4 00:23:34.215: %OSPF-4-ERRRCV: Received invalid packet:
mismatch area ID, from backbone area must be virtual-link but not
found from 192.168.125.5, FastEthernet0/2
R3(config-router)# area 21 virtual-link 2.2.2.2
R3(config-router)# area 21 stub

```

Refer to the exhibit. A network engineer is troubleshooting a failed link between R2 and R3. No traffic loss is reported from router R5 to HQ. Which command fixes the separated backbone?

- A. R3(config-router)#area 21 virtual-link 192.168.125.5
- B. R3(config-router)#no area 21 stub
- C. R2(config-router)#area 21 virtual-link 192.168.125.5
- D. R2(config-router)#no area 21 stub

Correct Answer: B

Section: (none)

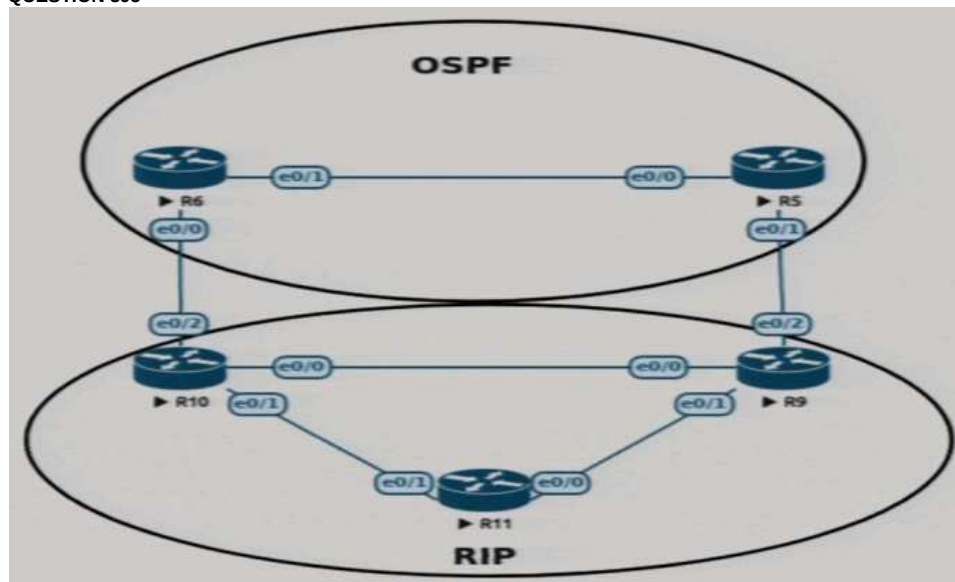
Explanation

Explanation/Reference:

Virtual links can be used to connect two parts of a partitioned backbone through a non-backbone transit area.

Although virtual link configuration is configured in both R2 and R3, problem occurs since the transit area "21" is a stub area which is not allowed for virtual link setup. Therefore the Stub setting configured in R3 has to be removed.

QUESTION 395



Refer to the exhibit. An engineer must configure OSPF with R9 and R10 and configure redistribution between OSPF and RIP, causing routing loop. Which

configuration on R9 and R10 meets this objective?

- A. `router ospf 1`
 `redistribute rip subnets tag 20`
 `!`
 `route-map deny_tag20 deny 10`
 `match tag 20`
 `route-map deny_tag20 permit 20`
 `!`
 `router ospf 1`
 `distribute-list route-map deny_tag20 in`
- B. `router ospf 1`
 `redistribute rip subnets tag 20`
 `!`
 `route-map deny_tag20 deny 10`
 `match tag 20`
 `route-map deny_tag20 permit 20`
 `!`
 `router rip 1`
 `distribute-list route-map deny_tag20 in`
- C. `router ospf 1`
 `redistribute rip subnets tag 20`
 `!`
 `route-map deny_tag20 permit 10`
 `match tag 20`
 `route-map deny_tag20 permit 20`
 `!`
 `router ospf 1`
 `distribute-list route-map deny_tag20 in`
- D. `router ospf 1`
 `redistribute rip subnets tag 20`
 `!`
 `route-map deny_tag20 deny 10`
 `match tag 20`
 `route-map deny_tag20 deny 20`
 `!`
 `router ospf 1`
 `distribute-list route-map deny_tag20 in`

Correct Answer: B

Section: (none)

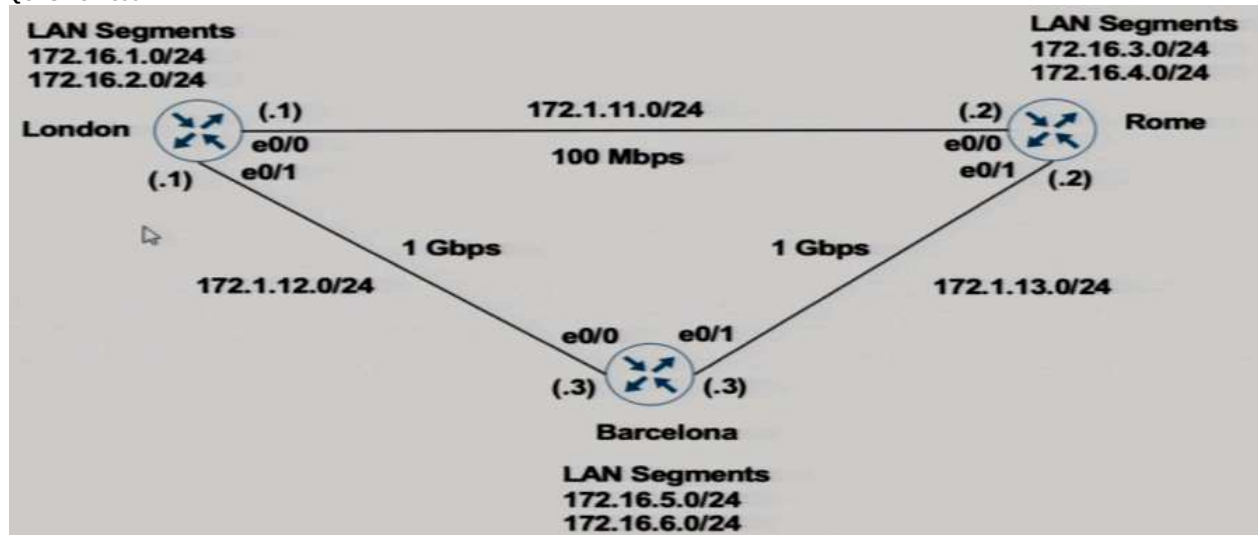
Explanation

Explanation/Reference:

One of the way to solve this is to tag the routes with e.g. 20 redistributed from RIP into OSPF.

Then when redistributing from OSPF into RIP, those routes originally sourced from RIP i.e. those tag with "20" should be denied.

QUESTION 396



```

London - "show ip route" output

Gateway of last resort is not set
C 172.1.12.0/24 is directly connected, Ethernet0/1
L 172.1.12.1/32 is directly connected, Ethernet0/1
D 172.1.13.0/24 [90/76800] via 172.1.11.2, 00:00:50, Ethernet0/0
  172.16.0.0/16 is variably subnetted, 8 subnets, 2 masks
C 172.16.1.0/24 is directly connected, Loopback0
L 172.16.1.1/32 is directly connected, Ethernet0/0
C 172.16.2.0/24 is directly connected, Loopback1
L 172.16.2.1/32 is directly connected, Loopback1
R 172.16.3.0/24 [120/1] via 172.1.11.2, 00:00:08, Ethernet0/0
R 172.16.4.0/24 [120/1] via 172.1.11.2, 00:00:08, Ethernet0/0
D 172.16.5.0/24 [90/156160] via 172.1.12.3, 00:00:50, Ethernet0/1
D 172.16.6.0/24 [90/156160] via 172.1.12.3, 00:00:50, Ethernet0/1

Rome - "show run | section router" output

router eigrp 111
 network 172.1.0.0
 redistribute connected
!
router rip
 version 2
 network 172.1.0.0
 network 172.16.0.0
 no auto-summary

```

Refer to the exhibits. London must reach Rome using a faster path via EIGRP if all the links are up, but it failed to take this path. Which action resolves the issue?

- A. Use the network statement on London to inject the 172.16.X.0/24 networks into EIGRP.
- B. Increase the bandwidth of the link between London and Barcelona
- C. Use the network statement on Rome to inject the 172.16.X.0/24 networks into EIGRP.
- D. Change the administrative distance of RIP to 150.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

London learns the 172.16.x.0/24 network through both EIGRP and RIP

- as EIGRP external route (due to redistribute connected) with AD 170

- as RIP route with AD 120.

Hence RIP routes are preferred. As RIP metric is based on hop count, London will send traffic for those networks to Rome directly through 100Mbps link instead of the path that goes through an additional router.

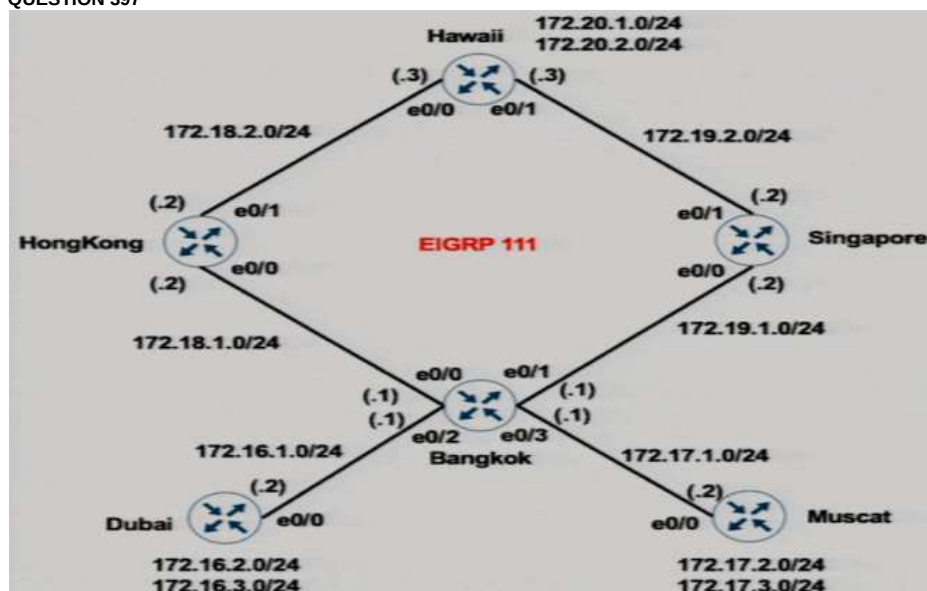
By injecting the 172.16.X.0/24 networks into EIGRP through network command, London then learns the 172.16.x.0/24 network

- as EIGRP internal route (due to redistribute connected) with AD 90

- as RIP route with AD 120.

Hence EIGRP routes are preferred. The minimum bandwidth will be considered and therefore London will send traffic for those networks to Barcelona taking advantages of the two 1Gbps links.

QUESTION 397



Refer to the exhibit. Bangkok is using ECMP to reach 172.20.2.0/24 network. The network administrator must configure it in such a way that traffic from 172.16.2.0/24 network uses the Singapore router as the preferred route. Which set of configurations accomplishes this task?

- A. Bangkok


```

access-list 101 permit ip 172.16.2.0 0.0.0.255 172.20.2.0 0.0.0.255
!
route-map PBR1 permit 10
 match ip address 101
 set ip next-hop 172.19.1.2
!
interface Ethernet0/2
 ip policy route-map PBR1

```
- B. Dubai

```

access-list 101 permit ip 172.16.2.0 0.0.0.255 172.20.2.0 0.0.0.255
!
route-map PBR1 permit 10
match ip address 101
set ip next-hop peer-address
!
interface Ethernet0/0
ip policy route-map PBR1

```

C. Dubai

```

access-list 101 permit ip 172.16.2.0 0.0.0.255 172.20.2.0 0.0.0.255
!
route-map PBR1 permit 10
match ip address 101
set ip next-hop 172.19.1.2
!
interface Ethernet0/0
ip policy route-map PBR1

```

D. Bangkok

```

access-list 101 permit ip 172.16.2.0 0.0.0.255 172.20.2.0 0.0.0.255
!
route-map PBR1 permit 10
match ip address 101
set ip next-hop 172.19.1.2
!
interface Ethernet0/1
ip policy route-map PBR1

```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

For routing based on information including source address, PBR should be used.

Since Bangkok should send the concerned traffic through Singapore router instead of the other equal cost multiple paths, PBR should be configured in Bangkok router.

Since the concerned traffic from network 172.16.2.0/24 enters Bangkok's e0/2, PBR setup should be applied to this interface.

QUESTION 398

Drag and drop the descriptions from the left onto the corresponding MPLS components on the right.

Select and Place:

FEC	routers in the core of the provider network known as P routers
LSP	all traffic to be forwarded using the same path and same label
LER	routers that connect to the customer routers known as PE routers
LSR	used for exchanging label mapping information between MPLS enabled routers
LDP	path along which the traffic flows across an MPLS network

Correct Answer:

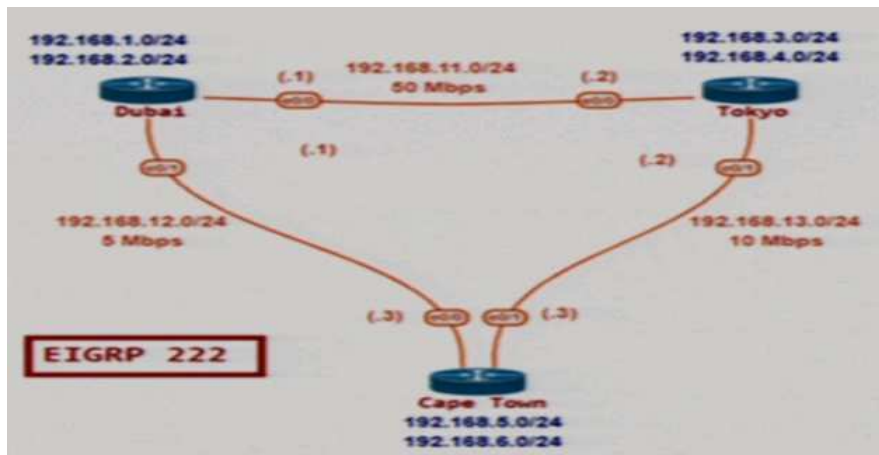
	LSR
	FEC
	LER
	LDP
	LSP

Section: (none)

Explanation

Explanation/Reference:

QUESTION 399



Routing table in Cape Town

```

Gateway of last resort is not set

D    192.168.1.0/24 [90/409600] via 192.168.12.1, 00:17:40, Ethernet0/0
D    192.168.2.0/24 [90/409600] via 192.168.12.1, 00:09:11, Ethernet0/0
D    192.168.3.0/24 [90/409600] via 192.168.13.2, 00:17:23, Ethernet0/1
D    192.168.4.0/24 [90/409600] via 192.168.13.2, 00:17:23, Ethernet0/1
D    192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.5.0/24 is directly connected, Loopback0
L    192.168.5.1/32 is directly connected, Loopback0
D    192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.6.0/24 is directly connected, Loopback1
L    192.168.6.1/32 is directly connected, Loopback1
D    192.168.11.0/24 [90/307200] via 192.168.13.2, 00:17:40, Ethernet0/1
    [90/307200] via 192.168.12.1, 00:17:40, Ethernet0/0
D    192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected, Ethernet0/0
L    192.168.12.3/32 is directly connected, Ethernet0/0
D    192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.13.0/24 is directly connected, Ethernet0/1
L    192.168.13.3/32 is directly connected, Ethernet0/1
  
```

Refer to the exhibit. The network administrator must configure Cape Town to reach Dubai via Tokyo based on the speeds provided by the service provider. It was noticed that Cape Town is reaching Dubai directly and failed to meet the requirement. Which configuration fixes the issue?

- A. **Cape Town**
`router eigrp 100`
`variance 2`
- B. **Cape Town**
`interface E0/0`
`bandwidth 5000`
`interface E0/1`
`bandwidth 10000`
- C. **Dubai**
`router eigrp 100`
`variance 2`
- D. **Cape Town**
`interface E0/0`
`bandwidth 5000`
`interface E0/1`
`bandwidth 10000`

Correct Answer: B

Section: (none)

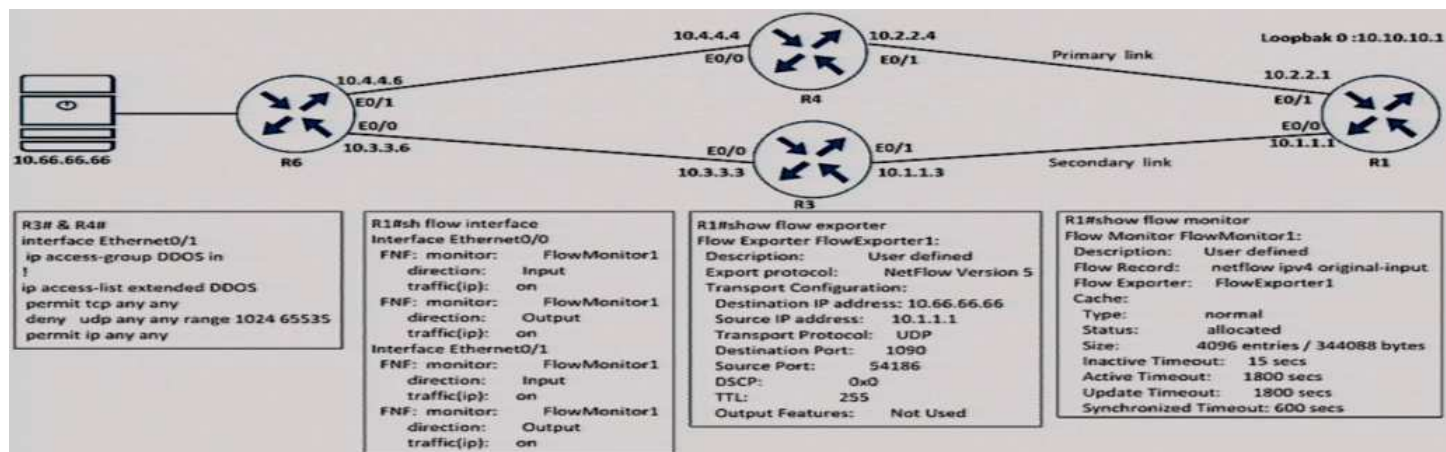
Explanation

Explanation/Reference:

From the EIGRP routes 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24 and 192.168.4.0/24 in Cape Town, all have the metric 409600. Since all networks are one-hop away, this means that the bandwidth settings configured for the two interfaces in Cape Town are the same (probably using the same default bandwidth of the Ethernet interface)

In order to use the bandwidth provided by the Service Provider for path selection, the bandwidth information should be configured in the interface which the routes are learnt. Since Dubai routes can be learnt via Tokyo and Tokyo routes can be learnt via Dubai, the appropriate bandwidth should be configured in all routers.

QUESTION 400



Refer to the exhibit. An engineer configured NetFlow but cannot receive the flows from R1. Which two configurations resolve the issue? (Choose two).

- A. R3(config)#ip access-list extended DDOS
 R3(config-ext-nacl)#5 permit udp any host 10.66.66.66 eq 1090
- B. R4(config)#ip access-list extended DDOS
 R4(config-ext-nacl)#5 permit udp any host 10.66.66.66 eq 1090
- C. R1(config)#flow exporter FlowExporter1
 R1(config-flow-exporter)#destination 10.66.66.66
- D. R3(config)#flow exporter FlowExporter1
 R3(config-flow-exporter)#destination 10.66.66.66
- E. R4(config)#flow exporter FlowExporter1
 R4(config-flow-exporter)#destination 10.66.66.66

Correct Answer: AB

Section: (none)

Explanation

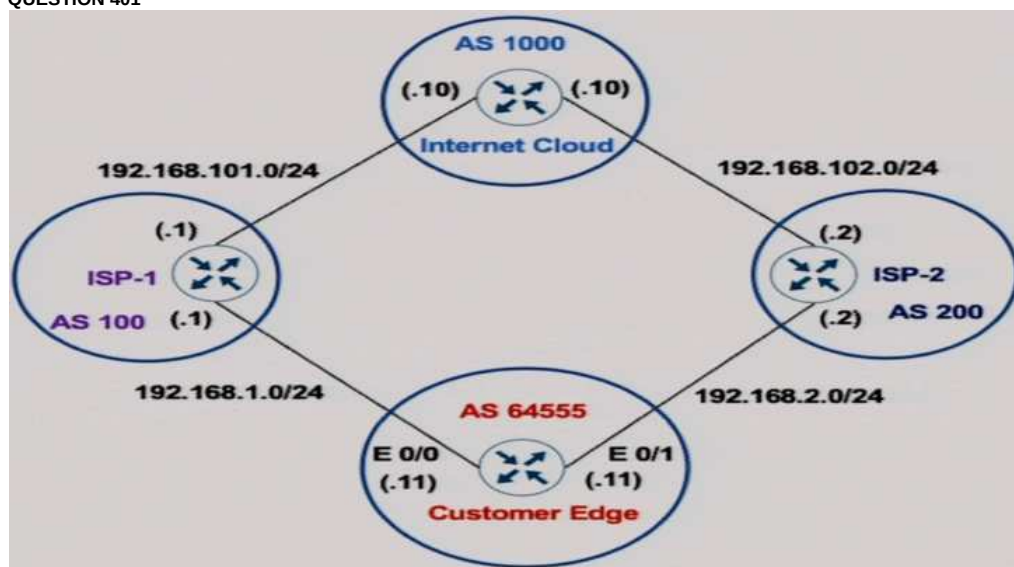
Explanation/Reference:

From the exhibit concerning R1:

- Exporter is configured properly with correct IP destination address.
- Monitor is configured properly with the above exporter
- Monitor is applied properly to the interfaces.

However, when Netflow traffic with destination port 1090 flows through R3 or R4, it is blocked by the ACL DDOS applied in e1/1 of R3 and R4. Hence you need to add a rule to permit the traffic in both routers.

QUESTION 401



Refer to the exhibit. The Customer Edge router wants to use AS 100 as the preferred ISP for all external routes and ISP-2 as a backup.

Customer-Edge

```
route-map SETAS
 set as-path prepend 111
```

```
router bgp 64555
 neighbor 192.168.1.1 remote-as 100
 neighbor 192.168.2.2 remote-as 200
 neighbor 192.168.2.2 route-map SETAS in
```

After this configuration, all the backup routes have disappeared from the BGP table on the Customer Edge router. Which set of configurations solves the issue on the Customer Edge router?

- A. route-map SETAS
 set as-path prepend 111

```

router bgp 64555
 neighbor 192.168.1.1 remote-as 100
 neighbor 192,168.2.2 remote-as 200
 neighbor 192168.2.2 route-map SETAS out
B. route-map SETAS
   set as-path prepend 200

router bgp 64555
 neighbor 192.168.1.1 remote-as 100
 neighbor 192,168.2.2 remote-as 200
 neighbor 192168.2.2 route-map SETAS in
C. route-map SETAS
   set as-path prepend 111

router bgp 64555
 neighbor 192.168.2.2 remote-as 100
 neighbor 192,168.1.1 remote-as 200
 neighbor 192168.2.2 route-map SETAS in
D. route-map SETAS
   set as-path prepend 200

router bgp 64555
 neighbor 192.168.1.1 remote-as 100
 neighbor 192,168.2.2 remote-as 200
 neighbor 192168.2.2 route-map SETAS out

```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

For the configuration in Customer Edge:

- If you want to use AS-path prepend to affect BGP route selection in the same router, the route-map should be applied as inbound direction.
- For forming neighbor with ISP-2 router, you need to setup 192.168.2.2 with remote AS number 200.

Based on the basic factor shown above, only one choice meets them.

Remarks:

Actually for the correct configuration mentioned in the question, there should be no problem i.e. BGP routes learnt from ISP-2 should be found in the BGP table and is not selected as best. There may be other factors not mentioned that cause the disappearing of those BGP routes. Since there is only one valid choice, it is chosen as suggested answer but this may or may not actually solve the problem of disappearing BGP routes.

On the other hand, prepending the AS number of the neighbor's AS is often preferred for inbound AS-path prepending.

QUESTION 402

A network administrator is troubleshooting a failed AAA login issue on a Cisco Catalyst c3560 switch. When the network administrator tries to log in through SSH using TACACS+ username and password credentials, the switch is no longer authenticating and is failing back to the local account. Which action resolves this issue?

- Configure ip tacacs-server source-interface GigabitEthernet 1/1.
- Configure ip tacacs source-ip 192.168.100.55.
- Configure ip tacacs-server source-ip 192.168.100.55.
- Configure ip tacacs source-interface GigabitEthernet 1/1.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Only one of them is a valid configuration command.

The help output for Cisco Catalyst c3560 switch is the same as the following router:

```
Router(config)#ip tacacs?
tacacs
```

```
Router(config)#ip tacacs ?
  source-interface Specify interface for source address in TACACS packets
```

QUESTION 403

A network engineer must configure a DMVPN network so that 3 spoke establishes a direct path to another spoke if the two must send traffic to each other. A spoke must send traffic directly to the hub if required. Which configuration meets this requirement?

- At the hub router:


```

interface tunnel10
 ip nhrp nhs multicast dynamic
 ip nhrp nhs shortcut
 tunnel mode gre multipoint

```

 On the spokes router:


```

interface tunnel10
 ip nhrp nhs multicast dynamic
 ip nhrp nhs redirect
 tunnel mode gre multipoint

```
- At the hub router:


```

interface tunnel10
 ip nhrp nhs dynamic multipoint
 ip nhrp nhs shortcut
 tunnel mode gre multicast

```

 On the spokes router:


```

interface tunnel10
 ip nhrp nhs multicast dynamic

```

```
ip nhrp nhs redirect
tunnel mode gre multicast
```

C. At the hub router:

```
interface tunnel10
ip nhrp map multicast dynamic
ip nhrp redirect
tunnel mode gre multipoint
```

On the spokes router:

```
interface tunnel10
ip nhrp map multicast dynamic
ip nhrp shortcut
tunnel mode gre multipoint
```

D. At the hub router:

```
interface tunnel10
ip nhrp map dynamic multipoint
ip nhrp redirect
tunnel mode gre multicast
```

On the spokes router:

```
interface tunnel10
ip nhrp nhs multicast dynamic
ip nhrp shortcut
tunnel mode gre multicast
```

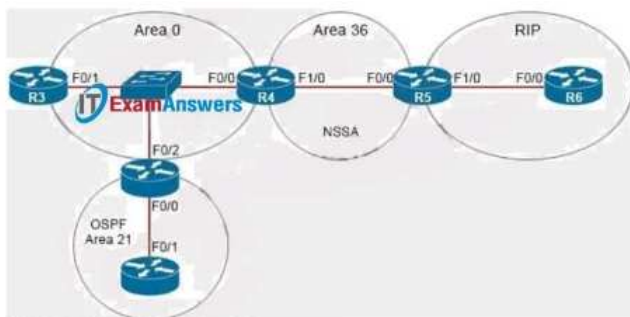
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 404



```
R5# show ip ospf 1 | begin Area 36
Area 36
Number of interfaces in this area is 2
It is a NSSA area
Area has no authentication
SPF algorithm last executed 00:32:46.376 ago
SPF algorithm executed 13 times
Area ranges are
172.16.0.0/16 Passive Advertise
```

Refer to the exhibit. The network engineer configured the summarization of the RIP routes into the OSPF domain on R5 but still sees four different 172.16.0.0/24 networks on R4. Which action resolves the issue?

- R4(config)#router ospf 1
R4(config-router)#no area
R4(config-router)#summary-address 172.16.0.0 255.255.252.0
- R4(config)#router ospf 99
R4(config-router)#network 172.16.0.0 0.255.255.255 area 56
R4(config-router)#summary-address 172.16.0.0 255.255.252.0
- R5(config)#router ospf 1
R5(config-router)#no area
R5(config-router)#summary-address 172.16.0.0 255.255.252.0
- R5(config)#router ospf 99
R5(config-router)#network 172.16.0.0 0.255.255.255 area 56
R5(config-router)#summary-address 172.16.0.0 255.255.252.0

Correct Answer: C

Section: (none)

Explanation

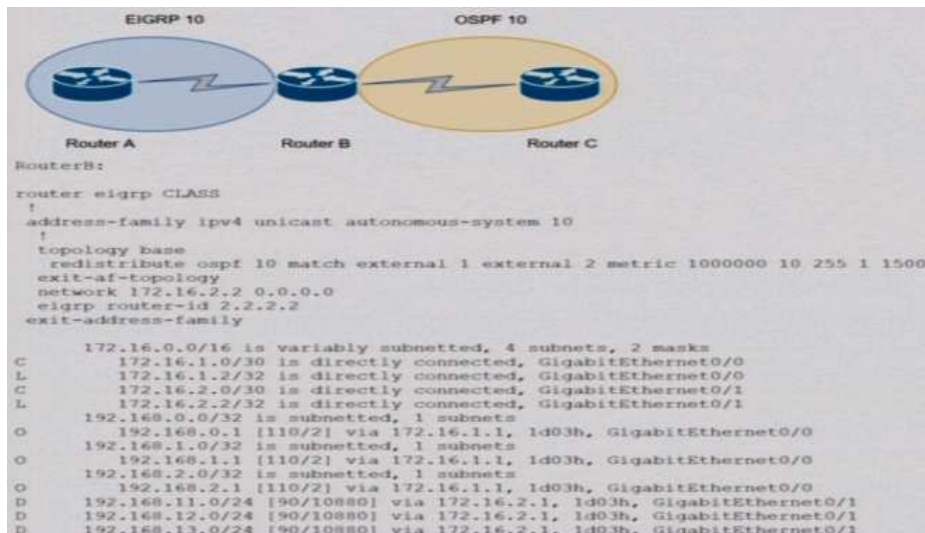
Explanation/Reference:

From the exhibit, R5 is configured with area ... range ... command to perform summarization for 172.16.0.0/16 but this cannot summarize route redistributed from RIP. You need to configure summary-address to perform the summarization.

Remarks :

Note that "no area" in the suggested answer is not a valid command. The valid command should be "no area 36" which can clear the area ... range ... configured in R5 as shown in the exhibit.

QUESTION 405



Refer to the exhibit. An engineer configured route exchange between two different companies for a migration project. EIGRP routes were learned in router C, but no OSPF routes were learned in router A. Which configuration allows router A to receive OSPF routes?

- A. (config-router-af-topology)#redistribute connected
- B. (config-router-af-topology)#redistribute ospf 10 metric 1000000 10 255 1 1500
- C. (config-router-af-topology)#no redistribute ospf 10 match external 1 external 2 metric 1000000 10 255 1 1500
- D. (config-router-af-topology)#redistribute ospf 10 1000000 10 255 1 1500

Correct Answer: B

Section: (none)

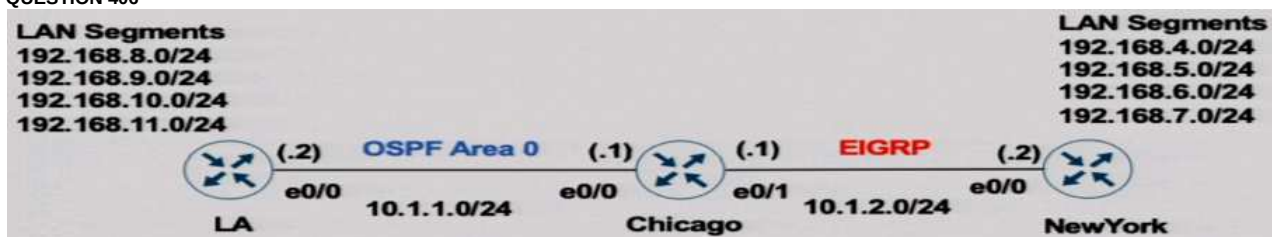
Explanation

Explanation/Reference:

Since router B is configured to redistribute OSPF routes into EIGRP, the internal OSPF routes shown in router B's routing table are not redistributed.

Note that as shown in the example in the exhibit, the redistribute command requires the keyword "metric" before entering the values.

QUESTION 406



Refer to the exhibit. The network administrator configures the Chicago router to mutually redistribute the LA and NewYork routes with EIGRP routes to be summarized as a single route in OSPF:

```

router eigrp 100
 redistribute ospf 1 metric 10 10 10 10 10
router ospf 1
 redistribute eigrp 100 subnets
 area 0 range 192.168.4.0 255.255.252.0

```

After the configuration, the LA router receives all the specific NewYork routes except the summary route. Which set of configurations resolve the issue on the Chicago router?

- A. router ospf 1
summary-address 192.168.4.0 255.255.252.0
- B. router ospf 1
area 0 range 192.168.4.0 255.255.0.0
- C. interface E 0/0
summary-address 192.168.4.0 255.255.252.0
- D. interface E 0/0
ip summary-address eigrp 100 192.168.4.0 255.255.252.0

Correct Answer: A

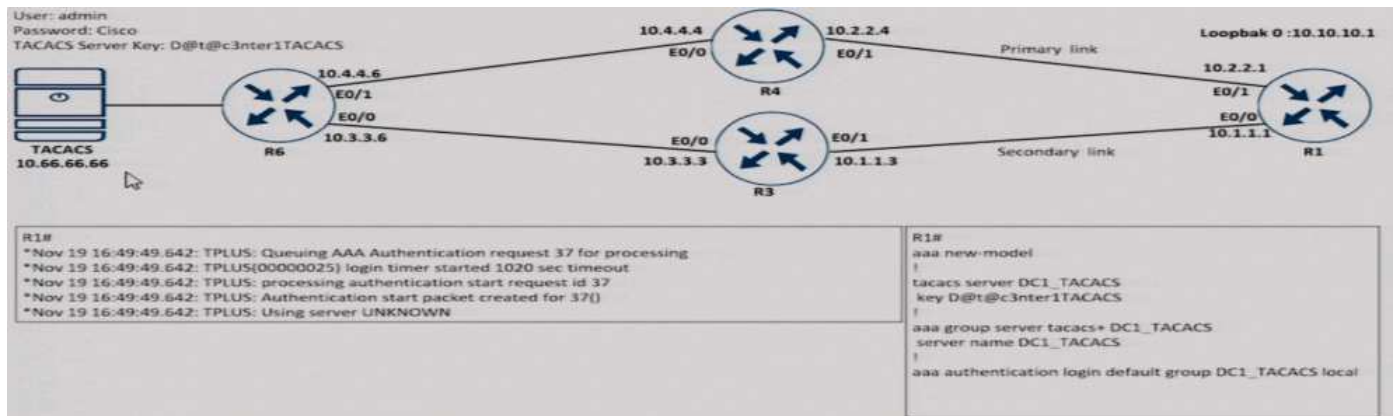
Section: (none)

Explanation

Explanation/Reference:

For summarizing OSPF routes redistributed in the ASBR, summary-address should be used instead of area ... range ...

QUESTION 407



Refer to the exhibit. R1 cannot authenticate via TACACS. Which configuration resolves the issue?

- A. tacacs server DC1_TACACS
address ipv4 10.60.66.66
key D@t@c3nter1TACACS
- B. aaa group server tacacs+ DC1_TACACS
server name DC_TACACS
- C. tacacs server DC1_TACACS
address ipv4 10.66.66.66
key D@t@c3nter1TACACS
- D. aaa group server tacacs+ DC_TACACS
server name DC_TACACS

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

For the R1's configuration in the exhibit, the tacacs server "DC1_TACACS" is missing the IP address "10.66.66.66".

QUESTION 408

What must be configured by the network engineer to circumvent AS_PATH loop prevention mechanism in IP/VPN Hub and Spoke deployment scenanos?

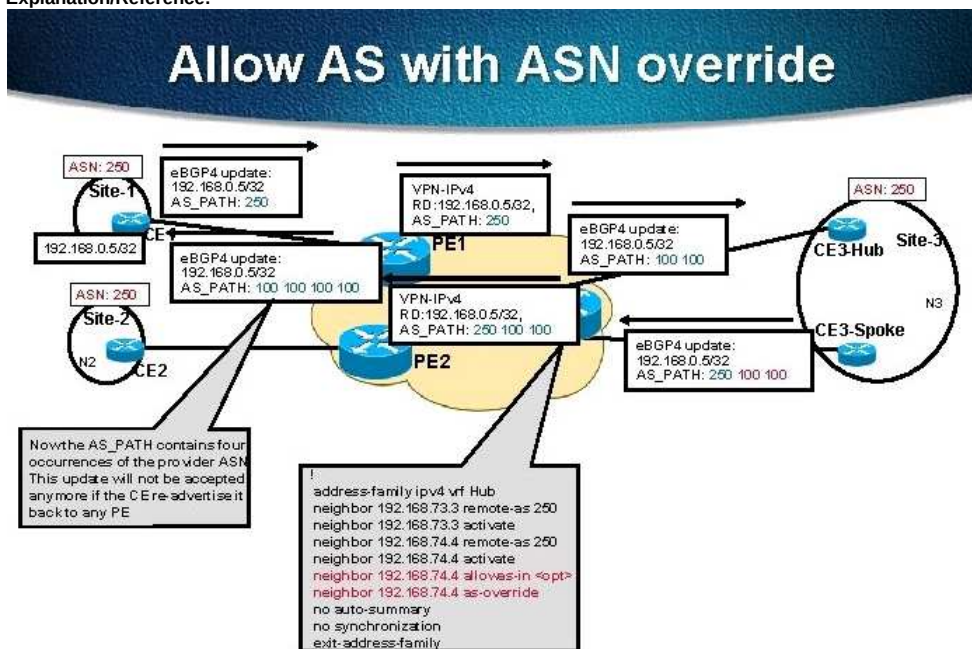
- A. Use allowas-in and as-override at all PEs
- B. Use allowas-in and as-override at the PE_Hub
- C. Use allowas at the PE_Hub
- D. Use as-override at the PE_Hub

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:



QUESTION 409


```

admin@linux:~$ telnet 198.51.100.64
Trying 198.51.100.64...
Connected to 198.51.100.64.
Escape character is '^]'.

User Access Verification

Password: admin
CPE> exit
Connection closed by foreign host.
admin@linux:~$ ssh 198.51.100.64
admin@198.51.100.64's password: admin
Permission denied, please try again.
admin@198.51.100.64's password: admin
Permission denied, please try again.
admin@198.51.100.64's password: admin
Connection closed by 198.51.100.64 port 22
admin@linux:~$

```

Refer to the exhibit. An administrator can log in to the device using Telnet, but the attempts to log in to the same device using SSH with the same credentials fail. Which action resolves this issue?

- A. Configure transport input all on the VTY lines to allow SSH.
- B. Configure the VTY lines with login local.
- C. Configure to use the Telnet user database for SSH as well.
- D. Configure SSH service on the router.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Unlike Telnet which can be logged in with password only, SSH must require username and password pair for login.

QUESTION 410

An engineer must override the normal routing behavior of a router. The engineer must send HTTP traffic that is destined to 10.100.100.100 from 10.1.1.0/24 via a next hop of 10.2.2.2, two hops away from the router that is connected to the 10.1.1.0/24 subnet. Which configuration reroutes traffic according to this requirement?

- A.

```
access-list 100 permit tcp 10.1.1.0 0.0.0.255 host 10.100.100.100 eq http
!
route-map POLICY permit 10
 match ip address 100
 set ip next-hop recursive 10.2.2.2
```
- B.

```
access-list 100 permit tcp 10.1.1.0 0.0.0.255 host 10.100.100.100 eq http
!
route-map POLICY permit 10
 match ip address 100
 set ip next-hop 10.2.2.2
route-map POLICY permit 20
```
- C.

```
access-list 100 permit tcp 10.1.1.0 0.0.0.255 host 10.100.100.100 eq http
!
route-map POLICY permit 10
 match ip address 100
 set ip next-hop recursive 10.2.2.2
route-map POLICY permit 20
```
- D.

```
access-list 100 permit tcp 10.1.1.0 0.0.0.255 host 10.100.100.100 eq http
!
route-map POLICY permit 10
 match ip address 100
 set ip next-hop 10.2.2.2
```

Correct Answer: A

Section: (none)

Explanation

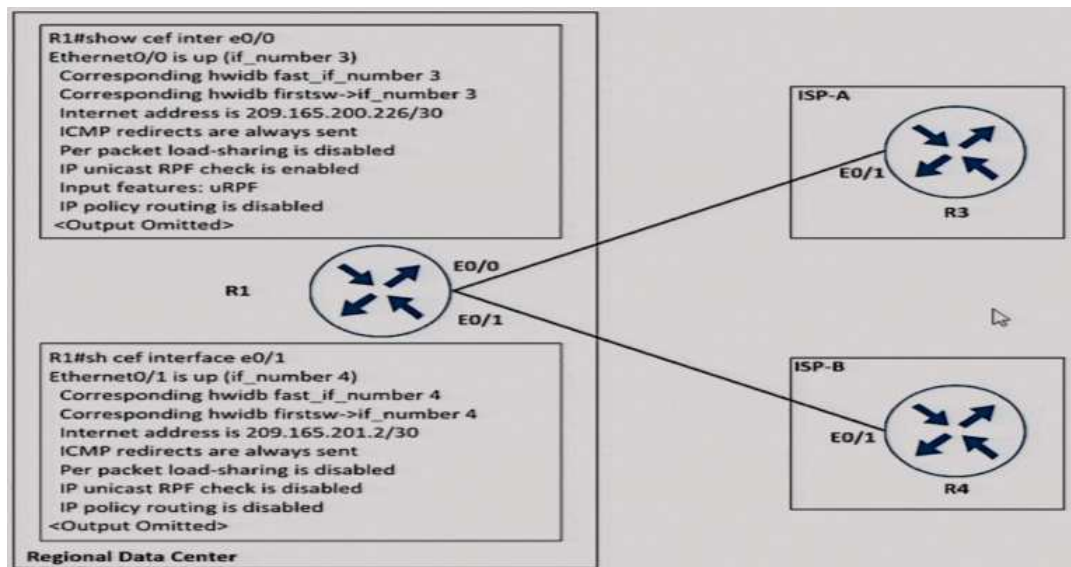
Explanation/Reference:

Since routing decision has to be based on source 10.1.1.0/24, PBR is required.

For configuring PBR with a next hop address that is not directly connected (i.e. 10.2.2.2 in this case), the keyword "recursive" is required.

Note that the rule 20 that match all for permitting must NOT be configured since this will perform PBR for all other traffic as well which is not intended.

QUESTION 411



Refer to the exhibit. The company implemented uRPF to address an antispoofing attack. A network engineer received a call from the IT security department that the regional data center is under an IP spoofing attack. Which configuration must be implemented on R1 to resolve this issue?

- interface ethernet0/1
ip unicast RPF check reachable-via any allow-default allow-self-ping
- interface ethernet0/0
ip verify unicast reverse-path
- interface ethernet0/1
ip verify unicast reverse-path
- interface ethernet0/0
ip unicast RPF check reachable-via any allow-default allow-self-ping

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

From the output in the exhibit:

- RPF check in e0/0 is enabled
- RPF check in e0/1 is disabled

Therefore RPF should be enabled in e0/1 using "ip verify unicast" with the option "reverse-path ..." or "source ...".

```
Router(config-if)#ip verify unicast ?
notification drop-rate notify
reverse-path Reverse path validation of source address (old command format)
source Validation of source address
```

QUESTION 412

What is a characteristic of IPv6 RA Guard?

- RA messages are allowed from the host port to the switch.
- It is unable to protect tunneled traffic
- It filters rogue RA broadcasts from connected hosts
- It is supported on the egress direction of the switch.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

The IPv6 RA Guard feature provides support for allowing the network administrator to **block or reject unwanted or rogue router advertisement (RA) guard messages** that arrive at the network device platform. RA messages

If a port is applied with a policy with device-role "host", all RA and router redirect messages are disallowed on the port. RA messages are only allowed for ports applied with a policy with device-role "router".

Restrictions for IPv6 RA Guard

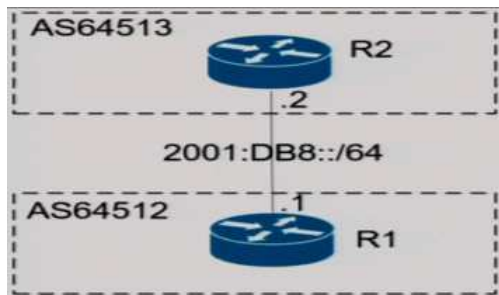
- The IPv6 RA Guard feature **does not offer protection in environments where IPv6 traffic is tunneled**.
- This feature is supported only in hardware when the ternary content addressable memory (TCAM) is programmed.
- This feature can be configured on a switch port interface in the ingress direction only (i.e. egress direction not supported)
- This feature is not supported on EtherChannel and EtherChannel port members.
- This feature is not supported on trunk ports with merge mode.

Actually it seems that two choices can be correct:

- It filters rogue RA broadcasts from connected hosts
- It is unable to protect tunneled traffic

C is chosen here since it is the main characteristics of IPv6 RA Guard. However, B is also correct.

QUESTION 413



```
R1#show ipv6 access-list
IPv6 access list inbound-acl
  permit tcp host 2001:DB8::2 eq bgp host 2001:DB8::1 (75 matches) sequence 20
  permit tcp host 2001:DB8::2 host 2001:DB8::1 eq bgp (17 matches) sequence 30
  deny ipv6 2001:DB8::/32 any (77 matches) sequence 40
  permit ipv6 any (20 matches) sequence 1000
R1#ping ipv6 2001:DB8::2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8::2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
R1#show ipv6 access-list
IPv6 access list inbound-acl
  permit tcp host 2001:DB8::2 eq bgp host 2001:DB8::1 (77 matches) sequence 20
  permit tcp host 2001:DB8::2 host 2001:DB8::1 eq bgp (19 matches) sequence 30
  deny ipv6 2001:DB8::/32 any (95 matches) sequence 40
  permit ipv6 any (23 matches) sequence 1000
R1#
```

Refer to the exhibit. An engineer applied an IPv6 traffic filter on R1. The interface flapped between R1 and R2 and clearing the BGP session did not restore the BGP session and failed. Which action must the engineer take to restore the BGP session from R2 to R1?

- A. Enable the BGP session, which went down when the session was cleared.
- B. Swap the source and destination IP addresses in the IPv6 traffic filter.
- C. Apply the IPv6 traffic filter in the outbound direction on the interface.
- D. ICMPv6 must be permitted by the IPv6 traffic filter.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Since the rules "deny ipv6 2001:DB8::/32 any" blocks all ICMPv6 messages sent by R2, Neighbor Solicitation / Advertisement messages will be blocked and this will prevent R1 / R2 to communicate with each other.

QUESTION 414

CPE#	show snmp mib ifmib ifindex detail					
Description		ifIndex	Active	Persistent	Saved	TrapStatus
Loopback1		8	yes	disabled	no	enabled
GigabitEthernet1		1	yes	disabled	no	enabled
GigabitEthernet3		3	yes	disabled	no	enabled
GigabitEthernet3.123		10	yes	disabled	no	disabled
VoIP-Null0		5	yes	disabled	no	enabled
Loopback0		7	yes	disabled	no	enabled
Null0		6	yes	disabled	no	enabled
Loopback2		9	yes	disabled	no	enabled
GigabitEthernet4		4	yes	disabled	no	enabled
GigabitEthernet2		2	yes	disabled	no	enabled

Refer to the exhibit. After reloading the router, an administrator discovered that the interface utilization graphs displayed inconsistencies with their previous history in the NMS. Which action prevents this issue from occurring after another router reload in the future?

- A. Rediscover all the router interfaces through SNMP after the router is reloaded.
- B. Save the router configuration to startup-config before reloading the router.
- C. Configure SNMP to use static OIDs referring to individual router interfaces.
- D. Configure SNMP interface index persistence on the router.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 415

```

R1(config)#ip prefix-list EIGRP seq 10 permit 10.0.0.0/8
R1(config)#ip prefix-list EIGRP seq 20 deny 0.0.0.0/0 le 32
R1(config)#router eigrp 10
R1(config-router)#distribute-list prefix EIGRP in Ethernet0/0

R1#show ip route eigrp | include 10.
D EX 10.0.0.0/8 [170/2665332] via 192.168.10.1, 00:00:10,
Ethernet0/0

```

Refer to the exhibit. An engineer applies a prefix-list filter that filters most of the network 10 prefixes instead of allowing them. Which action resolves the Issue?

- A. Modify the ip prefix-list EIGRP seq 10 permit 10.0.0.0/8 le 9 command.
- B. Modify the ip prefix-list EIGRP seq 20 permit 10.0.0.0/8 ge 9 command.
- C. Modify the ip prefix-list EIGRP seq 10 permit 10.0.0.0/8 le 32 command.
- D. Modify the ip prefix-list EIGRP seq 20 permit 10.0.0.0/8 le 32 command.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

For the rule sequence 10 in the prefix-list, it allows the 10.0.0.0 route with exactly prefix length of 8.

In order to allow subnets with 10.0.0.0/8, you need to configure:

- two rules that permit 10.0.0.0/8 and 10.0.0.0/8 ge 9
- one rule that permit 10.0.0.0/8 le 32

Among the choices, assume that you need to preserve the existing rule "deny 0.0.0.0/0 le 32" of sequence 20, you should configure a single rule to replace sequence 10. Hence 10.0.0.0/8 le 32 should be used.

Remarks:

Actually "seq 20 permit 10.0.0.0/8 ge 9" can be an answer since even the deny all in seq 20 is replaced, there is an implicitly deny in the prefix list that perform the same behavior. However "seq 10 permit 10.0.0.0/8 le 32" seems to be a better answer.

QUESTION 416

```

RtrA#show ip eigrp topology all-links
IP-EIGRP Topology Table for AS(1)/ID(10.1.6.1)
.....snip.....
P 10.200.1.0/24, 1 successors, FD is 21026560
via 10.1.1.2 (21026560/20514560), Serial1/0
via 10.1.2.2 (46740736/20514560), Serial1/1
via 10.1.3.2 (46740736/46228736), Serial1/2

```

Refer to the exhibit. Which action makes 10.1.3.2 the feasible successor to reach 10.200.1.0/24?

- A. Increase path bandwidth higher than 10.1.1.2 and lower than 10.1.2.2 between RtrA and the destination.
- B. Increase path bandwidth higher than 10.1.2.2 and lower than 10.1.1.2 between RtrA and the destination.
- C. Increase path bandwidth lower than 10.1.1.2 and lower than 10.1.2.2 between RtrA and the destination.
- D. Increase path bandwidth higher than 10.1.1.2 and higher than 10.1.2.2 between RtrA and the destination.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

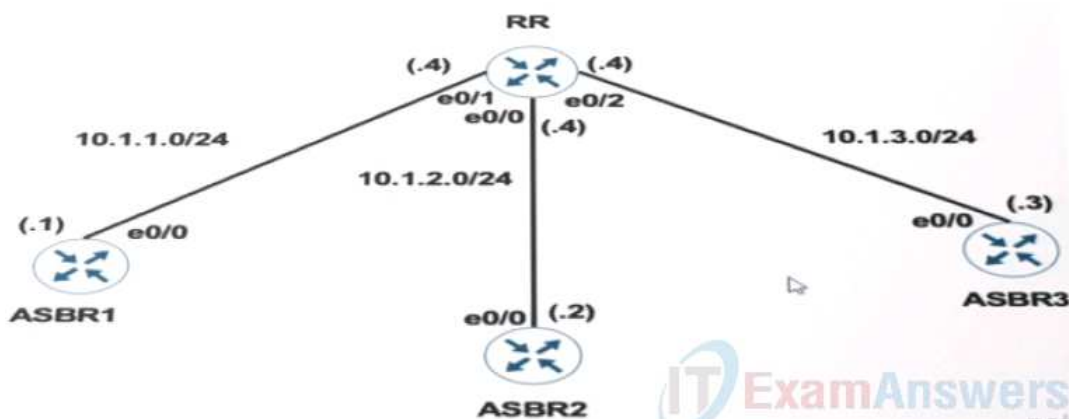
Assume all metric is mainly based on bandwidth:

- In order to be the feasible successor, the bandwidth should be lower than the successor 10.1.1.2. (i.e. preventing 10.1.3.2 becomes the successor).
- In order to be the feasible successor, the bandwidth should be higher than the successor 10.1.2.2.

Remarks :

Note that even after changes is made, 10.1.3.2 will NOT be the feasible succesor since its reported distance 46228736 is larger than the FD 21026560. For example you need to decrease the bandwidth between RtrA and 10.1.1.2 so that FD is greater than 46228736. However, since none of the choices deal with this issue, it is ignored in choosing the suggested answer.

QUESTION 417



Refer to the exhibit-

RR Configuration:

```

router bgp 100
 neighbor IBGP peer-group
 neighbor IBGP route-reflector-client
 neighbor 10.1.1.1 remote-as 100
 neighbor 10.1.2.2 remote-as 100
 neighbor 10.1.3.3 remote-as 100

```

The network administrator configured the network to establish connectivity between all devices and notices that the ASBRs do not have routes for each other. Which set of configurations resolves this issue?

- A.

```
router bgp 100
 neighbor 10.1.1.1 next-hop-self
 neighbor 10.1.2.2 next-hop-self
 neighbor 10.1.3.3 next-hop-self
```
- B.

```
router bgp 100
 neighbor IBGP next-hop-self
```
- C.

```
router bgp 100
 neighbor IBGP update-source Loopback0
```
- D.

```
router bgp 100
 neighbor 10.1.1.1 peer-group IBGP
 neighbor 10.1.2.2 peer-group IBGP
 neighbor 10.1.3.3 peer-group IBGP
```

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

The neighbors of RR has not been assigned as RR client since those neighbors have not been assigned to the peer-group "IBGP"

QUESTION 418

The network administrator configured CoPP so that all HTTP and HTTPS traffic from the administrator device located at 172.16.1.99 toward the router CPU is limited to 500 kbps. Any traffic that exceeds this limit must be dropped.

```
access-list 100 permit ip host 172.16.1.99 any
```

```
class-map CM-ADMIN
 match access-group 100
```

```
policy-map PM-COPP
 class CM-ADMIN
  police 500000 conform-action transmit
```

```
interface E0/0
 service-policy input PM-COPP
```

CoPP failed to capture the desired traffic and the CPU load is getting higher. Which two configurations resolve the issue? (Choose two.)

- A.

```
no access-list 100
 access-list 100 permit tcp host 172.16.1.99 any eq 80
 access-list 100 permit tcp host 172.16.1.99 any eq 443
```
- B.

```
no access-list 100
 access-list 100 permit tcp host 172.16.1.99 any eq 80
```
- C.

```
policy-map PM-COPP
 class CM-ADMIN
  no police 500000 conform-action transmit
  police 500 conform-action transmit
!
```
- D.

```
interface E0/0
 no service-policy input PM-COPP
!
```
- E.

```
policy-map PM-COPP
 class CM-ADMIN
  no police 500000 conform-action transmit
  police 500 conform-action transmit
```

Correct Answer: AD

Section: (none)

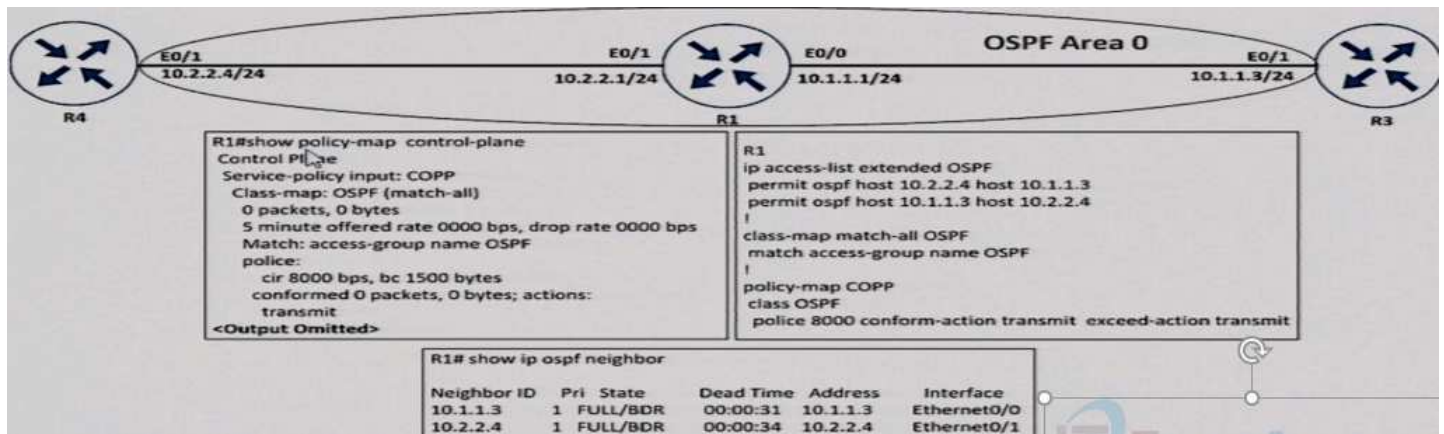
Explanation

Explanation/Reference:

In order to limit the traffic entering through any interface, you should configure CoPP which requires the apply of the policy map to "control-plane". Moreover, in order to meet the requirement, you also need to modify the ACL to match only HTTP and HTTPS traffic

Note that there is no need to change the policy map since the default unit for policing bandwidth is in kbps.

QUESTION 419



Refer to the exhibit. An engineer implemented CoPP but did not see OSPF traffic going through it. Which configuration resolves the issue?

- A. ip access-list extended OSPF
permit ospf any any
- B. policy-map COPP
class OSPF
police 8000 conform-action transmit exceed-action transmit violate-action drop
- C. class-map match-all OSPF
match access-group name OSPF
- D. control-plane
service-policy input COPP

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Based on the output, all except one is the current configuration settings.

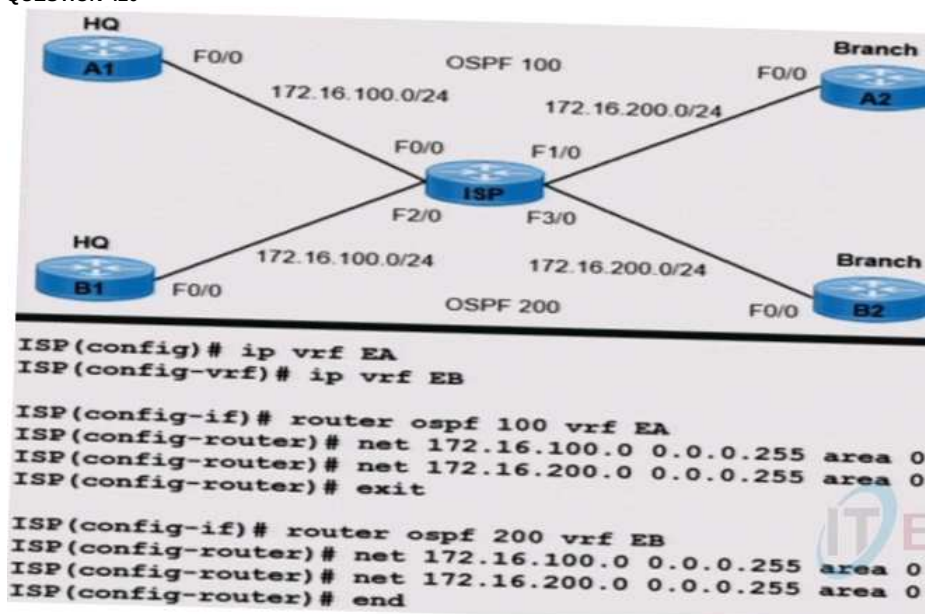
The one that is not currently configured is the permit of all OSPF traffic regardless of source / destination. Therefore it should be the answer.

The current configuration will have problem if there are other settings not shown in the policy map "COPP" that drops the remaining traffic. It is because OSPF may use multicast traffic for communication.

Remarks:

Note that without other settings in the policy map "COPP", since all remaining traffic will be allowed without bandwidth limitation, no problem should occur.

QUESTION 420



Refer to the exhibit A network engineer is provisioning end-to-end traffic service for two different enterprise networks with these requirements:

- The OSPF process must differ between customers on HQ and Branch office routers, and adjacencies should come up instantly.
- The enterprise networks are connected with overlapping networks between HQ and a Branch office.

Which configuration meets the requirements for a customer site?

- A. LSP(config)#int f3/0
LSP(config-if)#ip vrf forwarding EA
LSP(config-if)#description T0->EA2_Branch
LSP(config-if)#ip address 172.16.200.2 255.255.255.0
LSP(config-if)#no shut
- B. LSP(config)#int f2/0
LSP(config-if)#ip vrf forwarding EA
LSP(config-if)#description T0->EA1_HQ
LSP(config-if)#ip address 172.16.100.2 255.255.255.0


```

LSP(config-if)#no shut
C. LSP(config)#int f0/0
LSP(config-if)#ip vrf forwarding EB
LSP(config-if)#description T0->EB1_HQ
LSP(config-if)#ip address 172.16.100.2 255.255.255.0
LSP(config-if)#no shut
D. LSP(config)#int f1/0
LSP(config-if)#ip vrf forwarding EA
LSP(config-if)#description T0->EA2_Branch
LSP(config-if)#ip address 172.16.200.2 255.255.255.0
LSP(config-if)#no shut

```

Correct Answer: D

Section: (none)

Explanation

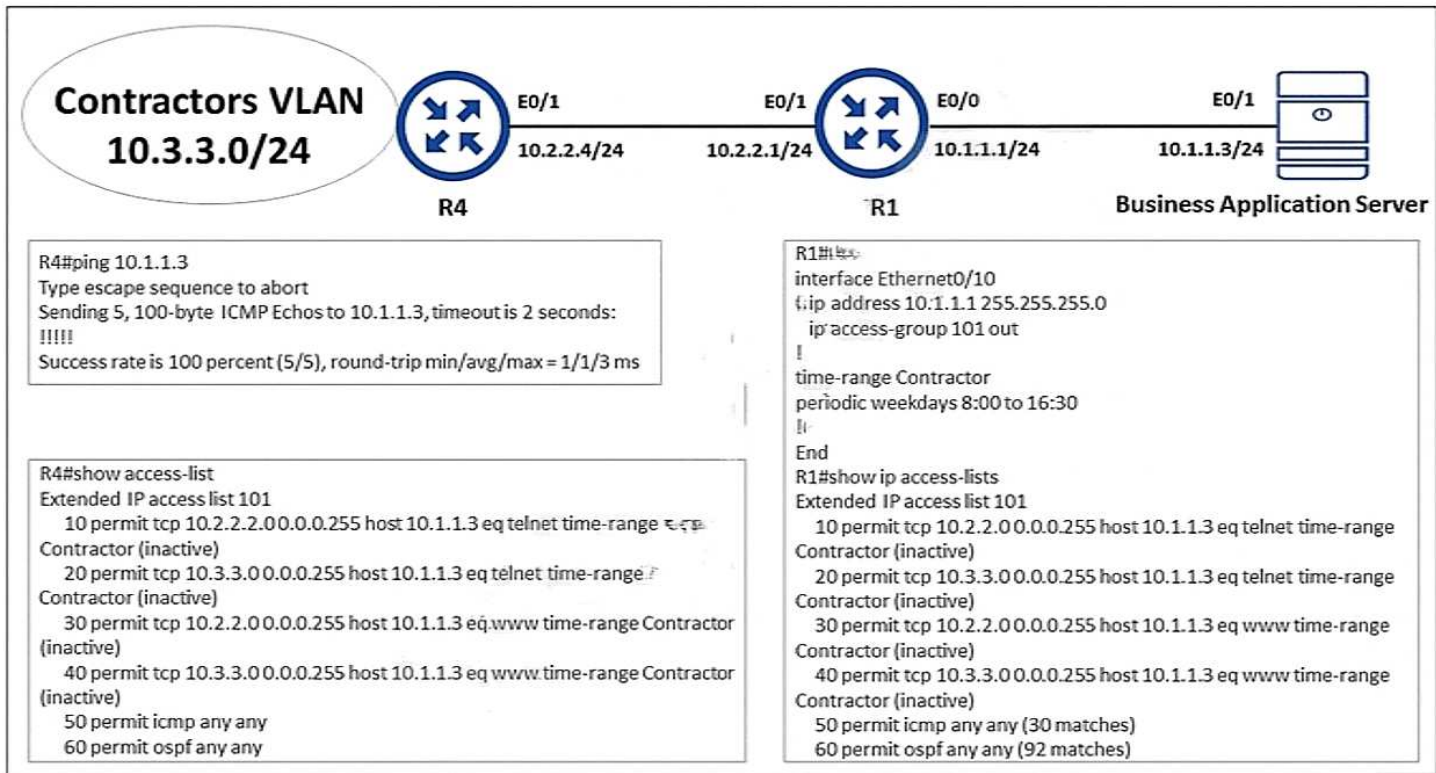
Explanation/Reference:

From the exhibit:

- OSPF process 100 is running for VRF EA. Since the diagram shows that ISP's f0/0 and f1/0 belongs to OSPF 100, therefore f0/0 and f1/0 should be assigned to VRF EA.
- OSPF process 200 is running for VRF EB. Since the diagram shows that ISP's f2/0 and f3/0 belongs to OSPF 200, therefore f2/0 and f3/0 should be assigned to VRF EB.

Only the choice in which f1/0 is being assigned to VRF EA meets the above two conditions.

QUESTION 421



Refer to the exhibit. An engineer is troubleshooting failed access by contractors to the business application server via Telnet or HTTP during the weekend. Which configuration resolves the issue?

- A. R1
no access-list 101 permit tcp 10.3.3.0 0.0.0.255 host 10.1.1.3 eq telnet time-range Contractor
- B. R4
no access-list 101 permit tcp 10.3.3.0 0.0.0.255 host 10.1.1.3 eq telnet time-range Contractor
- C. R1
time-range Contractor
no periodic weekdays 8:00 to 16:30
periodic daily 8:00 to 16:30
- D. R4
time-range Contractor
no periodic weekdays 8:00 to 16:30
periodic daily 8:00 to 16:30

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

In the current configuration in R1, access from contractors' networks are allowed only on certain time period in weekdays. In order to allow access in weekend, the time-range Contractor has to be changed in R1.

Remarks:

For R4, since the access list with rules using time-range is shown as inactive, the time-range "Contractor" should have been configured in R4 and is not matched at the time the output is shown. However, since you can only choose one answer in this question, the choice with time-range being modified at R1 is chosen. It is because, the question does not mention whether the ACL 101 has been applied to any interface in R4.

QUESTION 422

```
int GigabitEthernet0/0
no shut
int GigabitEthernet0/0.1
encapsulation dot1Q 1
ip address 10.1.1.1 255.255.255.0
int GigabitEthernet0/0.2
encapsulation dot1Q 2
ip address 10.2.2.1 255.255.255.0
```

Refer to the exhibit. Two routers are connected back to back via Gigabit Ethernet 0/0 interfaces. Which configuration provides VRF-Lite connectivity for two separate VRFs using the prefixes 10.1.1.0/24 for one VRF and 10.2.2.0/24 for the other VRF?

- A.

```
ip vrf 1
rd 65001:1
ip vrf 2
rd 65001:2
!
int GigabitEthernet0/0
no shut
!
int GigabitEthernet0/0.1
encapsulation dot1Q 1
ip vrf forwarding
ip address 10.1.1.1 255.255.255.0
!
int GigabitEthernet0/0.2
encapsulation dot1Q 2
ip vrf forwarding
ip address 10.2.2.1 255.255.255.0
```
- B.

```
ip vrf 1
ip vrf 2
!
int GigabitEthernet0/0
no shut
!
int GigabitEthernet0/0.1
encapsulation dot1Q 1
ip address 10.1.1.1 255.255.255.0
ip vrf forwarding 1
!
int GigabitEthernet0/0.2
encapsulation dot1Q 2
ip address 10.2.2.1 255.255.255.0
ip vrf forwarding 2
```
- C.

```
ip vrf 1
ip vrf 2
!
int GigabitEthernet0/0
no shut
!
int GigabitEthernet0/0.1
encapsulation dot1Q 1
ip vrf forwarding 1
ip address 10.1.1.1 255.255.255.0
!
int GigabitEthernet0/0.2
encapsulation dot1Q 2
ip vrf forwarding 2
ip address 10.2.2.1 255.255.255.0
```
- D.

```
ip vrf 1
rd 65001:1
ip vrf 2
rd 65001:2
!
int GigabitEthernet0/0
no shut
!
int GigabitEthernet0/0.1
encapsulation dot1Q 1
ip vrf forwarding 1
ip address 10.1.1.1 255.255.255.0
!
int GigabitEthernet0/0.2
encapsulation dot1Q 2
ip vrf forwarding 2
ip address 10.2.2.1 255.255.255.0
```

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

For configuring VRF:

- An interface / subinterface can be assigned to a VRF with the command "ip vrf forwarding" and must be followed by the VRF name.
- After assigning VRF to an interface / subinterface, the IP address setting will be cleared and must be configured again.

Unlike using VRF in VPNv4, the configuration of RD for each VRF is optional.

QUESTION 423

```

R2
interface LoopBack 0
  no ip address
  ipv6 address 2002::1/128
  ipv6 eigrp 1

interface FastEthernet0/1
  ipv6 address autoconfig
  ipv6 eigrp 1
!
ipv6 router eigrp 1
  stub summary
  no shutdown

```

Refer to the exhibit. R1 connecting to R2's f0/1 cannot receive the R2 prefix. What must be reconfigured to advertise R2 interfaces to R1?

- A. EIGRP process on R2 by removing the stub command keyword summary
- B. EIGRP process on R2 with the command stub summary connected
- C. interface FastEthernet0/1 on R2 with an EIGRP summary for the loopback prefix
- D. EIGRP process on R2 with the command stub summary receive-only

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since R2 is configured with "stub summary", only summarized route will be advertised. In order to advertise the connected network e.g. lo0 of R2 through EIGRP, you can:

- add the "connected" keyword i.e. "eigrp stub summary connected" OR
- just configure "eigrp stub" which includes connected network by default.

Therefore two choices can achieve the result.

However, the choice does not mention how to remove the keyword "summary". If you use the command "no eigrp stub summary", the entire "stub" settings will be removed. Hence the using of "eigrp stub summary connected" is more clear and therefore is selected as suggested answer.

Remarks:

Note that the question is wrong since the command "stub summary" shown in the exhibit is NOT valid. It should be "eigrp stub summary".

QUESTION 424

```

R1(config)#ip access-list standard EIGRP-FILTER
R1(config-std-nacl)#deny 10.10.10.0 0.0.0.0
R1(config-std-nacl)#permit 0.0.0.0 0.0.0.0
R1(config)#router eigrp 10
R1(config-router)#distribute-list route-map EIGRP in
!
R1(config)#route-map EIGRP permit 10
R1(config-route-map)#match ip address EIGRP-FILTER
!
R1#show ip route eigrp | include 10.10.10.
D      10.10.10.128/25

```

Refer to the exhibit. An engineer must filter EIGRP updates that are received to block all 10.10.10.0/24 prefixes. The engineer tests the distribute list and finds one associated prefix. Which action resolves the issue?

- A. There is a permit in the route map that allows this prefix. A deny 20 statement is required with no match condition to block the prefix.
- B. There is a permit in the route map that allows this prefix. A deny 20 statement is required with a match condition to match a new ACL that denies all prefixes.
- C. There is a permit in the ACL that allows this prefix into EIGRP. The ACL should be modified to deny 10.10.10.0 0.0.0.255.
- D. There is a permit in the ACL that allows this prefix into EIGRP. The ACL should be modified to deny 10.10.10.0 255.255.255.0.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

When using an standard ACL for filtering routes, only the network address part will be matched. In this case, 10.10.10.0 0.0.0.0 will match the route with network address which is exactly "10.10.10.0" regardless of prefix length.

Since the route "10.10.10.128/25" has the network address "10.10.10.128", it does not match "10.10.10.0" exactly and therefore is not filtered.

By changing the rules to "deny 10.10.10.0 0.0.0.255", it matches the network address "10.10.10.x" regardless of prefix length. Hence the route "10.10.10.128/25" can be filtered.

QUESTION 425

A network administrator must optimize the segment size of the TCP packet on the DMVPN IPsec protected tunnel interface, which carries application traffic from the head office to a designated branch. The TCP segment size must not overwhelm the MTU of the outbound link. Which configuration must be applied to the router to improve the application performance?

- A. interface tunnel30
 - ip mtu 1400
 - ip tcp max-segment 1360
 - !
 - crypto ipsec fragmentation before-encryption
- B. interface tunnel30
 - ip mtu 1400
 - ip tcp payload-size 1360
 - !
 - crypto ipsec fragmentation before-encryption
- C. interface tunnel30

```

ip mtu 1400
ip tcp adjust-mss 1360
!
crypto ipsec fragmentation before-encryption
D. interface tunnel30
ip mtu 1400
ip tcp packet-size 1360
!
crypto ipsec fragmentation before-encryption

```

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Among all the "ip tcp ..." commands, only "ip tcp adjust-mss ..." is the valid command.

Router(config-if)#ip tcp ?

adjust-mss Adjust the mss of transit packets

Router(config-if)#ip tcp

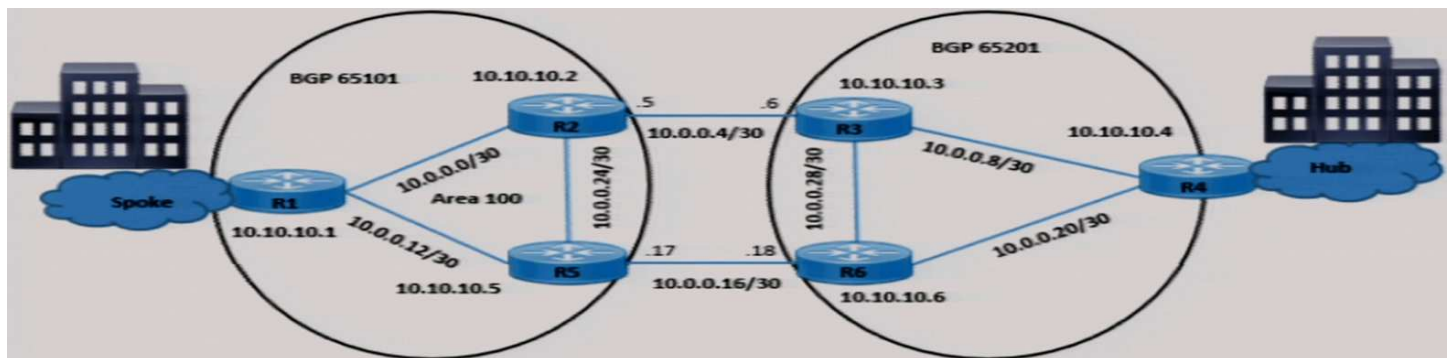
QUESTION 426

```

R6#
*Sep  5 05:31:58.891: BGP: 10.0.0.17 went from Idle to Active
*Sep  5 05:31:58.895: BGP: 10.0.0.17 open active, local address 10.0.0.18
*Sep  5 05:31:58.907: BGP: 10.0.0.17 read request no-op
*Sep  5 05:31:58.911: BGP: 10.0.0.17 went from Active to OpenSent
*Sep  5 05:31:58.911: BGP: 10.0.0.17 sending OPEN, version 4, my as: 65201, holdtime
180 seconds
*Sep  5 05:31:58.911: BGP: 10.0.0.17 send message type 1, length (incl. header) 53
*Sep  5 05:31:58.927: BGP: 10.0.0.17 remote close
*Sep  5 05:31:58.931: BGP: 10.0.0.17 -reset the session
*Sep  5 05:31:58.931: BGPNSF state: 10.0.0.17 went from nsf_not_active to
nsf_not_active

R5#
*Sep  5 05:34:22.063: BGP: 10.0.0.18 passive open to 10.0.0.17
*Sep  5 05:34:22.063: BGP: 10.0.0.18 passive open failed - 10.0.0.17 is not update-
source Loopback0's address (10.10.10.5)
*Sep  5 05:34:22.063: BGP: 10.0.0.18 remote connection attempt failed, local address
10.0.0.17

```



Refer to the exhibit. The traffic from spoke to hub is dropping. The operations team observes:

R2-R3 link is down due to the fiber cut.

R2 and R5 receive traffic from R1 in AS 65101.

R3 and R5 receive traffic from R4 in AS 65201.

Which configuration resolves the issue?

- A. R6(config)#router bgp 65201
R6(config-router)#neighbor 10.10.10.5 remote-as 65101
R6(config-router)#neighbor 10.10.10.5 update-source Loopback0
R6(config-router)#neighbor 10.10.10.5 ebgp-multihop 3
- B. R5(config)#router bgp 65101
R5(config-router)#neighbor 10.10.10.6 remote-as 65201
R5(config-router)#neighbor 10.10.10.6 update-source Loopback0
R5(config-router)#neighbor 10.10.10.6 ebgp-multihop 3
- C. R5(config)#router bgp 65101
R5(config-router)#no neighbor 10.0.0.18 update-source Loopback0
- D. R6(config)#router bgp 65101
R6(config-router)#no neighbor 10.0.0.17 update-source Loopback0

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

In R6's message, it shows that R6 is configured with the following but peering cannot be formed. There is no hint why the peering cannot be formed.

```
router bgp 65201
```

```
neighbor 10.0.0.17 remote-as 65101
```

In R5's message, other than the peering cannot be formed, the messages also complain that the neighbor router 10.0.0.18 (i.e R6) is NOT configured with a neighbor command with IP address of R5's loopback address i.e. 10.10.10.5.

10.0.0.18 passive open failed - 10.0.0.17 is not update-source loopback0's address (10.10.10.5)
This is because R5 is configured to form neighbor with 10.0.0.18 with update-source lo0 a follows and therefore it expects R6 to form neighbor with its lo0 i.e. 10.10.10.5.
router bgp 65101
neighbor 10.0.0.18 remote-as 65201
neighbor 10.0.0.18 update-source lo0 <-- this causes the problem

In order to solve this issue, either one of the following changes can be made:
- Modify all neighbor commands in both routers to form neighbor using loopback addresses.
- Remove the neighbor ... update-source lo0 command in R5. (i.e. the choice that remove this command in R5).

Remarks:
- Actually R5 should also have shown messages like the following two lines and this will show the problem more clearly. However, the question does not include them in the exhibit.

BGP: 10.0.0.18 active went from Idle to Active
BGP: 10.0.0.18 open active, local address 10.10.10.5

- For the observations mentioned in the question, the 3rd point should be:
R3 and R6 receive traffic from R4.

QUESTION 427

Which MPLS value is combined with the IP prefix to convert to a VPNv4 prefix?

- A. 16-byte Route Distinguisher
- B. 8-byte Route Distinguisher
- C. 16-byte Route Target
- D. 8-byte Route Target

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

A VPN-IPv4 address consists of 12 bytes. The first eight bytes represent the RD, followed by a four-byte IPv4 address prefix.

QUESTION 428

Which technique removes the outermost label of an MPLS-tagged packet before the packet is forwarded to an adjacent LER?

- A. label swap
- B. explicit-null
- C. PHP
- D. label imposition

Correct Answer: C

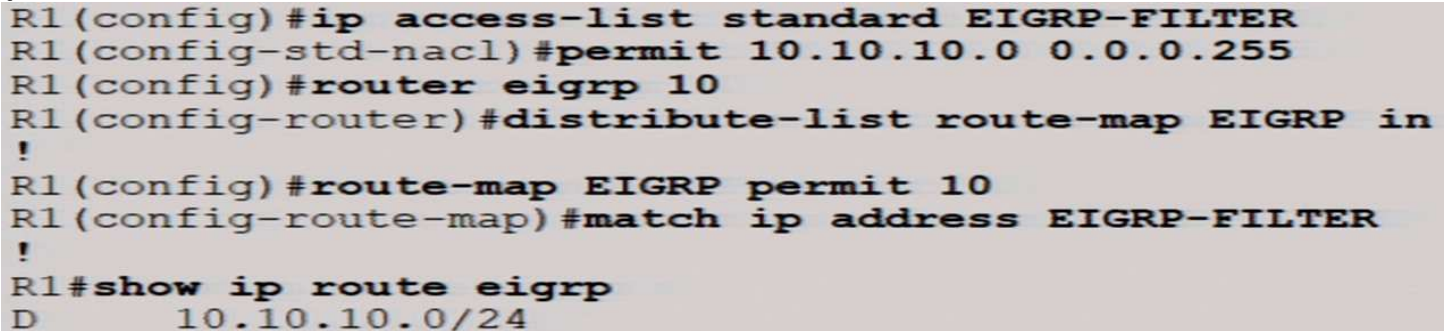
Section: (none)

Explanation

Explanation/Reference:

PHP refers to the process whereby the outermost label of an MPLS tagged packet is removed by a Label Switch Router (LSR) before the packet is passed to an adjacent Label Edge Router (LER).

QUESTION 429



```
R1(config)#ip access-list standard EIGRP-FILTER
R1(config-std-nacl)#permit 10.10.10.0 0.0.0.255
R1(config)#router eigrp 10
R1(config-router)#distribute-list route-map EIGRP in
!
R1(config)#route-map EIGRP permit 10
R1(config-route-map)#match ip address EIGRP-FILTER
!
R1#show ip route eigrp
D      10.10.10.0/24
```

Refer to the exhibit. An engineer must filter incoming EIGRP updates to allow only a set of specific prefixes. The distribute list is tested, and it filters out all routes except network 10.10.10.0/24. How should the engineer temporarily allow all prefixes to be learned by the router again without adjusting the existing access list?

- A. A permit any statement should be added before completing the ACL with the required prefixes, and then the permit any statement can be removed.
- B. A permit 20 statement should be added before completing the ACL with the required prefixes, and then the permit 20 statement can be removed.
- C. A continue statement should be added within the permit 10 statement before completing the ACL with the required prefixes, and then the continue statement can be removed.
- D. An extended access list must be used instead of a standard access list to accomplish the task.

Correct Answer: B

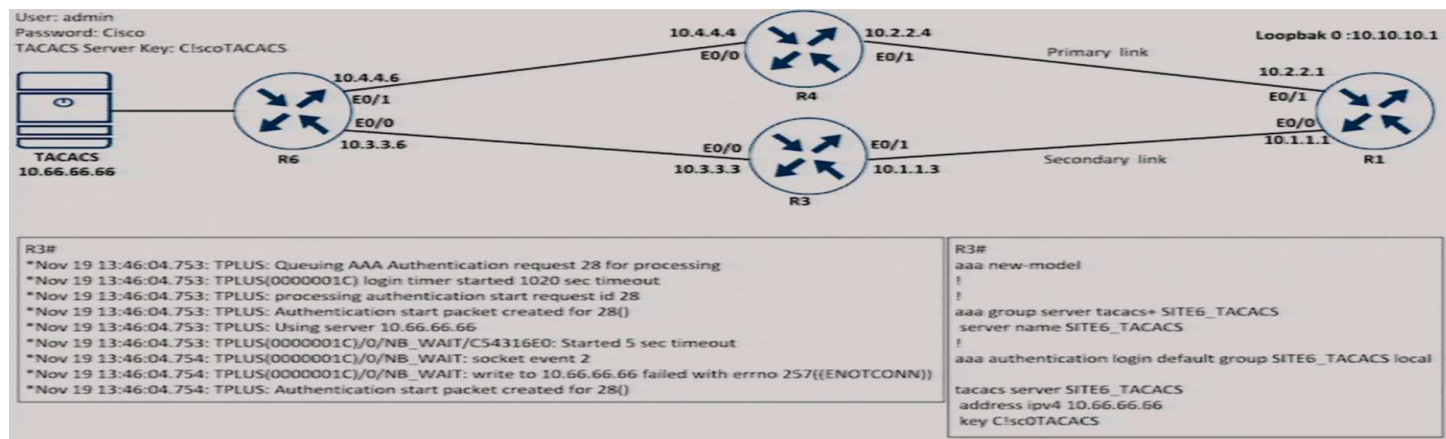
Section: (none)

Explanation

Explanation/Reference:

Since the question mentions that you should not change the existing ACL, you can add a "permit 20" statement in the route-map EIGRP to allow more prefixes. Actually, if you want to allow all prefixes, you can configure the additional "permit 20" statement without any need to specify a condition with "match ip address <new ACL>".

QUESTION 430



Refer to the exhibit. R3 cannot authenticate via TACACS. Which configuration resolves the issue?

- A. tacacs server SITE6_TACACS
address ipv4 10.66.66.66
key CiscoTACACS
- B. tacacs server SITE6_TACACS
key C!scoTACACS
- C. tacacs server SITE6_TACACS
key C!sc0TACACS
- D. tacacs server SITE6_TACACS
address ipv4 10.60.66.66
key C!scoTACACS

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since the messages in R3 show the word "timeout", this means that R3 cannot contact the TACACS server 10.66.66.66. Other than routing / networking problem, a reason that the TACACS server does not respond is that the wrong key is configured for the TACACS server.

In the diagram, the TACACS server is configured with the key C!scoTACACS but the key configured in R3 is C!sc0TACACS. Hence, you should change the uppercase "O" in the key configured in R3 to the lowercase "o".

QUESTION 431

The network administrator configured CoPP so that all SNMP traffic from Cisco Prime located at 192.168.1.11 toward the router CPU is limited to 1000 kbps. Any traffic that exceeds this limit must be dropped.

```
access-list 100 permit udp any any eq 161
```

```
class-map CM-SNMP
match access-group 100
```

```
policy-map PM-COPP
class CM-SNMP
police 1000 conform-action transmit
```

```
control-plane
service-policy input PM-COPP
```

The network administrator is not getting the desired result for the SNMP traffic and SNMP traffic is getting dropped frequently. Which set of configurations resolves the issue?

- A. policy-map PM-COPP
class CM-SNMP
no police 1000 conform-action transmit
police 1000000 conform-action transmit
- B. no access-list 100
access-list 100 permit udp host 192.168.1.11 any eq 161
!
policy-map PM-COPP
class CM-SNMP
no police 1000 conform-action transmit
police 1000000 conform-action transmit
!
control-plane
no service-policy input PM-COPP
!
interface E 0/0
service-policy input PM-COPP
!
interface E 0/1
service-policy input PM-COPP
- C. no access-list 100
access-list 100 permit tcp host 192.168.1.11 any eq 161
- D. no access-list 100
access-list 100 permit udp host 192.168.1.11 any eq 161
!
policy-map PM-COPP
class CM-SNMP
no police 1000 conform-action transmit
police 1000000 conform-action transmit

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

The command "police" expects traffic rate specified in bps.

Router(config-pmap-c)#police ?

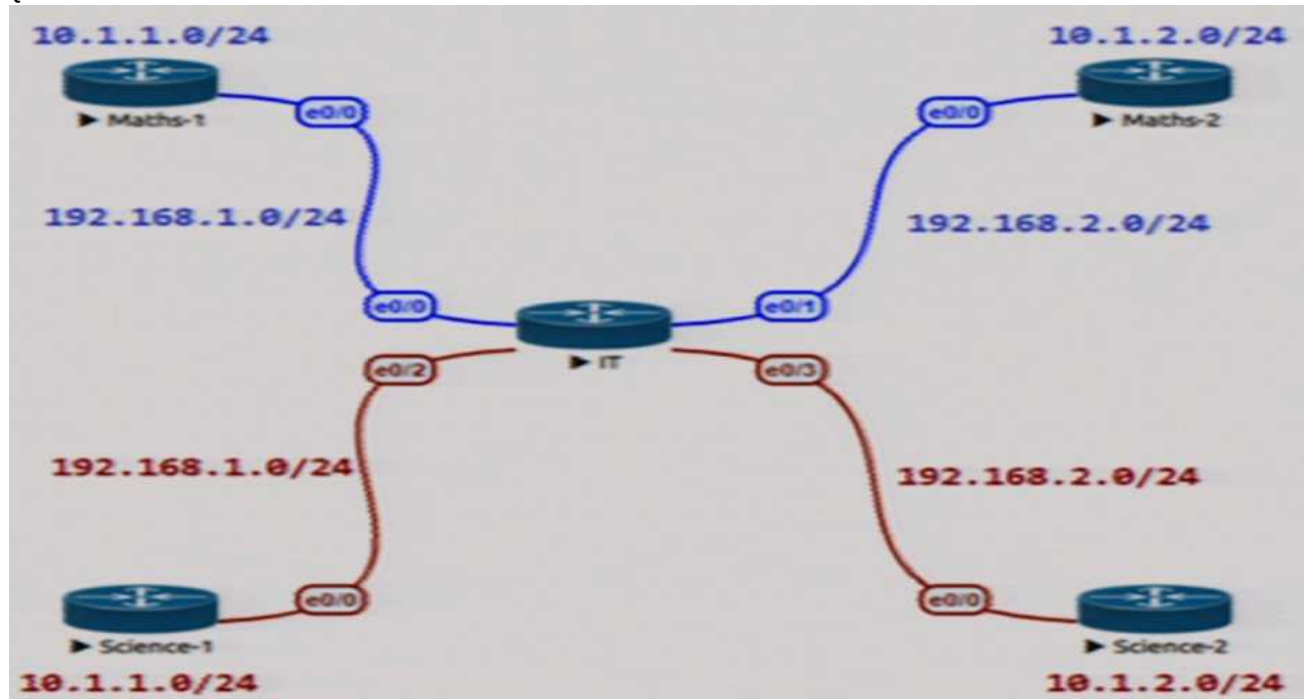
<8000-2000000000>	Target Bit Rate (bits per second) (postfix k, m, g optional; decimal point allowed)
cir	Committed information rate
rate	Specify police rate, PCR for hierarchical policies or SCR for single-level ATM 4.0 policer policies

Since the question mentions the rate limited should be 1000kbps, the correct command should be:

police 1000000 conform-action transmit

Regarding the ACL, since the question mentions about limit SNMP from Cisco Prime only, the rule should also be changed to having 192.168.1.11 as source. Without this, the desired result of dropping over-limit SNMP traffic from Cisco Prime only may not be achieved.

QUESTION 432



```
vrf definition Maths
 address-family ipv4
 !
 Interface E 0/0
  Vrf forwarding Maths
  Ip address 192.168.1.1 255.255.255.0
  No shut
 !
 Interface E 0/1
  Vrf forwarding Maths
  Ip address 192.168.2.1 255.255.255.0
  No shut
```

Refer to the exhibit. The IT router has been configured with the Maths VRF and the interfaces have been assigned to the VRF. Which set of configurations exchanges routes between the Maths-1 and Maths-2 routers using OSPF in area 0?

- A. router ospf 1 vrf Maths area 0
network 192.168.1.0
network 192.168.2.0
- B. router ospf 1 vrf Maths
network 192.168.1.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0
- C. router ospf 1
address-family ipv4 vrf Maths
network 192.168.1.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0
- D. router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Since the Math-1 and Math-2 routers are connected to the IT router through interfaces of the same vrf "Math", you can just configure OSPF for the vrf Math for route exchange.

Note that the correct way for setting up VRF in normal OSPF i.e. version 2 is "router ospf 1 vrf <name>"

```
Router(config)#router ospf 1 ?
      vrf  VPN Routing/Forwarding Instance
      <cr>
```

```
Router(config-router)#address-family ipv4 ?
      multicast  Address Family modifier
```

The "address-family ipv4 vrf <name>" is only used for setting up in OSPFv3 e.g. under "router ospfv3 1"

QUESTION 433

What is a function of the IPv6 DHCP Guard feature for DHCP messages?

- A. All client messages are always switched regardless of the device role.
- B. It blocks only DHCP request messages.
- C. If the device is configured as a DHCP server, no message is switched.
- D. Only access lists are supported for matching traffic.

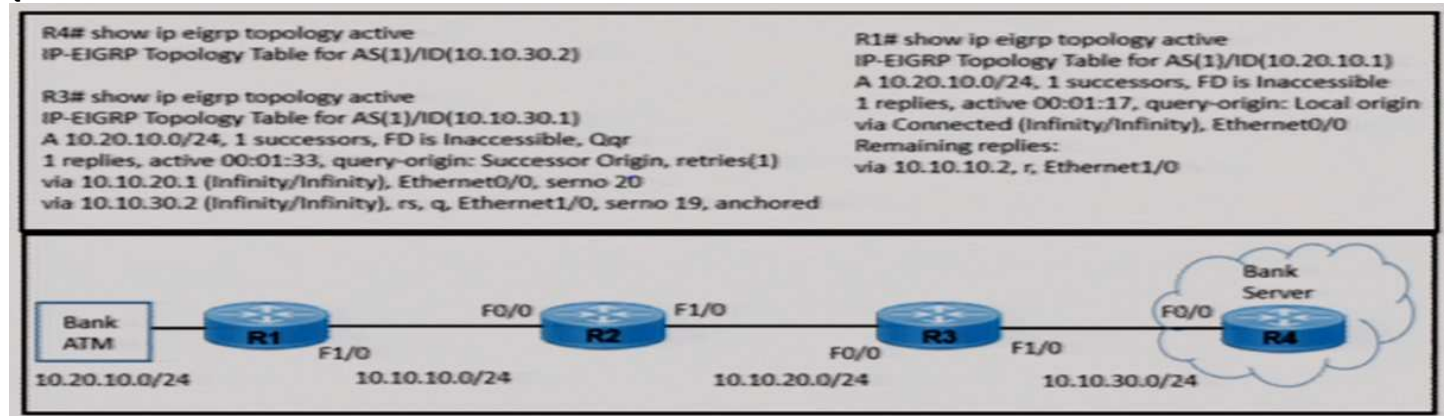
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The DHCPv6 Guard feature blocks reply and advertisement messages that come from unauthorized DHCP servers and relay agents (i.e. server messages received in ports not configured with "server" role policy). It does not block DHCP client messages.

QUESTION 434

Refer to the exhibit. A bank ATM site has difficulty connecting with the bank server. A network engineer troubleshoots the issue and finds that R4 has no active route to the bank ATM site. Which action resolves the issue?

- A. Advertise 10.10.30.0/24 subnet in R1 EIGRP AS.
- B. Advertise 10.10.30.0/24 subnet in R3 EIGRP AS.
- C. EIGRP peering between R3 and R4 to be fixed.
- D. EIGRP peering between R1 and R2 to be fixed.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Although R4 has no EIGRP route in active state, there is one for "10.20.10.0/24" in R3. Since R4 cannot only send traffic to Bank ATM in 10.20.10.0/24 through R3, there will be problem after the packets is sent to R3. Moreover, SIA for the same route is also found in R1.

The most probable reason is that R1 is disconnected with the network 10.20.10.0/24 and since there is no feasible successor, R1 therefore sends a query to R2. Since R2 also has no feasible successor for this prefix, it sends a query to R3. Since R3 also has no feasible successor, it sends a query to R4.

Normally after R4 discovered that it has no feasible successor for this prefix and it has no other neighbor, it will send back a reply about the inaccessible network to R3. This goes on in R2 and finally R1. After a reply is received, the router will remove the EIGRP route from the topology.

In the question, the problem is that the concerned route is Stuck In Active in R1, R3 (should also occur R2). This may due to the fact that R3 and R4 has problem in EIGRP communication so that R4 does not receive the query from R3 or R3 does not receive the reply from R4.

QUESTION 435

```

R1# show ip int br | ex una
Interface                IP-Address  OK?  Method Status  Protocol
Ethernet1/0              203.0.113.1 YES   manual up      up
Loopback1                 172.16.50.1 YES   manual up      up
Loopback2                 172.16.100.1 YES   manual up      up
Loopback3                 172.16.150.1 YES   manual up      up

R1# show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(1)
H  Address                  Interface Hold Uptime   SRTT RTO  Q  Seq
                               (sec)      (ms)  Cnt Num
0  203.0.113.2              Et1/0  14 00:31:16 1018 5000 0 24

R1# show ip eigrp topo all-links
EIGRP-IPv4 Topology Table for AS(1)/ID(172.16.10.1)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

P 192.168.10.0/24, 1 successors, FD is 409600, serno 34
   via 203.0.113.2 (409600/128256), Ethernet1/0
P 172.16.100.0/24, 1 successors, FD is 128256, serno 32
   via Connected, Loopback2
P 192.168.30.0/24, 1 successors, FD is 409600, serno 36
   via 203.0.113.2 (409600/128256), Ethernet1/0
P 203.0.113.0/24, 1 successors, FD is 281600, serno 33
   via Connected, Ethernet1/0
P 172.16.150.0/24, 1 successors, FD is 128256, serno 31
   via Connected, Loopback3
P 172.16.50.0/24, 1 successors, FD is 128256, serno 30
   via Connected, Loopback1
P 192.168.20.0/24, 1 successors, FD is 409600, serno 35
   via 203.0.113.2 (409600/128256), Ethernet1/0

```

Refer to the exhibit. Routers R1 and R2 have established a network adjacency using EIGRP, and both routers are advertising subnets to its neighbor. After issuing the show ip EIGRP topology all-links command in R1, some prefixes are not showing R2 as a successor. Which action resolves the issue?

- A. Enable split-horizon.
- B. Configure the network statement on the neighbor.
- C. Resolve the incorrect metric on the link.
- D. Rectify the incorrect router ID in R2.

Correct Answer: B

Section: (none)

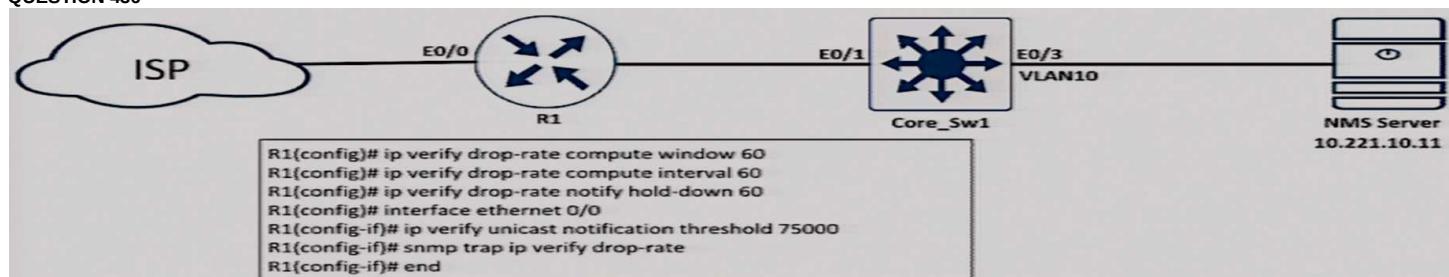
Explanation

Explanation/Reference:

In the output from "show ip eigrp topo all-links", some prefixes do not show R2 as successor. This is because those prefixes are the directly connected networks of R1.

Since there is no choice for the above, it probably means that some network prefix cannot be found in the EIGRP topology. Among all the choices, since the connection is through Ethernet (not a serial / tunnel connection) and R2 is the only EIGRP neighbor with R1, the most probable reason is that some "network" commands are missing in R2 and therefore those networks are not advertised by R2.

QUESTION 436



Refer the exhibit. An engineer configured SNMP traps to record spoofed packets drop of more than 48000 a minute on the ethernet0/0 interface. During an IP spoofing attack, the engineer noticed that no notifications have been received by the SNMP server. Which configuration resolves the issue on R1?

- A. ip verify unicast notification threshold 48000
- B. ip verify unicast notification threshold 800
- C. ip verify unicast notification threshold 80
- D. ip verify unicast notification threshold 8000

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

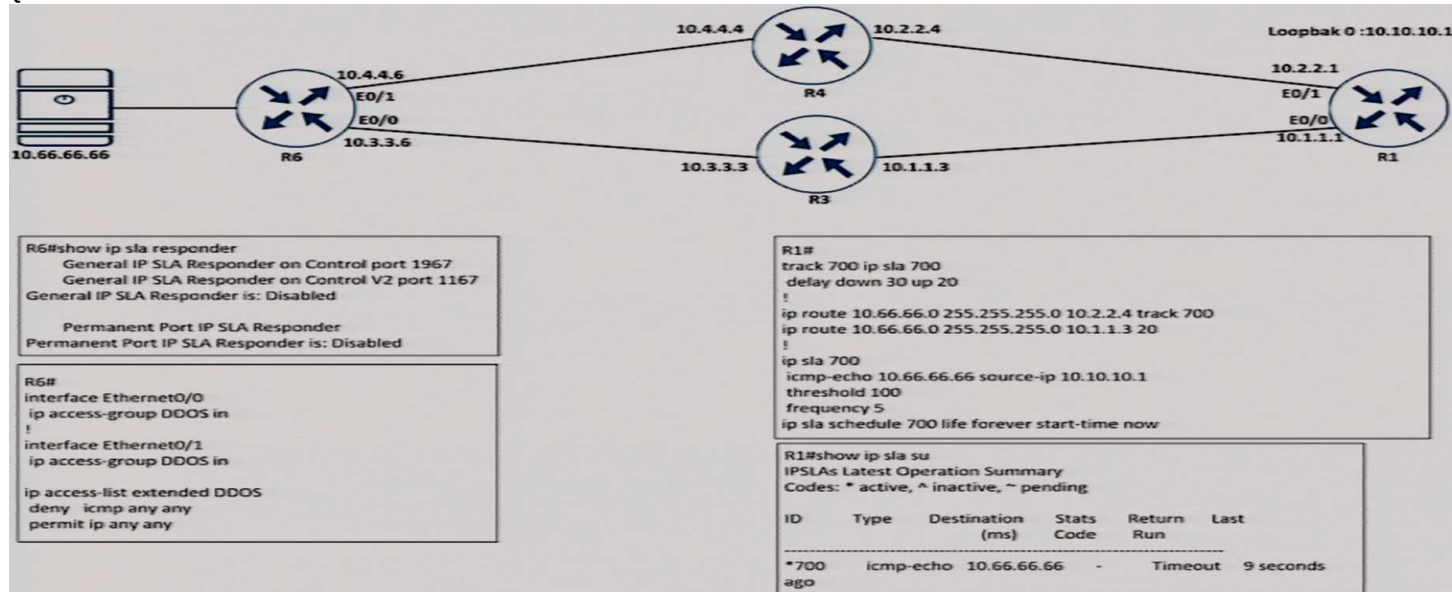
The command is configured using Packet Per Second (pps).

R6(config-if)#ip verify unicast notification threshold ?

<0-4294967295> Drop rate in pps triggering notify - 0 is any drops

Hence for 48000 per minute, it should be $48000 / 60 = 800$

QUESTION 437



Refer to the exhibit. A network administrator is trying to switch to the privileged EXEC level on R1 but failed. Which configuration resolves the issue?

- A. tacacs server enable-password Cisco@123
- B. tacacs-server enable-password Cisco@123
- C. enable-password Cisco@123
- D. enable password Cisco@123

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

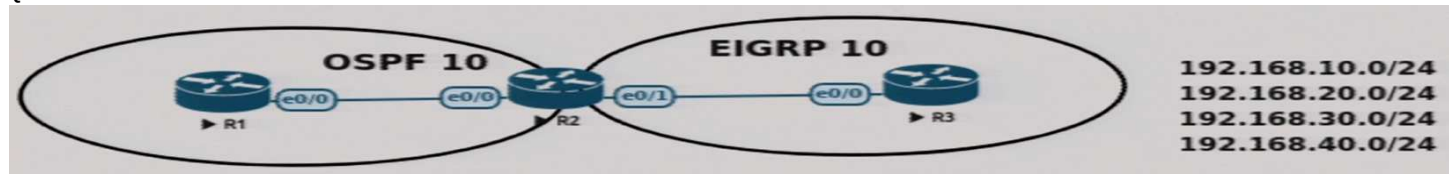
The answer is D since only "enable password <string>" is a valid command.

Remarks:

There is no relation between diagram and question. The answer is suggested by ignoring the diagram.

You cannot go to privileged mode through a vty connection if no enable password is set. You can set the enable password through console connection.

QUESTION 438



Refer to the exhibit. An engineer must redistribute networks 192.168.10.0/24 and 192.168.20.0/24 into OSPF from EIGRP, where the metric must be added when traversing through multiple hops to start an external route of 20. The engineer notices that the external metric is fixed and does not add at each hop. Which configuration resolves the issue?

- A. R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#access-list 10 permit 192.168.20.0 0.0.0.255
!
R1(config)#route-map RD permit 10
R1(config-route-map)#match ip address 10
R1(config-route-map)#set metric 20
R1(config-route-map)#set metric-type type-1
!
R1(config)#router ospf 10
R1(config-router)#redistribute eigrp 10 subnets route-map RD
- B. R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#access-list 10 permit 192.168.20.0 0.0.0.255
!
R1(config)#route-map RD permit 10
R1(config-route-map)#match ip address 10
R1(config-route-map)#set metric 20
R1(config-route-map)#set metric-type type-2
!
R1(config)#router ospf 10
R1(config-router)#redistribute eigrp 10 subnets route-map RD
- C. R2(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R2(config)#access-list 10 permit 192.168.20.0 0.0.0.255
!

```

R2(config)#route-map RD permit 10
R2(config-route-map)#match ip address 10
R2(config-route-map)#set metric 20
R2(config-route-map)#set metric-type type-1
!
R2(config)#router ospf 10
R2(config-router)#redistribute eigrp 10 subnets route-map RD
D. R2(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R2(config)#access-list 10 permit 192.168.20.0 0.0.0.255
!
R2(config)#route-map RD permit 10
R2(config-route-map)#match ip address 10
R2(config-route-map)#set metric 20
R2(config-route-map)#set metric-type type-2
!
R2(config)#router ospf 10
R2(config-router)#redistribute eigrp 10 subnets route-map RD

```

Correct Answer: C

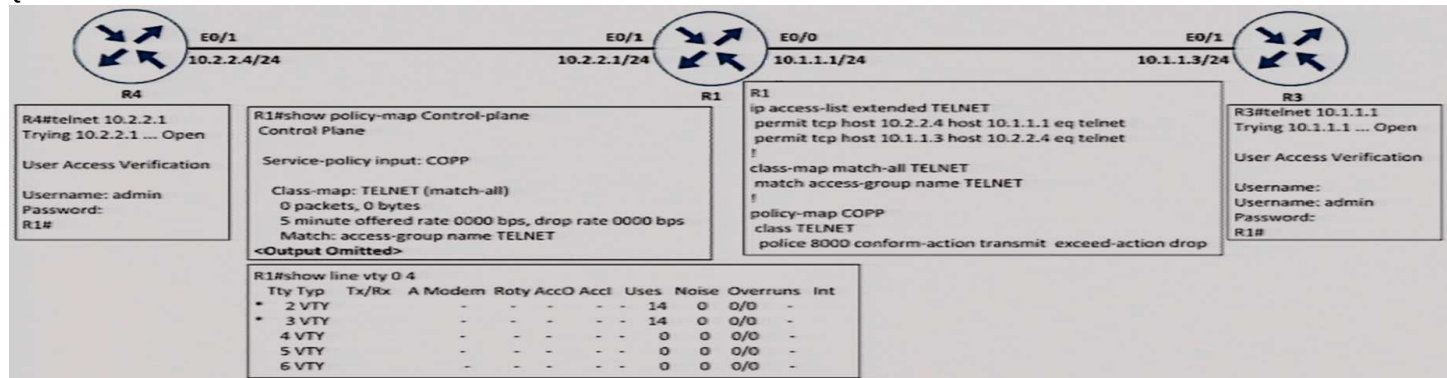
Section: (none)

Explanation

Explanation/Reference:

Since only R2 is running both EIGRP and OSPF, redistribution should be configured in R2. Moreover, the EIGRP routes should be redistributed as OSPF E1 route in order to include OSPF internal cost in the external OSPF routes.

QUESTION 439



Refer to the exhibit. An engineer implemented CoPP to limit Telnet traffic to protect the router CPU. It was noticed that the Telnet traffic did not pass through CoPP. Which configuration resolves the issue?

- policy-map COPP
class TELNET
police 8000 conform-action transmit exceed-action transmit violate-action drop
- ip access-list extended TELNET
permit tcp host 10.2.2.1 host 10.2.2.4 eq telnet
permit tcp host 10.1.1.1 host 10.1.1.3 eq telnet
- policy-map COPP
class TELNET
police 8000 conform-action transmit exceed-action transmit
- ip access-list extended TELNET
permit tcp host 10.2.2.4 host 10.2.2.1 eq telnet
permit tcp host 10.1.1.3 host 10.1.1.1 eq telnet

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Explanation:

From the telnet diagram in R4 and R3:

- the device 10.2.2.4 (source) is making telnet to 10.2.2.1 (destination) and
- the device 10.1.1.3 (source) is making telnet to 10.1.1.1 (destination)

The source and destination in the existing ACL "TELNET" are wrongly configured.

QUESTION 440

An engineer configures PBR on R5 and wants to create a policy that matches traffic destined toward 10.10.10.0/24 and forwards it toward 10.1.1.1. This traffic must also have its IP precedence set to 5. All other traffic should be forwarded toward 10.1.1.2 and have its IP precedence set to 0. Which configuration meets the requirements?

- access-list 1 permit 10.10.10.0 0.0.0.255
access-list 2 permit any
route-map CCNP permit 10
match ip address 1
set ip next-hop 10.1.1.1
set ip precedence 5
!
route-map CCNP permit 20
match ip address 2
set ip next-hop 10.1.1.2
set ip precedence 0
!
route-map CCNP permit 30
- access-list 1 permit 10.10.10.0 0.0.0.255
route-map CCNP permit 10

```

match ip address 1
set ip next-hop 10.1.1.1
set ip precedence 5
!
route-map CCNP permit 20
set ip next-hop 10.1.1.2
set ip precedence 0
C. access-list 100 permit 10.10.10.0 0.0.0.255
route-map CCNP permit 10
match ip address 100
set ip next-hop 10.1.1.1
set ip precedence 0
!
route-map CCNP permit 20
set ip next-hop 10.1.1.2
set ip precedence 5
!
route-map CCNP permit 30
D. access-list 100 permit ip any 10.10.10.0 0.0.0.255
route-map CCNP permit 10
match ip address 100
set ip next-hop 10.1.1.1
set ip precedence 5
!
route-map CCNP permit 20
set ip next-hop 10.1.1.2
set ip precedence 0

```

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Since you need to set PBR based on destination, an extended ACL is required.

Only one choice has the correct ACL and set IP precedence 5 for traffic to next hop 10.1.1.1 for destination 10.10.10.0/24.

QUESTION 441

The summary route is not shown in the Router_B routing table after this below configuration on Router_A:

```

interface ethernet 0
description location ID:T1344S9A34B19
ip address 192.168.3.1 255.255.255.0
ip summary-address eigrp 1 172.16.80.0 255.255.240.0

```

Which Router_A configuration resolves the issue by advertising the summary route to Router_B?

- A. interface loopback 0
ip address 172.18.81.1 255.255.255.0
interface Ethernet 0
ip address 192.168.3.1 255.255.255.0
ip summary-address eigrp 1 172.16.80.0 255.255.240.0
- B. interface loopback 0
ip address 172.16.96.1 255.255.255.0
interface Ethernet 0
ip address 192.168.3.1 255.255.255.0
ip summary-address eigrp 1 172.16.80.0 255.255.240.0
- C. interface loopback 0
ip address 172.16.81.1 255.255.255.0
interface Ethernet 0
ip address 192.168.3.1 255.255.255.0
ip summary-address eigrp 1 172.16.80.0 255.255.240.0
- D. interface loopback 0
ip address 172.16.79.1 255.255.255.0
interface Ethernet 0
ip address 192.168.3.1 255.255.255.0
ip summary-address eigrp 1 172.16.80.0 255.255.240.0

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

EIGRP summary route is only advertised if there is an individual EIGRP route falling in the range of the summary route.

Since there is no further information, we can therefore only assume that no individual EIGRP route within 192.168.80.0/20. Moreover, we also need to assume that the loopback0 address is included in the network command of EIGRP. Since only loopback with IP address 172.16.81.1/24 is within 192.168.80.0/20. The choice with this loopback address is the answer.

QUESTION 442

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

After redistributing the OSPF route for 192.168.5.16/28 from OSPF to RIP, only the RIP router R3 have route to 192.168.5.17 since traceroute can be sent. The RIP router R4 does not have the route and therefore traceroute cannot be sent. The reason is that the OSPF route for 192.168.5.16/28 is being redistributed to RIP with metric 15. After R3 receives it, it will not send to R4 since the metric becomes 16 (i.e. inaccessible). In order to let R3 send the redistributed RIP route to R4, the redistribution from OSPF into RIP must be set with a metric of 14 or lower.

QUESTION 443

```
ip sla 1
  icmp-echo 8.8.8.8
  threshold 1000
  timeout 2000
  frequency 5
ip sla schedule 1 life forever start-time now
!
track 1 ip sla 1
!
ip route 0.0.0.0 0.0.0.0 Ethernet0/0 203.0.113.1 name ISP1 track 1
ip route 0.0.0.0 0.0.0.0 Ethernet0/1 198.51.100.1 2 name ISP2
```

Refer to the exhibit. After recovering from a power failure, Ethernet0/1 stayed down while Ethernet0/0 returned to the up/up state. The default route through ISP1 was not reinstated in the routing table until Ethernet0/1 also came up. Which action resolves the issue?

- A. Configure the default route through ISP1 with a higher administrative distance than 2.
- B. Remove the references to the interface names from both static default routes.
- C. Add a static route to the 8.8.8.8/32 destination through the next hop 203.0.113.1.
- D. Reference the track object 1 in both static default routes.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

After the router has recovered:

It cannot add the static default route named ISP1 to the routing table since IP SLA has not been up yet.

It cannot add the static default route named ISP2 to the routing table since the specified interface in the route is down.

No default route is entered into the routing table and the router cannot access Internet.

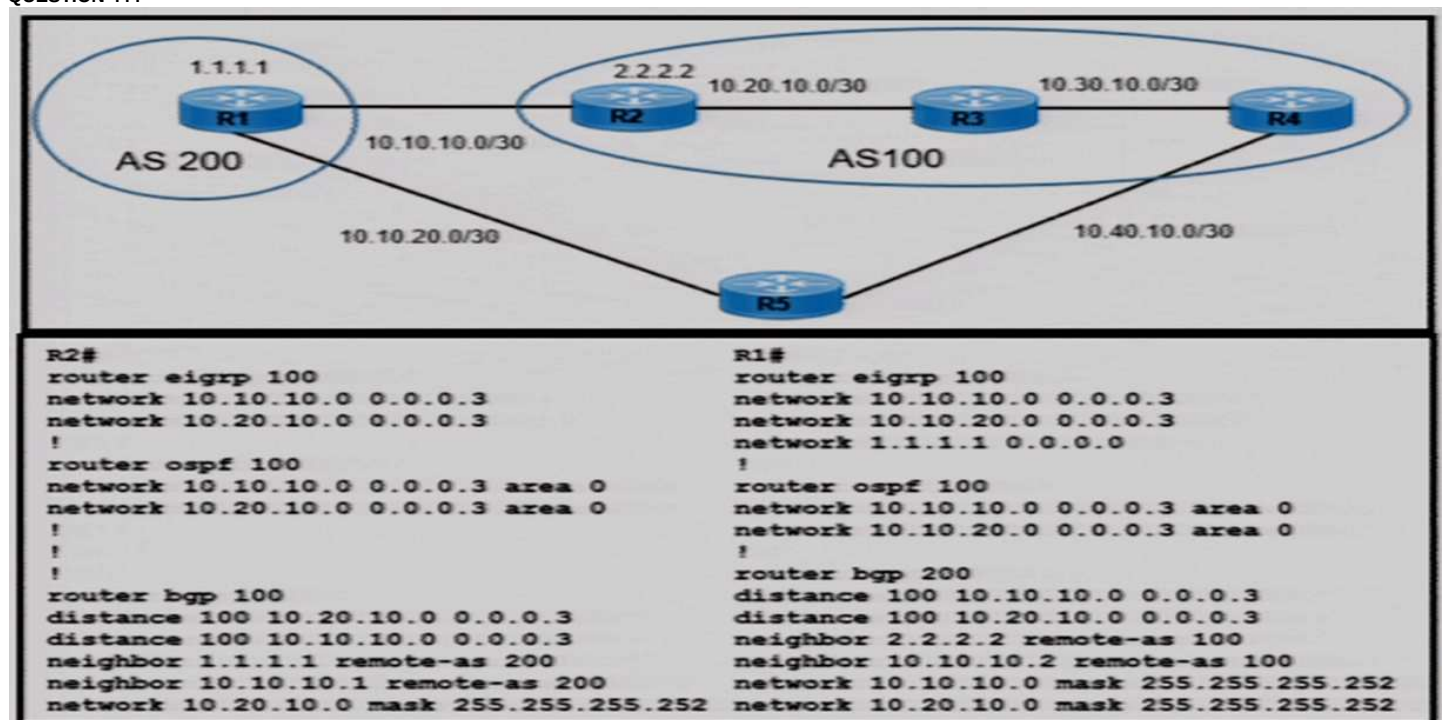
IP SLA remains down since the target 8.8.8.8 in the Internet cannot be reached.

By adding a static host route for 8.8.8.8/24 with next hop 203.0.113.1 (probably a ISP1's router), the IP SLA target can be reached through ISP1 even if there is no default route. After IP SLA becomes up, the static default route named ISP1 is entered into the routing table for accessing Internet through ISP1.

Remarks:

Just by removing the interface specified in the static routes probably does not help. For example for the static default route named ISP2, the next hop 198.51.100.1 is probably a ISP2's router connected by e0/1. If e0/1 is down, its connected network e.g. 198.51.100.0/24 will not be found in the routing table. Since the next hop address 198.51.100.1 is not accessible, the static default route named ISP2 will not be entered into the routing table.

QUESTION 444



Refer to the exhibit. R1 and R2 use IGP protocol to route traffic between AS 100 and AS 200 despite being configured to use BGP. Which action resolves the issue and ensures the use of BGP?

- A. Configure distance to 100 under the OSPF process of R1 and R2.
- B. Remove distance commands under BGP AS 100 and AS 200.
- C. Remove distance commands under BGP AS 100.
- D. Configure distance to 100 under the EIGRP process of R1 and R2.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

There are two ways to modify the AD of BGP routes:

distance bgp 20 200 200 (for settings AD for all external, internal and local BGP routes)

distance 100 10.10.10.0 0.0.0.3 (for settings AD 100 for all BGP routes learnt from neighbor in the network 10.10.10.0/30)

Since both R1 and R2 forms neighbor with each other using IP addresses in 10.10.10.0/30 and both have distance 100 10.10.10.0 0.0.0.3, both will set AD 100 to the route of each other. Since this will have an AD lower than EIGRP (AD 90), EIGRP routes instead of BGP routes will be preferred.

To use the BGP routes, since R1 and R2 learn each other routes as eBGP routes, you can remove the distance command in BGP in both routers so that the BGP routes will have the lowest AD i.e. 20.

Remarks :

If you change the AD of EIGRP in R1 and R2 to 100, although EIGRP and eBGP routes have the same AD 100, since the default AD (AD before change) of eBGP is 20 which is smaller than the default AD of EIGRP, eBGP routes will be preferred. However, the removing of distance command under BGP is therefore preferred as suggested answer.

Note that R1 and R2 will not form BGP neighboring through their loopback addresses (i.e. 1.1.1.1 and 2.2.2.2) since "update-source ..." "ebgp multihop" are not configured.

QUESTION 445

```
CPE# show ntp associations

address          ref clock      st   when   poll reach  delay
offset  disp
~10.1.255.40      .INIT.        16    -     64     0 0.000
0.000 15937.
* sys.peer, # selected, + candidate, - outlyer, x falseticker,
~ configured

CPE# debug ip icmp
*Feb 20 22:49:32.913: ICMP: dst (10.0.12.1) port unreachable rcv
from 10.1.255.40
*Feb 20 22:50:37.918: ICMP: dst (10.0.12.1) port unreachable rcv
from 10.1.255.40
*Feb 20 22:51:44.951: ICMP: dst (10.0.12.1) port unreachable rcv
from 10.1.255.40
```

Refer to the exhibit. An administrator is troubleshooting a time synchronization problem for the router's time to another Cisco IOS XE-based device that has recently undergone security hardening. Which action resolves the issue?

- A. NTP service is disabled and must be enabled on 10.1.255.40.
- B. Allow NTP in the ingress ACL on 10.1.255.40 by permitting TCP destined to port 123.
- C. Allow NTP in the ingress ACL on 10.1.255.40 by permitting UDP destined to port 123.
- D. Ensure that the CPE router has a valid route to 10.1.255.40 for NTP and rectify if not reachable.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

From the 1st command output, it shows that no association is formed by the CPE router with the NTP server 10.1.255.40 .

From the 2nd command output, it shows that 10.1.255.40 sends ICMP port unreachable (ICMP type 3 code 3) to the CPE router. It probably means that the server 10.1.255.40 is rejecting packets (the NTP packets sent from CPE router) since NTP service is not enabled in 10.1.255.40.

Note that, it also means that the NTP packets from CPE router can reach 10.1.255.40 although they are eventually rejected.

Remarks :

If the NTP packets from CPE router is blocked by ACL in 10.1.255.40, the ICMP type 3 code 13 message will be sent instead. The wording in the message will be "administratively prohibited unreachable" instead of the "port unreachable" shown in the question.

QUESTION 446

An engineer must establish a connection between two CE routers for two customers with overlapping IP addresses. Customer_a is connected to interfaces Gig0/0, and Customer_b is connected to interfaces Gig0/1. Routers CE1 and CE2 are configured as follows:

```
ip vrf customer_a
rd 1:1
route-target both 1:1
```

```
ip vrf customer_b
rd 2:2
route-target both 2:2
```

Drag and drop the code snippets from the right onto the boxes in the configuration to establish the needed connection. Snippets may be used more than once.

Select and Place:

<pre> CE1 Interface Gig0/0 ip vrf forwarding ip address Interface Gig0/1 ip vrf forwarding ip address </pre>	customer_a customer_b 192.168.1.1 255.255.255.0 192.168.1.2 255.255.255.0
<pre> CE2 Interface Gig0/0 ip vrf forwarding ip address Interface Gig0/1 ip vrf forwarding ip address </pre>	

Correct Answer:

<pre> CE1 Interface Gig0/0 ip vrf forwarding customer_a ip address 192.168.1.1 255.255.255.0 Interface Gig0/1 ip vrf forwarding customer_b ip address 192.168.1.1 255.255.255.0 </pre>	customer_a customer_b 192.168.1.1 255.255.255.0 192.168.1.2 255.255.255.0
<pre> CE2 Interface Gig0/0 ip vrf forwarding customer_a ip address 192.168.1.2 255.255.255.0 Interface Gig0/1 ip vrf forwarding customer_b ip address 192.168.1.2 255.255.255.0 </pre>	

Section: (none)

Explanation

Explanation/Reference:

Although there is no requirement mentioned in the question, using ".1" address for CE1 router and ".2" address for CE2 router is often preferred.

QUESTION 447

```

CPE(config)# lin c 0
CPE(config-line)# no exec
CPE(config-line)# end
CPE#
*Jan 31 23:07:22.655: %SYS-5-CONFIG_I: Configured from console
by console
CPE# wr
Building configuration...
[OK]
CPE# exit

CPE con0 is now available

Press RETURN to get started.

! Console stopped responding at this moment !

```

Refer to the exhibit. An administrator is attempting to disable the automatic logout after a period of inactivity. After logging out, the console stopped responding to all keyboard inputs. Remote access through SSH still works. Which action resolves the issue?

A. Configure the no exec-timeout command on line con 0.

- B. Configure the default exec-timeout command on line con 0.
- C. Configure the absolute-timeout command on line con 0.
- D. Configure the exec command on line con 0.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Although you can use "exec-timeout 0" to disable automatic logout, the major problem that is now having is that console does not enter "exec" mode any more after "no exec" is configured. The problem can be solved by entering the "exec" command under "line con 0" through a SSH session.

QUESTION 448

```
R3(config-ipv6-acl) #do show ipv6 access-list
IPv6 access list Block PC1 to PC2
permit ipv6 2001:DB8::/16 2001:DB8::/16 sequence 20
deny ipv6 host 2001:DB8:0:10::3B host 2001:DB8:A:A::19 sequence 30
```

Refer to the exhibit. An IPv6 ACL is applied to restrict PC1 from communicating with PC2 and allow all other traffic. Which configuration resolves the issue?

- A. R3(config-ipv6-acl)#no sequence 30
R3(config-ipv6-acl)deny ipv6 host 2001:DB8:0:10::3B any sequence 10
- B. R3(config-ipv6-acl)#no sequence 20
R3(config-ipv6-acl)deny ipv6 host 2001:DB8:0:10::3B any sequence 10
- C. R3(config-ipv6-acl)#no sequence 20
R3(config-ipv6-acl)deny ipv6 host 2001:DB8:0:10::3B host 2001:DB8:A:A::19 sequence 10
- D. R3(config-ipv6-acl)#no sequence 30
R3(config-ipv6-acl)deny ipv6 host 2001:DB8:0:10::3B host 2001:DB8:A:A::19 sequence 10

Correct Answer: D

Section: (none)

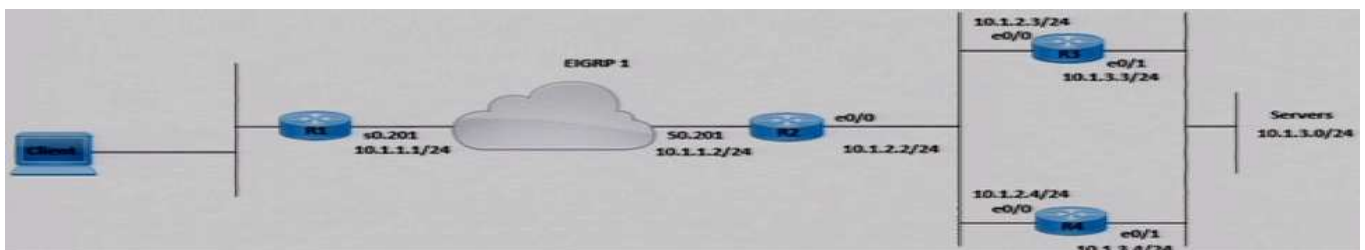
Explanation

Explanation/Reference:

The existing permit rule in sequence 20 allows communication between PC1 and PC2. The deny rule in sequence 30 will never be matched. By removing the deny rule in sequence 30 and configure it in sequence 10, it makes sure that the deny rule will be matched before the permit rule.

QUESTION 449

```
R2# show ip eigrp topology 10.1.3.0 255.255.255.0
IP-EIGRP (AS 1): topology entry for 10.1.3.0/24
State is Passive, Query origin flag is 1, 1 Successor(s), FD is 307200
Routing Descriptor Blocks:
 10.1.2.3 (Ethernet0), from 10.1.2.3, Send flag is 0x0
   Composite metric is (307200/281600), Route is Internal
   Vector metric:
     Minimum bandwidth is 10000 Kbit
     Total delay is 2000 microseconds
     Reliability is 255/255
     Load is 1/255
     Minimum MTU is 1500
     Hop count is 1
 10.1.2.4 (Ethernet0), from 10.1.2.4, Send flag is 0x0
   Composite metric is (312320/286720), Route is Internal
   Vector metric:
     Minimum bandwidth is 10000 Kbit
     Total delay is 2200 microseconds
     Reliability is 255/255
     Load is 1/255
     Minimum MTU is 1500
     Hop count is 1
```



Refer to the exhibit. A network is configured for EIGRP equal-cost load balancing, but the traffic destined to the servers is not load balanced. Link metrics from router R2 to R3 and R4 are the same. Which delay value must be configured to resolve the issue?

- A. 200 on R3 E0/1
- B. 120 on R3 E0/1
- C. 2200 on R4 E0/1
- D. 120 on R4 E0/1

Correct Answer: B

Section: (none)

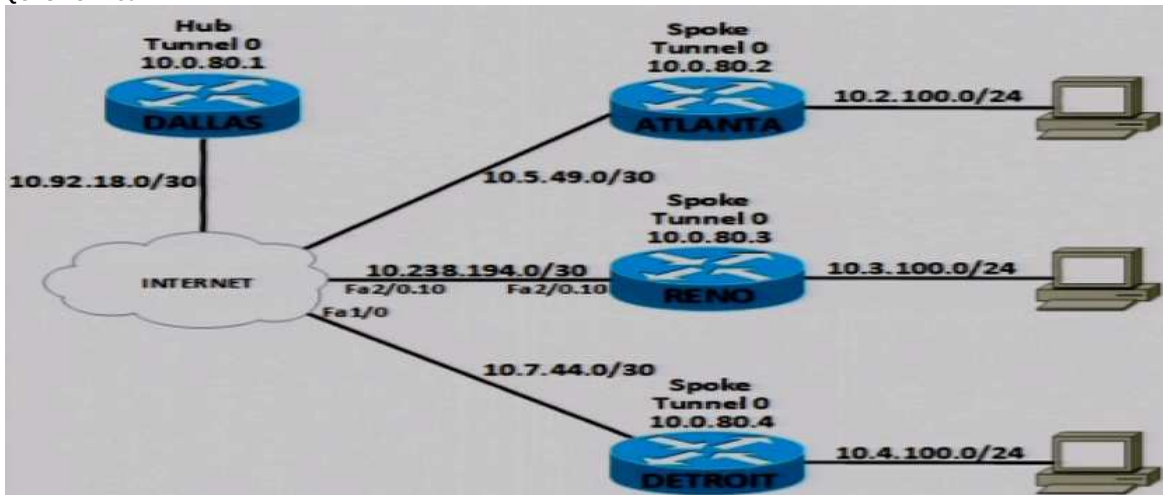
Explanation

Explanation/Reference:

Only one route is entered for equal-cost load balancing since the other route has a higher metric. From the exhibit, it shows that the route having higher metric (i.e. with next hop 10.1.2.4) has a larger delay value of 2200 (compared to 2000 in the first one). In order to change it, you need to modify the delay in the interface through which concerned traffic packets will be sent out. For increasing the delay by 200, you can change it in R3 e0/1. You cannot change it in R2's e0/0 which this will affect the delay of both routes instead of just one. Since packets from R2 to 10.1.3.0 goes through two interfaces R2's e0/0 and R3's e0/1, we assume that each of the interface has a delay of 1000 microseconds (therefore a total of 2000 microseconds in the best route as seen in the exhibit). Hence R3's e0/1 has to be configured with a delay of 1200 microseconds so that both routes will have the same metric.

Note that since the delay command in an interface is configured based on "tens of microseconds", you need to enter the command "delay 120" in R3's e0/1.

QUESTION 450



Refer to the exhibit. An engineer must connect the Reno and Detroit spokes using DMVPN phase 2. Hub tunnel configuration is:

Dallas:

```
interface Tunnel0
ip address 10.0.80.1 255.255.255.0
ip nhrp authentication cisco123
ip nhrp map multicast dynamic
ip nhrp network-id 5
tunnel source Serial0/0
tunnel mode gre multipoint
```

Which configuration accomplishes the task?

A. Reno:

```
interface Tunnel0
ip address 10.0.80.3 255.255.255.0
ip nhrp authentication cisco123
ip nhrp map multicast 10.92.18.2
ip nhrp map 10.0.80.1 10.92.18.2
ip nhrp network-id 5
ip nhrp nhs 10.0.80.1
tunnel source 10.238.194.2
tunnel mode gre multipoint
```

Detroit:

```
interface Tunnel0
ip address 10.0.80.4 255.255.255.0
ip nhrp authentication cisco123
ip nhrp map 10.0.80.1 10.92.18.2
ip nhrp map multicast 10.92.18.2
ip nhrp network-id 5
ip nhrp nhs 10.0.80.1
tunnel source 10.7.44.2
tunnel mode gre multipoint
```

B. Reno:

```
interface Tunnel0
ip address 10.0.80.3 255.255.255.0
ip nhrp authentication cisco123
ip nhrp map broadcast 10.92.18.2
ip nhrp map 10.0.80.1 10.92.18.2
ip nhrp network-id 5
ip nhrp nhs 10.0.80.1
tunnel source 10.238.194.2
tunnel mode gre multipoint
```

Detroit:

```
interface Tunnel0
ip address 10.0.80.4 255.255.255.0
ip nhrp authentication cisco123
ip nhrp map 10.0.80.1 10.92.18.2
ip nhrp map broadcast 10.92.18.2
ip nhrp network-id 5
ip nhrp nhs 10.0.80.1
tunnel source 10.7.44.2
tunnel mode gre multipoint
```

C. Reno:

```
interface Tunnel0
ip address 10.0.80.3 255.255.255.0
ip nhrp authentication cisco321
ip nhrp map multicast 10.92.18.2
ip nhrp map 10.0.80.1 10.92.18.2
ip nhrp network-id 5
ip nhrp nhs 10.0.80.1
tunnel source 10.238.194.2
tunnel mode gre multipoint
```

Detroit:

```
interface Tunnel0
ip address 10.0.80.4 255.255.255.0
ip nhrp authentication cisco321
ip nhrp map 10.0.80.1 10.92.18.2
ip nhrp map multicast 10.92.18.2
ip nhrp network-id 5
ip nhrp nhs 10.0.80.1
tunnel source 10.7.44.2
tunnel mode gre multipoint
```

D. Reno:

```
interface Tunnel0
ip address 10.0.80.3 255.255.255.0
ip nhrp authentication cisco123
ip nhrp map multicast 10.92.18.2
ip nhrp map 10.92.18.2 10.0.80.1
ip nhrp network-id 5
ip nhrp nhs 10.0.80.1
tunnel source 10.238.194.2
tunnel mode gre multipoint
```

Detroit:

```
interface Tunnel0
ip address 10.0.80.4 255.255.255.0
ip nhrp authentication cisco123
ip nhrp map 10.92.18.2 10.0.80.1
ip nhrp map multicast 10.92.18.2
ip nhrp network-id 5
ip nhrp nhs 10.0.80.1
tunnel source 10.7.44.2
tunnel mode gre multipoint
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

For phrase 2, instead of having a fixed destination configured in the tunnel interface, a spoke should be able to resolve the NBMA address of another spoke router in the following form:

```
ip nhrp nhs 10.0.80.1
ip nhrp map 10.0.80.1 10.92.18.x (where x is 1 or 2)
tunnel mode gre multipoint
```

The commands in two of the choices are wrong since there is no command for mapping "broadcast" (i.e. "ip nhrp map broadcast ..." is invalid command).

Among the remaining two, only one is properly configured with "ip nhrp map ..." command for the NHS in the format "ip nhrp map < tunnel IP> <NBMA address>".

QUESTION 451

What is considered the primary advantage of running BFD?

- A. reduction in time needed to detect Layer 2 switched neighbor failures
- B. reduction in CPU needed to detect Layer 2 switch neighbor failures
- C. reduction in CPU needed to detect Layer 3 routing neighbor failures
- D. reduction in time needed to detect Layer 3 routing neighbor failures

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

BFD works only for directly connected neighbors. BFD neighbors must be no more than one IP hop away. Therefore it detects layer 2 failure instead of layer 3 failure.

QUESTION 452

The network administrator deployed the Binding Table Recovery feature. Which two devices recover the missing binding table entries? (Choose two.)

- A. destination host
- B. source host
- C. DHCP server
- D. DHCP client
- E. DHCP relay agent

Correct Answer: AC

Section: (none)

Explanation

Explanation/Reference:

The IPv6 First-Hop Security Binding Table Recovery Mechanism feature recovers the missing binding table entries when the resolution for a destination address fails in the destination guard. Upon a failure, a binding table entry is recovered by querying the **DHCP server** or the **destination host** depending on the configuration.

QUESTION 453


```

Router#show ip route
<output omitted>
Gateway of last resort is not set

    3.0.0.0/32 is subnetted, 1 subnets
C       3.3.3.3 is directly connected, Loopback0
    192.168.1.0/32 is subnetted, 1 subnets
O       192.168.1.1 [110/21] via 192.168.3.1, 23:00:29, Ethernet0/1
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, Ethernet0/1
L       192.168.3.2/32 is directly connected, Ethernet0/1
Router#show ip bgp
BGP table version is 3, local router ID is 3.3.3.3
<output omitted>
   Network                Next Hop           Metric  LocPrf  Weight  Path
* i  192.168.2.2/32        209.165.200.225      0       100      0       ?
Router#show ip bgp summary
BGP router identifier 3.3.3.3, local AS number 65000
<output omitted>
Neighbor    V      AS      MsRcvd  MsgSent  Tblver  Up/Down  State/PfxRcd
192.168.1.1  4      65000          7         6         3    00:02:04        1
Router#

```

Refer to the exhibit. Which action installs route 192.168.2.2/32 in the routing table?

- A. Redistribute connected networks into BGP on the local router.
- B. Configure NAT on the local router to translate private IP addresses.
- C. Configure the next-hop-self attribute for the peering on the local router.
- D. Configure the next-hop-self attribute for the peering on the peer router.

Correct Answer: D

Section: (none)

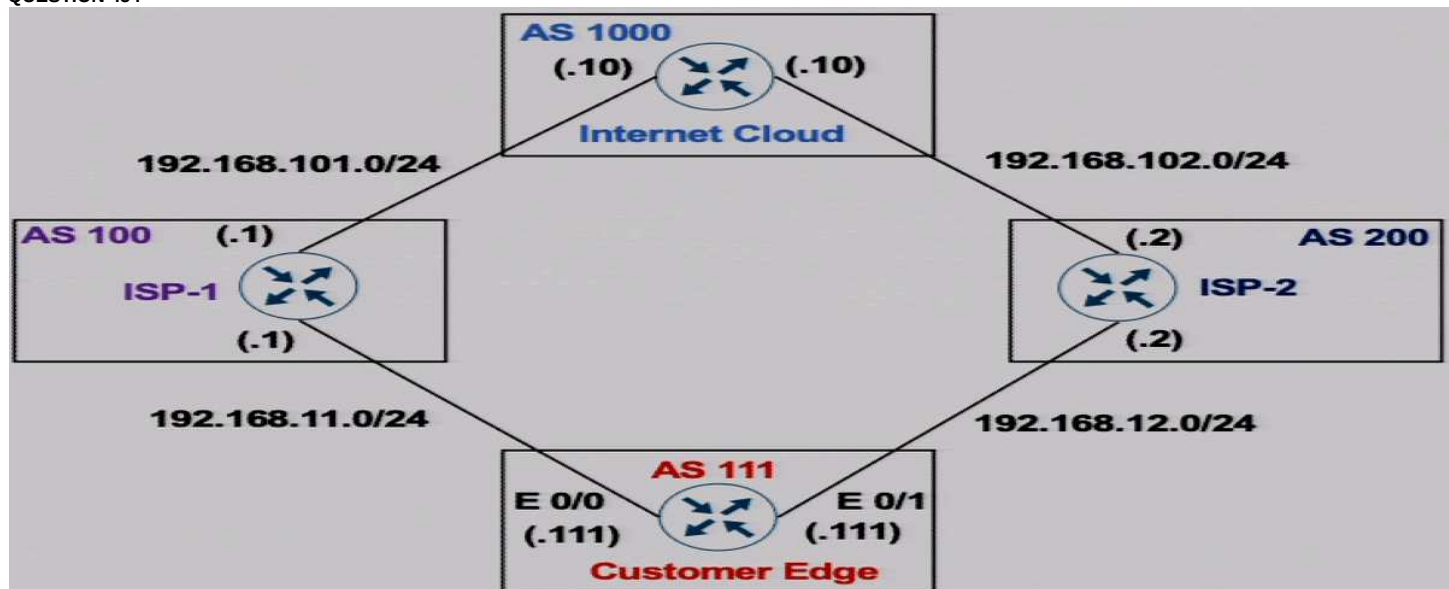
Explanation

Explanation/Reference:

The BGP route "192.168.2.2/32" cannot be entered into the routing table since its next hop "209.165.200.225" is not accessible by the router (i.e. the next hop address cannot be matched to any entry in the routing table).

Other than adding a routing entry for the next hop, you can change the BGP neighbor router's settings to advertise the route with its IP address (i.e. 192.168.1.1) as next hop (i.e. next-hop-self) attribute to this router.

QUESTION 454



ISP-1

```

ip as-path access-list 1 permit ^111
!
router bgp 100
 neighbor 192.168.101.10 remote-as 1000
 neighbor 192.168.11.111 remote-as 111
 neighbor 192.168.11.111 filter-list 1 in

```

Refer to the exhibit. AS 111 must not be used as a transit AS, but ISP-1 is getting ISP-2 routes from AS 111. Which configuration stops Customer AS from being used as a transit path on ISP-1?

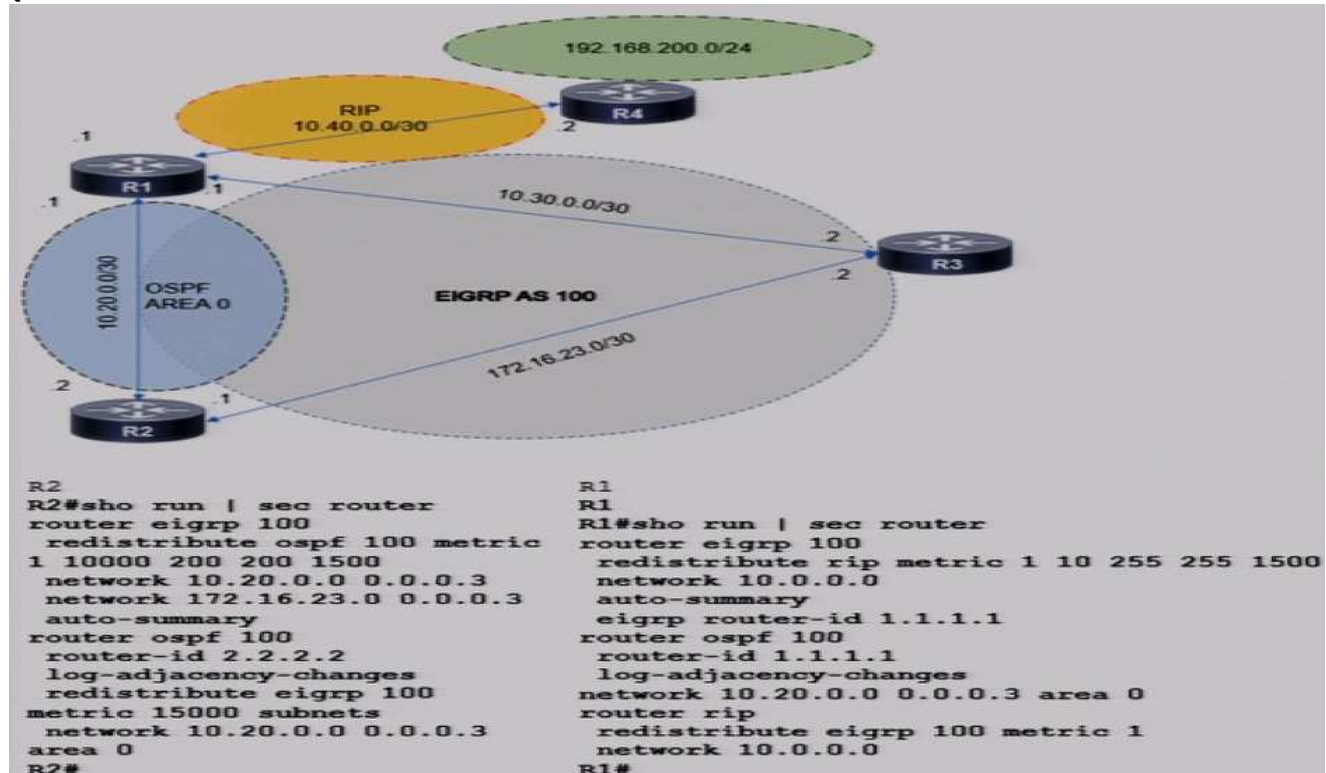
- A. ip as-path access-list 1 permit *

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

The current AS path list with “^111” permit all routes with path list starting “111” e.g. “111 200” for the route originated from AS 200 and learnt through AS 111. You need to change it to “^111\$” so that the path list can only have one AS “111” i.e. routes originated and learnt directly from AS 111.

QUESTION 455



Refer to the exhibit. The route to 192.68.200_0 is flapping between R1 and R2. Which set of configuration changes resolves the flapping route?

- A. R2(config)#router ospf 100
R2(config-router)#no redistribute eigrp 100
R2(config-router)#redistribute eigrp 100 metric 1 subnets
- B. R1(config)#no router rip
R1(config)#ip route 192.168.200.0 255.255.255.0 10.40.0.2
- C. R2(config)#router eigrp 100
R2(config-router)#no redistribute ospf 100
R2(config-router)#redistribute rip
- D. R1(config)#router ospf 100
R1(config-router)#redistribute rip metric 1 metric-type 1 subnets

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

Problem occurs since:

- R1 learns the RIP route 192.168.200.0/24 from R4.
- R1 redistributes the RIP route into EIGRP route and advertise to R3.
- R3 sends the EIGRP route to R2.
- R2 redistributes the EIGRP route into OSPF.
- R1 learns the OSPF route. Although it is an external OSPF route (E2 by default), it has a lower AD than RIP. Hence R1 will use this OSPF route instead of the RIP route learnt from R4.
- Flapping therefore occurs.

One of the ways to solve the problem is to let R1 redistribute the RIP route into OSPF as external E1 route. Since E1 route is preferred than E2 route redistributed by R2, R1 will not use the OSPF route redistributed by R2 and no flapping will occur.

QUESTION 456

Which table is used to map the packets in an MPLS LSP that exit from the same interface, via the same next hop, and have the same queuing policies?

- A. LDP
B. CEF
C. RIB
D. FEC

Correct Answer: D
Section: (none)
Explanation

Explanation/Reference:

Since queuing policies are involved, FEC should be the answer.

Forwarding Equivalence Class (FEC) is a set of prefixes treated in the same way. If you have packets that are sent: 1) via the same next hop, 2) exit interface, 3) and possibly with the same queuing policies, then those prefixes belong the same FEC, because they have the same treatment from the perspective of the LSRs that are processing these prefixes.

QUESTION 457

The network administrator is tasked to configure R1 to authenticate telnet connections based on Cisco ISE using RADIUS. ISE has been configured with an IP address of 192.168.1.5 and with a network device pointing towards R1 (192.168.1.1) with a shared secret password of Cisco123. If ISE is down, the administrator should be able to connect using the local database with a username and password combination of admin/cisco123.

The administrator has configured the following on R1:

```
aaa new-model
!
username admin password cisco123
!
radius server ISE1
address ipv4 192.168.1.5
key Cisco123
!
aaa group server tacacs+ RAD-SERV
server name ISE1
!
aaa authentication login RAD-LOCAL group RAD-SERV
```

ISE has gone down. The Network Administrator is not able to Telnet to R1 when ISE went down. Which two configuration changes will fix the issue? (Choose two.)

- A. line vty 0 4
login authentication RAD-SERV
- B. aaa authentication login RAD-LOCAL group RAD-SERV local
- C. aaa authentication login RAD-SERV group RAD-LOCAL local
- D. line vty 0 4
login authentication RAD-LOCAL
- E. line vty 0 4
login authentication default

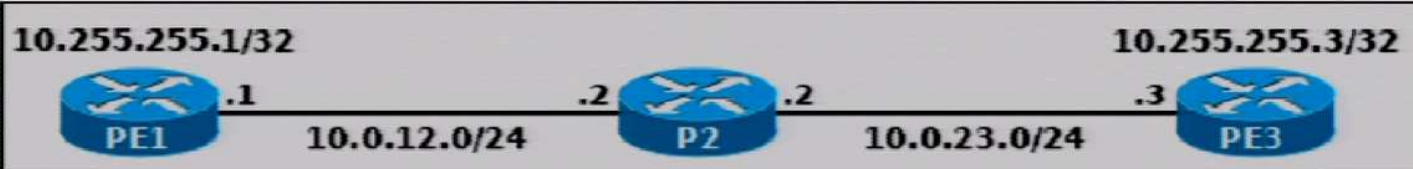
Correct Answer: BD

Section: (none)

Explanation

Explanation/Reference:

Note that in the question:
- "RAD-SERV" is a TACACS+ server group name.
- "RAD-LOCAL" is a AAA authentication list name.

QUESTION 458


```
PE1# show run | sec router bgp
router bgp 65000
  bgp log-neighbor-changes
  neighbor 10.255.255.3 remote-as 65000
  neighbor 10.255.255.3 update-source Loopback0
```

```
PE1# show ip bgp summary
BGP router identifier 10.255.255.1, local AS number 65000
BGP table version is 1, main routing table version 1

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ
OutQ Up/Down State/PfxRcd
10.255.255.3    4    65000      0        0         1    0
0 never Idle
```

```
PE1# ping 10.255.255.3 source lo0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.255.255.3, timeout is 2
seconds:
Packet sent with a source address of 10.255.255.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/1/1 ms
```



```

PE1# debug ip tcp transactions
PE1# debug ip icmp

[...snip...]
*Feb 22 14:04:12.374: TCP: sending SYN, seq 379810712, ack 0
*Feb 22 14:04:12.374: TCP0: Connection to 10.255.255.3:179,
advertising MSS 1460
*Feb 22 14:04:12.374: TCP0: state was CLOSED -> SYNSENT [21381 -
> 10.255.255.3(179)]
*Feb 22 14:04:12.375: ICMP: dst (10.255.255.1) administratively
prohibited unreachable rcv from 10.0.12.2
*Feb 22 14:04:12.375: TCP0: ICMP destination unreachable
received
*Feb 22 14:04:12.375: Released port 21381 in Transport Port
Agent for TCP IP type 1 delay 240000
*Feb 22 14:04:12.375: TCP0: state was SYNSENT -> CLOSED [21381 -
> 10.255.255.3(179)]
*Feb 22 14:04:12.375: TCB 0xE35A92B8 destroyed

```

Refer to the exhibit. The administrator is troubleshooting a BGP peering between PE1 and PE3 that is unable to establish. Which action resolves the issue?

- A. P2 must have a route to PE3 to establish a BGP session to PE1.
- B. Ensure that the PE3 loopback address is used as a source for BGP peering to PE1.
- C. Disable sending ICMP unreachables on P2 to allow PE1 to establish a session with PE3.
- D. Remove the traffic filtering rules on P2 blocking the BGP communication between PE1 and PE3.

Correct Answer: D

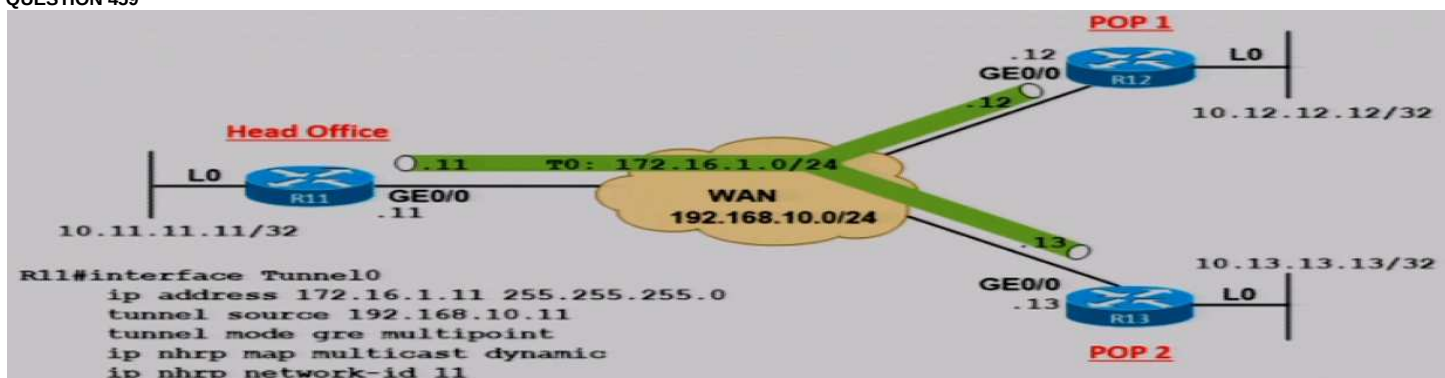
Section: (none)

Explanation

Explanation/Reference:

Problem is NOT related to "neighbor 10.255.255.3 ebgp-multihop" since PE1 and PE3 are in the same AS 65000. Therefore this command is not required. Since PE1 can use its loopback as source to ping 10.255.255.3, there is no routing problem. In the debugging messages, it shows that the TCP traffic (probably the BGP traffic) is blocked by "10.0.12.2" (i.e. P2). Hence you need to remove the filtering in P2.

QUESTION 459



Refer to the exhibit. A company builds WAN infrastructure between the head office and POPs using DMVPN hub-and-spoke topology to provide end-to-end communication. All POPs must maintain point-to-point connectivity with the head office. Which configuration meets the requirement at routers R12 and R13?

- A. Configure routers R12 and R13 as:

```

interface Tunnel0
ip nhrp map multicast 172.16.1.11
ip nhrp map 172.16.1.11 192.168.10.11
ip nhrp network-id 11
ip nhrp nhs 192.168.10.11

```

- B. Configure routers R12 and R13 as:

```

interface Tunnel0
ip nhrp map multicast 192.168.10.11
ip nhrp map 172.16.1.11 192.168.10.11
ip nhrp network-id 11
ip nhrp nhs 172.16.1.11

```

- C. R12#
interface Tunnel0
ip nhrp map multicast 172.16.1.11
ip nhrp map 172.16.1.11 192.168.10.11
ip nhrp network-id 12
ip nhrp nhs 192.168.10.11

```

R13#
interface Tunnel0
ip nhrp map multicast 172.16.1.11
ip nhrp map 172.16.1.11 192.168.10.11
ip nhrp network-id 13
ip nhrp nhs 192.168.10.11

```

- D. R12#
interface Tunnel0
ip nhrp map multicast 192.168.10.11

```
ip nhrp map 172.16.1.11 192.168.10.11
ip nhrp network-id 12
ip nhrp nhs 172.16.1.11
```

```
R13#
interface Tunnel0
ip nhrp map multicast 192.168.10.11
ip nhrp map 172.16.1.11 192.168.10.11
ip nhrp network-id 13
ip nhrp nhs 172.16.1.11
```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

Note that:

- All routers in the same DMVPN must have the same Network ID.

- The configuration commands for NHS in the Spoke routers are:

```
ip nhrp nhs <NHS tunnel IP>
```

```
ip nhrp map <NHS tunnel IP> <NHS NBMA Address>
```

```
ip nhrp map multicast <NHS NBMA Address>
```

QUESTION 460

An engineer configured VRF-Lite on a router for VRF blue and VRF red. OSPF must be enabled on each VRF to peer to a directly connected router in each VRF. Which configuration forms OSPF neighbors over the network 10.10.10.0/28 for VRF blue and 192.168.0.0/30 for VRF red?

- A. router ospf 1 vrf blue
network 10.10.10.0 0.0.0.240 area 0
router ospf 2 vrf red
network 192.168.0.0 0.0.0.252 area 0
- B. router ospf 1 vrf blue
network 10.10.10.0 0.0.0.3 area 0
router ospf 2 vrf red
network 192.168.0.0 0.0.0.15 area 0
- C. router ospf 1 vrf blue
network 10.10.10.0 0.0.0.252 area 0
router ospf 2 vrf red
network 192.168.0.0 0.0.0.240 area 0
- D. router ospf 1 vrf blue
network 10.10.10.0 0.0.0.15 area 0
router ospf 2 vrf red
network 192.168.0.0 0.0.0.3 area 0

Correct Answer: D

Section: (none)

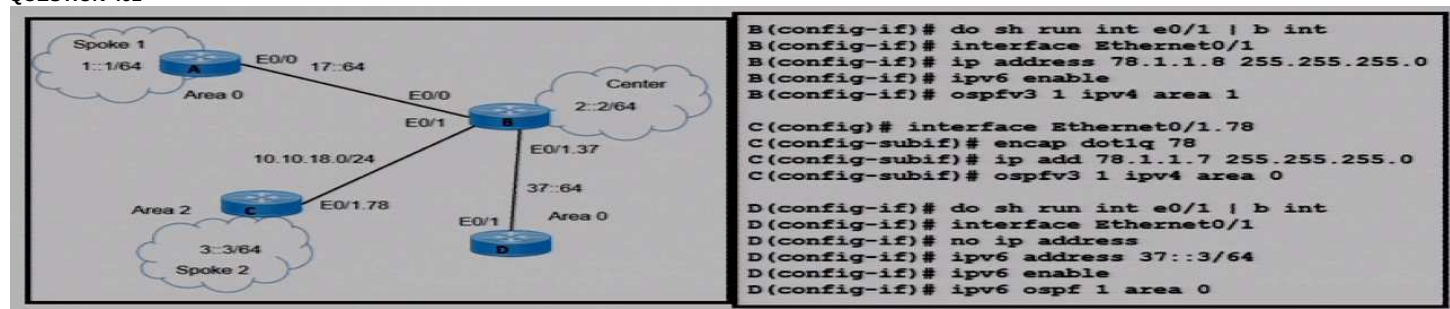
Explanation

Explanation/Reference:

The subnet mask for 10.10.10.0/28 is 255.255.255.240, the wildcard mask required is therefore 255 – 240 = 15.

The subnet mask for 192.168.0.0/30 is 255.255.255.252, the wildcard mask required is therefore 255 – 252 = 3.

QUESTION 461



Refer to the exhibit. A network engineer receives a report that Spoke 1 users can perform bank transactions with the server located at the Center site, but Spoke 2 users cannot. Which action resolves the issue?

- A. Configure encapsulation dot1q 78 on the router C interface.
- B. Configure OSPFv2 on the routers B and C interfaces.
- C. Configure IPv6 on the routers B and C interfaces.
- D. Configure the Spoke 2 users IP on the router B OSPF domain.

Correct Answer: C

Section: (none)

Explanation

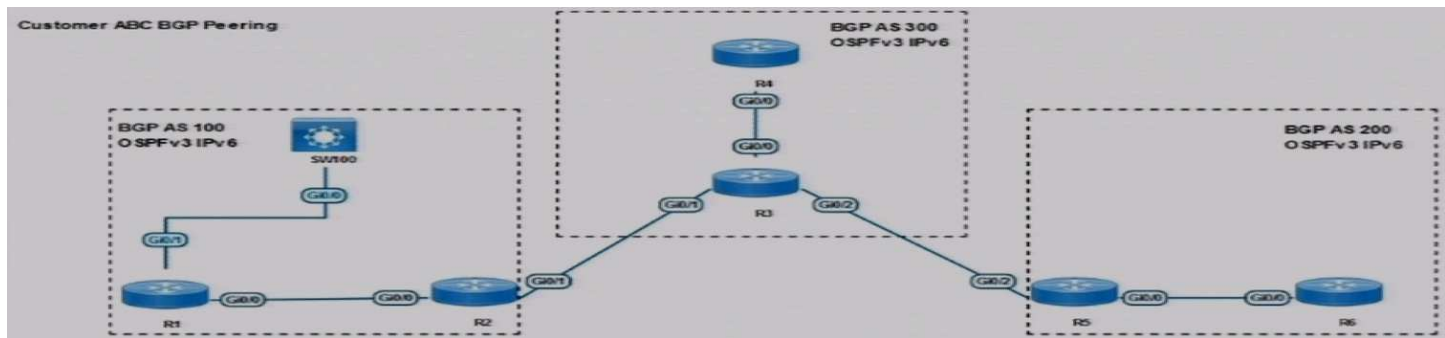
Explanation/Reference:

Since Spoke2 and Center is having IPv6 addresses, you need to setup IPv6 addresses in the link connecting between Router B and C in order to pass IPv6 traffic.

Remarks:

There may be other problem. For example, the OSPF area configured in B and C is the same.

QUESTION 462



```
R2#sh ip bgp ipv6 uni
BGP table version is 45, local router ID is 2.2.22.22
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop        Metric LocPrf Weight Path
*>  2001::5/128      2001::5              0         0 300 200 i
*>  2001::4/128      2001::4              0         0 300 i
*>  2002::2/128      :::                  0         0 32768 i

R2#sh run | section bgp
router bgp 100
  address-family ipv6
    neighbor 2001::4 route-map Filter in
  ip as-path access-list 1 permit _300_[0-9]
  route-map Filter permit 10
    match as-path 1
```

Refer to the exhibit. R2 has been receiving routes from R4 that originated outside AS300. A network engineer configured an AS-Path ACL to avoid adding these routes to the R2 BGP table, but the routes are still present in the R2 routing table. Which action resolves the issue?

- A. Replace as-path access-list 1 with the ip as-path access-list 1 permit ^300_ command.
- B. Replace as-path access-list 1 with the ip as-path access-list 1 permit ^300\$ command.
- C. Replace as-path access-list 1 with the ip as-path access-list 1 permit ^300.* command.
- D. Replace as-path access-list 1 with the ip as-path access-list 1 permit _300.* command.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

^300\$ is the required setting in order to accept BGP routes originated in AS 300 from AS 300.

QUESTION 463

The network administrator must implement IPv6 in the network to allow only devices that not only have registered IP addresses but are also connecting from assigned locations. Which security feature must be implemented?

- A. IPv6 Prefix Guard
- B. IPv6 Snooping
- C. IPv6 Router Advertisement Guard
- D. IPv6 Destination Guard

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 464

What are the two reasons for RD and VPNv4 addresses in an MPLS Layer 3 VPN? (Choose two.)

- A. They are used for a BGP session with the CE device.
- B. When the PE redistributes customer routes into MP-BGP they must be unique.
- C. They are on a CE device to use for static configuration.
- D. RD is prepended to each prefix to make routes unique.
- E. VPN RT communities are used to identify customer unique routes.

Correct Answer: BD

Section: (none)

Explanation

Explanation/Reference:

QUESTION 465


```

CPE# copy flash:packages.conf ftp://192.0.2.40/
Address or name of remote host [192.0.2.40]?
Destination filename [packages.conf]?
Writing packages.conf
%Error opening ftp://192.0.2.40/packages.conf (Incorrect
Login/Password)
CPE#

```

Refer to the exhibit. An administrator must upload the packages.conf file to an FTP server. However, the FTP server rejected anonymous service and required users to authenticate. What are the two ways to resolve the issue? (Choose two.)

- A. Use the copy flash:packages.conf ftp: command instead, and enter the FTP server credentials when prompted.
- B. Create a user on the router matching the username and password on the FTP server and log in before attempting the copy.
- C. Use the copy flash:packages.conf scp: command instead, and enter the FTP server credentials when prompted.
- D. Use ip ftp username and ip ftp password configuration commands to specify valid FTP server credentials.
- E. Enter the FTP server credentials directly in the FTP URL using the ftp://username:password@192.0.2.40/ syntax.

Correct Answer: DE

Section: (none)

Explanation

Explanation/Reference:

Method1:

```

Router#copy ftp://user1:password1@192.168.0.1/file1.txt flash:
Destination filename [file1.txt]?
Accessing ftp://*****:*****@192.168.0.1/file1.txt...
Erase flash: before copying? [confirm]n
Loading file1.txt
[OK - 4/4096 bytes]

```

```

Verifying checksum... OK (0x1826)
4 bytes copied in 0.104 secs (38 bytes/sec)

```

Router#

Method2:

```

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip ftp username user1
Router(config)#ip ftp password password1
Router(config)#^Z
Router#copy ftp://192.168.0.1/file2.txt flash:
Destination filename [file2.txt]?
Accessing ftp://192.168.0.1/file2.txt...
Erase flash: before copying? [confirm]n
Loading file2.txt
[OK - 4/4096 bytes]

```

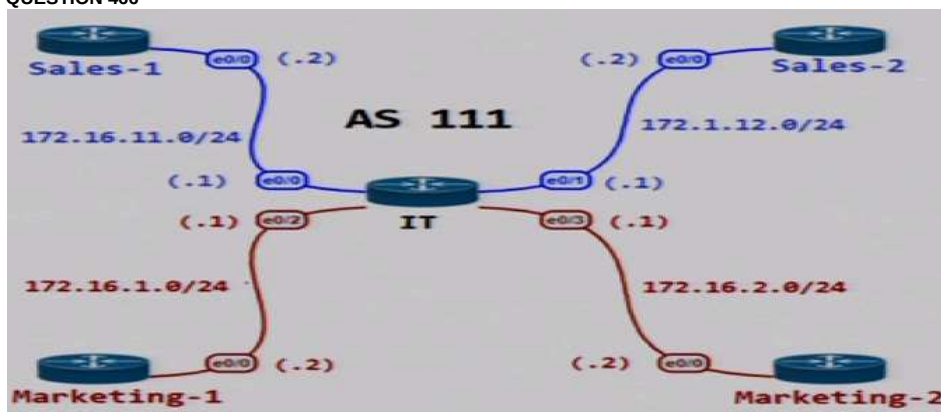
```

Verifying checksum... OK (0x1826)
4 bytes copied in 0.120 secs (33 bytes/sec)

```

Router#

QUESTION 466



```

vrf definition Marketing
rd 111:1
address-family ipv4
!
interface E 0/2
vrf forwarding Marketing
ip address 172.16.1.1 255.255.255.0
no shut
!
interface E 0/3
vrf forwarding Marketing
ip address 172.16.2.1 255.255.255.0
no shut
!
vrf definition Sales
rd 111:2
address-family ipv4
!
interface E 0/0
vrf forwarding Sales
ip address 172.16.11.1 255.255.255.0
no shut
!
interface E 0/1
vrf forwarding Sales
ip address 172.16.12.1 255.255.255.0
no shut

```

Refer to the exhibit. The IT router is connected with the Sales and Marketing departments. The interfaces have been assigned to their respective VRFs to keep the two department routes isolated. Which configuration set must the IT router use for BGP to distribute routes for each department that maintains their own routing table for network isolation?

- A. router bgp 111
 - address-family ipv4 vrf Marketing
 - neighbor 172.16.1.2 remote-as 111
 - neighbor 172.16.2.2 remote-as 111
 - !
 - address-family ipv4 vrf Sales
 - neighbor 172.16.11.2 remote-as 111
 - neighbor 172.16.12.2 remote-as 111
- B. router bgp 111
 - address-family ipv4 vrf Marketing
 - neighbor 172.16.1.2 remote-as 111
 - neighbor 172.16.1.2 Route-reflector-client
 - neighbor 172.16.2.2 remote-as 111
 - neighbor 172.16.2.2 Route-reflector-client
 - !
 - address-family ipv4 vrf Sales
 - neighbor 172.16.11.2 remote-as 111
 - neighbor 172.16.11.2 Route-reflector-client
 - neighbor 172.16.12.2 remote-as 111
 - neighbor 172.16.12.2 Route-reflector-client
- C. router bgp 111
 - neighbor 172.16.1.2 remote-as 111
 - neighbor 172.16.2.2 remote-as 111
 - neighbor 172.16.11.2 remote-as 111
 - neighbor 172.16.12.2 remote-as 111
- D. router bgp 111
 - address-family ipv4 unicast
 - neighbor 172.16.1.2 remote-as 111
 - neighbor 172.16.2.2 remote-as 111
 - neighbor 172.16.11.2 remote-as 111
 - neighbor 172.16.12.2 remote-as 111

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 467

```

R1(config)#ip access-list standard EIGRP-FILTER
R1(config-std-nacl)#deny 10.10.10.0 0.0.0.0
R1(config-std-nacl)#permit 0.0.0.0 0.0.0.0
R1(config)#router eigrp 10
R1(config-router)#distribute-list route-map EIGRP in
!
R1(config)#route-map EIGRP permit 10
R1(config-route-map)#match ip address EIGRP-FILTER
!
R1#show ip route eigrp | include 10.10.10.
D      10.10.10.128/25

```

Refer to the exhibit. An engineer must filter EIGRP updates that are received to block all 10.10.10.0/24 prefixes. The engineer tests the distribute list and finds one associated prefix. Which action resolves the issue?

- A. There is a permit in the ACL that allows this prefix into EIGRP. The ACL should be modified to deny 10.10.10.0 0.0.0.255.
- B. There is a permit in the route map that allows this prefix. A deny 20 statement is required with no match condition to block the prefix.
- C. There is a permit in the route map that allows this prefix. A deny 20 statement is required with a match condition to match a new ACL that denies all prefixes.
- D. There is a permit in the ACL that allows this prefix into EIGRP. The ACL should be modified to deny 10.10.10.0 255.255.255.0.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

In the current access list, the deny rule with "10.10.10.0 0.0.0.0" can only filter routes with prefix having 10.10.10.0 exactly. Therefore, it cannot filter route having prefix 10.10.10.128.

In order to filter prefix having 10.10.10.X, the deny rule in ACL should be "10.10.10.0 0.0.0.255"

QUESTION 468

Loopback1: 100A:0:100C::1/64
 Loopback2: 200A:0:200C::1/64
 Loopback3: 300A:0:300C::1/64
 Loopback4: 400A:0:400C::1/64

Loopback1: 1001:ABC:2011:7::1/64

```
R1#
!
line con 0
logging synchronous
line aux 0
line vty 0 4
password cisco@123
login
transport input ssh telnet
!
end
```

```
R3#ping ipv6 400A:0:400C::1 source lo0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 400A:0:400C::1, timeout is 2 seconds:
Packet sent with a source address of 1001:ABC:2011:7::1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 5/5/5 ms

R3#telnet 400A:0:400C::1 /source-interface lo0
Trying 400A:0:400C::1 ...
% Destination unreachable; gateway or host down
```

```
R1#ping ipv6 1001:ABC:2011:7::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1001:ABC:2011:7::1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/5/21 ms
```

Refer to the exhibit. An engineer is trying to log in to R1 via R3 loopback address. Which action resolves the issue?

- A. Remove the IPv6 traffic filter from R1, which is blocking the Telnet.
- B. Add transport input SCP.
- C. Add transport input none.
- D. Remove the IPv6 traffic filter from R1, which is blocking the SSH.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The error message "% Destination unreachable; gateway or host down" often appears in one of the following conditions:

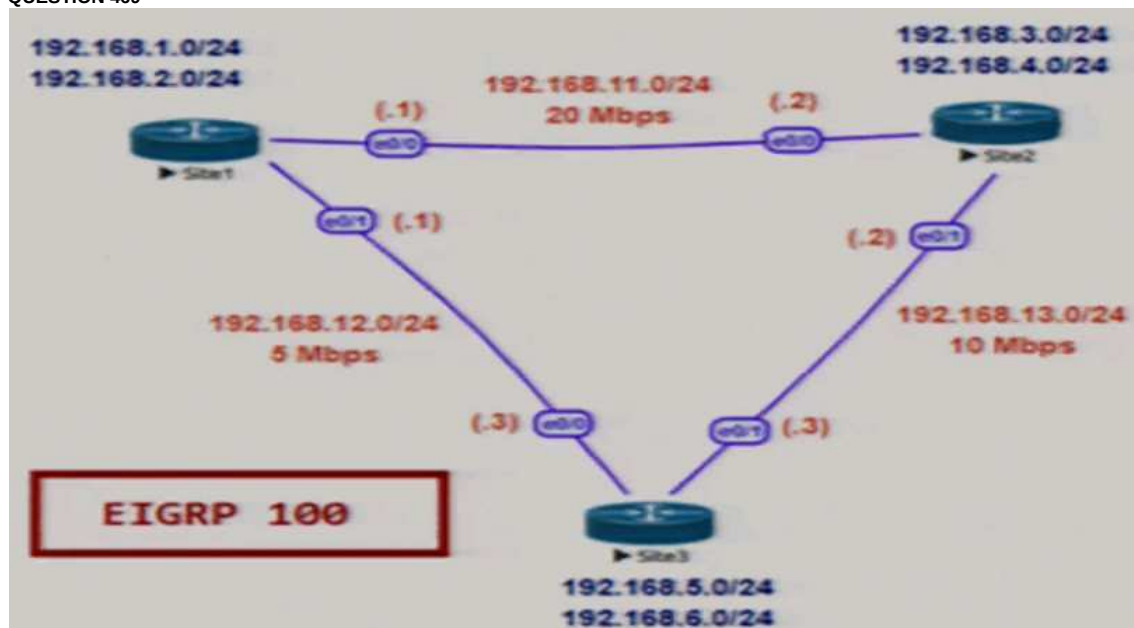
The telnet traffic is blocked (e.g. by packet filtering / firewall).

There is no route to the destination IP.

If the target is unresponsive e.g. no device with such IP, the message will be "% Connection timed out; remote host not responding".

If the target is not offering telnet service, the message will be "% Connection refused by remote host".

QUESTION 469



```

Site1 - Show ip route
Gateway of last resort is not set

  192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, Loopback0
L    192.168.1.1/32 is directly connected, Loopback0
D    192.168.3.0/24 [90/281600] via 192.168.11.2, 00:00:23, Ethernet0/0
D    192.168.4.0/24 [90/281600] via 192.168.11.2, 00:00:23, Ethernet0/0
D    192.168.5.0/24 [90/665600] via 192.168.12.3, 00:00:23, Ethernet0/1
    [90/435200] via 192.168.11.2, 00:00:23, Ethernet0/0
D    192.168.6.0/24 [90/665600] via 192.168.12.3, 00:00:23, Ethernet0/1
    [90/435200] via 192.168.11.2, 00:00:23, Ethernet0/0
  192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.11.0/24 is directly connected, Ethernet0/0
L    192.168.11.1/32 is directly connected, Ethernet0/0
  192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected, Ethernet0/1
L    192.168.12.1/32 is directly connected, Ethernet0/1
D    192.168.13.0/24 [90/563200] via 192.168.12.3, 00:00:23, Ethernet0/1
    [90/307200] via 192.168.11.2, 00:00:23, Ethernet0/0

```

```

Site1 - Show ip eigrp topology
P 192.168.3.0/24, 1 successors, FD is 230400
  via 192.168.11.2 (281600/128256), Ethernet0/0
  via 192.168.12.3 (691200/204800), Ethernet0/1
P 192.168.12.0/24, 1 successors, FD is 537600
  via Connected, Ethernet0/1
P 192.168.13.0/24, 2 successors, FD is 307200
  via 192.168.12.3 (563200/76800), Ethernet0/1
  via 192.168.11.2 (307200/281600), Ethernet0/0
P 192.168.1.0/24, 1 successors, FD is 128256
  via Connected, Loopback0
P 192.168.6.0/24, 2 successors, FD is 435200
  via 192.168.12.3 (665600/128256), Ethernet0/1
  via 192.168.11.2 (435200/409600), Ethernet0/0
P 192.168.4.0/24, 1 successors, FD is 230400
  via 192.168.11.2 (281600/128256), Ethernet0/0
  via 192.168.12.3 (691200/204800), Ethernet0/1
P 192.168.5.0/24, 2 successors, FD is 435200
  via 192.168.12.3 (665600/128256), Ethernet0/1
  via 192.168.11.2 (435200/409600), Ethernet0/0
P 192.168.11.0/24, 1 successors, FD is 153600
  via Connected, Ethernet0/0

```

```

Site1 - Show run | section router eigrp
router eigrp 100
 variance 2
 network 192.168.1.0
 network 192.168.2.0
 network 192.168.11.0
 network 192.168.12.0

```

Site1 must perform unequal cost load balancing toward the segments behind Site2 and Site3. Some of the routes are getting bad balanced but others are not. Which configuration allows Site1 to load balance toward all the LAN segments of the remote routers?

- A. Site2


```

router eigrp 100
 variance 3
      
```
- B. Site1


```

router eigrp 100
 variance 3
      
```
- C. Site2


```

router eigrp 100
 variance 2
      
```
- D. Site3


```

router eigrp 100
 variance 2
      
```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

From EIGRP topology, there are two paths for each of the networks 192.168.3.0/24, 192.168.4.0/24, 192.168.5.0/24 and 192.168.6.0/24. However, only one route shown for each of the network 192.168.3.0/24 and 192.168.4.0/24.

The two paths for either 192.168.3.0/24 and 192.168.4.0/24 are 281600 and 691200 where $691200 / 281600 = 2.45$. Hence, unequal cost load balancing will not be effective for them with "variance 2". You need at least "variance 3" in the Site 1 router (i.e. the router that you want to perform unequal cost load balancing).

QUESTION 470

What is the function of BFD?

- A. It negotiates to the highest version if the neighbor version differs.
- B. It provides uniform failure detection on the same media type.
- C. It provides uniform failure detection regardless of media type.
- D. It creates high CPU utilization on hardware deployments.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Cisco IOS Release 12.4(9)T supports BFD Version 1 as well as BFD Version 0. All BFD sessions come up as Version 1 by default and will be interoperable with Version 0. The system automatically performs BFD version detection, and BFD sessions between neighbors will run in the highest common BFD version between neighbors. For example, if one BFD neighbor is running BFD Version 0 and the other BFD neighbor is running Version 1, the session will run BFD Version 0. Since BFD uses UDP, it can work on all media types.

QUESTION 471

An engineer must override the normal routing behavior of a router for Telnet traffic that is destined to 10.10.10.10 from 10.10.1.0/24 via a next hop of 10.4.4.4, which is directly connected to the router that is connected to the 10.1.1.0/24 subset. Which configuration reroutes traffic according to this requirement?

- A.

```
access-list 100 permit tcp 10.10.1.0 0.0.0.255 host 10.10.10.10 eq 23
!
route-map POLICY permit 10
match ip address 100
set ip next-hop recursive 10.4.4.4
```
- B.

```
access-list 100 deny tcp 10.10.1.0 0.0.0.255 host 10.10.10.10 eq 23
!
route-map POLICY permit 10
match ip address 100
set ip next-hop 10.4.4.4
route-map POLICY permit 20
```
- C.

```
access-list 100 permit tcp 10.10.1.0 0.0.0.255 host 10.10.10.10 eq 23
!
route-map POLICY permit 10
match ip address 100
set ip next-hop 10.4.4.4
route-map POLICY permit 20
```
- D.

```
access-list 100 permit tcp 10.10.1.0 0.0.0.255 host 10.10.10.10 eq 23
!
route-map POLICY permit 10
match ip address 100
set ip next-hop recursive 10.4.4.4.
route-map POLICY permit 20
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The router needs to send traffic to 10.10.10.10 sourced from 10.10.1.0/24 to the next hop 10.4.4.4. Hence you need an access-list like the following:

```
access-list 100 permit tcp 10.10.1.0 0.0.0.255 host 10.10.10.10 eq 23
```

Moreover, the next hop router is directly connected to the router through the subnet 10.1.1.0/24. Since 10.4.4.4 is not the IP address of a directly connected network 10.1.1.0/24, you need to add the keyword "recursive" when setting "ip next-hop" command.

Among the choices, there are only two choices meeting the above two requirements. However, a match-all route-map statement at the end is not required since all un-matched packets will be routed through normal routing i.e. routing entries in routing table.

QUESTION 472

An administrator attempts to download the .pack NBAR2 file using TFTP from the CPE router to another device over the Gi0/0 interface. The CPE is configured as below:

hostname CPE

ip access-list extended WAN

<...>

Remark => All UDP rules below for WAN ID: D246P42F81E36

permit udp any eq domain any

permit udp any any eq tftp

deny udp any any

interface GigabitEthernet0/0

<...>

ip access-group WAN in

<...>

tftp-server flash:pp-adv-csr1000v-1608.2a-46-27.0.0.pack

The transfer fails. Which action resolves this issue?

- A. Change the WAN ACL to permit the entire UDP destination port range.
- B. Shorten the file name to the 8+3 naming convention.
- C. Change the WAN ACL to permit the UDP port 69 to allow TFTP.
- D. Make the permit udp any eq tftp any entry the last entry in the WAN ACL.

Correct Answer: A

Section: (none)

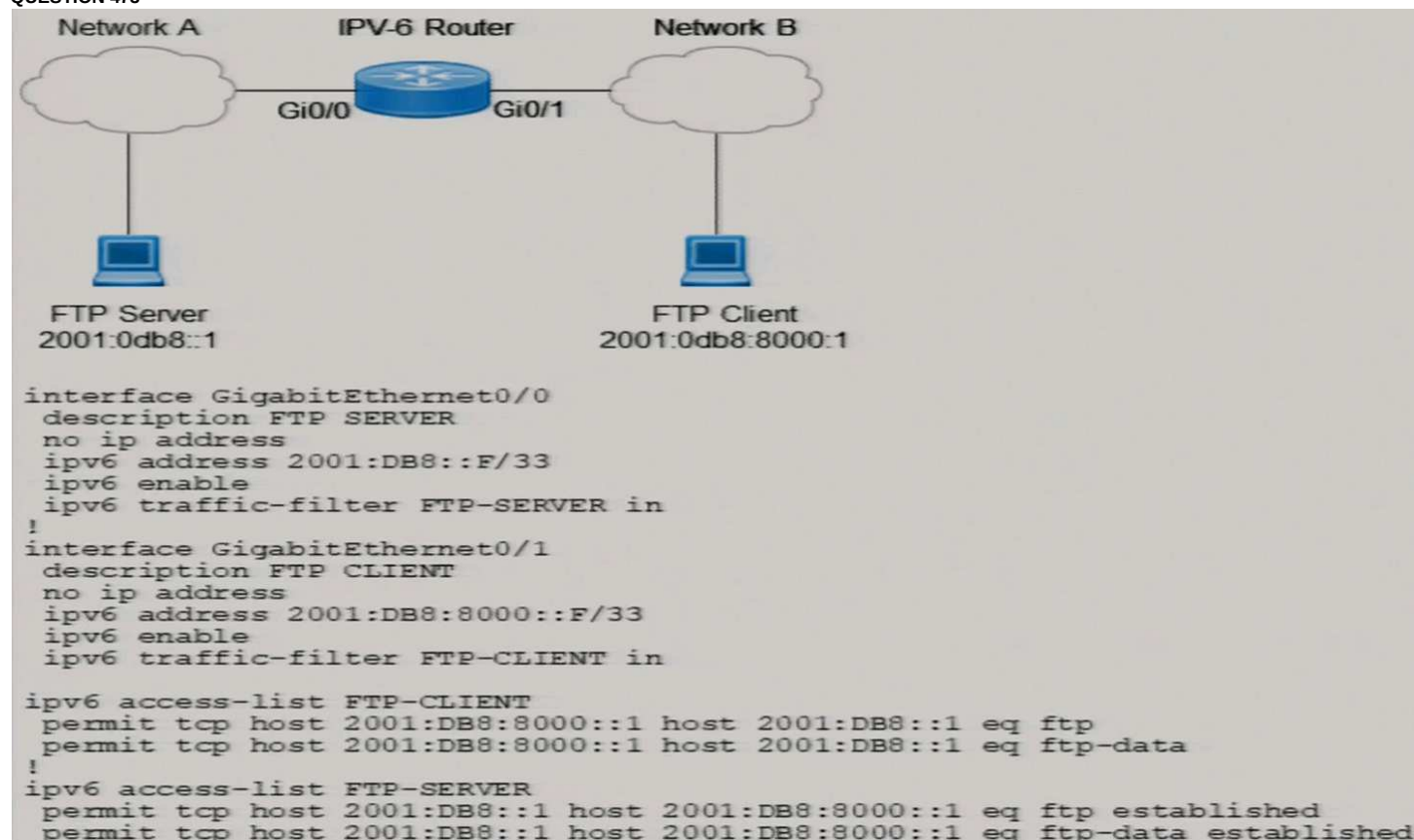
Explanation

Explanation/Reference:

TFTP is a UDP-based protocol. Servers listen on port 69 for the initial client-to-server packet to establish the TFTP session, then use a port above 1023 for all further

packets during that session. On the other hand, clients always use ports above 1023. Hence, just allowing destination UDP port 69 (i.e. the port name "ftp" shown in ACL) to the server is not enough.

QUESTION 473



When an FTP client attempts to use passive FTP to connect to the FTP server, the file transfers fail. Which action resolves the issue?

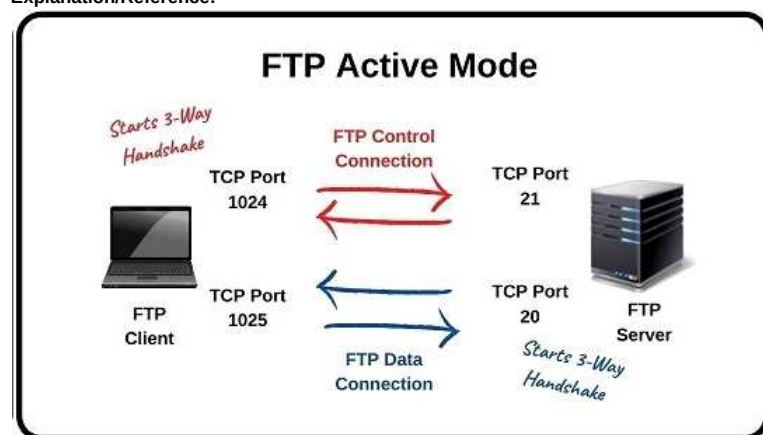
- A. Configure to permit TCP ports higher than 1023.
- B. Modify traffic filter **FTP-SERVER** in to the outbound direction.
- C. Configure active FTP traffic.
- D. Modify FTP-SERVER access list to remove **established** at the end.

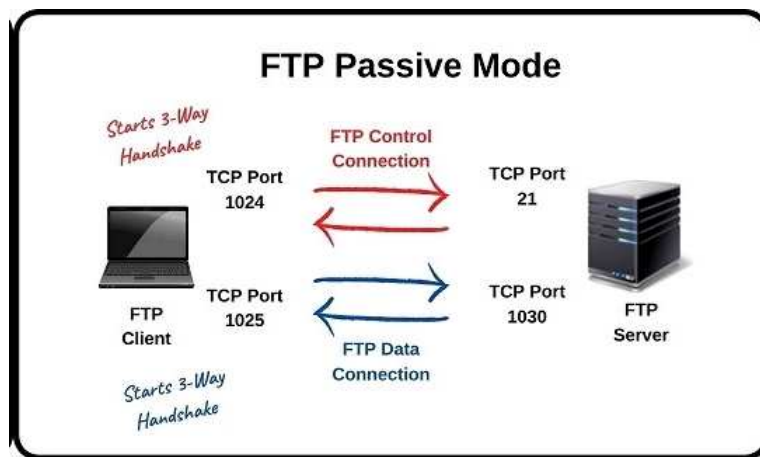
Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:





In the above diagrams, all port numbers > 1023 are dynamic ports which means that other port numbers may also be used instead of those shown in the diagram. For FTP Active mode:

- The inbound ACL "FTP-Client" using destination port "ftp" i.e. 21 and "ftp-data" i.e. "20" is correct.
- The inbound ACL "FTP-Server" using destination port "ftp" i.e. 21 and "ftp-data" i.e. "20" is **NOT** correct. You need to either:

* set the two ports as source ports and remove the "established" in the rule with port "ftp-data". OR
 * change the ACL to be applied in outbound direction and remove the "established" in the rule with port "ftp"

However, for this question which states the use of FTP passive mode, since dynamic ports are used in both FTP server and FTP client, you need to reconfigure both access list to allow the TCP port numbers higher than 1023.

Note that since the current configuration still needs changes in ACL "FTP-Server" if FTP Active mode is used instead. Hence, you cannot just change FTP to Active mode to solve the problem.

QUESTION 474

Configuration Output:

```
aaa new-model
aaa group server tacacs+ admin
server name admin
!
ip tacacs source-interface GigabitEthernet1
aaa authentication login admin group tacacs+ local enable
aaa session-id common
!
tacacs server admin
address ip 10.11.15.6
key 7 01150F165E1C07032D
!
line vty 0 4
login authentication admin
```

Debug Output:

```
Oct 22 12:38:57.587: AAA/BIND(0000001A): Bind w/
Oct 22 12:38:57.587: AAA/AUTHEN/LOGIN (0000001A): Pick method list 'admin'
Oct 22 12:38:57.587: AAA/AUTHEN/ENABLE(0000001A): Processing request action LOGIN
Oct 22 12:38:57.587: AAA/AUTHEN/ENABLE(0000001A): Done status GET_PASSWORD
Oct 22 12:39:02.327: AAA/AUTHEN/ENABLE(0000001A): Processing request action LOGIN
Oct 22 12:39:02.327: AAA/AUTHEN/ENABLE(0000001A): Done status FAIL - bad password
```

An administrator configured a Cisco router for TACACS authentication, but the router is using the local enable password instead. Which action resolves the issue?

- Configure the aaa authentication login default group admin local if-authenticated command instead.
- Configure the aaa authentication login admin group tacacs+ local enable none command instead.
- Configure the aaa authentication login admin group tacacs+ local if-authenticated command instead.
- Configure the aaa authentication login admin group admin local enable command instead.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

The tacacs+ group is setup with "aaa group server tacacs+ **admin**". Hence the group name for the tacacs+ server group is "**admin**". Hence, you need to change the authentication list "admin" to from using the server group name "tacacs+" to the group name "**admin**" i.e.

```
aaa authentication login admin group admin local enable
```

QUESTION 475

```

Router#show ip bgp vpnv4 rd 1100:1001 10.30.116.0/23
BGP routing table entry for 1100:1001:10.30.116.0/23, version 26765275
Paths: (9 available, best #6, no table)
  Advertised to update-groups:
    1      2      3
(65001 64955 65003) 65089, (Received from a RR-client)
  172.16.254.226 (metric 20645) from 172.16.224.236 (172.16.224.236)
    Origin IGP, metric 0, localpref 100, valid, confed-internal
    Extended Community: RT:1100:1001
    mpls labels in/out nolabel/362
(65008 64955 65003) 65089
  172.16.254.226 (metric 20645) from 10.131.123.71 (10.131.123.71)
    Origin IGP, metric 0, localpref 100, valid, confed-external
    Extended Community: RT:1100:1001
    mpls labels in/out nolabel/362
(65001 64955 65003) 65089
  172.16.254.226 (metric 20645) from 172.16.216.253 (172.16.216.253)
    Origin IGP, metric 0, localpref 100, valid, confed-external
    Extended Community: RT:1100:1001
    mpls labels in/out nolabel/362
(65001 64955 65003) 65089
  172.16.254.226 (metric 20645) from 172.16.216.252 (172.16.216.252)
    Origin IGP, metric 0, localpref 100, valid, confed-external
    Extended Community: RT:1100:1001
    mpls labels in/out nolabel/362
(64955 65003) 65089
  172.16.254.226 (metric 20645) from 10.77.255.57 (10.77.255.57)
    Origin IGP, metric 0, localpref 100, valid, confed-external
    Extended Community: RT:1100:1001
    mpls labels in/out nolabel/362
(64955 65003) 65089
  172.16.254.226 (metric 20645) from 10.57.255.11 (10.57.255.11)
    Origin IGP, metric 0, localpref 100, valid, confed-external, best
    Extended Community: RT:1100:1001
    mpls labels in/out nolabel/362

(64955 65003) 65089
  172.16.254.226 (metric 20645) from 172.16.224.253 (172.16.224.253)
    Origin IGP, metric 0, localpref 100, valid, confed-internal
    Extended Community: RT:1100:1001
    mpls labels in/out nolabel/362
(65003) 65089
  172.16.254.226 (metric 20645) from 172.16.254.234 (172.16.254.234)
    Origin IGP, metric 0, localpref 100, valid, confed-external
    Extended Community: RT:1100:1001
    mpls labels in/out nolabel/362
65089, (Received from a RR-client)
  172.16.228.226 (metric 20645) from 172.16.228.226 (172.16.228.226)
    Origin IGP, metric 0, localpref 100, valid, confed-internal
    Extended Community: RT:1100:1001
    mpls labels in/out nolabel/278

```

An engineer configured BGP and wants to select the path from 10.77.255.57 as the best path instead of current best path. Which action resolves the issue?

- A. Configure higher MED to select as the best path.
- B. Configure AS_PATH prepend for the current best path.
- C. Configure lower LOCAL_PREF to select as the best path.
- D. Configure AS_PATH prepend for the desired best path.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

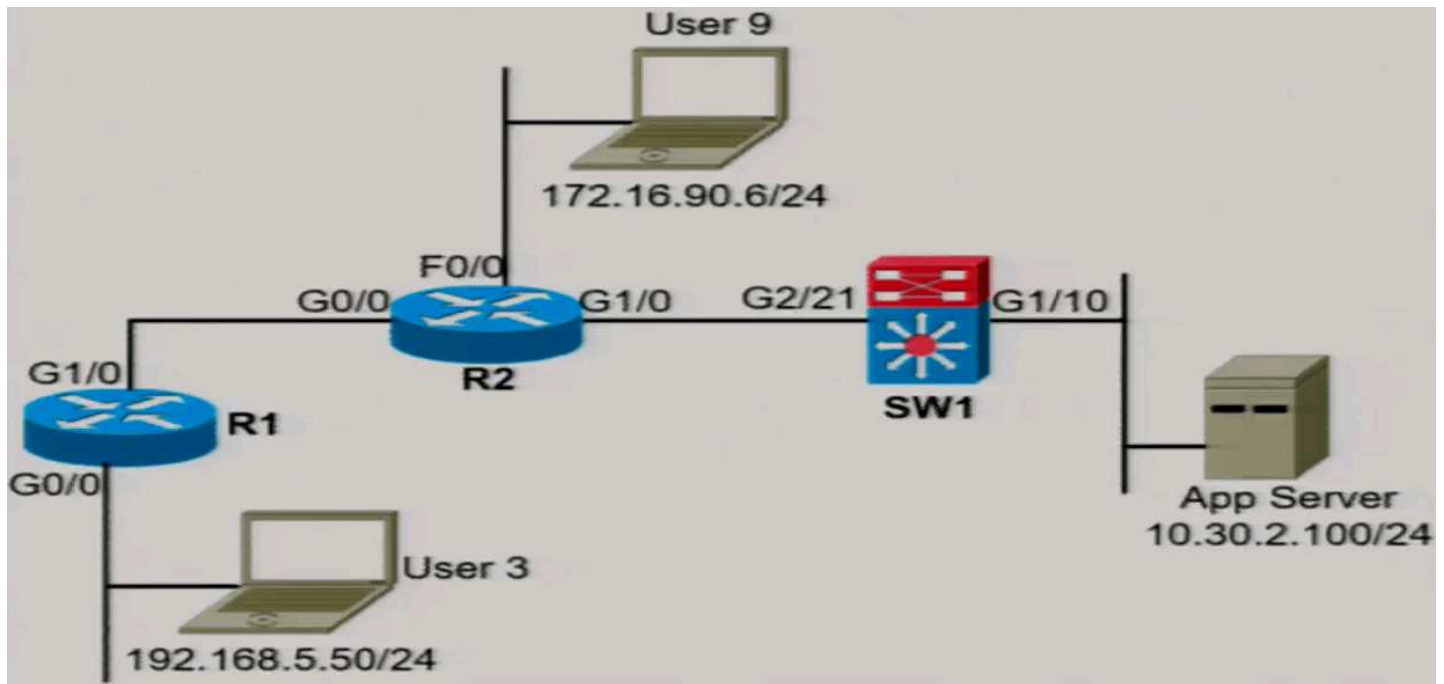
You can configure settings to choose an alternate desired path as the best path by either:

- Increase the Local Preference of the alternate desired path. OR
- Lower the Local Preference or use AS Prepending for the current best path.

Remarks :

MED only affects the path choosing in the neighbor AS' router.

QUESTION 476



A network administrator must block ping from user 3 to the App Server only. An inbound standard access list is applied to R1 interface G0/0 to block ping. The network administrator was notified that user 3 cannot even ping user 9 anymore. Where must the access list be applied in the outgoing direction to resolve the issue?

- A. SW1 interface G2/21
- B. R2 interface G1/0
- C. SW1 interface G1/10
- D. R2 interface G0/0

Correct Answer: C

Section: (none)

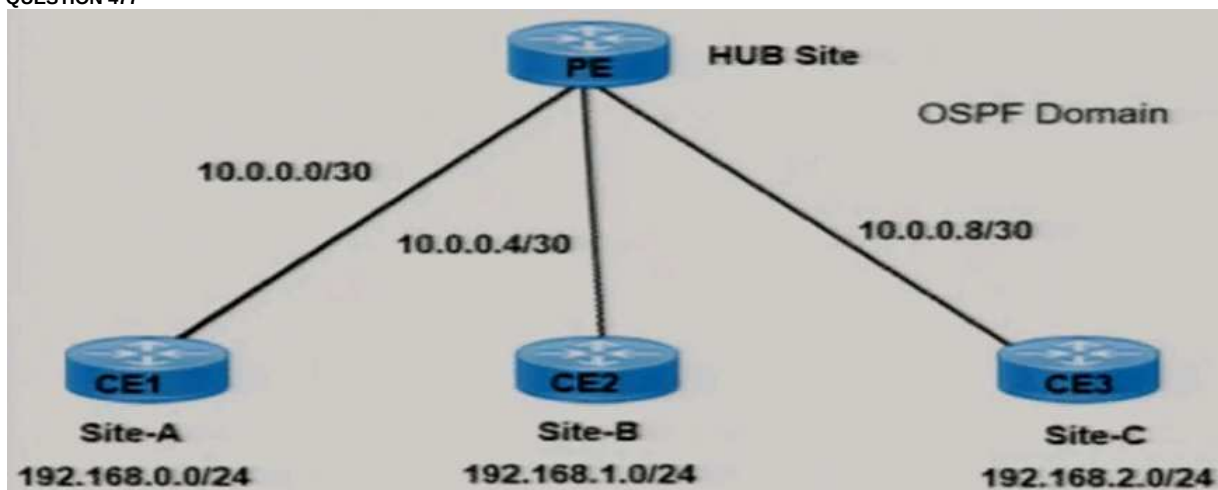
Explanation

Explanation/Reference:

The ping traffic from User 3 to User 9 will not go through the interfaces R2's G1/0, Sw1's G2/21 and Sw1's G1/10. Hence, applying ACL in those interfaces will never affect the ping traffic from User 3 to User 9.

However, since the question is using standard access list to block ping packets, no destination IP can be specified. For standard access list, it should be applied in an interface closest to the destination so as to avoid unintended blocking. Hence, it should be applied to the Layer3 switch Sw1's G1/0 outbound direction. This allows User3 to ping Sw1 if needed.

QUESTION 477



A network engineer must establish communication between three different customer sites with these requirements:

- Site-A: must be restricted to access to any users at Site-B or Site-C.
- Site-B and Site-C: must be able to communicate between sites and share routes using OSPF.

PE interface configuration:

```
interface FastEthernet0/0
ip vrf forwarding Site-A
```

```
interface FastEthernet0/1
ip vrf forwarding SharedSites
```

```
interface FastEthernet0/2
ip vrf forwarding SharedSites
```

Which configuration meets the requirements?

- A. PE(config)#router ospf 10 vrf Site-A
PE(config-router)#network 0.0.0.0 255.255.255.255 area 0
PE(config)#router ospf 10 vrf SharedSites
PE(config-router)#network 0.0.0.0 255.255.255.255 area 0
- B. PE(config)#router ospf 10 vrf Site-A
PE(config-router)#network 0.0.0.0 255.255.255.255 area 0
PE(config)#router ospf 20 vrf SharedSites
PE(config-router)#network 0.0.0.0 255.255.255.255 area 0
- C. PE(config)#router ospf 10 vrf Site-A
PE(config-router)#network 0.0.0.0 255.255.255.255 area 0
PE(config)#router ospf 20 vrf SharedSites
PE(config-router)#network 0.0.0.0 255.255.255.255 area 1
- D. PE(config)#router ospf 10 vrf Site-A
PE(config-router)#network 0.0.0.0 255.255.255.255 area 0
PE(config)#router ospf 10 vrf SharedSites
PE(config-router)#network 0.0.0.0 255.255.255.255 area 1

Correct Answer: B

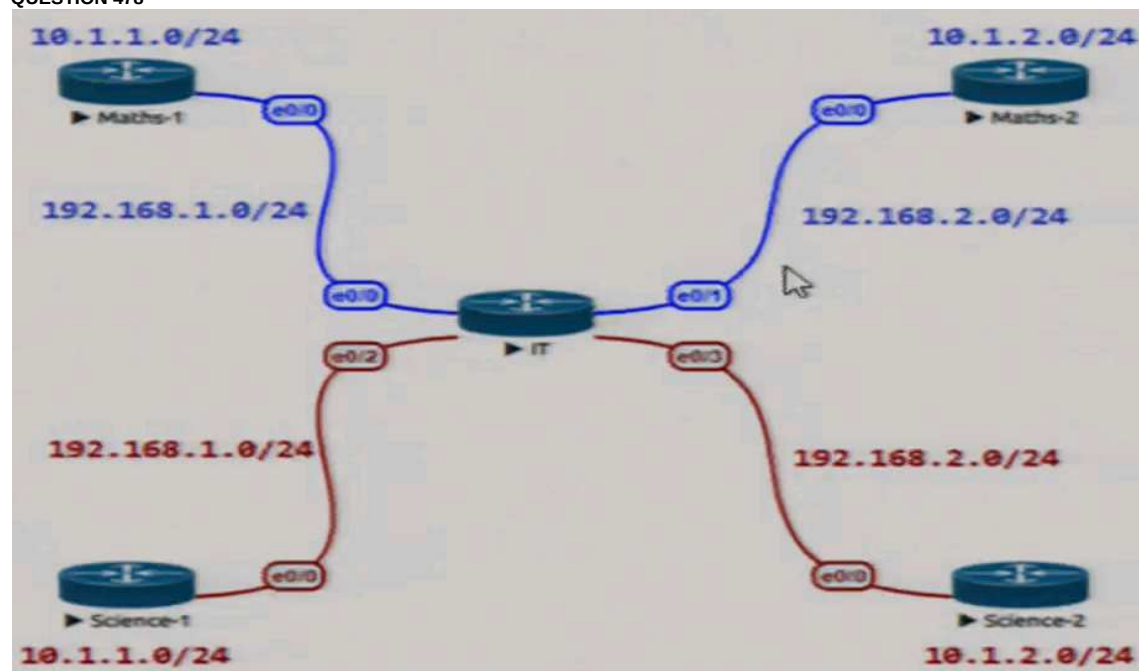
Section: (none)

Explanation

Explanation/Reference:

OSPF process of different VRFs should use different process numbers. Since OSPF process of different VRFs can use their own Area 0 for different networks, we shall use the choice that configure Area 0 in both VRFs. (Actually the choice of area depends on the configuration in CE2 and CE3 router).

QUESTION 478



IT Router

```
vrf definition Science
  address-family ipv4
!
Interface E 0/2
  Vrf forwarding Science
  Ip address 192.168.1.1 255.255.255.0
  No shut
!
Interface E 0/3
  Vrf forwarding Science
  Ip address 192.168.2.1 255.255.255.0
  No shut
```

The IT router has been configured with the Science VRF and the interfaces have been assigned to the VRF. Which set of configurations advertises Science-1 and Science-2 routes using EIGRP AS 111?

- A. router eigrp 1
 address-family ipv4 vrf Science autonomous-system 111
 network 192.168.1.0
 network 192.168.2.0
- B. router eigrp 111
 address-family ipv4 vrf Science
 network 192.168.1.0
 network 192.168.2.0
- C. router eigrp 111
 network 192.168.1.0
 network 192.168.2.0
- D. router eigrp 111
 address-family ipv4 vrf Science autonomous-system 1
 network 192.168.1.0
 network 192.168.2.0

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Note that when configure EIGRP for VRF, the command e.g. "route eigrp 1" uses the "1" as a name and is NOT an EIGRP AS number. Hence, you need to specify the AS 111 using "autonomous-system 111".

QUESTION 479



An engineer must configure PBR on R1 to reach to 10.2.2.0/24 via R3 AS64513 as the primary path and a backup route through default route via R2 AS64513. All BGP routes are in the routing table of R1, but a static default route overrides BGP routes. Which PBR configuration achieves the objective?

- A. access-list 100 permit ip 10.1.1.0 0.0.0.255 10.2.2.0 0.0.0.255
 !
 route-map PBR permit 10
 match ip address 100
 set ip next-hop 10.3.3.1

- B. access-list 100 permit ip 10.1.1.0 0.0.0.255 10.2.2.0 0.0.0.255
!
route-map PBR permit 10
match ip address 100
set ip next-hop recursive 10.3.3.1
- C. access-list 100 permit ip 10.1.1.0 255.255.255.0 10.2.2.0 255.255.255.0
!
route-map PBR permit 10
match ip address 100
set ip next-hop 10.3.3.1
- D. access-list 100 permit ip 10.1.1.0 255.255.255.0 10.2.2.0 255.255.255.0
!
route-map PBR permit 10
match ip address 100
set ip next-hop recursive 10.3.3.1

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

First, wildcard mask must be used in access list. Hence two to the choices are wrong since they are not using wildcard mask.

For the remaining two choices, since 10.3.3.1 (the IP address of R7) is not an IP address in the directly connected network of R1, the keyword "recursive" is needed.

QUESTION 480

```
ip sla 1
 icmp-echo 8.8.8.8
 threshold 1000
 timeout 2000
 frequency 5
ip sla schedule 1 life forever start-time now
!
track 1 ip sla 1
!
ip route 0.0.0.0 0.0.0.0 203.0.113.1 name ISP1 track 1
ip route 0.0.0.0 0.0.0.0 198.51.100.1 name ISP2 track 1
```

An administrator configures a router to stop using a particular default route if the DNS server 8.8.8.8 is not reachable through that route. However, this configuration did not work as desired and the default route still works even if the DNS server 8.8.8.8 is unreachable. Which two configuration changes resolve the issue? (Choose two.)

- A. Configure two static routes for the 8.8.8.8/32 destination to match the IP SLA probe for each ISP.
- B. Associate every IP SLA probe with the proper WAN address of the router.
- C. Use a separate IP SLA probe and track object for every static route.
- D. Use a separate track object to reference the existing IP SLA 1 probe for every static route.
- E. Reference the proper exit interfaces along with the next hops in both static default routes.

Correct Answer: BC

Section: (none)

Explanation

Explanation/Reference:

In the existing configuration, since both static routes are using the same track object i.e. the same IP SLA 1, they will be added or removed in the routing table at the same time.

To fix this, you need to configure two different IP SLAs, each for one static route. Moreover, each IP SLA should be configured to use the corresponding ISP (through "source-ip" or "source-interface") for sending probes.

QUESTION 481

What is LDP label binding?

- A. two routers with label distribution session
- B. neighboring router with label
- C. destination prefix with label
- D. source prefix with label

Correct Answer: C

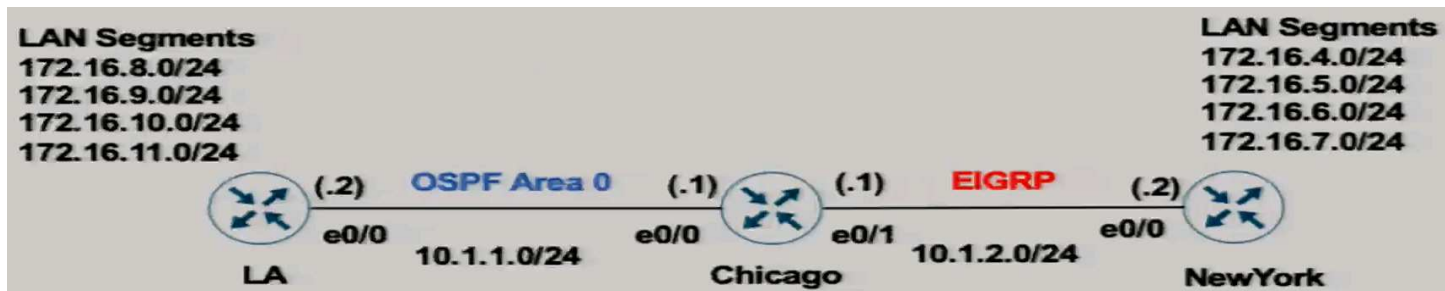
Section: (none)

Explanation

Explanation/Reference:

label binding—An association between a destination prefix and a label.

QUESTION 482



The network administrator configured the Chicago router to mutually redistribute the LA and NewYork routes with OSPF routes to be summarized as a single route in EIGRP using the longest summary mask:

```
router eigrp 100
 redistribute ospf 1 metric 10 10 10 10 10
router ospf 1
 redistribute eigrp 100 subnets
```

```
interface E 0/0
 ip summary-address eigrp 100 172.16.0.0 255.255.0.0
```

After the configuration, the New York router receives all the specific LA routes but the summary route. Which set of configurations resolves the issue on the Chicago router?

- A. interface E 0/1
ip summary-address eigrp 100 172.16.0.0 255.255.0.0
- B. router eigrp 100
summary-address 172.16.8.0 255.255.252.0
- C. router eigrp 100
summary-address 172.16.0.0 255.255.0.0
- D. interface E 0/1
ip summary-address eigrp 100 172.16.8.0 255.255.252.0

Correct Answer: D

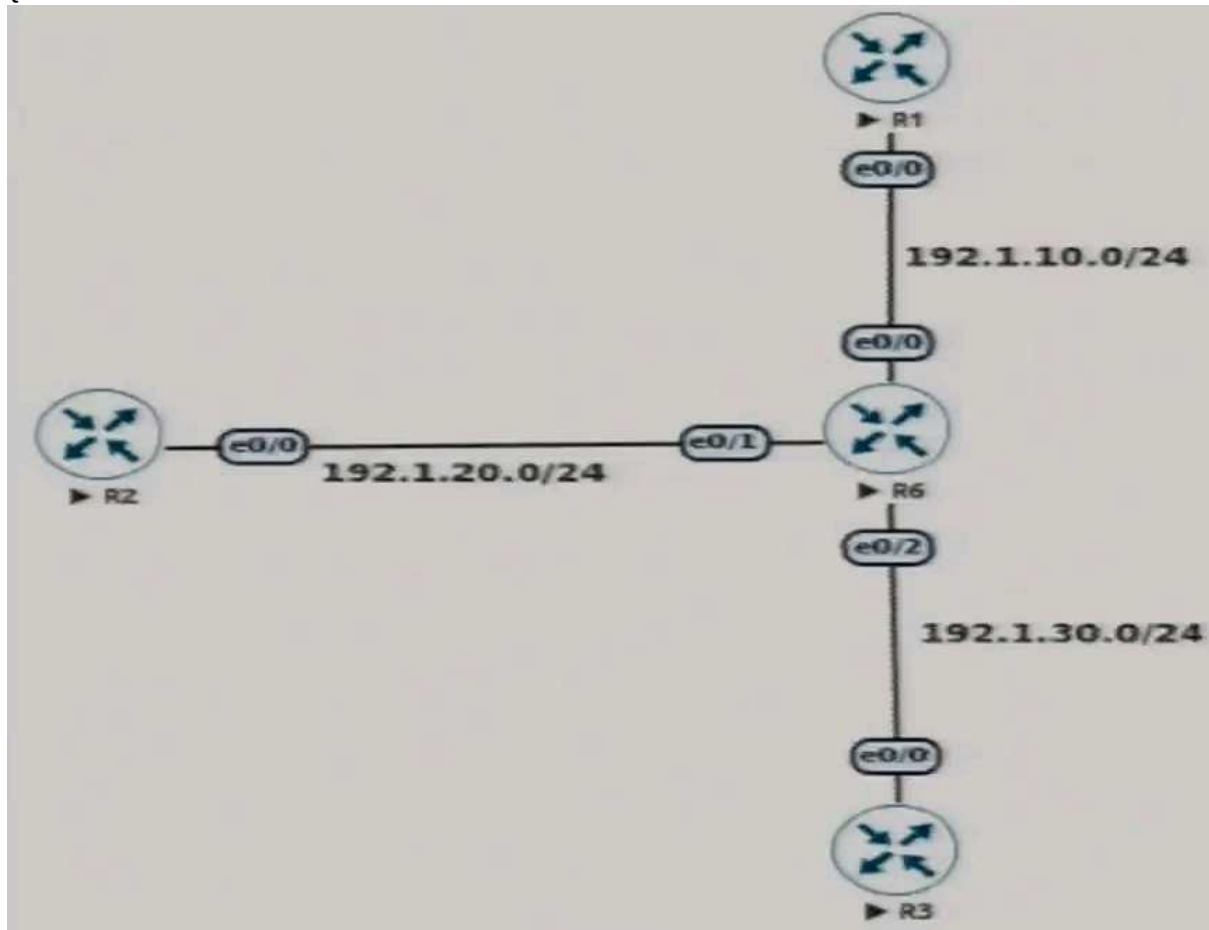
Section: (none)

Explanation

Explanation/Reference:

In order to summary routes for advertising in EIGRP, you need to configure the "ip summary-address" command in an EIGRP interface. The existing configuration is wrong since Chicago's E0/0 is an OSPF interface. Hence, you should configure this command in Chicago's E0/1. For summarizing 172.16.8.0/24 – 172.16.11.0/24 with longest summary mask, you should use 172.16.8.0/22 i.e. 172.16.8.0 255.255.252.0.

QUESTION 483



An engineer must configure DMVPN Phase 3 hub-and-spoke topology to enable a spoke-to-spoke tunnel. Which NHRP configuration meets the requirement on R6?

- A. Interface Tunnel 1
ip address 192.168.1.1 255.255.255.0
tunnel source e 0/1
tunnel mode gre multipoint
ip nhrp network-id 1
ip nhrp map 192.168.1.2 192.1.20.2
- B. Interface Tunnel 1
ip address 192.168.1.1 255.255.255.0
tunnel source e 0/0
tunnel mode gre multipoint
ip nhrp network-id 1
- C. interface Tunnel1
ip nhrp authentication Cisco123
ip nhrp map multicast dynamic
ip nhrp network-id 1
ip nhrp holdtime 300
ip nhrp shortcut
- D. interface Tunnel1
ip nhrp authentication Cisco123
ip nhrp map multicast dynamic
ip nhrp network-id 1
ip nhrp holdtime 300
ip nhrp redirect

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

For DMVPN phase 3, you need to configure "ip nhrp redirect" in the Hub router.

Remarks:

The question seems to have problems since R6 is connecting to the Spoke routers through multiple ethernet interfaces (e0/0, e0/1 and e0/2) instead of a single interface. Moreover, for the choice with "ip nhrp redirect", the tunnel interface is missing "IP address", "tunnel mode" and "tunnel source" settings. Some of these settings cannot be configured properly in the tunnel since 3 different ethernet interfaces are involved.

QUESTION 484

```

R1:
interface Loopback1
 no ip address
 ipv6 address 100A:0:100C::1/64
 ipv6 enable
 ipv6 ospf 10 area 0
!
interface Loopback4
 no ip address
 ipv6 address 400A:0:400C::1/64
 ipv6 enable
 ipv6 ospf 10 area 0
!
interface Serial1/0
 no ip address
 ipv6 address AB01:2011:7:100::/64 eui-64
 ipv6 enable
 ipv6 ospf network point-to-point
 ipv6 ospf 10 area 0
 ipv6 traffic-filter DENY_TELNET_Lo4 in
 serial restart-delay 0
 clock rate 64000
!
ipv6 router ospf 10
 router-id 1.1.1.1
 log-adjacency-changes
!
ipv6 access-list DENY_TELNET_L04
 sequence 20 deny tcp host 100:ABC:2011:7 host 400A:0:400C::1 eq telnet permit ipv6 any any
end

R2:
interface Loopback0
 no ip address
 ipv6 address 1001:ABC:2011:7::1/64
 ipv6 enable
 ipv6 ospf 10 area 0
!
interface Serial1/0
 no ip address
 ipv6 address AB01:2011:7:100::/64 eui-64
 ipv6 enable
 ipv6 ospf network point-to-point
 ipv6 ospf 10 area 0
 serial restart-delay 0
!
ipv6 router ospf 10
 router-id 2.2.2.2
 log-adjacency-changes
!
end

```

An engineer implemented an access list on R1 to allow anyone to Telnet except R2 Loopback0 to R1 Loopback4. How must sequence 20 be replaced on the R1 access list to resolve the issue?

- A. sequence 20 deny tcp host 1001:ABC:2011:7::1 host 400A:0:400C::1 eq telnet
- B. sequence 20 deny tcp host 400A:0:400C::1 host 1001:ABC:2011:7::1 eq telnet
- C. sequence 20 permit tcp host 1001:ABC:2011:7::1 host 400A:0:400C::1 eq telnet
- D. sequence 20 permit tcp host 400A:0:400C::1 host 1001:ABC:2011:7::1 eq telnet

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

In the IPv6 access list "DENY_TELNET_L04", the source IPv6 "100:ABC:2011:7" configured in the deny rule sequence number 20 is wrong. The R2's Lo0 address is actually "1001:ABC:2011:7::1".

Remarks :

The "sequence 20 " shown in the diagram should be content of two rules e.g.:

sequence 20 deny tcp host 100:ABC:2011:7 host 400A:0:400C::1 eq telnet

sequence 30 permit ipv6 any any

After changing ...

sequence 20 deny tcp host 1001:ABC:2011:7::1 host 400A:0:400C::1 eq telnet

sequence 30 permit ipv6 any any

QUESTION 485

An engineer notices that R1 does not hold enough log messages to identify the root cause during troubleshooting. Which command resolves this issue?

- A. (config)#logging buffered 16000 informational
- B. #logging buffered 4096 critical
- C. #logging buffered 16000 critical
- D. (config)#logging buffered 4096 informational

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

"4096" bytes is the minimum size that you can configured in the global configuration mode command "logging buffered". Hence you should configure a larger value.

QUESTION 486

```
R1#sh run | section eigrp
router eigrp 10
network 10.10.10.0 0.0.0.255
no auto-summary
neighbor 10.10.10.2 FastEthernet0/0
neighbor 10.10.10.3 FastEthernet0/0

R1#show ip eigrp neighbors
IP-EIGRP neighbors for process 10
H   Address                  Interface      Hold Uptime    SRTT    RTO    Q
Seq                                     (sec)          (ms)                  Cnt
Num
1   10.10.10.2                Fa0/0         10 00:01:01    42     232    0    6
0   10.10.10.3                Fa0/0         10 00:01:03    43     244    0    6
```

Refer to the exhibit. The remote branch locations have a static neighbor relationship configured to R1 only. R1 has successful neighbor relationships with the remote locations of R2 and R3, but the end users cannot communicate with each other. Which configuration resolves the issue?

- A. R1
interface FastEthernet0/0
no ip split-horizon eigrp 10
- B. R2
interface FastEthernet0/0.10
encapsulation dot1Q
ip address 10.10.10.2 255.255.255.0

R3
interface FastEthernet0/0.10
encapsulation dot1Q
ip address 10.10.10.3 255.255.255.0
- C. R2 and R3
interface FastEthernet0/0
no ip split-horizon eigrp 10
- D. R2
interface FastEthernet0/0.10
encapsulation dot1Q 10
ip address 10.10.10.2 255.255.255.0

R3
interface FastEthernet0/0.10
encapsulation dot1Q 10
ip address 10.10.10.3 255.255.255.0

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

QUESTION 487

Drag and drop the operations from the left onto the locations where the operations are performed on the right. Which are the operations of Label Switch Router (Choose two)?

- A. assigns labels to unlabeled packets
- B. performs penultimate hop popping
- C. handles traffic between multiple VPNs
- D. reads the labels and forwards the packet based on the labels

Correct Answer: BD

Section: (none)

Explanation

Explanation/Reference:

assigns labels to unlabeled packets	Label Switch Router
performs penultimate hop popping	
handles traffic between multiple VPNs	Label Edge Router
reads the labels and forwards the packet based on the labels	

QUESTION 488

Drag and drop the operations from the left onto the locations where the operations are performed on the right.
Which are the operations of Label Edge Router (Choose two)?

- A. assigns labels to unlabeled packets
- B. performs penultimate hop popping
- C. handles traffic between multiple VPNs
- D. reads the labels and forwards the packet based on the labels

Correct Answer: AC

Section: (none)

Explanation

Explanation/Reference:

assigns labels to unlabeled packets	Label Switch Router
performs penultimate hop popping	
handles traffic between multiple VPNs	Label Edge Router
reads the labels and forwards the packet based on the labels	

QUESTION 489



S1

10.0.0.2/30
E0/1

10.0.0.1/30
E0/1



R3

```

S1#ping 10.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms

S1#telnet 10.0.0.1
Trying 10.0.0.1 ... Open

[Connection to 10.0.0.1 closed by foreign host]

```

```

R3#
hostname R3
!
enable password cisco
!
no aaa new-model
!
username admin password 0 cisco
!
interface Ethernet0/1
ip address 10.0.0.1 255.255.255.252
!
line con 0
logging synchronous
line aux 0
line vty 0 4
password cisco
login
no exec
transport input all
!
end

```

A network engineer cannot remote access R3 using Telnet from switch S1. Which action resolves the issue?

- A. Allow the inbound connection via the exec command on R3.
- B. Add the transport input telnet command on R3.
- C. Allow to use the ssh -l admin 10.0.0.1 command on the switch.
- D. Add the login admin command on the switch.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Existing configuration in R3 under "line vty 0 4" contains "no exec" which prevents remote connection from getting command line interface. You need to enter line configuration mode command "exec" to remove that setting in order to restore the line configuration to its default setting i.e. allowing command line interface.

QUESTION 490

What is LDP used for in an LSR?

- A. to create a database of label bindings that allow for hop-by-hop forwarding
- B. to communicate the routes known for a specific interface
- C. to allow for a system-wide exchange of labels across MPLS network
- D. to create a label across the PE routers for end-to-end path assignment

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

MPLS Label Distribution Protocol (LDP) enables peer label switch routers (LSRs) in a Multiprotocol Label Switching (MPLS) network to exchange label binding information for supporting hop-by-hop forwarding in an MPLS network. Since labels are assigned by each router independent of other routers, the labels assigned are exchanged between adjacent peer routers and are not system-wide.

QUESTION 491

Which IPv6 feature enables a device to reject traffic when it is originated from an address that is not stored in the device binding table?

- A. IPv6 DAD Proxy
- B. IPv6 RA Guard
- C. IPv6 Source Guard
- D. IPv6 Snooping

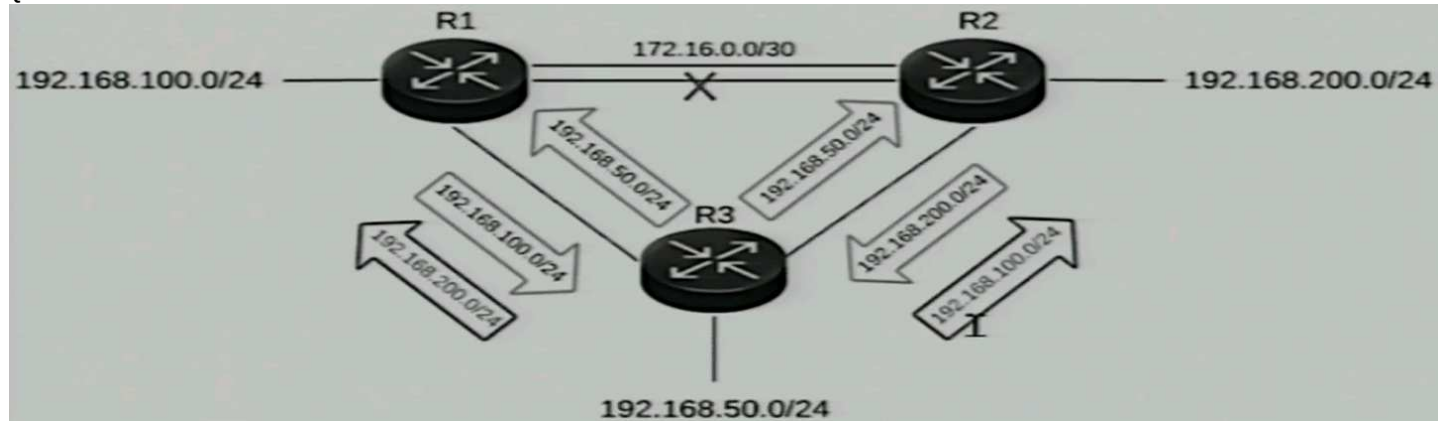
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 492



The primary link between R1 and R2 went down, but R3 is still advertising the 192.168.200.0/24 network to R1 and the 192.168.100.0/24 network to R2, which creates a loop. Which action resolves the issue?

- A. Configure the eigrp stub command under the EIGRP process on R2.
- B. Configure the summary-address 192.168.0.0 255.255.0.0 100 command on R3.
- C. Configure the eigrp stub leak-map command under the EIGRP process on R1.
- D. Configure the eigrp stub command under the EIGRP process on R3.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

If loops are created due to R3 advertising the learnt route 192.168.200.0/24 to R1 and the learnt route 192.168.100.0/24 to R2, the simplest way is to configure "eigrp stub" in R3 so that it will not advertise learnt routes.

Remarks :

The question does not mention the routing protocol currently being used by the routers. For the choice, the command "summary-address" is for OSPF external routes summarization while all others are for EIGRP. Since the question does not mentioned anything about redistribution or external routes, the command "summary-address" is not applicable. Moreover, sending the OSPF summary route may still cause the existing routing loop problem. Among the EIGRP commands, only the one configured in R3 can solve the problem.

QUESTION 493


```

R1#sh run | begin router eigrp 100
router eigrp 100
  distribute-list route-map ENT out GigabitEthernet0/1
  network 172.16.1.0 0.0.0.3
  network 172.16.250.0 0.0.0.3
  !
route-map ENT deny 10
  match ip address 20
  !
access-list 15 permit 172.16.1.15
access-list 20 permit 172.16.2.0 0.0.0.3
  !
end

```

```

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PDR

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 7 subnets, 3 masks
C       172.16.1.0/30 is directly connected, GigabitEthernet0/1
L       172.16.1.1/32 is directly connected, GigabitEthernet0/1
D       172.16.1.16/28
        [90/130816] via 172.16.1.2, 00:13:34, GigabitEthernet0/1
D       172.16.2.0/30
        [90/13072] via 172.16.250.2, 00:13:26, GigabitEthernet0/0
D       172.16.2.16/28
        [90/130872] via 172.16.250.2, 00:13:26, GigabitEthernet0/0
C       172.16.250.0/30 is directly connected, GigabitEthernet0/0
L       172.16.250.1/32 is directly connected, GigabitEthernet0/0

end

```

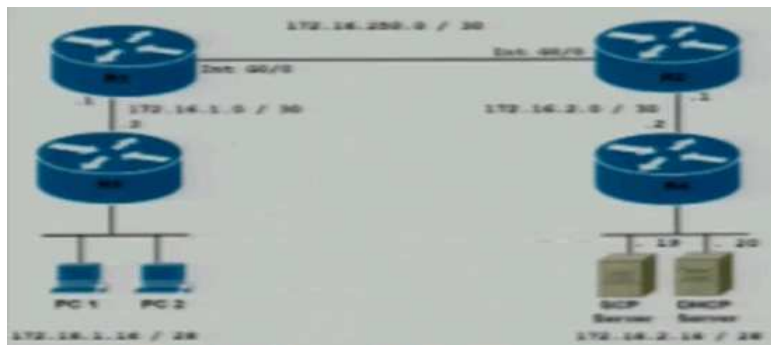
```

R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PDR

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
C       172.16.1.0/30 is directly connected, GigabitEthernet0/1
L       172.16.1.2/32 is directly connected, GigabitEthernet0/1
C       172.16.1.16/28 is directly connected, Loopback0
L       172.16.1.17/32 is directly connected, Loopback0
R3#sh run | begin router eigrp 100
router eigrp 100
  network 172.16.1.0 0.0.0.3
  network 172.16.1.16 0.0.0.15
  !
end

```

Which configuration command reestablishes the connectivity from 172.16.1.16/28 to the server subnet 172.16.2.16/28?

- A. R1(config)#route-map ENT permit 20
- B. R1(config)#access-list 20 permit 172.16.2.16 0.0.0.15
- C. R3(config-router)#no eigrp stub
- D. R1(config)#ip route 172.16.2.16 255.255.255.240 172.16.250.1 91

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

In the existing configuration, the PCs in the subnet 172.16.1.16/28 (bottom left of the diagram) cannot access the servers in the subnet 172.16.2.16/28 (bottom right of the diagram). This is because when the traffic from the PCs is sent to their connected router i.e. R3, the traffic is dropped since R3 does not have the route to the subnet 172.16.2.16/28.

R3 does not learn any EIGRP route since R1 blocks the sending of all EIGRP routes to R3 due to the route-map ENT with deny statements only. In order for R1 to deny specific EIGRP routes specified in the access-list 20 while sending all other EIGRP routes (including 172.16.2.16/28) to R3, you need to append a permit statement to the route-map ENT in R1 at the end.

Remarks:

Adding "172.16.2.16 0.0.0.15" to the access-list 20 does not work since this access-list is used by the route-map's seq 10 which has a deny action.

QUESTION 494

```
ip flow-export destination 203.0.113.254 9995
ip flow-export source loopback2
ip flow-export version 9
ip flow-cache timeout active 1
ip flow-cache timeout inactive 15
snmp-server ifindex persist
!
R1# show ip flow interface
Ethernet1/1
    ip flow ingress
Ethernet1/2
    ip flow ingress
!
R1# show ip flow export
Flow export v9 is enabled for main cache
Export source and destination details :
VRF ID : Default
    Source(1)      172.16.1.1 (Unknown)
    Destination(1) 203.0.113.254 (9995)
Version 9 flow records
0 flows exported in 0 udp datagrams
0 flows failed due to lack of export packet
0 export packets were sent up to process level
0 export packets were dropped due to no fib
0 export packets were dropped due to adjacency issues
0 export packets were dropped due to fragmentation failures
0 export packets were dropped due to encapsulation fixup
failures
```

It was noticed that after NetFlow is configured in the router, the collector stopped receiving flow information. Which action resolves the issue?

- A. Configure an IP address on the loopback2 interface to use as a source.
- B. Modify the source through the ip flow-export source loopback1 command.
- C. Change the IP address of the loopback 2 interface to a public IP address.
- D. Apply the ip flow egress command to the loopback2 interface.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

If everything is properly configured, the output from "show ip flow export" in R1 should contain the followings:

```
Source(1)      172.16.1.1 (Loopback2)
Destination(1) 203.0.113.254 (9995)
```

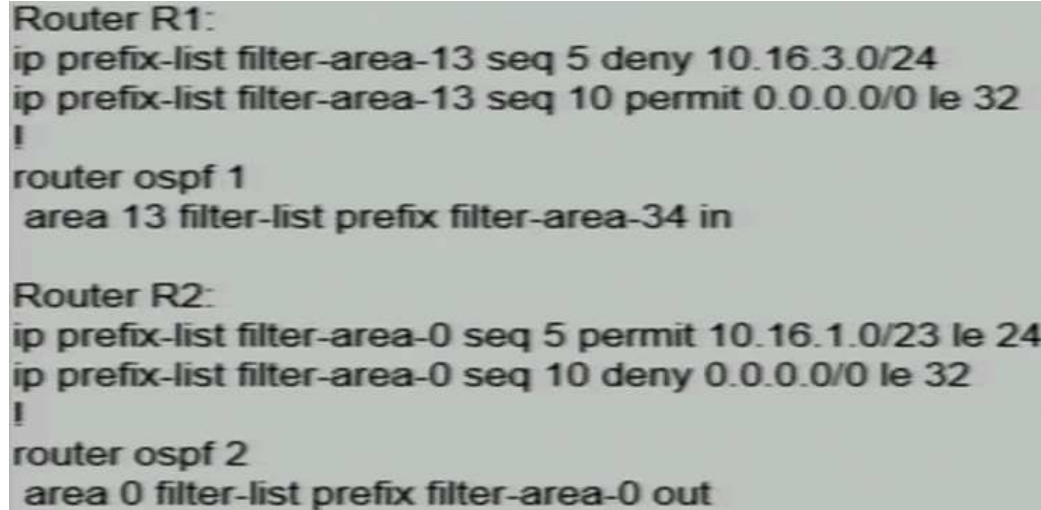
However, if there is an incorrect configuration change e.g. IP address is removed from the Loopback2, then the followings (i.e. the output shown in the question diagram) will be shown:

```
Source(1)      172.16.1.1 (Unknown)
Destination(1) 203.0.113.254 (9995)
```

After the incorrect change, the source IP will be unknown and R1 will not send flow information to the collector any more.

For solving the problem, you need to configure an IP address to Loopback2 so that R1 will send flow information to the collector again.

QUESTION 495



```
Router R1:
ip prefix-list filter-area-13 seq 5 deny 10.16.3.0/24
ip prefix-list filter-area-13 seq 10 permit 0.0.0.0/0 le 32
!
router ospf 1
 area 13 filter-list prefix filter-area-34 in

Router R2:
ip prefix-list filter-area-0 seq 5 permit 10.16.1.0/23 le 24
ip prefix-list filter-area-0 seq 10 deny 0.0.0.0/0 le 32
!
router ospf 2
 area 0 filter-list prefix filter-area-0 out
```

R1 should receive 10.16.2.0/24 from R2. Which action resolves the issue?

- A. Modify prefix-list seq 5 on R2 to permit 10.16.0.0/23.
- B. Add prefix-list seq 1 on R1 to permit 10.16.2.0/24.
- C. Add prefix-list seq 1 on R1 to permit 10.16.0.0/22.
- D. Modify prefix-list seq 5 on R2 to permit 10.16.0.0/22.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

In the existing configuration:

- R2 blocks the sending of the route 10.16.2.0/24 since it is not explicitly permitted and therefore is denied by the last rule in the prefix-list "filter-area-0". (The wrong network format 10.16.1.0/23 entered will be converted to 10.16.0.0/23 automatically in the running-config. However "10.16.0.0/23 le 24" includes only 10.16.0.0/24 and 10.16.1.0/24. Therefore it still does not include 10.16.2.0/24.)
- R1 does NOT block the receiving of the route 10.16.2.0/24 since it is not explicitly denied by the 1st rule of the filter-list "filter-area-13" and therefore is permitted by the last rule in the prefix-list.

Hence, in order for R1 to receive the route 10.16.2.0/24 from R2, you need to add a rule to permit 10.16.2.0/24 in the prefix-list of R2. However, since there is no choice for adding an additional rule in R2, you need to modify the existing rule in seq 5 to 10.16.0.0/22 (i.e. "permit 10.16.0.0/22 le 24") so that it can continue to permit 10.16.0.0/24 and 10.16.1.0/24 as well as permitting the additional networks 10.16.2.0/24 and 10.16.3.0/24.

QUESTION 496



```
!
summary-address 10.1.0.0 255.255.0.0
!
```

The non-area 0 routers in OSPF still receive more specific routes of 10.1.1.0, 10.1.2.0, 10.1.3.0 from area 0. Which action resolves the issue?

- A. Configure the summary-address 10.1.0.0 255.255.252.0 command under OSPF process.
- B. Configure route summarization on OSPF-enabled interfaces.
- C. Summarize by using the summary-address 10.1.0.0 255.255.252.0 command.
- D. Summarize by using the area range command on ABRs.

Correct Answer: D

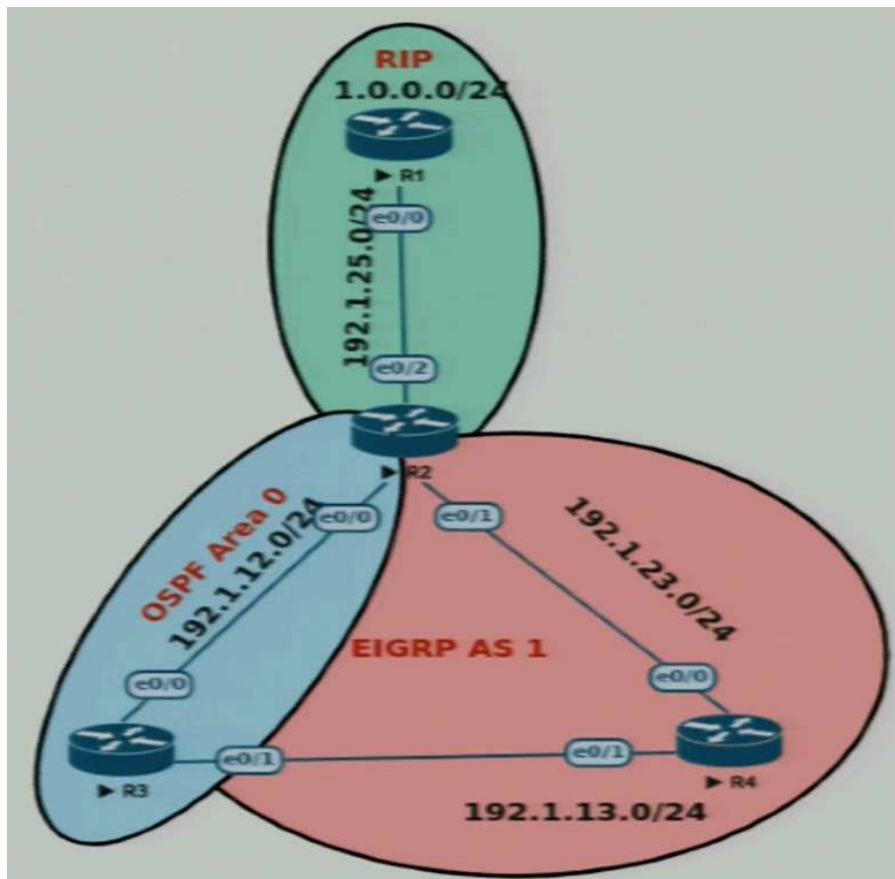
Section: (none)

Explanation

Explanation/Reference:

The "summary-address" command in OSPF can only summarize OSPF external routes at ASBR. For summarizing inter-area OSPF routes from an area (e.g. area 0 in this case) by ABR, you should use "area ... range ..." command.

QUESTION 497



R3 is learning the 1.0.0.0/24 route through OSPF instead of EIGRP. Which action causes R3 to choose EIGRP to reach the 1.0.0.0/24 network?

- A. Configure OSPF administrative distance to 120.
- B. Configure EIGRP administrative distance to 110.
- C. Configure OSPF administrative distance to 200.
- D. Configure EIGRP administrative distance to 120.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

Since all types of OSPF routes have an AD of 110 and external EIGRP routes have an AD of 170, external OSPF route is preferred. In order to prefer external EIGRP route, you can either:

- increase the AD of OSPF routes to a value more than 170.
- decrease the AD of external EIGRP routes to a value less than 110.

QUESTION 498

Which characteristic is representative of a hub-and-spoke topology between PE routers in a Layer 3 MPLS VPN network?

- A. The PE routers use different RTs to import and export M-BGP prefixes.
- B. Each PE router uses a different RD to identify all branches.
- C. The PE routers use different RDs for each VRF to import and export M-BGP prefixes.
- D. The PE routers are configured with multiple VRFs for all branches.

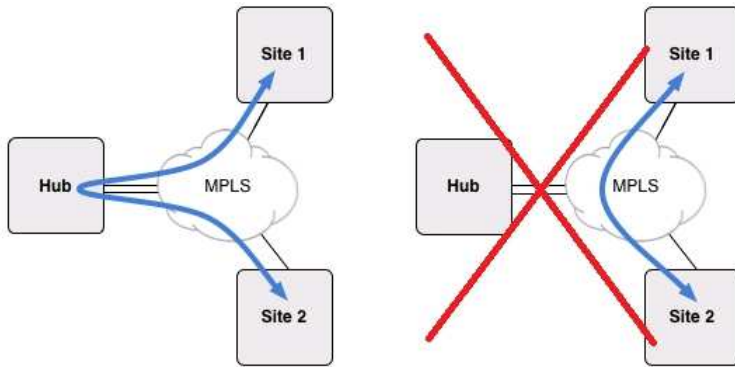
Correct Answer: A

Section: (none)

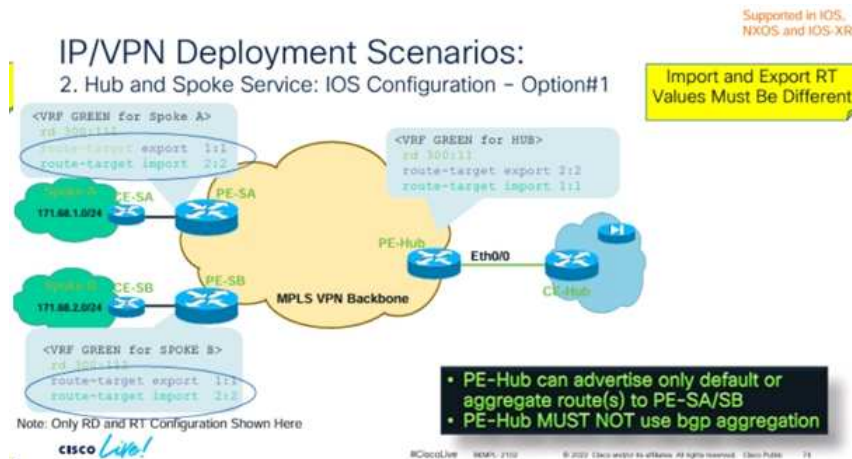
Explanation

Explanation/Reference:

MPS VPN hub-and-spoke means that the traffic from a spoke site / branch must be sent to the Hub site for forwarding i.e. the traffic cannot be sent from a spoke site to a spoke site directly.



The characteristics for such implementation is to configure different RTs for import and export i.e. the import RT is different from the export RT.



For other choices in question:

- The using of different RDs for different branches are required in some special setup but it not a requirement in all cases.
- RD is used for identifying vpnv4 routes and is NOT used for import / export.
- Multiple VRFs in a PE router is only required if that PE router is connected to multiple CE routers of different sites / branches.

QUESTION 499

How does MPLS Layer 3 VPN function?

- When an EIGRP internal route is redistributed into BGP by one PE and then back into EIGRP by another PE, the originating router ID for the route is changed to the router ID of the first PE.
- When a VPN route is learned from a CE device and injected into IGP, a VPN route distinguisher attribute is associated with it.
- When a destination PE device receives a labeled packet, it pops the label and uses it to forward the packet to the correct CE device.
- When a PE device forwards a packet received from a CE device across the provider network, it labels the packet with the label learned from the source PE device.

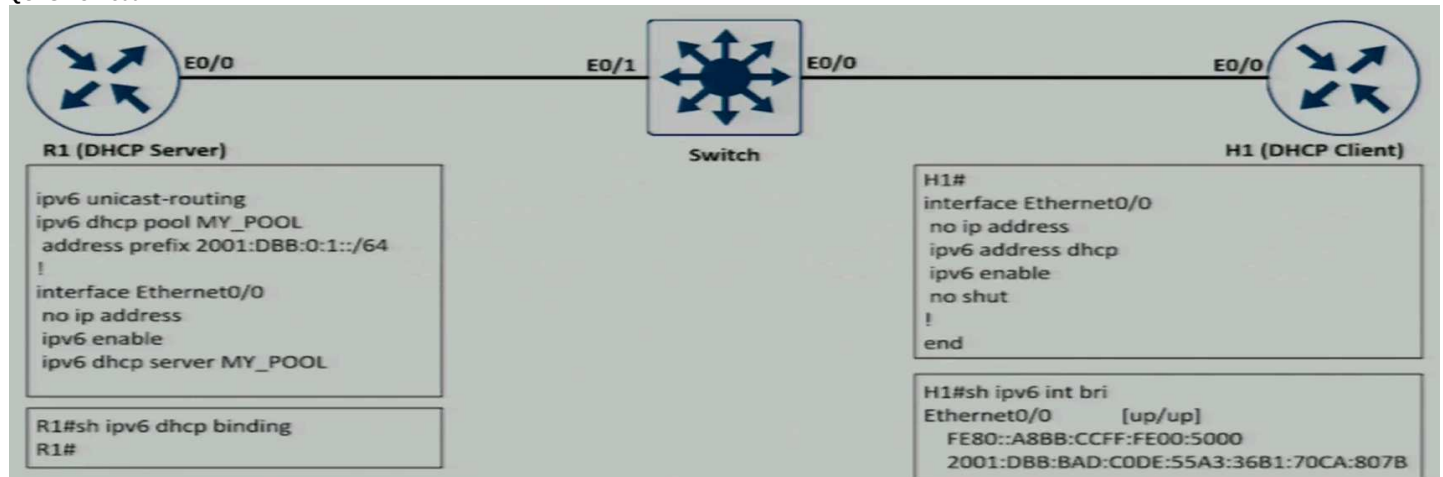
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 500



The client received the IPv6 address from the IPv6 DHCP server but the show command does not show the IPv6 DHCP bindings on the server. Which action resolves the issue?

- A. Configure authorized DHCP servers to avoid IPv6 addresses from a rogue DHCP server.
- B. Extend the DHCP lease time because R1 removed the IPv6 address earlier after the lease expired.
- C. Configure H1 as the DHCP client that manually assigns the IPv6 address on interface e0/0.
- D. Use the 2001:DBB:BAD:C0DE::/64 prefix for the DHCP pool on R1.

Correct Answer: A

Section: (none)

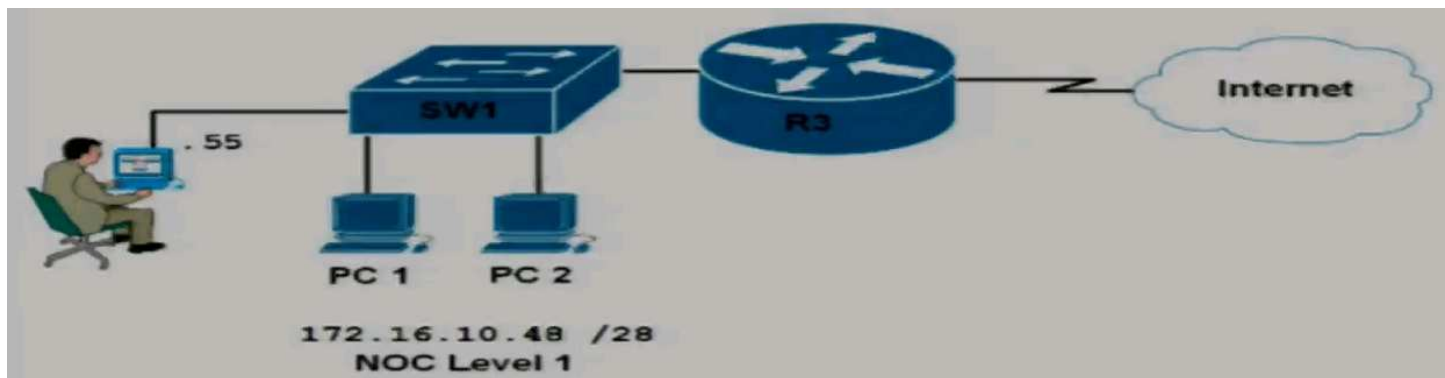
Explanation

Explanation/Reference:

Features e.g. DHCPv6 Guard can be implemented in the switch.

QUESTION 501

```
R3#sh run | begin ip http server
ip http server
ip http access-class 20
ip http authentication local
no ip http secure-server
ip http max-connections 2
!
access-list 20 permit 172.16.10.48 0.0.0.15
!
end
```



Which configuration allows the operation level 1 team of 10 engineers to log in at least three at a time to router R3 using network credentials over HTTP?

- A. R3(config)#ip http authentication aaa
R3(config)#no access-list 20 permit 172.16.10.48 0.0.0.15
R3(config)#access-list 20 permit 172.16.10.48 0.0.0.7
- B. R3(config)#ip http authentication aaa
R3(config)#ip http max-connections 3
- C. R3(config)#ip http max-connections 3
R3(config)#ip http accounting commands 3 default
- D. R3(config)#ip http authentication enable
R3(config)#no access-list 20 permit 172.16.10.48 0.0.0.15
R3(config)#access-list 20 permit 172.16.10.48 0.0.0.7

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

For the subnet "172.16.10.48/28", it is correctly configured as the access list 20 using "172.16.10.48 0.0.0.15". Therefore, no change is needed for the access list.

In order to allow 3 login at the same time, the command "ip http max-connection 3" is required to change the existing configuration. However, both of the choices having this command include another command:

- One includes "ip http authentication aaa". This requires that "aaa new-model" in the router. The AAA authentication list can use local database and/or an AAA server for performing authentication. In case of AAA server (which has to be properly configured in the router and users' network credentials has to be stored in the AAA server). The question does not mention anything about them.
- One include "ip http accounting commands 3 default". This requires that "aaa new-model" in the router and a default AAA accounting list to be configured in order to enter this command. Moreover, a properly setup AAA server and its settings configured in the router are required for AAA accounting to work properly. The command ask the router to send accounting information about privilege level 3 commands to the AAA server specified in the default AAA accounting list.

Comparing the above two, the one with "ip http authentication aaa" seems to be better especially if "network credentials" in this question means that the credentials are stored in the network e.g. AAA server. (Generally, "network credential" means "credentials for network access").

QUESTION 502

R1

```
Interface loopback1
no ip address
ipv6 address 100A:0:100C::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface Loopback2
no ip address
ipv6 address 200A:0:200C::1/64
ipv6 enable
ipv6 ospf 1 area 0
ipv6 traffic-filter DENY_TELNET_Lo2 in
!
interface GigabitEthernet0/0
no ip address
ipv6 address AB01:2011:8:100::/64 eui-64
ipv6 enable
ipv6 ospf network point-to-point
ipv6 ospf 1 area 0
!
ipv6 access-list DENY_TELNET_Lo2
sequence 20 deny tcp host 100B:1:310B::1 host 200A:0:210C::1 eq telnet
permit ipv6 any any
```



Routers R1 and R2 exchange routes to each other's loopback through OSPF. Telnet traffic must be blocked from R2 Lo0 to R1 Lo2. Which configuration resolves the issue?

A. R1

```
Interface loopback1
no ip address
ipv6 address 100A:0:100C::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface Loopback2
no ip address
ipv6 address 200A:0:200C::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/0
no ip address
ipv6 address AB01:2011:8:100::/64 eui-64
ipv6 enable
ipv6 ospf network point-to-point
ipv6 ospf 1 area 0
!
ipv6 access-list DENY_TELNET_Lo2
sequence 20 deny tcp host 100B:1:310B::1 host 200A:0:210C::1 eq telnet
permit ipv6 any any
```

B. R1

```
Interface loopback1
no ip address
ipv6 address 100A:0:100C::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface Loopback2
no ip address
ipv6 address 200A:0:200C::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/0
no ip address
```



```

ipv6 address AB01:2011:8:100::/64 eui-64
ipv6 enable
ipv6 ospf network point-to-point
ipv6 ospf 1 area 0
ipv6 access-class DENY_TELNET_Lo2 in
!
ipv6 access-list DENY_TELNET_Lo2
sequence 20 deny tcp host 100B:1:310B::1 host 200A:0:210C::1 eq telnet
permit ipv6 any any

```

C. R1

```

Interface loopback1
no ip address
ipv6 address 100A:0:100C::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface Loopback2
no ip address
ipv6 address 200A:0:200C::1/64
ipv6 enable
ipv6 ospf 1 area 0
ipv6 access-class DENY_TELNET_Lo2 in
!
interface GigabitEthernet0/0
no ip address
ipv6 address AB01:2011:8:100::/64 eui-64
ipv6 enable
ipv6 ospf network point-to-point
ipv6 ospf 1 area 0
!
ipv6 access-list DENY_TELNET_Lo2
sequence 20 deny tcp host 100B:1:310B::1 host 200A:0:210C::1 eq telnet
permit ipv6 any any

```

D. R1

```

Interface loopback1
no ip address
ipv6 address 100A:0:100C::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface Loopback2
no ip address
ipv6 address 200A:0:200C::1/64
ipv6 enable
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/0
no ip address
ipv6 address AB01:2011:8:100::/64 eui-64
ipv6 enable
ipv6 ospf network point-to-point
ipv6 ospf 1 area 0
ipv6 traffic-filter DENY_TELNET_Lo2 in
!
ipv6 access-list DENY_TELNET_Lo2
sequence 20 deny tcp host 100B:1:310B::1 host 200A:0:210C::1 eq telnet
permit ipv6 any any

```

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

In order to block specific telnet connection based on a configured IPv6 access-list source, you can:

- Configure "ipv6 traffic-filter" under int g0/0 to use that access-list for blocking inbound telnet traffic.
- Configure "ipv6 access-class" under line vty to use that access-list for blocking telnet connection.

There are no choices for configuring access-list under the line vty. Hence, you need to apply the IPv6 access-list to the interface g0/0 for filtering inbound traffic through "ipv6 traffic-filter" configured under the interface g0/0.

QUESTION 503

CISCO DNA DESIGN POLICY PROVISION ASSURANCE

OSPF Adjacency Failed on Device * 10.30.255.101 * Interface TenGigabitEthernet1/0/23 with Neighbor 10.30.255.2

Status: Open ▾ Last Occurred: Jan 11, 2018 9:4

2 Check OSPF neighbors.

Check OSPF neighbors
: show ip ospf neighbor Success

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.30.255.2	1	EXSTART/-	00:00:36	10.30.251.101	TenGigabitEthernet1/0/23

3 If the Neighbor is in "Init" state, Check if there is authentication configured using "show run | sec OSPF".
Authentication type and keys should match on both routers Run

An engineer is investigating an OSPF issue reported by the Cisco DNA Assurance Center. Which action resolves the issue?

- A. One of the neighbor links is down. Bring the interface up by running shut and no shut.
- B. An ACL entry blocking multicast on the interfaces. Allow multicast through the interface ACL.
- C. One of the interfaces is using the wrong authentication. Match interface authentication on both links.
- D. One of the interfaces is using the wrong MTU. Match interface MTU on both links.

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

Since the neighboring problem occurs in EXSTART state, it is usually related to MTU mismatched.

QUESTION 504

Which Layer 3 VPN attribute allows different customers to connect to the same MPLS network with overlapping IP ranges?

- A. MP-BGP
- B. RD
- C. RT
- D. VRF

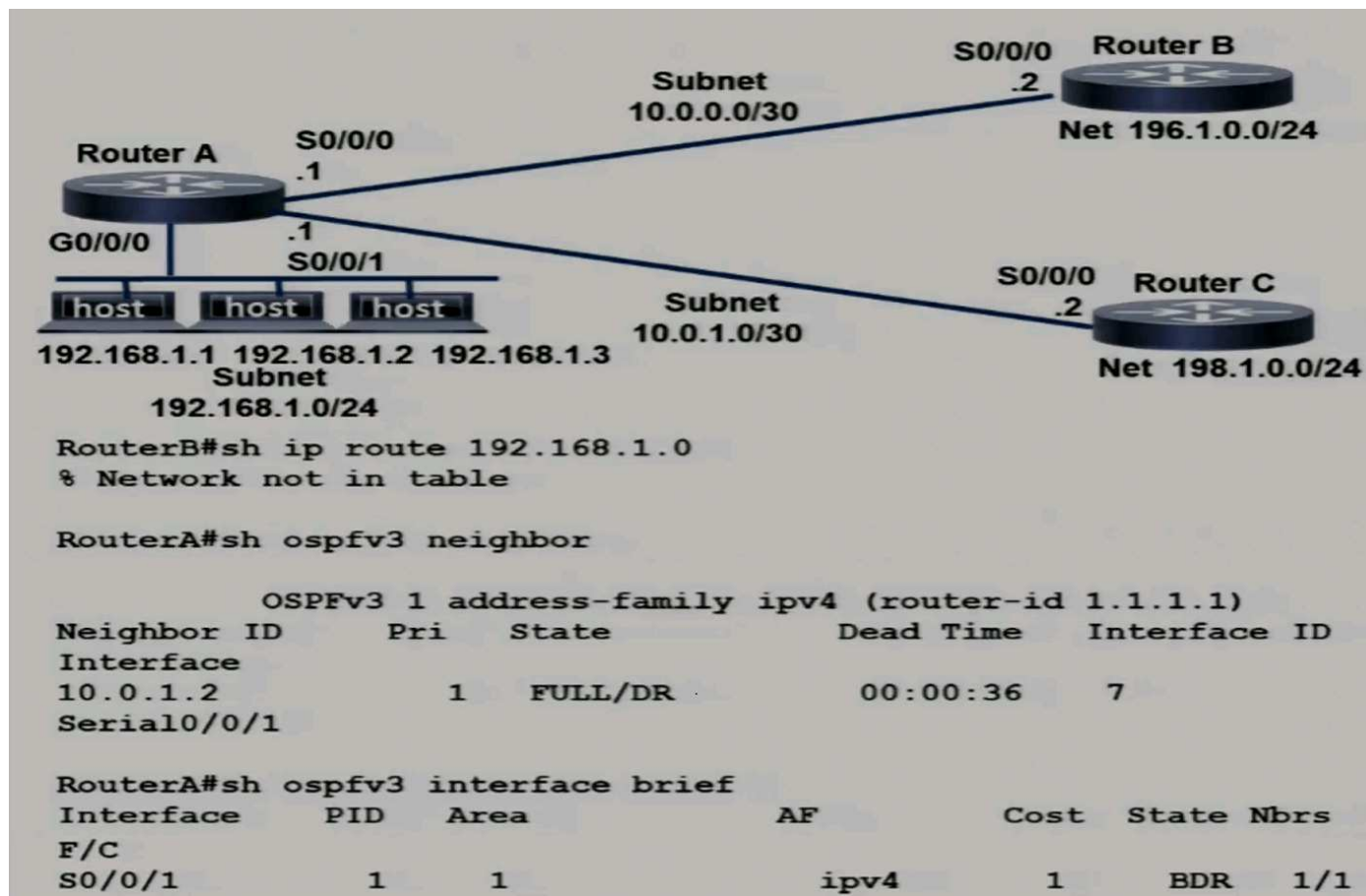
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 505



An engineer must advertise LAN network 192.168.1.0 of router A to router B through OSPF. The engineer notices that router B was configured, but the LAN network of router A is not in the routing table of router B. Which configuration on router A resolves the problem?

- A. interface Serial0/0/0
ip address 10.0.0.1 255.255.255.0
negotiation auto
ipv6 enable
ospfv3 1 ipv4 area 1

router ospfv3 1
address-family ipv4 unicast
area 1 range 192.168.1.0 255.255.255.0
router-id 1.1.1.1
exit-address-family
- B. interface GigabitEthernet0/0/0
ip address 192.168.1.254 255.255.255.0
negotiation auto
ipv6 enable

interface Serial0/0/0
ip address 10.0.0.1 255.255.255.0
negotiation auto
ipv6 enable
ospfv3 1 ipv4 area 1

router ospfv3 1
address-family ipv4 unicast
router-id 1.1.1.1
exit-address-family
- C. interface GigabitEthernet0/0/0
ip address 192.168.1.254 255.255.255.0
negotiation auto
ipv6 enable
ospfv3 1 ipv4 area 1

interface Serial0/0/0
ip address 10.0.0.1 255.255.255.0
negotiation auto
ipv6 enable

router ospfv3 1
address-family ipv4 unicast
router-id 1.1.1.1
exit-address-family
- D. interface GigabitEthernet0/0/0
ip address 192.168.1.254 255.255.255.0
negotiation auto
ipv6 enable
ospfv3 1 ipv4 area 1

```

interface Serial0/0/0
ip address 10.0.0.1 255.255.255.0
negotiation auto
ipv6 enable
ospfv3 1 ipv4 area 1

router ospfv3 1
address-family ipv4 unicast
router-id 1.1.1.1
exit-address-family

```

Correct Answer: D

Section: (none)

Explanation

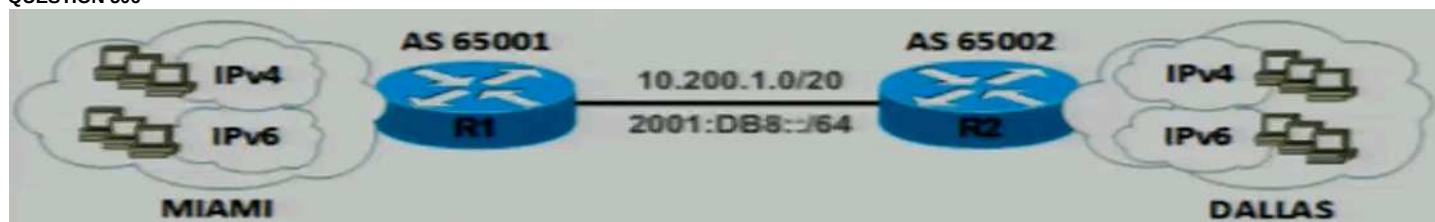
Explanation/Reference:

From the command output of the question, Router A only forms OSPF neighbor through s0/0/1 i.e. with Router C.

In order for Router A to form neighbor with Router B and advertise the network 192.168.1.0/24 connected by g0/0/0, you need to configure OSPF using the interface configuration mode commands "ospfv3" in the following two interfaces of Router A:

- Interface s0/0/0 which connects to Router B. This allows the forming of neighbor with Router B.
 - Interface g0/0/0 which connects to the network 192.168.1.0/24. This allows the advertising of the network 192.168.1.0/24.
- Only one of the choices configure the "ospfv3" command in both of the above interfaces.

QUESTION 506



A network engineer configured routers R1 and R2 with MP-BGP. The engineer noticed that the routers cannot exchange any IPv6 routes; however, the IPv4 neighbor relationship is working fine. Which configuration must the engineer apply to router R2 to exchange IPv6 routes?

- A. ipv6 unicast-routing
 ipv6 cef
 !
 interface Loopback100
 ipv6 address 2001:DB8:128::2/128
 !
 interface GigabitEthernet1/0
 ipv6 address 2001:DB8:1::2/64
 !
 router bgp 65002
 no bgp default ipv4-unicast
 neighbor 2001:DB8:1::1 remote-as 65001
 !
 address-family ipv6
 network 2001:DB8:128::2/128
 neighbor 2001:DB8:1::1 activate
- B. ipv6 unicast-routing
 ipv6 cef
 !
 interface Loopback100
 ipv6 address 2001:DB8:128::2/128
 !
 interface GigabitEthernet1/0
 ipv6 address 2001:DB8:1::2/64
 !
 router bgp 65002
 no bgp default ipv4-unicast
 neighbor 2001:DB8:1::1 remote-as 65001
 !
 address-family ipv6
 network 2001:DB8:128::2/128
- C. ipv6 cef
 !
 interface Loopback100
 ipv6 address 2001:DB8:128::2/128
 !
 interface GigabitEthernet1/0
 ipv6 address 2001:DB8:1::2/64
 !
 router bgp 65002
 no bgp default ipv4-unicast
 neighbor 2001:DB8:1::1 remote-as 65001
 !
 address-family ipv6
 network 2001:DB8:128::2/128
 neighbor 2001:DB8:1::1 activate
- D. ipv6 unicast-routing
 ipv6 cef
 !
 interface Loopback100
 ipv6 address 2001:DB8:128::2/128
 !
 interface GigabitEthernet1/0
 ipv6 address 2001:DB8:1::2/64

- Configure two static routes on R1, one pointing to ISP-B with more specific routes and another pointing to ISP-A with summary routes.
- Change the bandwidth of the interface on R1 so that interface to ISP-A has a higher value than the interface to ISP-B.
- Configure two static routes on R1, one pointing to ISP-A and another pointing to ISP-B with 222 admin distance.
- Configure OSPF between R1, ISP-A, and ISP-B for dynamic failover if any ISP link to R1 fails.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

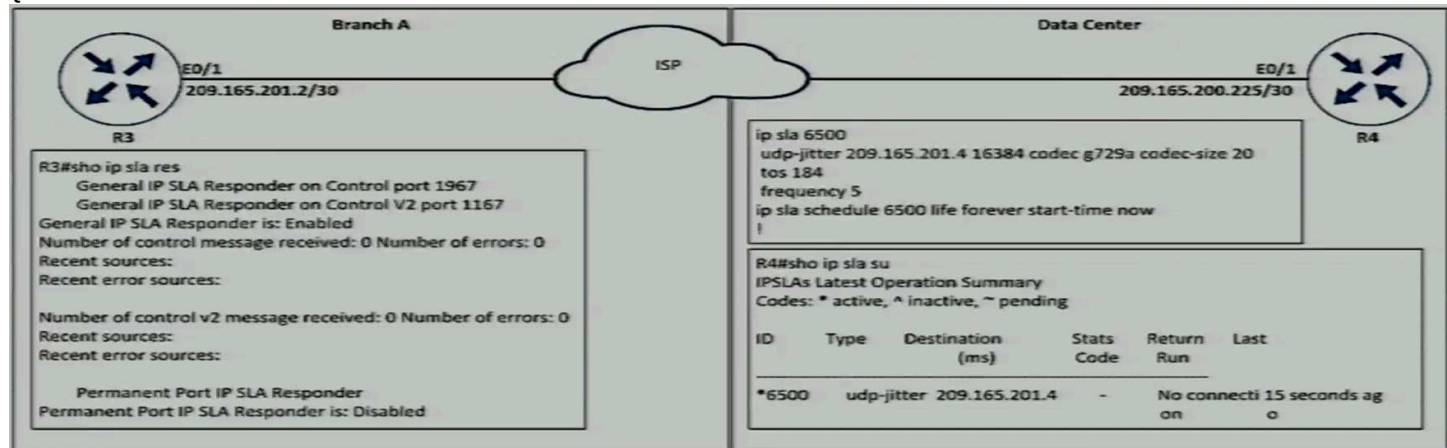
ISP-A will be preferred by changing both of the followings in R1:

- Change the use of static routes to OSPF (if the ISPs support it).
 - Change the bandwidth of the links to the ISPs so that link to ISP-A has a higher configured bandwidth.
- However, changing only one of the above alone does not solve the problem

Specific routes are preferred than a summary route for routing. Hence, configuring specific routes to ISP-B does not meet the requirement since ISP-B will be preferred.

Hence, the only choice meeting the requirement is to set a higher AD for the static route configured for ISP-B (e.g. 222).

QUESTION 509



Which action resolves the IP SLA for the UDP jitter problem between R4 and R3 Ethernet 0/1 IP addresses?

- Delete and configure the ip sla responder command with R4 E0/1 IP address.
- Delete and configure the ip sla 6500 command with R3 e0/1 IP address.
- Configure the ip sla 6500 command with R3 e0/1 IP address.
- Configure the ip sla responder command with R4 E0/1 IP address.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

For the IP SLA 6500 configured in R4, the IP address of R3's e0/1 (i.e. 209.165.201.2) is incorrectly entered as "udp-jitter 209.165.201.4 ...".

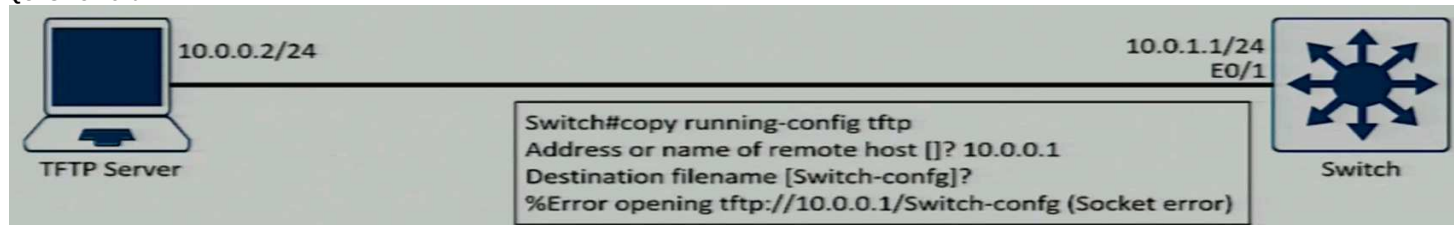
In order to solve it, you need to enter "ip sla 65000" and use the ip-sla-jitter configuration mode command "dest-ipaddr" to change it.

```

R4(config)#ip sla 65000
R4(config-ip-sla-jitter)#dest-ipaddr 209.165.201.2
R4(config-ip-sla-jitter)#
  
```

Although you can also delete the configuration for "ip sla 65000" and configure a new one from start again, modifying just the destination IP address is simpler.

QUESTION 510



Which action allows the engineer to successfully copy running-config to the TFTP server?

- Add a route in the switch to the TFTP server.
- Use file name Switch-config.txt.
- Add the TFTP server configuration in the switch.
- Use TFTP server IP address 10.0.1.1.

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

There are two problems in the diagram of this question which can cause the error:

- The TFTP server has an IP address of 10.0.0.2 but the "copy" command entered in the switch use "10.0.0.1" as the remote host.

- The IP address of the TFTP server and the IP address of the switch are in 2 different networks (10.0.0.0/24 and 10.0.1.0/24) but they seem to be connected directly without any router in between.

Three of the choices must be wrong since:

- You can use any file name you like for uploading to TFTP server and therefore you do not need to append the file extension ".txt".
- There is no need to pre-configure a TFTP server configuration setting in the switch in order to upload file to that TFTP server.
- The TFTP server's IP address is not "10.0.1.1" since it is the IP address of the switch itself.

Hence the only possible answer left is to add a route. However, this assumes that there are mistakes in the diagram:

- There is a router between the TFTP server and the switch.
- The TFTP server's IP address matches the one entered in the copy command e.g. both are "10.0.0.1".

QUESTION 511

```
R1# show route-map
route-map Redistribution_EIGRP, permit, sequence 10
  Match clauses:
    ip address (access-lists): 10
  Set clauses:
    tag 666
  Policy routing matches: 0 packets, 0 bytes
route-map Redistribution_EIGRP, permit, sequence 20
  Match clauses:
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes

R1# show access-lists
Standard IP access list 10
  10 permit 172.16.1.0, wildcard bits 0.0.0.255
  20 permit 172.16.2.0, wildcard bits 0.0.0.255
```

The router is redistributing a prefix 172.16.10.0/24 that should have been filtered. Which action resolves the issue?

- A. Match the tag 666 for the route in the route map.
- B. Add the route in access-list 10.
- C. Remove route-map sequence 20.
- D. Permit the route in route-map sequence 20.

Correct Answer: C

Section: (none)

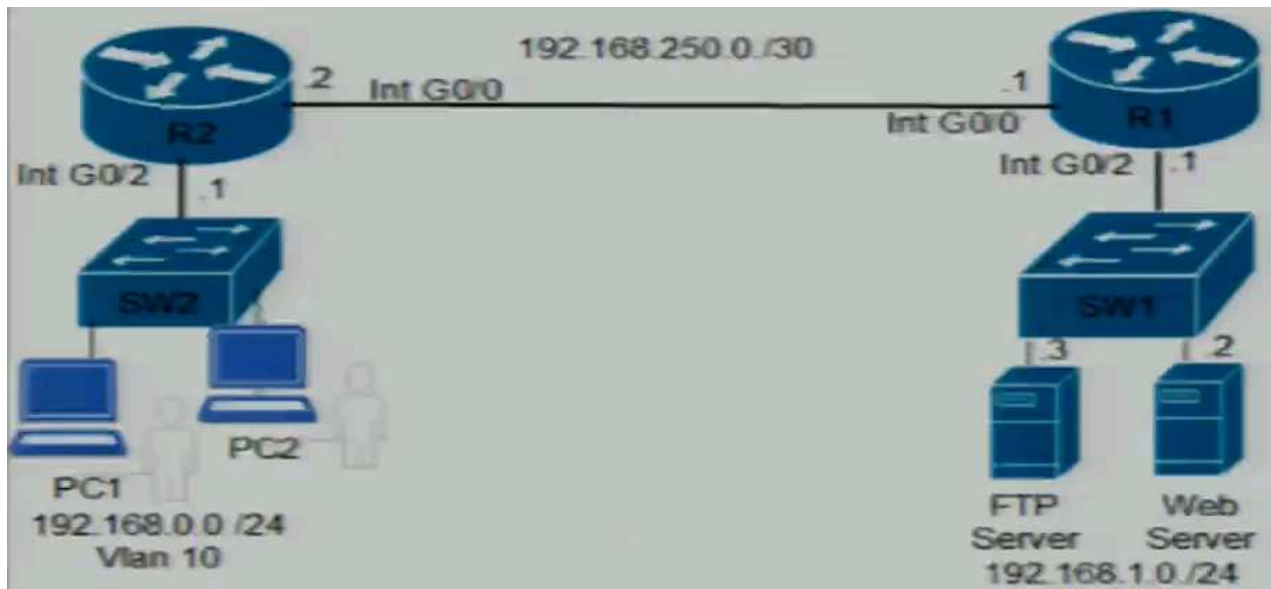
Explanation

Explanation/Reference:

The seq 10 of the route-map uses the access-list 10 to match the network address 172.16.1.0 and permits it. Then seq 20 of the route-map permits all other remaining unmatched network addresses. Hence, the prefix 172.16.10.0 is permitted by seq 20 of the route map.

In order to prevent the network 172.16.10.0 or other un-matched prefixes from redistribution, you can remove seq 20 of the route-map.

QUESTION 512



```
R2#debug ip dhcp server events
000249: *Jun 19 02:13:33.818: DHCPD: Sending notification of DISCOVER:
000250: *Jun 19 02:13:33.823: DHCPD: htype 1 chaddr 0c82.430d.db00
000251: *Jun 19 02:13:33.827: DHCPD: remote id 020a0000c0a8000100000000
000252: *Jun 19 02:13:33.830: DHCPD: circuit id 00000000
000253: *Jun 19 02:13:33.836: DHCPD: Seeing if there is an internally specified pool class:
000254: *Jun 19 02:13:33.840: DHCPD: htype 1 chaddr 0c82.430d.db00
000255: *Jun 19 02:13:33.843: DHCPD: remote id 020a0000c0a8000100000000
000256: *Jun 19 02:13:33.846: DHCPD: circuit id 00000000
000257: *Jun 19 02:13:33.851: DHCPD: subnet [192.168.0.1,192.168.0.2] in address pool WAN is empty.
000258: *Jun 19 02:13:33.853: DHCPD: Sending notification of ASSIGNMENT FAILURE:
000259: *Jun 19 02:13:33.857: DHCPD: htype 1 chaddr 0c82.430d.db00
000260: *Jun 19 02:13:33.861: DHCPD: remote id 020a0000c0a8000100000000
000261: *Jun 19 02:13:33.865: DHCPD: circuit id 00000000
000262: *Jun 19 02:13:33.870: DHCPD: Sending notification of ASSIGNMENT_FAILURE:
000263: *Jun 19 02:13:33.872: DHCPD: due to: POOL EXHAUSTED
000264: *Jun 19 02:13:33.877: DHCPD: htype 1 chaddr 0c82.430d.db00
000265: *Jun 19 02:13:33.879: DHCPD: remote id 020a0000c0a8000100000000
000266: *Jun 19 02:13:33.879: DHCPD: circuit id 00000000
000267: *Jun 19 02:13:36.860: DHCPD: Sending notification of DISCOVER:
000268: *Jun 19 02:13:36.862: DHCPD: htype 1 chaddr 0c82.430d.db00
```

Router R2 VLAN 10 users cannot get dynamic IP addresses from R1. Which action resolves the issue?

- A. Expand the address scope of VLAN 10.
- B. Configure the IP helper feature on the Interface GigabitEthernet 0/2 of router R2.
- C. Eliminate the port security feature on the ports of switch SW2.
- D. Identify the host with the duplicate IP address.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

In the question, the users' PCs should get IP address from the DHCP service running in R1. However, in the debug output of R2, the directly connected router R2 tries to assign IP address to the PCs but R2 finds that there is no address pool is configured for that subnet. In order for the PCs to use the DHCP service in the remote router R1 instead of the directly connected router R2, IP helper feature has to be configured in R2 g0/2 (i.e. the interface connected to the PCs).

QUESTION 513

Which two features are required for MPLS forwarding on which types of routers? (Choose two.)

- A. CEF on PE and CE routers
- B. LDP on PE and CE routers
- C. MPLS on PE and core routers
- D. MPLS on CE and core routers
- E. LDP on PE and core routers

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

For MPLS, no MPLS related feature is required on the customers' routers (i.e. CE routers).

QUESTION 514

A newly installed router starts establishing an LDP session from another MPLS router to which it is not directly connected. Which LDP message type responds by target router to the initiating router using UDP protocol?

- A. advertisement message
- B. notification message
- C. extended discovery message

D. session message

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

If a MPLS router is more than one hop from its neighbor, they are not directly connected. For these non-directly connected neighbors, a router need to send out a targeted Hello message as a unicast UDP packet. The other router will then respond to the targeted Hello message to establish an LDP session. This is called extended discovery.

QUESTION 515

OSPF Adjacency Failed on Device "CSR103.ap.com" GigabitEthernet2

Open ▾

Description

OSPF adjacency failed on device name:'CSR103.ap.com';
interface:'GigabitEthernet2' at site:'HQ' with neighbor
'172.16.100.5'

Last Occurred: Jan 11, 2022 9:28 PM

Syslog Events

Jan 10, 2022 9:34 PM to Jan 11, 2022 9:34 PM

```
GigabitEthernet2 is up, line protocol is up
Internet Address 172.16.1.42/30, Interface ID 8, Area 1
Attached via Network Statement
Process ID 1, Router ID 172.16.100.7, Network Type BROADCAST, Cost: 1
Topology-MTID      Cost      Disabled      Shutdown      Topology Name
0                  1          no            no            Base
Transmit Delay is 1 sec, State DR, Priority 1
Designated Router (ID) 172.16.100.7, Interface address 172.16.1.42
Backup Designated router (ID) 172.16.100.5, Interface address 172.16.1.41
Timer intervals configured, Hello 10, Dead 40, wait 40, Retransmit 5
oob-resync timeout 40
Hello due in 00:00:01

Neighbor Count is 1, Adjacent neighbor count is 1
Adjacent with neighbor 172.16.100.5 (Backup Designated Router)
Suppress hello for 0 neighbor(s)
Cryptographic authentication enabled
Sending SA: Key 1, Algorithm HMAC-SHA-256 - key chain ospf
Rollover in progress, 1 neighbor(s) using the old key(s):
key id 1 algorithm MD5

CSR103#
CSR103#
CSR103#sh ip ospf nei

Neighbor ID      Pri   State           Dead Time   Address        Interface
172.16.100.3     1     FULL/DR         00:00:30    172.16.1.25    GigabitEthernet3
172.16.100.5     1     FULL/BDR        00:00:16    172.16.1.41    GigabitEthernet2
CSR103#
CSR103#
*Jan 11 16:49:35.311: %SYS-6-LOGOUT: User admin has exited tty session 1(10.228.200.250)
*Jan 11 16:49:45.396: %OSPF-5-ADJCHG: Process 1, Nbr 172.16.100.5 on GigabitEthernet2 from
FULL to DOWN, Neighbor Down: Dead timer expired
```

Which configuration resolves the issue?

- A. int GigabitEthernet 2
ip ospf key 1 cisco
ip ospf authentication
- B. router ospf 1
area 1 authentication message-digest
int GigabitEthernet 2
ip ospf message-digest-key 1 md5 cisco
- C. key chain ospf
key 1
key-string 7 01041C321954
cryptographic-algorithm hmac-sha-1
interface GigabitEthernet2
ip ospf authentication key-chain ospf
- D. int GigabitEthernet 2
ip ospf message-digest-key 1 md5 cisco
ip ospf authentication message-digest

Correct Answer: C

Section: (none)

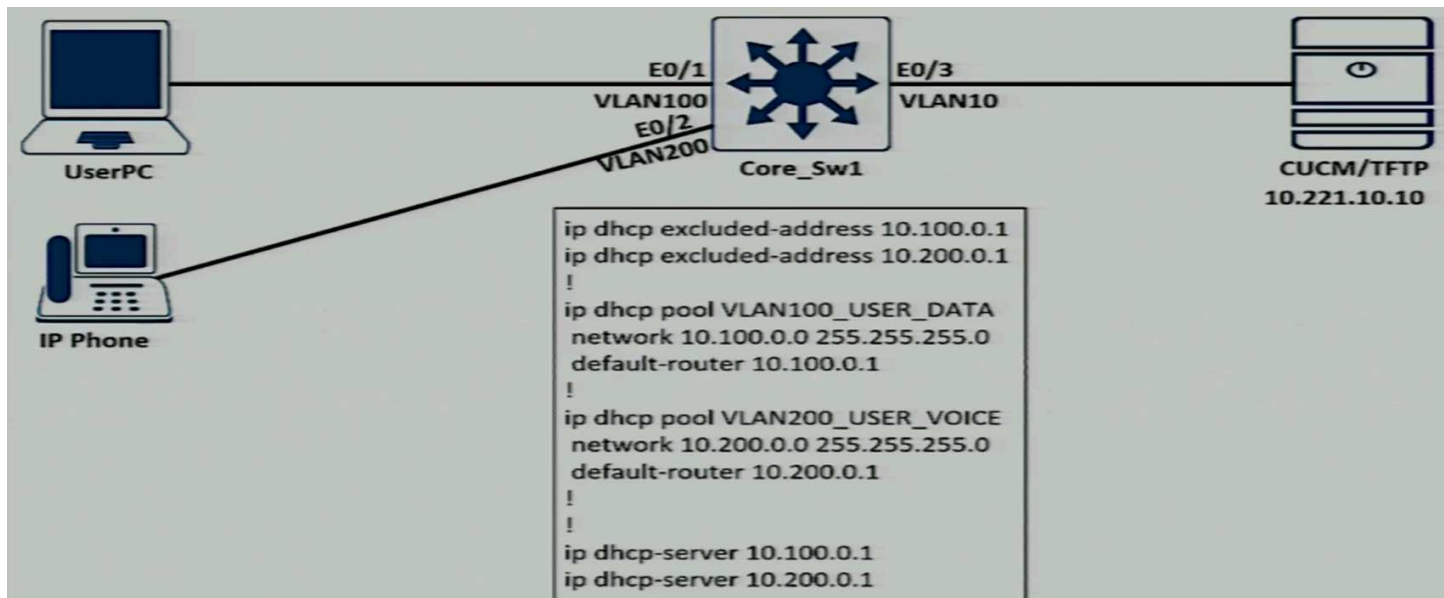
Explanation

Explanation/Reference:

A reason why a neighbor becomes DOWN state from FULL state is that you have changed the OSPF authentication string in just one of the neighbor routers (e.g. you changed the authentication string in a router and you are going to change the authentication string in the router router). In this case, you can solve this by changing the authentication string in the remaining router.

From the output in the diagram, the OSPF authentication is based on "HMAC-SHA-256" configured in the key chain "ospf". Hence, you should change the authentication string using the same algorithm.

QUESTION 516



User IP phone receives the IP address but does not register to the call manager. Which command in ip dhcp pool VLAN200_USER_VOICE resolves the issue?

- A. option 160 ip 10.221.10.10
- B. option 15 ip 10.221.10.10
- C. option 150 ip 10.221.10.10
- D. option 117 ip 10.221.10.10

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

For registration with CUCM (i.e. call manager), other than obtaining normal IP settings from the DHCP server, an IP phone needs CUCM's TFTP server IP address. This allows the IP phone to download its phone configuration from that TFTP server in order to complete the process of phone registration with CUCM. In order to provide TFTP server's IP address to an IP phone, the DHCP option 150 has to be configured with the IP address of the CUCM's TFTP server under the DHCP pool for IP phone (i.e. VLAN200_USER_VOICE).

QUESTION 517

What is an advantage of MPLS Layer 3 VPN deployment?

- A. Planning and modifications are required for the customer Intranet before migrating to Layer 3 VPN.
- B. Scalable VPNs are created using connection-oriented, point-to-point, or multipoint overlay connections.
- C. QoS provides performance with policy and support for a best-effort service level in an MPLS VPN.
- D. Security is provided at the edge of the provider network through encryption.

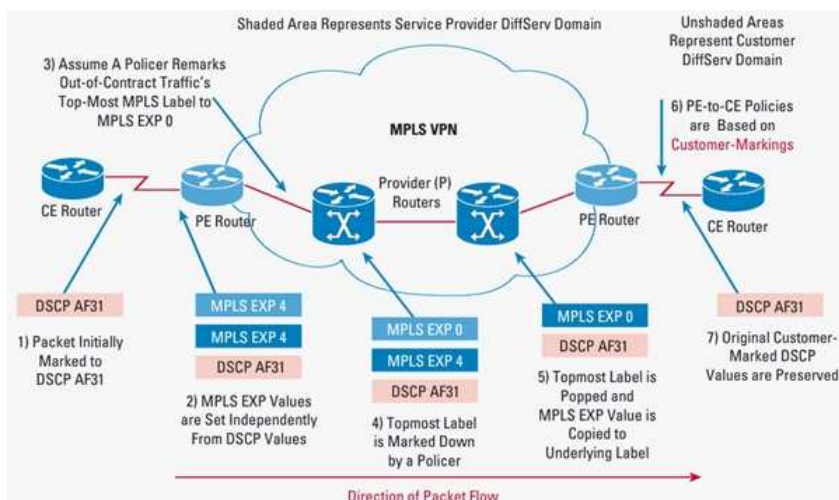
Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QoS can be supported since the IP precedence bits in the packets can be copied into the MPLS EXP bits in the label header.



Instead of DiffServ, MPLS TE can also be used to provide granular policy-based control over traffic flows.

Remarks :

MPLS is a **connectionless** VPN technology: a VPN user (customer equipment) does not have a direct relationship with any other VPN user; rather, it is connected to the MPLS service as a "cloud," which ensures that packets are forwarded correctly to the other VPN user site. Specifically, a VPN user does not have explicit knowledge of other VPN users. This is one of the key advantages of MPLS IP VPNs: they scale very well because less VPN information needs to be kept at the edge. Encryption is NOT provided in MPLS VPN implementation. Rather than relying on encryption algorithm, privacy is achieved by defining a single path within a provider's network.

QUESTION 518

Which feature is used by LDP in the forwarding path within the MPLS cloud?

- A. IP forwarding
- B. LSP
- C. TDP
- D. TTL

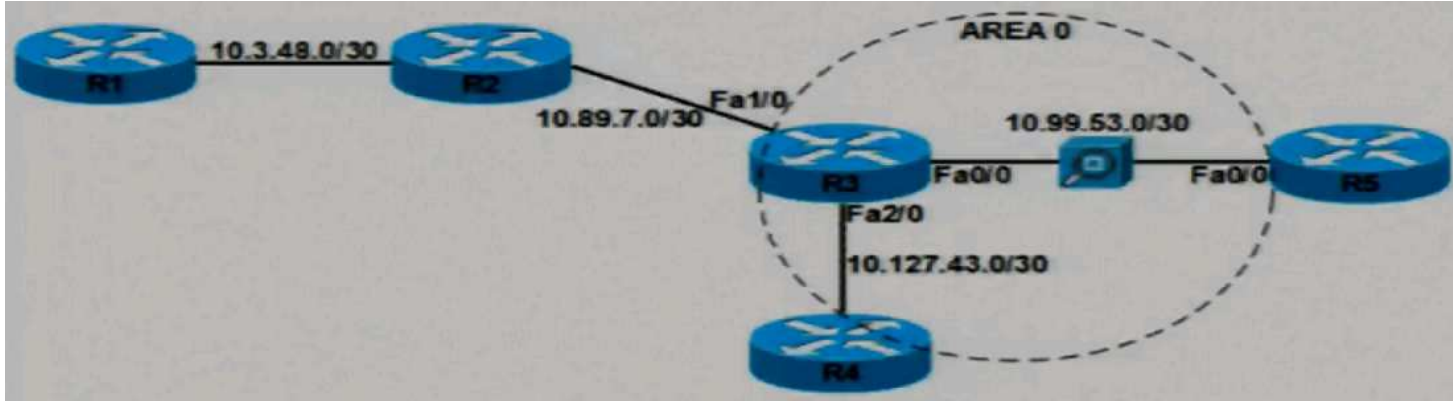
Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

LDP is used to build and maintain label-switched path (LSP) databases that are used to forward traffic through MPLS networks.

QUESTION 519

The security department recently installed a monitoring device between routers R3 and R5, which caused a loss of network connectivity for users connected to R5. Troubleshooting revealed that the monitoring device cannot forward multicast packets. The team already updated R5 with the correct configuration. Which configuration must be implemented on R3 to resolve the problem by ensuring R3 as the DR for the R3-R5 segment?

- A. interface FastEthernet0/0
ip address 10.99.53.1 255.255.255.252
ip access-group 122 in
ip ospf network non-broadcast
ip ospf priority 0
!
router ospf 10
router-id 10.10.3.255
network 10.99.53.0 0.0.0.3 area 0
neighbor 10.99.53.2
!
access-list 122 permit 89 host 10.99.53.2 host 10.99.53.1
access-list 122 deny 89 any any
access-list 122 permit tcp any any
access-list 122 permit udp any any
access-list 122 permit icmp any any
- B. interface FastEthernet0/0
ip address 10.99.53.1 255.255.255.252
ip access-group 122 in
ip ospf network non-broadcast
ip ospf priority 100
!
router ospf 10
router-id 10.10.3.255
network 10.99.53.0 0.0.0.3 area 0
neighbor 10.99.53.2
!
access-list 122 permit 89 host 10.99.53.2 host 10.99.53.1
access-list 122 deny 89 any any
access-list 122 permit tcp any any
access-list 122 permit udp any any
access-list 122 permit icmp any any
- C. interface FastEthernet0/0
ip address 10.99.53.1 255.255.255.252
ip access-group 122 in
ip ospf network point-to-point
ip ospf priority 100
!
router ospf 10
router-id 10.10.3.255
network 10.99.53.0 0.0.0.3 area 0
neighbor 10.99.53.2
!
access-list 122 permit 88 host 10.99.53.2 host 10.99.53.1
access-list 122 deny 88 any any
access-list 122 permit tcp any any
access-list 122 permit udp any any
access-list 122 permit icmp any any
- D. interface FastEthernet0/0
ip address 10.99.53.1 255.255.255.252
ip access-group 122 in
ip ospf network non-broadcast

```

ip ospf priority 0
!
router ospf 10
router-id 10.10.3.255
neighbor 10.99.53.2
!
access-list 122 permit 88 host 10.99.53.2 host 10.99.53.1
access-list 122 deny 88 any any
access-list 122 permit tcp any any
access-list 122 permit udp any any
access-list 122 permit icmp any any

```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

In order for R3 to be the DR for forming neighbor with R5 without multicast:

- The OSPF network type for R3's interface f0/0 should be "**non-broadcast**" and R5's IP address has to be specified in the OSPF's router configuration mode command "neighbor".

- The R3's interface f0/0 should have a priority larger than 1 (e.g. 100)

Among the choices, only one satisfy all of the above requirements.

QUESTION 520

```

*Sep 3 23:18:21.264: EIGRP: Neighbor (10.1.2.192) not yet found
*Sep 3 23:19:18.675: Going down: Peer 10.1.2.1 total=2 stub 0, iidb-stub=0 iid-all=0
*Sep 3 23:19:18.675: EIGRP: Handle deallocation failure [1]
*Sep 3 23:19:18.675: EIGRP: Neighbor 10.1.2.1 went down on Tunnel1.
*Sep 3 23:19:22.943: EIGRP: New peer 10.1.2.1.
*Sep 3 23:19:22.943: %DUAL-5-NBRCHANGE: EIGRP-IPv4 3111: Neighbor 10.1.2.1 (Tunnel1) is up: new adjacency

```

Which configuration command establishes an EIGRP neighbor adjacency between the hub and spoke?

- A. connected 10.1.2.192 command on spoke router
- B. neighbor 10.1.2.192 command on hub router
- C. eigrp-peer 10.1.2.192 command on the hub router
- D. network 10.1.2.192 command on spoke router

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

The hub and spoke network in the question probably does not support multicast and therefore there is problem in forming EIGRP neighbor.

In order to form neighbor successfully, unicast should be used and this can be configured through the EIGRP's router configuration mode command "neighbor".

QUESTION 521

What is the failure detection time with BFD?

- A. 1 second
- B. 3 seconds
- C. 2 seconds
- D. less than a second

Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 522

Network > Device 360

		DEVICE_AVAILABILITY:REACHABLE	Event	8:52:52.443 PM
●	Notice	DUAL:NBRCHANGE	Syslog	8:46:37.210 PM
●	Notice	DUAL:NBRCHANGE	Syslog	8:46:37.207 PM

● DUAL_NBRCHANGE Jan 11, 2022 8:46:37 PM

Detailed Information

Severity	Notice
Mnemonic	NBRCHANGE
Facility	DUAL
Message Text	682: "Jan 11 15:41:03.036: EIGRP-IPv4 88: Neighbor 172.16.33.2 (GigabitEthernet2.10) is down: authentication mode changed
Message Type	Syslog

R1 lost its directly connected EIGRP peer 172.16.33.2 (SW1). Which configuration resolves the issue?

- A. key chain EIGRP
key 1
key-string Cisco
!
interface GigabitEthernet 2.10
ip authentication mode eigrp 88 md5
ip authentication key-chain eigrp 88 EIGRP
- B. key chain EIGRP
key 1
key-string Cisco
!
interface GigabitEthernet 2
ip authentication mode eigrp 88 md5
ip authentication key-chain eigrp 88 Cisco
- C. key chain EIGRP
key 1
key-string Cisco
!
interface GigabitEthernet 2
ip authentication mode eigrp 88 md5
ip authentication key-chain eigrp 88 EIGRP
- D. key chain EIGRP
key 1
key-string Cisco
!
interface GigabitEthernet 2.10
ip authentication mode eigrp 88 md5
ip authentication key-chain eigrp 88 Cisco

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

When using key chain to setup EIGRP authentication, after the authentication string has been setup under a key in the keychain of a specific name, the interface configuration mode command "ip authentication key-chain" has to be configured as follows:

- Under the appropriate interface. For example in the diagram, it shows that the concerned interface is "GigabitEthernet2.10".

- The keychain name has to be specified. (For example if the keychain configured has the name "EIGRP", this name has to be specified in the command i.e. "ip authentication key-chain ... EIGRP" command).

Among the choices, only one satisfy all of the above requirements.

QUESTION 523

```

R1#show running-config | begin router eigrp
router eigrp 100
 network 172.16.250.0 0.0.0.3
 redistribute ospf 10 metric 1 1 1 1 1
!
router ospf 10
 redistribute eigrp 100 metric 100 subnets route-map CCNP
 network 172.16.1.0 0.0.0.3 area 0
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
!
route-map CCNP deny 10
 match route-type local
!
!
access-list 10 permit 172.16.2.32
!

```

```

R4#show running-config | begin router eigrp
router eigrp 100
 network 172.16.2.0 0.0.0.3
 network 172.16.2.16 0.0.0.15
 network 172.16.2.32 0.0.0.15
 redistribute static metric 100 1 1 1 1 route-map CCNP
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
ip route 172.16.2.48 255.255.255.240 172.16.2.1
!
!
route-map CCNP permit 10
 match ip address 10
 set tag 200
!
!
access-list 10 permit 172.16.2.48 0.0.0.15
!

```

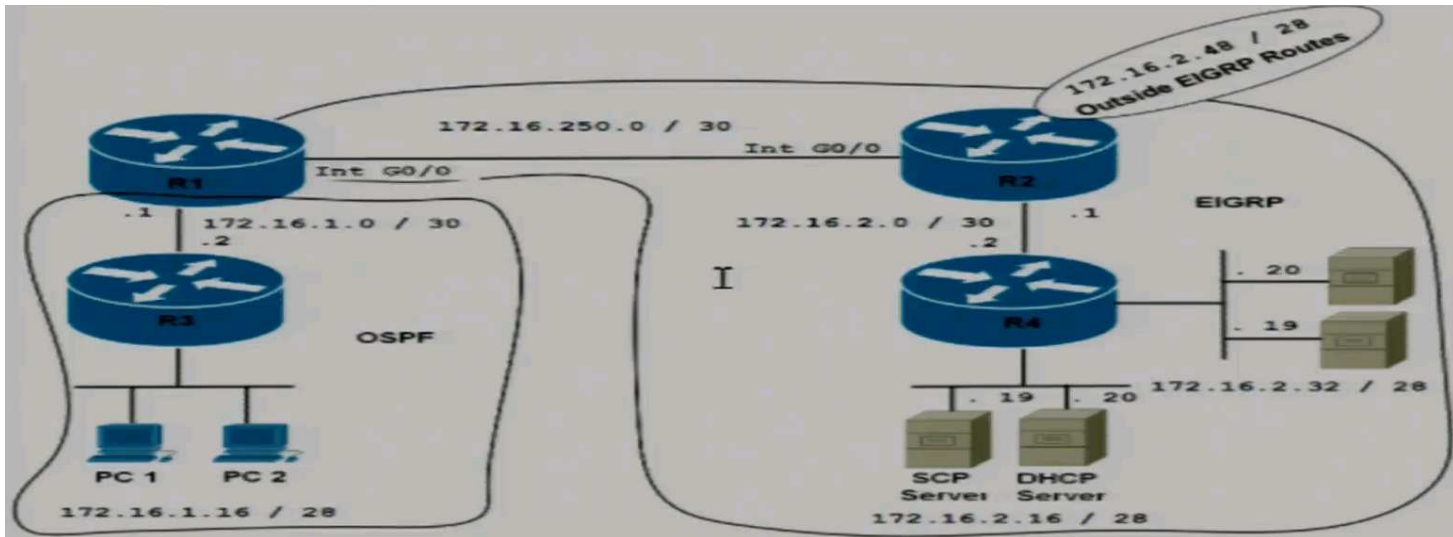
```

R3#sh ip route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 7 subnets, 3 masks
C       172.16.1.0/30 is directly connected, GigabitEthernet0/1
L       172.16.1.2/32 is directly connected, GigabitEthernet0/1
C       172.16.1.16/28 is directly connected, Loopback1
L       172.16.1.17/32 is directly connected, Loopback1
C       172.16.1.32/28 is directly connected, Loopback2
L       172.16.1.33/32 is directly connected, Loopback2
S       172.16.1.48/28 [1/0] via 172.16.1.18
R3#

```



Which configuration resolves the route filtering issue on R1 to redistribute all the routes except 172.16.2.48/28?

- R1(config)#route-map CCNP deny 10
R1(config-route-map)#no match route-type local
R1(config-route-map)#match route-type external
R1(config)#route-map CCNP permit 20
- R1(config)#route-map CCNP deny 10
R1(config-route-map)#no match route-type local
R1(config-route-map)#match route-type level-2
R1(config)#route-map CCNP permit 20
- R1(config)#route-map CCNP deny 10
R1(config-route-map)#no match route-type local
R1(config-route-map)#match route-type external type-2
R1(config)#route-map CCNP permit 20
- R1(config)#route-map CCNP deny 10
R1(config-route-map)#no match route-type local
R1(config-route-map)#match route-type external type-1
R1(config)#route-map CCNP permit 20

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

From the diagram, the route 172.16.2.16/28 is specified as "Outside EIGRP route". In the R4's configuration shown, the subnet 172.16.2.16/28 is configured as a static route which is then redistributed into EIGRP as an EIGRP external route.

In order to block this EIGRP external routes in R1, you should setup "match route-type external" in the deny statement of the route-map "CCNP" in R1.

Command	Redistribution Support						
	connected	static	RIP	EIGRP	OSPF	IS-IS	BGP
match route-type external	—	—	—	Yes	Yes	Yes	Yes
match route-type internal	—	—	—	Yes	Yes	—	Yes
match route-type local	—	—	—	—	—	—	Yes
match route-type nssa-external	—	—	—	—	Yes	—	—
match route-type {level-1 level-2}	—	—	—	—	—	Yes	—

Note that “type-1” or “type-2” is not needed in matching EIGRP routes for redistribution.

Note that a permit statement without condition has to be appended to the route-map “CCNP” in R1 for allowing all other EIGRP routes to be advertised to R3. The existing route-map “CCNP” lacks this permit statement and therefore no EIGRP route are redistributed into OSPF and therefore no OSPF route can be found in the output of R3’s “sh ip route”.

QUESTION 524

Which two reasons would cause an LSP to break between two PE routers? (Choose two.)

- A. MPLS not enabled
- B. matching labels
- C. lost LDP adjacency
- D. IGP hello adjacency
- E. prefix mismatch

Correct Answer: CE

Section: (none)

Explanation

Explanation/Reference:

Other than problem in LDP adjacency, the mismatch of network prefix will also cause LSP to break.

Remarks :

If label bindings are received from a downstream neighbor for prefixes (including subnet mask) which do not appear in a router's routing and CEF tables, these bindings will not be used. In a similar manner, if a router advertises labels for a subnet/subnet mask pair, which do not correspond to the routing updates also advertised by this router for the same subnet/subnet mask pair, these labels will not be used by upstream neighbors and the Label Switched Path (LSP) between these devices will fail.

QUESTION 525

What is the function of penultimate hop popping?

- A. The transport label is popped off at the egress LSR, and unlabeled traffic is forwarded toward the CE.
- B. The last P router in the path pops off the transport label before traffic is forwarded toward the PE.
- C. The VPN label is popped off at the egress LSR, and unlabeled traffic is forwarded toward the CE.
- D. The second to last P router in the path pops off the VPN label before traffic is forwarded to the last P router.

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

QUESTION 526

Which two technologies optimize MPLS infrastructure using bandwidth protection services when experiencing slow response? (Choose two.)

- A. SD-MPLS
- B. Fast-Reroute
- C. VPLS
- D. MPLS OAM
- E. IP LFA

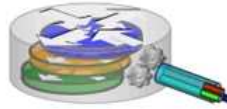
Correct Answer: BE

Section: (none)

Explanation

Explanation/Reference:

MPLS: The WAN Service Enabler



- L3 VPN Services
- L2 VPN Services
 - Layer-2 transport: Ethernet, ATM/FR, HDLC/PPP, interworking, VPLS
- Traffic Engineering - Explicit Path Routing
 - Traffic Engineering, disjointed paths, new attributes for best path (latency, packet loss)
 - Optimization of bandwidth and protection using Fast-ReRoute (FRR)
- Bandwidth Protection Services
 - Fast Re-Route, IP LFA (IP FRR)
- IP Multicast
- Interworking with new solutions - VXLAN → L2/L3 VPN
- Leverage Segment Routing for Next-Gen Scale, Central Control, optimized services
 - Offers an "SD-MPLS" solution moving forward

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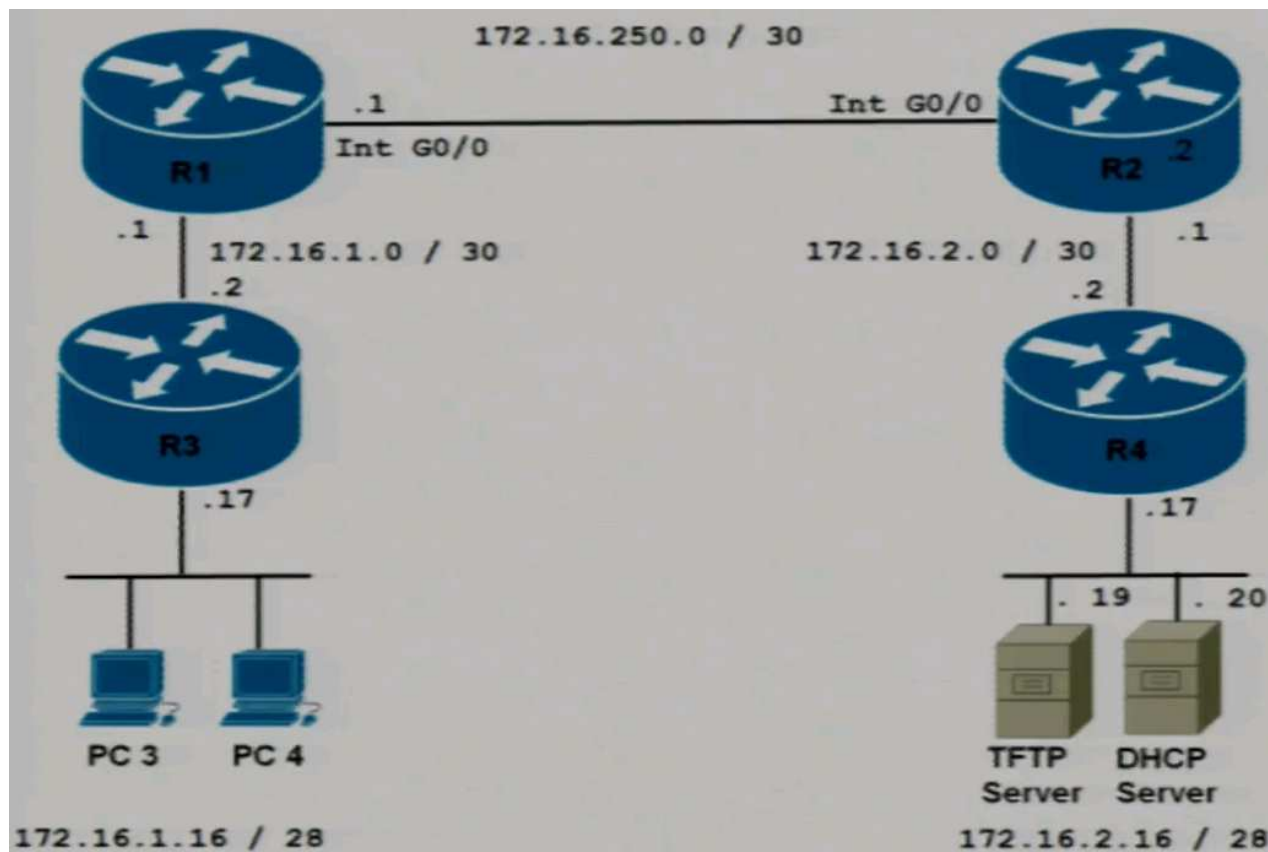
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QUESTION 527

```
R3#sh ip int brie
Interface                IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0      unassigned      YES NVRAM  administratively
down down
GigabitEthernet0/1      172.16.250.2    YES manual  up
up
GigabitEthernet0/2      172.16.250.14   YES manual  up
up
GigabitEthernet0/3      172.16.1.17     YES manual  up
up
R3#
```

```
R3#sh run | begin router eigrp
router eigrp 100
 network 172.16.1.0 0.0.0.3
 network 172.16.1.16 0.0.0.15
!
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0/3
!
line con 0
line aux 0
line vty 0 4
 login
 transport input none
!
```

```
R4#sh run
!
hostname R4
!
ip cef
!
interface GigabitEthernet0/0
 ip address 172.16.2.2 255.255.255.252
 ip access-group 120 in
!
interface GigabitEthernet0/1
 ip address 172.16.2.17 255.255.255.240
!
router eigrp 100
 network 172.16.2.0 0.0.0.3
 network 172.16.2.16 0.0.0.15
!
access-list 120 permit udp host 172.16.1.2 host 172.16.2.19 eq tftp
access-list 120 deny    udp any any eq tftp
access-list 120 permit tcp any any
```



The engineer is trying to transfer the new IOS file to the router R3 but is getting an error. Which configuration achieves the file transfer?

- A. R4(config)#no access-list 120 permit udp host 172.16.1.2 host 172.16.2.19 eq 69
R4(config)#no access-list 120 deny udp any any eq tftp
R4(config)#access-list 120 permit tcp any any
- B. R4(config)#no access-list 120 permit udp host 172.16.1.2 host 172.16.2.19 eq 69
R4(config)#access-list 120 permit udp host 172.16.1.17 host 172.16.2.19 eq 69
R4(config)#access-list 120 permit tcp any any
- C. R4(config)#no access-list 120 permit udp host 172.16.1.2 host 172.16.2.19 eq 69
R4(config)#access-list 120 permit tcp host 172.16.1.17 host 172.16.2.19 eq 69
R4(config)#access-list 120 permit tcp any any
- D. R4(config)#no access-list 120 permit udp host 172.16.1.2 host 172.16.2.19 eq 69
R3(config)#no ip tftp source-interface GigabitEthernet0/3

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

In R3's existing configuration, the interface g0/3 having the IP address 172.16.1.17 is configured as the source interface for accessing TFTP server by the global configuration mode command "ip tftp source-interface GigabitEthernet0/3". Hence, the IP 172.16.1.17 will be used as source for accessing TFTP server.

In R4's configuration, an access-list 120 is configured to filter g0/0 inbound traffic. It allows TFTP traffic only from source 172.16.1.2 (R3's interface connecting to R1). Hence R3 cannot access the TFTP server.

In order for R3 to download IOS file from the TFTP server successfully, either one of the following should be performed:

- Remove the command "ip tftp source-interface GigabitEthernet0/3" in R3 so that it will use the IP address 172.16.1.2 as source for accessing TFTP server; OR

- Modify the access-list 120 in R4 so that it allows TFTP traffic from source 172.16.1.17. Note that TFTP traffic is **UDP** traffic with destination port 69.

Among the choices, only one meets the above i.e. implementing one and only one of the above alternatives.

Note that for the command "no access-list 120", regardless of the condition specified in the rule, the entire access-list will be removed by the command. (You can only delete individual rule by modifying the access-list in the way used for modifying a named access-list i.e. using the number 120 as if it is a access-list's name and modify it through sequence number).

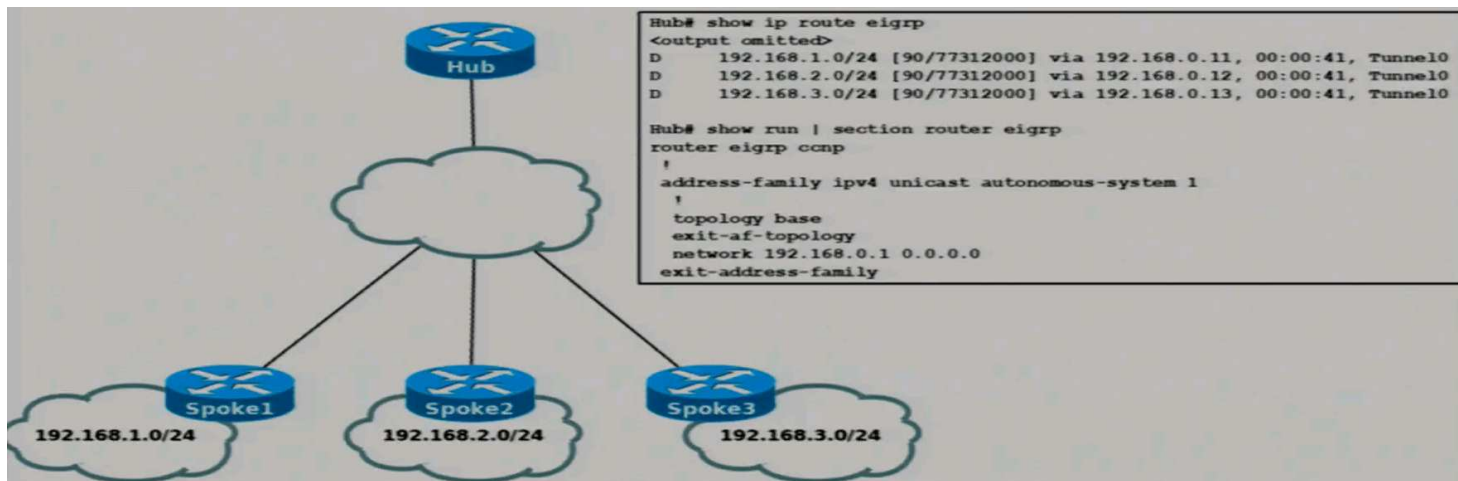
Therefore after a command "no access-list 120", you need to setup all rules of the entire access-list again by including at least two rules: one for allowing TFTP traffic with specific source 172.16.1.17 and one for allowing all TCP traffic. The rule for denying all other TFTP traffic is not required since it will be blocked by implicit deny.

Although the removing of R4's access-list 120 as well as R3' command "ip tftp source-interface GigabitEthernet0/3" at the same time can allow R3 to access TFTP server, it is not the best answer since undesired traffic (e.g. other TFTP traffic or UDP traffic) will be allowed.

Remarks :

You should ignore the IP addresses shown for R3's g0/1 and g0/2 in the output of "sh ip int brie" since R3 is not connecting to those networks in the diagram. For example: 172.16.250.2 shown for R3's g0/1 is actually the IP address of R2 g0/0. Moreover, the IP address of one of the R3's interface should be 172.16.1.2 but it is not shown in the output of "sh ip int brief".

QUESTION 528



Spoke routers do not learn about each other's routes in the DMVPN Phase2 network. Which action resolves the issue?

- A. Configure a static route in each spoke to establish a spoke-to-spoke tunnel.
- B. Rectify incorrect wildcard mask configured on the hub router network command.
- C. Remove default route from spoke routers to establish a spoke-to-spoke tunnel.
- D. Disable EIGRP split horizon on the Tunnel0 interface of the hub router.

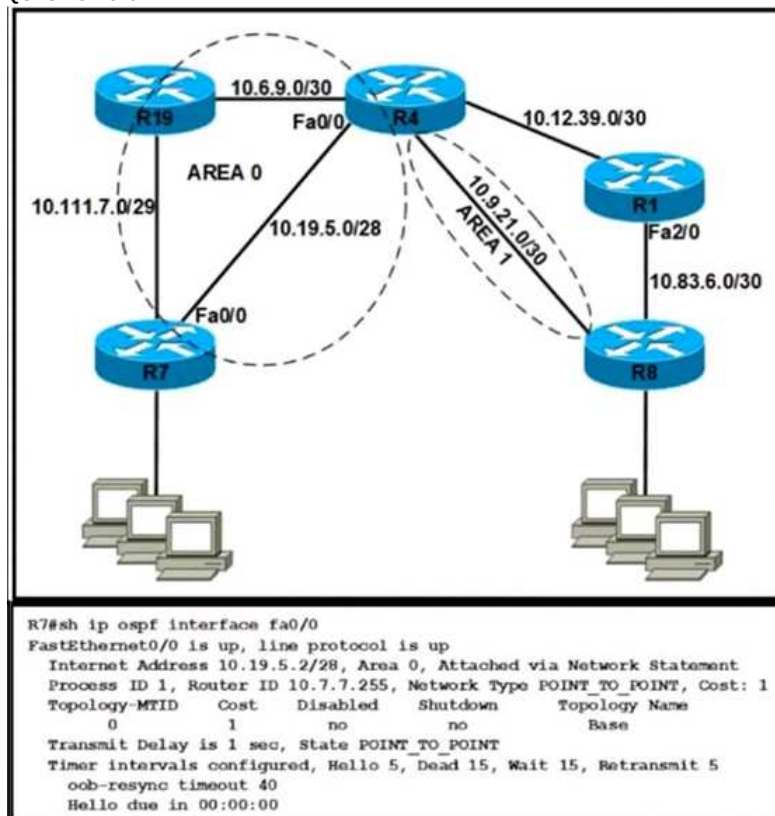
Correct Answer: D

Section: (none)

Explanation

Explanation/Reference:

QUESTION 529



Router R4 is configured correctly with default OSPF values. A network engineer configured R7 for OSPF. R7 must not be elected as a DR for the segment between R4-R7. The adjacency between R4 and R7 failed to form. Which configuration resolves the issue?

- A.


```

R7(config)#interface fa0/0
R7(config-if)#ip ospf priority 255
R7(config-if)#ip ospf hello-interval 10
R7(config-if)#ip ospf dead-interval 30
R7(config-if)#ip ospf network broadcast

```


- B.
- ```
R7(config)#interface fa0/0
R7(config-if)#ip ospf priority 255
R7(config-if)#ip ospf hello-interval 10
R7(config-if)#ip ospf dead-interval 40
R7(config-if)#ip ospf network non-broadcast
```
- C.
- ```
R7(config)#interface fa0/0
R7(config-if)#ip ospf priority 0
R7(config-if)#ip ospf hello-interval 10
R7(config-if)#ip ospf dead-interval 40
R7(config-if)#ip ospf network broadcast
```
- D.
- ```
R7(config)#interface fa0/0
R7(config-if)#ip ospf priority 0
R7(config-if)#ip ospf hello-interval 10
R7(config-if)#ip ospf dead-interval 30
R7(config-if)#ip ospf network non-broadcast
```

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

You can prevent R7 from participate in DR/BDR election by setting the OSPF priority to 0 in the concerned interface.

Between the two choices that set the OSPF priority to 0, the one that sets network type to "non-broadcast" requires additional configuration settings using "neighbor" commands. Since "neighbor" commands are probably not configured due to the existing network type "point-to-point", the network type "broadcast" should be the answer.

#### QUESTION 530

An engineer configured a router with this configuration:

```
ip access-list DENY_TELNET
10 deny tcp any any eq 23 log-input
```

The router console starts receiving log message "%SEC-6-IPACCESSLOGP: list DENY\_TELNET denied tcp192.168.1.10(10222) (FastEthernet1/0 E416.98ac.116e) ->192.168.2.20(23), 1 packet". Which action stops messages on the console while still denying Telnet?

- A. Remove log-input keyword from the access list.
- B. Configure a 20 permit ip any any log-input command.
- C. Configure a 20 permit ip any any command.
- D. Replace log-input keyword with the log keyword in the access list.

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The log message is shown since the deny rule configured in the access-list "DENY\_TELNET" is configured with the keyword "log -input" which means "Log matches against this entry, including input interface".

In order to stop the generating of these log messages, you need to remove the keyword "log -input" from the rule.

Remarks :

Replacing "log -input" with "log" will still generate log messages although the concerned input interface will not be included in the log message.

#### QUESTION 531

What are the two major components of an MPLS-based VPN? (Choose two.)

- A. VPN route distinguisher
- B. MP-BGP peering of VPN community CE devices
- C. MP-BGP peering of VPN community P devices
- D. MP-BGP peering of VPN community PE devices
- E. VPN route target communities
- F. VPN route reflectors

**Correct Answer:** DE

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 532

```

R1#show ipv6 ospf neighbor
R1#debug ipv6 ospf hello
OSPFv3 hello events debugging is on
OSPFv3: Rcv hello from 10.1.1.1 area 10 from FastEthernet0/1 FE80::C801:FFF:FE94:1C
interface ID 4
OSPFv3: Hello from FE80::C801:FFF:FE94:1C with mismatched NSSA option bit
OSPFv3: Send hello to FF02::5 area 10 on FastEthernet0/1 from FE80::C004:22FF:FE78:1
interface ID 5

```

Look at the exhibit, R1 is not forming an adjacency on a point to point interface, what can be determined from the debug output and what corrective action is required?

R1 is not forming adjacency on a point-to-point interface. Which action resolves the issue?

- A. The area numbers must be configured the same on each router.
- B. The area types must be configured the same on each router.
- C. The no-summary command must be included in the area configuration on R2.
- D. The no-summary command must be included in the area configuration on R1.

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

There is a debug message "OSPFv3:Hello from .... with mismatched NSSA option bit". This debug message appeared since the area types configured in the two routers are not the same.

#### QUESTION 533

How are LDP neighbors discovered?

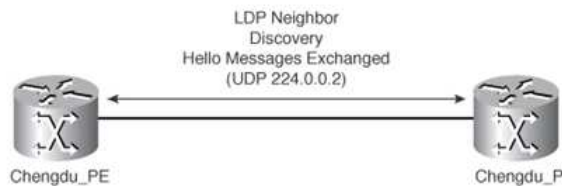
- A. Broadcasts hellos are sent to the 255.255.255.255 broadcast address
- B. Unicast hellos are sent to directly connected neighbors IP addresses
- C. Multicast hellos are sent to the 224.0.0.5 group address
- D. Multicast hellos are sent to the 224.0.0.2 group address

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**



#### QUESTION 534

**Company XYZ Network**

```

R101#sh run | section sla
ip sla 1
 tcp-connect 2.2.2.2 3000 source-ip 1.1.1.1
 threshold 1000
 timeout 1000
 frequency 10
ip sla schedule 1 life forever start-time now
ip sla 2
 icmp-jitter 2.2.2.2 source-ip 1.1.1.1 num-packets 100 interval 10
 threshold 1000
 timeout 1000
 frequency 10
ip sla schedule 2 life forever start-time now
R101#sh ip sla summary
IPSLAs Latest Operation Summary
Codes: * active, ^ inactive, ~ pending

```

| ID | Type        | Destination | Stats (ms) | Return Code   | Last Run       |
|----|-------------|-------------|------------|---------------|----------------|
| *1 | tcp-connect | 2.2.2.2     | -          | No connection | 33 seconds ago |
| *2 | icmp-jitter | 2.2.2.2     | RTT=4      | OK            | 3 seconds ago  |

While troubleshooting an issue on the network, an engineer notices that a TCP Connect operation failed on port 3000 between R101 and R201. Which command must be configured on R201 to respond to the R101 IP SLA configurations with a control connection on UDP port 1967?

- A. ip sla responder tcp-connect ipaddress 2.2.2.2 port 3001
- B. ip sla responder
- C. ip sla responder tcp-connect ipaddress 1.1.1.1 port 3000
- D. ip sla responder udp-echo ipaddress 1.1.1.1 port 1967

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Since R101 is configured with IP SLA for "tcp-connect" to R201's port 3000 and "control" is not disabled, R101 will send control message every time to tell R201 to response to packets for TCP port 3000. However, the return code "No connection" in the output means that R201 is not responding to the control message.

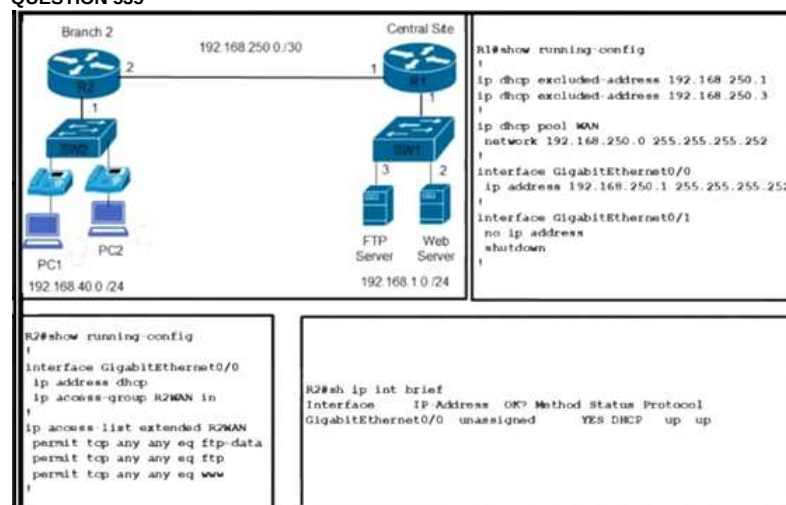
Hence, you should configure "ip sla responder" in R201 so that R201 can accept and response to R101's control message.

Remarks :

On the other hand, if "control disable" is configured in the R101's SLA 1 and R201 is not listening on TCP port 3000, the return code will be "Socket connect error" instead. In this case, since control message is not used by R101, you need to specify the port that has to be listened by R201 manually using the command "ip sla responder tcp-connect ipaddress 1.1.1.1 port 3000".

The UDP port 1967 mentioned in the question is the standard port used by IP SLA Responder for control messages. Therefore, no additional configuration is required for this port number.

#### QUESTION 535



Which configuration is required for R2 to get the IP address from the DHCP server.

- A. ip access-list extended R2WAN  
permit tcp any any eq 68
- B. ip access-list extended R2WAN  
permit udp any any eq 67
- C. interface GigabitEthernet0/0  
ip access-group R2WAN out
- D. ip access-list extended R2WAN  
permit udp any any eq 68

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The existing access list "R2WAN" configured in R2 is configured to permit certain TCP traffic only. Therefore, you have to add a rule to allow UDP port 68 (i.e. the port number used by DHCP client).



#### QUESTION 536

What is the downstream unsolicited distribution method in MPLS?

- A. It advertises labels to peers only when the peer requests.
- B. It sends a unicast hello message to a specific LSR.

- C. It sends a unicast hello message to a specific LER.
- D. It advertises labels to peers without peer request.

**Correct Answer:** D

**Section:** (none)

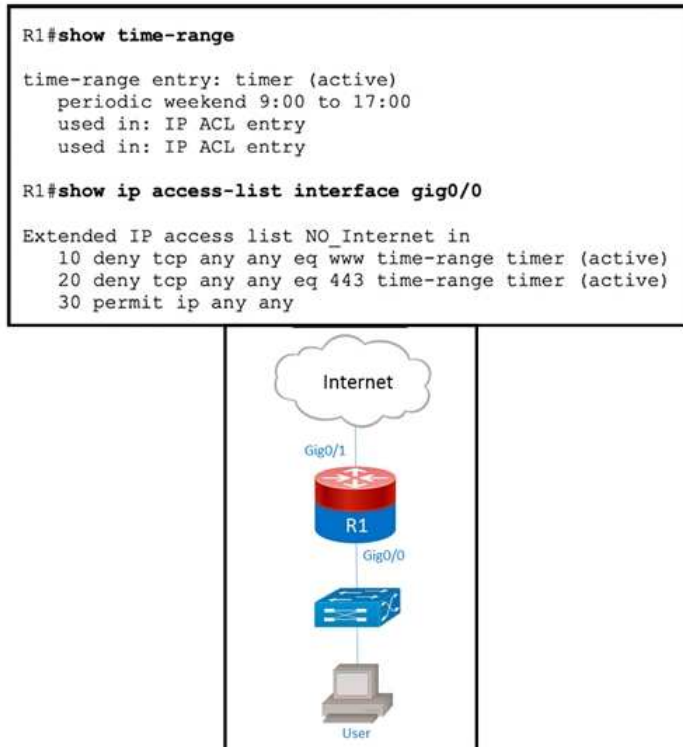
**Explanation**

**Explanation/Reference:**

After the LDP TCP connection session is established, the LSRs negotiate the session parameters, including the method of label distribution to be used. Two methods are available:

- Downstream Unsolicited: An LSR advertises label mappings to peers without being asked to.
- Downstream on Demand: An LSR advertises label mappings to a peer only when the peer asks for them.

#### QUESTION 537



Users on a call center report that they cannot browse the internet on Saturdays during the afternoon. Which configuration resolves the issue?

- A. time-range timer  
no periodic weekend 9:00 to 17:00  
periodic weekend 17:00 to 23:59
- B. no time-range timer
- C. interface gig0/0  
ip access-group NO\_Internet out
- D. ip access-list extended NO\_Internet  
15 permit tcp any any eq www

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Among the choices, all except the inserting of the rule "15 permit tcp any any eq www" will let users to browse Internet (since the rule the rule "15 permit tcp any any eq www" is inserted after the deny rule for www traffic.)

However:

- The choice of reversing the direction for applying of the access list so that it is applied to the wrong direction does not seems to be the logical way to allow users to browse Internet.
- Although the removing of the time-range "timer" will make the time-range condition always be met, it is also not logical since it is better to reconfigure the ACL rules so that no any time range is specified.

For the existing time-range "timer", users can browse Internet between 9:00-17:00 in weekend. If users cannot browse the Internet in Saturday's afternoon, they are probably browsing the Internet at e.g. "17:01". Hence, reconfigure the time-range to 17:00-23:59 allows users to browse Internet.

Note that this is not a perfect solution based on the wordings in the question since users will then not be able to browse Internet in Saturday's afternoon at other time e.g. "16:49". However, this is the best answer when comparing with other choices.

#### QUESTION 538



An engineer configured user login based on authentication database on the router, but no one can log into the router. Which configuration resolves the issue?

- A. aaa authentication login default local
- B. aaa authorization network default local
- C. aaa authentication login default enable
- D. aaa authorization exec default local

**Correct Answer:** A

**Section:** (none)

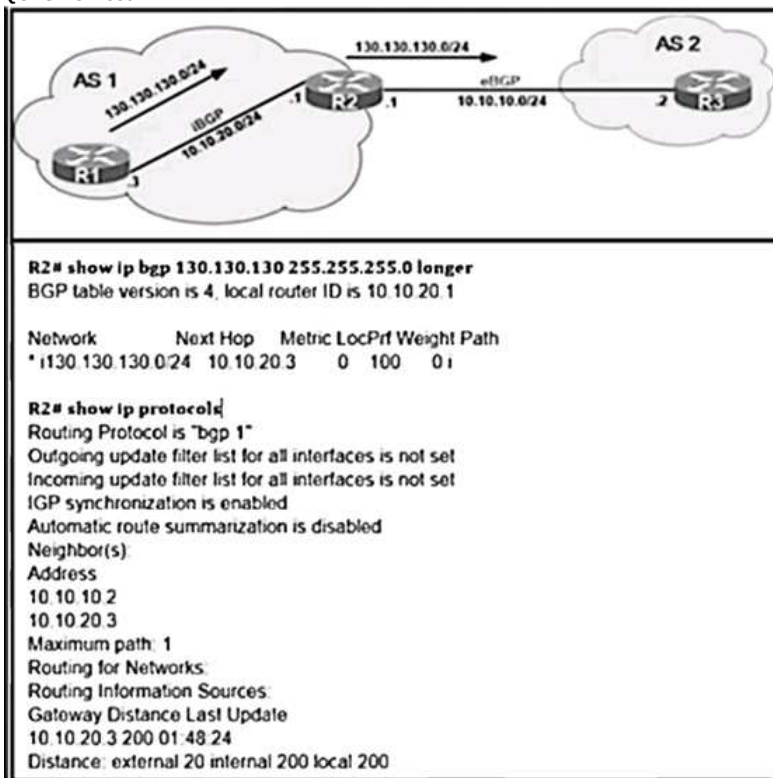
**Explanation**

**Explanation/Reference:**

Remarks :

In practice, after "aaa new-model" is configured, the line configuration mode command "login local" will be removed automatically. However, you can configure a default authentication list using "local" (i.e. local database) so that user can continue to login using the username / password configured in the router.

#### QUESTION 539



The 130.130.130.0/24 route shows in the R2 routing table but is getting filtering toward R3. Which action resolves the issue?

- A. Automatic route summarization must be enabled on R2.
- B. The outgoing filter list for all interfaces must be set on R2.
- C. The incoming filter list for all interfaces must be set on R2.
- D. IGP synchronization must be disabled on R2.

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

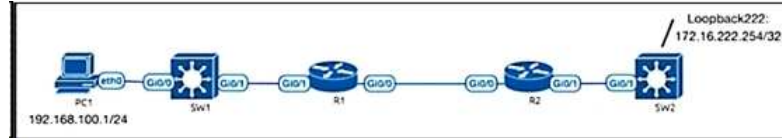
From the command output, automatic route summarization is disabled. However, IGP synchronization is enabled.

Hence, the learnt BGP 130.130.130.0/24 route from R1 will not be advertised by R2 to other routers if this route is not learnt through IGP. This can be solved by:

- disabling IGP synchronization in R2's BGP configuration OR
- configure R1 to also advertise the route through IGRP.



#### QUESTION 540



R2 can reach Loopback222, but R1, SW1, and PC1 cannot communicate with 172.16.222.254. R1 and R2 configurations are shown here:

```
R1#show run | sec router eigrp
router eigrp VR1
!
address-family ipv4 unicast autonomous-system 1
!
topology base
exit-af-topology
network 172.16.1.1 0.0.0.0
network 192.168.100.0
network 192.168.200.0
network 192.168.255.91 0.0.0.0
exit-address-family
```

```
R2(config)#do show run | sec router eigrp
router eigrp 1
network 172.16.1.2 0.0.0.0
network 172.16.222.0 0.0.0.255
network 192.168.222.254 0.0.0.0
```

Which EIGRP configuration command resolves the issue?

- A. R1(config-router)# redistribute static
- B. R2(config-router)# redistribute static
- C. R1(config-router)# network 172.16.222.254 0.0.0.0
- D. R1(config-router)# network 172.16.222.254 255.255.255.255

**Correct Answer:** B

**Section:** (none)

**Explanation**

#### Explanation/Reference:

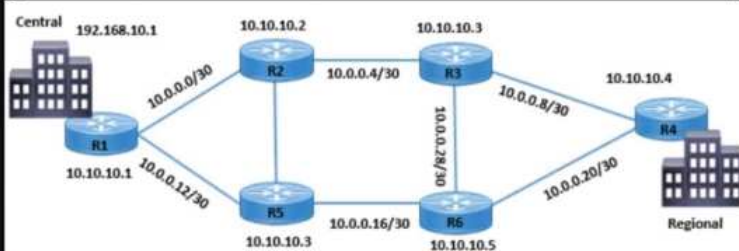
Since 172.16.222.254 is a loopback interface 222 of Sw2 i.e. not a directly connected network of R2, the "network 172.16.222.0 0.0.0.255" command in R2 will not advertise the route 172.16.222.254/32.

Assume that Sw2 is NOT configured with any routing protocol, since R2 can reach Sw2's Loopback222, a static route for 172.16.222.254/32 with Sw2 as next hop is properly configured in R2. In this case, you can configure "redistribute static" in the R2's EIGRP configuration so that this static route can be advertised as EIGRP external route to R1.

#### QUESTION 541

```
R3#show ip sla statistics
IPSLAs Latest Operation Statistics
IPSLA operation id: 10
Type of operation: icmp-echo
Latest RTT: 24 milliseconds
Latest operation start time: *21:26:43.211 UTC Sat Sep 18 2021
Latest operation return code: OK
Number of successes: 75
Number of failures: 0
Operation time to live: Forever

IPSLA operation id: 20
Type of operation: icmp-echo
Latest RTT: NoConnection/Busy/Timeout
Latest operation start time: *21:26:47.499 UTC Sat Sep 18 2021
Latest operation return code: No connection
Number of successes: 128
Number of failures: 459
Operation time to live: Forever
```



Traffic from R3 to the central site does not use alternate paths when R3 cannot reach 10.10.10.2. Traffic on R3 destined to R4 takes an alternate route via 10.10.10.6 when 10.10.10.4 is not accessible from R3. Which configuration switches traffic destined to 10.10.10.2 from R3 on the alternate path?

- A. R3(config)#ip route 192.168.10.1 255.255.255.255 10.10.10.2 track 20
- B. R6(config)#ip route 10.10.10.3 255.255.255.255 10.0.0.30



- C. R3(config)#track 20 ip sla 20 reachability  
D. R2(config)#ip route 10.10.10.3 255.255.255.255 10.0.0.6

**Correct Answer:** A  
**Section:** (none)  
**Explanation**

**Explanation/Reference:**

- In order to configure R3 to use an alternate path e.g. R6 to reach central site if R2 (10.10.10.2) cannot be reached, the followings are needed:
- Set up an IP SLA to check whether 10.10.10.2 i.e. R2 is reachable. (The IP SLA 20 shown in the command output is probably configured for this).
  - Set up a track object e.g. 20 to monitor the reachability or state of the above IP SLA 20.
  - Set up the static route to central site using R2 as next hop. The route should include "track" specifying the track object number e.g. 20 above.
  - Set up a floating route (i.e. a route with a large AD distance e.g. 254) for the central site in R3 using R6 as next hop.

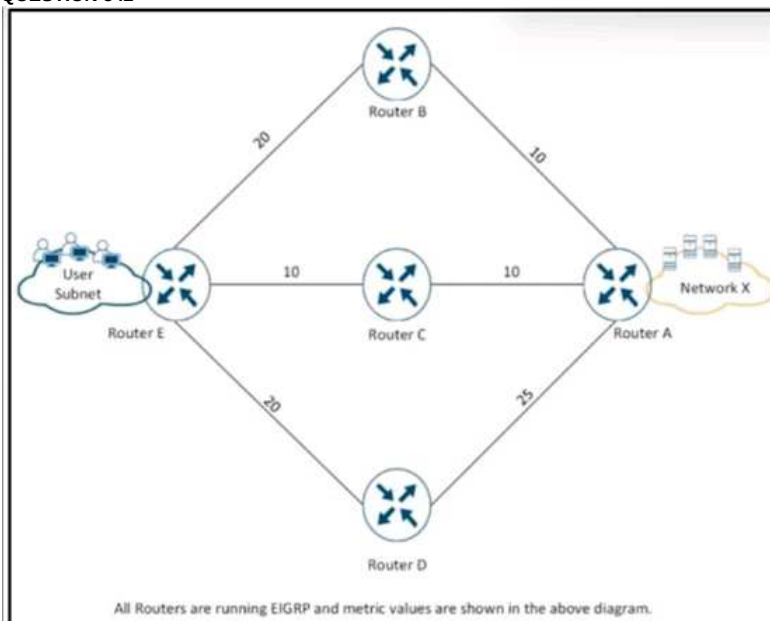
There are two choices (one is the setup of static route with track and the other is the command creating the track object) that corresponds to the commands required in step ii and iii. If this question asks you to choose two answers, then both should be selected as answers. However, this question allows you to select one single answer only, without further information, it is hard to tell which one of these two commands is missing in R3.

Since the question asks which configuration switches traffic, the suggested answer is the command that related directly with the performing of switching by disabling the route with R2 as next hop

**Remarks:**

The IP addresses of R5 and R6 are wrong in the diagram. They should be 10.10.10.5 and 10.10.10.6 respectively.

**QUESTION 542**



The IT manager received reports from users about slow applications through network x. Which action resolves the issue?

- Upgrade the IOS on router E.
- Move the servers into the users subnet.
- Increase the bandwidth from the service provider.
- Use the variance 2 command to enable load balancing.

**Correct Answer:** D  
**Section:** (none)  
**Explanation**

**Explanation/Reference:**

The best answer is to use the command "variance 2" so that more than one route can be used for routing traffic from user subnet to network x. The using of more links effectively increases the bandwidth available for routing the concerned traffic.

**QUESTION 543**

What is a limitation of IPv6 RA Guard?

- It does not offer protection in environments where IPv6 traffic is tunneled.
- It cannot be configured on a switch port interface in the ingress direction.
- Packets that are dropped by IPv6 RA Guard cannot be spanned.
- It is not supported in hardware when TCAM is programmed.

**Correct Answer:** A  
**Section:** (none)  
**Explanation**

**Explanation/Reference:**

The IPv6 RA Guard feature does not offer protection in environments where IPv6 traffic is tunneled.

All other choices are incorrect about IPv6 Guard since IPv6 Guard has the following characteristics:

- This feature can be configured on a switch port interface in the ingress direction.
- Packets dropped by the IPv6 RA Guard feature can be spanned.
- This feature is supported only in hardware when the ternary content addressable memory (TCAM) is programmed.

**QUESTION 544**

How is the LDP router ID used in an MPLS network?

- A. The force keyword changes the router ID to the specified address without causing any impact.
- B. The loopback with the highest IP address is selected as the router ID.
- C. The MPLS LDP router ID must match the IGP router ID.
- D. If not configured, the operational physical interface is chosen as the router ID even if a loopback is configured.

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Remarks:

By default, loopback with the highest IP address is selected by LDP as the router ID.

Instead of using loopback with the highest IP address, you can specify a router ID with "mpls ldp router-id" command. However, the configured router ID will only be effective when the next time it is necessary to select an LDP router ID, which is typically the next time the interface is shut down or the address is configured.

You can use "force" keyword in the above command to make the change of router ID to be effective immediately. However, this tears down any existing LDP sessions, releases label bindings learned via the LDP sessions, and interrupts MPLS forwarding activity associated with the bindings.

#### QUESTION 545

```
interface loopback0
ip address 4.4.4.4 255.255.255.0
!
interface FastEthernet1/0
Description **** WAN link ****
ip address 10.0.0.1 255.255.255.0
!
interface FastEthernet1/1
Description **** LAN Network ****
ip address 192.168.1.1 255.255.255.0
!
!
router ospf 1
router-id 4.4.4.4
log-adjacency-changes
network 4.4.4.4 0.0.0.0 area 0
network 10.0.0.1 0.0.0.0 area 0
network 192.168.1.1 0.0.0.0 area 10
!
```

Which set of commands restore reachability to loopback0?

- A. interface loopback0  
ip address 4.4.4.4 255.255.255.0  
ip ospf network point-to-point
- B. interface loopback0  
ip address 4.4.4.4 255.255.255.0  
ip ospf interface area 10
- C. interface loopback0  
ip address 4.4.4.4 255.255.255.0  
ip ospf network broadcast
- D. interface loopback0  
ip address 4.4.4.4 255.255.255.0  
ip ospf interface type network

**Correct Answer:** A

**Section:** (none)

**Explanation**

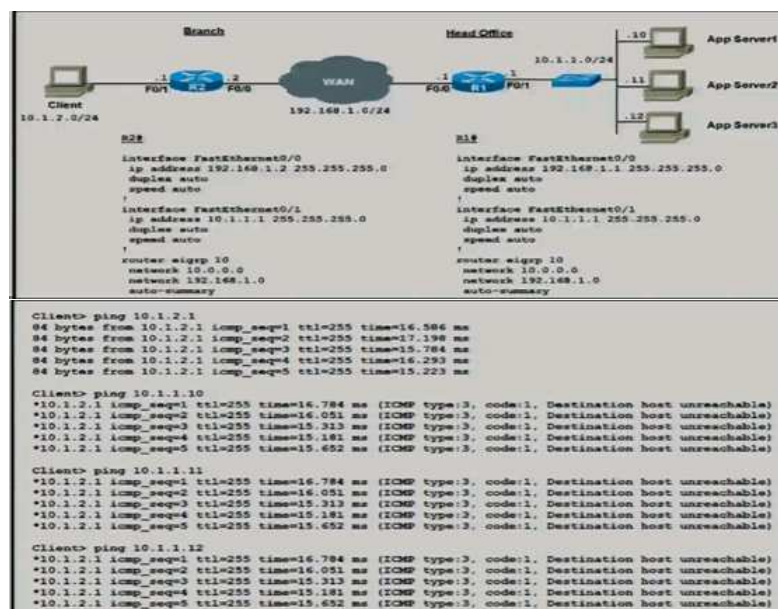
**Explanation/Reference:**

With the existing configuration, loopback0 should be reachable although the concerned network is advertised as 4.4.4.4/32.

Among the choices, only changing the network type to "point-to-point" has effect since the router will then advertise 4.4.4.0/24 instead of 4.4.4.4/32. Therefore, it is chosen as suggest answer.

All other choices either have no effect (e.g. the network type of loopback is by default "broadcast") or the commands are wrong (commands starting with "ip ospf interface ...." are wrong since there is no interface configuration mode command starting with "ip ospf interface ....").

#### QUESTION 546



The EIGRP neighborhood between the routers is up. However, the app servers are unreachable from the client PC. Which action must the engineer take to resolve the issue?

- A. Configure the network 10.1.1.0 command in the EIGRP protocol on R1 only
- B. Configure the neighbor 10.1.2.0 FastEthernet 0/0 command in the EIGRP protocol on R2 only
- C. Configure the no auto-summary command in the EIGRP protocol on both routers
- D. Configure the default network 10.1.2.0.0.0.255 command in the EIGRP protocol on R2 only

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Since EIGRP configuration in both routers are configured with "auto-summary", the subnet routes 10.1.1.0/24 and 10.1.2.0/24 will be advertised as 10.0.0.0/8 from the routers.

Since both routers will also generate a null route for the summarized route 10.0.0.0/8 with an AD of 5, traffic for remote subnets (i.e. subnets that are not directly connected) of the network 10.0.0.0/8 cannot be reached by the two routers.

#### QUESTION 547

High input/output error on Switch interfaces > Issue Instance

### High input/output error on interface 'GigabitEthernet0/2'

[Open](#) ▾

#### Description

High Input/Output error Rx 1.75% Tx 0.0% on interface 'GigabitEthernet0/2'

[Go to SW1.ap.com](#) ↗

Last Occurred: Jan 12, 2022 12:53 AM

```

GigabitEthernet0/3 is up, line protocol is up (connected)
Hardware is iGbE, address is 502c.4801.2a03 (bia 502c.4801.2a03)
Internet address is 172.16.1.25/30
MTU 1500 bytes, BW 1000000 kbit/sec, DLY 10 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Auto Duplex, Auto Speed, link type is auto, media type is RJ45
output flow-control is unsupported, input flow-control is unsupported
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:06, output 00:00:00, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 272000 bits/sec, 48 packets/sec
5 minute output rate 236000 bits/sec, 40 packets/sec
365738 packets input, 100001249 bytes, 0 no buffer
Received 2 broadcasts (0 IP multicasts)
8568 runts, 0 giants, 0 throttles
8568 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
0 watchdog, 0 multicast, 0 pause input
196663 packets output, 27428090 bytes, 0 underruns
0 output errors, 0 collisions, 5 interface resets
0 unknown protocol drops
0 babbles, 0 late collision, 0 deferred
1 lost carrier, 0 no carrier, 0 pause output
0 output buffer failures, 0 output buffers swapped out

```

Which action resolves the issue?

- A. Upgrade the hardware to increase the interface input buffers
- B. Implement traffic policing to prevent the interface input traffic being exceeded
- C. Resolve the misconfigured QoS parameters
- D. Replace the faulty cable

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

From the output, all input errors are runts. Runt packets are packets that are less than the minimum packet size (64 bytes for Ethernet). They often occur due to collisions caused by mis-matched duplex or the hardware (e.g. network card / cable) is faulty.

**QUESTION 548**



The engineer is reviewing the logs on the DENVER router and notices that this error message repeats constantly Jun 12 13:42:03.399: %TCP-6-BADAUTH: No MD5 digest from 10.40.1.1(27174) to 10.40.1.2(179) tableid - 0

Which action resolves the issue?

- A. Configure BGP authentication on the router with IP address 10.40.1.2
- B. Configure OSPF link authentication on the router with IP address 10.40.1.1
- C. Configure NTP authentication on the router with IP address 10.40.1.1
- D. Configure BGP authentication on the router with IP address 10.40.1.1

**Correct Answer:** D

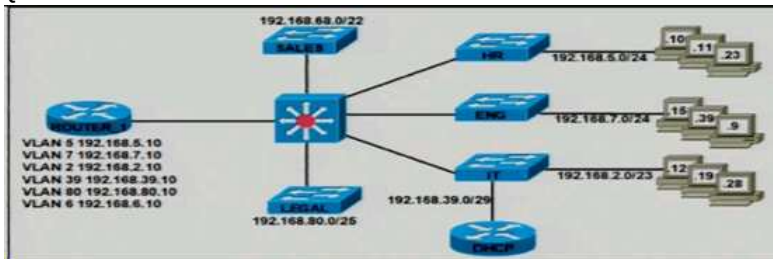
**Section:** (none)

**Explanation**

**Explanation/Reference:**

The error message appears when there is a problem concerning BGP neighbor authentication. The error message shown in the question means that MD5 digest information is missing for the BGP packets sent from router "10.40.1.1" to the router "10.40.1.2". Hence, the router "10.40.1.1" is probably missing configuration setting concerning BGP authentication.

**QUESTION 549**



After an engineer configured a new cisco router as a DHCP server, users reported two primary issues:

- Devices in the HR subnet have intermittent connectivity problems
- Workstations in the LEGAL subnet cannot obtain IP addresses

Which configurations must the engineer apply to ROUTER\_1 to restore connectivity for the affected devices?

- A. 

```
interface GigabitEthernet0/0.5
 encapsulation dot1Q 5
 ip address 192.168.5.10 255.255.255.0
 ip helper-address 192.168.39.100
 !
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 192.168.80.10 255.255.255.128
 ip helper-address 192.168.39.100
 !
ip dhcp excluded-address 192.168.5.1 192.168.5.10
ip dhcp excluded-address 192.168.80.1 192.168.80.10
 !
ip dhcp pool LEGAL
 network 192.168.80.0 255.255.255.128
 default-router 192.168.80.10

ip dhcp pool HR
 network 192.168.5.0 255.255.255.0
 default-router 192.168.5.10
```
- B. 

```
interface GigabitEthernet0/0.5
 encapsulation dot1Q 5
 ip address 192.168.5.10 255.255.255.0
 ip helper-address 192.168.39.100
 !
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 192.168.80.10 255.255.255.128
 ip helper-address 192.168.39.100
 !
ip dhcp excluded-address 192.168.80.1 192.168.80.10
 !
ip dhcp pool LEGAL
 network 192.168.80.0 255.255.255.128
```

```

 default-router 192.168.80.10
 !
ip dhcp pool HR
network 192.168.5.0 255.255.255.0
default-router 192.168.5.10
C. interface GigabitEthernet0/0.5
 encapsulation dot1Q 5
 ip address 192.168.5.10 255.255.255.0
 ip helper-address 192.168.93.100
 !
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 192.168.80.10 255.255.255.128
 ip helper-address 192.168.39.100
 !
ip dhcp excluded-address 192.168.5.1 192.168.5.1
ip dhcp excluded-address 192.168.80.1 192.168.80.10
 !
ip dhcp pool LEGAL
network 192.168.80.0 255.255.255.128
default-router 192.168.80.10
 !
ip dhcp pool HR
network 192.168.5.0 255.255.255.0
default-router 192.168.5.10
D. interface GigabitEthernet0/0.5
 encapsulation dot1Q 5
 ip address 192.168.5.10 255.255.255.0
 ip helper-address 192.168.39.100
 !
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 192.168.80.10 255.255.255.128
 ip helper-address 192.168.39.100
 !
ip dhcp excluded-address 192.168.5.1 192.168.5.5
ip dhcp excluded-address 192.168.80.1 192.168.80.110
 !
ip dhcp pool LEGAL
network 192.168.80.0 255.255.255.128
default-router 192.168.80.10
 !
ip dhcp pool HR
network 192.168.5.0 255.255.255.0
default-router 192.168.5.10

```

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Other choices are wrong / not recommended since they have one or both of the following incorrect settings:

- The helper-address is not correctly configured using 192.168.39.100 e.g. "ip helper-address 192.168.93.100"

- For one or both networks, the reserved address configured with "ip dhcp excluded-address" is missing or its excluded range does not cover ".10" which is the host address used by "ROUTER\_1".

#### QUESTION 550

What is a function of IPv6 ND inspection?

- A. It learns and secures bindings for stateless autoconfiguration addresses in Layer 2 neighbor tables
- B. It learns and secures bindings for stateful autoconfiguration address in Layer 2 neighbor tables
- C. It learns and secures bindings for stateless autoconfiguration addresses in Layer 3 neighbor tables
- D. It learns and secures bindings for stateful autoconfiguration addresses in Layer 3 neighbor tables

**Correct Answer:** A

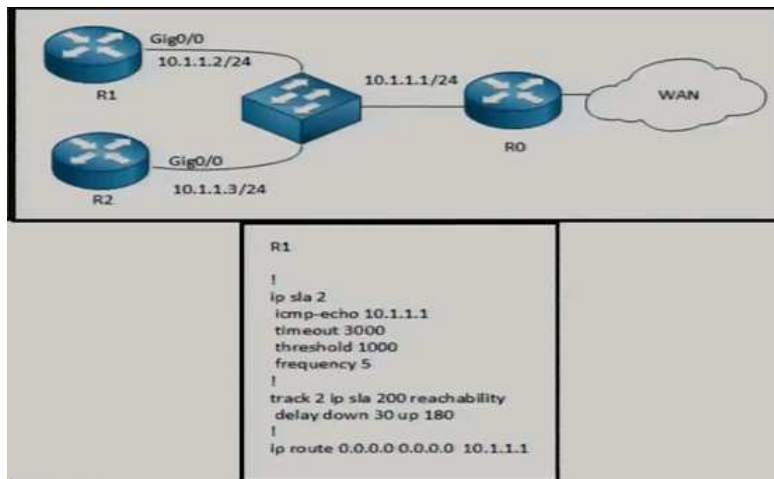
**Section:** (none)

**Explanation**

**Explanation/Reference:**

IPv6 ND inspection learns and secures bindings for stateless autoconfiguration addresses in Layer 2 neighbor tables.

#### QUESTION 551



The default route from R1 must be withdrawn from the routing table if R1 cannot ping 10.1.1.1, but it is not working correctly. Which configuration resolves the issue?

- A. 

```
ip sla 2
icmp-echo 10.1.1.1
timeout 3000
threshold 1000
frequency 5
!
ip sla schedule 2 life forever start-time now
!
track 2 ip sla 200 reachability
delay down 30 up 180
!
ip route 0.0.0.0 0.0.0.0 10.1.1.1 track 2 250
```
- B. 

```
ip sla 2
icmp-echo 10.1.1.1
timeout 3000
threshold 1000
frequency 5
!
track 2 ip sla 2 reachability
delay down 30 up 180
!
ip route 0.0.0.0 0.0.0.0 10.1.1.1 track 2
```
- C. 

```
ip sla 2
icmp-echo 10.1.1.1
timeout 3000
threshold 1000
frequency 5
!
ip sla schedule 2 life forever start-time now
!
track 2 ip sla 2 reachability
delay down 30 up 180
!
ip route 0.0.0.0 0.0.0.0 10.1.1.1 track 2
```
- D. 

```
ip sla 2
icmp-echo 10.1.1.1
timeout 3000
threshold 1000
frequency 5
!
ip sla schedule 2 life forever start-time now
!
track 2 ip sla 200 reachability
delay down 30 up 180
!
ip route 0.0.0.0 0.0.0.0 10.1.1.1
```

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The existing configuration shown in the output is missing settings / incorrect settings:

- The configuration of a schedule to start the IP SLA operation
- The track object 2 is wrongly configured to use IP SLA 200 (the correct IP SLA number is 2).
- The inclusion of "track" in the static default route.

Only one of the choices configured all of the above 3 settings correctly.

**QUESTION 552**



```

R3#show cef interface gi0/3
GigabitEthernet0/3 is up (if_number 5)
Corresponding hwidb fast_if_number 5
Corresponding hwidb firstsw->if_number 5
Internet address is 172.16.4.253/30
ICMP redirects are never sent
Per packet load-sharing is disabled
IP unicast RPF check is enabled
Input features: uRPF
IP policy routing is disabled
BGP based policy accounting on input is disabled
BGP based policy accounting on output is disabled
Hardware idb is GigabitEthernet0/3
Fast switching type 1, interface type 27
IP CEF switching enabled
IP CEF switching turbo vector
IP prefix lookup IPv4 mtrie 8-8-8 optimized
Input fast flags 0x4000, Output fast flags 0x0
ifindex 5(5)
Slot Slot unit 3 VC -1
IP MTU 1500
R3#show run int gi0/3
Building configuration...

Current configuration : 162 bytes
!
interface GigabitEthernet0/3
ip address 172.16.4.253 255.255.255.252
ip verify unicast source reachable-via rx
duplex auto
speed auto
media-type rj45
end

```

An engineer implements uRPF to increase security and stop incoming spoofed IP packets. Same asymmetrically routed packets are also blocked after the configuration. Which command resolves the issue?

- A. ip verify unicast source reachable-via any
- B. ip verify unicast source reachable-via rx
- C. ip verify unicast reverse-path
- D. ip verify unicast reverse-path any

**Correct Answer:** A  
**Section:** (none)  
**Explanation**

**Explanation/Reference:**  
uRPF loose mode removes the match requirement on the specific ingress interface so that asymmetric routing can be supported.

#### QUESTION 553

How do devices operate in MPLS L3VPN topology?

- A. CE routers connect to the provider network and perform LSP functionality
- B. P and associated PE routers with IGP populate the VRF table in different VPNs
- C. P routers support PE to PE VPN tunnel without LSP functionality
- D. P routers provide connectivity between PE devices with MPLS switching

**Correct Answer:** D  
**Section:** (none)  
**Explanation**

**Explanation/Reference:**

#### QUESTION 554

```

ip access-list extended CoPP-ICMP
 permit icmp any any echo
!
ip access-list extended CoPP-BGP
 permit tcp any eq bgp any established
!
ip access-list extended CoPP-EIGRP
 permit eigrp any host 224.0.0.10
!
Class-map match-all CoPP-CLASS
 match access-group name CoPP-ICMP
 match access-group name CoPP-BGP
 match access-group name CoPP-EIGRP
!

```

A CoPP policy is implemented to allow specific control traffic, but the traffic is not matching as expected and is getting unexpected behavior of control traffic. Which action resolves the issue?

- A. Use match-any instruction in class-map.
- B. Create a separate class map against each ACL.
- C. Create a separate class map for ICMP traffic.
- D. Use default-class to match ICMP traffic.

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The class-map "CoPP-CLASS" configured with multiple ACLs. Since a packet cannot be a packet of the BGP, ICMP and EIGRP protocols at the same time, the configured class map will never match any packet. Therefore, "match-any" should be configured for the class map (instead of the default "match-all")

#### QUESTION 555

Which tag is used by the PE router to forward the packet to the correct customer?

- A. Extended-community
- B. VNI
- C. RT
- D. RD

**Correct Answer:** B

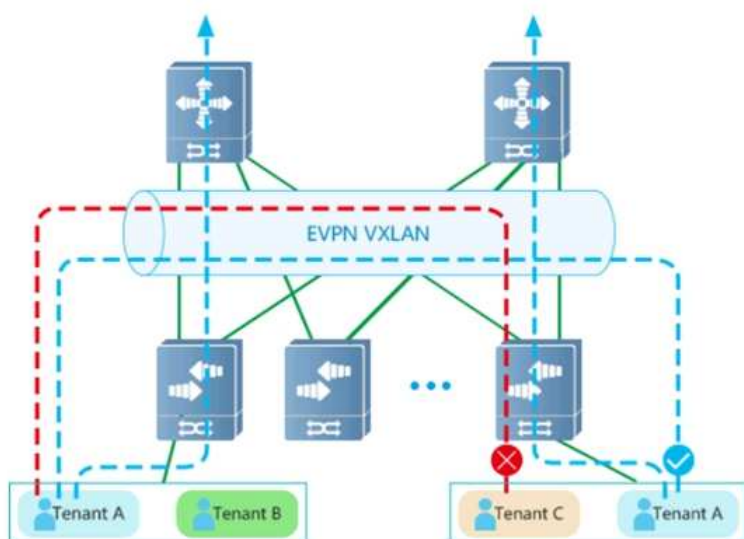
**Section:** (none)

**Explanation**

**Explanation/Reference:**

For all other choices, those tags are used to mark / identify BGP routes. Hence, those tags are not used directly for the forwarding of packet to the correct customer.

VNI (or VNID) can be used to provide multitenancy since it can provide isolation so that hosts of a subscriber of a Provider Service are isolated from hosts of other subscribers. Hence, this tag can be used to forward the packet to the correct customer. However, you have to assume that VXLAN (e.g. EVPN VXLAN) is being used in the provider network.



#### QUESTION 556

A network engineer must configure an EIGRP stub router at a site that advertises only connected and summary routes. Which configuration performs this task?

- A. router eigrp 100  
eigrp stub connected
- B. router eigrp 100  
eigrp stub
- C. router eigrp 100  
eigrp stub summary
- D. router eigrp 100  
eigrp stub redistribute

**Correct Answer:** B

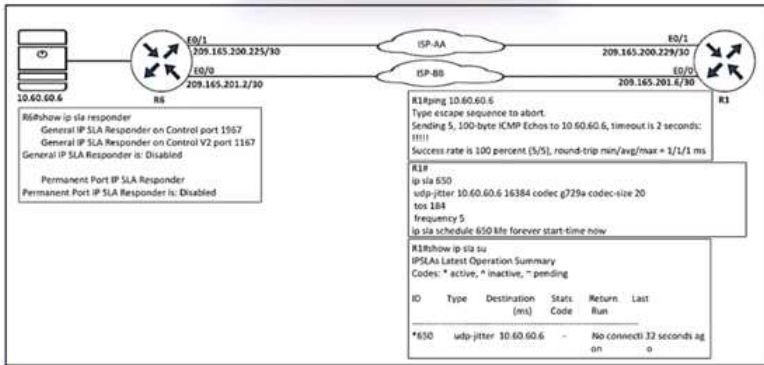
**Section:** (none)

**Explanation**

**Explanation/Reference:**

With "eigrp stub" command alone without additional options, the router advertises connected and summary routes by default.

#### QUESTION 557



Which configuration resolves the IP SLA issue from R1 to the server?

- A. R6(config)#ip sla responder
- B. R6(config)#ip sla 650  
R6(config-ip-sla)#udp-jitter 10.60.60.6
- C. R6(config)#ip sla responder udp-echo ipaddress 10.60.60.6 po 5000
- D. R6(config)#ip sla schedule 10 life forever start-time now

**Correct Answer: A**

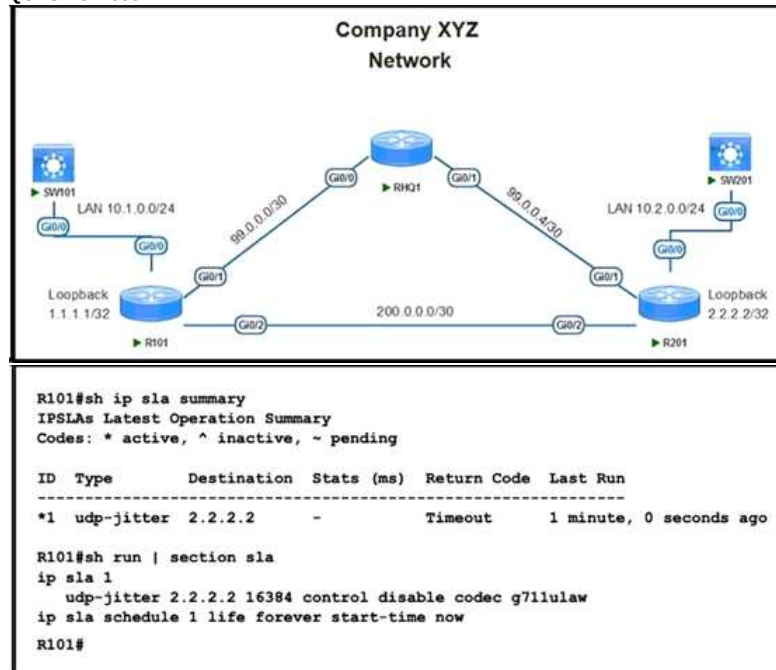
**Section: (none)**

**Explanation**

**Explanation/Reference:**

Since the output of "show ip sla summary" in R1 shows "No connection" in column "Return Code", it means that there is no response to the control message sent by R1. Therefore, you need to setup an IP SLA Responder in R6 using the "ip sla responder" command.

#### QUESTION 558



Jitter on the link between R101 and R201 was tested for voice traffic over port 16384 without the control communication on port 1967. Which command enables R201 to receive RTT for the configured IP SLA?

- A. ip sla responder tcp-connect port 1967
- B. ip sla responder udp-echo ipaddress 2.2.2.2 port 16384
- C. ip sla responder udp-echo ipaddress 1.1.1.1 port 16384
- D. ip sla responder auto-register 1.1.1.1

**Correct Answer: B**

**Section: (none)**

**Explanation**

**Explanation/Reference:**

Since control is disabled, you can setup "ip sla responder <specifying address and port info>" so that the router will response to IP SLA packets targeted to the specified IP address and port.

In this question, R101's IP SLA is setup for sending udp-echo to "2.2.2.2" and port 16384 with control disable. Hence, you can setup IP SLA responder with that IP address and port number in R201.

#### QUESTION 559

| Prefix             | Next Hop     | Interface          |
|--------------------|--------------|--------------------|
| 0.0.0.0/0          | no route     |                    |
| 0.0.0.0/8          | drop         |                    |
| 0.0.0.0/32         | receive      |                    |
| 127.0.0.0/8        | drop         |                    |
| 172.16.1.0/30      | 172.16.3.254 | GigabitEthernet0/2 |
|                    | 172.16.4.254 | GigabitEthernet0/3 |
| 172.16.3.252/30    | attached     | GigabitEthernet0/2 |
| 172.16.3.252/32    | receive      | GigabitEthernet0/2 |
| 172.16.3.253/32    | receive      | GigabitEthernet0/2 |
| 172.16.3.254/32    | attached     | GigabitEthernet0/2 |
| 172.16.3.255/32    | receive      | GigabitEthernet0/2 |
| 172.16.4.252/30    | attached     | GigabitEthernet0/3 |
| 172.16.4.252/32    | receive      | GigabitEthernet0/3 |
| 172.16.4.253/32    | receive      | GigabitEthernet0/3 |
| 172.16.4.254/32    | attached     | GigabitEthernet0/3 |
| 172.16.4.255/32    | receive      | GigabitEthernet0/3 |
| 172.16.222.254/32  | 172.16.4.254 | GigabitEthernet0/3 |
| 192.168.100.0/24   | 172.16.3.254 | GigabitEthernet0/2 |
| 192.168.200.0/24   | 172.16.3.254 | GigabitEthernet0/2 |
| 192.168.222.0/24   | 172.16.4.254 | GigabitEthernet0/3 |
| 224.0.0.0/4        | drop         |                    |
| 224.0.0.0/24       | receive      |                    |
| Prefix             | Next Hop     | Interface          |
| 240.0.0.0/4        | drop         |                    |
| 255.255.255.255/32 | receive      |                    |

An engineer recently implemented uRPF by configuring the ip verify unicast source reachable-via rx command on interface gi0/3. The engineer noticed right after implementing uRPF that an inbound packet on the gi0/3 interface with a source address of 172.16.3.251 was dropped. Which action resolves the issue?

- A. Configure uRPF loose mode to forward the packet.
- B. Permit the 172.16.3.251 in the inbound ACL on interface gi0/3.
- C. Remove inbound ACL from the interface gi0/3 to allow 172.16.3.251.
- D. Configure uRPF strict mode to forward the packet.

**Correct Answer:** A

**Section:** (none)

**Explanation**

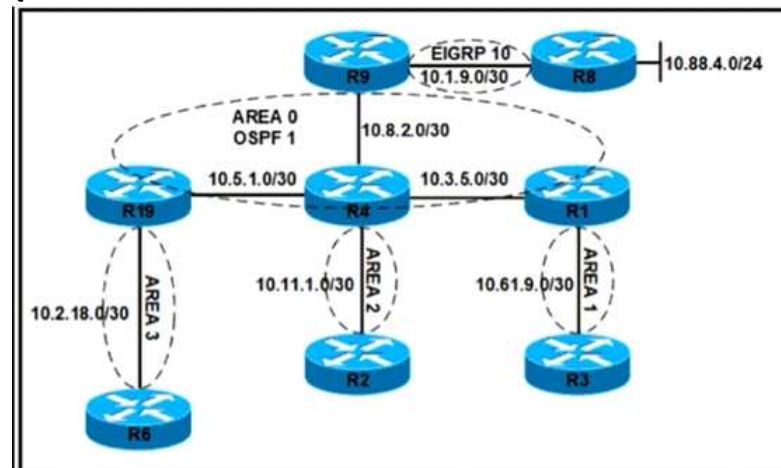
**Explanation/Reference:**

Among all the choices, only configuring uRPF loose mode may solve issues similar to that mentioned in the question. Hence it is selected as the suggested answer.

**Remarks :**

From the output, the corresponding interface for the network 172.16.3.252/30 is g0/2. However, packets with source 172.16.3.251 are not matched to that network or any other prefix shown in the output. Therefore, configuring uRPF loose mode actually cannot resolve the issue in this question.

#### QUESTION 560



An engineer recently modified the configuration for area 3 to accept only type 1, 2, and 3 LSAs. Immediately after the changes, users connected to router R6 began to report connectivity issues. Which configuration restores connectivity to R6 and meets the requirement?

- A. R19#  
 router ospf 1  
 area 3 stub no-summary  
 network 10.2.18.0 0.0.0.3 area 3  
 network 10.5.1.0 0.0.0.3 area 0
- R6#  
 router ospf 1  
 network 10.2.18.0 0.0.0.3 area 3  
 network 10.6.6.255 0.0.0.0 area 3

- B. R19#  
router ospf 1  
area 3 stub  
network 10.2.18.0 0.0.0.3 area 3  
network 10.5.1.0 0.0.0.3 area 0
- R6#  
router ospf 1  
network 10.2.18.0 0.0.0.3 area 3  
network 10.6.6.255 0.0.0.0 area 3
- C. R19#  
router ospf 1  
area 3 nssa  
network 10.2.18.0 0.0.0.3 area 3  
network 10.5.1.0 0.0.0.3 area 0
- R6#  
router ospf 1  
area 3 nssa  
network 10.2.18.0 0.0.0.3 area 3  
network 10.6.6.255 0.0.0.0 area 3
- D. R19#  
router ospf 1  
area 3 stub  
network 10.2.18.0 0.0.0.3 area 3  
network 10.5.1.0 0.0.0.3 area 0
- R6#  
router ospf 1  
router-id 10.6.6.255  
area 3 stub  
network 10.2.18.0 0.0.0.3 area 3  
network 10.6.6.255 0.0.0.0 area 3

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Only D is the proper configuration in which:

- Both R6 and R19 must be configured with the same area types for area 3.
- Area 3 is configured as stub area that allows only type 1, 2, and 3 LSAs as required in the question.

Remarks:

Although a stub area only advertises type 1, 2, and 3 LSAs, a default route is also generated so that hosts connected to R6 can still access external networks connected by R8 through the default route.

**QUESTION 561**

Which routing protocol is used by the PE router to advertise routes to a CE router without redistribution or static after removing the RD tag from the P router?

- A. OSPF
- B. BGP IPv4
- C. IS-IS
- D. MP-BGP

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Since MP-BGP is used for exchanging routes between PE routers, BGP routes learnt can be advertise to CE router directly without the need to perform redistribution.

**QUESTION 562**

### Configuration

```
flow exporter Flow-to-collector
destination 192.168.100.17 vrf Mgmt-intf
transport udp 2601
export-protocol netflow-v5
!
flow monitor My-netflow
exporter Flow-to-collector
record netflow ipv4 original-input
!
! and the management-interface is configured as follows:
interface GigabitEthernet0
description Management-Interface
vrf forwarding Mgmt-intf
ip address 192.168.100.50 255.255.255.0
negotiation auto

router#sh flow exporter statis
Flow Exporter Flow-to-collector:
 Packet send statistics (last cleared 1w4d ago):
 Successfully sent: 0 (0 bytes)
 Reason not
given: 8696868 (11473678976 bytes)
 Client send statistics:
 Client: Flow Monitor OeKB-netflow
 Records added: 256783312
 - failed to send: 256783312
 Bytes added: 2783766384
 - failed to send: 2783766384
router#
```

A network administrator configured NetFlow data but the data is not visible at the NetFlow collector. Which configuration allows the router to send the records?

- A. Configure a different interface to send the records.
- B. Configure the management interface in the global routing table to send the records.
- C. Configure the NetFlow collector to listen at export-protocol netflow-v5.
- D. Rectify NetFlow collector reachability from the management interface.

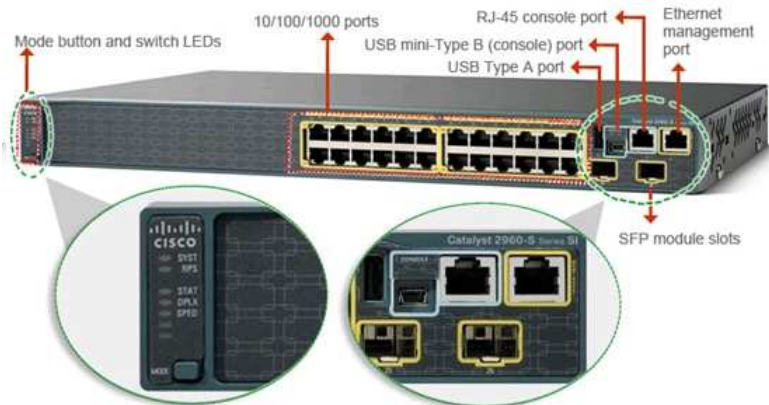
**Correct Answer: A**

**Section: (none)**

**Explanation**

#### Explanation/Reference:

Ethernet Management Port (the one in top right corner in the following diagram) may exist in some Cisco devices. It is usually named as g0 or g0/0.

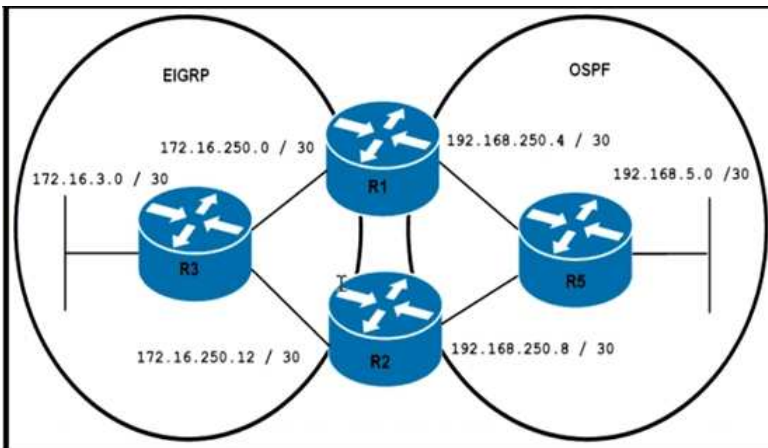


Many features are not supported by this management port. For example, Flexible NetFlow export is not supported on the Ethernet management port. The interface GigabitEthernet0 in this question is probably a management port.

**QUESTION 563**



|                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> R1#show running-config   begin router eigrp router eigrp 100  network 172.16.250.0 0.0.0.255  redistribute ospf 1 metric 1 1 1 1 1 ! router ospf 1  redistribute eigrp 100 subnets  network 192.168.250.0 0.0.0.255 area 0 </pre>                | <pre> R5#traceroute 172.16.3.1 Type escape sequence to abort. Tracing the route to 172.16.3.1 VRF info: (vrf in name/id, vrf out name/id)  1 192.168.250.9 66 msec  192.168.250.6 6 msec  192.168.250.9 8 msec  2 172.16.250.2 33 msec  172.16.250.14 88 msec  172.16.250.2 11 msec R5# </pre> |
| <pre> R2#show runn   begin router eigrp router eigrp 100  network 172.16.250.0 0.0.0.255  redistribute ospf 1 metric 1 1 1 1 1 ! router ospf 1  redistribute eigrp 100 subnets  network 192.168.250.0 0.0.0.255 area 0 ! ip forward-protocol nd </pre> |                                                                                                                                                                                                                                                                                                |



An engineer is troubleshooting a routing loop on the network to reach the 172.16.3.0/16 from the OSPF domain. Which configuration on router R1 resolves the issue?

- A. 

```

router ospf 1
 redistribute eigrp 100 subnets route-map LOOPFILT
!
route-map LOOPFILT deny 10
 match ip address 15
!
route-map LOOPFILT permit 20
!
access-list 15 permit 172.16.0.0 0.0.255.255

```
- B. 

```

router eigrp 100
 redistribute ospf 1 metric 1 1 1 1 1 route-map LOOPFILT
!
route-map LOOPFILT deny 10
 match ip address 15
!
access-list 15 permit 172.16.0.0 0.0.255.255

```
- C. 

```

router ospf 1
 redistribute eigrp 100 route-map LOOPFILT
!
route-map LOOPFILT deny 10
 match ip address 15
!
access-list 15 permit 172.16.0.0 0.0.255.255

```
- D. 

```

router eigrp 100
 redistribute ospf 1 metric 1 1 1 1 1 route-map LOOPFILT
!
route-map LOOPFILT deny 10
 match ip address 15
!
route-map LOOPFILT permit 20
!
access-list 15 permit 172.16.0.0 0.0.255.255

```

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

For networks running both OSPF and EIGRP, routing loops occurs when an External EIGRP route (as route with AD 170) is redistributed into OSPF (as route with AD 110) in one router and that OSPF route is then redistributed back to EIGRP (as route with AD 170) in another router. This external EIGRP route redistributed from OSPF may replace the original external EIGRP in some routers causing routing loops.

In this case, you need to prevent the above route from being redistributed back from OSPF in the EIGRP configuration. Hence, D is the suggested answer.

Note that the other choice that block redistribution from OSPF into EIGRP is wrong since it is missing the permit all at the end of the route map (seq 20 of the route

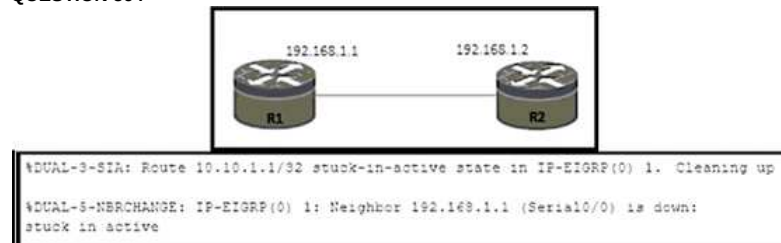
map). Hence, other normal OSPF route cannot be redistributed into EIGRP.

#### Remarks:

There are various problems in this question:

- The command in the choices block redistribution of the network 172.16.0.0/16 while the question mentions 172.16.3.0/16 (this is an invalid) not reachable but the diagram shows 172.16.3.0/30.
- From the output of the "traceroute", there is no sign of a routing loop. It is because if there is a routing loop, the output will show many hops and it ends only after it reaches the default hop count limit (30 by default).
- Routing loop only occurs if the original route 172.16.0.0/16 is redistributed into EIGRP as external route. However, the question does not mention how the network is originally advertised in EIGRP.
- Even if the original route 172.16.0.0/16 is redistributed into EIGRP as external route by R3, no routing loop will occur in the network topology shown in the diagram. Routing loop only occurs if the external route is learnt by R3 e.g. the route is redistributed from an additional EIGRP router connected on the left of R3.

#### QUESTION 564



An engineer notices a connectivity problem between routers R1 and R2. The frequency of this problem is high during peak business hours. Which action resolves the issue?

- A. Increase the available bandwidth between R1 and R2.
- B. Decrease the EIGRP keepalive and hold down timers on R1 and R2.
- C. Increase the MTU on the interfaces that connect R1 and R2.
- D. Set static EIGRP neighborship between R1 and R2.

**Correct Answer: A**

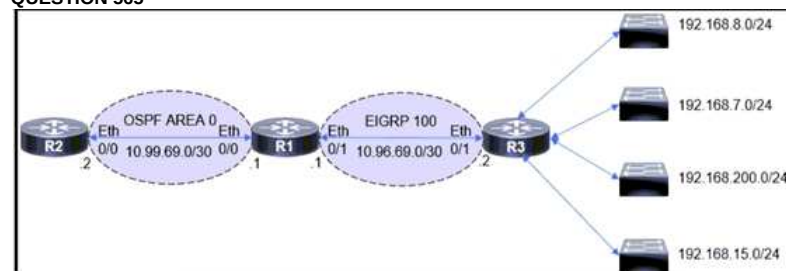
**Section: (none)**

**Explanation**

#### Explanation/Reference:

Stuck in active occurs if the router does not receive a reply to the outstanding EIGRP query that it has sent. When the network is busy, the reply may have been dropped due to network congestion. Hence, increasing bandwidth is probably a solution.

#### QUESTION 565



```

R1#show route-map
route-map FROM->EIGRP, permit, sequence 10
 Match clauses:
 ip address (access-lists): 10
 Set clauses:
 Policy routing matches: 0 packets, 0 bytes
R1#show run | sec router
router eigrp 100
 network 10.96.69.0 0.0.0.3
 no auto-summary
 eigrp router-id 1.1.1.1
router ospf 100
 router-id 1.1.1.1
 log-adjacency-changes
 redistribute eigrp 100 subnets route-map FROM->EIGRP
 network 10.99.69.0 0.0.0.3 area 0
R1#show ip access-list
Standard IP access list 10
 10 permit 192.168.16.0, wildcard bits 0.0.3.255
 11 permit 192.168.0.0, wildcard bits 0.0.7.255
 20 deny any

```

The engineer configured route redistribution in the network but soon received reports that R2 cannot access 192.168.7.0/24 and 192.168.15.0/24 subnets. Which configuration resolves the issue?

- A.
- ```

R1(config)#ip access-list standard 10
R1(config-std-nacl)#no 10 permit
R1(config-std-nacl)#no 11 permit
R1(config-std-nacl)#10 permit 192.168.0.0 0.0.3.255
R1(config-std-nacl)#11 permit 192.168.8.0 0.0.3.255

```
- B.
- ```

R1(config)#ip access-list standard 10
R1(config-std-nacl)#no 10 permit
R1(config-std-nacl)#no 11 permit
R1(config-std-nacl)#10 permit 192.168.4.0 0.0.3.255
R1(config-std-nacl)#11 permit 192.168.12.0 0.0.3.255

```
- C.
- ```

R1(config)#ip access-list standard 10
R1(config-std-nacl)#no 10 permit
R1(config-std-nacl)#no 11 permit
R1(config-std-nacl)#10 permit 192.168.0.0 0.0.3.255
R1(config-std-nacl)#11 permit 192.168.8.0 0.0.7.255

```
- D.
- ```

R1(config)#ip access-list standard 10
R1(config-std-nacl)#no 10 permit
R1(config-std-nacl)#no 11 permit
R1(config-std-nacl)#10 permit 192.168.0.0 0.0.7.255
R1(config-std-nacl)#11 permit 192.168.8.0 0.0.3.255

```

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

In the question, a route map is configured to allow only specific routes to be redistributed from EIGRP to OSPF. In order to allow 192.168.7.0/24 and 192.168.15.0/24, the two routes must be matched by the access list being used by the route map.

Among the choices, the one with the followings in the access list meets the requirement.  
 192.168.4.0 0.0.3.255 à 192.168.4.X to 192.168.7.X

192.168.12.0 0.0.3.255 à 192.168.12.X to 192.168.15.X  
Other choices can match only one of the two routes.

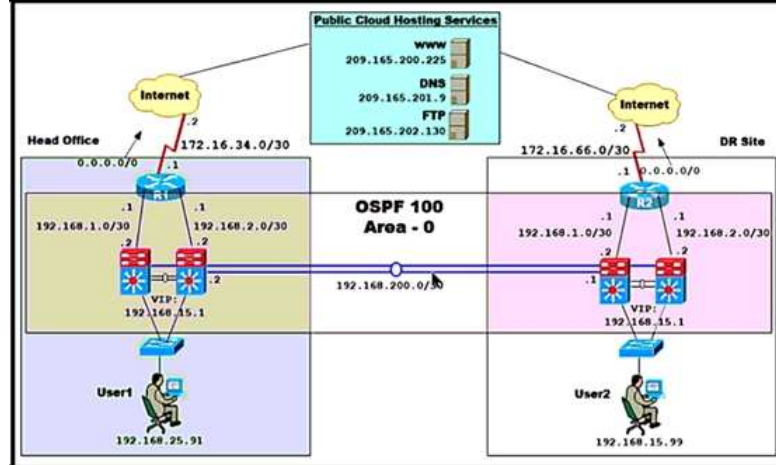
Remarks:

There is something wrong in the question. In the original access list, there is a rule with the following:

192.168.0.0 0.0.7.255 à 192.168.0.X to 192.168.7.X

Hence, the route 192.168.7.0/24 should be redistributed and there should be no problem in accessing the network 192.168.7.0/24.

#### QUESTION 566



Router R1 (head office) is not always preferred for all users to access hosting services R1 should be primary, and the DR site should only be used if router R1 or its uplink fails.

Which configuration resolves the issue?

- A.
- ```
R1
router ospf 100
default-information originate
redistribute static subnets metric-type 1

R2
router ospf 100
default-information originate
redistribute static subnets metric-type 2
```
- B.
- ```
R1
router ospf 100
default auto-cost reference-bandwidth 10000

R2
router ospf 100
auto-cost reference-bandwidth 1000
```
- C.
- ```
R1
router ospf 100
default-information originate
summary-address 192.168.0.0 255.255.0.0 tag 100

R2
router ospf 100
default-information originate
summary-address 192.168.0.0 255.255.0.0 tag 200
```
- D.
- ```
R1
router ospf 100
default-metric 10000

R2
router ospf 100
default-metric 1000
```

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

The setting up of "auto-cost reference-bandwidth" in a router or the adding of tag to OSPF route does not affect metric of the OSPF routes learnt by other routers and therefore cannot solve the problem.

The setting of default-metric can affect the cost of E2 external routes redistributed by "redistribute" command but cannot change the cost the OSPF E2 external default route generated by "default-information originate". The latter always has a cost of 1.

The best way to solve the problem in this question is to set metric or metric-type in the command "default-information originate".

```
R1(config-router)#default-information originate ?
 always Always advertise default route
 metric OSPF default metric
 metric-type OSPF metric type for default routes
 route-map Route-map reference
 <cr>
```

However, there is no choice that change the options of that command.

The remaining choice i.e. the one with "default-information originate" and "redistribute static ..." is chosen as suggested answer but it cannot solve the problem directly. In order for this choice to solve the problem, a static route e.g. "209.165.200.0/22" must be added in both routers. Then since E1 route for "209.165.200.0/22" advertised by R1 is preferred than E2 route for "209.165.200.0/22" advertised by R2, users will send traffic to hosting services through R1 instead of R2.

Note that "redistribute static" in OSPF cannot redistribute static default route. Therefore, the default route generated by "default-information originate" is not affected and both routers will advertise a E2 default route. Hence, when users are accessing Internet other than the hosting services, R1 will NOT be the preferred router.

#### QUESTION 567

Which control plane process allows the MPLS forwarding state to recover when a secondary RP takes over from a failed primary RP?

- A. LDP uses SSO to recover from disruption in control plane service.
- B. MP-BGP uses control plane services for label prefix bindings in the MPLS forwarding table.
- C. FEC uses a control plane service to distribute information between primary and secondary processors.
- D. LSP uses NSF to recover from disruption in control plane service.

**Correct Answer:** D

**Section:** (none)

**Explanation**

#### Explanation/Reference:

LDP NSF allows an RP to recover from disruption in service without losing its MPLS forwarding state.

An LDP Restart occurs after an SSO event interrupts LDP communication with all LDP neighbors. If the RPs are configured with LDP NSF, the backup RP retains the MPLS forwarding state and reestablishes communication with the LDP neighbors. Then the RP ensures that the MPLS forwarding state is recovered.

#### QUESTION 568

What is a MPLS PHP label operation?

- A. Downstream node signals to remove the label.
- B. It uses implicit-NULL for traffic congestion from source to destination forwarding.
- C. PE removes the outer label before sending to the P router.
- D. It improves P router performance by not performing multiple label lookup.

**Correct Answer:** A

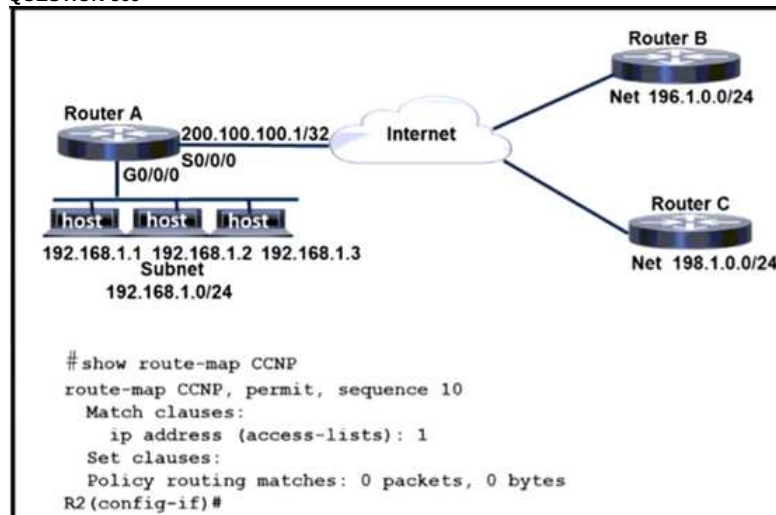
**Section:** (none)

**Explanation**

#### Explanation/Reference:

A downstream PE router can use implicit-NULL to inform the upstream P router telling the P router to remove the MPLS label before forwarding packet to the PE router. This can enhance the performance of the PE router. This operation is the MPLS PHP label operation.

#### QUESTION 569



An engineer configures router A to mark all inside to outside traffic from network 192.168.1.0. except from host 192.168.1.1. with critical IP precedence. The policy did not work as expected. Which configuration resolves the issue?



- A. RouterA(config)#access-list 1 deny host 192.168.1.1  
RouterA(config)#access-list 1 permit any any  
RouterA(config)#route-map CCNP permit 10  
RouterA(config)#match ip address 1  
RouterA(config)#set ip precedence critical  
RouterA(config)#route-map CCNP permit 20  
RouterA(config)#set ip precedence critical  
RouterA(config)# interface g0/0/0  
RouterA(config-if)#ip address 192.168.1.4 255.255.255.0  
RouterA(config-if)#ip policy route-map CCNP
- B. RouterA(config)#access-list 1 deny host 192.168.1.1  
RouterA(config)#route-map CCNP permit 10  
RouterA(config)#match ip address 1  
RouterA(config)#set ip precedence critical  
RouterA(config)#route-map CCNP permit 20  
RouterA(config)# interface g0/0/0  
RouterA(config-if)#ip address 192.168.1.4 255.255.255.0  
RouterA(config-if)#ip policy route-map CCNP
- C. RouterA(config)#access-list 1 deny host 192.168.1.1  
RouterA(config)#access-list 1 permit any any  
RouterA(config)#route-map CCNP deny 10  
RouterA(config)#match ip address 1  
RouterA(config)#set ip precedence critical  
RouterA(config)#route-map CCNP permit 20  
RouterA(config)# interface g0/0/0  
RouterA(config-if)#ip address 192.168.1.4 255.255.255.0  
RouterA(config-if)#ip policy route-map CCNP
- D. RouterA(config)#access-list 1 deny host 192.168.1.1  
RouterA(config)#access-list 1 permit any any  
RouterA(config)#route-map CCNP permit 10  
RouterA(config)#match ip address 1  
RouterA(config)#set ip precedence critical  
RouterA(config)# interface g0/0/0  
RouterA(config-if)#ip address 192.168.1.4 255.255.255.0  
RouterA(config-if)#ip policy route-map CCNP

Correct Answer: D

Section: (none)

Explanation

#### Explanation/Reference:

For configuring policy for an interface, the route-map does not need an empty permit statement at the end for handling traffic not matched by previous statement. The unmatched traffic will be handled / routed normally.

The "permit any" is required as the second rule in the access list so that it matches the traffic (except host 192.168.1.1) that needs to send with critical IP Precedence. Remarks:

Note that in the choices, the rule is wrongly written as "permit any any".

#### QUESTION 570

```

RouterA#sh ip eigrp interfaces gigabitEthernet 0/0/1
EIGRP-IPv4 Interfaces for AS(1)

```

| Interface                                                 | Peers | Un/Reliable | Un/Reliable | SRRT | Mean | Pacing Time | Multicast | Pending |
|-----------------------------------------------------------|-------|-------------|-------------|------|------|-------------|-----------|---------|
| EIGRP-IPv4 VR(CCNP) Address-Family Interfaces for AS(100) |       |             |             |      |      |             |           |         |
| Interface                                                 | Peers | Un/Reliable | Un/Reliable | SRRT | Mean | Pacing Time | Multicast | Pending |
| Gi0/0/1                                                   | 0     | 0/0         | 0/0         | 0    | 0/0  | 50          | 0         | 0       |

RouterA#

EIGRP adjacency between router A and router C is not working as expected. Which two configurations resolve the issue? (Choose two.)



- A. **Router C**  
 router eigrp CCNP  
 address-family ipv4 unicast autonomous-system 100  
 af-interface GigabitEthernet0/0/0  
 hold-time 90  
 exit-af-interface  
 topology base  
 exit-af-topology  
 exit-address-family
- B. **Router A**  
 router eigrp CCNP  
 address-family ipv4 unicast autonomous-system 10  
 af-interface GigabitEthernet0/0/1  
 hello-interval 15  
 hold-time 90  
 exit-af-interface  
 topology base  
 exit-af-topology  
 network 198.18.133.0  
 exit-address-family
- C. **Router C**  
 router eigrp CCNP  
 address-family ipv4 unicast autonomous-system 100  
 topology base  
 exit-af-topology  
 network 198.18.133.0  
 exit-address-family
- D. **Router A**  
 router eigrp CCNP  
 address-family ipv4 unicast autonomous-system 100  
 topology base  
 exit-af-topology  
 network 198.18.133.0  
 exit-address-family
- E. **Router A**  
 router eigrp CCNP  
 address-family ipv4 unicast autonomous-system 100  
 af-interface GigabitEthernet0/0/1  
 hello-interval 15  
 topology base  
 exit-af-topology  
 network 192.18.133.0  
 exit-address-family

Correct Answer: CD

Section: (none)

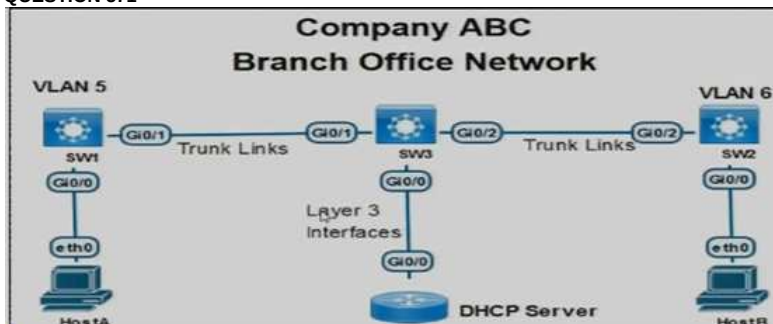
Explanation

Explanation/Reference:

To resolve the EIGRP adjacency issue between Router A and Router C, the following configurations are required:

- Correct Autonomous System: Both routers must be configured with the same EIGRP autonomous system number.
- Network Statements: The network commands must include the correct IP address range to ensure that the interfaces are included in EIGRP.

#### QUESTION 571



```

SW3#sh run
!
interface GigabitEthernet0/0
ip address 1.0.0.1 255.255.255.252
interface Vlan5
ip address 10.5.5.1 255.255.255.0
interface Vlan6
ip address 10.6.6.1 255.255.255.0
end

*Feb 10 00:34:30.978: DHCP_SNOOPING_SW: bridge packet get invalid mat entry:
FFFF.FFFF.FFFF, packet is flooded to ingress VLAN: (5)
*Feb 10 00:34:30.979: DHCP_SNOOPING_SW: bridge packet send packet to cpu port: Vlan5.
*Feb 10 00:36:16.243: DHCP_SNOOPING_SW: bridge packet get invalid mat entry:
FFFF.FFFF.FFFF, packet is flooded to ingress VLAN: (6)
*Feb 10 00:36:16.244: DHCP_SNOOPING_SW: bridge packet send packet to cpu port: Vlan6.

```

HostA and HostB cannot receive IP addresses from the DHCP server. The switches are configured with the DHCP snooping. Which configuration on SW3 resolves the issue?

- A. ip dhcp relay information option-insert
- B. no ip dhcp snooping information option
- C. ip dhcp server use subscriber-id client-id
- D. ip helper-address 1.0.0.2

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 572

Drag and drop the terminology from the left onto the corresponding definitions on the right.

**Select and Place:**

|                                      |                                                                                        |
|--------------------------------------|----------------------------------------------------------------------------------------|
| Multiprotocol Label Switching (MPLS) | set of packets with similar characteristics that might be bound to the same MPLS label |
| Label Distribution Protocol (LDP)    | data-carrying mechanism that is independent of any data link layer protocol            |
| Forwarding Equivalence Class (FEC)   | router that functions as the ingress and/or egress router to the MPLS domain           |
| Provider Edge (PE)                   | route through an MPLS network, defined by a signaling protocol such as LDP or BGP      |
| Label Switched Path (LSP)            | mechanism by which two routers exchange label mapping information                      |

**Correct Answer:**

|  |                                      |
|--|--------------------------------------|
|  | Forwarding Equivalence Class (FEC)   |
|  | Multiprotocol Label Switching (MPLS) |
|  | Provider Edge (PE)                   |
|  | Label Switched Path (LSP)            |
|  | Label Distribution Protocol (LDP)    |

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 573

Which two label distribution methods are used by routers in MPLS? (Choose two)

- A. Downstream unsolicited
- B. LDP session protection message
- C. LDP discovery hello message
- D. Downstream on demand
- E. Targeted hello message

**Correct Answer:** AD

**Section:** (none)

**Explanation**

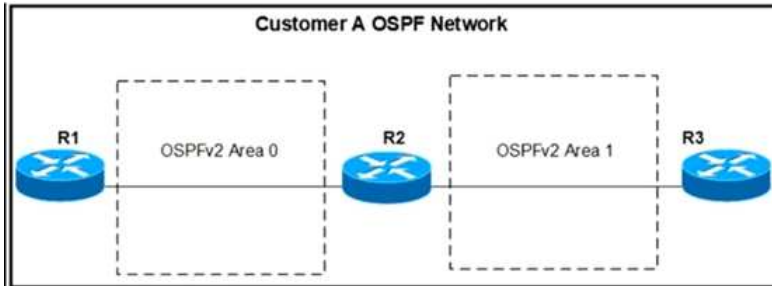
**Explanation/Reference:**

After the LDP TCP connection session is established, the LSRs negotiate the session parameters, including the method of label distribution to be used. Two methods are available:

Downstream Unsolicited: An LSR advertises label mappings to peers without being asked to.

Downstream on Demand: An LSR advertises label mappings to a peer only when the peer asks for them.

#### QUESTION 574



An engineer must ensure that R3 sees only type 1 and 2 LSAs in area 1. Which command must the engineer apply on R2?

- A. area 1 nssa no-summary
- B. area 1 stub
- C. area 1 stub nssa
- D. area 1 stub no-summary

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Use the OSPF router configuration area xx stub no-summary command in order to define a totally stub area. In this area, the routers will NOT receive any OSPF external routes through type 5 LSAs or any summary LSA through type 3 LSAs from the backbone.

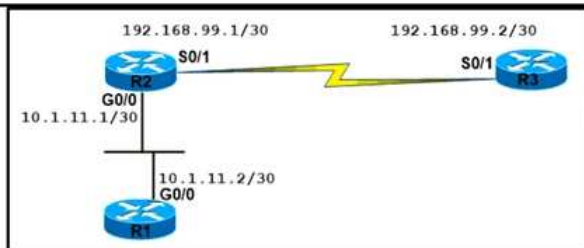
#### QUESTION 575

```

R2# show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
192.168.99.2 1 EXCHANGE/ 00:00:36 192.168.99.1 Serial0/1
router-6#

R3# show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
192.168.99.1 1 EXSTART/ 00:00:33 192.168.99.2 Serial0/1

```



An OSPF neighbor relationship between R2 and R3 is showing stuck in EXCHANGE/EXSTART state. The neighbor is established between R1 and R2. The network engineer can ping from R2 to R3 and vice versa, but the neighbor is still down. Which action resolves the issue?

- A. Administrative "shut then no shut" both router interfaces.
- B. Enable OSPF on the interface, which is required.
- C. Restore the Layer 2/Layer 3 connectivity issue in the ISP network.
- D. Match MTU on both router interfaces or ignore MTU.

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 576

Users report web connectivity problems on the server (10.1.1.10). Which IP SLA configuration captures the failure details through the network to resolve the issue?

A.

```

ip sla 5
icmp-echo 10.1.1.10
frequency 1
timeout 3
ip sla schedule 5 life forever start-time now

```

B.

```

ip sla 5
http get url http://10.1.1.10
frequency 60
timeout 3000
!
ip sla schedule 5 life forever start-time now

```

C.

```

ip sla 5
icmp-echo 10.1.1.10
frequency 60
timeout 3000
ip sla schedule 5 life forever start-time now

```

D.

```

ip sla 5
http get url http://10.1.1.10
frequency 1
timeout 3
!
ip sla schedule 5 life forever start-time now

```

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Since failure details about web connectivity is required, the HTTP operation of IP SLA should be used in the IP SLA. This operation provides the HTTP server response time measurements of three types:

DNS lookup - RTT taken to perform domain name lookup.

TCP Connect - RTT taken to perform a TCP connection to the HTTP server.

HTTP transaction time - RTT taken to send a request and get a response from the HTTP server. The operation retrieves only the home HTML page.

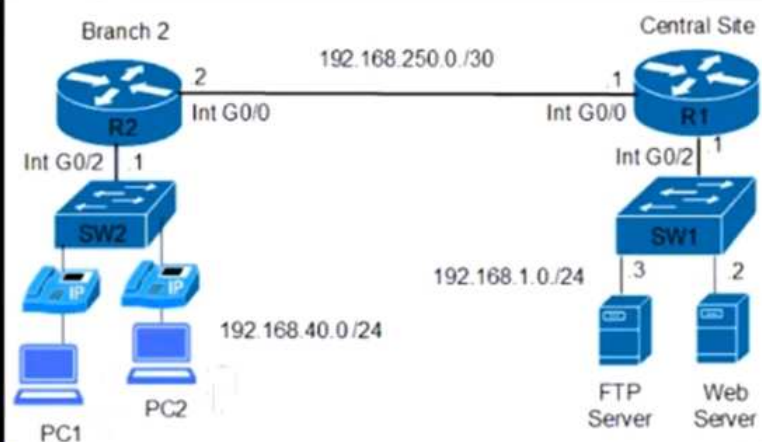
Note that the default and minimum frequency value for an IP SLAs HTTP operation is 60 seconds.

**QUESTION 577**

```

R2
!
interface GigabitEthernet0/0
ip address 192.168.250.2 255.255.255.252
ip helper-address 192.168.250.1
!
interface GigabitEthernet0/2
ip address 192.168.40.1 255.255.255.0

```



```

R1
!
ip dhcp excluded-address 192.168.40.1 192.168.40.10
!
ip dhcp pool Branch 2
network 192.168.40.0 255.255.255.0
default-router 192.168.40.1
dns-server 192.168.1.40

```

Branch 2 hosts cannot receive dynamic IP addresses. Which action resolves the issue?

- A. Configure the ip helper command on the Layer 2 switch SW2 interfaces.
- B. Configure the ip helper command on the Interface GigabitEthernet 0/0 of the DHCP router.
- C. Configure the ip helper command on the Interface GigabitEthernet 0/2 of the DHCP router.
- D. Configure the ip helper command on the interface GigabitEthernet 0/2 of the R2 router.

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The "ip helper" command should be configured on the interface that is directly connected to the concerned DHCP client hosts. In this case, the command should be configured under R2's g0/2 according to the diagram.

#### QUESTION 578

Which feature filters the IPv6 traffic on Layer 2 untrusted ports?

- A. Binding Table Recovery
- B. DHCPv6 Guard
- C. IPv6 Source Guard
- D. IPv6 RA Guard

**Correct Answer:** C

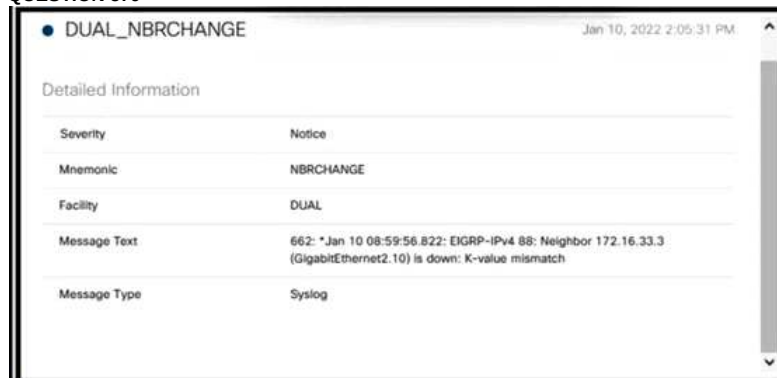
**Section:** (none)

**Explanation**

**Explanation/Reference:**

IPv6 Source Guard (SG) is a security feature that filters the IPv6 traffic on Layer 2 ports that are not trusted.

#### QUESTION 579



EIGRP peering was lost.

Which configuration resolves the issue?

- A.

```
router EIGRP 88
metric weights 1 0 1 0 10
```
- B.

```
router EIGRP 88
metric weights 1 1 1 0 0 0
```
- C.

```
router EIGRP 88
metric weights 0 1 1 0 0 1
```
- D.

```
router EIGRP 88
metric weights 0 1 1 1 0 0
```

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

When configuring EIGRP metric settings under normal EIGRP configuration, 6 parameters are required. (For named mode, you can configure 6 or 7 parameters). There are only two choices with 6 parameters.

However, the 1st parameter is the ToS value but it is always 0.

```
R5(config-router)#metric weights ?
<0-8> Type (Only TOS 0 supported)
```

If you enter any value other than 0 in the 1st parameter as the ToS value, it will change to 0 automatically.

```
R5(config-router)#metric weights 1 1 1 1 1 1
R5(config-router)#do sh run | i metric
metric weights 0 1 1 1 1 1
```

Hence, the choice with 6 parameters with the first parameter being 0 should be the best answer.



## QUESTION 580

|                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> R1#show run   begin router eigrp 100 router eigrp 100  network 172.16.250.0 0.0.0.3  redistribute ospf 10 metric 1 1 1 1 ! router ospf 10  network 192.168.1.0 0.0.0.3 area 0 ! ip forward-protocol nd ! no ip http server </pre> | <pre> R1#show ip route Gateway of last resort is not set  C    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks L    192.168.1.0/30 is directly connected, GigabitEthernet0/1 L    192.168.1.2/32 is directly connected, GigabitEthernet0/1 C    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks L    192.168.2.0/24 is directly connected, Loopback2 L    192.168.2.32/32 is directly connected, Loopback2 C    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks L    192.168.3.0/24 is directly connected, Loopback1 L    192.168.3.17/32 is directly connected, Loopback1 R1# </pre> |
| <pre> R1#traceroute 172.16.2.48 Type escape sequence to abort. Tracing the route to 172.16.2.48 VRF info: (vrf in name/id, vrf out name/id)  0  *  *  *  1  *  *  *  2  *  *  *  3  *  *  * </pre>                                      | <pre> R4#show running-config   begin router eigrp router eigrp 100  network 172.16.2.0 0.0.0.3  network 172.16.2.16 0.0.0.15  network 172.16.2.32 0.0.0.15  redistribute static metric 100 1 1 1 route-map CCNP ! ip forward-protocol nd ! no ip http server no ip http secure-server ip route 172.16.2.48 255.255.255.240 172.16.2.34 ! route-map CCNP permit 10  match ip address 10  set tag 200 ! ! access-list 10 permit 172.16.2.48 0.0.0.15 </pre>                                                                                                                                                        |

An engineer must troubleshoot a connectivity issue impacting the redistribution of the subnet 172.16.2.48 / 28 into the OSPF domain. Which configuration on router R1 advertises this subnet into the OSPF domain?

- A. R1(config)#route-map CCNP permit 10  
R1(config-route-map)#match tag 200  
R1(config-route-map)#exit  
R1(config)#router ospf 10  
R1(config-router)#redistribute eigrp 100 subnets route-map CCNP
- B. R1(config)#route-map CCNP permit 10  
R1(config-route-map)#match route-type internal  
R1(config)#router ospf 10  
R1(config-router)#redistribute eigrp 100 subnets route-map CCNP
- C. R1(config)#route-map CCNP permit 10  
R1(config-route-map)#match route-type level-2  
R1(config)#router ospf 10  
R1(config-router)#redistribute eigrp 100 subnets route-map CCNP
- D. R1(config)#route-map CCNP deny 10  
R1(config-route-map)#match tag 200  
R1(config)#route-map CCNP permit 10  
R1(config)#router ospf 10  
R1(config-router)#redistribute eigrp 100 subnets route-map CCNP

**Correct Answer:** A

**Section:** (none)

**Explanation**

### Explanation/Reference:

According to the running configuration in the EIGRP router R4, the static route 172.16.2.48/28 is redistributed into EIGRP and is marked with the tag number 200. In order for R1 to redistribute this route into OSPF, you can configure a route map that matches this tag number 200 and use this route map for redistribution from EIGRP in the OSPF configuration.

## QUESTION 581

```

*Mar 10 20:13:58.156: AAA/BIND(00000055): Bind i/f
*Mar 10 20:13:58.156: AAA/AUTHN/LOGIN (00000055): Pick method list 'default'
*Mar 10 20:13:58.156: TAC+: Queuing AAA Authentication request 85 for processing
*Mar 10 20:13:58.156: TAC+: (00000055) login timer started 1020 sec timeout
*Mar 10 20:13:58.156: TAC+: processing authentication start request id 85
*Mar 10 20:13:58.156: TAC+: Authentication start packet created for 85()
*Mar 10 20:13:58.156: TAC+: Using server 10.106.60.182
*Mar 10 20:13:58.156: TAC+: (00000055)/O/NB_WAIT/225FE2DC: Started 5 sec timeout
*Mar 10 20:13:58.156: TAC+: (00000055)/O/NB_WAIT: socket event 2
*Mar 10 20:13:58.156: TAC+: (00000055)/O/NB_WAIT: wrote entire 38 bytes request
*Mar 10 20:13:58.156: TAC+: (00000055)/O/READ: socket event 1
*Mar 10 20:13:58.156: TAC+: (00000055)/O/READ: Would block while reading
*Mar 10 20:13:58.156: TAC+: (00000055)/O/READ: socket event 1
*Mar 10 20:13:58.156: TAC+: (00000055)/O/READ: read entire 12 header bytes (expect 6 bytes data)
*Mar 10 20:13:58.156: TAC+: (00000055)/O/READ: socket event 1
*Mar 10 20:13:58.156: TAC+: (00000055)/O/READ: read entire 18 bytes response
*Mar 10 20:13:58.156: TAC+: (00000055)/O/225FE2DC: Processing the reply packet
*Mar 10 20:13:58.156: TAC+: received bad AUTHN packet: length = 6, expected 43974
*Mar 10 20:13:58.156: TAC+: Invalid AUTHN packet (check keys).

```

An engineer must troubleshoot an issue with the aaa authentication that affected the user's login to router R1. Which command allows the configured user to authenticate?

- A. aaa authentication login default group tacacs+



- B. aaa authentication login default group radius
- C. aaa authentication login default group radius tacacs+
- D. aaa authentication login default group radius local

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

From the output, there is a problem in the key configured for the TACACS+ server. However, there is no choice that change this key.

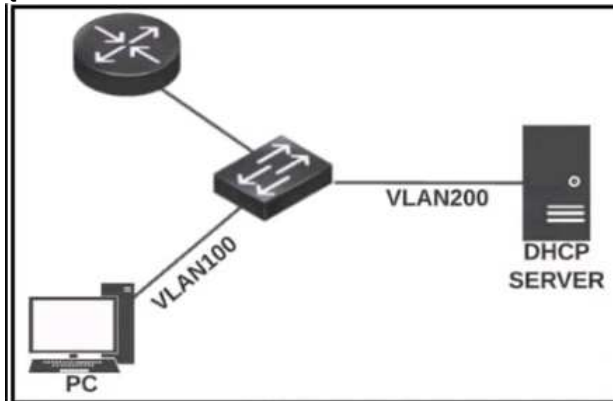
Assume that the user name/password is also configured in the router's configuration, the only possible answer is the choice with "local".

Remarks :

The choices with "radius" in the command may not help in authentication since there is no information whether there is a RADIUS server in the network and whether that RADIUS is properly setup with its IP address and key in the router.

The choice that configures the authentication default list to use "tacacs+" does not help since you must change the TACACS server key to a correct one before the router can use it to authenticate user.

**QUESTION 582**



A PC is configured to obtain an IP address automatically, but it receives an IP address only from the 169.254.0.0 subnet. The DHCP server logs contained no DHCPDISCOVER message from the MAC address of the PC. Which action resolves the issue?

- A. Configure a DHCP reservation on the server for the PC.
- B. Configure an ip helper-address on the router to forward DHCP messages to the server.
- C. Configure DHCP Snooping on the switch to forward DHCP messages to the server.
- D. Configure a static IP address on the PC and exclude it from the DHCP pool.

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Since the PC and the DHCP server are in different VLANs, they cannot send broadcast to each other since they are in different networks. The PC therefore needs a router configured with "ip helper-address" on the same VLAN to help it to forward the DHCP Discover message to the DHCP server.

**QUESTION 583**

Which protocol must be secured with MD-5 authentication across the mpls cloud to prevent hackers from introducing bogus routers?

- A. LDP
- B. LSP
- C. RSVP
- D. MP-BGP

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The Multiprotocol Label Switching (MPLS)—Label Distribution Protocol (LDP) Message Digest 5 (MD5) Global Configuration feature provides enhancements to the LDP implementation of the MD5 password. The MPLS—LDP MD5 Global Configuration feature allows you to enable LDP MD5 globally instead of on a per peer basis. Using this feature, you can set up password requirements for a set of LDP neighbors to help prevent unauthorized peers from establishing LDP sessions and to block spoofed TCP messages.

**QUESTION 584**

An IPv6 network, the network administrator restricted remote access of the core router to a single user. The administrator configured the IPv6 access list and applied it on the WAN interface of the core router. The administrator also wanted to investigate who else is trying to access the router and added a permit host entry with log statement in the access list but could not find any details. Which action does the network administrator take to resolve the issue?

- A. Apply the access list on line vty using the ipv6 access-class out command
- B. Apply the access list on all interfaces using the ipv6 access-class in command
- C. Apply the access list on line vty using the ipv6 access-class in command
- D. Apply the access list on the WAN interface using the ipv6 traffic-filter out command

**Correct Answer:** C

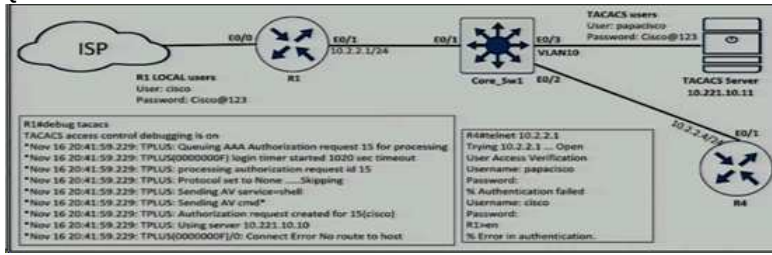
**Section:** (none)

**Explanation**

**Explanation/Reference:**

For logging remote access to the router, you should apply an IPv6 access list to the line vty through the command "ipv6 access-class <your access list> in". Unlike the using of the existing IPv6 access list applied to the WAN interface, the access list configured in line vty can also log remote access from the LAN interface.

# QUESTION 585



A network administrator is trying to switch to the privileged EXEC level on R1 but failed. Which configuration resolves the issue?

- A. tacacs-server enable-password Cisco@123
- B. enable-password Cisco@123
- C. enable password Cisco@123
- D. tacacs server enable-password Cisco@123

**Correct Answer: C**

**Section: (none)**

**Explanation**

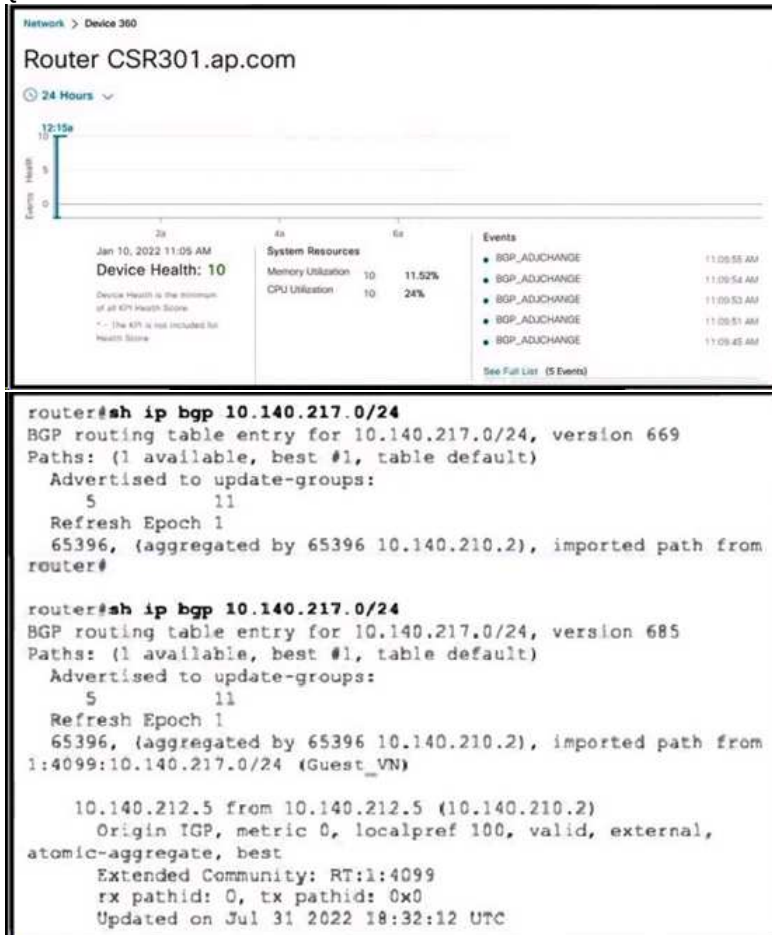
**Explanation/Reference:**

From the debugging messages, the Tacacs server cannot be contacted. Among the choices, only "enable password ..." is a valid command.

**Remarks :**

From the diagram showing the telnet session, it shows that the user trying to login with a username configured in Tacas server but fails. Then he successfully uses the username "cisco" to login and it is probably a local user configured in the router. In order to let the local user not configured with privileged level 15 to switch to privileged EXEC level, you need to configure the enable password / secret in the router.

# QUESTION 586



In Cisco DNA Center, a network engineer identifies that BGP-learned networks are repeatedly withdrawn from peers. Which configuration must the engineer apply to resolve the issue?

- A. router bgp 100  
bgp dampening
- B. router bgp 100  
bgp graceful-restart
- C. route-map Dampening permit 10  
set dampening 15 750 2000 60  
router bgp 100  
neighbor 10.140.212.5 route-map Dampening out
- D. route-map Dampening permit 10

```
set dampening 15 750 2000 60
router bgp 100
neighbor 10.140.212.5 route-map Dampening in
```

**Correct Answer:** D  
**Section:** (none)  
**Explanation**

**Explanation/Reference:**

In order to prevent adverse effect due to the flapping of route, you can configure BGP dampening. You can configure dampening in various ways under the BGP configuration:

- Enable dampening globally with "bgp dampening".
- Enable dampening for specific routes matched by network prefix or AS number through a route map with "bgp dampening route-map XXX". (where "set dampening" is configured in the route map XXX)
- Enable dampening for routes learnt from a specific neighbor through a route map with "neighbor ... route-map XXX in" (where "set dampening" is configured in the route map XXX)

For this question, since the route 10.140.217.0/24 shown in the diagram has been learnt from the neighbor 10.140.212.5 (having BGP router ID 10.140.210.1) and the question mentions problem from that specific neighbor only. Dampening should be configured in a route map applied to the neighbor 10.140.212.5 for the inbound direction.

Remarks :

Some older IOS versions do NOT support the applying of a route map configured with "set dampening" to the neighbor command.

```
R2(config-router)#nei 192.168.0.1 route-map dmap in
% "dmap" used as BGP inbound route-map, set dampening not supported
R2(config-router)#
```

**QUESTION 587**

| Router# show ip interface brief |            |     |        |        |          |
|---------------------------------|------------|-----|--------|--------|----------|
| Interface                       | IP-Address | OK? | Method | Status | Protocol |
| Ethernet0/0                     | 90.0.1.1   | YES | NVRAM  | up     | up       |
| Ethernet1/0                     | 118.8.10.1 | YES | NVRAM  | up     | up       |
| Loopback6                       | 10.1.7.6   | YES | NVRAM  | up     | up       |
| Loopback7                       | 10.1.7.7   | YES | NVRAM  | up     | up       |

| Router# show ip bgp                                                             |              |        |        |        |      |
|---------------------------------------------------------------------------------|--------------|--------|--------|--------|------|
| BGP table version is 10, local router ID is 10.1.7.7                            |              |        |        |        |      |
| Status codes: s suppressed, d damped, h history, * valid, > best, i - internal, |              |        |        |        |      |
| r RIB-failure, S Stale, m multipath, b backup-path, x best-external             |              |        |        |        |      |
| Origin codes: i - IGP, e - EGP, ? - incomplete                                  |              |        |        |        |      |
| Network                                                                         | Next Hop     | Metric | LocPrf | Weight | Path |
| *> 10.0.0.0                                                                     | 0.0.0.0      | 0      | 32768  | ?      |      |
| *> 10.1.7.7/32                                                                  | 0.0.0.0      | 0      | 32768  | i      |      |
| *> 90.0.0.0                                                                     | 0.0.0.0      | 0      | 32768  | ?      |      |
| r> 90.0.1.0/24                                                                  | 100.0.1.254  | 0      |        | 0      | 6 ?  |
| *> 118.0.0.0                                                                    | 0.0.0.0      | 0      | 32768  | ?      |      |
| r>i118.8.10.0/24                                                                | 118.8.10.254 | 0      | 100    | 0      | ?    |

Which action adds the 10.1.7.6/32 route to the BGP table?

- A. Add a static route for the 10.1.7.6/32 network.
- B. Add summary-address 10.1.7.6 255.255.255.0.
- C. Add the network 10.1.7.6 mask 255.255.255.255 backdoor command.
- D. Add the network 10.1.7.6 mask 255.255.255.255 command.

**Correct Answer:** D  
**Section:** (none)  
**Explanation**

**Explanation/Reference:**

Since 10.1.7.6 is a directly connected network of the router, you can just advertise it through BGP by the "network" command.

Remarks :

The command "network ... mask ... backdoor" is a special command. Instead of advertising a connected network, the command matches learnt eBGP routes from its eBGP neighbor and then set an AD of 200 to the learnt route (instead of AD 20 from a normal eBGP route).

For example, you have a private link running OSPF and it connects to the network "X.X.X.0/24" of another AS. In your router, you can configure "network X.X.X.0 mask 255.255.255.0 backdoor" and the eBGP route "X.X.X.0/24" learnt by your router will be assigned with AD 200 which is higher (less preferred) than the OSPF route for that network learnt through the private link. Hence, the OSPF route for that network (using the private link) will be entered into the routing table instead of the eBGP route. Traffic to "X.X.X.0/24" will therefore go through the private link instead of the Internet. However, this command is not applicable to this question.

**QUESTION 588**

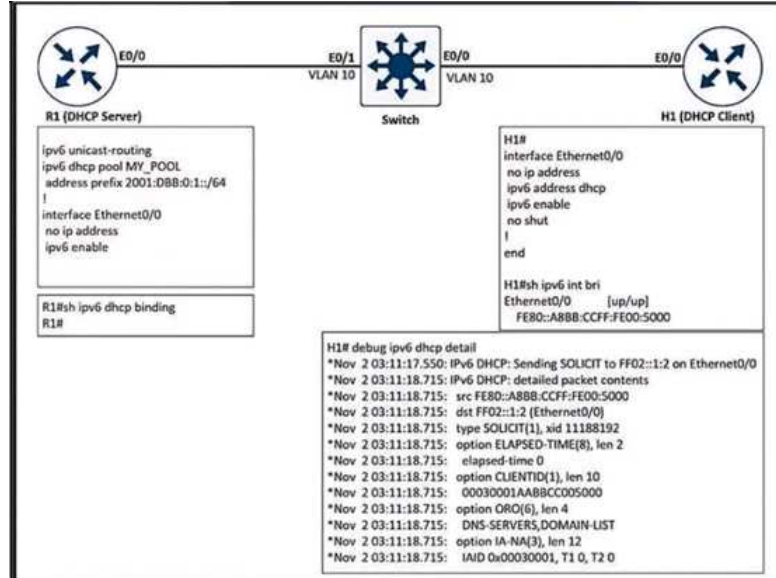
Which router translates the customer routing information into VPNv4 routes to exchange VPNv4 routes with other devices through MP-BGP?

- A. CE
- B. PE
- C. VPNv4 RR
- D. P

**Correct Answer:** B  
**Section:** (none)  
**Explanation**

**Explanation/Reference:**

# QUESTION 589



After the network administrator rebuilds the IPv6 DHCP server, clients are not getting the IPv6 address lease. Which action resolves the issue?

- A. Add ipv6 dhcp server MY\_POOL under the interface ethernet 0/0 on H1.
- B. Remove FE80::A8BB:CCFF:FE00:5000 assigned by the IPv6 DHCP server.
- C. Configure FF02::1:2 to discover all IPv6 DHCP clients.
- D. Add ipv6 dhcp server MY\_POOL under the interface ethernet 0/0 on R1.

**Correct Answer: D**

**Section: (none)**

**Explanation**

**Explanation/Reference:**

In the router acting as IPv6 DHCP server, you have to add the "ipv6 dhcp server ..." command under interface facing the DHCPv6 clients so that the router knows what pool to use for those clients.

# QUESTION 590

An engineer received a ticket about a router that has reloaded. The monitoring system graphs show different traffic patterns between logical and physical interfaces when the router is rebooted. Which action resolves the issue?

- A. Configure the snmp ifindex persist command on the physical interfaces.
- B. Clear the logical interfaces with snmp ifindex clear command.
- C. Trigger a new snmpwalk from the monitoring system to synchronize interface OIDs.
- D. Configure the snmp ifindex persist command globally.

**Correct Answer: A**

**Section: (none)**

**Explanation**

**Explanation/Reference:**

Since the command "snmp ifindex persist" is an interface configuration mode command for preserving index value for a particular interface, it should be configured on an interface (instead of globally).

Remarks :

Normally, the global configuration mode command "snmp-server ifindex persist" is configured for this purpose and should be the best answer for this question.

Alternatively, you can configure the "snmp ifindex persist" command for each physical and logical interfaces that need to be monitored.

However, this is a problem in the available choices of this question. Without a choice for configuring the "snmp ifindex persist" command in the logical interfaces e.g. tunnel interfaces, there will be still be problem in the index values for those logical interfaces.

# QUESTION 591

```

enable secret 5 <password>
username cisco privilege 15 secret 5 <password>
username operator password 7 <password>
line vty 0 4
session-timeout 240
password 7 <password>
transport input telnet

```

The authentication is not working as desired and the user drops into user-exec mode. Which configuration resolves the issue?

- A. aaa new-model
  - aaa authentication common-id default local
  - aaa authorization exec default local
  - !
  - line vty 0 4

```
login authentication default
authorization exec default
```

- B. `aaa new-model`  
`aaa authentication login default local`  
`aaa authorization priv default 15`  
`!`  
`line vty 0 4`  
`login authentication default`  
`authorization exec priv15`
- C. `aaa new-model`  
`aaa authentication login local`  
`aaa authorization exec local`  
`!`  
`line vty 0 4`  
`login authentication local`  
`authorization exec default`
- D. `aaa new-model`  
`aaa authentication login default local`  
`aaa authorization exec default local`  
`!`  
`line vty 0 4`  
`login authentication default`  
`authorization exec default`

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 592

```
R1#debug ip ospf adj
23:42:08.259: OSPF: Send DBD to 2.2.2.2 on Ethernet0/0 seq u opt 0x52 flag 0x7 len 32
23:42:08.339: OSPF: Rcv DBD from 2.2.2.2 on Ethernet0/0 seq 0x836 opt 0x52 flag 0x7 len
32 mtu 1532 state EXSTART

R2#debug ip ospf adj
23:42:08.423: OSPF: Send DBD to 1.1.1.1 on Ethernet0/0 seq 0x836 opt 0x52 flag 0x7 len 32
23:42:08.423: OSPF: First DBD and we are not SLAVE
23:42:08.511: OSPF: Rcv DBD from 1.1.1.1 on Ethernet0/0 seq 0x836 opt 0x52 flag 0x2 len
52 mtu 1500 state EXSTART
```



R1 cannot establish a neighbor relationship with R2. Which action resolves the issue?

- A. Configure the `mtu ignore` command on the interfaces of R1 and R2.
- B. Configure the `ip ospf network point-to-point` command on the interfaces of R1 and R2.
- C. Configure the `neighbor 2.2.2.2` command on R1 under the OSPF process.
- D. Configure the `ip ospf network broadcast` command on the interfaces of R1 and R2.

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 593

```
RouterA#show snmp community
Community name: ILMI
Community Index: ILMI
Community SecurityName: ILMI
storage-type: read-only active

Community name: ccnp
Community Index: ccnp Community SecurityName: ccnp
storage-type: nonvolatile active access-list: 4

RouterA#show ip access-lists
Standard IP access list 4
10 permit 172.16.1.1
20 permit 172.16.2.2
30 permit 172.16.3.3
Extended IP access list BRANCHES
10 permit ip 172.16.4.4 any (95 matches)
20 deny ip any any (95 matches)
```

The SNMP server with IP address 172.16.4.4 cannot access host router A. Which configuration command on router A resolves the issue?

- A. access-list 4 permit 172.16.4.0 0.0.0.3
- B. access-list 4 permit host 172.16.4.4
- C. snmp-server host 172.16.4.4 ccnp
- D. snmp-server community ccnp

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

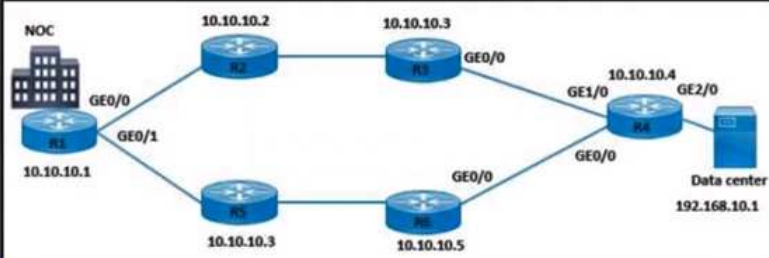
Remarks:

The command "snmp-server host ..." is used by the router to send SNMP Traps notification.

#### QUESTION 594

```
R4#show ip flow export
Flow export v9 is enabled for main cache
Export source and destination details :
VRF ID : Default
Source(1) 10.0.0.10 (GigabitEthernet2/0)
Destination(1) 192.168.10.1 (656)
Version 9 flow records
254 flows exported in 41 udp datagrams
0 flows failed due to lack of export packet
0 export packets were sent up to process level
41 export packets were dropped due to no fib
0 export packets were dropped due to adjacency issues
0 export packets were dropped due to fragmentation failures
0 export packets were dropped due to encapsulation fixup failures

R4#show ip flow interface
GigabitEthernet2/0
ip flow ingress
```



An enterprise operations team must monitor all application server traffic in the data center. The team finds that traffic coming from the hub site from R3 and R6 is monitored successfully, but traffic destined to the application server is not monitored. Which action resolves the issue?

- A. R4(config)#int gigabitEthernet 1/0  
R4(config-if)#ip flow ingress
- B. R3(config)#int gigabitEthernet 0/0  
R3(config-if)#ip flow egress
- C. R4(config)#int gigabitEthernet 2/0  
R4(config-if)#ip flow egress
- D. R1(config)#int gigabitEthernet 0/0  
R1(config-if)#ip flow egress

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The output of "show ip flow interface" in R4 shows that only the command "ip flow ingress" is configured under the interface g2/0. The traffic destined to the application server in the data center is the outbound traffic of R4's g2/0. Therefore, in order to monitor this traffic direction, you also need to configure "ip flow egress" in that interface.

#### QUESTION 595

What must a network architect consider for RTs when planning for a single customer full-mesh VPN in an MPLS Layer 3 network?

- A. Each RT value must be identical to an RD value within the same VPN.
- B. RT must be globally identical within the same VPN.
- C. RT values must be different from the RD values in the same VPN.
- D. RT must be globally unique within the same VPN.

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Since it is a full-mesh VPN, all routes from all sites should be learnt by each other. Hence, the same Route Target for import / export should be configured in all sites using this VPN.

#### QUESTION 596

Which router attaches the VPN label to incoming packets from CE routing?

- A. PE router
- B. P router
- C. CE router



D. core router

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 597

Which router takes an active role between two LDP neighbors when initiating LDP session negotiation and LDP TCP connection establishment?

- A. with one interface in the MPLS backbone
- B. with the larger number of LDP TCP neighbors
- C. with the lowest IP address
- D. with the higher IP address

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

To initiate an LDP session between routers, the routers determine which router will take the active role and which router will take the passive role. The router that takes the active role establishes the LDP TCP connection session and initiates the negotiation of the LDP session parameters. To determine the roles, the two routers compare their transport addresses. The router with the higher IP address takes the active role and establishes the session.

#### QUESTION 598

```
*Mar 10 20:13:58.156: AAA/BIND(00000055): Bind i/f
*Mar 10 20:13:58.156: AAA/AUTHEN/LOGIN (00000055): Pick method list 'default'
*Mar 10 20:13:58.156: TAC+: Queuing AAA Authentication request 85 for processing
*Mar 10 20:13:58.156: TAC+: (00000055) login timer started 1020 sec timeout
*Mar 10 20:13:58.156: TAC+: processing authentication start request id 85
*Mar 10 20:13:58.156: TAC+: Authentication start packet created for 85()
*Mar 10 20:13:58.156: TAC+: Using server 10.106.60.182
*Mar 10 20:13:58.156: TAC+: (00000055)/0/NB_WAIT/225FE2DC: Started 5 sec timeout
*Mar 10 20:13:58.156: TAC+: (00000055)/0/NB_WAIT: socket event 2
*Mar 10 20:13:58.156: TAC+: (00000055)/0/NB_WAIT: wrote entire 38 bytes request
*Mar 10 20:13:58.156: TAC+: (00000055)/0/READ: socket event 1
*Mar 10 20:13:58.156: TAC+: (00000055)/0/READ: Would block while reading
*Mar 10 20:13:58.156: TAC+: (00000055)/0/READ: socket event 1
*Mar 10 20:13:58.156: TAC+: (00000055)/0/READ: read entire 12 header bytes (expect 6 bytes data)
*Mar 10 20:13:58.156: TAC+: (00000055)/0/READ: socket event 1
*Mar 10 20:13:58.156: TAC+: (00000055)/0/READ: read entire 18 bytes response
*Mar 10 20:13:58.156: TAC+: (00000055)/0/225FE2DC: Processing the reply packet
*Mar 10 20:13:58.156: TAC+: received bad AUTHEN packet: length = 6, expected 43974
*Mar 10 20:13:58.156: TAC+: Invalid AUTHEN packet (check keys).
```

An engineer must troubleshoot an issue affecting the communication from router R2 to the TACACS server. Which configuration resolves the issue?

- A. R1(config)#tacacs-server packet maxsize 43974
- B. R2(config)#tacacs-server packet maxsize 43974
- C. R2(config)#tacacs server advrt  
R2(config-server-tacacs)#key xyz123
- D. R1(config)#tacacs server advrt  
R1(config-server-tacacs)#key xyz123

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 599

```
Router# show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited, 0 flushes, 0
overruns, xml disabled, filtering disabled)

No Active Message Discriminator.
No Inactive Message Discriminator.

Console logging: level debugging, 8 messages logged, xml disabled,
filtering disabled
Monitor logging: level debugging, 0 messages logged, xml disabled,
filtering disabled
Buffer logging: level debugging, 8 messages logged, xml disabled,
filtering disabled
Exception Logging: size (5192 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled
```

A network engineer lost remote access to the router due to a network problem. The engineer used the console to access the router and noticed continuous logs on the console terminal. Which configuration limits the number of log messages on the console to critical and higher severity level messages?

- A. term no monitor
- B. no logging console
- C. logging console 2
- D. logging console 5

**Correct Answer:** C

**Section:** (none)

## Explanation

### Explanation/Reference:

Levels include emergencies 0, alerts 1, critical 2, errors 3, warnings 4, notifications 5, informational 6, and debugging 7

### QUESTION 600

What are the two benefits of using BFD? (Choose two.)

- A. supports all routing protocols
- B. supports UDLD failure
- C. forwarding path failure detection
- D. subsecond failure detection
- E. synchronous path determination

**Correct Answer:** CD

**Section:** (none)

### Explanation

### Explanation/Reference:

BFD is a detection protocol designed to provide fast forwarding path failure detection times for all media types, encapsulations, topologies, and routing protocols.

However, the routing protocols supported by BFD depend on the IOS version. Routing protocols such as RIPv1 and IGRP are not supported.

### QUESTION 601



A prefix of 10.1.1.0/31 is assigned for the DWDM circuit between R1 and R2 to connect the two attached VRFs on each router. Which configuration provides the required connectivity on R1?

- A. 

```
interface Ethernet0/0
no ip address
!
interface Ethernet0/0.10
encapsulation dot1Q 10
ip vrf forwarding CCNP
ip address 10.1.1.0 255.255.255.252
!
interface Ethernet0/0.20
encapsulation dot1Q 20
ip vrf forwarding CCNA
ip address 10.1.1.0 255.255.255.252
```
- B. 

```
interface Ethernet0/0
ip address 10.1.1.0 255.255.255.254
!
Interface Loopback 0
ip vrf forwarding CCNP
!
Interface Loopback 1
ip vrf forwarding CCNA
!
interface Tunnel0
tunnel source Loopback 0
tunnel mode gre IP
ip vrf forwarding CCNP
tunnel destination 10.1.1.1
!
interface Tunnel1
tunnel source Loopback 1
tunnel mode gre IP
ip vrf forwarding CCNA
tunnel destination 10.1.1.1
```
- C. 

```
interface Ethernet0/0
no ip address
!
interface Ethernet0/0.10
encapsulation dot1Q 10
ip vrf forwarding CCNP
ip address 10.1.1.0 255.255.255.254
!
interface Ethernet0/0.20
encapsulation dot1Q 20
ip vrf forwarding CCNA
ip address 10.1.1.0 255.255.255.254
```
- D. 

```
interface Ethernet0/0
ip address 10.1.1.0 255.255.255.252
!
Interface Loopback 0
ip vrf forwarding CCNP
!
Interface Loopback 1
ip vrf forwarding CCNA
!
interface Tunnel0
tunnel source Loopback 0
```

```

tunnel mode gre IP
ip vrf forwarding CCNP
tunnel destination 10.1.1.1
!
interface Tunnel1
tunnel source Loopback 1
tunnel mode gre IP
ip vrf forwarding CCNA
tunnel destination 10.1.1.1

```

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

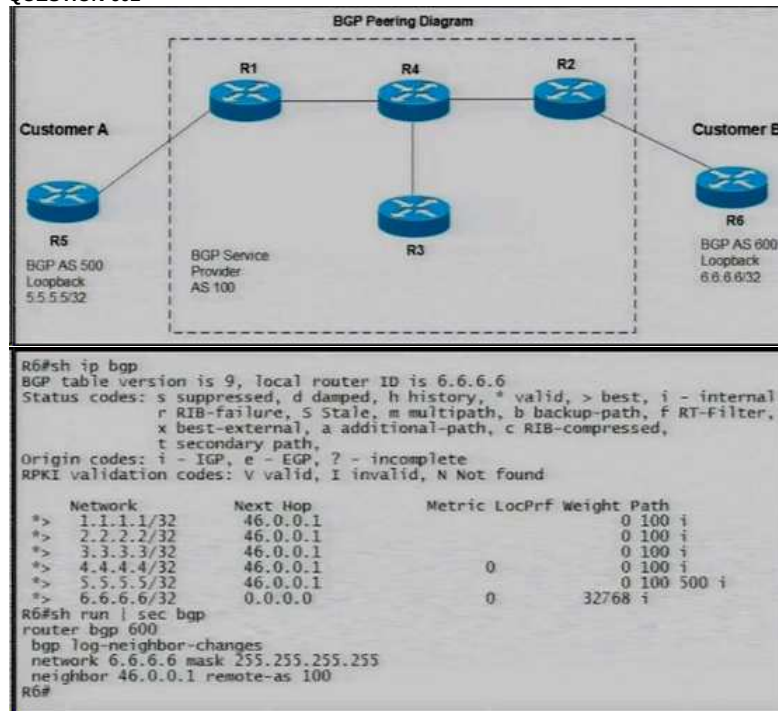
Answer can be found without knowing the details about DWDM. It is because only the configuration commands in one of the choices only are valid since:

- 10.1.1.0/31 uses the subnet mask 255.255.255.254 (not 255.255.255.252). It includes the IP address 10.1.1.0 and 10.1.1.1
- For the choice with tunnel configuration, the configuration is wrong since IP addresses are missing in the loopback interfaces acting as tunnel sources. (Actually, tunnel should not be used in this case).

Remarks:

Dense wavelength division multiplexing (DWDM) employs multiple light wavelengths to transmit signals over a single optical fiber. Today, DWDM is a crucial component of optical networks because it maximizes the use of installed fiber cable and allows new services to be quickly and easily provisioned over existing infrastructure. Flexible add/drop modules allow individual channels to be dropped and inserted along a route. An open-architecture system allows a variety of devices to be connected, including SONET terminals, ATM switches, and IP routers.

#### QUESTION 602



Customer B has decided not to receive any routes from R1 that originated outside the AS 100. Which AS path access list must the engineer choose to meet this requirement?

- ip as-path access-list 1 permit \_100\_
- ip as-path access-list 1 permit \_100\$
- ip as-path access-list 1 permit ^100\$
- ip as-path access-list 1 permit ^10[0-9]\*\$

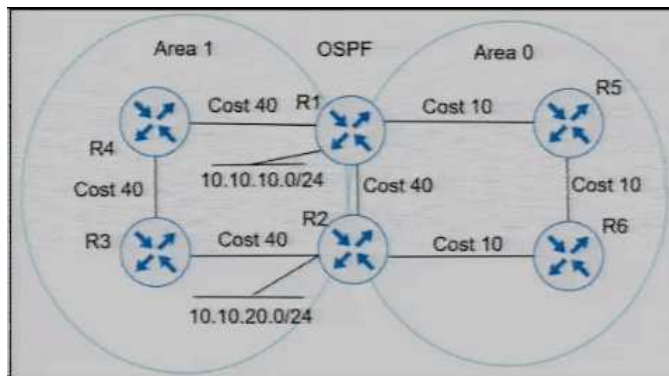
**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 603



Which action ensures that 10.10.10.0/24 reaches 10.10.20.0/24 through the direct link between R1 and R2?

- A. configure OSPF link cost to 1 between R1 and R2
- B. configure R1 and R2 links under area 1
- C. configure R1 and R2 LAN links as nonpassive
- D. configure OSPF path cost to 3 between R1 and R2

**Correct Answer:** B

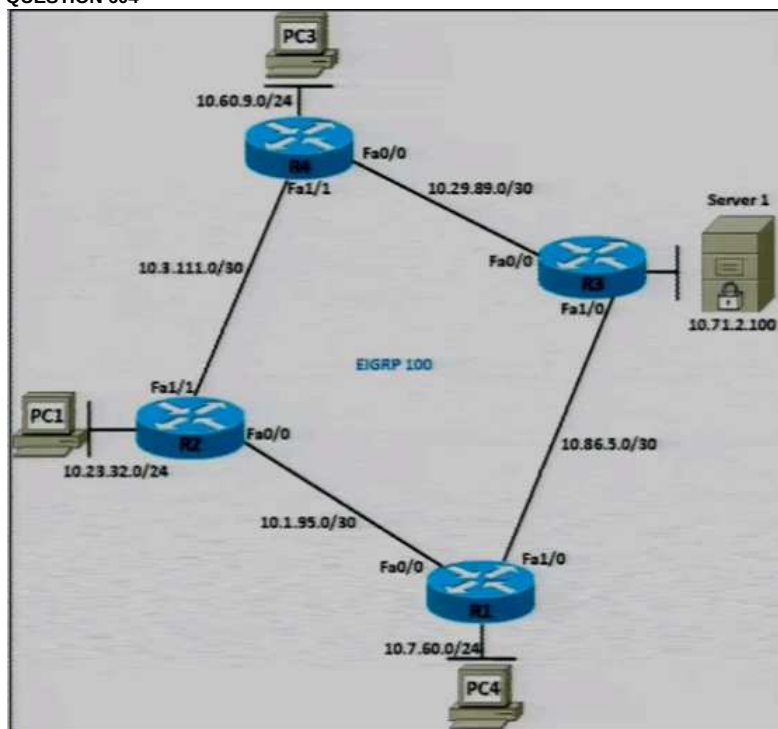
**Section:** (none)

**Explanation**

**Explanation/Reference:**

Since both networks are in Area 1, the link between R1 and R2 should also be in Area 1 since intra-area route is preferred even if the inter-area route has a lower cost.

#### QUESTION 604



The network operation team opened a trouble ticket stating that users connected to R2 and R4 cannot reach server 1. Some partial configuration and show commands are included in the ticket.

```

R4#sh ip route 10.71.2.100
Routing entry for 10.71.2.0/24
Known via "eigrp 100", distance 90, metric 35840, type internal
Redistributing via eigrp 100
Last update from 10.3.111.2 on FastEthernet1/1, 00:01:14 ago
Routing Descriptor Blocks:
* 10.3.111.2, from 10.3.111.2, 00:01:14 ago, via FastEthernet1/1

R2#sh ip route 10.71.2.0
Routing entry for 10.71.2.0/24
Known via "eigrp 100", distance 90, metric 33280, type internal
Redistributing via eigrp 100
Last update from 10.1.95.2 on FastEthernet0/0, 00:12:58 ago
Routing Descriptor Blocks:
* 10.1.95.2, from 10.1.95.2, 00:12:58 ago, via FastEthernet0/0

R2#sh run | i route
ip route 10.71.2.100 255.255.255.255 Null0 name SERVER1

R2#sh run | s eigrp
router eigrp 100
network 10.1.95.0 0.0.0.3
network 10.2.2.2 0.0.0.0
network 10.3.111.0 0.0.0.3
network 10.23.32.0 0.0.0.255

R4#sh run | i route
ip route 10.71.2.0 255.255.254.0 Null0 name SERVER1
!

R4#sh run | s eigrp
router eigrp 100
network 10.3.111.0 0.0.0.3
network 10.4.4.4 0.0.0.0
network 10.29.89.0 0.0.0.3
network 10.60.9.0 0.0.0.255
distance 200 10.29.89.2 0.0.0.0 10
!
access-list 10 permit 10.71.2.0 0.0.0.255

```

Which configuration resolves the issue?

- A. R2#  
router eigrp 100  
distance 90 10.1.95.2 0.0.0.0 10  
!  
no ip route 10.71.2.100 255.255.255.0 Null0 name SERVER1
- B. R4#  
router eigrp 100  
distance 160 10.29.89.2 0.0.0.0 10  
!  
no ip route 10.71.2.0 255.255.254.0 Null0 name SERVER1
- C. R4#  
router eigrp 100  
no distance 200 10.29.89.2 0.0.0.0 10  
!  
no access-list 10 permit 10.71.2.0 0.0.0.255
- D. R2#  
router eigrp 100  
distance 170 10.1.95.2 0.0.0.0 10  
!  
no ip route 10.71.2.100 255.255.255.255 Null0 name SERVER1

**Correct Answer:** CD

**Section:** (none)

**Explanation**

**Explanation/Reference:**

In R2, there is a static null0 host route "10.71.2.100 255.255.255.255". Therefore, packet to Server1 10.71.2.100 received by R2 will be match by this null0 route and discarded.

In R4, the configuration "distance 200 10.29.89.2 0.0.0.0 10" prevents it from entering the route "10.71.2.0/24" from R3 into its routing table. Hence, R4 enter the route learnt from R2 in the routing table. Packets to Server1 10.71.2.100 will be sent by R4 to R2 which will discard the packet as mentioned above.

In order to solve the problem:

- For R4, this can be solved by removing the distance command i.e. "no distance 200 10.29.89.2 0.0.0.0 10".

- For R2, this can be solved by removing the null0 host route i.e. "no ip route 10.71.2.100 255.255.255.255 Null0 name SERVER1". (one of the choices has the wrong subnet mask).

#### QUESTION 605

How does LDP operate in an MPLS network?

- A. when an adjacent LSR receives LDP discovery messages, TCP two-way handshake ensures that the LDP session has unidirectional connectivity.
- B. when topology changes occur such as a router failure, LDP generates peer discovery messages that terminate the LDP session to propagate an LSP change.
- C. LDP notification messages allow LERs to exchange label information to determine the next hops within a particular LSP.
- D. Peer routers establish the LDP session, and the LDP neighbors maintain and terminate the session by exchanging messages.

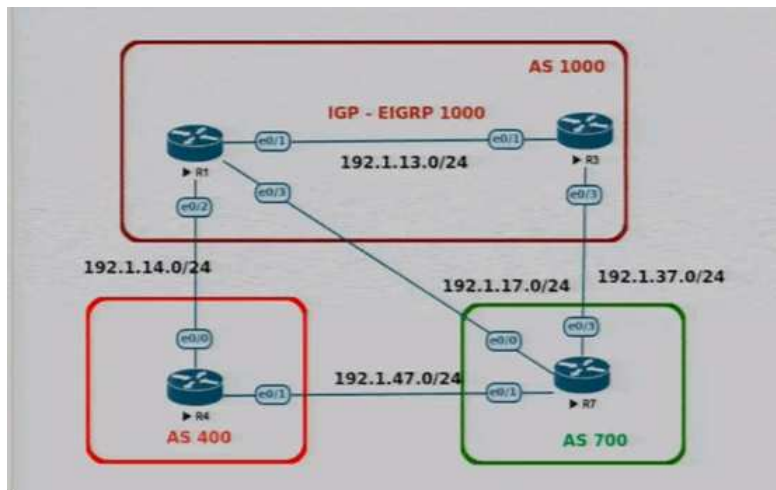
**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 606



```
R1#sh ip bgp
BGP table version is 9, local router ID is 11.11.11.11
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
 r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
 x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

 Network Next Hop Metric LocPrf Weight Path
*> 1.0.0.0 0.0.0.0 0 32768 i
*> 4.0.0.0 192.1.14.4 0 0 400 i
*> 11.11.11.0/24 0.0.0.0 0 32768 i
*> 44.44.44.0/24 192.1.14.4 0 0 400 i
*> 77.77.77.0/24 192.1.17.7 0 0 700 i
*> 77.77.78.0/24 192.1.17.7 0 0 700 ?
*> 77.77.79.0/24 192.1.17.7 0 0 700 ?
*> 77.77.80.0/24 192.1.17.7 0 0 700 ?
```

R1 should have BGP routes as "i". Which configuration resolves the issue?

- A. R7
  - route-map RC permit 10
  - match ip address 1
  - set origin igp
- B. R1
  - route-map RC permit 10
  - match ip address 1
  - set origin igp
- C. R1
  - route-map RC permit 10
  - match ip address 1
  - set metric 1
- D. R7
  - route-map RC permit 10
  - match ip address 1
  - set metric 1

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Although those routes with "?" start with the octet "77", the question does not mention that they are from R7. Moreover, the question does not mention whether these routes should also have origin "i" in other BGP routers.

Hence, in order for R1 to have these routes with origin "i", the safest way is to set them in R1 using "set origin igp" in route map.

**QUESTION 607**

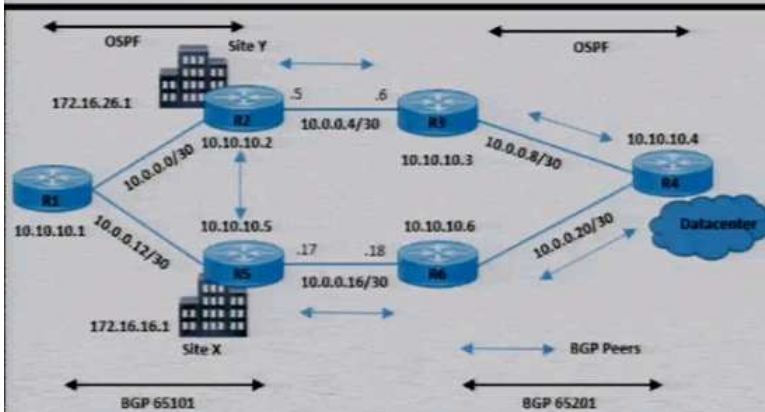


```

R5#
*Sep 19 08:29:51.088: BGP: 10.10.10.2 open active, local address 10.0.0.14
*Sep 19 08:29:51.120: BGP: 10.10.10.2 read request no-op
*Sep 19 08:29:51.124: BGP: 10.10.10.2 open failed: Connection refused by
remote host, open active delayed 12988ms (20000ms max, 60% jitter)

R2#show ip bgp neighbors 10.10.10.5
BGP neighbor is 10.10.10.5, remote AS 65101, internal link
 BGP version 4, remote router ID 0.0.0.0
 BGP state = Active
 Last read 00:01:18, last write 00:01:18, hold time is 15, keepalive
interval is 3 seconds
 Configured hold time is 15, keepalive interval is 3 seconds
 Minimum holdtime from neighbor is 0 seconds
 Address tracking is enabled, the RIB does have a route to 10.10.10.5
 Connections established 13; dropped 13
 Last reset 00:01:18, due to User reset
 Transport(tcp) path-mtu-discovery is enabled
 No active TCP connection

```



A customer reported a failure and intermittent disconnection between two office buildings, site X and site Y. the network team finds that site X and site Y are exchanging email application traffic with the data center network. Which configuration resolves the issue between site X and site Y?

- A. R2(config)#router bgp 65101  
R2(config-router)#no timers bgp 3 15
- B. R2(config)#router bgp 65101  
R2(config-router)#neighbor 10.10.10.5 update-source loopback 0
- C. R5(config)#router bgp 65101  
R5(config-router)#neighbor 10.10.10.2 next-hop-self
- D. R2(config)#router bgp 65101  
R2(config-router)#neighbor 10.10.10.5 next-hop-self

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

From the output message, the forming of BGP neighbor is rejected. "10.10.10.2" and "10.10.10.5" are the loopback addresses of R2 and R5 respectively (since the two routers are not directly connected) and therefore they are using the loopback addresses to form BGP neighbor.

Since the two routers cannot form BGP neighbor, the problem is likely due to the missing of "update - source loopback 0" in the neighbor command in those routers.

**Remarks :**

"next-hop-self" does not affect the forming of BGP neighbor.

**QUESTION 608**



The R2 OSPF route 10.2.2.0/24 shows in the R1 EIGRP routing table without route redistribution performed between OSPF and EIGRP routing protocols. Which configuration is required on router R2 to resolve the issue?

- A. add the passive-interface FastEthernet 1/0 command in EIGRP 100
- B. add the no auto-summary command in EIGRP 100
- C. replace the network 10.0.0.0 command with FastEthernet0/0 network in EIGRP 100
- D. add the passive-interface FastEthernet 0/0 command in OSPF 1

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

In the diagram, R1 learns the EIGRP internal route "10.2.2.0/24". A possible reason is that both R3 and R2 is also using EIGRP in the network "10.2.2.0/24" (the network should use OSPF only according to diagram). To solve the problem in R2, you can disable EIGRP in R2 or prevent R2 from forming EIGRP neighbor in its F1/0 interface.

#### QUESTION 609

A switch has been configured to provide DHCP relay on VLAN100 to a server with an IP address of 10.1.1.1. The DHCP server is sending syslog reports of multiple TFTP requests that also originate from the switch. As a result, the server CPU exceeded a configured threshold. Which action does the network administrator recommend to bring the server CPU threshold down?

- A. Configure the switch with no ip forward-protocol udp 69 on VLAN100
- B. Configure the switch with an access list on VLAN100 to deny TFTP.
- C. Configure the switch with ip forward-protocol udp 67 globally
- D. Configure the switch with a VACL on VLAN100 to deny TFTP

**Correct Answer:** A

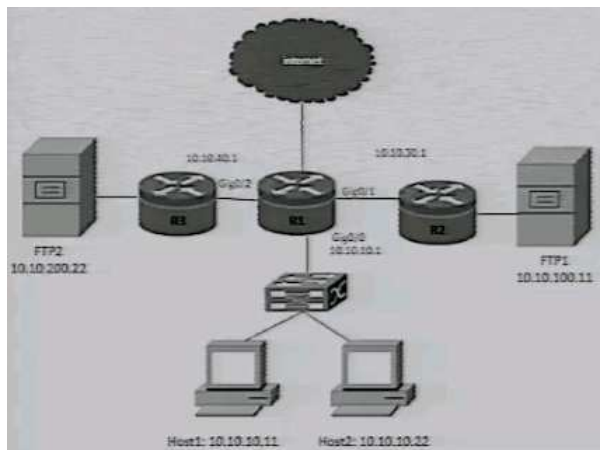
**Section:** (none)

**Explanation**

**Explanation/Reference:**

The switch probably configured to forward TFTP to the helper address (i.e. the DHCP server) and therefore TFTP requests are sent by the switch to the DHCP server. This can be solved by disabling the forwarding of TFTP traffic (UDP port 69) to the helper address.

#### QUESTION 610



The R1 routing table has the prefixes for the FTP1 and FTP2 file servers. A network engineer must configure the R1 with these requirements:

- Host1 must use the FTP1 fileserver
- Host2 must use the FTP2 fileserver

Which configuration meets the requirement on R1?

A.

```
ip access-list extended FTP1_R1
 permit ip host 10.10.10.11 host 10.10.100.11
ip access-list extended FTP2_R1
 permit ip host 10.10.10.22 host 10.10.200.22
!
route-map PBR_FTP permit 10
 match ip address FTP1_R1
 set ip next-hop 10.10.30.1
!
route-map PBR_FTP permit 20
 match ip address FTP2_R1
 set ip next-hop 10.10.40.1
!
interface GigabitEthernet 0/0
 ip policy route-map PBR_FTP
```

B.

```
ip access-list extended FTP1_R1
 permit ip host 10.10.10.11 any
ip access-list extended FTP2_R1
 permit ip host 10.10.10.22 any
route-map PBR_FTP permit 10
 match ip address FTP1_R1
 set ip next-hop 10.10.30.1
!
route-map PBR_FTP permit 20
 match ip address FTP2_R1
 set ip next-hop 10.10.40.1
!
interface GigabitEthernet 0/0
 ip policy route-map PBR_FTP
```

C.

```
ip access-list extended FTP1_R1
 permit ip host 10.10.10.11 host 10.10.100.11
ip access-list extended FTP2_R1
 permit ip host 10.10.10.22 host 10.10.200.22
!
route-map PBR_FTP permit 10
 match ip address FTP1_R1
 set ip next-hop 10.10.30.1
!
route-map PBR_FTP permit 20
 match ip address FTP2_R1
 set ip next-hop 10.10.40.1
!
ip local policy route-map PBR_FTP
```

D.

```
ip access-list extended FTP1_R1
 permit ip host 10.10.10.11 host 10.10.100.11
ip access-list extended FTP2_R1
 permit ip host 10.10.10.22 host 10.10.200.22
!
route-map PBR_FTP permit 10
 match ip address FTP1_R1
 set ip next-hop 10.10.40.1
route-map PBR_FTP permit 20
 match ip address FTP2_R1
 set ip next-hop 10.10.30.1
!
ip local policy route-map PBR_FTP
```

Correct Answer: B

Section: (none)

Explanation

Explanation/Reference:

For PBR using ACL with source "10.10.10.11" and destination "10.10.100.11", it only forces traffic from "10.10.10.11" to "10.10.100.11" to use the specific next hop. It still allows traffic from "10.10.10.11" to "10.10.200.22" to be routed through normal routing.

Therefore, you should use a PBR using ACL with source "10.10.10.11" and destination "any" to force all traffic from Host 1 "10.10.10.11" to a specific next hop. Host2

"10.10.10.22" should be configured similarly.

#### QUESTION 611

```

R1#show ip route
Gateway of last resort is not set

172.16.0.0/16 is variably subnetted, 8 subnets, 3 masks
C 172.16.3.0/30 is directly connected, GigabitEthernet0/1
L 172.16.2.2/32 is directly connected, GigabitEthernet0/1
C 172.16.2.16/28 is directly connected, Loopback1
L 172.16.2.17/32 is directly connected, Loopback1
C 172.16.2.32/28 is directly connected, Loopback2
L 172.16.2.33/32 is directly connected, Loopback2
S 172.16.2.48/28 [1/0] via 172.16.2.34
D 172.16.250.0/30 [90/3072] via 172.16.2.1, 1000h, GigabitEthernet0/1

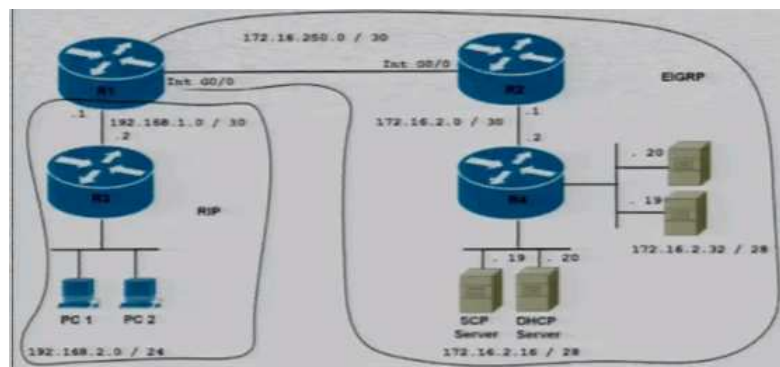
R1#show ip route
Gateway of last resort is not set

R 172.16.0.0/16 [120/18] via 192.168.1.1, 00:00:03, GigabitEthernet0/1
C 192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/30 is directly connected, GigabitEthernet0/1
L 192.168.1.2/32 is directly connected, GigabitEthernet0/1
C 192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.2.0/24 is directly connected, Loopback2
L 192.168.2.32/28 is directly connected, Loopback2
L 192.168.2.33/32 is directly connected, Loopback2
C 192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
L 192.168.3.0/24 is directly connected, Loopback1
L 192.168.3.17/32 is directly connected, Loopback1

R1#show running-config | begin router eigrp
router eigrp 100
network 172.16.250.0 0.0.0.3
redistribute rip
!
router rip
redistribute eigrp 100 metric 10
network 192.168.1.0
!
ip forward-protocol nd
!
route-map REDIST permit 10
match ip address 15
!
route-map CCM deny 10
match route-type external
!
route-map CCM permit 20
!
access-list 15 permit 192.168.0.0 0.0.0.255

R1#tracert 172.16.2.33
Type escape sequence to abort.
Tracing the route to 172.16.2.33
VRF info: (vrf in name/id, vrf out name/id)
 0 192.168.1.1 27 msec 31 msec 16 msec
 1 * * *
 2 * * *
 3 * * *
 4 * * *
R1#

```



Users from the 192.168.2.0/24 network cannot connect to the 172.16.2.32/28 network. Which configuration resolves the issue?

- A. R1(config)#route-map REDIST permit 10  
R1(config-route-map)#match ip address 15  
R1(config-route-map)#set metric 1000 10 255 1 1500  
R1(config-route-map)#exit  
R1(config)# access-list 15 permit 192.168.2.0 0.0.255.255
- B. R4(config)#ip route 0.0.0.0 0.0.0.0 172.16.2.1
- C. R1(config)#router eigrp 100  
R1(config-router)#network 192.168.0.0
- D. R1(config-router)#router eigrp 100  
R1(config-router)#redistribute rip  
R1(config-router)#default-metric 10000 100 255 100 1500

**Correct Answer:** D

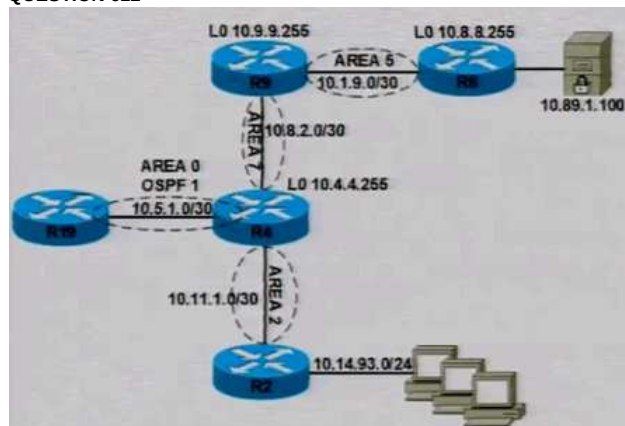
**Section:** (none)

**Explanation**

**Explanation/Reference:**

In the output, R4 is missing route to 192.168.2.0/24. Therefore users can send traffic to 172.16.2.32/28 but returned traffic cannot be sent back to the users. In R1 configuration, although mutual redistribution between RIP and EIGRP is configured, the command for redistribution from RIP to EIGRP is missing the metric settings and therefore no route from RIP is redistributed into EIGRP. One of the ways to solve this is to set the default metric for redistributing into EIGRP.

#### QUESTION 612



```

R9#
router ospf 1
 router-id 10.9.9.9
 network 10.1.9.0 0.0.0.3 area 5
 network 10.8.2.0 0.0.0.3 area 7
 network 10.9.9.255 0.0.0.0 area 7

R4#
router ospf 1
 router-id 10.4.4.4
 network 10.4.4.255 0.0.0.0 area 0
 network 10.8.2.0 0.0.0.3 area 7
 network 10.11.1.0 0.0.0.3 area 2

```

A network administrator added area 5 to the OSPF network, and now users connected to router R2 cannot access the server connected to R8. Which configuration resolves the issue?

- A. R9#
- ```

router ospf 1
  router-id 10.9.9.9
  area 7 virtual-link 10.4.4.4
  network 10.1.9.0 0.0.0.3 area 5
  network 10.8.2.0 0.0.0.3 area 7

R4#
router ospf 1
  router-id 10.4.4.4
  area 7 virtual-link 10.9.9.9
  network 10.11.1.0 0.0.0.3 area 2
  network 10.8.2.0 0.0.0.3 area 7

```
- B. R9#
- ```

router ospf 1
 router-id 10.9.9.9
 area 5 virtual-link 10.4.4.255
 network 10.1.9.0 0.0.0.3 area 5
 network 10.8.2.0 0.0.0.3 area 7

R4#
router ospf 1
 router-id 10.4.4.4
 area 5 virtual-link 10.9.9.255
 network 10.8.2.0 0.0.0.3 area 7
 network 10.11.1.0 0.0.0.3 area 2

```
- C. R9#
- ```

router ospf 1
  router-id 10.9.9.9
  area 7 virtual-link 10.4.4.4
  network 10.1.9.0 0.0.0.3 area 7
  network 10.8.2.0 0.0.0.3 area 5

R4#
router ospf 1
  router-id 10.4.4.4
  area 7 virtual-link 10.9.9.9
  network 10.8.2.0 0.0.0.3 area 2
  network 10.11.1.0 0.0.0.3 area 7

```
- D. R9#
- ```

router ospf 1
 router-id 10.9.9.9
 area 5 virtual-link 10.4.4.4
 network 10.1.9.0 0.0.0.3 area 5
 network 10.8.2.0 0.0.0.3 area 7

R4#
router ospf 1
 router-id 10.4.4.4
 area 5 virtual-link 10.9.9.9
 network 10.8.2.0 0.0.0.3 area 7
 network 10.11.1.0 0.0.0.3 area 2

```

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Problem occurs since Area 5 is not physically connected to Area 0. In order to solve this, a virtual link over Area 7 is required.

Remarks:

In another choice with virtual link over Area 7, the area numbers are wrongly configured in the network command for both routers.

**QUESTION 613**

```

track 11 ip sla 1
!
track 12 ip sla 2
!
track 13 ip sla 3
!
track 14 list boolean and
 object 11
 object 12
 object 13
!
ip route 0.0.0.0 0.0.0.0 203.0.113.1 name ISP1 track 14
ip route 0.0.0.0 0.0.0.0 198.51.100.1 14 name ISP2

```

The IP SLA probes 1, 2 and 3 verify the reachability of three distinct DNS servers hosted by ISP1. If all DNS servers are unreachable, the router should failover to ISP2 for Internet connectivity. However, the configuration is causing the router to failover to ISP2 even if some of the DNS servers are still reachable. Which action resolves the issue?

- A. Rewrite the track object 14 to use the "or" operator
- B. Reference the track object 14 to the IP SLA probes 1, 2, and 3 directly
- C. Use the administrative distance of 255 for the default route through ISP2
- D. Use the delay command in the track object 14

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

When using "and", all three track objects must be success in order to keep the static route to ISP1. If the static route should be kept if any track object success, "or" should be used.

#### QUESTION 614

```

R1#sh run | s bgp
router bgp 65001
no synchronization
bgp router-id 10.100.1.50
bgp log-neighbor-changes
network 10.1.1.0 mask 255.255.255.252
network 10.1.1.12 mask 255.255.255.252
network 10.100.1.50 mask 255.255.255.255
timers bgp 20 60
neighbor R2 peer-group
neighbor R4 peer-group
neighbor 10.1.1.2 remote-as 65001
neighbor 10.1.1.2 peer-group R2
neighbor 10.1.1.14 remote-as 65001
neighbor 10.1.1.14 peer-group R4
no auto-summary

```

While troubleshooting a BGP route reflector configuration, an engineer notices that reflected routes are missing from neighboring routers. Which two BGP configurations are needed to resolve the issue? (Choose two.)

- A. neighbor R2 route-reflector-client
- B. neighbor 10.1.1.2 allowas-in
- C. neighbor 10.1.1.2 route-reflector-client
- D. neighbor 10.1.1.14 route-reflector-client
- E. neighbor R4 route-reflector-client

**Correct Answer:** AE

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The peer-group R2 and R4 are configured with no settings. Hence nothing is applied to the neighbor 10.1.1.2 and 10.1.1.14 for configuring them as route reflector client.

Remarks:

If a neighbor is configured as a member of a peer-group, you cannot set the neighbor as route reflector client directly. You must set route reflector client through the peer group:

```

R1(config-router)#neighbor 10.1.1.2 route-reflector-client
% Invalid command for a peer-group member
R1(config-router)#

```

#### QUESTION 615



Compliance Summary > Startup vs Running Configuration

Change History (Running Config)

● In Sync ● Out Of Sync

☐ Show difference from Startup ☒ Show difference from previous Running

| Running Config (338 Lines) - January 07, 2022 05:14 AM | Running Config (342 Lines) - January 07, 2022 05:27 AM |
|--------------------------------------------------------|--------------------------------------------------------|
| 25 no mac syslog                                       | 25 no mac syslog                                       |
| 26 interface GigabitEthernet0                          | 26 interface GigabitEthernet0                          |
| 27 ip address 172.16.1.42 255.255.255.252              | 27 ip address 172.16.1.42 255.255.255.252              |
| 28 negotiation auto                                    | 28 negotiation auto                                    |
| 29 ipw enable                                          | 29 ipw enable                                          |
| 30 mpls l2 ipw area 0                                  | 30 mpls l2 ipw area 0                                  |
| 312 ip permit tcp any any eq 8443                      | 312 ip permit tcp any any eq 8443                      |
| 362 ip deny udp any any eq domain                      | 362 ip deny udp any any eq domain                      |
| 363 ip deny udp any eq bootpc any eq bootps            | 364 ip deny udp any eq bootpc any eq bootps            |
| 364 ip radius source-interface loopback0               | 365 ip radius source-interface loopback0               |
| 365 logging source-interface loopback0                 | 365 logging source-interface loopback0                 |
| 366 logging host 10.228.208.230                        | 366 logging host 10.228.208.230                        |
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```
R2(config-flow-monitor)#no exporter CCNPEXP1
R2(config-flow-monitor)#flow exporter CCNPEXP1
R2(config-flow-exporter)export-protocol ipfix
D. R2(config)#flow exporter CCNPEXP1
R2(config-flow-exporter)export-protocol netflow-v9
```

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Flow exporter are configured directly without entering the flow monitor configuration mode first.

```
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#flow exporter CCNPEXP1
R2(config-flow-exporter)#export-protocol netflow-v9
R2(config-flow-exporter)#end
R2#
```

#### QUESTION 617

```
R1 (config)#interface GigabitEthernet 0/0
R1 (config-if)#ip address 10.10.10.10 255.255.255.252
R1 (config-if)#ospfv3 1 ipv4 area 0

R2 (config)#interface GigabitEthernet 0/0
R2 (config-if)#ip address 10.10.10.11 255.255.255.252
R2 (config-if)#ospfv3 10 ipv4 area 0
R2 (config-if)#ospfv3 network broadcast
```

An engineering is troubleshooting an OSPF adjacency issue between directly connected routers R1 and R2. Which configuration resolves the issue?

- A. R2(config)#interface GigabitEthernet 0/0  
R2(config-if)#ip address 10.10.10.9 255.255.255.252
- B. R1(config)#interface GigabitEthernet 0/0  
R1(config-if)#ospfv3 10 ipv4 area 0
- C. R1(config)#interface GigabitEthernet 0/0  
R1(config-if)#ospfv3 network broadcast
- D. R2(config)#interface GigabitEthernet 0/0  
R2(config-if)#ospfv3 network broadcast

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

For 255.255.255.252, subnets 10.10.10.8 contains the assignable IP address ".9" and ".10". Hence, ".10" and ".11" are of different subnets and therefore causes problem in OSPF adjacency. Hence, you need to change the IP address of the interface in one of the router.

Remarks:

For Ethernet interface, "ospfv3 network broadcast" is the default setting.

#### QUESTION 618

```
CPE# telnet 198.51.100.64
% telnet connections not permitted from this terminal
CPE# ssh -l admin 198.51.100.64
% ssh connections not permitted from this terminal
CPE#
```

After a security hardening was performed on a router, the administrator cannot access the command line of any remote device. Which action resolves the issue?

- A. modify the AAA policy to allow the user to run the telnet and SSH commands.
- B. remove the transport output none command from the line con 0 section.
- C. move Telnet and SSH commands to a nonzero privilege level.
- D. remove the egress ACL blocking Telnet and SSH on the router interfaces.

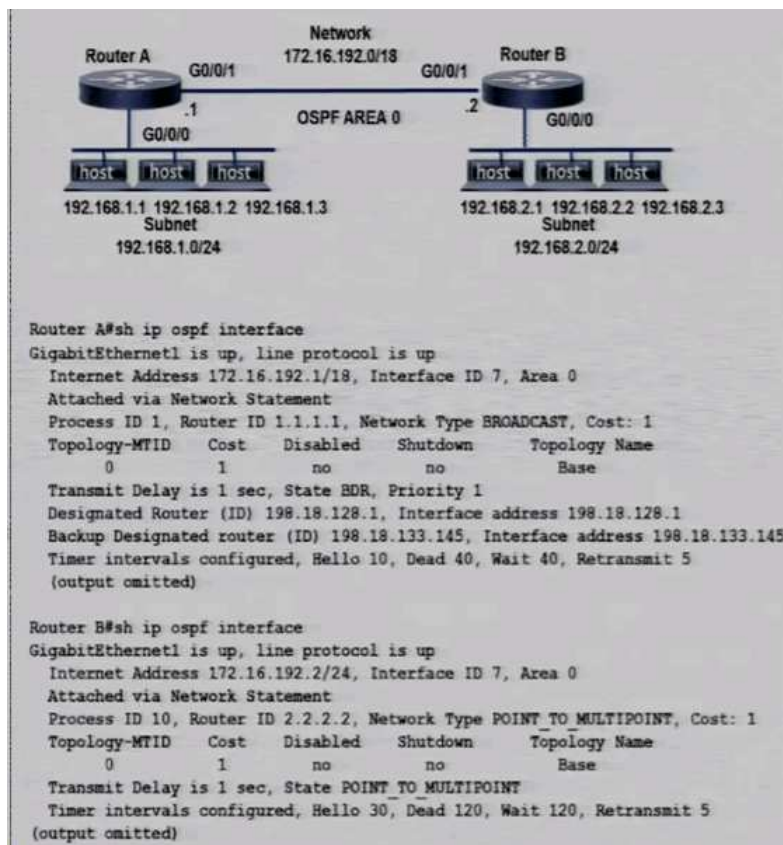
**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 619



OSPF adjacency is broken between router A and router B. Which configuration on router B resolves the issue?

- A. Router B(config)#int gigabitEthernet 1  
Router B(config-if)#ip ospf hello-interval 40  
Router B(config-if)#router ospf 10  
Router B(config-router)#router-id 1.1.1.1
- B. Router B(config)#int gigabitEthernet 1  
Router B(config-if)#ip ospf hello-interval 10  
Router B(config-if)#ip ospf cost 1000
- C. Router B(config)#int gigabitEthernet 1  
Router B(config-if)#ip address 172.16.192.2 255.255.255.192  
Router B(config-if)#ip ospf network broadcast  
Router B(config-if)#exit
- D. Router B(config)#int gigabitEthernet 1  
Router B(config-if)#ip address 172.16.192.2 255.255.192.0  
Router B(config-if)#ip ospf network broadcast  
Router B(config-if)#exit

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

For the network connecting Router A and Router B, the two routers have different settings:

- Router A is having subnet mask /18 and Router B is having the subnet mask /24.
- Router A is having broadcast network type and Router B is having the point-to-multipoint network type (therefore their default timer values are also different).

In order to fix the above in RouterB, the following changes are needed:

- You need to change the subnet mask to /18 i.e. 255.255.192.0.
- You need to change the network type to broadcast.

#### QUESTION 620

```

!
ip sla 1
icmp echo 192.168.2.1 source-interface GigabitEthernet0/0/1
timeout 1000
threshold 1000
frequency 30
ip sla schedule 1 life forever start-time now
!
track 1 ip sla 1 reachability

```

An engineer observes that every time the ICMP packet is lost at a polling interval, track 1 goes down, which causes unnecessary disruption and instability in the network. The engineer does not want the traffic to be rerouted if the loss of ICMP packets is negligible. If the packet loss is persistent for a longer duration, the track must go down and the traffic must be rerouted. Which action resolves the issue?

- A. Change the IP SLA schedule to run only at certain intervals.
- B. Increase the timeout value from 1000 to 1500.
- C. Define a delay timer under track 1.
- D. Increase the threshold value from 1000 to 1500.

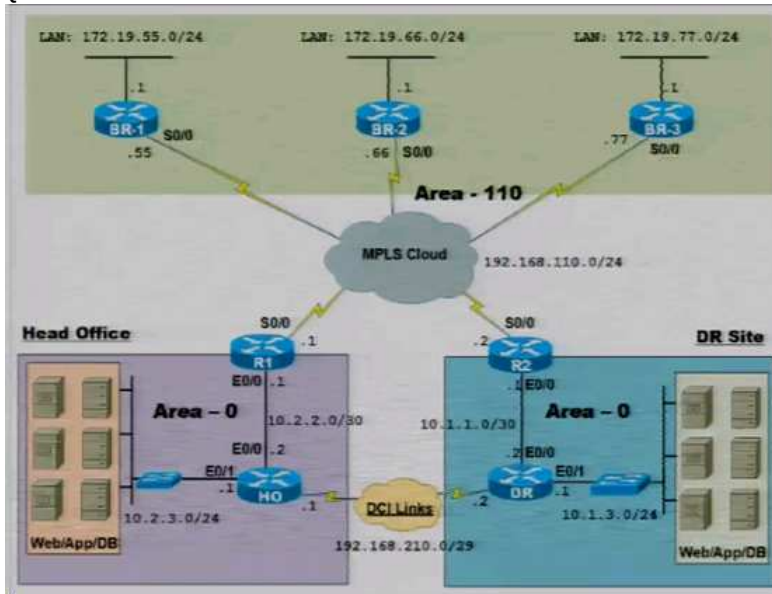
**Correct Answer:** C

Section: (none)  
Explanation

Explanation/Reference:  
The following show an example:

```
R1(config)#track 1 ip sla 1 reachability
R1(config-track)#delay down 90
R1(config-track)#
```

#### QUESTION 621



During an unannounced link-maintenance window at the ISP, the DCI link went down, which caused a significant service outage. What action must the network engineer take at the head office to ensure Area-0 connectivity without intervention from the ISP?

- A. Create a virtual link in Area-0 between HO E0/0 and DR E0/0.
- B. Create a virtual link in Area-110 between R1 S0/0 and R2 S0/0.
- C. Create a GRE tunnel interface in Area-110 between R1 S0/0 and R2 S0/0.
- D. Create a GRE tunnel interface in Area-0 between HO E0/0 and DR E0/0.

Correct Answer: B  
Section: (none)  
Explanation

Explanation/Reference:

After the DCI link connecting the Head Office and the DR site is down, a virtual link can be setup over Area 110 in R1 and R2 to connect the two Area 0 again.

Remarks:

Note that since all BR router interfaces connecting to MPLS Cloud are assigned to Area 110 in the above diagram, R1's s0/0 and R2's s0/0 must also be assigned to Area 110.

#### QUESTION 622

```
R1# show ip route static
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
S 10.100.13.0/24 is directly connected, Loopback10
R1# show run | begin router ospf 1
router ospf 1
 redistribute static
 network 10.100.11.0 0.0.0.255 area 1
 network 10.100.12.0 0.0.0.255 area 1
 network 10.100.14.0 0.0.0.255 area 1

R2# show ip route 10.100.13.0
% Subnet not in table
R2# show ip ospf neighbor | include 10.100.14.1
10.100.14.1 1 FULL/DR 00:00:39 10.100.14.1
GigabitEthernet0/0
```

R1 is directly connected to R2 over network 10.100.14.0/24. An engineer configures R1 to advertise a static route that is connected to a local loopback for network 10.100.13.0/24. The network is not in the routing table of R2. Which action resolves the issue?

- A. R2 must use a different OSPF process number and should be changed to ospf 1 to match R1.
- B. The redistribution command is incorrect in R1. The keyword subnets should be included with the redistribution command.
- C. The Loopback interface on R1 is administratively down. The interface should be enabled with the no shutdown command.
- D. The redistribution command is incorrect on R1. The default metric metric 200 should be included with the redistribution command.

Correct Answer: B  
Section: (none)  
Explanation

Explanation/Reference:

#### QUESTION 623

```

R2#show run | begin bgp

router bgp 64512
neighbor 10.1.1.1 remote-as 1
network 10.10.10.0 mask 255.255.255.0 route-map COMM

route-map COMM permit 10
set community 64512:600

R2#show ip bgp 10.10.10.0
BGP routing table entry for 10.10.10.0/24, version 1320 Paths: (1 available, best #1)
Local 0.0.0.0 from 0.0.0.0 (1.1.1.1)
Origin IGP, metric 0, localpref100, weight 32768,
valid, sourced, local, best
Community 64512:600

```

The eBGP neighbor 10.1.1.1 of router R2 receives and installs the prefix, but the community value is not found in the BGP routing table. How must this issue be resolved?

- A. R1 must have a route map assigned to neighbor R2, which is not matching the community values.
- B. R1 must have a route map assigned to neighbor R2, which is clearing the community values.
- C. R2 is not sending the community value to neighbor R1, configure R2 to send community.
- D. Community values are nontransitive and R2 is an eBGP peer, configure R2 to send community.

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The following shows an example:

```

R2(config)#router bgp 64512
R2(config-router)#neighbor 10.1.1.1 send-community
R2(config-router)#

```

#### QUESTION 624

What is a role of route distinguishers in an MPLS network?

- A. Route distinguishers define which prefixes are imported and exported on the edge router.
- B. Route distinguishers allow multiple instances of a routing table to coexist within the edge router.
- C. Route distinguishers are used for label bindings.
- D. Route distinguishers make a unique VPNv4 address across the MPLS network.

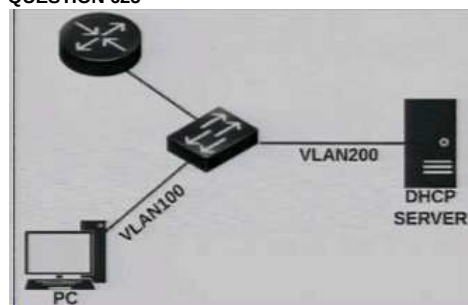
**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 625



A PC is configured to obtain an IP address automatically, but it receives an IP address only from the 169.254.0.0 subnet. The DHCP server logs contained no DHCPDISCOVER message from the MAC address of the PC. Which action resolves the issue?

- A. Configure a DHCP reservation on the server for the PC.
- B. Configure a static IP address on the PC and exclude it from the DHCP pool.
- C. Configure an ip helper-address on the router to forward DHCP messages to the server.
- D. Configure DHCP Snooping on the switch to forward DHCP messages to the server.

**Correct Answer:** C

**Section:** (none)

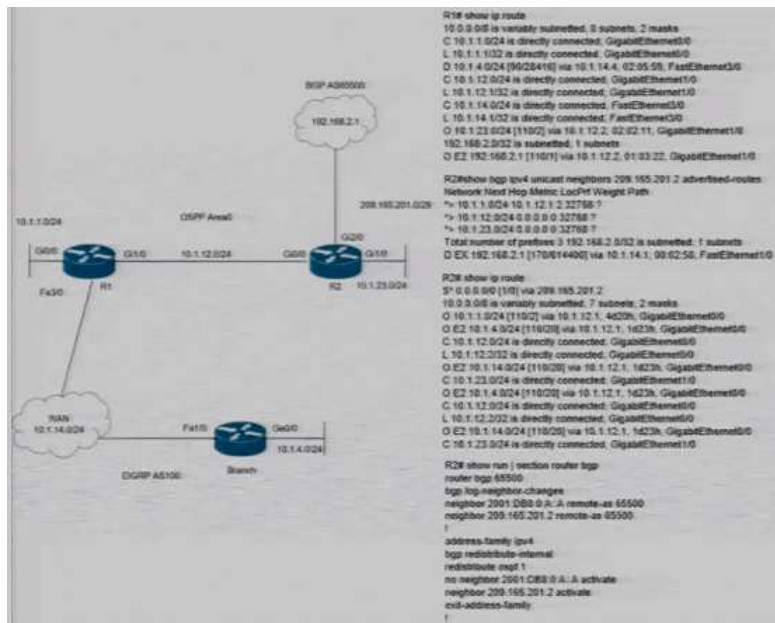
**Explanation**

**Explanation/Reference:**

Since the PC and the DHCP server are connected to different VLANs, "ip helper-address" has to be configured in a router to relay the DHCPDISCOVER message from the PC to the DHCP server.

#### QUESTION 626





Users in BGP autonomous system 65500 cannot access IPv4 resources in the EIGRP IPv4 autonomous system 100. Which action resolves the issue?

- A. Redistribute EIGRP routes into OSPF.
- B. Redistribute OSPF internal routes into BGP.
- C. Redistribute OSPF external routes into BGP.
- D. Redistribute BGP routes into OSPF.

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

In the BGP router R2's routing table, the networks in EIGRP AS 100 i.e. 10.1.14.0/24 and 10.1.4.0/24 can be found as OSPF E2 external routes. In order to advertise them to the BGP 65500, those OSPF E2 external routes have to be redistributed into BGP.

#### QUESTION 627

Which type of BFD session is created on a broadcast Ethernet interface with OSPF?

- A. with the DR and BDR, but not any DROTHER routers
- B. with all OSPF routers
- C. only on P2P Ethernet interfaces
- D. only with the DR

**Correct Answer:** A

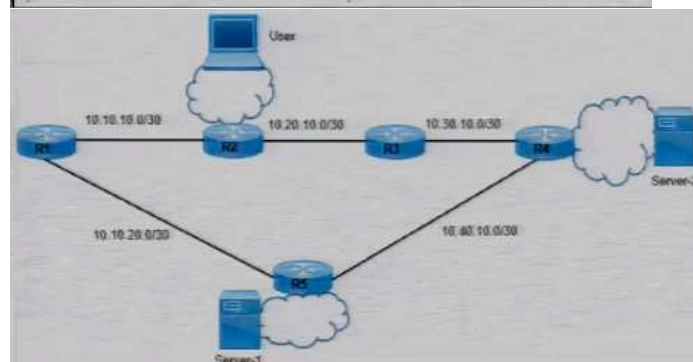
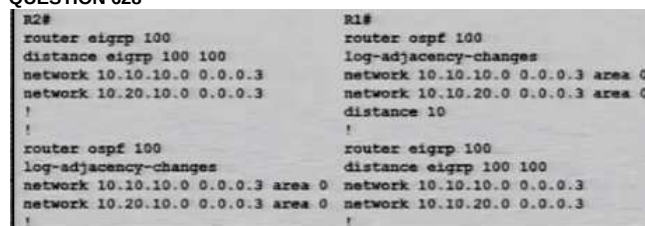
**Section:** (none)

**Explanation**

**Explanation/Reference:**

For broadcast interfaces, OSPF establishes a BFD session only with the designated router (DR) and backup designated router (BDR). The session is not established between any two devices in a DROTHER state.

#### QUESTION 628



The user experiences random packet drops when accessing Server-1 but does not have any issues when reaching out to Server-2. Which action resolves the issue?



- A. Configure the distance under the EIGRP process to 10 on R1.
- B. Modify the distance 10 under the OSPF process on R1 to 90.
- C. Configure the distance under the EIGRP process to 10 on R2.
- D. Remove the distance 10 under the OSPF process on R1.

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Since there is no information about how the routes to Server-1 and Server-2 are advertised, we shall focus on which choice can be a valid answer.

In R1, both internal and external EIGRP routes are set to an AD of 100 which is larger than the AD 10 manually set to OSPF (i.e. OSPF is preferred). Hence:

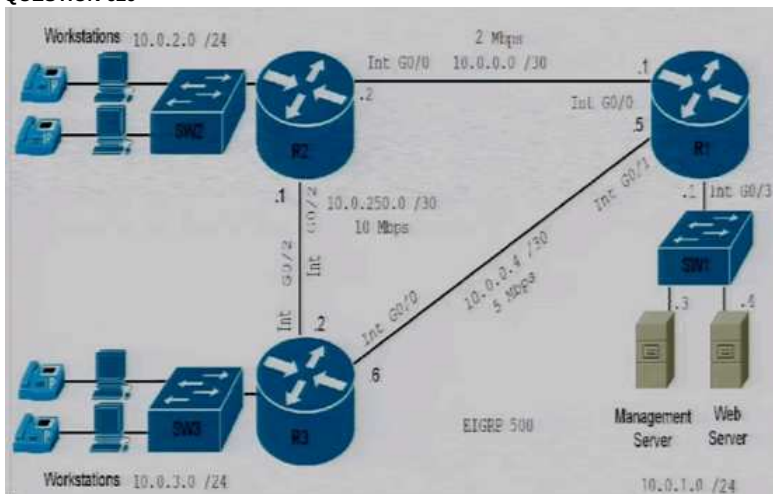
- for the choice that changes the AD of EIGRP to 10, although both EIGRP and OSPF will then have the same AD, since the metric value of OSPF is usually smaller than the metric value of EIGRP, therefore OSPF is still preferred and therefore there will be no difference.
- for the choice that changes the AD of OSPF to 90, it will make no difference since EIGRP still has an AD larger than OSPF.

In R2, both internal and external EIGRP routes are set to have an AD of 100 which is smaller than the default AD 110 of OSPF (i.e. EIGRP is preferred). Hence:

- for the choice that change the AD of EIGRP to 10 in R2, it will make no difference since EIGRP still has an AD smaller than OSPF.

On the other hand, the choice that removes AD 10 from OSPF in R1 will revert OSPF to use its default AD 110. Hence, this will make a difference in R1 since EIGRP (having AD 100) is now preferred than OSPF.

**QUESTION 629**



An engineer must route the management server traffic to the network 10.0.250.0 over the 2 Mbps link. Which configuration accomplishes this task on R1?

- A. 

```
route-map advrt permit 10
match ip address 110
set ip next-hop 10.0.0.1
!
access-list 110 permit ip host 10.0.1.3 10.0.250.0 0.0.0.3
!
interface GigabitEthernet0/3
ip policy route-map advrt
```
- B. 

```
route-map advrt permit 10
match ip address 110
set ip next-hop 10.0.0.2
!
access-list 110 permit ip host 10.0.1.3 10.0.250.0 0.0.0.3
!
interface GigabitEthernet0/3
ip policy route-map advrt
```
- C. 

```
interface GigabitEthernet0/1
bandwidth 20000
```
- D. 

```
interface GigabitEthernet0/0
bandwidth 20000
```

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

In order to send traffic from the management server 10.0.1.3 to the network 10.0.250.0/30, there are two ways:

- i. Setup PBR in R1 so that traffic from "host 10.0.1.3" to network "10.0.250.0 0.0.0.3" received by its g0/3 interface will be sent to the next hop "R2" i.e. 10.0.0.2. However, this method will use more resources in R1.
- ii. Modify R1's outgoing interface g0/0 to have a larger bandwidth (e.g. 20Mbps) than the 5Mbps link. Since the minimum bandwidth from R1 to 10.0.250.0/30 through R2 becomes larger, this route will have a lower metric. Hence, EIGRP will then prefer the route 10.0.250.0/30 learnt from R2. However, this method requires that the interfaces R1's g0/1 and R3's g0/0 are probably configured with the "bandwidth 5000" command (i.e. not using the default 1000Mbps bandwidth of the Gigabit Ethernet interface). Moreover, this method will have a side effect: all traffic from management server will prefer the 2Mbps link i.e. the traffic to workstations 10.0.2.0/24 and workstations 10.0.3.0/24 will also be sent over the 2Mbps link.

Due to the requirement and side effect of "ii", the method "i" is chosen as the suggested answer.

### QUESTION 630

## How does BFD protocol work?

- A. BFD operates on the route processor module and impacts the route processor CPU utilization.
- B. BFD control packets are sent via UDP port 3784 to the destination router.
- C. BFD echo packets are sent to the same source IP and different destination IP with TCP port of 3786.
- D. When BFD declares a failure on the primary IGP path, the router on the peer router chooses to use the secondary path.

**Correct Answer: B**

**Section: (none)**

### Explanation

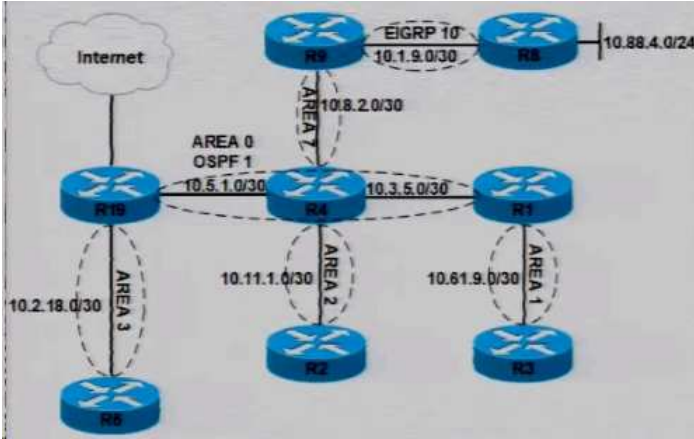
**Explanation/Reference:**

BFD control packets are sent to UDP 3784 and BFD echo packets are sent to UDP 3785.

Remarks:

If BFD in a router detects a failure, the routing protocol in that router (not the peer router) takes action to bypass the failed peer.

### QUESTION 631



After an engineer modified the configuration for area 7 to permit type 1, 2, and 7 LSAs only, users connected to router R9 reported that they could no longer access the Internet. Which configuration restores Internet access to users on R9 and permits only LSA type 1, 2, and 7?

- A. R4#  
router ospf 1  
area 7 nssa no-summary  
network 10.5.1.0 0.0.0.3 area 0  
network 10.8.2.0 0.0.0.3 area 7
- R9#  
router ospf 1  
area 7 nssa  
redistribute eigrp 10 subnets  
network 10.8.2.0 0.0.0.3 area 7
- B. R4#  
router ospf 1  
area 7 nssa  
network 10.5.1.0 0.0.0.3 area 0  
network 10.8.2.0 0.0.0.3 area 7
- R9#  
router ospf 1  
area 7 nssa  
redistribute eigrp 10 subnets  
network 10.8.2.0 0.0.0.3 area 7
- C. R4#  
router ospf 1  
area 0 area 7 stub no-summary  
network 10.5.1.0 0.0.0.3 area 0  
network 10.8.2.0 0.0.0.3 area 7
- R9#  
router ospf 1  
area 7 stub  
redistribute eigrp 10 subnets  
network 10.8.2.0 0.0.0.3 area 7
- D. R4#  
router ospf 1  
area 0 nssa default-information-originate  
network 10.5.1.0 0.0.0.3 area 0  
network 10.8.2.0 0.0.0.3 area 7
- R9#  
router ospf 1  
area 7 nssa  
redistribute eigrp 10 subnets  
network 10.8.2.0 0.0.0.3 area 7

**Correct Answer: A**

**Section: (none)**

### Section (c) Explanation

**Explanation/Reference:**

In order to allow type 1, 2 and 7 LSAs in area 7, the area should be setup as NSSA (type 7 LSA is needed since R9 is a ASBR importing routes from EIGRP) in all routers connecting to this area i.e. R4 and R9.

In order for R9 get a default route so that R9 can access Internet, you need to add the additional keyword "no-summary" i.e. the command "area 7 nssa no-summary" in the ABR router in the NSSA area i.e. R4.

**Remarks:**

Although using "area 7 nssa default-information-originate" in R4 can also provide a default route in area 7, however, the choice with the keyword "default-information-originate" is configuring it in the wrong area. It should be area 7 instead of area 0.

**QUESTION 632**

```
R1
!
router bgp 200
no synchronization
bgp log-neighbor-changes
neighbor 192.168.200.6 remote-as 100
neighbor 192.168.200.6 update-source Loopback0
no auto-summary
!
ip route 192.168.200.6 255.255.255.255 192.168.100.1
!
R1#show ip bgp neighbor 192.168.200.6
BGP neighbor is 192.168.200.6, remote AS 100, external link
 BGP version 4, remote router ID 0.0.0.0
 BGP state = Idle
 Last read 00:00:00, last write 00:00:00, hold time is 180, keepalive interval is
 60 seconds
!
For address family: IPv4 Unicast
 BGP table version 1, neighbor version 0/0
 Output queue size: 0
 Index 1, Offset 0, Mask 0x2
 !
 Connections established 0; dropped 0
 Last reset never
 No active TCP connection
```

The BGP neighbor is not coming up. Which action resolves the issue?

- A. Enable synchronization between the neighbors to bring the neighborship up.
- B. The route map on eBGP sessions must allow the prefixes from the neighbor.
- C. Configure the ebgp-multihop 2 command on R1 toward the neighbor.
- D. Configure a valid router ID on the neighbor that shows an invalid router ID of 0.0.0.0.

**Correct Answer: C**

**Section: (none)**

**Explanation**

**Explanation/Reference:**

**QUESTION 633**

```
R1#
interface ethernet 0
ip address 10.168.2.1 255.255.255.0
ip summary-address EIGRP 1 172.18.80.0 255.255.240.0
interface Serial 0
ip address 10.168.1.2 255.255.255.0
router EIGRP 1
network 10.168.1.0
network 10.168.2.0

R1# show ip route
```

```
C 10.168.1.0/24 is directly connected, ethernet 0
C 10.168.2.0/24 is directly connected, Serial 1
D 172.18.99.0/24 [90/409600] via 10.168.1.1, Serial 0
D 172.18.97.0/24 [90/409600] via 10.168.1.1, Serial 0
D 172.18.79.0/24 [90/409600] via 10.168.1.1, Serial 0
D 172.18.70.0/24 [90/409600] via 10.168.1.1, Serial 0
D 172.18.103.0/24 [90/409600] via 10.168.1.1, Serial 0
D 172.18.76.0/24 [90/409600] via 10.168.1.1, Serial 0
D 172.18.98.0/24 [90/409600] via 10.168.1.1, Serial 0
```

The neighboring router does not show any summarized prefixes. How must the configured summarization be seen on the neighboring router?

- A. There is no network statement for the summary prefix within EIGRP. The network 172.18.0.0 command must be configured under the EIGRP process.
- B. The summary mask is incorrect. Create a loopback interface within the summary range and enable within the EIGRP process with the network command to generate the summary.
- C. The auto summarization feature will be summarizing all routes into a single update and must be disabled by use of the no auto-summary command.

D. There is no network statement for the summary prefix within EIGRP. The network 172.18.80.0 0.0.15.0 command must be configured under the EIGRP process.

**Correct Answer:** B

**Section:** (none)

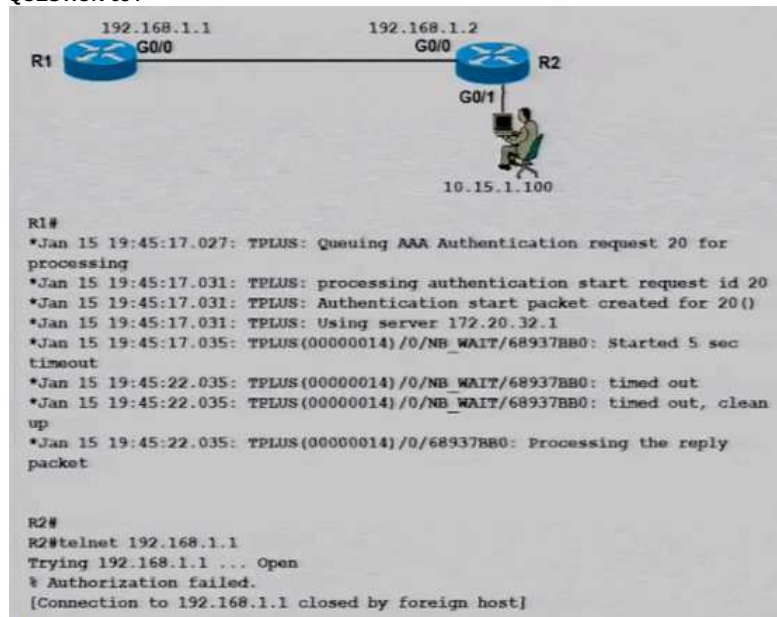
**Explanation**

**Explanation/Reference:**

Since R1 has no EIGRP route within the summary range e.g. 172.18.80.0/24 – 172.18.95.0/24, no summary route will be advertised.

In order to generate the summary route without any learnt EIGRP route within the summary range, you can configure a loopback interface with an IP address within that range and then use the “network” command to advertise it.

#### QUESTION 634



A network engineer is troubleshooting an AAA authentication issue for R1 from R2. When an engineer tries to open a telnet connection to R1, it opens the connection but shows a %Authorization failed error message on the terminal and closes the connection silently. Which action resolves the issue?

- A. Configure authorization commands in the tacacs+ server for the R1 router.
- B. Configure the tacacs+ server host IP on the R1 router.
- C. Resolve tacacs+ server host IP authentication miss configuration on the R1 router.
- D. Resolve tacacs+ server reachability from the R1 router.

**Correct Answer:** D

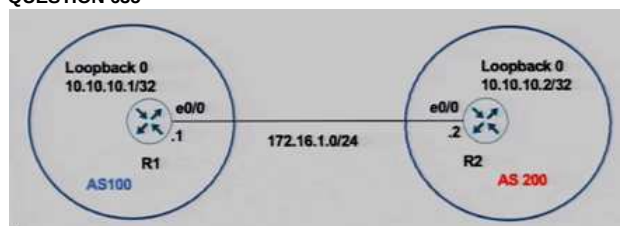
**Section:** (none)

**Explanation**

**Explanation/Reference:**

The IP address 172.20.32.1 is already configured as the tacacs+ server. Therefore the problem is probably that R1 cannot reach the tacacs+ server.

#### QUESTION 635



R1 and R2 have been configured where the neighbor relationship must be authenticated using MD5:

```
R1
router bgp 100
neighbor 172.16.1.2 remote-as 200
neighbor 172.16.1.1 password cisco123
```

```
R2
router bgp 200
neighbor 172.16.1.1 remote-as 100
neighbor 172.16.1.1 password cisco
```

The neighbor relationship is not coming up. Which configuration resolves the issue?

- A. R2

```
router bgp 200
neighbor 172.16.1.1 password cisco123
```
- B. R1

```
router bgp 100
neighbor 172.16.1.2 password MD5 cisco123
```

R2

```
router bgp 200
neighbor 172.16.1.1 password MD5 cisco123
```

C. R1

```
router bgp 100
neighbor 172.16.1.2 password MD5 cisco
```

R2

```
router bgp 100
neighbor 172.16.1.1 password MD5 cisco
```

D. R1

```
router bgp 100
neighbor 172.16.1.2 password cisco
```

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

There are two problems in the configuration:

- The IP addresses of the "neighbor ... password ..." command in R1 is wrong.
- The passwords in the two routers do not match.

The simplest way is to change the IP address of the "neighbor ... password ..." command in R1 to "172.16.1.2" and also change the password to "cisco" (i.e. the password configured in R2).

Remarks:

The "neighbor ... password ..." command does not accept the keyword "MD5".

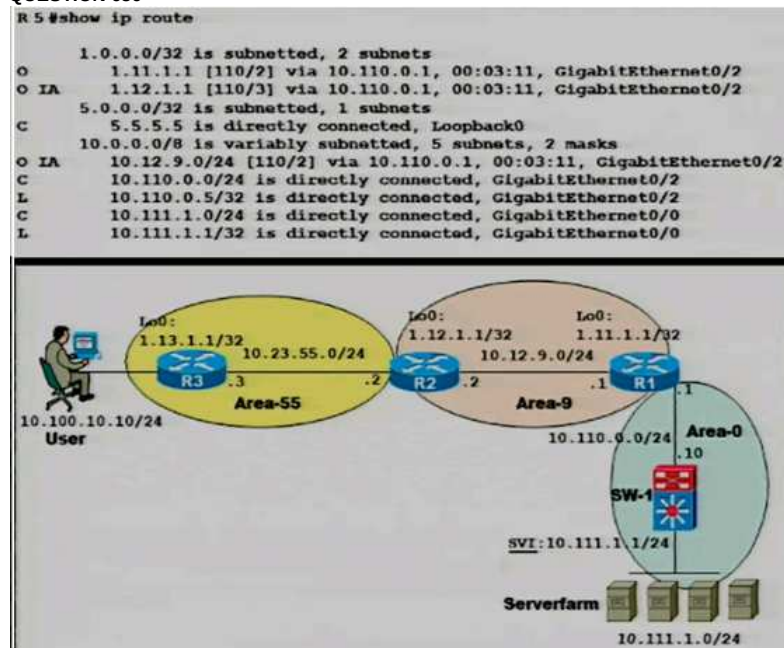
R1(config)#router bgp 100

R1(config-router)#neighbor 172.16.1.2 password ?

<0-7> Encryption type (0 to disable encryption, 7 for proprietary)  
LINE The password

R1(config-router)#

#### QUESTION 636



A user from Area-55 cannot access the HTTPS services running on the Area-0 server farm. An engineer identifies that the routing table of SW-1 is not learning the Area-55 routes. Which configuration resolves the issue?

A. R1

```
router ospf 100
area 9 virtual-link 10.12.9.1
```

R3

```
router ospf 100
area 9 virtual-link 1.23.55.2
```

B. R3

```
router ospf 100
area 9 virtual-link 1.13.1.1
```

R1

```
router ospf 100
area 9 virtual-link 1.12.1.1
```

C. R2

```
router ospf 100
area 9 virtual-link 1.11.1.1
```

```

R1
router ospf 100
area 9 virtual-link 1.12.1.1
D. R1
router ospf 100
area 9 virtual-link 10.110.0.1

R2
router ospf 100
area 9 virtual-link 1.12.9.1

```

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The OSPF router-id is used for configuring virtual link. Since loopback address is used by default as the OSPF router-id, the remote router's loopback address is used in the "area 9 virtual-link <remote router-id>" command.

#### QUESTION 637

```

R1#ping ipv6 ff02::a
Output Interface: fastethernet1/0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to FF02::A, timeout is 2 seconds:
Packet sent with a source address of FE80::C801:FFF:FE94:1C%FastEthernet1/0

Reply to request 0 received from FE80::C004:22FF:FE78:1, 40 ms
Request 1 timed out
Request 2 timed out
Request 3 timed out
Reply to request 4 received from FE80::C004:22FF:FE78:1, 56 ms
Success rate is 40 percent (2/5), round-trip min/avg/max = 40/48/56 ms

```

An engineer investigates an IPv6 EIGRP neighbor adjacency issue that sees the neighbors flapping and issued a ping from R1 to its directly connected neighbor. The link between the switches is stable at Layer 2, and other connected devices are also functioning. Which action resolves the issue?

- A. The switch between the two neighbors is not configured for IPv6 multicast and must be configured for IPv6 multicast.
- B. The switch between the two neighbors is not IPv6 compatible and must be replaced with an IPv6 compatible device.
- C. Multicast packets are not reliably transmitted over the link, and the switch must be replaced.
- D. Data is not reliably transmitted between the dynamically assigned link-local addresses of the routers and must be manually assigned.

**Correct Answer:** C

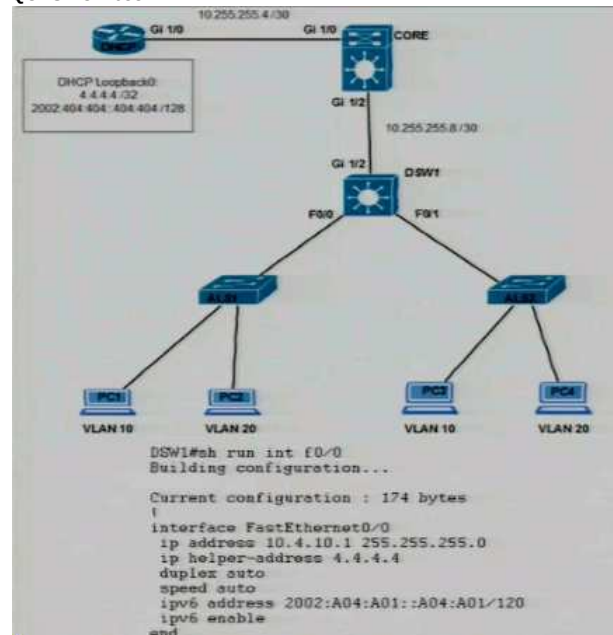
**Section:** (none)

**Explanation**

**Explanation/Reference:**

Since only 2 out of 4 ping is successful, there may be some problem in the link or switch.

#### QUESTION 638



Clients on ALS2 receive IPv4 and IPv6 addresses, but clients on ALS1 receive only IPv4 addresses and not IPv6 addresses. Which action on DSW1 allows clients on ALS1 to receive IPv6 addresses?

- A. Configure DSW1(config-if)#ipv6 dhcp relay destination 2002:404:404::404:404 GigabitEthernet1/2
- B. Configure DSW1(dhcp-config)#default-router 2002:A04:A01::A04:A01
- C. Configure DSW1(config-if)#ipv6 helper address 2002:404:404::404:404
- D. Configure DSW1(config)#ipv6 route 2002:404:404::404:404/128 FastEthernet1/0

**Correct Answer:** A



Section: (none)

Explanation

Explanation/Reference:

Remarks:

The configuration of the DSW1 interface f0/1 (not shown in the diagram) connecting to ALS2 probably has the command "ipv6 dhcp relay destination 2002:404:404::404:404 GigabitEthernet1/2" configured. Therefore clients on ALS2 receive IPv4 and IPv6 addresses

QUESTION 639

The network administrator configured CoPP so that EIGRP traffic toward the device is limited to 2 mbps. Any traffic that exceeds this limit must also be allowed at this point for traffic analysis.

```
access-list 100 permit EIGRP any any
!
class-map CM-EIGRP
match access-group 100
!
policy-map PM-COPP
class CM-EIGRP
police 2000000 conform-action transmit
!
control-plane
service-policy output PM-COPP
```

No EIGRP traffic is going through CoPP. Which configuration resolves the issue?

- A. policy-map PM-COPP  
class CM-EIGRP  
police 2000000 conform-action transmit exceed-action transmit  
!  
control-plane  
no service-policy output PM-COPP  
service-policy input PM-COPP
- B. policy-map PM-COPP  
class CM-EIGRP  
police 2000000 conform-action drop exceed-action drop  
!  
control-plane  
no service-policy output PM-COPP  
service-policy input PM-COPP
- C. policy-map PM-COPP  
class CM-EIGRP  
police 2000000 conform-action transmit exceed-action transmit
- D. control-plane  
no service-policy output PM-COPP  
service-policy input PM-COPP

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Other than setting "transmit" for the exceed-action, you also need to change the service-policy to "input".

QUESTION 640

```
R1
access-list 100 deny icmp any any echo-reply
access-list 100 permit ip any any
!
interface FastEthernet0/0
ip address 10.1.13.1 255.255.255.0
ip access-group 100 in
ip verify unicast source reachable-via rx
!
interface FastEthernet0/1
ip address 10.1.12.1 255.255.255.0
ip access-group 100 in
ip verify unicast source reachable-via rx
!
interface FastEthernet0/2
ip address 10.1.1.1 255.255.255.0
!
router eigrp 100
network 10.0.0.0
!
router rip
version 2
network 10.0.0.0

R2
interface FastEthernet0/0
ip address 10.1.12.2 255.255.255.0
!
interface FastEthernet0/1
ip address 10.1.23.2 255.255.255.0
!
router eigrp 100
network 10.0.0.0

R3
interface FastEthernet0/0
ip address 10.1.13.3 255.255.255.0
!
interface FastEthernet0/1
ip address 10.1.23.3 255.255.255.0
!
router rip
version 2
network 10.0.0.0
```

The diagram illustrates a network topology with three routers (R1, R2, R3) and a central switch (SW1). R1 is connected to SW1 via Fa0/1 and Fa0/2. R2 is connected to SW1 via Fa0/0. R3 is connected to SW1 via Fa0/0. Host A (GW: R2) is connected to R2. Host B (GW: R3) is connected to R3. Host C (GW: R1) is connected to R1 via Fa0/2.

Host A successfully pings host C, but host B fails to ping host C. Which action resolves this issue?

- A. Redistribute EIGRP 100 into RIP on R1.
- B. Disable auto-summary in RIP and EIGRP.
- C. Enable loose mode uRPF.
- D. Change the first statement in access list 100 to permit access.

Correct Answer: C

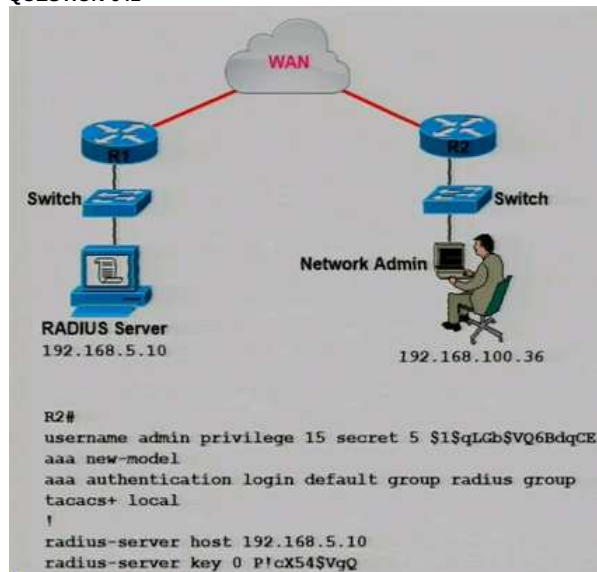
Section: (none)

## Explanation

### Explanation/Reference:

Since R1 learns the EIGRP route "10.1.23.0/24" (the networks connected by Host A and Host B) from R2 and enter it into its routing table instead of the RIP route from R3, therefore traffic from Host B to Host C going through R3 will fail the uRPF strict mode checking in R1.

## QUESTION 641



A network administrator is troubleshooting an authentication failure issue during login into R2 using a RADIUS server user account. The network administrator can ping the RADIUS server, but the user access to SSH is denied on R2. Which action resolves the issue?

- A. Enable AAA authorization to allow SSH connections.
- B. Modify the aaa command to replace TACACS+ with RADIUS.
- C. Configure the username in RADIUS server database.
- D. Remove "local" from the aaa authentication command.

**Correct Answer:** C

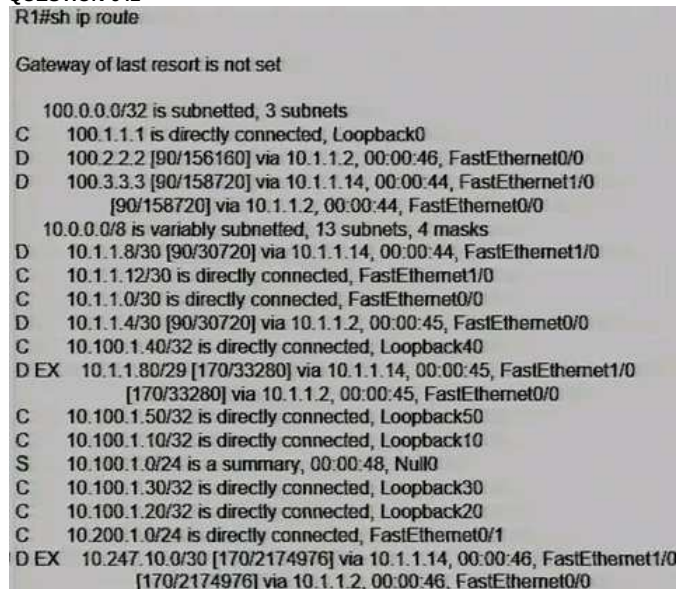
**Section:** (none)

**Explanation**

### Explanation/Reference:

Since R2's AAA configuration settings are properly configured, the only reason is that the username / password has not been configured in the RADIUS database.

## QUESTION 642



R1 must advertise all loopback interfaces IP addresses to neighbors, but EIGRP neighbors receive a summary route. Which action resolves the issue?

- A. Disable auto summarization on R1.
- B. Redistribute connected routes into EIGRP.
- C. Enable EIGRP on loopback interfaces.
- D. Remove the 10.100.1.0/24 static route.

**Correct Answer:** C

**Section:** (none)

**Explanation**

### Explanation/Reference:

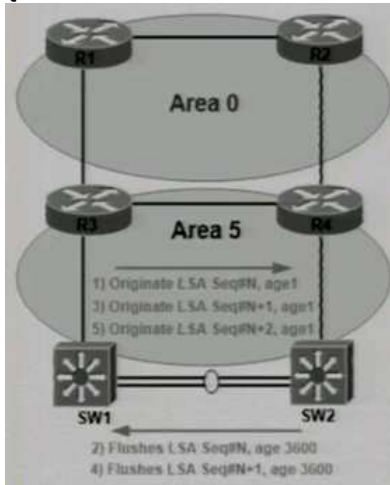
Since there is no EIGRP null0 route for 10.0.0.0/8 shown in R1's routing table, auto-summary is not enabled in R1.

Since the static null0 route "10.100.1.0/24" is "directly connected", it can be advertised by the network command e.g. "network 10.100.1.0 0.0.0.255" or "network 10.100.1.0 0.0.0.0". This EIGRP route "10.100.1.0/24" learnt by the neighbors is considered as summary route mentioned in the question.

Note that if the network command is "network 10.100.1.0 0.0.0.255", the loopback networks for lo10, lo20 ... etc should already be advertised to the neighbors. Hence, the network command configured in R1 is probably "network 10.100.1.0 0.0.0.0". Note that removing this static null0 route will not advertise the loopback network automatically.

In order to advertise the loopback's network, you can either redistribute connected or enable EIGRP on the loopback interfaces through "network" command. Among these two, "network" command is preferred.

#### QUESTION 643



An error message "an OSPF-4-FLOOD\_WAR" is received on SW2 from SW1. SW2 is repeatedly receiving its own link-state advertisement and flushes it from the network. Which action resolves the issue?

- A. Change area 5 to a normal area from a nonstub area.
- B. Resolve different subnet mask issue on the link.
- C. Resolve duplicate IP address issue in the network.
- D. Configure Layer 3 port channel on interfaces between switches.

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The OSPF flood war occurs when the router repeatedly receives its own Link State Advertisement (LSA) and flushes it from the network or sends a new version of it. This is meant to detect issues with Type-2 LSAs when duplicate IP addresses are present in the network, or with Type-5 LSAs when there is a duplicate router ID in different OSPF Areas.

#### QUESTION 644

```
Router A
line con 0
 exec-timeout 60 0
 logout-warning 15
 logging synchronous
 login
 transport output all
 stopbits 1
```

After a misconfiguration by a junior engineer, the console access to router A is not working. Which configuration allows access to router A?

- A. RouterA(config)#line console 0  
RouterA(config-line)#password cisco  
RouterA(config)#end
- B. RouterA(config)#line console 0  
RouterA(config-line)#password cisco  
RouterA(config-line)#login local  
RouterA(config)#end
- C. RouterA(config)#aaa new-model  
RouterA(config)#line console 0  
RouterA(config-line)#login authentication CONSOLE
- D. RouterA(config)#aaa new-model  
RouterA(config)#aaa authentication login my-auth-list tacacs+

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 645

Customer ABC BGP Peering

```

R1#sh ip bgp ipv6 unicast
BGP table version is 38, local router ID is 1.1.1.1
 Network Next Hop Metric LocPrf Weight Path
* i 2001::4/128 2001::4 0 100 0 300 i
*>i 2002::2/128 2001::2 0 100 0 i

```

```

R1#sh ipv6 route
IPv6 Routing Table - default - 7 entries
C 2001::/64 [0/0]
 via Loopback0, directly connected
L 2001::1/128 [0/0]
 via Loopback0, receive
O 2001::2/128 [110/1]
 via FE80::5200:C3FF:FE01:E600, GigabitEthernet0/0
C 2001:ABC:AABB:1122::/64 [0/0]
 via GigabitEthernet0/0, directly connected
L 2001:ABC:AABB:1122:1111:2222:AAA1/128 [0/0]
 via GigabitEthernet0/0, receive
B 2002::2/128 [200/0]
 via 2001::2
L FF00::/8 [0/0]
 via Null0, receive
R1#

```

R1 cannot reach the R4 loopback address (2001::4/128). Which configuration provides the most optimal and scalable resolution to the issue?

- A. R2  
router bgp 100  
address-family ipv6  
no synchronization
- B. R2  
router bgp 100  
address-family ipv6  
neighbor 2001::4 next-hop-unchanged  
redistribute connected
- C. R2  
router bgp 100  
address-family ipv6  
neighbor 2001::1 next-hop-self
- D. R1  
router bgp 100  
address-family ipv6  
no synchronization

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 646

What is the role of LDP in MPLS networks?

- A. It disables label binding information to exchange with peer LSRs
- B. It creates MPLS packet forwarding along with the IGP routes
- C. It enables label binding that exchanges route descriptors
- D. It enables label binding information to exchange with peer LSRs

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 647

```

interface Tunnel0
 ip address 172.23.5.10 255.255.255.0
 no ip redirects
 ip mtu 1420
 ip nhrp authentication C@trts81
 ip nhrp map multicast 192.168.200.1
 ip nhrp map 172.23.5.1 192.168.200.1
 ip nhrp network-id 10
 ip nhrp holdtime 300
 ip nhrp shortcut
 ip ospf network broadcast
 ip ospf priority 0
 tunnel source 192.168.100.146
 tunnel mode gre multipoint
 tunnel key 100

```

A network engineer is adding a new spoke router into an existing DMVPN Phase 3 tunnel with a hub router to provide secure communication between sites. Which additional configuration must the engineer apply to enable the tunnel to come up?

- A. ip nhrp registration no-unique
- B. ip nhrp nhs 172.23.5.1
- C. ip nhrp responder tunnel 0
- D. ip nhrp server-only non-caching

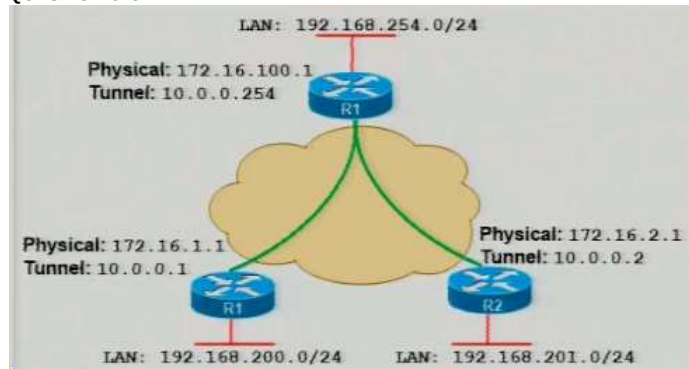
**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 648



A network engineer must provision two DMVPN spokes sites with a zero-touch deployment model. The spoke must communicate through hub with other spokes. Which configuration provisions these sites?

- A. interface tunnel10  
ip address 10.0.0.1 255.255.255.0  
ip nhrp network-id 100  
ip nhrp nhs registration 10.0.0.254 no-unique  
tunnel source G0/1  
tunnel destination 172.16.100.1  
tunnel mode mgre multipoint
- B. interface tunnel10  
ip address 10.0.0.1 255.255.255.0  
ip nhrp network-id 100  
ip nhrp nhs 10.0.0.254 nbma 172.16.100.1 broadcast  
tunnel source G0/1  
tunnel destination 172.16.100.1
- C. interface tunnel10  
ip address 10.0.0.1 255.255.255.0  
ip nhrp network-id 100  
ip nhrp 10.0.0.254 multicast dynamic  
tunnel source G0/1  
tunnel destination 172.16.100.1  
tunnel key 100
- D. interface tunnel10  
ip address 10.0.0.1 255.255.255.0  
ip nhrp network-id 100  
ip nhrp nhs 10.0.0.254 nbma 172.16.100.1 multicast  
tunnel source G0/1  
tunnel destination 172.16.100.1  
tunnel key 100

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

R1(config-if)#ip nhrp nhs ?  
A.B.C.D Protocol IP address of NHS  
cluster NHS Redundancy cluster configurations

```

dynamic NHS protocol address dynamically learnt
fallback NHS Redundancy Fallback time

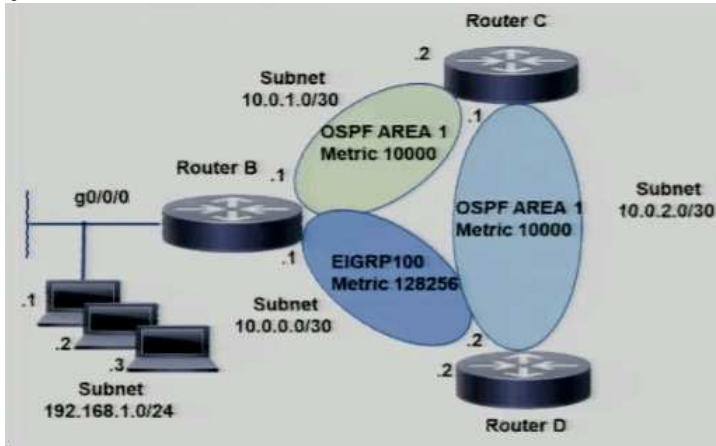
R1(config-if)#ip nhrp nhs 10.0.0.254 ?
cluster NHS cluster, don't specify for default cluster
nbma NBMA of NHS
priority NHS priority, don't specify for default priority
<cr>

R1(config-if)#ip nhrp nhs 10.0.0.254 nbma ?
A.B.C.D NBMA IP address of NHS
WORD NHS FQDN string

R1(config-if)#ip nhrp nhs 10.0.0.254 nbma 172.16.100.1 ?
cluster NHS cluster, don't specify for default cluster
multicast Use this NBMA mapping for broadcasts/multicasts
priority NHS priority, don't specify for default priority
<cr>

```

#### QUESTION 649



An engineer configures router B to direct all traffic from host 192.168.1.3 to router C. All other traffic must be routed through normal routing-protocol operations. Which configuration accomplishes the task?

- A. interface g0/0/0  
ip address 192.168.1.254 255.255.255.0  
ip policy route-map CCNP  
!  
access-list 101 permit ip host 192.168.1.3 any  
!  
route-map CCNP permit 10  
match ip address 101  
set ip next-hop 10.0.1.2
- B. interface g0/0/0  
ip address 192.168.1.254 255.255.255.0  
!  
access-list 101 permit ip host 192.168.1.3 any  
access-list 101 permit ip any any  
!  
route-map CCNP permit 10  
match ip address 101  
set ip next-hop 10.0.1.2
- C. interface g0/0/0  
ip address 192.168.1.254 255.255.255.0  
ip policy route-map CCNP  
!  
access-list 101 permit ip host 192.168.1.3 any  
!  
route-map CCNP permit 10  
match ip address 101  
set ip next-hop 10.0.2.1
- D. interface g0/0/0  
ip address 192.168.1.254 255.255.255.0  
ip policy route-map CCNP  
!  
access-list 101 permit ip any host 192.168.1.3  
!  
route-map CCNP permit 10  
match ip address 101  
set ip next-hop 10.0.1.2

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The access-list must only match 192.168.1.3 as source. The next hop router must be 10.0.1.2.

#### QUESTION 650

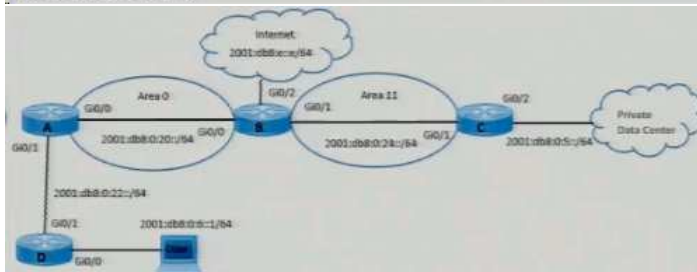


```

B# show run | section router ospfv3
router ospfv3 20
 area 11 stub
 (
 address-family ipv4 unicast
 passive-interface default
 no passive-interface GigabitEthernet0/0
 no passive-interface GigabitEthernet1/0
 default-information originate
 router-id 10.10.10.10
 exit-address-family
 !
 address-family ipv6 unicast
 passive-interface default
 no passive-interface GigabitEthernet0/0
 no passive-interface GigabitEthernet1/0
 router-id 10.10.20.20
 exit-address-family
)

D# traceroute 2001:db8::e
Type escape sequence to abort.
Tracing the route to 2001:db8::e
 0 2001:db8:0:22::1 10 10 10
C# show ipv6 route 2001:db8::e
* Route not found

```



An enterprise user reports an access issue with IPv6 content on the Internet. The user can access IPv4 content that is at the data center. Which action resolves the issue with IPv6 content?

- Advertise the 2001:db8::e/64 route on the router D OSPF process.
- Enable default information to originate in IPv6 AF On router B.
- Change Area 11 to Area 0 between routers B and C.
- Add a static route for 2001:db8:0:6::1/64 on router B RIB.

**Correct Answer: B**

**Section: (none)**

**Explanation**

**Explanation/Reference:**

In order to access Internet, you need a default route instead of a route for a specific IPv6 network.

#### QUESTION 651

```

router eigrp 1
 variance 2

R1#show ip eigrp topology 172.16.100.5 255.255.255.255
IP-EIGRP (AS 1): Topology entry for 172.16.100.3/32
 State is Passive, Query origin flag is 1, 1 Successor(s), FD is 409600
 Routing Descriptor Blocks:
 10.4.1.5 (Ethernet1/0), from 10.4.1.5, Send flag is 0x0
 Composite metric is (409600/128256), Route is Internal
 Vector metric:
 Minimum bandwidth is 10000 Kbit
 Total delay is 6000 microseconds
 Reliability is 255/255
 Load is 1/255
 Minimum MTU is 1500
 Hop count is 1
 10.3.1.6 (Serial2/0), from 10.3.1.6, Send flag is 0x0
 Composite metric is (435200/409600), Route is Internal
 Vector metric:
 Minimum bandwidth is 10000 Kbit
 Total delay is 7000 microseconds
 Reliability is 255/255
 Load is 1/255
 Minimum MTU is 1500
 Hop count is 2

```

A network engineer troubleshooting a packet drop problem for the host 172.16.100.5 notices that only one link is used and installed on the routing table, which saturates the bandwidth. Which action must the engineer take to resolve the high bandwidth utilization problem and share the traffic toward this host between the two available links?

- Change the EIGRP delay metric meet the feasibility condition.
- Set the eigrp variance equal to 4 to install a second route with a metric not larger than 4 times of the best metric.
- Set the eigrp variance equal to 3 to install a second route with a metric not larger than 3 times of the best metric.

D. Disable the eigrp split horizon loop protection mechanism.

**Correct Answer:** A

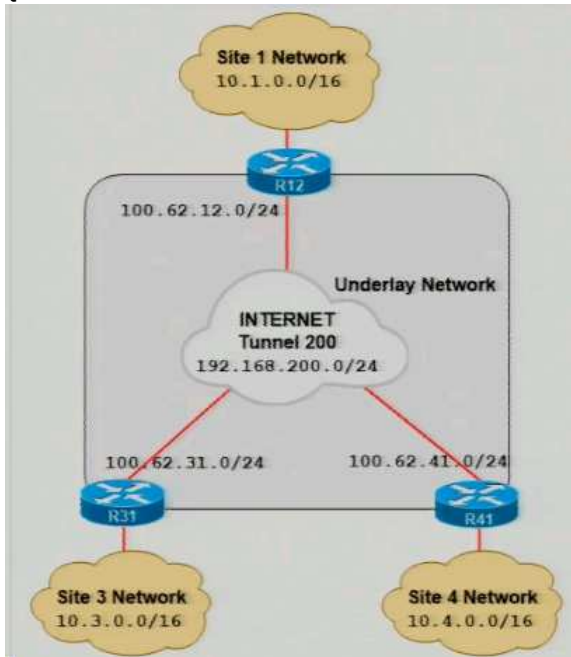
**Section:** (none)

**Explanation**

**Explanation/Reference:**

If the feasibility condition is not met, the route will never be used no matter what variance is configured.

#### QUESTION 652



The IPsec profile configuration must be optimal to address the extra bytes overhead per packet. Which configuration must be applied by the engineer on all routers to meet the requirement?

- A. crypto ipsec transform-set T-SET esp-aes 128 esp-sha-hmac  
mode mGRE
- B. crypto ipsec transform-set T-SET esp-aes 128 esp-sha-hmac  
mode tunnel
- C. crypto ipsec transform-set T-SET esp-aes 128 esp-sha-hmac  
mode ipsec ipv4
- D. crypto ipsec transform-set T-SET esp-aes 128 esp-sha-hmac  
mode transport

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

R31(config)#crypto ipsec transform-set T-SET esp-aes 128 esp-sha-hmac

R31(cfg-crypto-trans)#mode ?

transport transport (payload encapsulation) mode

tunnel tunnel (datagram encapsulation) mode

R31(cfg-crypto-trans)#

Although the underlay is Internet, the IPsec connection is formed within the tunnel of network 192.168.200.0/24. Hence, you can use IPsec transport mode to reduce the bytes overhead.

#### QUESTION 653

A company is redesigning WAN infrastructure so that all branch sites must communicate via the head office and the head office can directly communicate with each site independently. The network engineer must configure the head office router by considering zero-touch technology when adding new sites in the same WAN infrastructure. Which configuration must be applied to the head office router to meet this requirement?

- A. interface Tunnel0  
tunnel mode ip  
ip nhrp map multicast dynamic
- B. interface Tunnel0  
tunnel mode dvmrp  
ip nhrp redirect
- C. interface Tunnel0  
tunnel mode gre multipoint  
ip nhrp map multicast dynamic
- D. interface Tunnel0  
tunnel mode ip  
ip nhrp redirect

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

For using zero-touch technology, you can configure DMVPN. The tunnel mode for DMVPN is "gre multipoint".

#### QUESTION 654

```
!
router eigrp 1
 redistribute ospf 100 route-map ospf-to-eigrp
 default-metric 20000 2000 255 1 1500

!
Output suppressed.

!
route-map ospf-to-eigrp deny 10
 match tag 6
 match route-type external type-2
!
route-map ospf-to-eigrp permit 20
 match ip address prefix-list pfx
!
route-map ospf-to-eigrp permit 30
 set tag 8
```

Which action fixes the OSPF routes redistribution into EIGRP?

- A. Set a default metric in the route map
- B. Set tags before matching into EIGRP
- C. Match external type to type-1
- D. Match OSPF and EIGRP IDs

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The configuration is correct and there is nothing to fix. The answer is suggested just by eliminating incorrect choices. Setting tags in sequence 20 is recommended if there is another router redistributing EIGRP routes back to OSPF.

Other choices are incorrect since:

"default-metric" is already configured in Router Configuration mode, there is no need to configure / change the metric in the route map.

Since there is no information about the OSPF routes learnt by this router, therefore preventing the redistributing of OSPF E2 routes (instead of E1 routes) can be a valid setup

There is no need to match the OSPF process ID and EIGRP AS number.

**Remarks:**

The configuration is correct since similar configuration appears in Cisco documentation:

<https://www.cisco.com/c/en/us/support/docs/ip/border-gateway-protocol-bgp/491111-route-map-bestp.html#toc-h1d-351514541>

#### QUESTION 655

How are CE advertised routes segmented from other CE routers on an MPLS PE router?

- A. with a combination of VRF-Lite and MP-BGP
- B. by enabling multiple instances of BGP, one for each CE router
- C. by pushing MPLS labels advertised by LDP on customer routes
- D. by assigning CE-facing interfaces to different VRFs

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 656

### R3 BGP Configs

```
router bgp 300
no synchronization
bgp always-compare med
bgp log-neighbor-changes
neighbor 1.1.1.1 remote-as 100
neighbor 1.1.1.1 route-map R1 in
neighbor 2.2.2.2 remote-as 200
neighbor 2.2.2.2 route-map R2 in
no auto-summary
```



R3 is dual-homed to two service providers for traffic redundancy. R3 prefers its outbound traffic via R2. Which set of configurations achieves this goal?

- A. route-map R1 permit 10  
set metric 200  
!  
route-map R2 permit 10  
set metric 100
- B. route-map R1 permit 10  
set AS-Path Prepend 200  
!  
route-map R2 permit 10  
set AS-Path Prepend 100
- C. route-map R1 permit 10  
set metric 100  
!  
route-map R2 permit 10  
set metric 200
- D. route-map R1 permit 10  
set AS-Path Prepend 100  
!  
route-map R2 permit 10  
set AS-Path Prepend 200

**Correct Answer:** A

**Section:** (none)

**Explanation**

#### Explanation/Reference:

Since the path with the lower MED is preferred, the router will prefer routes from R2 if they have a MED of 100 and the routes from R1 have a MED of 200. Note that although the routes are from different ASes, this works since "bgp always-compare-med" has been included in the BGP configuration. Note that prepending the same amount of AS number to the routes from both routers has no effect on path selection.

### QUESTION 657

An engineer must configure encrypted packets for a single router OSPF neighborhood. Which configuration meets this requirement?

- A. interface Ethernet0/2  
ip ospf authentication-key exam
- B. router ospf 100  
area 0 authentication  
!  
interface Ethernet0/2  
ip ospf authentication-key exam
- C. interface Ethernet0/2  
ip ospf authentication message-digest  
ip ospf message-digest-key 1 md5 exam
- D. router ospf 100  
area 0 authentication message-digest-key 1 md5 exam  
!  
interface Ethernet0/2  
ip ospf message-digest-key 1 md5 exam

**Correct Answer:** C

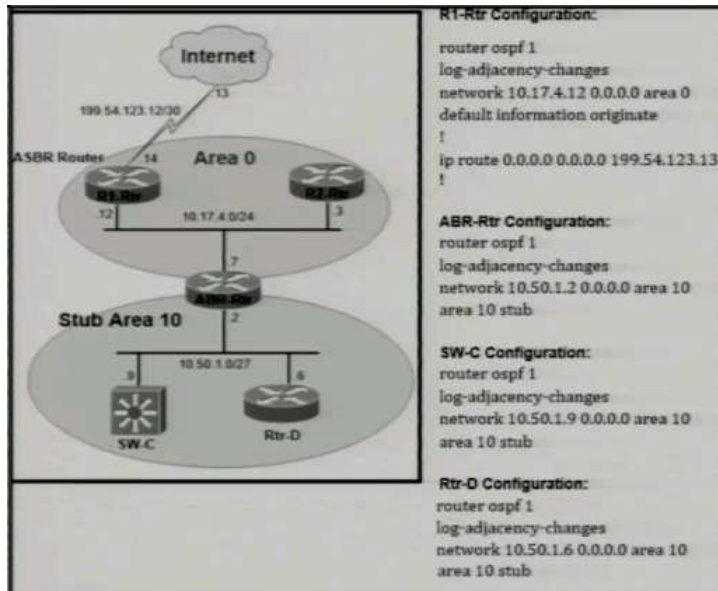
**Section:** (none)

**Explanation**

#### Explanation/Reference:

Since the question mentions "single router OSPF neighborhood", it should be configured in interface only instead of configuring authentication in the whole area 0.

### QUESTION 658



Router ABR-Rtr is not propagating the Internet routes in OSPF area 10, which causes Internet reachability problems in the area. Which action resolves the issue?

- A. ABR-Rtr must advertise the 0.0.0.0/0 default route in area 10.
- B. ABR-Rtr must configure the **area 10 stub no-summary** command.
- C. ABR-Rtr must configure the **default-information originate always** command.
- D. ABR-Rtr network type must be broadcast network

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

By default, the ABR will advertise a default route to the stub area. However, according to the configuration shown, ABR-Rtr is not configured as a ABR since it has no setup for Area 0. In order for ABR-Rtr to advertise the default route, the command "network 10.17.4.7 0.0.0.0 area 0" needs to be configured.

#### QUESTION 659

While BGP Internet routes are redistributed to a lower class of router via RIP, packets are being dropped and routes are failing to be distributed in RIP. Which action resolves the issue?

- A. Use WFQ in the output queue of the high-performance router.
- B. Use RIP V2 to be able to use classless networks from BGP.
- C. Use the **input-queue** command to prevent the loss of packets.
- D. Use OSPF instead of RIP to accept all BGP routes.

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 660

Which IP precedence value does BFD use to prioritize traffic within an infrastructure device?

- A. 4
- B. 5
- C. 6
- D. 7

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

BFD packets carries precedence of 7, and get prioritized by default.

#### QUESTION 661

```

R1
ip as-path access-list 1 deny 65412_$
ip as-path access-list 1 permit .*

router bgp 64560
 neighbor 10.10.10.10 remote-as 64570
 neighbor 10.10.10.10 route-map FILTER in

route-map FILTER permit 10
 match as-path 1

R1#show ip bgp

```

| Network        | Next Hop    | Metric | LocPrf | Weight | Path          |
|----------------|-------------|--------|--------|--------|---------------|
| *> 10.0.0.0/8  | 10.10.10.10 | 0      | 100    | 0      | 64570 i       |
| *> 10.1.0.0/16 | 10.10.10.10 | 0      | 100    | 0      | 64570 i       |
| *> 10.1.1.0/24 | 10.10.10.10 | 0      | 100    | 0      | 64570 65412 i |
| *> 10.1.2.0/24 | 10.10.10.10 | 0      | 100    | 0      | 64570 65412 i |
| *> 10.1.3.0/24 | 10.10.10.10 | 0      | 100    | 0      | 64570 65412 i |
| *> 10.1.4.0/24 | 10.10.10.10 | 0      | 100    | 0      | 64570 65412 i |
| *> 10.1.5.0/24 | 10.10.10.10 | 0      | 100    | 0      | 64570 65412 i |
| *> 10.1.6.0/24 | 10.10.10.10 | 0      | 100    | 0      | 64570 65412 i |

An engineer must filter prefixes that originate from AS65412, but it is not working correctly. Which configuration must the engineer apply to R1 to resolve the issue?

- A. no ip as-path access-list 1  
ip as-path access-list 1 deny 65412\_  
ip as-path access-list 1 permit .\*
- B. no ip as-path access-list 1  
ip as-path access-list 1 deny \_65412\$  
ip as-path access-list 1 permit .\*
- C. route-map FILTER permit 10  
match as-path 1  
route-map FILTER permit 20
- D. router bgp 64560  
neighbor 10.10.10.10 route-map FILTER out

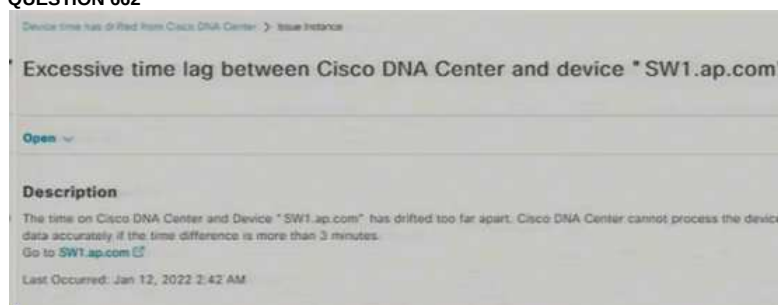
**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

#### QUESTION 662



Which action resolves the issue?

- A. Configure the local time on Cisco Catalyst Center (formerly DNA Center).
- B. Establish connectivity between the NTP server and the switch.
- C. Configure the local time on the SW1 device.
- D. Establish connectivity between the NTP server and Cisco Catalyst Center (formerly DNA Center).

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The reason for the issue is due to the time in the switch "SW1" is wrong, you should configure and check the NTP settings / connectivity in the switch. Note that if the time in the DNA is having problem, time lag errors will occur for many devices.

#### QUESTION 663

```

R1#show ip rip database
10.0.0.0/8 auto-summary
10.1.1.0/24 directly connected, GigabitEthernet0/0
10.1.3.0/24
[2] via 10.1.12.2, 00:00:03, GigabitEthernet1/0
10.1.12.0/24 directly connected, GigabitEthernet1/0
10.1.23.0/24
[1] via 10.1.12.2, 00:00:03, GigabitEthernet1/0

```

A customer reports that networks in the 10.0.1.0/24 space do not appear in the RIP database. What action resolves the issue?



- A. Permit 10.0.1.0/24 address in the ACL.
- B. Configure 10.0.1.0/24 network under RIP.
- C. Remove summarization of 10.0.0.0/8.
- D. Remove ACL on R1 blocking 10.0.1.0/24 network.

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

Since other routes for /24 subnets of 10.0.0.0/8 can be found in the database, summarization is NOT enabled. the reason may be the route is blocked by the "distribute-list" command. In this case, you should add a permit rule to allow it in the ACL specified by the "distribute-list" command.

Remarks:

Removing the entire ACL is wrong.

#### QUESTION 664

Which type of ports are protected by IPv6 Source Guard?

- A. Layer 3 ports
- B. access ports
- C. trunk ports
- D. Layer 2 ports

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

IPv6 Source Guard and Prefix Guard are layer 2 features that are supported only on access or trunk ports, and L2 VLAN configurations.

#### QUESTION 665

```
flow exporter EXPORTER-1
 destination 172.16.10.2
 export-protocol netflow-v9
 transport udp 90
 exit
!
flow record v4_r1
 match ipv4 tos
 match ipv4 protocol
 match ipv4 source address
 match ipv4 destination address
 match transport source-port
 match transport destination-port
 collect counter bytes long
 collect counter packets long
!
flow monitor FLOW-MONITOR-1
 record v4_r1
!
ip cef
!
interface GigabitEthernet 0/0/0
 ip address 172.16.6.2 255.255.255.0
 ip flow monitor FLOW-MONITOR-1 input
!
```

An engineer configured NetFlow to capture traffic information through the router, but it is not working as expected. Which action captures the flow information from this router to the collector?

- A. Change the flow exporter transport protocol from UDP to TCP.
- B. Configure a flow exporter under flow monitor FLOW-MONITOR-1.
- C. Change the interface configuration FLOW-MONITOR-1 from input to output.
- D. Configure more than one flow exporter destination addresses.

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The following shows an example:

```
flow monitor FLOW-MONITOR-1
 record v4_r1
 exporter EXPORTER-1
```

#### QUESTION 666

Which feature is required for IPv6 Source Guard to block traffic arriving on the server interface from unknown sources?

- A. Dynamic ARP Inspection
- B. IPv6 RA Guard
- C. DHCPv6 Guard
- D. IPv6 ND Inspection

**Correct Answer:** D

**Section:** (none)

**Explanation**

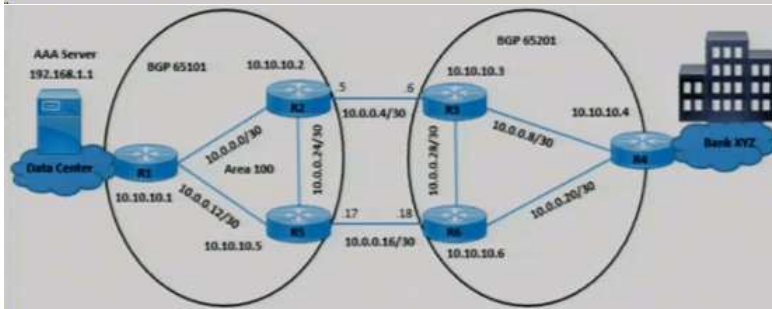
**Explanation/Reference:**

IPv6 source guard does not inspect ND or DHCP packets; rather, it works in conjunction with IPv6 neighbor discovery (ND) inspection or IPv6 address glean, both of which detect existing addresses on the link and store them into the binding table.

#### QUESTION 667

```
R1#show ip bgp 10.10.10.4/32
BGP routing table entry for 10.10.10.4/32, version 21
Paths: (2 available, best #2, table Default-IP-Routing-Table)
Not advertised to any peer
65201
 10.10.10.5 (metric 2) from 10.10.10.5 (10.10.10.5)
 Origin IGP, metric 0, localpref 100, valid, internal
65201
 10.10.10.2 (metric 2) from 10.10.10.2 (10.10.10.2)
 Origin IGP, metric 0, localpref 100, valid, internal, best

R4#show ip bgp 192.168.1.1/32
BGP routing table entry for 192.168.1.1/32, version 24
Paths: (2 available, best #1, table Default-IP-Routing-Table)
Not advertised to any peer
65101
 10.10.10.6 (metric 156160) from 10.10.10.6 (10.10.10.6)
 Origin IGP, metric 0, localpref 100, weight 32769, valid, internal, best
65101
 10.10.10.3 (metric 2) from 10.10.10.3 (10.10.10.3)
 Origin IGP, metric 0, localpref 100, valid, internal
```



A customer reports that user traffic of bank XYZ to the AAA server is not using the primary path via the R3-R2 link. The network team observes:

- No fiber is cut on links R2 and R3.
- AS101 and AS 201 routers established BGP peering.

Which configuration resolves the issue?

- A. R6(config)#router bgp 65201  
R6(config-router)#no neighbor 10.10.10.5 weight 32769
- B. R4(config)#router bgp 65201  
R4(config-router)#no neighbor 10.10.10.6 weight 32769
- C. R1(config)#route-map BGP-Path permit 10  
R1(config-route-map)# set local-preference 200  
R1(config)#router bgp 65101  
R1(config-router)# neighbor 10.10.10.2 route-map BGP-Path out
- D. R2(config)#route-map BGP-Path permit 10  
R2(config-route-map)# set metric 200  
R2(config)#router bgp 65101  
R2(config-router)# neighbor 10.10.10.3 route-map BGP-Path out

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

In the output of R4, the routes learnt from R6 has a weight 32769 and therefore the path through R6 is preferred. Since weight is not advertised, this weight setting must be configured in R4. When there is no further information about the BGP configuration in other routers, removing the weight setting for R6 in R4 router seems to be the logical solution.

Remarks:

Setting Local Preference in R1 does not help since it only affects the routes in the routers in AS65101.

Setting Metric 200 in R2 does not help. Since 200 is higher than the default metric of 100 and this makes routers in AS65201 to avoid the using of the link to R2.

#### QUESTION 668

```

Nov 30 10:19:29.595: IP: tableid=0, s=172.17.132.95 (GigabitEthernet0/0), d=192.168.1.1 (FastEthernet0/0), routed via RIB
Nov 30 10:19:29.595: IP: s=172.17.132.95 (GigabitEthernet0/0), d=192.168.1.1 (FastEthernet0/0), len 92, rcvd 3
Nov 30 10:19:29.603: IP: tableid=0, s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), routed via FIB
Nov 30 10:19:29.603: IP: s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), len 92, sending
Nov 30 10:19:29.607: IP: tableid=0, s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), routed via FIB
Nov 30 10:19:29.611: IP: s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), len 92, sending
Nov 30 10:19:29.615: IP: tableid=0, s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), routed via FIB
Nov 30 10:19:29.615: IP: s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), len 92, sending
Nov 30 10:19:29.723: IP: tableid=0, s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), routed via FIB
Nov 30 10:19:29.727: IP: s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), len 100, sending
Nov 30 10:19:29.739: IP: tableid=0, s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), routed via FIB
Nov 30 10:19:29.743: IP: s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), len 40, sending
Nov 30 10:19:29.807: IP: tableid=0, s=172.17.132.95 (GigabitEthernet0/0), d=192.168.1.1 (FastEthernet0/0), routed via RIB
Nov 30 10:19:29.807: IP: s=172.17.132.95 (GigabitEthernet0/0), d=192.168.1.1 (FastEthernet0/0), len 40, rcvd 3
Nov 30 10:19:29.815: IP: tableid=0, s=172.17.132.95 (GigabitEthernet0/0), d=192.168.1.1 (FastEthernet0/0), routed via RIB
Nov 30 10:19:29.815: IP: s=172.17.132.95 (GigabitEthernet0/0), d=192.168.1.1 (FastEthernet0/0), len 40, rcvd 3
Nov 30 10:19:29.819: IP: tableid=0, s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), routed via FIB
Nov 30 10:19:29.829: IP: s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), len 40, sending
Nov 30 10:19:29.827: IP: tableid=0, s=172.17.132.95 (GigabitEthernet0/0), d=192.168.1.1 (FastEthernet0/0), routed via RIB
Nov 30 10:19:29.831: IP: s=172.17.132.95 (GigabitEthernet0/0), d=192.168.1.1 (FastEthernet0/0), len 40, rcvd 3
Nov 30 10:19:29.831: IP: tableid=0, s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), routed via FIB
Nov 30 10:19:29.835: IP: s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0), len 40, sending

```

Users are experiencing slow response times from critical application servers connected to the network. The engineer determined from the console log that there is a DOS attack on the interface GigabitEthernet0/0. Which action must the engineer take to resolve the issue?

- A. Restrict CoPP traffic on the management interface from 172.17.132.95 and 192.168.1.1.
- B. Configure an access group on GigabitEthernet0/0 to block traffic from 192.168.1.1.
- C. Configure QoS on FastEthernet0/0 to restrict bandwidth usage for 172.17.132.95 and 192.168.1.1.
- D. Configure an access group on GigabitEthernet0/0 to block traffic from 172.17.132.95.

**Correct Answer:** D

**Section:** (none)

**Explanation**

**Explanation/Reference:**

From the console log:

For the 2nd entry "s=172.17.132.95 (GigabitEthernet0/0), d=192.168.1.1 (FastEthernet0/0)" having the word "rcvd", it means that a packet with source IP "192.168.0.2" is received by the router's interface g0/0. The packet is targeted to the IP address 192.168.1.1 which is an IP address of the router's f0/0 interface.

For the 3rd entry "s=192.168.1.1 (local), d=172.17.132.95 (GigabitEthernet0/0)" having the word "sending", it means that the router is sending the reply packet with source IP "192.168.1.1" which is the local IP address of the router. The reply packet is sending to the destination "172.17.132.95" out through the interface g0/0.

Hence, the DOS attack is from "172.17.132.95" entering the router through g0/0. Hence, it can be stopped by blocking the traffic from this IP on the interface g0/0.

#### QUESTION 669

```

R1#
interface Loopback0
ip address 100.1.1.1 255.255.255.255
|
interface Loopback10
ip address 10.100.1.10 255.255.255.255
|
interface Loopback20
ip address 10.100.1.20 255.255.255.255
|
interface Loopback30
ip address 10.100.1.30 255.255.255.255
|
interface Loopback40
ip address 10.100.1.40 255.255.255.255
|
interface Loopback50
ip address 10.100.1.50 255.255.255.255
|
interface FastEthernet0/0
ip address 10.1.1.1 255.255.255.252
ip authentication mode eigrp 100 md5
ip authentication key-chain eigrp 100 EIGRPKEY
ip summary-address eigrp 100 10.100.1.0 255.255.255.0 5 leak-map LOOPBACK50
|
router eigrp 100
network 10.1.1.0 0.0.0.3
network 10.1.1.12 0.0.0.3
neighbor 10.1.1.2 FastEthernet0/0
neighbor 10.1.1.14 FastEthernet1/0
default metric 1000000 10 255 1 1500

no auto-summary
|
ip prefix-list LOOPBACK50 seq 5 permit 10.100.1.50/32
|
route-map LOOPBACK50 permit 10
match ip address prefix-list LOOPBACK50

```

An engineer configured summarization for the R1 loopback addresses and failed. Which action permits the successful advertisement of the R1 loopback addresses?

- A. Configure EIGRP 100 with a network statement for loopback 0 and redistribute the connected route for loopback 50.
- B. Configure EIGRP 100 with a network statement for loopback 0 and remove the leak map for loopback 50.
- C. Configure 100.1.1.1 permit statement in the prefix list LOOPBACK50 and redistribute the connected route for loopback 50.
- D. Configure EIGRP 100 with a network statement for loopback 0 and configure and redistribute the static route for loopback 50.

**Correct Answer:** A

**Section:** (none)

**Explanation**

**Explanation/Reference:**

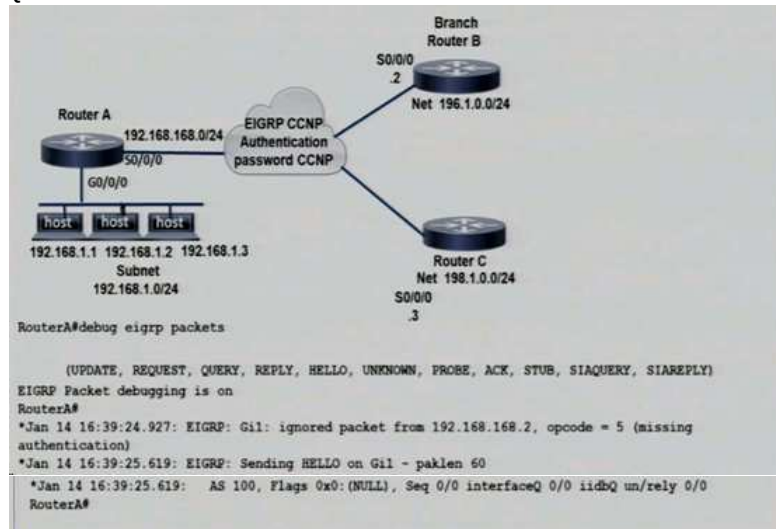
In order to advertise all loopback addresses of R1, the followings are required:

- For loopback 0, you need to include it by using network command or redistribute connected route for it into EIGRP.

- For sending the configured summary route "10.100.1.0/24", you need to include at least one of the other loopbacks by using network command or redistribute

connected route for it into EIGRP. (However, since there is a leak map for leaking lo5, the loopback chosen should be lo5). Only one choice satisfies both requirements.

#### QUESTION 670



The services at branch B are down. An engineer notices that router A and router B are not exchanging any routes. Which configuration resolves the issue on router B?

- A. router eigrp 100  
network 192.168.168.0  
  
key chain EIGRP  
key 1  
key-string CCNP  
interface serial0/0/0  
ip address 192.168.168.2 255.255.255.0  
ip authentication mode eigrp 100 md5  
negotiation auto
- B. router eigrp 100  
network 192.168.168.0  
  
key chain EIGRP  
key 1  
key-string CCNP  
interface serial0/0/0  
ip address 192.168.168.2 255.255.255.0  
ip authentication mode eigrp 100 md5  
ip authentication key-chain eigrp 100 EIGRP  
negotiation auto
- C. router eigrp 100  
network 192.168.168.0  
  
key chain CCNP  
key 1  
key-string EIGRP  
  
interface serial0/0/0  
ip address 192.168.168.2 255.255.255.0  
ip authentication mode eigrp 100 md5  
ip authentication key-chain eigrp 100 EIGRP  
negotiation auto
- D. router eigrp 100  
network 192.168.168.0  
  
key chain EIGRP  
key 1  
key-string CCNP  
  
interface serial0/0/0  
ip address 192.168.168.2 255.255.255.0  
ip authentication key-chain eigrp 100 EIGRP  
negotiation auto

**Correct Answer:** B

**Section:** (none)

**Explanation**

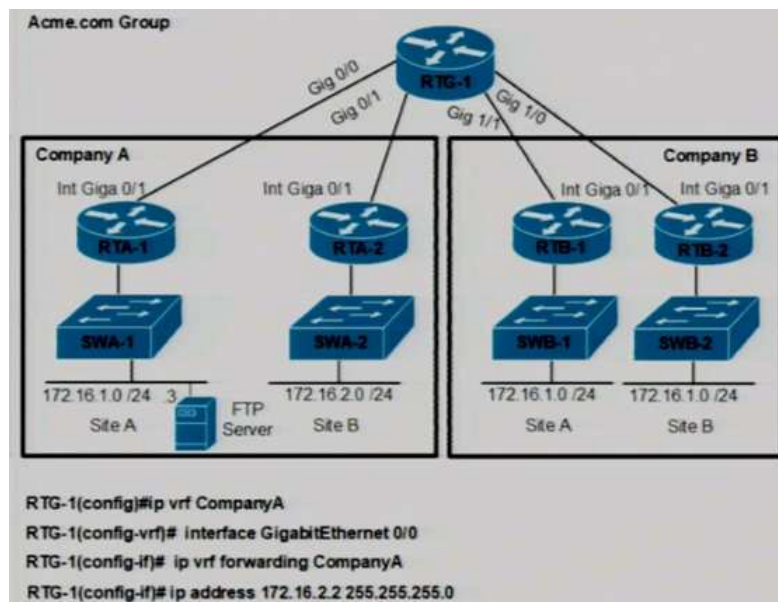
#### Explanation/Reference:

The correct choice should include the followings:

- A key chain that is configured with "key-string CCNP"
- The interface should be configured to use the name of the key chain (i.e. not the key string CCNP) for authentication.
- The interface should be enabled to use EIGRP MD5 authentication (plain text authentication is not supported in EIGRP).

Only one choice meets all the 3 requirements mentioned above.

#### QUESTION 671



An engineer must configure a per VRF for TACACS+ for company A. Which configuration on RTG-1 accomplishes the task?

- A. `aaa new-model`  
`aaa group server tacacs+ Tacacscluster`  
`server-private 172.16.1.3 port 49 key routing`  
`ip tacacs source-interface GigabitEthernet 0/0`  
`ip vrf CompanyA`
- B. `aaa new-model`  
`aaa group server tacacs+ Tacacscluster`  
`server-private 172.16.1.1 port 49 key routing`  
`ip tacacs source-interface GigabitEthernet 0/0`  
`ip vrf forwarding CompanyA`
- C. `aaa new-model`  
`aaa group server tacacs+ Tacacscluster`  
`server-private 172.16.1.3 port 49 key routing`  
`ip tacacs source-interface GigabitEthernet 0/1`  
`ip vrf forwarding CompanyA`
- D. `aaa new-model`  
`aaa group server tacacs+ Tacacscluster`  
`server-private 172.16.1.1 port 49 key routing`  
`ip tacacs source-interface GigabitEthernet 0/1`  
`ip vrf CompanyA`

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**

The command for configuring an AAA server for "per VRF AAA services" is "ip vrf forwarding <VRF name>". Moreover, the interface assigned to that VRF should be configured as the source interface.

Since the interface g0/0 is assigned to the VRF is "CompanyA", therefore the following two are required in the answer:

```

ip tacacs source-interface GigabitEthernet 0/0
ip vrf forwarding CompanyA

```

#### QUESTION 672

How many labels are present in an MPLS Layer 3 packet traversing through the network without traffic engineering?

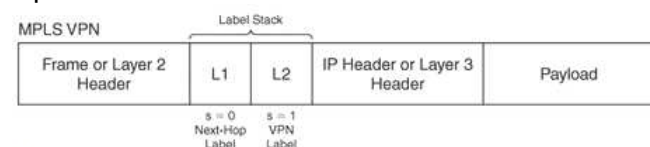
- A. 1
- B. 2
- C. 3
- D. 4

**Correct Answer:** B

**Section:** (none)

**Explanation**

**Explanation/Reference:**



#### QUESTION 673

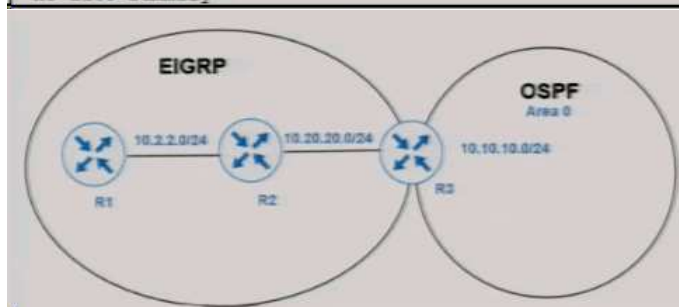


```

R2#show ip eigrp topology 10.10.10.0 255.255.255.0
IP-EIGRP (AS 1): Topology entry for 10.10.10.0/24
 State is Passive, Query origin flag is 1, 1 Successor(s), FD
 is 256005120
 Routing Descriptor Blocks:
 10.20.20.3 (FastEthernet0/1), from 10.20.20.3, Send flag is
 0x0
 Composite metric is (256005120/256002560), Route is
 External
 Vector metric:
 Minimum bandwidth is 10 Kbit
 Total delay is 200 microseconds
 Reliability is 10/255
 Load is 10/255
 Minimum MTU is 10
 Hop count is 1
 External data:
 Originating router is 10.1.1.1
 AS number of route is 1
 External protocol is OSPF, external metric is 0
 Administrator tag is 0 (0x00000000)

R1#sh run | s eigrp
router eigrp 1
router-id 10.1.1.1
network 10.2.2.0 0.0.0.255
no auto-summary

```



An engineer configured router R3 to redistribute the prefix 10.10.10.0/24 from OSPF into EIGRP. R1 has no connectivity to the prefix. Which action enables receipt of prefixes on R1?

- A. R1 does not have a neighbor relationship with R2. The EIGRP should be clear on R1.
- B. Duplicate router IDs on R1 and R3. R1 should modify its router ID.
- C. R3 is advertising the 10.20.20.0/24 prefix with a TTL of 1. R3 must set the TTL to 2 for this prefix.
- D. R1 is not receiving the next-hop IP address of R3. R2 must enable the network 10.20.20.0/24 within EIGRP.

**Correct Answer:** B

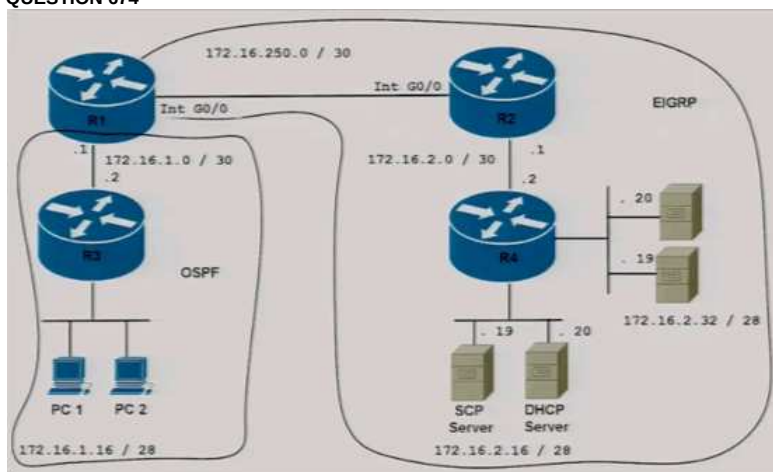
**Section:** (none)

**Explanation**

**Explanation/Reference:**

From the output for the route 10.10.10.0/24 in R2's topology, the originating router is 10.1.1.1 but this route is received from 10.20.20.3 (i.e. R3). This means that R3 has a router ID "10.1.1.1" which is the same as that of R1 (shown in the output of "sh run" in R1).

#### QUESTION 674





```

R1#show run | begin router eigrp 100
router eigrp 100
network 172.16.250.0 0.0.0.3
redistribute ospf 10
!
router ospf 10
redistribute eigrp 100 subnets
network 172.16.1.0 0.0.0.3 area 0
!
ip forward-protocol nd
!
!
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static
 route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 * - replicated route, % - next hop override, p - overrides from PF
Gateway of last resort is not set

 172.16.0.0/16 is variably subnetted, 13 subnets, 3 masks
C 172.16.1.0/30 is directly connected, GigabitEthernet0/1
L 172.16.1.2/32 is directly connected, GigabitEthernet0/1
L 172.16.1.16/28 is directly connected, Loopback1
L 172.16.1.17/32 is directly connected, Loopback1
C 172.16.1.32/28 is directly connected, Loopback2
L 172.16.1.33/32 is directly connected, Loopback2
S 172.16.1.48/28 [1/0] via 172.16.1.18
O E2 172.16.2.0/30 [110/20] via 172.16.1.1, 00:12:20, GigabitEthernet0/1
O E2 172.16.2.16/28 [110/20] via 172.16.1.1, 00:12:20, GigabitEthernet0/1
O E2 172.16.2.32/28 [110/20] via 172.16.1.1, 00:12:20, GigabitEthernet0/1
O E2 172.16.2.48/28 [110/20] via 172.16.1.1, 00:12:20, GigabitEthernet0/1
O E2 172.16.250.0/30 [110/20] via 172.16.1.1, 00:12:20, GigabitEthernet0/1
R#

```

```

R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PF
Gateway of last resort is not set

 172.16.0.0/16 is variably subnetted, 8 subnets, 3 masks
C 172.16.2.0/30 is directly connected, GigabitEthernet0/1
L 172.16.2.2/32 is directly connected, GigabitEthernet0/1
C 172.16.2.16/28 is directly connected, Loopback1
L 172.16.2.17/32 is directly connected, Loopback1
C 172.16.2.32/28 is directly connected, Loopback2
L 172.16.2.33/32 is directly connected, Loopback2
S 172.16.2.48/28 [1/0] via 172.16.2.34
D 172.16.250.0/30 [90/3072] via 172.16.2.1, 1d03h, GigabitEthernet0/1
R4#

```

The users from subnet 172.16.1.16/28 are experiencing connectivity issues to subnet 172.16.2.32/48. Which configuration resolves the issue?

- A. R1(config)#route-map REDIST permit 10  
R1(config-route-map)#match ip address 15  
R1(config-route-map)#exit  
R1(config)# access-list 15 permit 172.16.0.0 0.0.255.255  
R1(config-router)#router ospf 1  
R1(config-router)#redistribute eigrp 100 subnets route-map REDIST
- B. R1(config-router)#router ospf 1  
R2(config-router)#redistribute eigrp 100 subnets metric 100
- C. R1(config-router)#router eigrp 100  
R1(config-router)#redistribute ospf 1 metric 1000000 1 1 1 1
- D. R1(config)#route-map REDIST permit 10  
R1(config-route-map)#match ip address 15  
R1(config-route-map)#exit  
R1(config)# access-list 15 permit 172.16.0.0 0.0.255.255  
R1(config-router)#router eigrp 100  
R1(config-router)#redistribute ospf 1 subnets route-map REDIST

**Correct Answer:** C

**Section:** (none)

**Explanation**

**Explanation/Reference:**

For the traffic from "172.16.1.16/28" à "172.16.2.32", there should be no problem since the route "172.16.2.32" is found in the routing table of R3. For the return traffic from "172.16.2.32" à "172.16.1.16/28", there is a problem since the route "172.16.1.16" is NOT found in the routing table of R4.

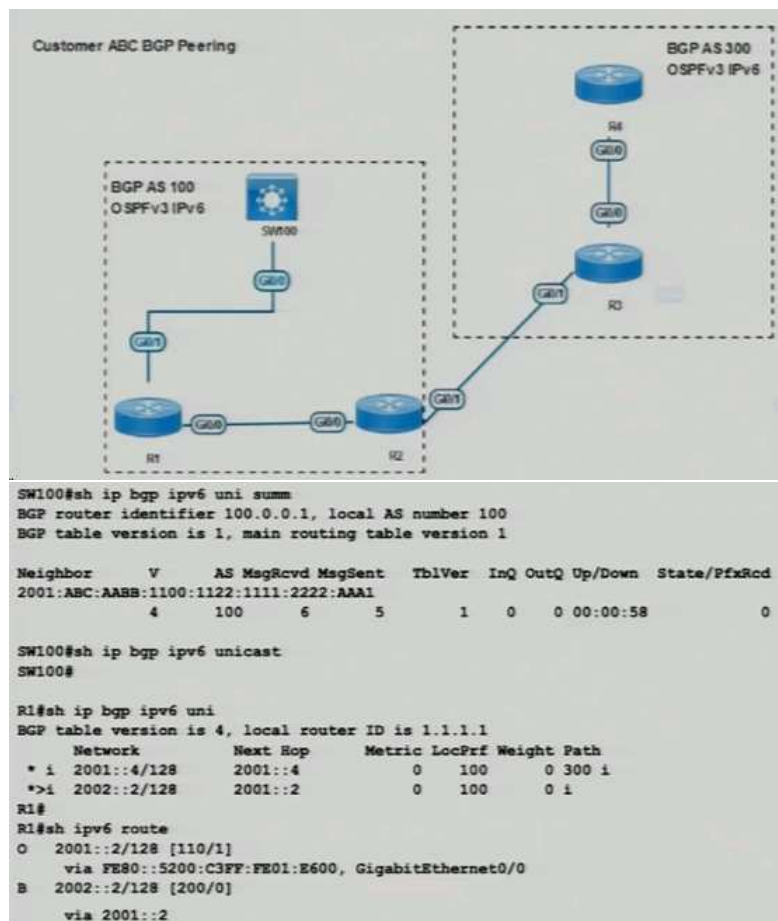
From the "sh run" of R1, under "router eigrp 100", it shows that the configuration settings that redistribute OSPF routes into EIGRP has no metric configured and there is no default metric. Hence, routes (e.g. "172.16.1.16/28") from OSPF cannot be redistributed into EIGRP. you need to configure metric for the redistributing OSPF into EIGRP in R1.

**Remarks:**

The "172.16.2.32/48" in the question text should be "172.16.2.32/28".

The IP addresses of the PCs in the subnet "172.16.2.32/28" within the diagram are wrong. They should be e.g. ".34" and ".35".

**QUESTION 675**



- ```
A. R1
router bgp 100
  address-family ipv6
    redistribute connected

R2
router bgp 100
  address-family ipv6
    redistribute connected

B. R1
router bgp 100
  address-family ipv6
    neighbor 2001::2 route-reflector-client
    neighbor 2001:ABC:AABB:1100:1122:1111:2222:AAA2 route-reflector-client

R2
router bgp 100
  address-family ipv6
    neighbor 2001::2
    neighbor 2001::1 as-override

C. R1
router bgp 100
  address-family ipv6
    neighbor 2001::2 route-reflector-client
    neighbor 2001:ABC:AABB:1100:1122:1111:2222:AAA2 route-reflector-client

R2
router bgp 100
  address-family ipv6
    neighbor 2001::2
    neighbor 2001::1 next-hop-self

D. R1
router bgp 100
  address-family ipv6
    no synchronization

R2
router bgp 100
  address-family ipv6
    no synchronization

SW100
router bgp 100
  address-family ipv6
    no synchronization
```

Section: (none)

Explanation

Explanation/Reference:

In the output of "sh ip bgp" in R1, a route "2001::4/128" originated from AS 300 is learnt. However, this route is not entered into the routing table of R1 and is not learnt by SW100.

The route is not entered into R1's routing table since R1 has no route to the next hop "2001::4" of the route. This can be solved by configuring R2 to send the route to R1 with "next-hop-self".

Moreover, the route will not be learnt by SW100 even if R2 is configured with "next-hop-self" since R1 learns the route from an iBGP peer (i.e. R2) and therefore will not send it to another iBGP peer (i.e. SW100). This can be solved by configuring R1 to be a BGP route reflector.

Therefore both configurations are required.

QUESTION 676

```
R1#show bgp ipv6 unicast 2001:db8::1/128
BGP routing table entry for 2001:db8::1/128, version 3
Paths: (1 available, best #1, table Global-IPv6-Table)
Not advertised to any peer
Local
2001:db8:33:33::33 (metric 128) from 2001:db8:11:11::11 (1.1.1.1)
Origin IGP, metric 0, localpref 100, valid, internal, best
Originator: 3.3.3.3, Cluster list: 1.1.1.1
```

An engineer examines the BGP update for the IPv6 prefix 2001:db8::1/128, which should have been summarized into a /64 prefix. Which sequence of actions achieves the summarization?

- A. R1 is a route reflector with a router ID of 1.1.1.1, and the originator of the prefix is a route reflector client, which has a router ID of 3.3.3.3. Both routers belong to the same AS. Configure an aggregate address on the router with ID 1.1.1.1 for the prefix.
- B. R1 is a route reflector client of a RR with a router ID of 1.1.1.1, and the originator of the prefix has a router ID of 3.3.3.3. Both routers belong to different ASs. The prefix is not advertised to any peer and must be advertised using the network statement on R3.
- C. R1 is a route reflector client of a RR with a router ID of 1.1.1.1, and the originator of the prefix has a router ID of 3.3.3.3. Both routers belong to the same AS. Configure an aggregate address on the router with ID 3.3.3.3 for the prefix.
- D. R1 is a route reflector with a router ID of 3.3.3.3, and the originator of the prefix is a route reflector client, which has a router ID of 3.3.3.3. Both routers belong to the same AS. Configure an aggregate address on the router with ID 1.1.1.1 for the prefix.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

In the output:

All RR and its clients must be in the same AS.

"Cluster list" shows the ID of the route reflector (RR). Hence, the RR has the ID "1.1.1.1".

"Originator" shows the ID of the route reflector client that generates the route. Hence the originating route reflector client is "3.3.3.3".

In order to aggregate the route for all other routers, the best location is the router that originates the route i.e. the reflector client i.e. "3.3.3.3"

Remarks:

R1 is NOT the RR since the word "Cluster list" is only shown in the output of a RR's client. Hence, in this question, R1 is NOT having the router ID "1.1.1.1" and is NOT the RR.

QUESTION 677

What are two features of BFD? (Choose two.)

- A. scalable
- B. reliable
- C. replaces hello messages
- D. requires routing protocols.
- E. intensive on CPU for Layer 2 links

Correct Answer: AB

Section: (none)

Explanation

Explanation/Reference:

BFD can also be used for static route or HSRP and therefore it does not require the running of a routing protocol in order to work.

QUESTION 678

```
R1#sh ip route
C    192.168.10.0/24 is directly connected, Serial1/0
C    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C    172.16.160.0/19 is directly connected, Loopback1
C    172.16.128.0/19 is directly connected, Loopback0
C    172.16.224.0/19 is directly connected, Loopback3
C    172.16.192.0/19 is directly connected, Loopback2
B    172.16.0.0/16 is a summary, 00:01:27, Null0
```

Which configuration advertises more specific routes to R1 without sending a BGP summary route?

- A. R2#configure terminal
R2 (config)#router BGP 100
R2 (config-router)#auto-summary

- B. R1#configure terminal
R1 (config)#router BGP 100
R1 (config-router)#auto-summary
- C. R2#configure terminal
R2 (config)#router BGP 100
R2 (config-router)#no auto-summary
- D. R1#configure terminal
R1 (config)#router BGP 100
R1 (config-router)#no auto-summary

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

In order to disable the sending of summary route automatically, you need "no auto-summary".

For the question text, it said "advertises more specific routes to R1". Hence, the command "no auto-summary" needs to be configured in a router other than R1.

QUESTION 679

```
R1(config)#ipv6 prefix-list PRE-PEND-PREFIX permit 2001:db8:0:a::/64
R1(config)#route-map PRE-PEND permit 10
R1(config-route-map)#match ipv6 address prefix-list PRE-PEND-PREFIX
R1(config-route-map)#set as-path prepend 65412
R1(config)#router bgp 65412
R1(config-router)#address-family ipv6
R1(config-router-af)#neighbor 2001:db8:0:20::2 route-map PRE-PEND out
```

R1 has a route map configured, which results in a loss of partial IPv6 prefixes for the BGP neighbor, resulting in service degradation. How can the full service be restored?

- A. The route map requires a deny 20 statement without set conditions, and this will allow additional prefixes to be advertised.
- B. The neighbor requires a soft reconfiguration, and this will clear the policy without resetting the BGP TCP connection.
- C. The route map requires a permit 20 statement without set conditions, and this will allow additional prefixes to be advertised.
- D. The prefix list requires all prefixes that R1 is advertising to be added to it, and this will allow additional prefixes to be advertised.

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

QUESTION 680

```
R10#ping 10.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/99/152 ms
R10#copy tftp://10.1.1.1/c890-universal-mz.154-2.M3.bin flash:
Destination filename [c890-universal-mz.154-2.M3.bin]?
Accessing tftp://10.1.1.1/c890-universal-mz.154-2.M3.bin...
%Error opening tftp://10.1.1.1/c890-universal-mz.154-2.M3.bin (Timed out)

R11#sh run interface GigabitEthernet 0/0
interface GigabitEthernet0/0
ip vrf forwarding TFTP
ip address 10.1.1.1 255.255.255.0
speed auto
duplex auto
end
```

R10 attempts to copy an image file from R11 using TFTP, and the operation is timing out. Which action resolves the issue?

- A. R11 requires the TFTP server source-interface to be set to GigabitEthernet0/0.
- B. R10 must be configured to belong to the same VRF TFTP.
- C. Change R10 duplex to auto to resolve the duplex mismatch between routers R10 and R11.
- D. R11 requires the TFTP server command for the image to be set to the VRF TFTP.

Correct Answer: A

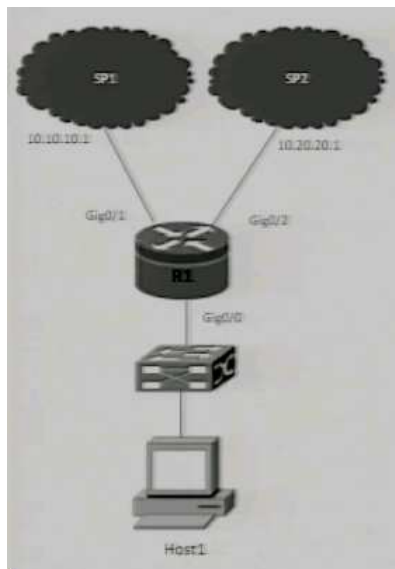
Section: (none)

Explanation

Explanation/Reference:

In order to allow TFTP traffic for a VRF, you need to configure the TFTP source interface with the interface assigned to that VRF.

QUESTION 681



R1 uses SP1 as the primary path. A network engineer must force all SSH traffic generated from R1 toward SP2. Which configuration accomplishes the task?

- A.

```
ip access-list extended match_SSH
permit tcp any any eq 22
!
route-map PBR_SSH permit 10
match ip address match_SSH
set ip next-hop 10.20.20.1
!
interface Gig0/0
ip policy route-map PBR_SSH
```
- B.

```
ip access-list extended match_SSH
permit tcp any any eq 22
!
route-map PBR_SSH permit 10
match ip address match_SSH
set ip next-hop 10.20.20.1
!
ip local policy route-map PBR_SSH
```
- C.

```
ip access-list extended match_SSH
permit tcp any any eq 22
!
route-map PBR_SSH permit 10
match ip address match_SSH
set ip next-hop 10.10.10.1
!
ip local policy route-map PBR_SSH
```
- D.

```
ip access-list extended match_SSH
permit tcp any any eq 22
!
route-map PBR_SSH permit 10
match ip address match_SSH
set ip next-hop 10.20.20.1
!
interface Gig0/1
ip policy route-map PBR_SSH
```

Correct Answer: B

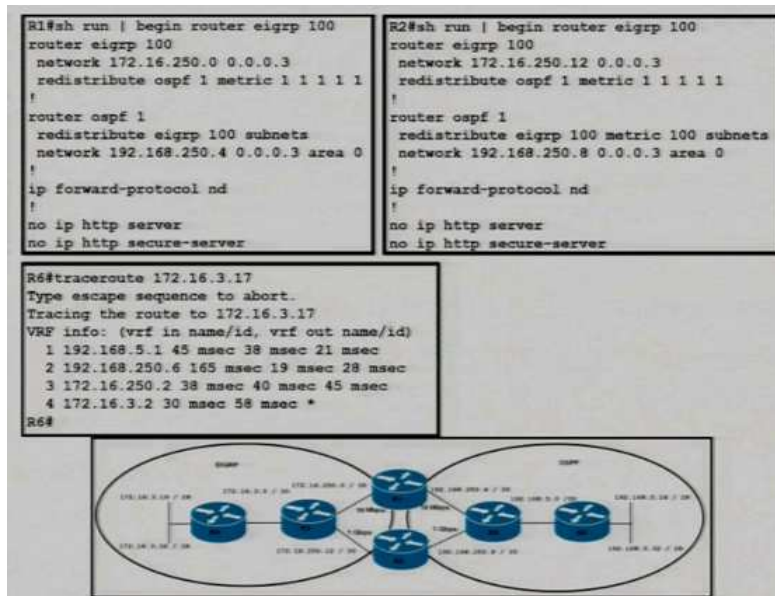
Section: (none)

Explanation

Explanation/Reference:

Since the question specifies the use of SP2 (i.e. the next hop 10.20.20.1) for the SSH traffic generated by R1 itself locally, the command "ip local policy" should be configured with route-map having "set ip next-hop 10.20.20.1".

QUESTION 682



An engineer is troubleshooting suboptimal communication from the 192.168.5.32/28 subnet to the 172.16.3.16/28 segment using the slowest links. Which configuration resolves the suboptimal routing issue?

- A. R1(config-router)#router eigrp 100
R1(config-router)#redistribute ospf 1 metric 1000000 1 1 1 1
- B. R2(config-router)#router ospf 1
R2(config-router)#default-metric 1
R1(config-router)#router ospf 1
R1(config-router)#default-metric 10
- C. R2(config-router)#router eigrp 100
R2(config-router)#redistribute ospf 1 metric 1000000 1 1 1 1
- D. R2(config-router)#router ospf 1
R2(config-router)#default-metric 10
R1(config-router)#router ospf 1
R1(config-router)#default-metric 1

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

From the output of traceroute, the traffic from 192.168.5.32/28 to 172.16.3.16/28 goes through the upper router R1 which links have 10Mbps only. (192.168.250.6 is an IP address of the subnet 192.168.250.4/30 between R1 and R5).

When redistributing from the OSPF route "172.16.3.16/28" into EIGRP, the metric set for the redistributed EIGRP route in both R1 and R2 is 1kbps (i.e. the 1st number "1" after the metric keyword in "redistribute ospf 1 ..." command). Since "1Kbps" is smaller than 10Mbps and 1Gbps, it is used as the minimum bandwidth for calculating the cost and therefore both redistributed routes "172.16.3.16/28" from R1 and R2 has equal cost in R6. Hence, there is a chance that R6 sends traffic to "172.16.3.16/28" through R1.

In order to solve this, R2 should redistribute the OSPF route "172.16.3.16/28" into EIGRP with a higher bandwidth value as metric.

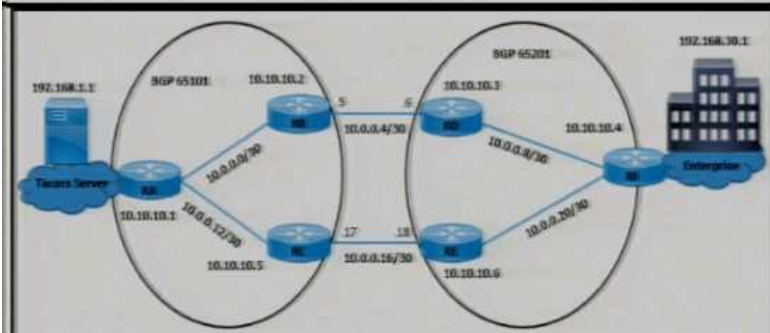
QUESTION 683


```

RF#traceroute 192.168.1.1
 1 10.0.0.9 40 msec 28 msec 24 msec
 2 * * *
 3 * * *
RE#show ip prefix-list detail
Prefix-list with the last deletion/insertion: Customer
ip prefix-list Customer:
  count: 2, range entries: 1, sequences: 5 - 10, refcount: 3
  seq 5 deny 192.168.1.1/32 (hit count: 5, refcount: 1)
  seq 10 permit 0.0.0.0/0 le 32 (hit count: 26, refcount: 1)

RC#show ip prefix-list detail
Prefix-list with the last deletion/insertion: Customer
ip prefix-list Customer:
  count: 1, range entries: 1, sequences: 10 - 10, refcount: 4
  seq 10 permit 0.0.0.0/0 le 32 (hit count: 7, refcount: 1)

```



The enterprise users fail to authenticate with the TACACS server when a direct fiber link fails between RB and RD. The NOC team observes:

- Users connected on AS65201 fail to authenticate with TACACS server 192.168.1.1.
- Users connected on AS65101 successfully authenticate with TACACS server 192.168.1.1.
- All AS65101 and AS65201 users are configured to authenticate with the TACACS server.

Which configuration resolves the issue?

- A. RF(config)#no ip prefix-list Customer seq 5 deny 192.168.1.1/32
- B. RF(config)#router bgp 65201
RF(config-router)# neighbor 10.0.0.17 prefix-list Customer out
- C. RC(config)#ip prefix-list Customer seq 5 permit 192.168.30.1/32
- D. RC(config)#router bgp 65101
RC(config-router)# neighbor 10.0.0.18 prefix-list Customer in

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

A lot of assumptions are required for this question:

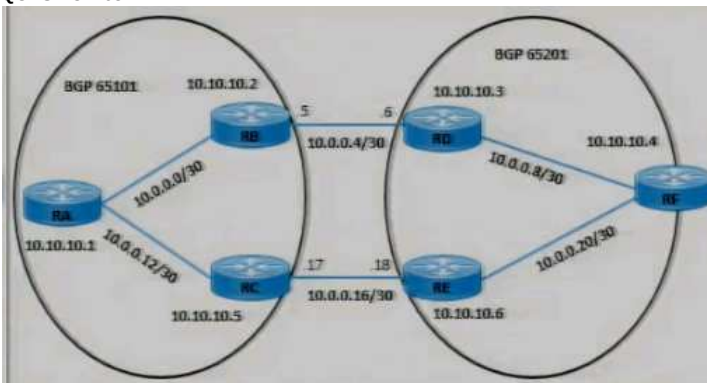
- assume RE use the prefix list to filter routes learnt from RC or advertised to RF. The seq 5 in the prefix list blocks the route 192.168.1.1/32.
- assume RC use the prefix list to filter routes learnt from RA or advertised to RE. This does not block anything.

Hence, you need to remove the seq 5 in the prefix list in RE. However, the only choice of removing seq 5 is in RF. Since all other choices have no effect on the blocking. We just assume that the choice intends to remove seq 5 in RE.

Remarks:

A hint about the choices with "RF" actually refer to the router "RE" is that the IP address "10.0.0.17" (i.e. IP address of "RC") in the "neighbor 10.0.0.17" in one of the choices is actually connected by "RE" instead of "RF".

QUESTION 684



```

RB#
*Sep 19 00:53:43.002: BGPNSF state: 10.10.10.3 went from nsf_not_active to
nsf not active
*Sep 19 00:53:43.006: BGP: 10.10.10.3 went from Established to Idle
*Sep 19 00:53:43.006: %BGP-5-ADJCHANGE: neighbor 10.10.10.3 Down User reset
*Sep 19 00:53:43.006: BGP: 10.10.10.3 closing
*Sep 19 00:53:43.106: BGP_Router: unhandled major event code 128, minor 0

RD#show ip bgp neighbors 10.10.10.2
BGP neighbor is 10.10.10.2, remote AS 65101, external link
  BGP version 4, remote router ID 0.0.0.0
  BGP state = Active
  Last read 00:01:35, last write 00:01:35, hold time is 180, keepalive
interval is 60 seconds
Default minimum time between advertisement runs is 30 seconds
Address tracking is enabled, the RIB does have a route to 10.10.10.2
  Connections established 11; dropped 11
  Last reset 00:01:36, due to Peer closed the session
  External BGP neighbor may be up to 3 hops away.
  Transport(tcp) path-mtu-discovery is enabled
  No active TCP connection

```

A NOC team receives a ticket that data traffic from RA to RF is not forwarded when the link between the RC-RE path goes down. All routers learn loopback IP through the IGP protocol. Which configuration resolves the issue?

- A. RD(config)#router bgp 65201
RD(config-router)#neighbor 10.10.10.2 ebgp-multihop 3
- B. RB(config)#router bgp 65101
RB(config-router)#neighbor 10.10.10.3 update-source loopback 0
- C. RB(config)#router bgp 65101
RB(config-router)#neighbor 10.10.10.3 ebgp-multihop 3
- D. RD(config)#router bgp 65201
RD(config-router)#neighbor 10.10.10.2 update-source loopback 0

Correct Answer: C

Section: (none)

Explanation

Explanation/Reference:

After RC-RE path is down, traffic cannot be sent through RB-RD path since there is a problem in RB-RD peering. It is proved by the output in RB.

From the output of RD, the line "External BGP neighbor may be up to 3 hops away" means that the command "nei 10.10.10.2 ebgp-multihop 3" has been configured in RD.

Hence, the most probable reason is that RB is missing similar "ebgp-multihop" command.

QUESTION 685

A network engineer wants to redirect HTTP traffic originating from subnet 192.168.3.1/24 to the next-hop IP address of 10.2.3.1. This configuration was implemented:

```

route-map PBR permit 10
match ip address 100
set ip next-hop 10.2.3.1
!
interface GigabitEthernet0/0
description - To ISP1 link ID 64:f9bf:3421:5f1f:49ff:9f47:0f9f
ip policy route-map PBR

```

Which command completes the configuration?

- A. Router(config)#access-list 100 permit tcp 192.168.3.0 0.0.0.255 any eq 80
- B. Router(config)#access-list 100 permit tcp 192.168.3.0 0.0.0.255 eq any 80
- C. Router(config)#access-list 100 permit tcp any 192.168.3.0 0.0.0.255 eq 80
- D. Router(config)#access-list 100 permit tcp any eq 80 192.168.3.0 0.0.0.255

Correct Answer: A

Section: (none)

Explanation

Explanation/Reference:

Simulation Questions

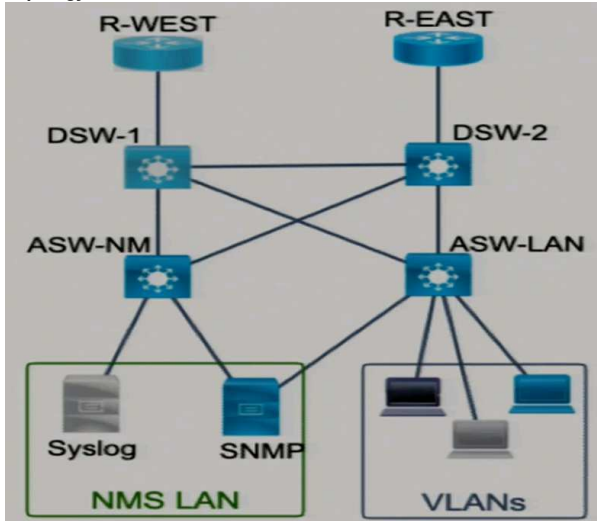
QUESTION 1

Guidelines:

This is a lab item in which tasks will be performed on virtual devices.

- Refer to the Tasks tab to view the tasks for this lab item.
- Refer to the Topology tab to access the device console(s) and perform the tasks.
- Console access is available for all required devices by clicking the device icon or using the tab(s) above the console window.
- All necessary preconfigurations have been applied.
- Do not change the enable password or hostname for any device.
- Do not replace existing routing policies or configurations.
- **Save your configurations to NVRAM before moving to the next item.**
- Click Next at the bottom of the screen to submit this lab and move to the next question.
- When Next is clicked, the lab closes and cannot be reopened.

Topology:



Tasks:

Troubleshoot R-WEST to achieve the desired results:

1. All the commands should be locally saved to the router as well as sent to the Syslog server except passwords.
2. All the Cisco OSPF LSA traps should be sent to the SNMP server.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

R-WEST

```
en
config t
archive
  log config
  logging enable
  hidekeys
  notify syslog
  exit
exit
snmp-server enable traps ospf lsa
end
copy run start
```

Explanations:

The answer is based on the following existing settings found in R-WEST. You should use "sh run" in the router to check them.

R-WEST

```
logging host 10.1.100.14 sequence-num-session
snmp-server community public R0
snmp-server host 10.2.100.161 public
```

Hence, both the Syslog server and the SNMP server host have been configured.

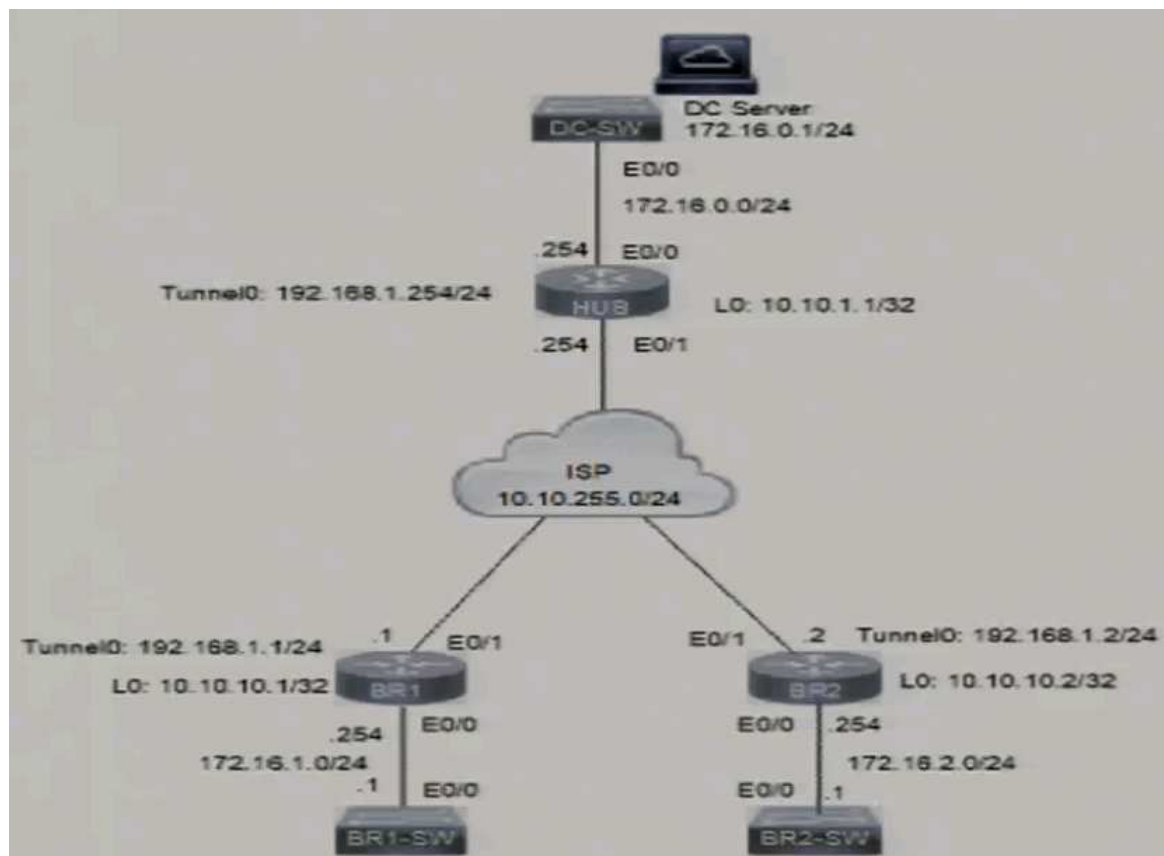
Remarks:

Router(config)#snmp-server enable traps ospf ?

```
cisco-specific  Cisco specific traps
errors          Error traps
lsa             Lsa related traps
rate-limit      Trap rate limit values
retransmit      Packet retransmit traps
state-change    State change traps
<cr>
```

QUESTION 2

Topology:



Tasks:

A DMVPN network is preconfigured with tunnel 0 IP address 192.168.1.254 on the HUB, IP connectivity, crypto policies, profiles, and EIGRP AS 100. The NHRP password is **ccnp123**, and the network ID and tunnel key is **EIGRP ASN**. Do not introduce a static route. Configure DMVPN connectivity between routers BR1 and BR2 to the HUB router using physical interface as the tunnel source to achieve these goals:

1. Configure NHRP authentication, static IP-to-NBMA address maps, hold time 5 minutes, network ID, and server on branch router BR1.
2. Configure NHRP authentication, static IP-to-NBMA address maps, hold time 5 minutes, network ID, and server on branch router BR2.
3. Ensure that packet fragmentation is done before encryption to account for GRE and IPsec header and allow a maximum TCP segment size of 1360 on an IP MTU of 1400 on the tunnel interfaces of both branch routers.
4. Apply an IPsec profile to the tunnel. Verify that direct spoke-to-spoke tunnel is functional between branch routers BR1 and BR2 by using traceroute to Ethernet0/0 IP address to get a full score.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

Task 1:

```
BR1
en
config t
int tu 0
 tunnel mode gre multipoint
 tunnel source e0/1
 tunnel key 100
ip nhrp authentication ccnp123
ip nhrp map 192.168.1.254 10.10.255.254
ip nhrp map multicast 10.10.255.254
ip nhrp holdtime 300
ip nhrp network-id 100
ip nhrp nhs 192.168.1.254
end
copy run start
```

Task 2:

```
BR2
en
config t
int tu 0
 tunnel mode gre multipoint
 tunnel source e0/1
 tunnel key 100
ip nhrp authentication ccnp123
ip nhrp map 192.168.1.254 10.10.255.254
ip nhrp map multicast 10.10.255.254
ip nhrp holdtime 300
ip nhrp network-id 100
```

```
ip nhrp nhs 192.168.1.254
end
copy run start
```

Task 3:

```
BR1
en
config t
int tu 0
ip mtu 1400
ip tcp adjust-mss 1360
end
copy run start
```

```
BR2
en
config t
int tu 0
ip mtu 1400
ip tcp adjust-mss 1360
end
copy run start
```

Task 4:

```
BR1
en
config t
int tu 0
tunnel protection ipsec profile ccnp
ip nhrp shortcut
end
copy run start
```

```
BR2
en
config t
int tu 0
tunnel protection ipsec profile ccnp
ip nhrp shortcut
end
copy run start
```

```
BR1
traceroute 172.16.2.254
```

```
BR2
traceroute 172.16.1.254
```

Explanations:

You should use sh run in the routers to check if the existing configuration settings are the same as those described below:
IPsec settings have been configured in BR1 and BR2.

For BR1 and BR2, the interface Tunnel 0 is setup but the only configuration under the interface is the IP address settings.

```
BR1
int Tunnel 0
ip addr 192.168.1.1 255.255.255.0
```

```
BR2
int Tunnel 0
ip addr 192.168.1.2 255.255.255.0
```

EIGRP configuration in BR1 and BR2 are as follows:

```
BR1
router eigrp 100
network 10.10.10.0 0.0.0.255
network 172.16.1.0 0.0.0.255
network 192.168.1.0
```

```
BR2
router eigrp 100
network 10.10.10.0 0.0.0.255
network 172.16.2.0 0.0.0.255
network 192.168.1.0
```

For Task 1 & 2, since the general requirement asks you to use physical interface as tunnel source, therefore you must use "tunnel source e0/1". You should check the following through "sh run" in the routers and the information mentioned in the general requirements:

Use "sh run" in BR1 and BR2 to check whether EIGRP AS number is "100".

Whether the question specifies that NHRP password is "ccnp123".

Use "sh run" in BR1 and BR2 to check whether the IPSec profile in each router is "ccnp".

For Task 4, we have already configured "tunnel mode gre multipoint" in task 1 and 2. "ip nhrp shortcut" is configured so that a Spoke router can send

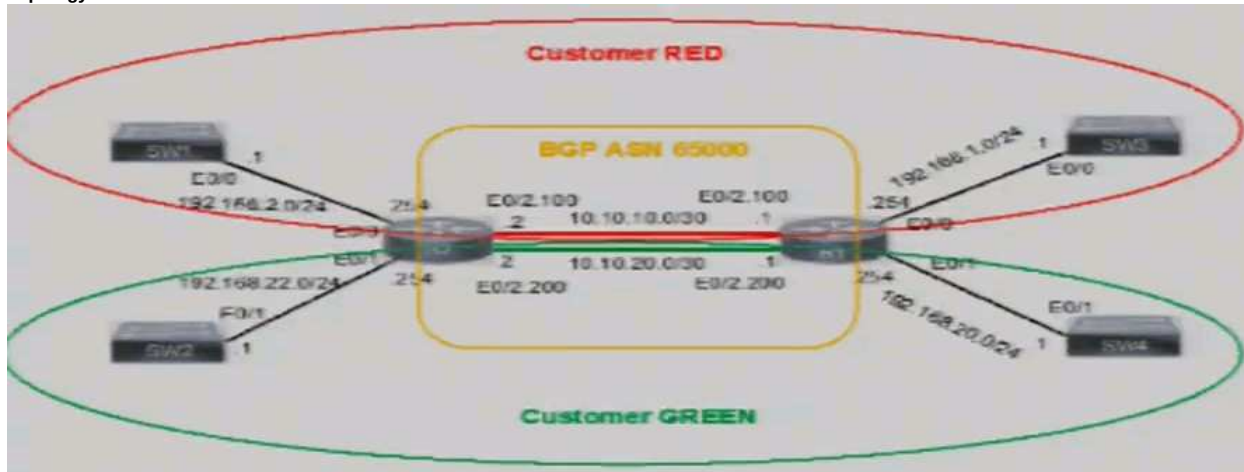
packet directly to the another Spoke router.

For troubleshooting, you can use the following commands:

```
sh ip nhrp
sh ip route
sh ip route eigrp
sh crypto ipsec sa
ping 192.168.1.254
ping 192.168.1.1
ping 192.168.1.2
```

QUESTION 3

Topology:



Tasks:

Configure individual VRFs for each customer according to the topology to achieve these goals:

1. VRF "cu-red" has interfaces on routers R1 and R2. Both routers are preconfigured with IP addressing, VRFs, and BGP. Do not use the BGP network statement for advertisement.
2. VRF "cu-green" has interfaces on routers R1 and R2.
3. BGP on router R1 populates VRF routes between router R1 and R2.
4. BGP on router R2 populates VRF routes between router R1 and R2.
5. LAN to LAN is reachable between SW1 and SW3 for VRF "cu-red" and between SW2 and SW4 for VRF "cu-green". All switches are preconfigured.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

Task 1:

```
R1
en
config t
int e0/0
ip vrf forwarding cu-red
ip address 192.168.1.254 255.255.255.0
int e0/2.100
ip vrf forwarding cu-red
ip addr 10.10.10.1 255.255.255.252
end
copy run start
```

```
R2
en
config t
int e0/0
ip vrf forwarding cu-red
ip address 192.168.2.254 255.255.255.0
int e0/2.100
ip vrf forwarding cu-red
ip addr 10.10.10.2 255.255.255.252
end
copy run start
```

Task 2:

```
R1
en
config t
int e0/1
ip vrf forwarding cu-green
ip address 192.168.20.254 255.255.255.0
int e0/2.200
ip vrf forwarding cu-green
ip addr 10.10.20.1 255.255.255.252
```



```
end
copy run start
```

```
R2
en
config t
int e0/1
    ip vrf forwarding cu-green
    ip address 192.168.22.254 255.255.255.0
int e0/2.200
    ip vrf forwarding cu-green
    ip addr 10.10.20.2 255.255.255.252
end
copy run start
```

Task 3:

```
R1
en
config t
router bgp 65000
    address-family ipv4 vrf cu-red
        neighbor 10.10.10.2 remote-as 65000
        redistribute connected
    exit-address-family
    address-family ipv4 vrf cu-green
        neighbor 10.10.20.2 remote-as 65000
        redistribute connected
    exit-address-family
end
copy run start
```

Task 4:

```
R2
en
config t
router bgp 65000
    address-family ipv4 vrf cu-red
        neighbor 10.10.10.1 remote-as 65000
        redistribute connected
    exit-address-family
    address-family ipv4 vrf cu-green
        neighbor 10.10.20.1 remote-as 65000
        redistribute connected
    exit-address-family
end
copy run start
```

Task 5:

```
SW1
ping 192.168.1.1
```

```
SW3
ping 192.168.2.1
```

```
SW2
ping 192.168.20.1
```

```
SW4
ping 192.168.22.1
```

Explanations:

Since R1 and R2 is using a sub-interface for each VRF, the setup is using VRF-Lite with BGP for VRFs as routing protocol i.e. The setup required does NOT use VPNv4 of MP-BGP.

You should use sh run in the routers to check if the existing configuration settings are the same as those described below:

```
R1
ip vrf cu-green
    rd 65000:200
ip vrf cu-red
    rd 65000:100
int Loopback0
    ip addresss 10.10.1.1 255.255.255.255
int Ethernet0/0
    ip address 192.168.1.254 255.255.255.0
int Ethernet0/1
    ip address 192.168.20.254 255.255.255.0
int Ethernet0/2
    no ip address
int Ethernet0/2.100
    encapsulation dot1Q 100
    ip addr 10.10.10.1 255.255.255.252
int Ethernet0/2.200
    encapsulation dot1Q 200
    ip addr 10.10.20.1 255.255.255.252
router bgp 65000
    bgp log-neighbor-changes
    no bgp default ipv4-unicast
```

```
R2
ip vrf cu-green
```

```

rd 65000:200
ip vrf cu-red
rd 65000:100
int Loopback0
ip addresss 10.10.2.2 255.255.255.255
int Ethernet0/0
ip address 192.168.2.254 255.255.255.0
int Ethernet0/1
ip address 192.168.22.254 255.255.255.0
int Ethernet0/2
no ip address
int Ethernet0/2.100
encapsulation dot1Q 100
ip addr 10.10.10.2 255.255.255.252
int Ethernet0/2.200
encapsulation dot1Q 200
ip addr 10.10.20.2 255.255.255.252
router bgp 65000
bgp log-neighbor-changes
no bgp default ipv4-unicast

```

Task1:

This asks you to configure VRF "cu-red" for the two routers according to the network diagram. Therefore you need to configure VRF "cu-red" for:

In R1, e0/2.100 and e0/0
In R2, e0/2.100 and e0/0

Task2:

This asks you to configure VRF "cu-green" for the two routers according to the network diagram. Therefore you need to configure VRF "cu-green" for:

In R1, e0/2.200 and e0/1
In R2, e0/2.200 and e0/1

Since R1 and R2 is using a sub-interface for each VRF, the setup is VRF-Lite using normal BGP as routing protocol i.e. NOT using vpnv4 of MP-BGP.

Task3:

This asks you to configure BGP for the 2 VRFs in R1 i.e. for the ipv4 address family of the two VRFs. According to requirement mentioned in task1, BGP "network" command is not allowed.

Task4:

This asks you to configure BGP for the 2 VRFs in R2 i.e. for the ipv4 address family of the two VRFs. According to requirement mentioned in task1, BGP "network" command is not allowed.

Task5:

SW1 and SW3 should be able to ping each other.
SW2 and SW4 should be able to ping each other.

For troubleshooting, you can use the following commands:

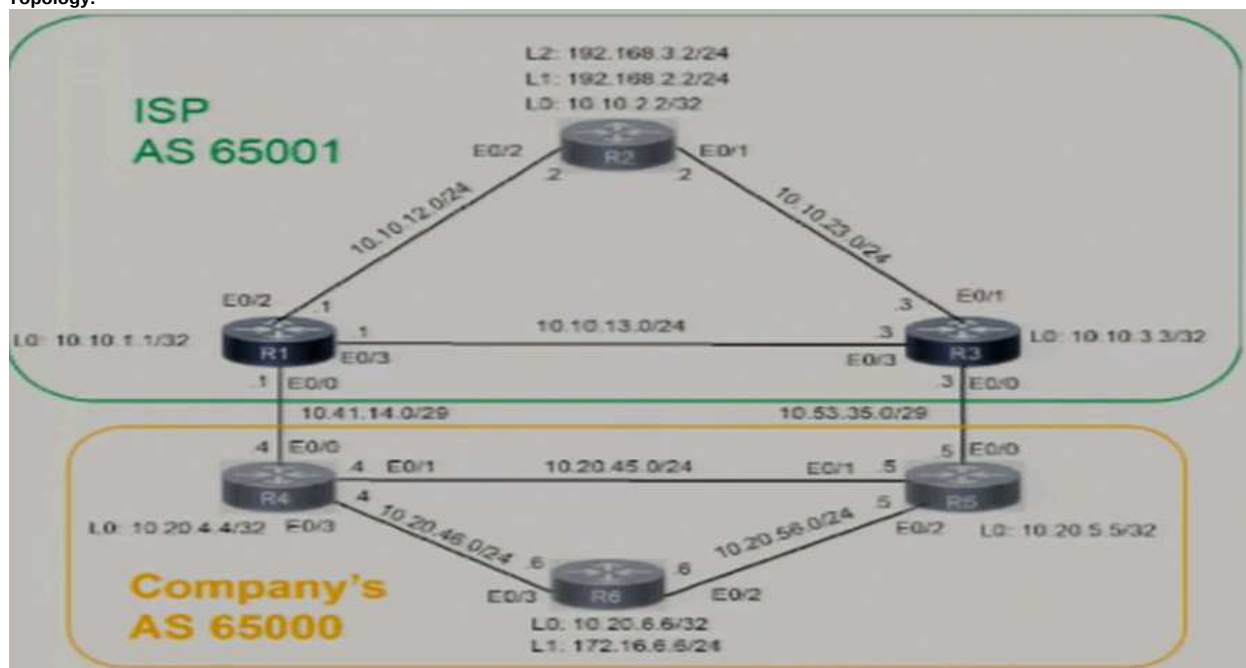
```

sh ip route vrf cu-red
sh ip route vrf cu-green
sh bgp vrf cu-red
sh bgp vrf cu-green

```

QUESTION 4

Topology:



Tasks:

A company is connected to an ISP and some of the networks between the ISP and the company are not reachable. Troubleshoot and resolve the issues to achieve these goals:

1. A single /16 is advertised for all infrastructure-connected interfaces that belong to the 10.20.x.x network using BGP network commands from border routers connected to the ISP. Configuration modification is allowed in R4 and R5 to achieve the results. Do not use the BGP aggregate command.
2. R6 receives the ISP R2 Loopback2 from R4 and receives a summary address for both Loopbacks of ISP R2 from R4 or R5. Use BGP attribute local-preference, add <default value + router number>, for example, for R6, use "default+6 = value to be used". Use the existing prefix lists or route maps with the sequence numbering starting at 10 and added in increments of 10.
3. R6 receives the ISP R2 Loopback1 from R5 and receives a summary address for both Loopbacks of ISP R2 from R4 or R5 using the same guidelines.
4. R6 advertises its Loopback1 /24 address through BGP.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

Task 1:

```
R4
en
config t
ip route 10.20.0.0 255.255.0.0 null0
end
copy run start
```

```
R5
en
config t
ip route 10.20.0.0 255.255.0.0 null0
end
copy run start
```

Task 2 and 3:

```
R4
en
config t
ip prefix-list AS65001-in seq 20 permit 192.168.3.0/24
router bgp 65000
  bgp default local-preference 104
  address-family ipv4
    aggregate-address 192.168.2.0 255.255.254.0
no route-map AS65001-in
route-map AS65001-in permit 10
  match ip address prefix-list AS65001-in
end
copy run start
```

```
R5
en
config t
ip prefix-list AS65001-in seq 20 permit 192.168.2.0/24
router bgp 65000
  bgp default local-preference 105
  address-family ipv4
    aggregate-address 192.168.2.0 255.255.254.0
no route-map AS65001-in
route-map AS65001-in permit 10
  match ip address prefix-list AS65001-in
end
copy run start
```

Task 4:

```
R6
en
config t
router bgp 65000
  address-family ipv4
    no network 172.16.6.0
    network 172.16.6.0 mask 255.255.255.0
end
copy run start
```

Explanations:

You should use sh run in the routers to check if the existing configuration settings are the same as those described for each task:

Task 1:

For the existing BGP settings in both R4 and R5, there is a network command under BGP as follows:

```
router bgp 65000
  address-family ipv4
    network 10.20.0.0 mask 255.255.0.0
```

However, since their connected networks are 10.20.x.0/24, there is no matching IGP route for 10.20.0.0/16 in the two routers and therefore the BGP route 10.20.0.0/16 will not be advertised. The simplest way to setup a matching route in the two routers is to add a static route for 10.20.0.0/16.

Task 2 & 3:

For the existing settings in R4 and R5, both are configured with similar configuration as follows:

```

R4
router bgp 65000
  address-family ipv4
    neighbor 10.41.14.1 route-map AS65001-in in

ip prefix-list AS65001-in seq 10 deny 192.168.2.0/24

route-map AS65001-in deny 10
  match ip address prefix-list AS65001-in

```

```

R5
router bgp 65000
  address-family ipv4
    neighbor 10.53.35.3 route-map AS65001-in in

ip prefix-list AS65001-in seq 10 deny 192.168.3.0/24

route-map AS65001-in deny 10
  match ip address prefix-list AS65001-in

```

Since the route-map have no permit statement, all BGP routes learnt from AS65001 will be denied. For the requirement, R4 needs to advertise R2's loopback2 i.e. 192.168.3.0/24 and R5 needs to advertise R2's loopback1 i.e. 192.168.2.0/24. There are many ways to configure this requirement (depending on whether AS65001 sends other BGP routes i.e. routes other than R2's loopback2 and loopback3 to AS65000 and whether they should be permitted or denied). The following assumes that only 2 BGP routes (i.e. R2's loopback2 and loopback3) are sent from AS65001.

In R4, add a seq 20 rule for permitting 192.168.3.0/24 in the prefix-list and change the route-map statement 10 from deny to permit. Then only the route permitted in the prefix list will be accepted. Hence, 192.168.2.0/24 will be denied but 192.168.3.0/24 will be permitted in the route map.

In R5, add a seq 20 rule for permitting 192.168.2.0/24 in the prefix-list and change the route-map statement 10 from deny to permit. Then only the route permitted in the prefix list will be accepted. Hence, 192.168.3.0/24 will be denied but 192.168.2.0/24 will be permitted in the route map.

For sending summary route for R2's loopback1 and loopback2, you can configure "aggregate-address" in both R4 and R5.

Normally, in order to set local preference, you can use route-map. However, in order to set the local preference for the summary route generated by R4 and R5 themselves, you need an additional route map for the neighbor R6 in outbound direction. Since the question only requires you to use existing prefix-list or route-map, therefore you need to configure a default local-preference. The default local-preference will be added for all routes sending to other BGP routers within the same AS. Hence both the R2' loopback and the summary route will be set with the local preference.

Task 4:

For the existing BGP settings in R6, there is a network command under BGP as follows:

```

router bgp 65000
  address-family ipv4
    network 172.16.6.0

```

Since the loopback 1 of R6 is 172.16.6.6/24 which is not a classful network, you need to remove the above "network" command and then replace it with a network command with a suitable "mask" i.e. 255.255.255.0.

For troubleshooting, you can use "clear ip bgp *" to make the changes effective. Then after the neighbors are formed again, you can use the following commands:

```

sh ip route
sh ip route bgp
sh ip bgp nei
sh ip bgp

```

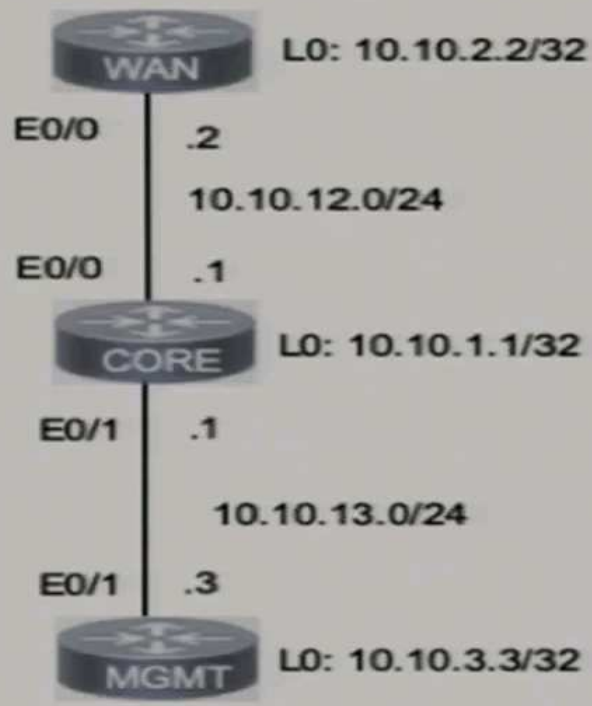
Remarks:

Although you cannot enter privileged mode in R2, the router is configured so that you can enter show commands in user mode.

QUESTION 5

Topology:

EIGRP ASN 101



Tasks:

A network is configured with CoPP to protect the CORE router route processor for stability and DDoS protection. As a company policy, a class named class-default is preconfigured and must not be modified or deleted. Troubleshoot CoPP to resolve the issues introduced during the maintenance window to ensure that:

- Dynamic routing policies are under CoPP-CRITICAL and are allowed only from the 10.10.x.x range.
- Telnet, SSH, and ping are under CoPP-IMPORTANT and are allowed strictly to/from 10.10.x.x to the CORE router (Hint: you can verify using Loopback1).
- All devices ping (UDP) any CORE router interface successfully to/from the 10.10.x.x range and do not allow any other IP address.
- All devices run a successful traceroute (UDP) to any interface on the CORE router to/from the 10.10.x.x range, are under CoPP-NORMAL, and do not allow any other IP address (make sure default traceroute TTL is accounted for). The traceroute is to be under CoPP-NORMAL (Hint: Traceroute port range 33434 33464).

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

Task 1:

CORE

```

en
config t
access-list 120 permit eigrp 10.10.0.0 0.0.255.255 any
end
copy run start
  
```

Task 2:

CORE

```

en
config t
access-list 121 permit icmp 10.10.0.0 0.0.255.255 any echo
access-list 121 permit icmp 10.10.0.0 0.0.255.255 any echo-reply
access-list 121 permit tcp 10.10.0.0 0.0.255.255 any eq 22
access-list 121 permit tcp 10.10.0.0 0.0.255.255 eq 22 any
access-list 121 permit tcp 10.10.0.0 0.0.255.255 any eq 23
access-list 121 permit tcp 10.10.0.0 0.0.255.255 eq 23 any
end
copy run start
  
```

Task 3:

CORE

```

config t
access-list 121 permit udp 10.10.0.0 0.0.255.255 any
access-list 121 permit icmp 10.10.0.0 0.0.255.255 any port-unreachable
end
copy run start
  
```

Task 4:

CORE

```
en
config t
access-list 122 permit udp 10.10.0.0 0.0.255.255 any range 33434 33464
access-list 122 permit icmp 10.10.0.0 0.0.255.255 any ttl-exceeded
end
copy run start
```

Explanations:

You should use `sh run` in the CORE router to check if the existing configuration settings are the same as those described below:

```
class-map match-all CoPP-CRITICAL
  match access-group 120
class-map match-all CoPP-NORMAL
  match access-group 122
class-map match-all CoPP-IMPORTANT
  match access-group 121

policy-map CoPP
  class CoPP-CRITICAL
    police 1000000 50000 50000 conform-action transmit exceed-action drop
  class CoPP-IMPORTANT
    police 100000 20000 20000 conform-action transmit exceed-action drop
  class CoPP-NORMAL
    police 64000 6400 6400 conform-action transmit exceed-action drop
  class class-default
    police 8000 1500 1500 conform-action drop exceed-action drop

control-plane
  service-policy input CoPP
```

The access lists 120, 121 and 122 are pre-configured with a remark line only.

Task 1:

EIGRP is configured in CORE router. Since CoPP-CRITICAL is configured to use the ACL 120, rule has to be added to the access-list 120 for permitting EIGRP traffic from 10.10.0.0/16.

Task 2:

Since CoPP-IMPORTANT is configured to use the ACL 121, rule has to be added to the access-list 121 for permitting Telnet SSH and ping traffic from/to 10.10.0.0/16. Each of them should have two rules (i.e. for traffic initiated from both directions). As it is a CoPP for inbound, all rules should be set with 10.10.0.0/16 as source.

Task 3:

Although it does not mention, it seems that rules for UDP ping should also configured in CoPP-IMPORTANT i.e. in the ACL 121. For UDP ping from devices 10.10.0.0/16, you need to allow UDP traffic generated by them. For UDP ping from CORE router (although most Cisco devices do not support UDP ping), you need to allow the returned ICMP port unreachable from those devices.

Task 4:

Since CoPP-NORMAL is configured to use the ACL 122. For traceroute from devices 10.10.0.0/16, you need to allow UDP traffic with destination port 33434-33464 generated by them. For traceroute from CORE router, you need to allow the returned ICMP time exceeded traffic or ICMP port-unreachable traffic. However, ICMP port-unreachable has already been allowed in Task 2.

For reference only :

After configuring all required settings, the following problems may happen:

- The EIGRP neighbor flaps (i.e. up and down) from time to time.
- You cannot ping CORE from WAN or MGMT.

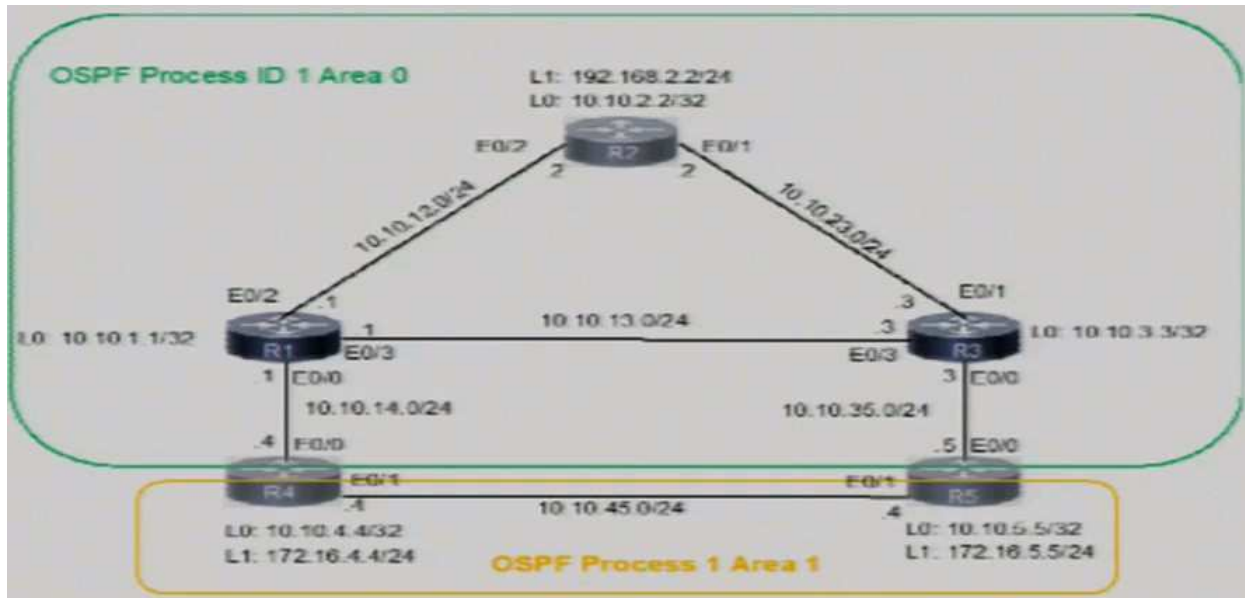
If the above happens, the version of the question you are encountering has a bug.

The policy for the class "class-default" is configured to drop all packets and therefore CORE will drop all ARP requests sent by WAN and MGMT. Hence, WAN and MGMT cannot get the MAC address of the CORE's interfaces. This causes the above problems to happen.

There is no easy fix about this through the modification of rules in the access-list since you cannot configure the matching of ARP protocol in the IP extended access list. The only way to fix it without changing the policy of the class "class-default" is to change one of the class to "match-any" and then, other than matching an access-list, you need to add an additional "match protocol arp" in that class.

QUESTION 6

Topology:



Tasks:

A network is configured with IP connectivity, and the routing protocol between devices started having problems right after the maintenance window to implement network changes. Troubleshoot and resolve to a fully functional network to ensure that:

1. Inter-area links have link authentication (not area authentication) using MD5 with the key 1 string CCNP.
2. R3 is a DR regardless of R2 status while R1 and R2 establish a DR/BDR relationship.
3. OSPF uses the default cost on all interfaces. Network reachability must follow OSPF default behavior for traffic within an area over intra-area VS inter-area links.
4. The OSPF external route generated on R4 adds link cost when traversing through the network to reach R2. A network command to advertise routes is not allowed.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

Task 1:

```
R4
en
config t
key chain CCNP
  key 1
    key-string CCNP
end
copy run start
```

Task 2:

```
R2
en
config t
interface Ethernet0/1
  ip ospf priority 0
end
copy run start
```

Task 3:

```
R5
en
config t
interface Ethernet0/1
  no ip ospf cost
copy run start
```

Task 4:

```
R4
en
config t
router ospf 1
  no redistribute connected
  redistribute connected metric-type 1 subnets route-map to-ospf
copy run start
```

Explanations:

You should use sh run in the routers to check if the existing configuration settings are the same as those described below:

The following existing settings can be found in R2, R4 and R5:

```
R2
interface Loopback0
 ip addr 10.10.2.2 255.255.255.255
 ip ospf 1 area 0
interface Loopback1
 ip addr 192.168.2.2 255.255.255.0
 ip ospf 1 area 0
interface Ethernet0/1
 ip addr 10.10.23.2 255.255.255.0
 ip ospf 1 area 0
interface Ethernet0/2
 ip addr 10.10.12.2 255.255.255.0
 ip ospf 1 area 0
router ospf 1
 passive-interface default
 no passive-interface Ethernet0/0
 no passive-interface Ethernet0/1

R4
key chain CCNP
 key 1
  key-string ccnp
  cryptographic-algorithm md5
interface Loopback0
 ip addr 10.10.4.4 255.255.255.255
 ip ospf 1 area 1
interface Loopback1
 ip addr 172.16.4.4 255.255.255.0
interface Ethernet0/0
 ip addr 10.10.14.4 255.255.255.0
 ip ospf authentication key-chain CCNP
 ip ospf 1 area 0
interface Ethernet0/1
 ip addr 172.16.45.4 255.255.255.0
 ip ospf 1 area 1
router ospf 1
 redistribute connected subnets route-map to-ospf
 passive-interface default
 no passive-interface Ethernet0/0
 no passive-interface Ethernet0/1
route-map to-ospf permit 10
 match interface Loopback1

R5
key chain CCNP
 key 1
  key-string CCNP
  cryptographic-algorithm md5
interface Loopback0
 ip addr 10.10.5.5 255.255.255.255
 ip ospf 1 area 1
interface Loopback1
 ip addr 172.16.5.5 255.255.255.0
interface Ethernet0/0
 ip addr 10.10.35.5 255.255.255.0
 ip ospf authentication key-chain CCNP
 ip ospf 1 area 0
interface Ethernet0/1
 ip addr 172.16.45.5 255.255.255.0
 ip ospf 1 area 1
 ip ospf cost 60
router ospf 1
 redistribute connected subnets route-map to-ospf
 passive-interface default
 no passive-interface Ethernet0/0
 no passive-interface Ethernet0/1
route-map to-ospf permit 10
 match interface Loopback1
```

Task 1:

The task requires you to use interface authentication and use key 1 with string "CCNP". Actually this is already configured but R4 has an incorrect string "ccnp". Hence R5 can form neighbor with R3 but R4 cannot form neighbor with R1. You need to change the key 1 string in R4. After the configuration, you can use "sh ip ospf nei" in R4 to find that it forms neighbor with R1.

Task 2:

In order to make R3 always be a DR, the easiest way is to disable R2's E0/1 from participating in the election of DR by setting priority 0 in the interface. Since the task only requires R1 and R2 to form DR/BDR relationship, the link between R2 and R3 can have a DR only (i.e. no BDR elected in the link).

Task 3:

Since R5's E0/1 is configured with a cost, you must remove it.

Task 4:

Loopback1 of R4 is redistributed as the E2 route by default in R4. In order to add link cost, you need to change it so that it will be redistributed as E1 route.

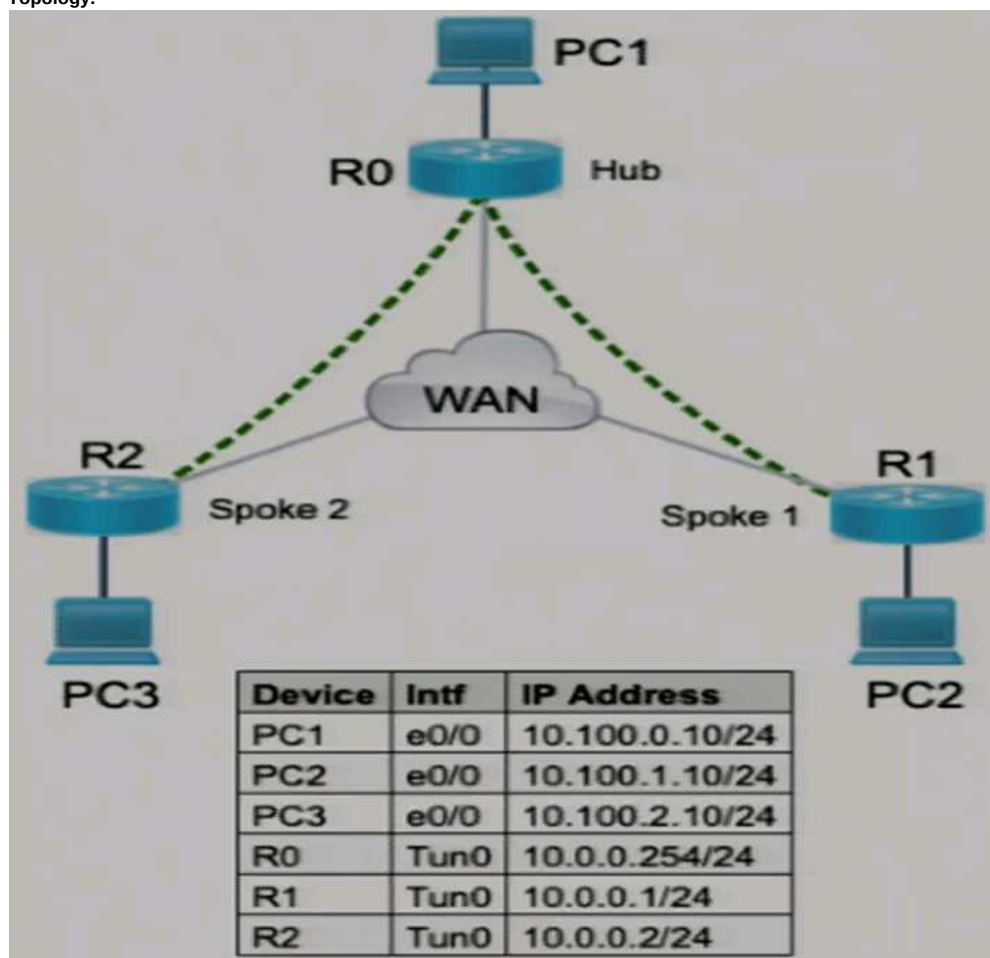
After the configuration, you can use "sh ip route ospf" in R2 to find that the route "172.16.4.0/24" is a "O E1" route.

For troubleshooting, you can use the following commands:

```
sh ip route
sh ip route ospf
sh ip ospf nei
```

QUESTION 7

Topology:



Tasks:

Configure IPSec security policy on tunnel interfaces to ensure data confidentiality and integrity where mGRE tunnels are up and running between HUB and SPOKE routers.

- Configure the ISAKMP policy parameters with the following attributes:
 - AES256
 - SHA256
 - Group2
 - lifetime 86400
- Ensure that GRE IP Header should be encrypted inside the IPSec packet. Verify IPSec security association and ISAKMP encrypted key. Use ISAKMP key "abc123".
- Configure a flexible ISAKMP Policy on the HUB to add peers that have the dynamic IP addresses where SPOKES must add HUB IP static entry using an encrypted key. Use a single command to configure it. Use IPSec phase-2 transform-set name as 'T-SET' and IPSec Profile name as 'IPSEC-PROFILE'.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

```
R0
en
config t
crypto isakmp policy 10
  hash sha256
  encryption aes 256
  group 2
  lifetime 86400
  authentication pre-share
crypto isakmp key abc123 address 0.0.0.0
crypto ipsec transform-set T-SET esp-des esp-md5-hmac
mode tunnel
crypto ipsec profile IPSEC-PROFILE
set transform-set T-SET
interface tu0
  tunnel protection ipsec profile IPSEC-PROFILE
end
copy run start
```

```
R1
en
config t
crypto isakmp policy 10
  hash sha256
  encryption aes 256
```

```

group 2
lifetime 86400
authentication pre-share
crypto isakmp key abc123 address 192.168.1.2
crypto ipsec transform-set T-SET esp-des esp-md5-hmac
mode tunnel
crypto ipsec profile IPSEC-PROFILE
set transform-set T-SET
interface tu0
tunnel protection ipsec profile IPSEC-PROFILE
end
copy run start

```

```

R2
en
config t
crypto isakmp policy 10
hash sha256
encryption aes 256
group 2
lifetime 86400
authentication pre-share
crypto isakmp key abc123 address 192.168.1.2
crypto ipsec transform-set T-SET esp-des esp-md5-hmac
mode tunnel
crypto ipsec profile IPSEC-PROFILE
set transform-set T-SET
interface tu0
tunnel protection ipsec profile IPSEC-PROFILE
end
copy run start

```

Explanations:

You should use sh run in the routers to check if the existing configuration settings are the same as those described below:

```

R0
interface Tunnel0
ip address 10.0.0.254 255.255.255.0
no ip redirects
ip nhrp network-id 10
ip ospf network broadcast
ip ospf priority 255
tunnel source 192.168.1.2
tunnel mode gre multipoint
interface Ethernet0/0
ip address 192.168.1.2 255.255.255.252
interface Ethernet0/1
ip address 10.100.0.1 255.255.255.0
router ospf 1
network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.4 255.255.255.252 192.168.1.1
ip route 192.168.1.8 255.255.255.252 192.168.1.1

```

```

R1
interface Tunnel0
ip address 10.0.0.1 255.255.255.0
ip nhrp map multicast 192.168.1.2
ip nhrp map 10.0.0.254 192.168.1.2
ip nhrp network-id 10
ip nhrp nhs 10.0.0.254
ip ospf network broadcast
ip ospf priority 0
tunnel source 192.168.1.6
tunnel destination 192.168.1.2
interface Ethernet0/0
ip address 192.168.1.6 255.255.255.252
interface Ethernet0/1
ip address 10.100.1.1 255.255.255.0
router ospf 1
network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.0 255.255.255.252 192.168.1.5
ip route 192.168.1.8 255.255.255.252 192.168.1.5

```

```

R2
interface Tunnel0
ip address 10.0.0.2 255.255.255.0
ip nhrp map multicast 192.168.1.2
ip nhrp map 10.0.0.254 192.168.1.2
ip nhrp network-id 10
ip nhrp nhs 10.0.0.254
ip ospf network broadcast
ip ospf priority 0
tunnel source 192.168.1.10
tunnel destination 192.168.1.2
interface Ethernet0/0
ip address 192.168.1.10 255.255.255.252
interface Ethernet0/1
ip address 10.100.2.1 255.255.255.0
router ospf 1
network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.0 255.255.255.252 192.168.1.9
ip route 192.168.1.4 255.255.255.252 192.168.1.9

```

Remarks:

With the existing configuration, the Spoke routers cannot communicate directly with each other. You should not change this since the IPSec configuration task do not

ask you to build IPSec between the two Spoke routers.

Task 1

For task 1, you need to configure the following ISAKMP policy in all 3 routers:

```
crypto isakmp policy 10
  hash sha256
  encryption aes 256
  group 2
  lifetime 86400
```

Task 2 & 3

For task 2 & 3, since GRE IP Header must be encrypted, you need to use tunnel mode. This can be configured in a transform set which has to be named "T-SET" as required by the question. Since the task does not mention which ESP / AH encryption / authentication should be used, you can use any but you must configure the same in all 3 routers.

GRE over IPsec Packet Format

Tunnel Mode

ESP IP Header	ESP Header	GRE IP Header	GRE	IP Header	TCP Header	Data	ESP Trailer
---------------	------------	---------------	-----	-----------	------------	------	-------------

Transport Mode

GRE IP Header	ESP Header	GRE	IP Header	TCP Header	Data	ESP Trailer
---------------	------------	-----	-----------	------------	------	-------------

```
crypto ipsec transform-set T-SET esp-des esp-md5-hmac
mode tunnel
```

The ISAKMP key required is "abc123". Hub must be configured so that more Spoke routers with dynamic IPs may be added and therefore "address 0.0.0.0" should be configured for the key. However, Spoke is required to configure the static IP address of the Hub router's physical interface and therefore "address 192.168.1.2" should be configured for the key in the Spoke routers. Since key is used for IKE authentication, remember of adding "authentication pre-share" to the IKE policy configured in task 1.

R0

```
crypto isakmp key abc123 address 0.0.0.0
crypto isakmp policy 10
  authentication pre-share
```

R1

```
crypto isakmp key abc123 address 192.168.1.2
crypto isakmp policy 10
  authentication pre-share
```

R2

```
crypto isakmp key abc123 address 192.168.1.2
crypto isakmp policy 10
  authentication pre-share
```

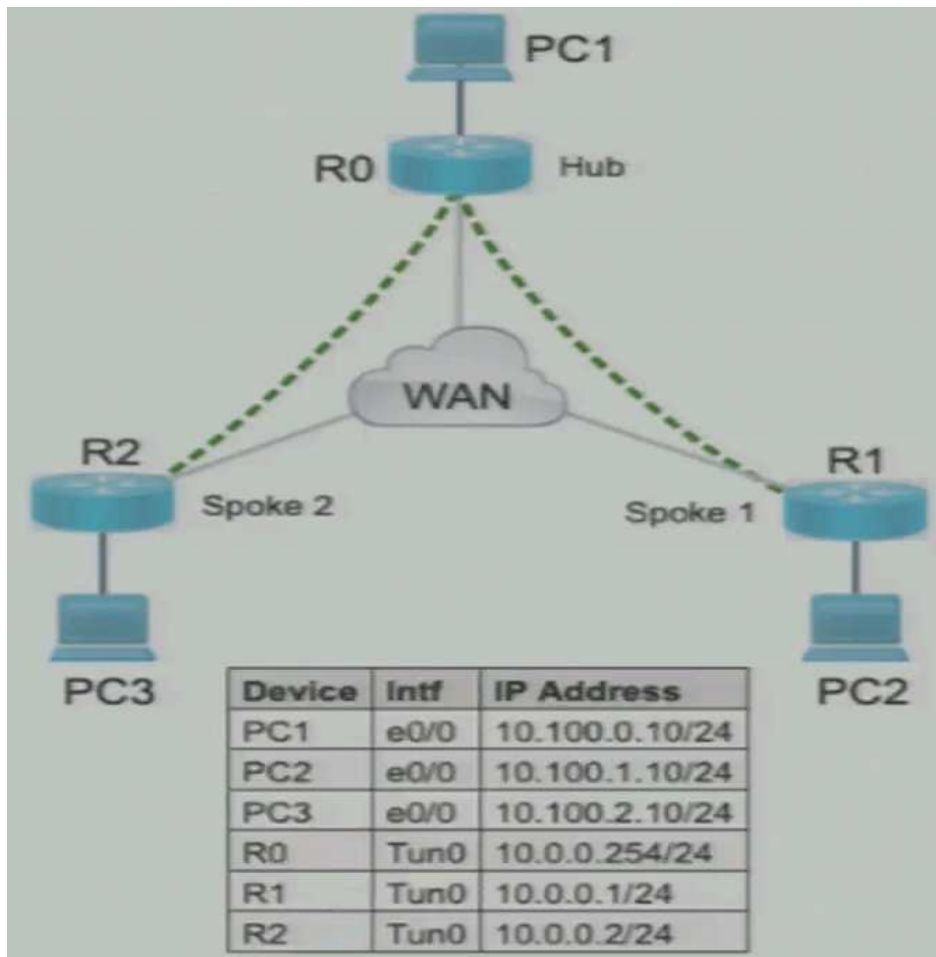
Finally, configure the IPSec Profile with the name "IPSEC-PROFILE " as required by the question, set it to use the transform set "T-SET" and then apply it to the DMVPN tunnel interface for all three routers.

```
crypto ipsec profile IPSEC-PROFILE
  set transform-set T-SET
interface tu0
  tunnel protection ipsec profile IPSEC-PROFILE
```

For verifying if IPSec is configured properly, you can check if R0 can form OSPF neighbor with R1 and R2. Finally, you can check if the PCs can ping each other using the IP addresses of the PCs shown in the diagram.

QUESTION 8

Topology:



Tasks:

Configure HUB and SPOKE routers according to the topology to achieve these goals:

1. Configure mGRE neighborhood to provide end-to-end reachability between Hub and Spokes.
2. Configure NHRP authentication using password "C!\$c0123". Use 180 sec hold time for NHRP members where NHS should maintain next hop client NBMA registration messages for 60 sec. Verify configurations with Ping from PC1 to PC2 and PC3.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

```
R0
en
config t
interface tu0
    tunnel mode gre multipoint
    ip nhrp authentication C!$c0123
end
copy run start
```

```
R1
en
config t
interface tu0
    ip nhrp authentication C!$c0123
    ip nhrp holdtime 180
    ip nhrp registration timeout 60
end
copy run start
```

```
R2
en
config t
interface tu0
    ip nhrp authentication C!$c0123
    ip nhrp holdtime 180
    ip nhrp registration timeout 60
end
copy run start
```

For verification, in PC1

```
ping 10.100.1.10
Ping 10.100.2.10
```


Explanations:

You should use `sh run` in the routers to check if the existing configuration settings are the same as those described below:

R0

```
interface Tunnel0
 ip address 10.0.0.254 255.255.255.0
 ip nhrp network-id 10
 ip ospf network broadcast
 ip ospf priority 255
 tunnel source 192.168.1.2
interface Ethernet0/0
 ip address 192.168.1.2 255.255.255.252
interface Ethernet0/1
 ip address 10.100.0.1 255.255.255.0
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.4 255.255.255.252 192.168.1.1
ip route 192.168.1.8 255.255.255.252 192.168.1.1
```

R1

```
interface Tunnel0
 ip address 10.0.0.1 255.255.255.0
 ip nhrp map multicast 192.168.1.2
 ip nhrp map 10.0.0.254 192.168.1.2
 ip nhrp network-id 10
 ip nhrp nhs 10.0.0.254
 ip ospf network broadcast
 ip ospf priority 0
 tunnel source 192.168.1.6
 tunnel destination 192.168.1.2
interface Ethernet0/0
 ip address 192.168.1.6 255.255.255.252
interface Ethernet0/1
 ip address 10.100.1.1 255.255.255.0
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.0 255.255.255.252 192.168.1.5
ip route 192.168.1.8 255.255.255.252 192.168.1.5
```

R2

```
interface Tunnel0
 ip address 10.0.0.2 255.255.255.0
 ip nhrp map multicast 192.168.1.2
 ip nhrp map 10.0.0.254 192.168.1.2
 ip nhrp network-id 10
 ip nhrp nhs 10.0.0.254
 ip ospf network broadcast
 ip ospf priority 0
 tunnel source 192.168.1.10
 tunnel destination 192.168.1.2
interface Ethernet0/0
 ip address 192.168.1.10 255.255.255.252
interface Ethernet0/1
 ip address 10.100.2.1 255.255.255.0
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.0 255.255.255.252 192.168.1.9
ip route 192.168.1.4 255.255.255.252 192.168.1.9
```

Task 1

For Task1, you need to configure `tunnel mode gre multipoint` in the tunnel interface of the Hub router R0.

Remarks :

Actually, the task does not specifically mention Spoke to Spoke communication, therefore the suggested answer do not remove tunnel destination "no tunnel destination" and configure "tunnel mode gre multipoint" in the Spoke routers.

Task 2

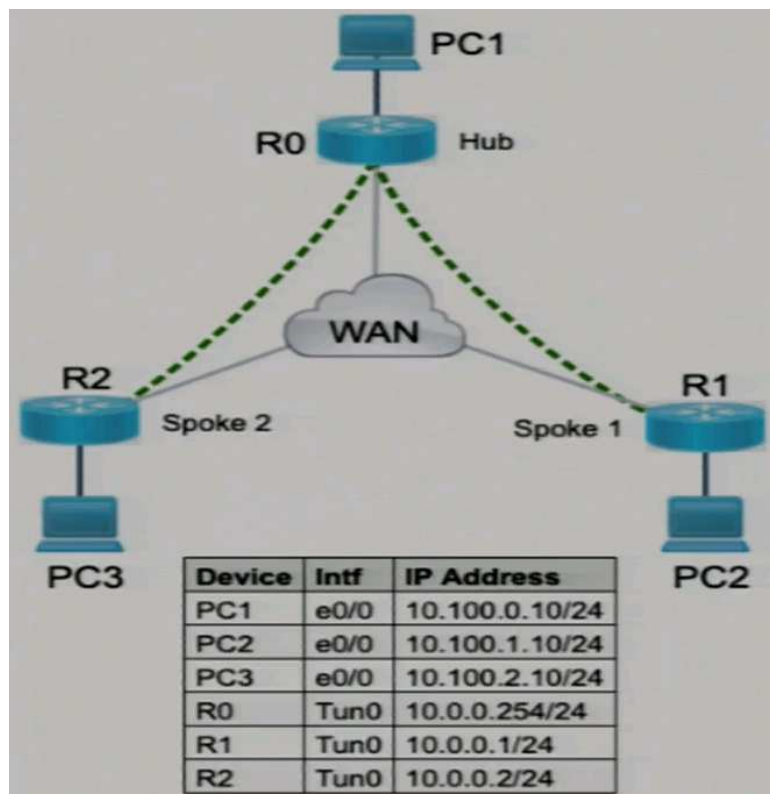
For Task2, you need to configure NHRP authentication using the command "`ip nhrp authentication C!$c0123`" in all 3 routers.

```
interface Tunnel0
 ip nhrp authentication C!$c0123
```

Moreover, you also need to configure the following NHRP hold time and registration timeout settings in the 2 Spoke routers:

```
interface tu0
 ip nhrp holdtime 180
 ip nhrp registration timeout 60
```

QUESTION 9**Topology:**



Tasks:

Configure IPsec security policy on tunnel interfaces to ensure data confidentiality and integrity where mGRE tunnels are up and running between HUB and SPOKE routers.

1. Configure the ISAKMP policy parameters with the following attributes:

- AES128
- MD5
- Group2
- lifetime 86400

2. Ensure that GRE IP Header should not be encrypted inside the IPsec packet.

3. Configure a flexible ISAKMP Policy to add peers that have the dynamic IP addresses. Use a single command to configure it. Use IPsec phase-2 transform-set name as 'T-SET' and IPsec Profile name as 'T-SET-PROFILE'. Use ISAKMP key "abc123". Verify configuration with Ping from PC1 to PC2 and PC3.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

```
R0
en
config t
crypto isakmp policy 10
  hash md5
  encryption aes 128
  group 2
  lifetime 86400
  authentication pre-share
crypto isakmp key abc123 address 0.0.0.0
crypto ipsec transform-set T-SET esp-des esp-md5-hmac
  mode transport
crypto ipsec profile T-SET-PROFILE
  set transform-set T-SET
interface Tunnel0
  tunnel protection ipsec profile T-SET-PROFILE
end
copy run start
```

```
R1
en
config t
crypto isakmp policy 10
  hash md5
  encryption aes 128
  group 2
  lifetime 86400
  authentication pre-share
crypto isakmp key abc123 address 0.0.0.0
crypto ipsec transform-set T-SET esp-des esp-md5-hmac
  mode transport
crypto ipsec profile T-SET-PROFILE
  set transform-set T-SET
interface Tunnel0
  tunnel protection ipsec profile T-SET-PROFILE
end
```

copy run start

```
R2
en
config t
crypto isakmp policy 10
  hash md5
  encryption aes 128
  group 2
  lifetime 86400
  authentication pre-share
crypto isakmp key abc123 address 0.0.0.0
crypto ipsec transform-set T-SET esp-des esp-md5-hmac
mode transport
crypto ipsec profile T-SET-PROFILE
set transform-set T-SET
interface Tunnel0
tunnel protection ipsec profile T-SET-PROFILE
end
copy run start
```

For verification, in PC1

```
ping 10.100.1.10
Ping 10.100.2.10
```

Explanations:

You should use sh run in the routers to check if the existing configuration settings are the same as those described below:

```
R0
interface Tunnel0
ip address 10.0.0.254 255.255.255.0
ip nhrp network-id 10
ip ospf network broadcast
ip ospf priority 255
tunnel source 192.168.1.2
tunnel mode gre multipoint
interface Ethernet0/0
ip address 192.168.1.2 255.255.255.252
interface Ethernet0/1
ip address 10.100.0.1 255.255.255.0
router ospf 1
network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.4 255.255.255.252 192.168.1.1
ip route 192.168.1.8 255.255.255.252 192.168.1.1
```

```
R1
interface Tunnel0
ip address 10.0.0.1 255.255.255.0
ip nhrp map multicast 192.168.1.2
ip nhrp map 10.0.0.254 192.168.1.2
ip nhrp network-id 10
ip nhrp nhs 10.0.0.254
ip ospf network broadcast
ip ospf priority 0
tunnel source 192.168.1.6
tunnel destination 192.168.1.2
interface Ethernet0/0
ip address 192.168.1.6 255.255.255.252
interface Ethernet0/1
ip address 10.100.1.1 255.255.255.0
router ospf 1
network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.0 255.255.255.252 192.168.1.5
ip route 192.168.1.8 255.255.255.252 192.168.1.5
```

```
R2
interface Tunnel0
ip address 10.0.0.2 255.255.255.0
ip nhrp map multicast 192.168.1.2
ip nhrp map 10.0.0.254 192.168.1.2
ip nhrp network-id 10
ip nhrp nhs 10.0.0.254
ip ospf network broadcast
ip ospf priority 0
tunnel source 192.168.1.10
tunnel destination 192.168.1.2
interface Ethernet0/0
ip address 192.168.1.10 255.255.255.252
interface Ethernet0/1
ip address 10.100.2.1 255.255.255.0
router ospf 1
network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.0 255.255.255.252 192.168.1.9
ip route 192.168.1.4 255.255.255.252 192.168.1.9
ip route 192.168.1.4 255.255.255.252 192.168.1.9
```

This question is similar to another simulation question but the requirements are different!!

Task 1

For task 1, you need to configure the following ISAKMP policy in all 3 routers:

```
crypto isakmp policy 10
  hash md5
  encryption aes 128
  group 2
```

lifetime 86400

Task 2 & 3

For task 2 & 3, since GRE IP Header should **NOT** be encrypted, you need to use **transport** mode. This can be configured in a transform set with the name "T-SET" as required by the question. The task does not mention which ESP / AH encryption / authentication should be used. Therefore, you can use any settings but you have to make sure that they are the same in the 3 routers.

GRE over IPsec Packet Format

Tunnel Mode

ESP IP Header	ESP Header	GRE IP Header	GRE	IP Header	TCP Header	Data	ESP Trailer
---------------	------------	---------------	-----	-----------	------------	------	-------------

Transport Mode

GRE IP Header	ESP Header	GRE	IP Header	TCP Header	Data	ESP Trailer
---------------	------------	-----	-----------	------------	------	-------------

```
crypto ipsec transform-set T-SET esp-des esp-md5-hmac
mode transport
```

The ISAKMP key should be "abc123" with dynamic IP addresses. Hence, all are assumed to be configured with "address 0.0.0.0" (note that this requirement is different other the other question). However, other than the single command to configure the key, you also need to set the authentication to pre-share in the IKE policy configured in task 1 so that the key will be used for authentication. Hence, the following commands are required in all 3 routers.

```
crypto isakmp key abc123 address 0.0.0.0
crypto isakmp policy 10
authentication pre-share
```

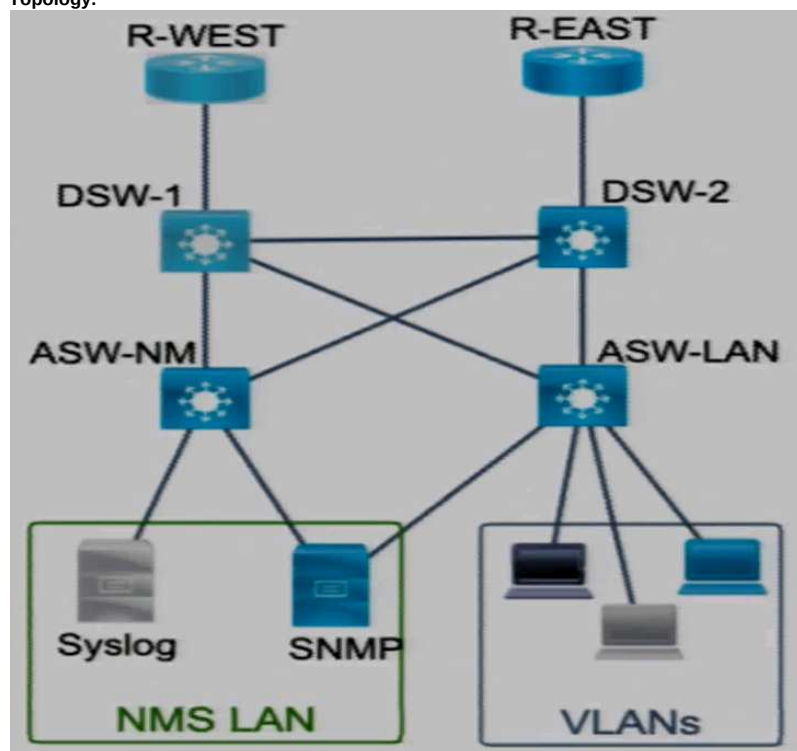
Finally configure the IPsec Profile with the name "T-SET-PROFILE" as required by the question, set it to use the transform set "T-SET" and then apply it to the DMVPN tunnel interface for all three routers.

```
crypto ipsec profile T-SET-PROFILE
set transform-set T-SET
interface Tunnel0
tunnel protection ipsec profile T-SET-PROFILE
```

For verifying, you can check if R0 can form OSPF neighbor with R1 and R2. Moreover, you should be able to ping from PC1 to PC2 and P3 using the IP addresses shown in the diagram.

QUESTION 10

Topology:



Tasks:

Troubleshoot R-WEST to achieve the desired results:

1. The locally generated logs should have sequence numbers, date and time.
2. The SNMP traps related to OSPF and participating interface state changes utilizing RFC1253-MIB OSPFv2 should be send to SNMP server.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

R-WEST
en

```

config t
service sequence-numbers
service timestamps log datetime
snmp-server enable traps ospf state-change if-state-change
end
copy run start

```

Explanations:

For locally generated logs, you should configure "service sequence-numbers" if you want sequence number.

```

R1(config)#service ?
.....
.... skipped ....
.....
sequence-numbers      Stamp logger messages with a sequence number
slave-log             Enable log capability of slave IPs
tcp-keepalives-in     Generate keepalives on idle incoming network
                      connections
tcp-keepalives-out    Generate keepalives on idle outgoing network
                      connections
tcp-small-servers     Enable small TCP servers (e.g., ECHO)
telnet-zeroidle       Set TCP window 0 when connection is idle
timestamps            Timestamp debug/log messages
udp-small-servers     Enable small UDP servers (e.g., ECHO)

```

For locally generated logs, you should configure "service timestamps log datetime" if you want date and time.

```

R1(config)#service timestamps log ?
datetime      Timestamp with date and time
uptime       Timestamp with system uptime
<cr>

```

Note that:

- Since the question does not specify local time, we do NOT add "localtime" after "service timestamps log datetime".
- Since the question does not specify the showing of date and time in debug messages, we do NOT include "service timestamps debug datetime".

For SNMP traps about OSPF interface state changes, you should configure "snmp-server enable traps ospf state-change if-state-change".

```

R1(config)#snmp-server enable traps ospf ?
cisco-specific  Cisco specific traps
errors          Error traps
lsa            Lsa related traps
rate-limit     Trap rate limit values
retransmit     Packet retransmit traps
state-change   State change traps
<cr>

```

```

R1(config)#snmp-server enable traps ospf state-change ?
if-state-change      Non-virtual interface state changes
neighbor-state-change Non-virtual neighbor state changes
virtif-state-change  Virtual interface state changes
virtneighbor-state-change Virtual neighbor state changes
<cr>

```

After configuring the settings about log, you should find the sequence number and date / time similar to the following in the R-WEST router when you end the Global Configuration mode.

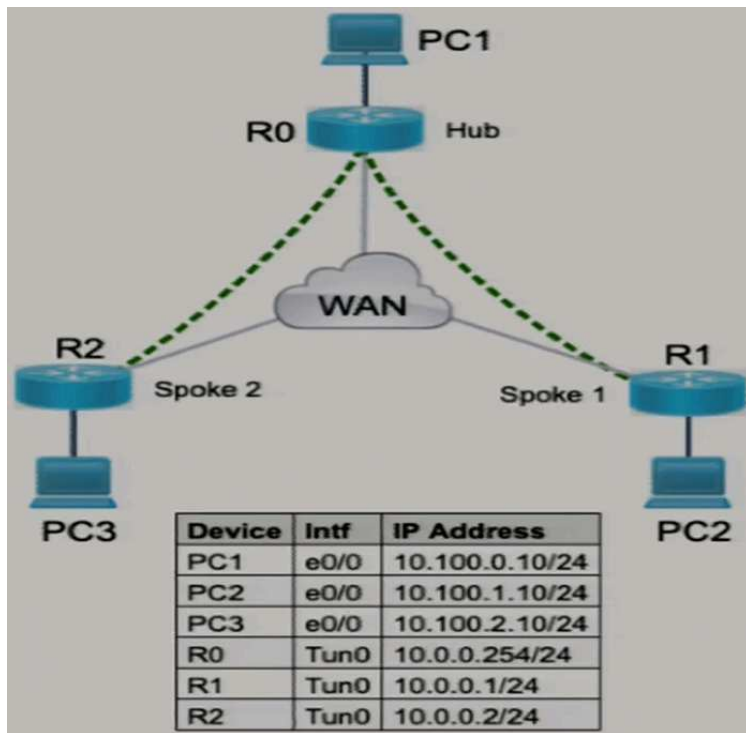
```

R1(config)#end
R1#
000035: *Apr 28 10:10:00.391: %SYS-5-CONFIG_I: Configured from console by console
R1#

```

QUESTION 11

Topology:



Tasks:

HUB and SPOKE routers are configured using mGRE tunnels according to the topology. Configure HUB and SPOKE routers to achieve these goals:

1. Configure NHRP authentication using password "Ci\$c0123".
2. Configure SPOKE routers to establish direct tunnels between them where LAN routes should be learned from the tunnel interfaces.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

Task 1:

```
R0
en
config t
int tu 0
  ip nhrp authentication Ci$c0123
end
copy run start
```

```
R1
en
config t
int tu 0
  ip nhrp authentication Ci$c0123
end
copy run start
```

```
R2
en
config t
int tu 0
  ip nhrp authentication Ci$c0123
end
copy run start
```

Task 2:

```
R1
en
config t
int tu 0
  no tunnel destination
  tunnel mode gre multipoint
end
copy run start
```

```
R2
en
config t
int tu 0
  no tunnel destination
  tunnel mode gre multipoint
end
```


copy run start

Explanations:

You should use `sh run` in the routers to check if the pre-configured settings are the same as the followings since the suggested answer is based on these configuration settings:

```
R0
interface Tunnel0
 ip address 10.0.0.254 255.255.255.0
 no ip redirects
 ip nhrp network-id 10
 ip ospf network broadcast
 ip ospf priority 255
 tunnel source 192.168.1.2
 tunnel mode gre multipoint
interface Ethernet0/0
 ip address 192.168.1.2 255.255.255.252
interface Ethernet0/1
 ip address 10.100.0.1 255.255.255.0
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.4 255.255.255.252 192.168.1.1
ip route 192.168.1.8 255.255.255.252 192.168.1.1
```

```
R1
interface Tunnel0
 ip address 10.0.0.1 255.255.255.0
 ip nhrp map multicast 192.168.1.2
 ip nhrp map 10.0.0.254 192.168.1.2
 ip nhrp network-id 10
 ip nhrp nhs 10.0.0.254
 ip ospf network broadcast
 ip ospf priority 0
 tunnel source 192.168.1.6
 tunnel destination 192.168.1.2
interface Ethernet0/0
 ip address 192.168.1.6 255.255.255.252
interface Ethernet0/1
 ip address 10.100.1.1 255.255.255.0
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.0 255.255.255.252 192.168.1.5
ip route 192.168.1.8 255.255.255.252 192.168.1.5
```

```
R2
interface Tunnel0
 ip address 10.0.0.2 255.255.255.0
 ip nhrp map multicast 192.168.1.2
 ip nhrp map 10.0.0.254 192.168.1.2
 ip nhrp network-id 10
 ip nhrp nhs 10.0.0.254
 ip ospf network broadcast
 ip ospf priority 0
 tunnel source 192.168.1.10
 tunnel destination 192.168.1.2
interface Ethernet0/0
 ip address 192.168.1.10 255.255.255.252
interface Ethernet0/1
 ip address 10.100.2.1 255.255.255.0
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.0 255.255.255.252 192.168.1.9
ip route 192.168.1.4 255.255.255.252 192.168.1.9
```

The subnets "192.168.1.X/30" are the networks configured for the proper routing across the WAN and are not shown in the diagram.

In the above configurations, both NHRP and OSPF are properly preconfigured. You should be able to find the OSPF routes for the LAN networks of other routers in each of the router e.g.

```
R1#sh ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O       10.100.0.0/24 [110/1001] via 10.0.0.254, 00:00:36, Tunnel0
O       10.100.2.0/24 [110/1001] via 10.0.0.2, 00:00:26, Tunnel0
```

R1#

Hence, instead of configuring NHRP and OSPF from start, you just need to modify the configurations according to the requirements specified in the tasks.

Task 1:

When configuring NHRP authentication, you need to configure the interface configuration command "`ip nhrp authentication Ci$c0123`" under the tunnel interfaces of all 3 routers.

If you just configure the setting in 1 router e.g. R0, although there seems to have no adverse effect initially, it will no longer communicate with other routers through NHRP and the concerned NHRP entries in R0 will expire within 10 minutes and then traffic cannot be sent from R0 to other routers through the tunnel afterwards.

Task 2:

In the NHRP setting pre-configured in the Spoke routers, a fixed tunnel destination is specified. This is the setup of NHRP phase 1 and therefore spoke-to-spoke tunnel cannot be formed. All traffic from Spoke to Spoke will pass through the Hub router. You can use the following to prove that the traffic from R1 to R2's LAN interface passes through R0.

```
R1#traceroute 10.100.2.1 numeric
Type escape sequence to abort.
Tracing the route to 10.100.2.1
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.0.254 6 msec 6 msec 6 msec
 2 10.0.0.2 7 msec * 9 msec
R1#
```

In order to form spoke-to-spoke tunnel i.e. NHRP phase 2, you need to remove the tunnel destination and setup the tunnel to use "multipoint GRE" with the interface configuration mode commands "no tunnel destination" and "tunnel mode gre multipoint" in the tunnel interfaces of both Spoke routers (i.e. R1 and R2).

Note that depending on the routing protocols being used, extra commands may be needed:

- For EIGRP, "no ip split-horizon eigrp <AS number>" is need in the tunnel interface of the Hub router (this is not applicable for this question since OSPF is used).
- For OSPF with network type "broadcast", "ip ospf priority 0" is need in the tunnel interface of each Spoke router (this is not needed in this question since it is already pre-configured).

After the configuring the Spoke routers, the OSPF neighbor in the Spoke router will become down and neighbor will be formed again after a short while.

Then you can try the following to prove the forming of spoke-to-spoke tunnel. The first traceroute triggers the setup of the spoke-to-spoke tunnel and the second traceroute proves that the traffic from R1 to R2's LAN interface can now be sent directly.

```
R1#traceroute 10.100.2.1 numeric
Type escape sequence to abort.
Tracing the route to 10.100.2.1
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.0.254 11 msec 7 msec 7 msec
 2 10.0.0.2 8 msec * 5 msec
R1#
```

```
R1#traceroute 10.100.2.1 numeric
Type escape sequence to abort.
Tracing the route to 10.100.2.1
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.0.2 6 msec * 5 msec
R1#
```

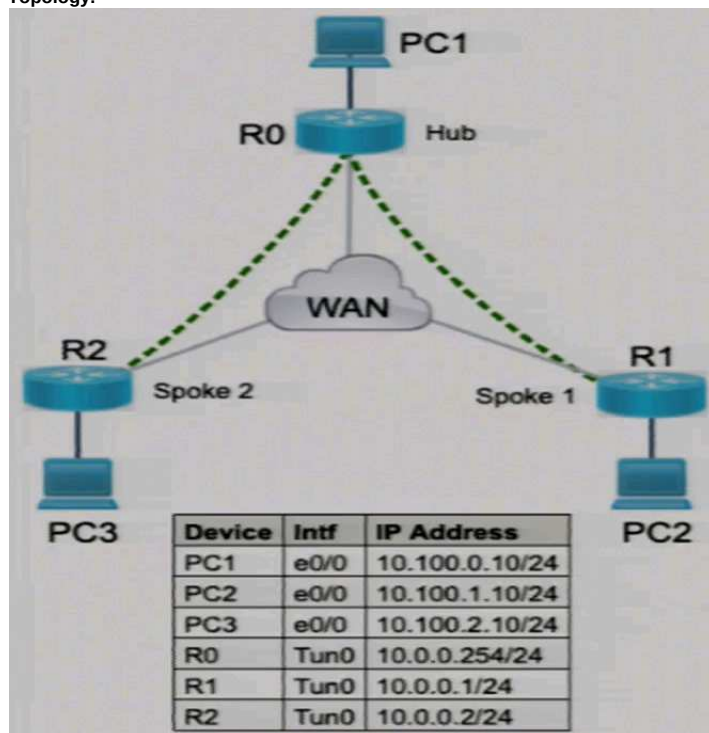
Remarks :

In this question, since do not need to optimize the traffic for any summarised route advertised by the Hub router, therefore the setup for DMVPN phase 3 using commands "ip nhrp redirect" (for Hub router) and "ip nhrp shortcut" (for Spoke routers) are not required.

However, if the question is using EIGRP as the routing protocol and the Hub router is advertising a summarized router for the LANs of the spoke routers, the task in the question may require you to optimize the traffic from a Spoke router to the LAN of another Spoke router using the Spoke-to-Spoke tunnel. In this case, the above two commands will be needed.

QUESTION 12

Topology:



Tasks:

Configure HUB and SPOKE routers according to the topology to achieve these goals:

1. configure mGRE neighborship by using physical interface IP addresses to provide end-to-end reachability. Verify all router's tunnel interfaces accordingly.
2. Configure NHRP on HUB router as multipoint dynamic IP to NBMA mapping. SPOKE1 and SPOKE2 routers should be configured by using static NHS and multicast mapping for HUB. Use network-id 2 for NHRP. Verify NHRP neighborship between routers and verify results using ping from PC1 to PC2 and PC3 accordingly.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

```
R0
en
config t
interface Tunnel0
 ip address 10.0.0.254 255.255.255.0
 ip nhrp network-id 2
 tunnel source 192.168.1.2
 tunnel mode gre multipoint
end
copy run start
```

```
R1
en
config t
interface Tunnel0
 ip address 10.0.0.1 255.255.255.0
 ip nhrp map multicast 192.168.1.2
 ip nhrp map 10.0.0.254 192.168.1.2
 ip nhrp network-id 2
 ip nhrp nhs 10.0.0.254
 tunnel source 192.168.1.6
 tunnel mode gre multipoint
end
copy run start
```

```
R2
en
config t
interface Tunnel0
 ip address 10.0.0.2 255.255.255.0
 ip nhrp map multicast 192.168.1.2
 ip nhrp map 10.0.0.254 192.168.1.2
 ip nhrp network-id 2
 ip nhrp nhs 10.0.0.254
 tunnel source 192.168.1.10
 tunnel mode gre multipoint
end
copy run start
```

For verification, in PC1

```
ping 10.100.1.10
Ping 10.100.2.10
```

Explanations:

You should use sh run in the routers to check if the existing configuration settings are the same as those described below:

```
R0
interface Tunnel0
 no ip address
 ip ospf network broadcast
 ip ospf priority 255
interface Ethernet0/0
 ip address 192.168.1.2 255.255.255.252
interface Ethernet0/1
 ip address 10.100.0.1 255.255.255.0
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.4 255.255.255.252 192.168.1.1
ip route 192.168.1.8 255.255.255.252 192.168.1.1
```

```
R1
interface Tunnel0
 no ip address
 ip ospf network broadcast
 ip ospf priority 0
interface Ethernet0/0
 ip address 192.168.1.6 255.255.255.252
interface Ethernet0/1
 ip address 10.100.1.1 255.255.255.0
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.0 255.255.255.252 192.168.1.5
ip route 192.168.1.8 255.255.255.252 192.168.1.5
```

```
R2
interface Tunnel0
 no ip address
 ip ospf network broadcast
 ip ospf priority 0
interface Ethernet0/0
 ip address 192.168.1.10 255.255.255.252
interface Ethernet0/1
 ip address 10.100.2.1 255.255.255.0
router ospf 1
 network 10.0.0.0 0.255.255.255 area 0
ip route 192.168.1.0 255.255.255.252 192.168.1.9
```

```
ip route 192.168.1.4 255.255.255.252 192.168.1.9
```

Task 1

For task 1, you need to tunnel IP addresses of the 3 routers according to the diagram. Moreover, you need to find the IP address of the physical interface connecting to WAN and use it to configure as tunnel source. (Note that the IP address "10.100.X.10" belongs to the network connecting to the PCs according to the diagram. Hence, the IP address "192.168.1.X" should be the one connecting to the WAN")

Since mGRE neighborhood is required, you need to configure GRE Multipoint as the tunnel mode.

R0

```
interface Tunnel0
 ip address 10.0.0.254 255.255.255.0
 tunnel source 192.168.1.2
 tunnel mode gre multipoint
```

R1

```
interface Tunnel0
 ip address 10.0.0.1 255.255.255.0
 tunnel source 192.168.1.6
 tunnel mode gre multipoint
end
copy run start
```

R2

```
interface Tunnel0
 ip address 10.0.0.2 255.255.255.0
 tunnel source 192.168.1.10
 tunnel mode gre multipoint
```

Task 2

For the Hub router, you only need to configure the network ID "2".

R0

```
interface Tunnel0
 ip nhrp network-id 2
```

For the Spoke routers, you need to configure the followings:

- Configure a mapping to send multicast traffic to Hub router's physical IP address "192.168.1.2"
- A static mapping between the Hub router's tunnel IP address "10.0.0.254" and physical IP address "192.168.1.2"
- Configure the NHS by specifying the Hub router's tunnel IP address i.e. "10.0.0.254"
- Configure the network ID "2".

R1

```
interface Tunnel0
 ip nhrp map multicast 192.168.1.2
 ip nhrp map 10.0.0.254 192.168.1.2
 ip nhrp nhs 10.0.0.254
 ip nhrp network-id 2
```

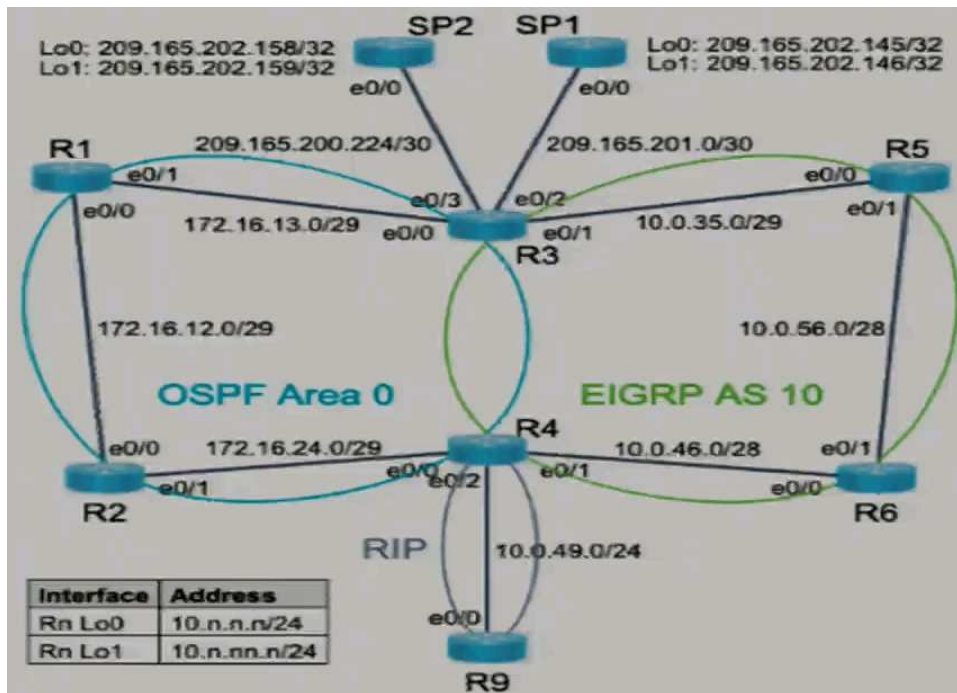
R2

```
interface Tunnel0
 ip nhrp map multicast 192.168.1.2
 ip nhrp map 10.0.0.254 192.168.1.2
 ip nhrp nhs 10.0.0.254
 ip nhrp network-id 2
```

For verifying, you can check if R0 can form OSPF neighbor with R1 and R2. Moreover, you should be able to ping from PC1 to PC2 and P3 using the IP addresses shown in the diagram.

QUESTION 13

Topology:



Tasks:

Troubleshoot and resolve the issues to achieve these goals:

1. Ensure that R5 can ping the prefix 10.9.99.9. Manipulate the first basic routing decision-making criteria that if a router learns a route from different routing protocols the admin distance can be changed without using a route-map. Use decimal value of 85 if required to accomplish this.
2. Ensure that R5 loopback 0 reaches to R2's loopback 0 by following the path through R5, R6, R4 to R2 and vice versa. Use metric values K1=100000, K2=1, K3=255, K4=10, K5=1500 to modify the default metric in EIGRP if required. Do not add or modify the default-metric command under router eigrp 10. Do not use a route-map.
3. Ensure that on R3, prefix 10.0.0.0/8 uses the SP2 to route to the Internet, whereas prefix 172.16.0.0/12 uses the SP1 to route to the Internet. Do not use BGP to accomplish this. Use the pre-configured route-maps SP1 and SP2 and modify to accomplish the task if required. Use the ping and trace commands from R5 and R1 to verify results.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

```
R3
en
config t
route-map SP1 permit 10
  set ip next-hop 209.165.201.2
route-map SP2 permit 10
  set ip next-hop 209.165.200.226
int e0/0
  ip policy route-map SP1
int e0/1
  ip policy route-map SP2
end
copy run start

R4
en
config t
router rip
  distance 85 10.0.49.0 0.0.0.255
router eigrp 10
  redistribute ospf 10 metric 100000 1 255 10 1500
end
copy run start
```

Explanations:

You should use sh run in the routers to check if the existing configuration settings are the same as those described below:

```
R3
interface Loopback0
  ip address 10.3.3.3 255.255.255.0
!
interface Loopback1
  ip address 10.3.33.3 255.255.255.0
!
interface Ethernet0/0
  ip address 172.16.13.3 255.255.255.248
  ip nat inside
!
interface Ethernet0/1
  ip address 10.0.35.3 255.255.255.248
  ip nat inside
```

```

!
interface Ethernet0/2
 ip address 209.165.201.1 255.255.255.252
 ip nat outside
!
interface Ethernet0/3
 ip address 209.165.200.225 255.255.255.252
 ip nat outside
!
router eigrp 10
 default-metric 1 1 1 1 1
 network 10.0.35.0 0.0.0.7
 network 10.3.33.0 0.0.0.255
 redistribute ospf 10
 redistribute static
 eigrp router-id 10.3.33.3
!
router ospf 10
 router-id 10.3.3.3
 redistribute static subnets
 redistribute eigrp 10 subnets
 network 10.3.3.0 0.0.0.255 area 0
 network 172.16.13.0 0.0.0.7 area 0
 default-information originate metric 100 metric-type 1
!
ip route 0.0.0.0 0.0.0.0 Null0
!
route-map SP2 permit 10
 match ip address 110
!
route-map SP1 permit 10
 match ip address 120
!
access-list 10 permit 10.0.0.0 0.255.255.255
access-list 20 permit 172.16.0.0 0.15.255.255
access-list 110 deny ip 10.0.0.0 0.255.255.255 172.16.0.0 0.15.255.255
access-list 110 permit ip 10.0.0.0 0.255.255.255 any
access-list 120 deny ip 172.16.0.0 0.15.255.255 10.0.0.0 0.255.255.255
access-list 120 permit ip 172.16.0.0 0.15.255.255 any

```

```

R4
interface Loopback0
 ip address 10.4.4.4 255.255.255.0
!
interface Loopback1
 ip address 10.4.44.4 255.255.255.0
!
interface Ethernet0/0
 ip address 172.16.24.4 255.255.255.248
!
interface Ethernet0/1
 ip address 10.0.46.4 255.255.255.240
!
interface Ethernet0/2
 ip address 10.0.49.4 255.255.255.0
!
router eigrp 10
 default-metric 1 1 1 1 1
 network 10.0.46.0 0.0.0.15
 network 10.4.46.0 0.0.0.255
 redistribute ospf 10
 redistribute rip
 eigrp router-id 10.4.44.4
!
router ospf 10
 router-id 10.4.4.4
 redistribute eigrp 10 subnets
 network 10.4.4.0 0.0.0.255 area 0
 network 172.16.24.0 0.0.0.27 area 0
!
router rip
 version 2
 redistribute eigrp 10 metric 6
 network 10.0.0.0

```

Task 1

In this network topology, routing loops occur for the networks of the R9's loopback interfaces. Take the network of R9's lo1 i.e. 10.9.99.0/24 as an example:

1. R9 --- RIP (AD 120) --> R4
2. Redistributed from RIP to EIGRP in R4
3. R4 --- External EIGRP (AD 170) --> R6
4. R6 --- External EIGRP (AD 170) --> R5
5. R5 --- External EIGRP (AD 170) --> R3
6. Redistributed from EIGRP to OSPF in R3
7. R3 --- OSPF E2 (AD 110) --> R1
8. R1 --- OSPF E2 (AD 110) --> R2
9. R2 --- OSPF E2 (AD 110) --> R4
10. In R4's routing table, the RIP route is replaced by the OSPF E2 due to lower AD (110 < 120)
11. Redistributed from OSPF to EIGRP in R4
12. The steps 3 – 9 and 11 are repeated.

In R4's routing table, you can find the following route using R2 as next hop and this proves that the step 10 mentioned (i.e. OSPF route replacing RIP route) has occurred.

```
0 E2      10.9.99.0/24 [110/20] via 172.16.24.2, 00:00:02, Ethernet0/0
```

You can also use "tracert 10.9.99.9 numeric" command in R5 to prove that routing loop is formed for the route 10.9.99.0/24:

```
R5#tracert 10.9.99.9 numeric
```



```
Type escape sequence to abort.
Tracing the route to 10.9.99.9
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.56.6 6 msec 3 msec 4 msec
 2 10.0.46.4 4 msec 5 msec 4 msec
 3 172.16.24.2 7 msec 6 msec 6 msec
 4 172.16.12.1 9 msec 5 msec 6 msec
 5 172.16.13.3 7 msec 6 msec 8 msec
 6 10.0.35.5 6 msec 5 msec 5 msec
 7 10.0.56.6 11 msec 7 msec 9 msec
 8 10.0.46.4 9 msec 8 msec 8 msec
 9 172.16.24.2 12 msec 9 msec 12 msec
10 172.16.12.1 10 msec 10 msec 11 msec
11 172.16.13.3 9 msec 10 msec 11 msec
12 10.0.35.5 8 msec 8 msec 8 msec
13 10.0.56.6 12 msec 11 msec 12 msec
14 10.0.46.4 10 msec 13 msec 13 msec
15 172.16.24.2 13 msec 14 msec 15 msec
16 172.16.12.1 13 msec 16 msec 13 msec
17 172.16.13.3 14 msec 14 msec 12 msec
18 10.0.35.5 13 msec 13 msec 15 msec
19 10.0.56.6 14 msec 14 msec 17 msec
20 10.0.46.4 16 msec 16 msec 17 msec
21 172.16.24.2 17 msec 18 msec 17 msec
22 172.16.12.1 18 msec 19 msec 18 msec
23 172.16.13.3 16 msec 17 msec 17 msec
24 10.0.35.5 17 msec 17 msec 17 msec
25 10.0.56.6 19 msec 20 msec 18 msec
26 10.0.46.4 24 msec 21 msec 18 msec
27 172.16.24.2 22 msec 22 msec 22 msec
28 172.16.12.1 21 msec 21 msec 21 msec
29 172.16.13.3 22 msec 21 msec 19 msec
30 10.0.35.5 22 msec 20 msec 20 msec
R5#
```

In order to prevent step 10 that causes the loop to occur, you can set a lower AD for the RIP route using the AD 85 mentioned in the task.

This can be done through the "distance" command under RIP configuration in R4 by specifying the IP / network of the RIP neighbor R9 e.g. 10.0.49.9/32 or 10.0.49.0/24 as follows:

```
Configuration required to be set in R4
router rip
 distance 85 10.0.49.0 0.0.0.255
```

After the above configuration, when R4 receives the RIP route from R9 again (may take a while), since the RIP routes now has an AD of 85, they will not be replaced by the external OSPF routes in R4's routing table. The route will become:

```
R          10.9.99.0/24 [85/1] via 10.0.49.9, 00:00:04, Ethernet0/2
```

After the above RIP route is found in R4's routing table, you can also use ping or "**traceroute 10.9.99.9 numeric**" command again in R5 to prove that R5 can now access R9's lo1:

```
R5#traceroute 10.9.99.9 numeric
Type escape sequence to abort.
Tracing the route to 10.9.99.9
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.56.6 3 msec 5 msec 3 msec
 2 10.0.46.4 5 msec 5 msec 5 msec
 3 10.0.49.9 5 msec * 5 msec
R5#
```

Task 2

In this network topology, R5 and R2 can reach each other lo0. However, the path taken may not be the one required by the task.

The following command shows that R5's lo0 reaches R2's lo0 via R3, R1 and R2 (note that the 4th octet of the IP address shows the router number). This does NOT meet the requirement.

```
R5#traceroute 10.2.2.2 source 10.5.5.5 numeric
Type escape sequence to abort.
Tracing the route to 10.2.2.2
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.35.3 3 msec 6 msec 5 msec
 2 172.16.13.1 5 msec 5 msec 4 msec
 3 172.16.12.2 7 msec * 4 msec
R5#
```

You can also check the routing entry for R2's lo0 in R5's routing table to prove that R5 will send the packet through R3.

```
D EX          10.2.2.2/32
               [170/2560000512] via 10.0.35.3, 00:39:16, Ethernet0/0
```

The following command shows that, in the reverse direction, R2's lo0 reaches R5's lo0 via R4, R6 and R5. This meets the requirement.

```
R2#traceroute 10.5.5.5 source 10.2.2.2 numeric
Type escape sequence to abort.
Tracing the route to 10.5.5.5
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.24.4 8 msec 4 msec 4 msec
 2 10.0.46.6 5 msec 5 msec 5 msec
 3 10.0.56.5 7 msec * 6 msec
R2#
```

Hence, you need to change the path taken by R5 to reach R2.

In the present configuration, both R3 and R4 redistribute OSPF routes into EIGRP using the metric 1 (bandwidth), 1 (delay), 1 (reliability), 1 (load), 1 (mtu) due to the command "default-metric 1 1 1 1" under their EIGRP configuration. Since both R3 and R4 uses the same metric for redistributing OSPF routes, R5 will

prefer the ASBR which is nearest i.e. R3. Hence, R5 reaches R2 through R3.

The question requires you to use the new metric 100000 (bandwidth), 1 (delay), 255 (reliability), 10 (load), 1500 (mtu). This metric should be preferred than the metric set by "default-metric 1 1 1 1". Hence, if you want R5 to reach R2 through R4 i.e. the lower path in the diagram. You should set the new metric in R4 for redistributing OSPF routes into EIGRP. This makes the external EIGRP routes redistributed by R4 to have better metrics than the external EIGRP routes redistributed by R3.

Since you are not allowed to change the "default-metric" command, you can set the new metrics for redistributing OSPF routes (since R2 advertises its loopback through OSPF) in R4 as follows:

Configuration required to be set in R4

```
router eigrp 10
 redistribute ospf 10 metric 100000 1 255 10 1500
```

After the above configuration, R5 will use the path through R4 to reach R2's lo0. You can prove this by traceroute:

```
R5#traceroute 10.2.2.2 source 10.5.5.5 numeric
Type escape sequence to abort.
Tracing the route to 10.2.2.2
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.56.6 4 msec 4 msec 6 msec
 2 10.0.46.4 5 msec 6 msec 5 msec
 3 172.16.24.2 9 msec * 6 msec
R5#
```

Or the routing entry in R5

```
D EX 10.2.2.2/32 [170/26368] via 10.0.56.6, 00:01:34, Ethernet0/1
```

Task 3

In R3, two route maps SP1 and SP2 are preconfigured. Their configurations are:

- SP1 uses ACL 120 which matches traffic from 172.16.0.0/12.
- SP2 uses ACL 110 which matches traffic from 10.0.0.0/8

In order to route traffic based on source address, you can setup PBR. Since the route map is already configured, you just need to add "set ip next-hop" command in the route map and apply the route maps as PBR to the appropriate interfaces.

Since route map SP1 matches source from 172.16.0.0/12:

- you need to use SP1's e0/0 IP address for the "set ip next-hop" command. (You can check it in SP1's router or calculate the address using R3's e0/2).
- you need to apply it to R3's e0/0 that connects to the OSPF network having IP "172.16.X.X".

Since route map SP2 matches source from 10.0.0.0/8:

- you need to use SP2's e0/0 IP address for the "set ip next-hop" command. (You can check it in SP2's router or calculate the address using R3's e0/3).
- you need to apply it to R3's e0/1 that connects to the EIGRP network having IP "10.X.X.X"

Configuration required to be set in R3

```
route-map SP1 permit 10
 set ip next-hop 209.165.201.2

route-map SP2 permit 10
 set ip next-hop 209.165.200.226

int e0/0
 ip policy route-map SP1
int e0/1
 ip policy route-map SP2
```

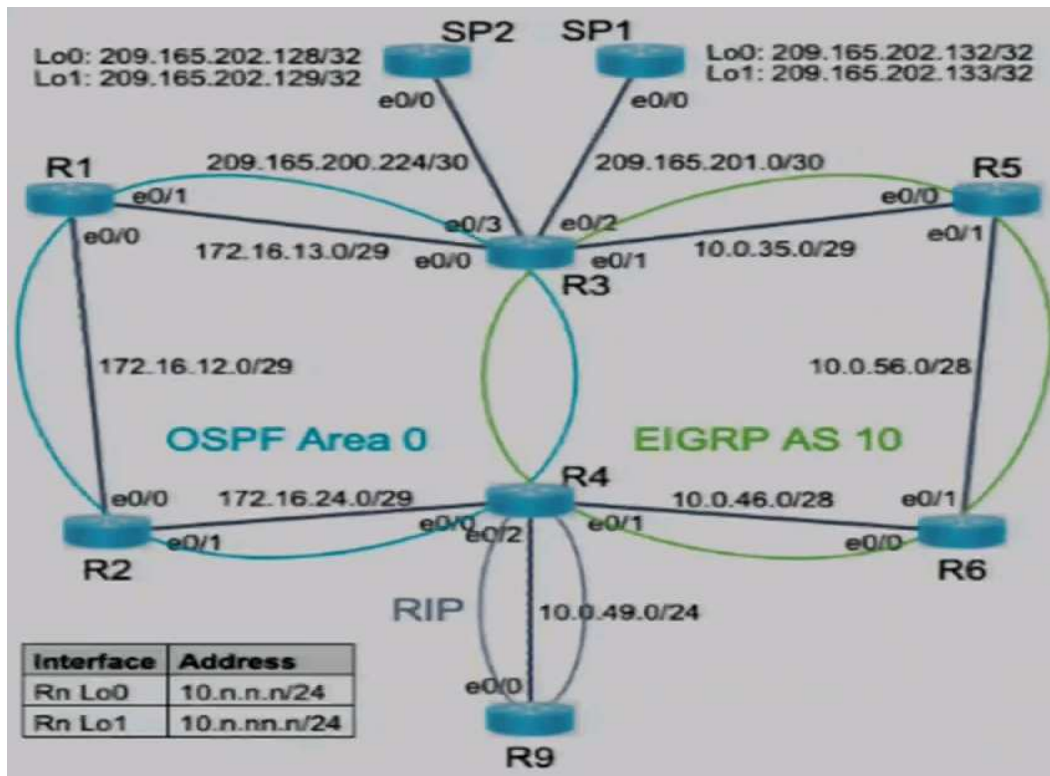
Remarks :

The IP address of SP1 and SP2 can be calculated as follows:

- For the network 209.165.201.0/30 shown in the diagram, since the IP address of R3's e0/2 is 209.165.201.1, the IP address of SP1's e0/0 is 209.165.201.2.
- For the network 209.165.200.224/30 shown in the diagram, since the IP address of R3's e0/3 is 209.165.200.225, the IP address of SP2's e0/0 is 209.165.200.226.

QUESTION 14

Topology:



Tasks:

Troubleshoot and resolve the issues to achieve these goals:

1. Ensure that R2 reaches the prefix 10.5.55.5 without any single point of failure in the path. Do not use a static route or a route-map to accomplish this.
2. Ensure that R1 loopback 0 reaches to R6's loopback 0 by following the path through R1, R5 to R6 and vice versa. Use metric values K1= 100000, K2=1, K3=255, K4=10, K5=1500 to modify the default metric in EIGRP if required. Do not use a route-map.
3. Ensure that on R3, prefix 10.0.0.0/8 uses the SP1 to route to the Internet, whereas prefix 172.16.0.0/12 uses the SP2 to route to the Internet. Do not use BGP to accomplish this. Use the pre-configured route-maps SP1 and SP2 and modify to accomplish the task if required. Use the ping and trace commands from R5 and R1 to verify results.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

```

R3
en
config t
router eigrp 10
  no distance 255 0.0.0.0 255.255.255.255 5
  redistribute ospf 10 metric 100000 1 255 10 1500
!
route-map SP1 permit 10
  set ip next-hop 209.165.201.2
!
route-map SP2 permit 10
  set ip next-hop 209.165.200.226
!
int e0/0
  ip policy route-map SP2
int e0/1
  ip policy route-map SP1
!
end
copy run start

```

Explanations:

You should use sh run in the routers to check if the existing configuration settings are the same as those described below:

```

R3
interface Loopback0
  ip address 10.3.3.3 255.255.255.0
!
interface Loopback1
  ip address 10.3.33.3 255.255.255.0
!
interface Ethernet0/0
  ip address 172.16.13.3 255.255.255.248
  ip nat inside
!
interface Ethernet0/1
  ip address 10.0.35.3 255.255.255.248
  ip nat inside
!

```

```

interface Ethernet0/2
 ip address 209.165.201.1 255.255.255.252
 ip nat outside
!
interface Ethernet0/3
 ip address 209.165.200.225 255.255.255.252
 ip nat outside
!
router eigrp 10
 default-metric 1 1 1 1 1
 network 10.0.35.0 0.0.0.7
 network 10.3.33.0 0.0.0.255
 redistribute ospf 10
 redistribute static
 distance 255 0.0.0.0 255.255.255.255 5
 eigrp router-id 10.3.33.3
!
router ospf 10
 router-id 10.3.3.3
 redistribute static subnets
 redistribute eigrp 10 subnets
 network 10.3.3.0 0.0.0.255 area 0
 network 172.16.13.0 0.0.0.7 area 0
 default-information originate metric 100 metric-type 1
!
ip route 0.0.0.0 0.0.0.0 Null0
!
route-map SP2 permit 10
 match ip address 120
!
route-map SP1 permit 10
 match ip address 110
!
access-list 5 permit 10.5.55.0
access-list 10 permit 10.0.0.0 0.255.255.255
access-list 20 permit 172.16.0.0 0.15.255.255
access-list 110 deny ip 10.0.0.0 0.255.255.255 172.16.0.0 0.15.255.255
access-list 110 deny ip 10.0.0.0 0.255.255.255 10.0.0.0 0.255.255.255
access-list 110 permit ip 10.0.0.0 0.255.255.255 any
access-list 120 deny ip 172.16.0.0 0.15.255.255 10.0.0.0 0.255.255.255
access-list 120 deny ip 172.16.0.0 0.15.255.255 172.16.0.0 0.15.255.255
access-list 120 permit ip 172.16.0.0 0.15.255.255 any

```

```

R4
interface Loopback0
 ip address 10.4.4.4 255.255.255.0
!
interface Loopback1
 ip address 10.4.44.4 255.255.255.0
!
interface Ethernet0/0
 ip address 172.16.24.4 255.255.255.248
!
interface Ethernet0/1
 ip address 10.0.46.4 255.255.255.240
!
interface Ethernet0/2
 ip address 10.0.49.4 255.255.255.0
!
router eigrp 10
 default-metric 1 1 1 1 1
 network 10.0.46.0 0.0.0.15
 network 10.4.46.0 0.0.0.255
 redistribute ospf 10
 redistribute rip
 eigrp router-id 10.4.44.4
!
router ospf 10
 router-id 10.4.4.4
 redistribute eigrp 10 subnets
 network 10.4.4.0 0.0.0.255 area 0
 network 172.16.24.0 0.0.0.27 area 0
!
router rip
 version 2
 redistribute eigrp 10 metric 6
 network 10.0.0.0

```

Task 1

In the existing network topology, R2 can reach R5's lo1 10.5.55.5 through R4 --> R6 --> R5 as proved by the following traceroute command in R2:

```

R2#traceroute 10.5.55.5 numeric
Type escape sequence to abort.
Tracing the route to 10.5.55.5
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.24.4 3 msec 2 msec 5 msec
 2 10.0.46.6 5 msec 5 msec 5 msec
 3 10.0.56.5 6 msec * 7 msec
R2#

```

The task required that R2 can reach R5's lo1 without any single point of failure in the path. This means that if the above path fails (e.g. R4 is dead), R2 should be able to reach R5's lo1 i.e. through the path R1 --> R3 --> R5.

Normally, since both R3 and R4 redistribute EIGRP routes including R5's lo1 into OSPF, there should be no problem. However, if you check the route 10.5.55.0/24 in the routing table of the router along the alternate path i.e. R1 and R3, you find that both uses the same path as R2 i.e. R3 --> R1 --> R2 --> R4 --> R5.

Routing Entry in R1

```
0 E2      10.5.55.0/24 [110/20] via 172.16.12.2, 00:02:51, Ethernet0/0
```

Routing Entry in R3

```
0 E2      10.5.55.0/24 [110/20] via 172.16.13.1, 00:03:24, Ethernet0/0
```

From the routing entry in R3, it seems that R3 does not learn the EIGRP route for R5's lo1 from its neighbor R5. If the EIGRP route is learnt, the internal EIGRP route with AD 90 learnt directly from R5 should exist in the routing table instead of the OSPF E2 route with AD 110.

More hints can be found through route for the R5's lo0 which is also a route advertised by R5. This route for R5's lo1 is normal since it is a EIGRP route in R3's routing table.

```
D          10.5.5.0/24 [90/130816] via 10.0.35.5, 00:00:07, Ethernet0/1
```

By checking the EIGRP configuration in R3, the following command is found:

Existing configuration in R3

```
router eigrp 10
  distance 255 0.0.0.0 255.255.255.255 5
!
access-list 5 permit 10.5.55.0
```

The above changes the AD of the EIGRP route learnt from any EIGRP neighbor for the routes specified in access list 5 i.e. the route 10.5.55.0. Since the AD set is 255 which means inaccessible, the route 10.5.55.0 learnt from R5 will not be entered into the R3's routing table. Hence, R5 uses the OSPF route learnt from R1 instead. On the other hand, the route for R5's lo0 i.e. 10.5.5.0 is not matched by the access list, the EIGRP route 10.5.5.0/24 is entered normally in R3's routing table.

In order to solve the problem, you can just remove the "distance" command shown above i.e.:

Configuration required to be set in R3

```
router eigrp 10
  no distance 255 0.0.0.0 255.255.255.255 5
```

After removing the above, the EIGRP route for R5's lo1 should be found in the R3's routing table and the R1 should learnt the redistributed OSPF route from R3.

Routing Entry in R1

```
0 E2      10.5.55.0/24 [110/20] via 172.16.13.3, 00:01:25, Ethernet0/1
```

Routing Entry in R3

```
D          10.5.55.0/24 [90/130816] via 10.0.35.5, 00:01:59, Ethernet0/1
```

After the modification, if R4 is dead, R2 should learn the route from R1 and can send packets to R5's lo1 through R1 --> R3 --> R5.

Task 2

In this network topology, R1 and R6 can reach each other lo0. However, the path taken may not be the one required by the task.

The following command shows that R1's lo0 reaches R6's lo0 via R3 --> R5 --> R6 (note that the 4th octet of the IP address shows the router number). This meets the requirement.

```
R1#traceroute 10.6.6.6 source 10.1.1.1 numeric
Type escape sequence to abort.
Tracing the route to 10.6.6.6
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.13.3 6 msec 3 msec 2 msec
 2 10.0.35.5 5 msec 5 msec 6 msec
 3 10.0.56.6 7 msec * 5 msec
R1#
```

You can also check the routing entry for R6's lo0 in R1's routing table to prove that R1 will send the packet through R3.

```
0 E2      10.6.6.0/24 [110/20] via 172.16.13.3, 00:01:25, Ethernet0/1
```

The following command shows that R6's lo0 reaches R1's lo0 via R4 --> R2 --> R1 (The task requires the R6's lo0 reaches R1's lo0 to following the path involving R5 due to the word "vice versa"). This does NOT meet the requirement.

```
R6#traceroute 10.1.1.1 source 10.6.6.6 numeric
Type escape sequence to abort.
Tracing the route to 10.1.1.1
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.46.4 6 msec 4 msec 2 msec
 2 172.16.24.2 5 msec 7 msec 4 msec
 3 172.16.12.1 6 msec * 5 msec
R6#
```

Hence, you need to change the path taken by R6 to reach R1.

In the present configuration, both R3 and R4 redistribute OSPF routes into EIGRP using the metric 1 (bandwidth), 1 (delay), 1 (reliability), 1 (load), 1 (mtu) due to the command "default-metric 1 1 1 1 1" under their EIGRP configuration. Since both R3 and R4 uses the same metric for redistributing OSPF routes, R6 will prefer the ASBR which is nearest i.e. R4. Hence, R6 reaches R1 through R4.

The question required you to use the new metric 100000 (bandwidth), 1 (delay), 255 (reliability), 10 (load), 1500 (mtu). This metric should be preferred than the metric set by "default-metric 1 1 1 1 1". Hence, if you want R6 to reach R1 through R3 i.e. the upper path in the diagram. You should set the new metric in R3 for redistribution so that its EIGRP routes redistributed from OSPF will be preferred by the EIGRP routers including R6.

Since you are not allowed to change the "default-metric" command, you can set the new metrics for redistributing OSPF routes (since R1 advertises its loopback through OSPF) into EIGRP as follows:

Configuration required to be set in R3

```
router eigrp 10
  redistribute ospf 10 metric 100000 1 255 10 1500
```

After the above configuration, R6 will use the path through R5 --> R3 --> R1 to reach R1's lo0. You can prove this by traceroute:

```
R6#traceroute 10.1.1.1 source 10.6.6.6 numeric
Type escape sequence to abort.
```

```

Tracing the route to 10.1.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.0.56.5 4 msec 3 msec 3 msec
  2 10.0.35.3 6 msec 6 msec 5 msec
  3 172.16.13.1 6 msec * 9 msec
R6#

```

Or the routing entry in R6

```

D EX    10.1.1.1/32 [170/26368] via 10.0.56.5, 00:01:32, Ethernet0/1

```

Task 3

In R3, two route map SP1 and SP 2 is preconfigured. Their configurations are:

- SP1 uses ACL 110 which matches traffic from 10.0.0.0/8.
- SP2 uses ACL 120 which matches traffic from 172.16.0.0/12.

In order to routing traffic according to source, you can setup PBR. Since the route map is already configured, you just need to add "set ip next-hop" command in the PBR and apply the route maps to the appropriate interfaces.

Since route map SP1 matches source from 10.0.0.0/8:

- you need to use SP1's e0/0 IP address for the "set ip next-hop" command. (You can check it in SP1's router or calculate the address using R3's e0/2).
- you need to apply it to R3's e0/1 that connects to the EIGRP network having IP "10.0.0.0/8".

Since route map SP2 matches source from 172.16.0.0/12:

- you need to use SP2's e0/0 IP address for the "set ip next-hop" command. (You can check it in SP2's router or calculate the address using R3's e0/3).
- you need to apply it to R3's e0/0 that connects to the OSPF network having IP "172.16.X.X"

Configuration required to be set in R3

```

route-map SP1 permit 10
  set ip next-hop 209.165.201.2

```

```

route-map SP2 permit 10
  set ip next-hop 209.165.200.226

```

```

int e0/0
  ip policy route-map SP2
int e0/1
  ip policy route-map SP1

```

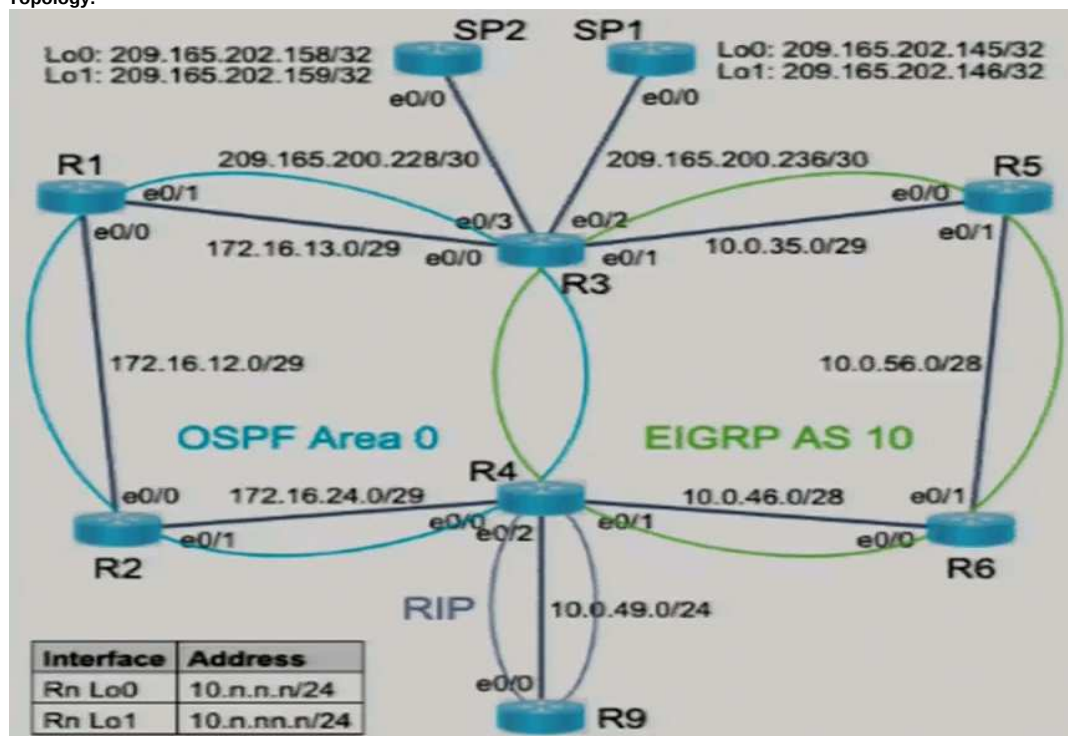
Remarks :

The IP address of SP1 and SP2 can be calculated as follows:

- For the network 209.165.201.0/30 shown in the diagram, since the IP address of R3's e0/2 is 209.165.201.1, the IP address of SP1's e0/0 is 209.165.201.2.
- For the network 209.165.200.224/30 shown in the diagram, since the IP address of R3's e0/3 is 209.165.200.225, the IP address of SP2's e0/0 is 209.165.200.226.

QUESTION 15

Topology:



Tasks:

Troubleshoot and resolve the issues to achieve these goals:

1. Ensure that R6 can ping the prefix 10.9.99.9. Manipulate the first basic routing decision-making criteria of longest prefix match that if a router learns a route from different routing protocols, the longest matched prefix can be changed. Use decimal value of 75 if required to accomplish this. Do not use a route-map.
2. Ensure that R2 loopback 1 reaches to R5's loopback 1 by following the path through R2, R4, R6 to R5 and R5 loopback 1 reaches R2's loopback 1 by following the path through R5, R6, R4 to R2. Use metric values K1= 100000, K2=1, K3=255, K4=10, K5=1500 to modify the default metric in EIGRP if required. Do not add or modify the default-metric command under router eigrp 10. Do not use a route-map to set metrics.
3. Ensure that on R3, prefix 10.0.56.6/32 uses the SP1 to route to the Internet, whereas prefix 172.16.12.2/32 uses the SP2 to route to the Internet. Do not use BGP to accomplish this. Use the pre-configured route-maps INTERNET1 and INTERNET2 and modify to accomplish the task if required. Use the ping and trace commands from R6 and R2 to prefixes 209.165.202.146 and 209.165.202.158, respectively to verify the results.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

```
R3
en
config t
route-map INTERNET1 permit 10
    set ip next-hop 209.165.200.237
route-map SP2 permit 10
    set ip next-hop 209.165.200.229
int e0/0
    ip policy route-map INTERNET2
int e0/1
    ip policy route-map INTERNET1
end
copy run start
```

```
R4
en
config t
router rip
    distance 75 10.0.49.0 0.0.0.255
router eigrp 10
    no distance 255 0.0.0.0 255.255.255.255 5
    redistribute ospf 10 route-map default metric 100000 1 255 10 1500
end
copy run start
```

Explanations:

You should use sh run in the routers to check if the existing configuration settings are the same as those described below:

```
R3
interface Loopback0
    ip address 10.3.3.3 255.255.255.0
!
interface Loopback1
    ip address 10.3.33.3 255.255.255.0
!
interface Ethernet0/0
    ip address 172.16.13.3 255.255.255.248
    ip nat inside
!
interface Ethernet0/1
    ip address 10.0.35.3 255.255.255.248
    ip nat inside
!
interface Ethernet0/2
    ip address 209.165.200.238 255.255.255.252
    ip nat outside
!
interface Ethernet0/3
    ip address 209.165.200.230 255.255.255.252
    ip nat outside
!
router eigrp 10
    default-metric 1 1 1 1 1
    network 10.0.35.0 0.0.0.7
    network 10.3.33.0 0.0.0.255
    redistribute ospf 10
    redistribute static
    eigrp router-id 10.3.33.3
!
router ospf 10
    router-id 10.3.3.3
    redistribute static subnets
    redistribute eigrp 10 subnets
    network 10.3.3.0 0.0.0.255 area 0
    network 172.16.13.0 0.0.0.7 area 0
    default-information originate metric 100 metric-type 1
!
ip nat inside source list 10 interface Ethernet1/0 overload
ip nat inside source list 20 interface Ethernet1/1 overload
ip route 0.0.0.0 0.0.0.0 Null0
!
route-map INTERNET2 permit 10
    match ip address 120
!
route-map INTERNET1 permit 10
    match ip address 110
!
access-list 10 permit 10.0.0.0 0.255.255.255
access-list 20 permit 172.16.0.0 0.15.255.255
access-list 110 deny ip 10.0.0.0 0.255.255.255 172.16.0.0 0.15.255.255
access-list 110 deny ip 10.0.0.0 0.255.255.255 10.0.0.0 0.255.255.255
access-list 110 permit ip host 10.0.56.6 any
access-list 120 deny ip 172.16.0.0 0.15.255.255 10.0.0.0 0.255.255.255
access-list 120 permit ip host 172.16.12.1 0.15.255.255 any
```

```
R4
interface Loopback0
    ip address 10.4.4.4 255.255.255.0
!
```

```

interface Loopback1
 ip address 10.4.44.4 255.255.255.0
!
interface Ethernet0/0
 ip address 172.16.24.4 255.255.255.248
!
interface Ethernet0/1
 ip address 10.0.46.4 255.255.255.240
!
interface Ethernet0/2
 ip address 10.0.49.4 255.255.255.0
!
router eigrp 10
 default-metric 1 1 1 1 1
 network 10.0.46.0 0.0.0.15
 network 10.4.46.0 0.0.0.255
 redistribute rip
 redistribute ospf 10 route-map default
 distance 255 0.0.0.0 255.255.255.255 5
 eigrp router-id 10.4.44.4
!
router ospf 10
 router-id 10.4.4.4
 redistribute eigrp 10 subnets
 network 10.4.4.0 0.0.0.255 area 0
 network 172.16.24.0 0.0.0.27 area 0
!
router rip
 version 2
 redistribute eigrp 10 metric 6
 network 10.0.0.0
!
route-map default deny 10
 match ip address 7
!
route-map default permit 20
!
access-list 7 permit 0.0.0.0

```

Task 1

In this network topology, routing loops occur for the networks of R9's loopback interfaces. Take the network of R9's lo1 i.e. 10.9.99.0/24 as an example:

1. R9 --- RIP (AD 120) --> R4
2. Redistributed from RIP to EIGRP in R4
3. R4 --- External EIGRP (AD 170) --> R6
4. R6 --- External EIGRP (AD 170) --> R5
5. R5 --- External EIGRP (AD 170) --> R3
6. Redistributed from EIGRP to OSPF in R3
7. R3 --- OSPF E2 (AD 110) --> R1
8. R1 --- OSPF E2 (AD 110) --> R2
9. R2 --- OSPF E2 (AD 110) --> R4
10. In R4's routing table, the RIP route is replaced by the OSPF E2 due to lower AD (110 < 120)
11. Redistributed from OSPF to EIGRP in R4
12. The steps 3 – 9 and 11 are repeated.

In R4's routing table, you can find the route similar as follows and this proves that the step 10 mentioned by occurred.

```
0 E2      10.9.99.0/24 [110/20] via 172.16.24.2, 00:00:02, Ethernet0/0
```

You can also use "**traceroute 10.9.99.9 numeric**" command in R6 to prove that routing loop is formed for the route 10.9.99.0/24:

```

R6#traceroute 10.9.99.9 numeric
Type escape sequence to abort.
Tracing the route to 10.9.99.9
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.46.4 9 msec 3 msec 2 msec
 2 172.16.24.2 4 msec 4 msec 4 msec
 3 172.16.12.1 5 msec 5 msec 5 msec
 4 172.16.13.3 7 msec 7 msec 6 msec
 5 10.0.35.5 4 msec 6 msec 6 msec
 6 10.0.56.6 5 msec 5 msec 4 msec
 7 10.0.46.4 8 msec 9 msec 10 msec
 8 172.16.24.2 8 msec 8 msec 9 msec
 9 172.16.12.1 10 msec 10 msec 10 msec
10 172.16.13.3 9 msec 9 msec 10 msec
11 10.0.35.5 11 msec 10 msec 9 msec
12 10.0.56.6 10 msec 7 msec 9 msec
13 10.0.46.4 12 msec 10 msec 11 msec
14 172.16.24.2 13 msec 12 msec 12 msec
15 172.16.12.1 15 msec 14 msec 12 msec
16 172.16.13.3 14 msec 11 msec 13 msec
17 10.0.35.5 13 msec 12 msec 14 msec
18 10.0.56.6 13 msec 11 msec 14 msec
19 10.0.46.4 14 msec 14 msec 14 msec
20 172.16.24.2 15 msec 15 msec 17 msec
21 172.16.12.1 17 msec 17 msec 17 msec
22 172.16.13.3 16 msec 16 msec 19 msec
23 10.0.35.5 17 msec 18 msec 16 msec
24 10.0.56.6 16 msec 17 msec 15 msec
25 10.0.46.4 18 msec 18 msec 17 msec
26 172.16.24.2 20 msec 20 msec 19 msec
27 172.16.12.1 22 msec 21 msec 22 msec
28 172.16.13.3 22 msec 20 msec 20 msec
29 10.0.35.5 20 msec 21 msec 19 msec
30 10.0.56.6 20 msec 20 msec 19 msec
R6#

```

In order to prevent step 10 that causes the loop to occur, you can set a lower AD for the RIP routes e.g. the AD 75 mentioned in the task.

This can be done by the "distance" command under RIP configuration in R4 through specifying the IP / network of the RIP neighbor R9 i.e. 10.0.49.9/32 or

10.0.49.0/24 as follows:

```
Configuration required to be set in R4
router rip
 distance 75 10.0.49.0 0.0.0.255
```

When R4 receives the RIP route from R9 again (it may take a while), since the RIP routes now has an AD of 75, they will not be replaced by the external OSPF routes in R4's routing table.

```
R      10.9.99.0/24 [75/1] via 10.0.49.9, 00:00:29, Ethernet0/2
```

After the above RIP route is found in R4's routing table, you can also use ping or "**traceroute 10.9.99.9 numeric**" command again in R6 to prove that R6 can now access R9's lo1:

```
R6#traceroute 10.9.99.9 numeric
Type escape sequence to abort.
Tracing the route to 10.9.99.9
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.46.4 6 msec 4 msec 2 msec
 2 10.0.49.9 6 msec * 5 msec
R6#
```

Task 2

In this network topology, R2 and R5 can reach each other lo1. However, the path taken may not be the one required by the task.

The following command shows that R2's lo1 reaches R5's lo1 via R1 --> R3 --> R5 (note that the 4th octet of the IP address shows the router number). This does NOT meet the requirement.

```
R2#traceroute 10.5.55.5 source 10.2.22.2 numeric
Type escape sequence to abort.
Tracing the route to 10.5.55.5
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.12.1 8 msec 3 msec 5 msec
 2 172.16.13.3 5 msec 4 msec 5 msec
 3 10.0.35.5 6 msec * 7 msec
R2#
```

You can also check the routing entry for R5's lo1 in R2's routing table to prove that R5 will send the packet through R3.

```
0 E2      10.5.55.0/24 [110/20] via 172.16.12.1, 00:17:29, Ethernet0/0
```

The following command shows that, in the reverse direction, R5's lo1 reaches R2's lo1 via R3--> R1 --> R2. This also does NOT meet the requirement.

```
R5#traceroute 10.2.22.2 source 10.5.55.5 numeric
Type escape sequence to abort.
Tracing the route to 10.2.22.2
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.35.3 6 msec 3 msec 8 msec
 2 172.16.13.1 4 msec 6 msec 5 msec
 3 172.16.12.2 6 msec * 5 msec
R5#
```

Hence, you need to change the path taken by both directions.

Under normal configuration, since both use external routes (OSPF / EIGRP) to reach each other loopback, the router usually take the path through the nearest ASBR.

However, for the traffic from R2's lo1 to R5's lo1, R2 send the traffic through the ASBR R3 instead of the directly connected ASBR R4. Hence, there must be some special settings pre-configured in the ASBRs R3 and/or R4 for this special behaviour. Since traffic of both directions can use the ASBR R3, the abnormal configuration is probably in R4. This can be further proved by the fact that although internal EIGRP route has a lower AD than OSPF routes, the routes for the loopback of R5 and R6 are OSPF external routes in R4's routing table.

```
0 E2      10.5.5.0/24 [110/20] via 172.16.24.2, 00:37:57, Ethernet0/0
0 E2      10.5.55.0/24 [110/20] via 172.16.24.2, 00:37:57, Ethernet0/0
0 E2      10.6.6.0/24 [110/20] via 172.16.24.2, 00:37:57, Ethernet0/0
0 E2      10.6.66.0/24 [110/20] via 172.16.24.2, 00:37:57, Ethernet0/0
```

By checking the configuration in R4, the following special pre-configured settings are found:

```
Existing Configuration in R4
router eigrp 10
 redistribute ospf 10 route-map default
 distance 255 0.0.0.0 255.255.255.255 5
!
route-map default deny 10
 match ip address 7
!
route-map default permit 20
!
access-list 7 permit 0.0.0.0
```

For the command "distance 255 0.0.0.0 255.255.255.255 5", since the access list 5 is not configured, the command is the same as "distance 255 0.0.0.0 255.255.255.255" i.e. making all route learnt from any EIGRP neighbor (0.0.0.0 255.255.255.255) to become inaccessible (due to AD 255). Without entering the EIGRP route for R5's lo1 in R4's routing table, R4 will not redistribute the EIGRP route to OSPF and therefore R2 only receive the route to R5's lo1 from R3's redistribution.

By removing the "distance" command, R2 should be able to use the path via R4 to reach R5's lo1.

For the reverse direction, in order for the traffic from R5 to R2 to use the ASBR R4 (instead of the ASBR R3 nearer to R5), R4 should redistribute the OSPF route with a better metric than that redistributed by R3. This can be achieved by using the metric required in the task. (It is a better metric than the default metric 1 1 1 1 pre-configured in R3 and R4)

Since you are not allowed to change the "default-metric" command, you can set the new metrics for redistributing OSPF routes (since R2 advertises its loopback through OSPF) as follows:

Configuration required to be set in R4

```
router eigrp 10
  no distance 255 0.0.0.0 255.255.255.255 5
  redistribute ospf 10 route-map default metric 100000 1 255 10 1500
```

Remarks :

The route map configured in the "redistribute ospf" prevents the route specified in the access list 7 from being redistributed. Since the access list 7 specifies "host 0.0.0.0", it blocks the redistribution of OSPF default route from into EIGRP. In the topology, R3 has a static default null0 route which is redistributed to OSPF and EIGRP through "redistribute static". Hence, R5 and R6 can still have the default route learnt from EIGRP advertised by R3 without the need to redistribute the OSPF default route by R4. Actually, the blocking of this redistribution can avoid R6 to use the longer path via R4 instead of the shorter path R5 to access Internet, we shall keep the route map in our configuration.

After the above configuration, R2 will use the path through R4 to reach R5's lo1. You can prove this by traceroute:

```
R2#traceroute 10.5.55.5 source 10.2.22.2 numeric
```

```
Type escape sequence to abort.
Tracing the route to 10.5.55.5
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.24.4 6 msec 4 msec 3 msec
 2 10.0.46.6 5 msec 5 msec 6 msec
 3 10.0.56.5 7 msec * 7 msec
```

R2#

R5 will also use the path through R4 to reach R2's lo1.

```
R5#traceroute 10.2.22.2 source 10.5.55.5 numeric
```

```
Type escape sequence to abort.
Tracing the route to 10.2.22.2
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.56.6 6 msec 5 msec 4 msec
 2 10.0.46.4 5 msec 5 msec 6 msec
 3 172.16.24.2 6 msec * 9 msec
```

R5#

Task 3

In R3, two route map INTERNET1 and INTERNET2 is preconfigured. Their configurations are:

- INTERNET1 uses ACL 110 which matches traffic from 10.0.0.0/8.
- INTERNET2 uses ACL 120 which matches traffic from 172.16.0.0/12

In order to routing traffic according to source, you can setup PBR. Since the route map is already configured, you just need to add "set ip next-hop" command in the PBR and apply the route maps to the appropriate interfaces.

Since route map INTERNET1 matches source from 10.0.56.6/32:

- you need to use SP1's e0/0 IP address for the "set ip next-hop" command. (You can check it in SP1's router or calculate the address using R3's e0/2).
- you need to apply it to R3's e0/1 that connects to the EIGRP network.

Remarks :

The route map used in the command "redistribute ospf 10 route-map default" in R4 ensures that R6 will access Internet through R3's e0/1 for PBR instead of going through the OSPF network. Hence, you should not remove the route map "default" in the previous configuration for task 2.

Since route map INTERNET2 matches source from 172.16.12.2/32:

- you need to use SP2's e0/0 IP address for the "set ip next-hop" command. (You can check it in SP2's router or calculate the address using R3's e0/3).
- you need to apply it to R3's e0/0 that connects to the OSPF network.

Configuration required to be set in R3

```
route-map INTERNET1 permit 10
  set ip next-hop 209.165.200.237

route-map INTERNET2 permit 10
  set ip next-hop 209.165.200.229

int e0/0
  ip policy route-map INTERNET2
int e0/1
  ip policy route-map INTERNET1
```

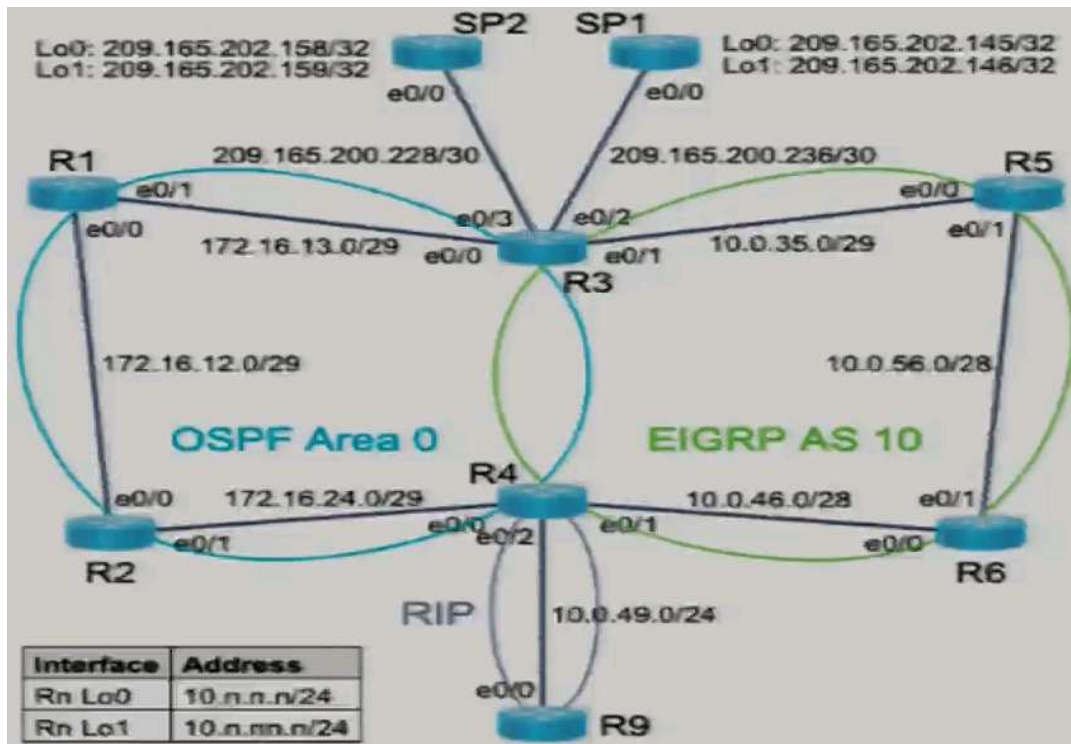
Remarks :

The IP address of SP1 and SP2 can be calculated as follows:

- From the diagram, the network between R3 and SP1 is 209.165.200.236/30, since R3's e0/2 is using ".238", the other IP left to be used by SP1 is ".237".
- From the diagram, the network between R3 and SP2 is 209.165.200.228/30, since R3's e0/3 is using ".230", the other IP left to be used by SP1 is ".229".

QUESTION 16

Topology:



Tasks:

Troubleshoot and resolve the issues to achieve these goals:

1. Ensure that R1 reaches R5 and R6 loopback1's without any single point of failure in the path. Do not use a static route or policy-based routing to accomplish this.
2. Ensure that R1 loopback 1 reaches to R6's loopback 1 by following the path through R1, R3, R5 to R6 and vice versa. Use metric values K1= 100000, K2=1, K3=255, K4=10, K5=1500 to modify the default metric in EIGRP if required. Do not add or modify the default-metric command under router eigrp 10. Do not use a route-map to set metrics.
3. Ensure that on R3, prefix 10.0.56.0/32 uses the SP1 to route to the Internet, whereas prefix 172.16.12.0/32 uses the SP2 to route to the Internet. Do not use BGP to accomplish this. Use the pre-configured route-maps INTERNET1 and INTERNET2 and modify to accomplish the task if required. Use the ping and trace commands from R6 and R2 to prefixes 209.165.202.146 and 209.165.202.158, respectively to verify the results.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

```
R3
en
config t
router eigrp 10
  no distance 255 0.0.0.0 255.255.255.255
router eigrp 10
  redistribute ospf 10 metric 100000 1 255 10 1500
route-map INTERNET1 permit 10
  set ip next-hop 209.165.200.237
route-map INTERNET2 permit 10
  set ip next-hop 209.165.200.229
int e0/0
  ip policy route-map INTERNET2
int e0/1
  ip policy route-map INTERNET1
end
copy run start
```

Explanations:

You should use sh run in the routers to check if the existing configuration settings are the same as those described below:

```
R3
interface Loopback0
  ip address 10.3.3.3 255.255.255.0
!
interface Loopback1
  ip address 10.3.33.3 255.255.255.0
!
interface Ethernet0/0
  ip address 172.16.13.3 255.255.255.248
  ip nat inside
!
interface Ethernet0/1
  ip address 10.0.35.3 255.255.255.248
  ip nat inside
!
interface Ethernet0/2
  ip address 209.165.200.238 255.255.255.252
  ip nat outside
!
```

```

interface Ethernet0/3
 ip address 209.165.200.230 255.255.255.252
 ip nat outside
!
router eigrp 10
 default-metric 1 1 1 1 1
 network 10.0.35.0 0.0.0.7
 network 10.3.33.0 0.0.0.255
 redistribute ospf 10
 redistribute static
 distance 255 0.0.0.0 255.255.255.255
 eigrp router-id 10.3.33.3
!
router ospf 10
 router-id 10.3.3.3
 redistribute static subnets
 redistribute eigrp 10 subnets
 network 10.3.3.0 0.0.0.255 area 0
 network 172.16.13.0 0.0.0.7 area 0
 default-information originate metric 100 metric-type 1
!
ip nat inside source list 10 interface Ethernet1/0 overload
ip nat inside source list 20 interface Ethernet1/1 overload
ip route 0.0.0.0 0.0.0.0 Null0
!
route-map INTERNET2 permit 10
 match ip address 120
!
route-map INTERNET1 permit 10
 match ip address 110
!
access-list 10 permit 10.0.0.0 0.255.255.255
access-list 20 permit 172.16.0.0 0.15.255.255
access-list 110 deny ip 10.0.0.0 0.255.255.255 172.16.0.0 0.15.255.255
access-list 110 deny ip 10.0.0.0 0.255.255.255 10.0.0.0 0.255.255.255
access-list 110 permit ip host 10.0.56.6 any
access-list 120 deny ip 172.16.0.0 0.15.255.255 10.0.0.0 0.255.255.255
access-list 120 permit ip host 172.16.12.1 0.15.255.255 any

```

```

R4
interface Loopback0
 ip address 10.4.4.4 255.255.255.0
!
interface Loopback1
 ip address 10.4.44.4 255.255.255.0
!
interface Ethernet0/0
 ip address 172.16.24.4 255.255.255.248
!
interface Ethernet0/1
 ip address 10.0.46.4 255.255.255.240
!
interface Ethernet0/2
 ip address 10.0.49.4 255.255.255.0
!
router eigrp 10
 default-metric 1 1 1 1 1
 network 10.0.46.0 0.0.0.15
 network 10.4.46.0 0.0.0.255
 redistribute rip
 redistribute ospf 10 route-map default
 eigrp router-id 10.4.44.4
!
router ospf 10
 router-id 10.4.4.4
 redistribute eigrp 10 subnets
 network 10.4.4.0 0.0.0.255 area 0
 network 172.16.24.0 0.0.0.27 area 0
!
router rip
 version 2
 redistribute eigrp 10 metric 6
 network 10.0.0.0
!
route-map default deny 10
 match ip address 7
!
route-map default permit 20
!
access-list 7 permit 0.0.0.0

```

Task 1

The following uses traceroute to check the path used by R1 to reach R5's lo1 and R6's lo1. Both uses the path through R4 to reach the destination.

```

R1#traceroute 10.5.55.5 numeric
Type escape sequence to abort.
Tracing the route to 10.5.55.5
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.12.2 5 msec 6 msec 4 msec
 2 172.16.24.4 6 msec 3 msec 4 msec
 3 10.0.46.6 7 msec 7 msec 4 msec
 4 10.0.56.5 8 msec * 5 msec
R1#

```

```

R1#traceroute 10.6.66.6 numeric
Type escape sequence to abort.
Tracing the route to 10.6.66.6
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.12.2 2 msec 3 msec 5 msec

```



```
2 172.16.24.4 3 msec 5 msec 4 msec
3 10.0.46.6 6 msec * 5 msec
R1#
```

There is a simple way to perform preliminary check whether R1 can use another path if R4 or any related link are dead: you can check the routing table of the router along the alternate path i.e. through R3. The routing entries for R5's lo1 and R6's lo1 in R3's routing table are:

```
Routing Entries in R3
0 E2 10.5.55.0/24 [110/20] via 172.16.13.1, 00:17:24, Ethernet0/0
0 E2 10.6.66.0/24 [110/20] via 172.16.13.1, 00:17:24, Ethernet0/0
```

Since the next hops of the two routes are R1, R3 cannot send the traffic to R5's lo1 and R6's lo1 directly to the EIGRP network. Instead, the traffic also travels along the path through R4. Hence, if R4 is dead, R1 does not have another path to reach to R5's lo1 and R6's lo1 i.e. there is now a single point of failure.

Normally, R3 should be able to learn internal EIGRP route of R5's lo1 and R6's lo1 and since they should have AD 90, they should be entered into the R3's routing table instead of the OSPF E2 routes as shown above.

After checking the EIGRP configuration in R3, the following is found:

```
Existing configuration in R3
router eigrp 10
 distance 255 0.0.0.0 255.255.255.255
```

Since the setting of AD 255 (meaning inaccessible) to internal EIGRP routes learnt from any EIGRP neighbor, R3 can only enter the routes of R5's lo1 and R6's lo1 learnt through OSPF.

Therefore, we can remove this "distance" command to let R3 to enter the EIGRP routes for R5's lo1 and R6's lo1 in its routing table again.

```
Configuration required to be set in R3
router eigrp 10
 no distance 255 0.0.0.0 255.255.255.255
```

After making the above configuration, since R1 will use the nearest ASBR to access external OSPF E2 route of the same cost, it will send the packet to R5's lo1 and R6's lo1 through R3.

```
R1#traceroute 10.5.55.5 numeric
Type escape sequence to abort.
Tracing the route to 10.5.55.5
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.13.3 4 msec 2 msec 6 msec
 2 10.0.35.5 5 msec * 4 msec
R1#
```

```
R1#traceroute 10.6.66.6 numeric
Type escape sequence to abort.
Tracing the route to 10.6.66.6
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.13.3 5 msec 2 msec 3 msec
 2 10.0.35.5 5 msec 5 msec 4 msec
 3 10.0.56.6 6 msec * 6 msec
R1#
```

However, if R3 is dead, R1 should be able use the previous path through R4. This can be indirectly proved by the routing table in R2.

```
Routing Entries in R2
0 E2 10.5.55.0/24 [110/20] via 172.16.24.4, 01:33:30, Ethernet0/1
0 E2 10.6.66.0/24 [110/20] via 172.16.24.4, 01:33:30, Ethernet0/1
```

Task 2

In this network topology, R1 and R6 can reach each other lo1. However, the path taken may not be the one required by the task.

The following command shows that R1's lo1 reaches R6's lo1 via R3 --> R5 --> R6 (note that the 4th octet of the IP address shows the router number). This meets the requirement.

```
R1#traceroute 10.6.66.6 numeric
Type escape sequence to abort.
Tracing the route to 10.6.66.6
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.13.3 5 msec 2 msec 4 msec
 2 10.0.35.5 5 msec 5 msec 3 msec
 3 10.0.56.6 6 msec * 4 msec
R1#
```

The following command shows that, in the reverse direction, R6's lo1 reaches R1's lo1 via R4 --> R2 --> R1. This also does NOT meet the requirement.

```
R6#traceroute 10.1.11.1 source 10.6.66.6 numeric
Type escape sequence to abort.
Tracing the route to 10.1.11.1
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.46.4 6 msec 3 msec 4 msec
 2 172.16.24.2 5 msec 6 msec 6 msec
 3 172.16.12.1 7 msec * 6 msec
R6#
```

Hence, you need to change the path taken by R6's lo1 to R1's lo1.

Under normal configuration, R6 prefers the nearest ASBR R4 and therefore the traffic is sent through R4. In order to change this, you need to make the external EIGRP routes of R1's lo1 redistributed by R3 from OSPF to be preferred than that redistributed by R4.

This can be achieved by using the better metric shown in the task for redistributing OSPF routes into EIGRP in R3.

Since you are not allowed to change the "default-metric" command, you can set the new metrics for redistributing OSPF routes (since R1 advertises its loopback through OSPF) as follows:

```
Configuration required to be set in R3
router eigrp 10
```

```
redistribute ospf 10 metric 100000 1 255 10 1500
```

After the above configuration, R6 will use the path through R3 to reach R1's lo1. You can prove this by traceroute:

```
R6#traceroute 10.1.11.1 source 10.6.66.6 numeric
Type escape sequence to abort.
Tracing the route to 10.1.11.1
VRF info: (vrf in name/id, vrf out name/id)
 1 10.0.56.5 3 msec 4 msec 4 msec
 2 10.0.35.3 6 msec 4 msec 5 msec
 3 172.16.13.1 7 msec * 6 msec
R6#
```

Task 3

In R3, two route maps INTERNET1 and INTERNET2 is preconfigured. Their configurations are:

- INTERNET1 uses ACL 110 which matches traffic from 10.0.0.0/8.
- INTERNET2 uses ACL 120 which matches traffic from 172.16.0.0/12

In order to routing traffic according to source address, you can setup PBR. Since the route map is already configured, you just need to add "set ip next-hop" command in the PBR and apply the route maps to the appropriate interfaces.

Since route map INTERNET1 matches source from 10.0.56.6/32:

- you need to use SP1's e0/0 IP address for the "set ip next-hop" command. (You can check it in SP1's router or calculate the address using R3's e0/2).
- you need to apply it to R3's e0/1 that connects to the EIGRP network.

Since route map INTERNET2 matches source from 172.16.12.2/32:

- you need to use SP2's e0/0 IP address for the "set ip next-hop" command. (You can check it in SP2's router or calculate the address using R3's e0/3).
- you need to apply it to R3's e0/1 that connects to the OSPF network.

Configuration required to be set in R3

```
route-map INTERNET1 permit 10
  set ip next-hop 209.165.200.237
```

```
route-map INTERNET2 permit 10
  set ip next-hop 209.165.200.229
```

```
int e0/0
 ip policy route-map INTERNET2
int e0/1
 ip policy route-map INTERNET1
```

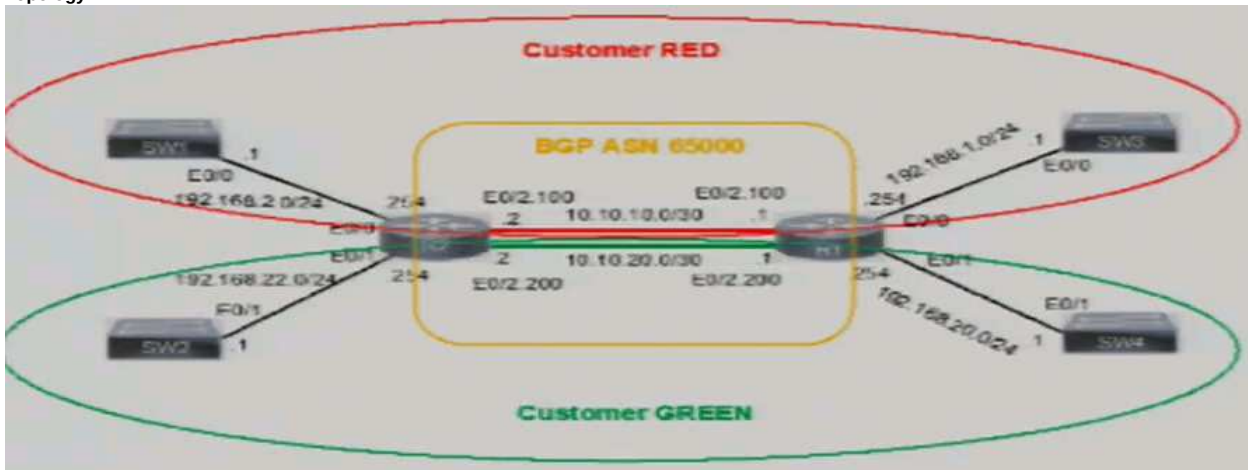
Remarks :

The IP address of SP1 and SP2 can be calculated as follows:

- From the diagram, the network between R3 and SP1 is 209.165.200.236/30, since R3's e0/2 is using ".238", the other IP left to be used by SP1 is ".237".
- From the diagram, the network between R3 and SP2 is 209.165.200.228/30, since R3's e0/3 is using ".230", the other IP left to be used by SP1 is ".229".

QUESTION 17

Topology:



Tasks:

Configure individual VRFs for each customer according to the topology to achieve these goals:

1. VRF "cu-red" has interfaces on routers R1 and R2. Both routers are preconfigured with IP addressing, VRFs, and BGP. Do not use the BGP network statement for advertisement.
2. VRF "cu-green" has interfaces on routers R1 and R2.
3. BGP on router R1 populates VRF routes between router R1 and R2.
4. BGP on router R2 populates VRF routes between router R1 and R2.
5. LAN to LAN is reachable between SW1 and SW3 for VRF "cu-red" and between SW2 and SW4 for VRF "cu-green". All switches are preconfigured.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

Task 1:

```

R1
en
config t
int e0/0
    ip vrf forwarding cu-red
    ip address 192.168.1.254 255.255.255.0
int e0/2.100
    ip vrf forwarding cu-red
    ip addr 10.10.10.1 255.255.255.252
end
copy run start

```

```

R2
en
config t
int e0/0
    ip vrf forwarding cu-red
    ip address 192.168.2.254 255.255.255.0
int e0/2.100
    ip vrf forwarding cu-red
    ip addr 10.10.10.2 255.255.255.252
end
copy run start

```

Task 2:

```

R1
en
config t
int e0/1
    ip vrf forwarding cu-green
    ip address 192.168.20.254 255.255.255.0
int e0/2.200
    ip vrf forwarding cu-green
    ip addr 10.10.20.1 255.255.255.252
end
copy run start

```

```

R2
en
config t
int e0/1
    ip vrf forwarding cu-green
    ip address 192.168.22.254 255.255.255.0
int e0/2.200
    ip vrf forwarding cu-green
    ip addr 10.10.20.2 255.255.255.252
end
copy run start

```

Task 3 & 4:

```

R1
en
config t
router bgp 65000
    address-family ipv4 vrf cu-red
        neighbor 10.10.10.2 remote-as 65000
        redistribute connected
    exit-address-family
    address-family ipv4 vrf cu-green
        neighbor 10.10.20.2 remote-as 65000
        redistribute connected
    exit-address-family
end
copy run start

```

```

R2
en
config t
router bgp 65000
    address-family ipv4 vrf cu-red
        neighbor 10.10.10.1 remote-as 65000
        redistribute connected
    exit-address-family
    address-family ipv4 vrf cu-green
        neighbor 10.10.20.1 remote-as 65000
        redistribute connected
    exit-address-family
end
copy run start

```

Task 5:

In SW1 ping SW3
ping 192.168.1.1

In SW3 ping SW1
ping 192.168.2.1

In SW2 ping SW4
ping 192.168.20.1

In SW4 ping SW2

```
ping 192.168.22.1
```

Explanations:

R1 and R2 is using a sub-interface for each VRF and the setup is using VRF-Lite with BGP as the routing protocol i.e. This setup does NOT use VPNv4 or MP-BGP.

You should use `sh run` in the routers to check if the existing configuration settings are the same as those described below:

```
R1
ip vrf cu-green
  rd 65000:200
ip vrf cu-red
  rd 65000:100
int Loopback0
  ip address 10.10.1.1 255.255.255.255
int Ethernet0/0
  ip address 192.168.1.254 255.255.255.0
int Ethernet0/1
  ip address 192.168.20.254 255.255.255.0
int Ethernet0/2
  no ip address
int Ethernet0/2.100
  encapsulation dot1Q 100
  ip addr 10.10.10.1 255.255.255.252
int Ethernet0/2.200
  encapsulation dot1Q 200
  ip addr 10.10.20.1 255.255.255.252
router bgp 65000
  bgp log-neighbor-changes
  no bgp default ipv4-unicast
```

```
R2
ip vrf cu-green
  rd 65000:200
ip vrf cu-red
  rd 65000:100
int Loopback0
  ip address 10.10.2.2 255.255.255.255
int Ethernet0/0
  ip address 192.168.2.254 255.255.255.0
int Ethernet0/1
  ip address 192.168.22.254 255.255.255.0
int Ethernet0/2
  no ip address
int Ethernet0/2.100
  encapsulation dot1Q 100
  ip addr 10.10.10.2 255.255.255.252
int Ethernet0/2.200
  encapsulation dot1Q 200
  ip addr 10.10.20.2 255.255.255.252
router bgp 65000
  bgp log-neighbor-changes
  no bgp default ipv4-unicast
```

Note that the two VRF "cu-red" and "cu-green" have already been defined in the two routers.

Task 1:

This asks you to configure VRF "cu-red" for the two routers according to the network diagram. According to the diagram, you need to configure VRF "cu-red" for:

- In R1, e0/2.100 and e0/0
- In R2, e0/2.100 and e0/0

Although IP addresses have been pre-configured in the sub-interfaces, since assigning VRFs to the interfaces / subinterfaces will remove the IP addresses from the interfaces, you need to configure the IP addresses back after assigning VRFs.

Configuration required in R1

```
int e0/0
  ip vrf forwarding cu-red
  ip address 192.168.1.254 255.255.255.0
int e0/2.100
  ip vrf forwarding cu-red
  ip addr 10.10.10.1 255.255.255.252
```

Configuration required in R2

```
int e0/0
  ip vrf forwarding cu-red
  ip address 192.168.2.254 255.255.255.0
int e0/2.100
  ip vrf forwarding cu-red
  ip addr 10.10.10.2 255.255.255.252
```

Task 2:

This asks you to configure VRF "cu-green" for the two routers according to the network diagram. Therefore you need to configure VRF "cu-green" for:

- In R1, e0/2.200 and e0/1
- In R2, e0/2.200 and e0/1

Similar configuration settings like Task 1 are needed:

Configuration required in R1

```
int e0/1
  ip vrf forwarding cu-green
  ip address 192.168.20.254 255.255.255.0
int e0/2.200
  ip vrf forwarding cu-green
  ip addr 10.10.20.1 255.255.255.252
```

Configuration required in R2

```
int e0/1
 ip vrf forwarding cu-green
 ip address 192.168.22.254 255.255.255.0
int e0/2.200
 ip vrf forwarding cu-green
 ip addr 10.10.20.2 255.255.255.252
```

Task 3 & 4:

This asks you to configure BGP for the 2 VRFs in R1 using the ipv4 address family of the two VRFs (You need address-family to configure BGP for specific VRF). According to requirement mentioned in task1, since BGP "network" command is not allowed, "redistribute connected" is used instead.

Configuration required in R1

```
router bgp 65000
 address-family ipv4 vrf cu-red
  neighbor 10.10.10.2 remote-as 65000
  redistribute connected
exit-address-family
 address-family ipv4 vrf cu-green
  neighbor 10.10.20.2 remote-as 65000
  redistribute connected
exit-address-family
```

Configuration required in R2

```
router bgp 65000
 address-family ipv4 vrf cu-red
  neighbor 10.10.10.1 remote-as 65000
  redistribute connected
exit-address-family
 address-family ipv4 vrf cu-green
  neighbor 10.10.20.1 remote-as 65000
  redistribute connected
exit-address-family
```

Task 5:

No configuration is needed for this task. This task just verify the configuration settings that you have made for the previous tasks.

- SW1 and SW3 should be able to ping each other to ensure the settings VRF "cu-red" is properly configured.
- SW2 and SW4 should be able to ping each other to ensure the settings VRF "cu-red" is properly configured.

IP address for the four switches can be found in the diagram as follows:

```
SW1: 192.168.2.1
SW3: 192.168.1.1
SW2: 192.168.22.1
SW4: 192.168.20.1
```

Therefore, the ping commands required are:

```
In SW1 ping SW3
ping 192.168.1.1
```

```
In SW3 ping SW1
ping 192.168.2.1
```

```
In SW2 ping SW4
ping 192.168.20.1
```

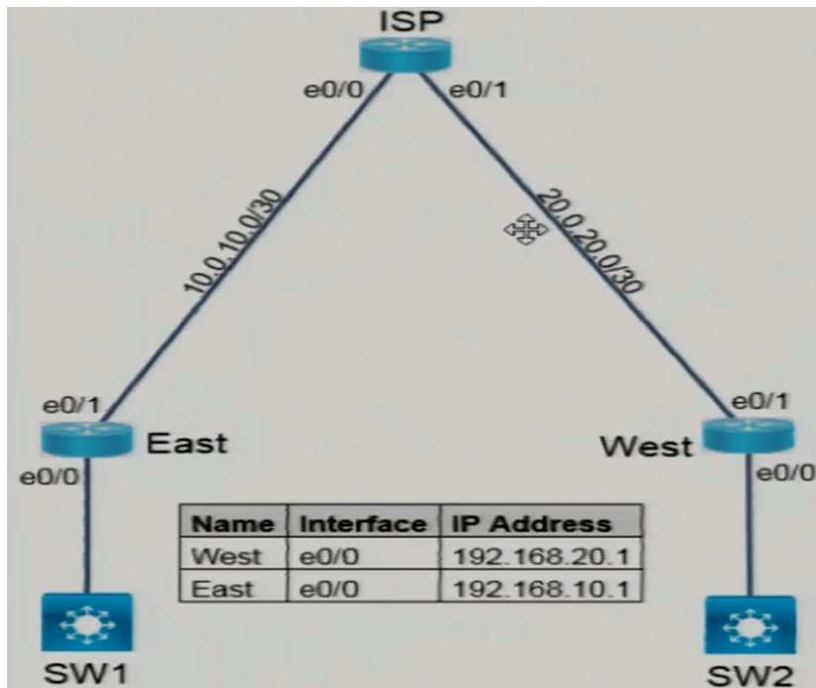
```
In SW4 ping SW2
ping 192.168.22.1
```

For troubleshooting, you can use the following commands:

```
sh ip route vrf cu-red
sh ip route vrf cu-green
sh bgp vrf cu-red
sh bgp vrf cu-green
```

QUESTION 18

Topology:



Tasks:

Troubleshoot and resolve the issues on West and East routers to achieve these goals. The username/password and enable password is 'cisco' on all devices.

1. Ensure that the west router authorizes remote users over ssh to use a local database matching username as "westadmin" and the password is "westadminpass". Generate RSA 2048 bit key. Modify the existing aaa policy name for configuration.
2. Ensure that the ping is permitted to the east router daily from 9:00 AM to 6:00 PM only. Complete the task using a preconfigured access list; do not add any deny entries.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

WEST

```
en
config t
crypto key generate rsa modulus 2048
username westadmin password westadminpass
no aaa authentication login SSH_Connections local none
aaa authentication login SSH_Connections local
line vty 0 4
  login authentication SSH_Connections
end
copy run start
```

EAST

```
en
config t
time-range blkicmp
  periodic daily 0:00 to 8:59
ip access-list extended icmp-rule
  permit ip any any
end
copy run start
```

Explanations:

You should use sh run in the routers to check if the existing configuration settings are the same as those described below:

EAST

```
interface Ethernet0/0
  ip address 192.168.10.1 255.255.255.0
!
interface Ethernet0/1
  ip address 10.0.10.2 255.255.255.252
  ip access-group icmp-rule in
!
ip route 0.0.0.0 0.0.0.0 10.0.10.1
!
ip access-list extended icmp-rule
  deny icmp any any time-range blkicmp
!
line con 0
  exec-timeout 0 0
  logging synchronous
line vty 0 4
  exec-timeout 5 0
  password 7 xxxxxxxxxxxxxx
  login local
  transport input all
```



```

!
time-range blkicmp
  periodic daily 18:01 to 23:59

WEST
ip domain name cisco.com
!
aaa new-model
aaa authentication login default local-case none
aaa authentication login SSH_Connections local none
!
interface Ethernet0/0
  ip address 192.168.20.1 255.255.255.0
!
interface Ethernet0/1
  ip address 20.0.20.2 255.255.255.252
!
ip route 0.0.0.0 0.0.0.0 20.0.20.1
!
line con 0
  exec-timeout 0 0
  logging synchronous
  stopbits 1
line vty 0 4
  exec-timeout 5 0
  transport input all

```

Note that for some devices, you may need to use "cisco" as the username / password or enable password in order to access them.

Task 1

In order to meet the requirement, you need to perform the followings in the router "WEST":

- Generate the RSA 2048 bit key to enable SSH.
- Create the username and password required by the task.

Configuration required in the router WEST
 crypto key generate rsa modulus 2048
 username westadmin password westadminpass

There are two existing AAA authentication list pre-configured in the router:
 aaa authentication login default local-case none
 aaa authentication login SSH_Connections local none

Without any AAA authentication setting preconfigured in line vty, the default authentication list shown above will be used. However, other than entering the correct username and password, a user can also grant access if he has typed a wrong username due to the method "none" included at the end of the default authentication list.

The pre-configured authentication list "SSH_Connections" also include "none". Therefore, you cannot use it directly in line vty since the task asks you to ensure that west router authorizes remote users.

Since the pre-configured authentication list "SSH_Connections" seems to be pre-configured for the purpose of SSH connection, you can make changes to the list by removing "none" from it as follows:

Configuration required in the router WEST
 no aaa authentication login SSH_Connections local none
 aaa authentication login SSH_Connections local

Then, you can configure line vty to use the modified authentication list "SSH_Connections" as follows:

Configuration required in the router WEST
 line vty 0 4
 login authentication SSH_Connections

Remarks:

For verification, since there is an access list applied to the interface e0/1 of the router EAST, you need to make the SSH connection from the router ISP for testing.

Task 2

In the router "EAST", the pre-configured access list that has been applied to the interface e0/1 and the time range involved are as follows:

```

ip access-list extended icmp-rule
  deny icmp any any time-range blkicmp
!
time-range blkicmp
  periodic daily 18:01 to 23:59

```

In above pre-configured settings:

- ICMP traffic between 18:01 – 23:59 daily will be denied.
- ICMP traffic in other period as well as all other traffic at any time will be denied by implicit deny.

In order to meet the requirement of task 2, there are two methods:

1. Create a new time range with the allowed period 9:00-18:00 and then insert a new permit rule (e.g. using seq 5) for ICMP traffic using the new time range at the top of the pre-configured access list "icmp-rule".
2. Modify the existing time range "blkicmp" to include the remaining period when ICMP is not allowed (i.e. 0:00 – 8:59) so that the existing rule will block ICMP when the time is outside of the allowed period. Then append a permit rule at the end of the pre-configured access list "icmp-rule" to allow ICMP traffic in the period not matched by the time range. (the task only disallows you to add deny entry)

Remarks:

Note that both methods use the pre-configured access list "icmp-rule" (instead of creating a new access list) since the task asks you to use it.

Since most simulation question usually wants you to modify the existing configuration instead of adding new configuration settings of your own, the suggested answer will use the method 2 as follows:

Configuration required in the router EAST

```

time-range blkicmp
  periodic daily 0:00 to 8:59
ip access-list extended icmp-rule
  permit ip any any

```

After entering the above commands, the time range and the access list in router EAST will become:

```

EAST#sh time-range
time-range entry: blkicmp (inactive)
  periodic daily 18:01 to 23:59
  periodic daily 0:00 to 8:59
  used in: IP ACL entry
EAST #

EAST#sh access-list
Extended IP access list icmp-rule
  10 deny icmp any any time-range blkicmp (inactive) (10 matches)
  20 permit ip any any (883 matches)
EAST#

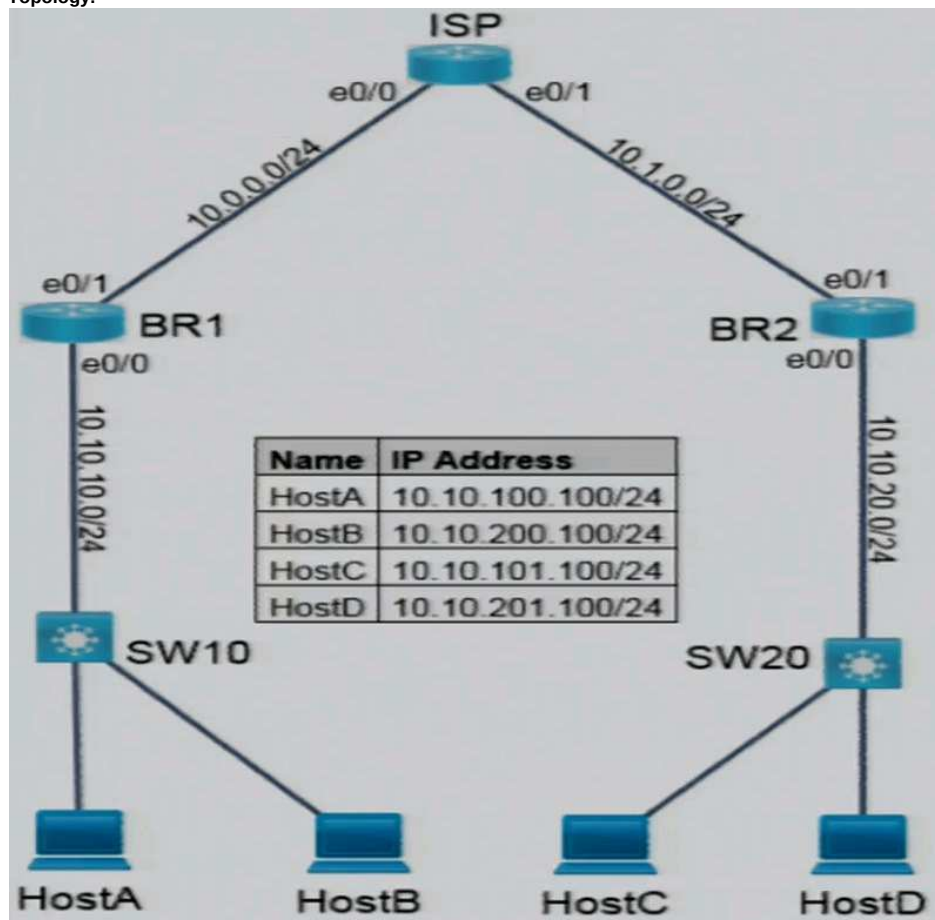
```

Remarks :

For the new permit rule appended, you may prefer to use "permit icmp any any" instead of "permit ip any any". It is because the task does not mention whether other traffic has to be pass between SW1 and ISP router. If you think that those traffic is not required, then you may configure "permit icmp any any" instead.

QUESTION 19

Topology:



Tasks:

Troubleshoot and resolve the issues on BR1 and BR2 to achieve these goals:

1. Ensure that BR1 console access is provided without any username and password, and telnet access for all sessions is provided upon entering the correct username and passwords. The users should be able to login according to locally defined privileges.

- BR1

- Username = cisco
- Privilege level = 15
- Password = cisco

- The enable password and username is 'Cisco' on all devices.

2. Ensure that all traffic from ISP sourced with RFC1918 IP addresses is blocked without affecting BGP adjacency between ISP and BR1 and BR2. Use the access list name "RFC1918". Ensure to allow traffic sourced from public IP addresses.

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

```

BR1
en
config t

```

```

username cisco privilege 15 password cisco
aaa authentication login telnetlogin local
aaa authorization exec telnetexec local
line vty 0 4
    login authentication telnetlogin
    authorization exec telnetexec
ip access-list standard BLK1918
    5 permit host 10.0.0.1
    no 10
    10 deny 10.0.0.0 0.255.255.255
    no 20
    20 deny 172.16.0.0 0.15.255.255
    permit any
end
copy run start

```

```

BR2
en
config t
ip access-list standard BLK1918
    5 permit host 10.1.0.1
    no 10
    10 deny 10.0.0.0 0.255.255.255
    no 20
    20 deny 172.16.0.0 0.15.255.255
end
copy run start

```

Explanations:

You should use `sh run` in the routers to check if the existing configuration settings are the same as those described below:

```

BR1
aaa new-model
!
aaa authentication login default none
!
username netadmin password 0 Cisco
username admin privilege 15 password 0 Cisco
username ops privilege 0 password 0 Cisco
!
interface Ethernet0/0
    ip address 10.10.10.1 255.255.255.0
!
interface Ethernet0/1
    ip address 10.0.0.2 255.255.255.0
    ip access-group BLK1918 in
!
router bgp 100
    redistribute static
    neighbor 10.0.0.1 remote-as 10
!
ip route 0.0.0.0 0.0.0.0 10.0.0.1
ip route 10.10.100.0 255.255.255.0 10.10.10.2
ip route 10.10.200.0 255.255.255.0 10.10.10.2
!
ip access-list standard BLK1918
    deny 10.0.0.0 0.0.255.255
    deny 172.16.0.0 0.0.255.255
    deny 192.168.0.0 0.0.255.255
!
line con 0
    exec-timeout 0 0
    logging synchronous
    login authentication none
!
line vty 0 4
    login authentication CONSOLE
    transport input all

```

```

BR2
enable secret 5 xxxxxxxxxxxxxxxxxxxx
!
no aaa new-model
!
aaa authentication login default none
!
username netadmin password 0 Cisco
username admin privilege 15 password 0 Cisco
username ops privilege 0 password 0 Cisco
!
interface Ethernet0/0
    ip address 10.10.20.1 255.255.255.0
!
interface Ethernet0/1
    ip address 10.1.0.2 255.255.255.252
    ip access-group BLK1918 in
!
router bgp 100
    redistribute static
    neighbor 10.1.0.1 remote-as 10
!
ip route 10.10.101.0 255.255.255.0 10.10.20.2
ip route 10.10.201.0 255.255.255.0 10.10.20.2
!
ip access-list standard BLK1918
    deny 10.0.0.0 0.0.255.255
    deny 172.16.0.0 0.0.255.255

```

```
deny 192.168.0.0 0.0.255.255
permit any
!
line con 0
exec-timeout 0 0
logging synchronous
login authentication none
!
line vty 0 4
login
transport input none
```

Task 1

BR1 currently does not require entering any username / password for console access. Therefore, we need to configure the telnet access without affecting the console access.

In order for users to be able to login using username / password and get access according to locally defined privileges, you need to:

- Configure a login authentication list with "local" and apply it to line vty.
- Configure an exec authorization list with "local" and apply it to line vty.

For the login authentication list, a non-exist login authentication list "CONSOLE" has already been configured to line vty, it is up to you if you want to configure the login authentication list for telnet access using this name. However, since this name seems like a list for console access, the suggested answer will create a new list with a different name as the login authentication list required by the task.

The task also specifies the need of a non-existing user "cisco". Therefore, we also need to create this user with the password and privilege level specified in the task.

The following shows the configuration settings for BR1:

```
BR1
username cisco privilege 15 password cisco
aaa authentication login telnetlogin local
aaa authorization exec telnetexec local
line vty 0 4
login authentication telnetlogin
authorization exec telnetexec
```

Remarks :

The task also includes a 2nd point about enable password and username is "Cisco". However, you cannot create the local usernames "cisco" and "Cisco" at the same time within a router. Hence, it contradicts with the 1st point in the task. Therefore, it probably just means that you may need to use "Cisco" as enable password or login for accessing all other devices. Although you cannot create another username "Cisco" in BR1, you may configure "enable secret cisco" in BR1 in addition to the suggested answer if you think that the 2nd point wants you to configure something in BR1.

You may verify by making a telnet connection to BR1 from SP1 only after task 2 has completed.

Task 2

In each of the two routers BR1 and BR2, there is a pre-configured access list "RFC1918" that has already been applied to their interfaces e0/1 connecting to the ISP router as follows:

```
Preconfigured in BR1
ip access-list standard BLK1918
deny 10.0.0.0 0.0.255.255
deny 172.16.0.0 0.0.255.255
deny 192.168.0.0 0.0.255.255
```

```
Preconfigured in BR2
ip access-list standard BLK1918
deny 10.0.0.0 0.0.255.255
deny 172.16.0.0 0.0.255.255
deny 192.168.0.0 0.0.255.255
permit any
```

Although blocking is being performed in both routers, there are some problems:

- Instead of all IP ranges specified in RFC1918, the range "10.1.0.0 – 10.255.255.255" and the range "172.17.0.0 – 172.31.255.255" are not blocked by the access lists in both routers.
- The access list in BR1 blocks all other non-RFC1918 traffic from Internet due to the implicit deny. (BR2 has no such problem since there is a "permit any" at the end of the access list).
- Since the ISP router uses "10.0.0.1" as the source IP for BGP packets sending to BR1 and uses "10.1.0.1" as the source IP for BGP packets sending to BR2 (you can check those IP addresses of the ISP router inside the ISP router or the neighbor command in the BGP configuration of BR1 and BR2), you need to allow the concerned address of the ISP router in the access lists in order for ISP to form neighbor and exchange routes with BR1 and BR2.

Assuming the result of "sh access-list" in the two routers are as follows:

```
Output in BR1
BR1#sh access-list
Standard IP access list BLK1918
10 deny 10.0.0.0, wildcard bits 0.0.255.255
20 deny 172.16.0.0, wildcard bits 0.0.255.255
30 deny 192.168.0.0, wildcard bits 0.0.255.255
BR1#
```

```
Output in BR2
BR2#sh access-list
Standard IP access list BLK1918
10 deny 10.0.0.0, wildcard bits 0.0.255.255
20 deny 172.16.0.0, wildcard bits 0.0.255.255
30 deny 192.168.0.0, wildcard bits 0.0.255.255
40 permit any
BR2#
```

The following configuration settings are required:

- A seq 5 rule is added so that the router ISP can communicate with BR1 / BR2.
- The seq 10 rule is changed to include the whole range 10.0.0.0/8 defined in RFC1918.
- The seq 20 rule is changed to include the whole range 172.16.0.0/12 defined in RFC1918.
- For BR1, a permit rule is appended at the end to allow Internet traffic.

Configuration required in BR1

```
ip access-list standard BLK1918
 5 permit host 10.0.0.1
 no 10
 10 deny 10.0.0.0 0.255.255.255
 no 20
 20 deny 172.16.0.0 0.15.255.255
 permit any
```

Configuration required in BR2

```
ip access-list standard BLK1918
 5 permit host 10.1.0.1
 no 10
 10 deny 10.0.0.0 0.255.255.255
 no 20
 20 deny 172.16.0.0 0.15.255.255
```

Remarks :

1. The following shows the wildcard mask required for 172.16.0.0 – 172.31.255.255:

```
Address: 172.16.0.0      10101100.0001 0000.00000000.00000000
Netmask: 255.240.0.0 = /12 11111111.1111 0000.00000000.00000000
Wildcard: 0.15.255.255    00000000.0000 1111.11111111.11111111
=>
Network: 172.16.0.0/12    10101100.0001 0000.00000000.00000000 (Class B)
Broadcast: 172.31.255.255 10101100.0001 1111.11111111.11111111
```

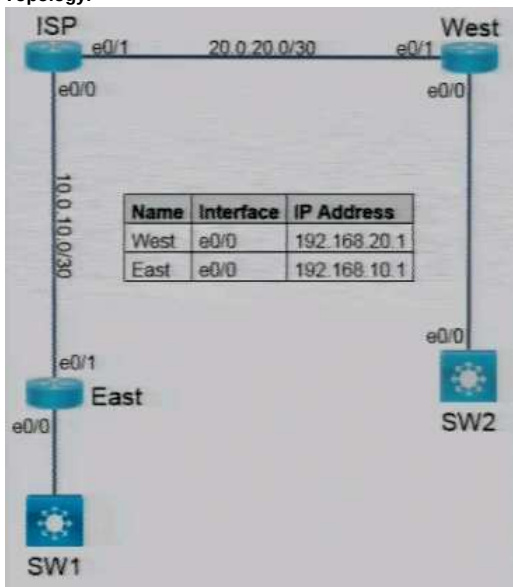
2. A better way to allow BGP is to use extended access-list. However, since the task asks you to configure the settings using the existing access list "RFC1918" which is a standard access list, therefore the suggested answer just allow any traffic from the IP address of the ISP router. Hence, you can telnet from SP1 to BR1 for verifying the configuration in task 1.

3. Since the interface e0/1 of both BR1 and BR2 are using RFC1918, the suggested answer does not allow communication (including telnet) between the two branches i.e. between BR1 / SW1 and BR2 / SW2). Hence, you can NOT telnet from BR2 to BR1 for verifying the configuration in task 1 (do not waste time trying to telnet BR1 from BR2 !!). No rule is added to allow communication between BR1 and BR2 since the task does NOT mention that this is required.

4. After configuration, although ISP can form BGP neighbor BR1 as well as BR2. BR1 and BR2 cannot learn BGP routes of each other through the BGP router ISP since both BR1 and BR2 are of the same AS number 100. Therefore, they will reject each other's routes since the AS path of the routes contain AS number 100. The task in the question do NOT mention that you need to solve it.

QUESTION 20

Topology:



Tasks:

Troubleshoot and resolve the issues on West and East routers to achieve these goals:

- ISP should be able to telnet SW2 using preconfigured authentication list and credentials configured locally on SW2. Any modification made to existing ACL must also include logging for the rule.
- ISP router should only allow traffic sourced from major network 192.168.0.0/16 inbound on its WAN interfaces. Use the preconfigured ACL on ISP router to meet this requirement and enable logging when new rules are added or existing are modified. Ensure SW2 can ping East Router's e0/0.

You must remove wrong preconfigs that have impact on tasks you are performing to fix issues.

Enable password is 'Cisco' on all devices.

SW2: Local username is "SW2" and password is "Cisco"

Correct Answer:

Section: (none)

Explanation

Explanation/Reference:

Ans:

SW2

```
en
config t
line vty 0 4
 transport input ssh telnet
 login authentication TELNET
```

```
end
copy run start
```

ISP

```
en
config t
ip access-list extended ACL
  no 10
  10 permit tcp host 192.168.20.2 eq telnet host 20.0.20.1 log
  no 20
  20 permit ip 192.168.0.0 0.0.255.255 any log
end
copy run start
```

For verification

In ISP, telnet 192.168.20.2 (username: SW2, password: Cisco)
In SW2, ping 192.168.10.1

Explanations:

You should use `sh run` in the routers to check if the existing configuration settings are the same as those described below:

ISP

```
interface Ethernet0/0
  ip address 10.0.10.1 255.255.255.252
  ip access-group ACL in
!
interface Ethernet0/0
  ip address 20.0.20.1 255.255.255.252
  ip access-group ACL in
!
ip route 192.168.10.0 255.255.255.0 10.0.10.2
ip route 192.168.20.0 255.255.255.0 20.0.20.2
!
ip access-list extended ACL
  permit tcp host 192.168.20.2 eq telnet host 20.0.20.1 eq telnet
  permit ip 192.168.0.0 0.0.255.255 255.255.255.0 0.0.0.255
  deny ip any any log
```

East

```
interface Ethernet0/0
  ip address 192.168.10.1 255.255.255.0
!
interface Ethernet0/1
  ip address 10.0.10.2 255.255.255.0
!
ip route 0.0.0.0 0.0.0.0 10.0.10.1
```

West

```
interface Ethernet0/0
  ip address 192.168.20.1 255.255.255.0
!
interface Ethernet0/1
  ip address 20.0.20.2 255.255.255.0
!
ip route 0.0.0.0 0.0.0.0 20.0.20.1
```

SW2

```
enable secret Cisco (password Cisco is encrypted)
username SW2 secret Cisco (password Cisco is encrypted)
!
aaa new-model
!
aaa authentication login TELNET local
!
interface Ethernet0/0
  ip address 192.168.20.2 255.255.255.0
!
ip route 0.0.0.0 0.0.0.0 192.168.20.1
!
line vty 0 4
  exec-timeout 0 0
  transport input ssh
```

Task 1

For task 1, several settings prevent the making of telnet connection from ISP to SW2 with the use of the preconfigured authentication list:

- Under line vty 0 4 in SW2, "telnet" is not included in "transport input".
- Under line vty 0 4 in SW2, the preconfigured authentication list "TELNET" is not configured for "login authentication".
- The inbound access list ACL in ISP does not allow returned telnet traffic from SW2 to ISP.

In order to allow telnet from ISP to SW2:

- The following configuration settings are required in SW2

Configuration settings required in SW2

```
line vty 0 4
  transport input ssh telnet
  login authentication TELNET
```

- Assume the access list ACL preconfigured required in ISP is as follows:

ISP#sh access-list

```
Extended IP access list ACL
  10 permit tcp host 192.168.20.2 eq telnet host 20.0.20.1 eq telnet
  20 permit ip 192.168.0.0 0.0.255.255 255.255.255.0 0.0.0.255
```



```
30 deny ip any any log
ISP#
```

The following change is required in rule 10 of the access list. Note that since logging is required for the new modified rule, the keyword "log" is included.

Configuration settings required in ISP

```
config t
ip access-list extended ACL
no 10
10 permit tcp host 192.168.20.2 eq telnet host 20.0.20.1 log
end
```

Remarks :

Since the router ISP acting as client will use a random port for making telnet connection, no destination port number should be specified in the rule for the returned telnet traffic.

After entering the above configuration settings, you should be able to telnet from ISP to SW2 using "telnet 192.168.20.2" with the username "SW2" and the password "Cisco" in ISP.

Task 2

For the router ISP, both interfaces "e0/0" and "e0/1" are its WAN interfaces. Although there seems to be a rule 20 to allow traffic from "192.168.0.0/16" only, the destination specified in the rule is strange since it is "255.255.255.0 0.0.0.255". This prevents SW2 from pinging East router's e0/0 interface.

```
20 permit ip 192.168.0.0 0.0.255.255 255.255.255.0 0.0.0.255
```

In order to allow normal traffic from "192.168.0.0/16", you need to change the destination of this rule to "any". Moreover, since logging is required for the new modified rule, the keyword "log" is needed.

Configuration settings required in ISP

```
config t
ip access-list extended ACL
no 20
20 permit ip 192.168.0.0 0.0.255.255 any log
End
```

After the change since the IP addresses of both SW2 and the e0/0 of East is within 192.168.0.0/16, the echo request and echo reply packets are allowed by the modified rule and can be routed through the ISP router.

After entering the above configuration settings, you should be able to ping from SW2 to East's e0/0 using "ping 192.168.10.1" in SW2.